Anti-Torque Force Gradient - Inspection

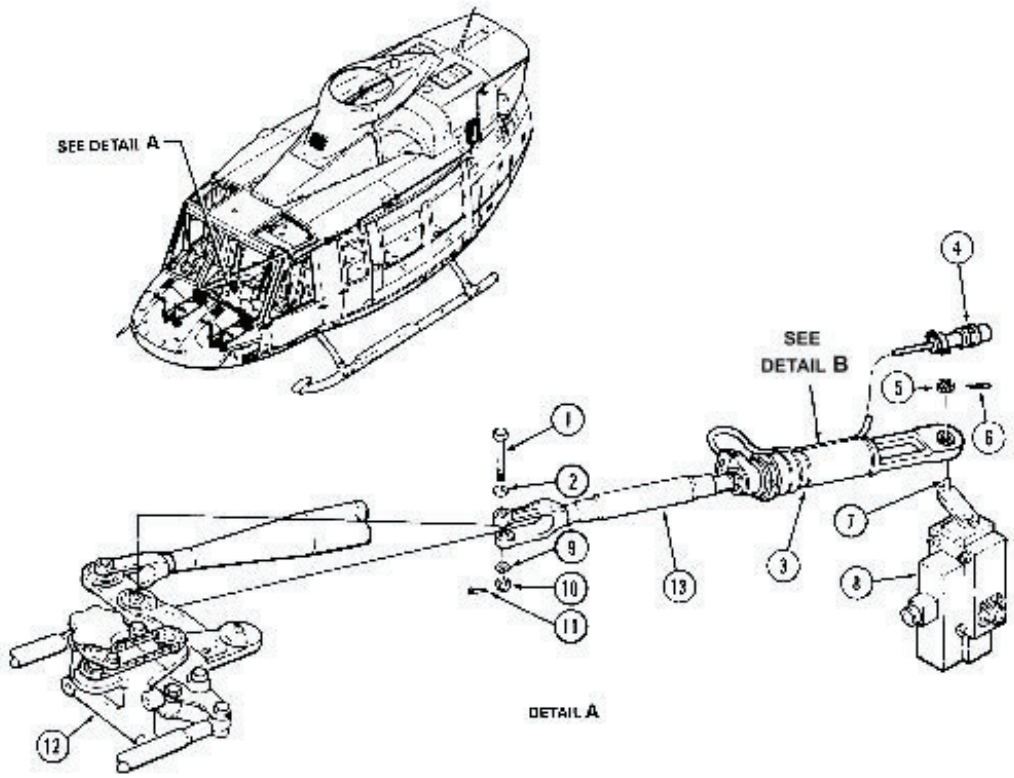
Procedure

NOTE

The following inspection procedures are applicable to anti-torque force gradients (incorporating a detent switch) used when the optional 4-axis AFCS/Flight Director kit is installed. For anti-torque force gradients not incorporating a detent switch, refer to 412-A-67-20-00-10A-280A-A / 01529.

1. Inspect force gradient (3, Figure 1) for corrosion and mechanical damage (BHT-412-CR&O). Superficial mechanical damage is acceptable without repair. Mechanical damage in excess of superficial and/or corrosion damage is not acceptable.

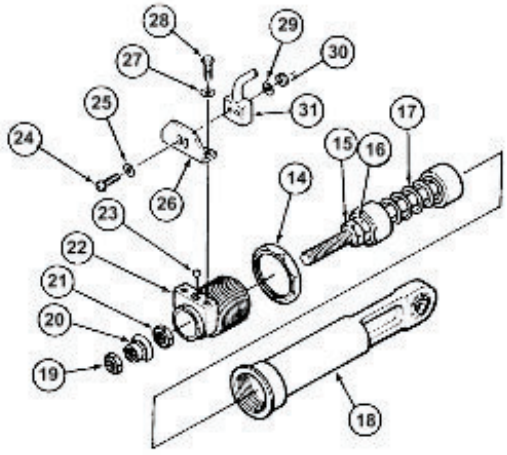
2. Inspect bearing for secure installation in housing and for wear (BHT-ALL-SPM, Standard Practices Manual).



1. Bolt
2. Aluminum washer
3. Force gradient
4. Electrical connector
5. Nut
6. Cotter pin
7. Stud
8. Rotary actuator
9. Aluminum washer
10. Nut
11. Cotter pin
12. Pedal adjuster
13. Clevis

ICN-412-A-671202-B-97499-00003-A-001-01

Figure 1. Anti-Torque Force Gradient (4-Axis Configuration) (Sheet 1 of 2)



DETAIL B

14. Jam nut
15. Outboard nut
16. Inboard nut
17. Spring assembly
18. Cylinder
19. Nut
20. Actuator
21. Nut
22. Cap
23. Ball
24. Screw
25. Washer
26. Retainer
27. Washer
28. Screw
29. Washer
30. Nut
31. Cable assembly (switch)

ICN-412-A-671202-B-97499-00004-A-001-01

Figure 1. Anti-Torque Force Gradient (4-Axis Configuration) (Sheet 2 of 2)

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Anti-Torque Force Gradient - Repair

Procedure

NOTE

The following repair procedure is applicable to anti-torque force gradients (incorporating a detent switch) used when the optional 4-axis AFCS/Flight Director kit is installed. For anti-torque force gradients not incorporating a detent switch, refer to 412-A-67-20-00-10A-685A-A / 01530.

1. Repair of force gradient consists of replacement of unserviceable components and adjustment of spring preload in accordance with 412-A-67-12-01-04A-720A-A / 01566.

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Anti-Torque Force Gradient - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Lockwire	C-405	AR	

Procedure



ENSURE THAT FORCE GRADIENT LENGTH IS ADJUSTED TO RECORDED LENGTH PRIOR TO INSTALLATION ON MAGNETIC BRAKE.

NOTE

The following installation procedures are applicable to anti-torque force gradients (incorporating a detent switch) used when the optional 4-axis AFCS/Flight Director kit is installed. For anti-torque force gradients not incorporating a detent switch, refer to 412-A-67-20-00-10A-720A-A / 01531.

NOTE

Force gradient is similar in appearance to the force gradients used in the cyclic controls but has a different part number. Ensure that correct part number force gradient is installed for the anti-torque control system. Refer to BHT-412 IPB.

1. If not previously accomplished, adjust spring preload of force gradient (3, Figure 1) as follows:

1.1. Cut and remove lockwire, loosen jam nut (14), and remove cap (22).

1.2. Remove spring assembly (17) from cylinder (18).

1.3. Adjust inboard nut (16) to provide 18 to 20 pounds (80.06 to 88.96 N) preload against spring assembly (17). Tighten outboard nut (15) against inboard nut (16).

1.4. Position spring assembly (17) in cylinder (18) and install cap (22) with jam nut (14) until all end play is eliminated. Tighten jam nut (14) and secure with lockwire (C-405) to cylinder (18).

2. If not previously accomplished, adjust actuator (20) on force gradient as follows:

2.1. Loosen nuts (19 and 21) on shaft of spring assembly (17).

2.2. Make sure retainer (26) is secured by screws (28) and washers (27).

2.3. Align actuator (20) on shaft of spring assembly (17) with ball (23) and plunger of cable assembly (switch) (31).

2.4. Loosen screws (24), washers (25 and 29), and nut (30) cable assembly (switch) (31) to retainer (26).

2.5. Push and hold actuator (20) down (180°) against cap (22) at point opposite location of cable assembly (31). Hold this position to initially adjust switch (31), then adjust switch to actuate within ±0.050 inch (1.27 mm) linear travel of spring assembly (17). Secure switch (31) to retainer (26) with screws (24), washers (25 and 29), and nuts (30). Secure actuator (20) with nut (21).

3. Install clevis (13) on shaft of spring assembly (17), adjust to previously recorded length, and secure with nut (19).

4. After assembly of force gradient, prior to installation, verify spring assembly (17) preload of 18 to 20 pounds (80.06 to 88.96 N).

5. Position force gradient (3) on pedal adjuster (12).

6. Install bolt (1), aluminum washers (2 and 9), nut (10), and cotter pin (11).

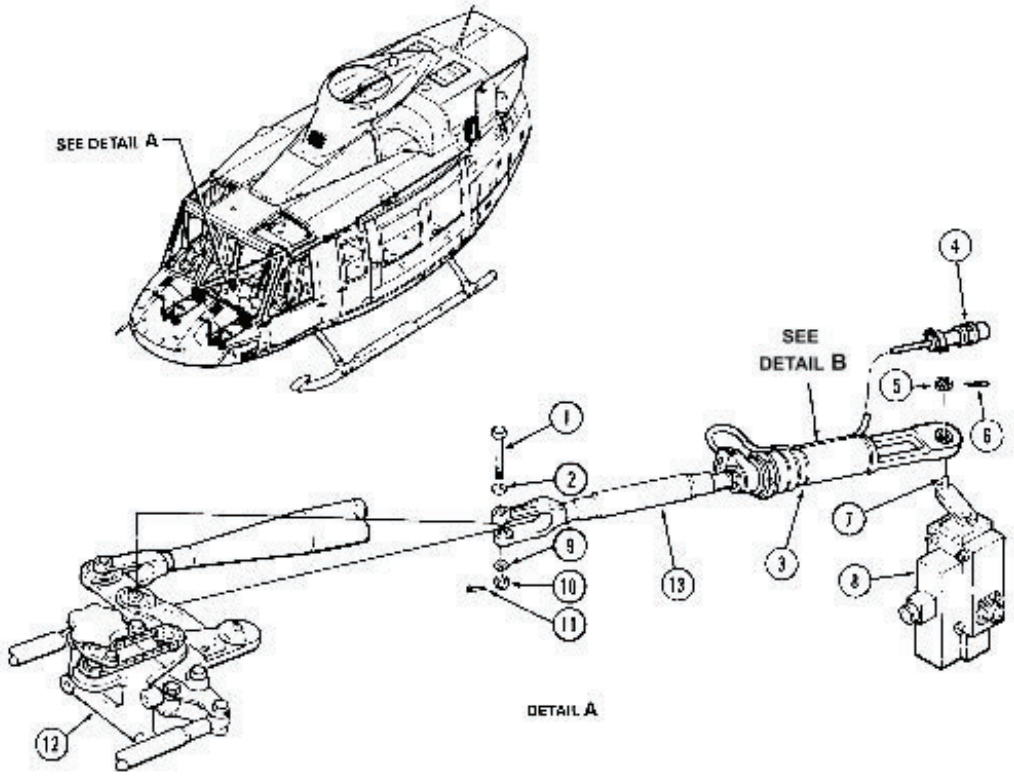
7. Position force gradient (3) on stud (7) of magnetic brake (8).

8. Install nut (5) and cotter pin (6).

9. Move anti-torque controls through full travel and check for obstruction and/or binding.

10. Confirm that anti-torque control rigging is within limits. Refer to 412-A-67-01-00-11A-276A-A / 01551.

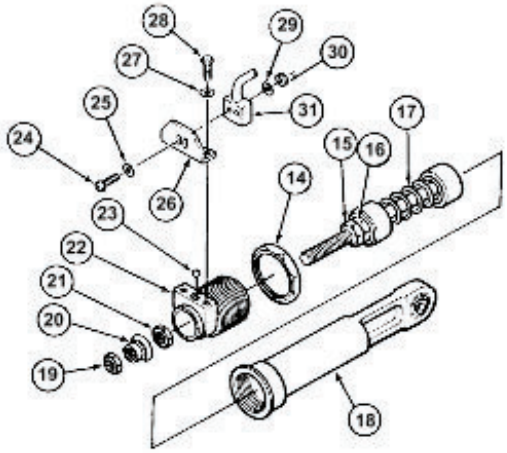
11. Install access panel on cabin floor under pilot seat (412-A-52-40-00-02A-720A-A / 00840).



1. Bolt
2. Aluminum washer
3. Force gradient
4. Electrical connector
5. Nut
6. Cotter pin
7. Stud
8. Rotary actuator
9. Aluminum washer
10. Nut
11. Cotter pin
12. Pedal adjuster
13. Clevis

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Figure 1. Anti-Torque Force Gradient (4-Axis Configuration) (Sheet 1 of 2)



DETAIL B

14. Jam nut
15. Outboard nut
16. Inboard nut
17. Spring assembly
18. Cylinder
19. Nut
20. Actuator
21. Nut
22. Cap
23. Ball
24. Screw
25. Washer
26. Retainer
27. Washer
28. Screw
29. Washer
30. Nut
31. Cable assembly (switch)

ICN-412-A-671202-B-97499-00004-A-001-01

Figure 1. Anti-Torque Force Gradient (4-Axis Configuration) (Sheet 2 of 2)

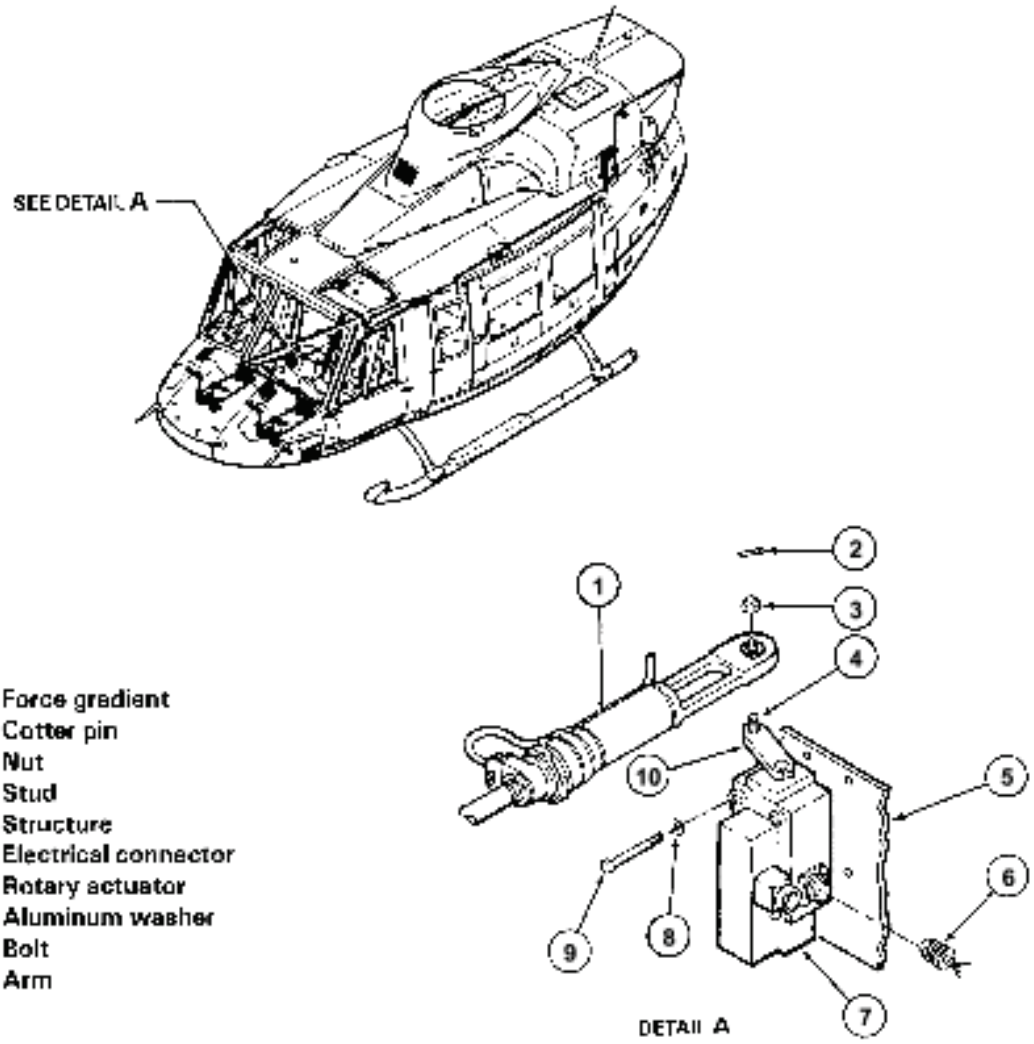
Anti-Torque Rotary Actuator - Removal

Procedure

NOTE

The following removal procedures are applicable to anti-torque rotary actuators used when the optional 4-axis AFCS/Flight Director kit is installed. For magnetic brakes used in the anti-torque system, refer to 412-A-67-20-00-08A-520A-A / 01520.

1. Remove access panel from cabin floor under pilot seat (412-A-52-40-00-02A-520A-A / 00837).
2. Disconnect electrical connector (6, Figure 1).
3. Remove cotter pin (2) and nut (3) from stud (4).
4. Lift force gradient (1) off arm (10).
5. Remove bolts (9) and aluminum washers (8).
6. Remove rotary actuator (7) from structure (5).



ICN-412-A-671202-B-97499-00005-A-001-01

Figure 1. Anti-Torque Rotary Actuator (4-Axis Configuration) (Sheet 1 of 1)

Anti-Torque Rotary Actuator - Inspection

Procedure

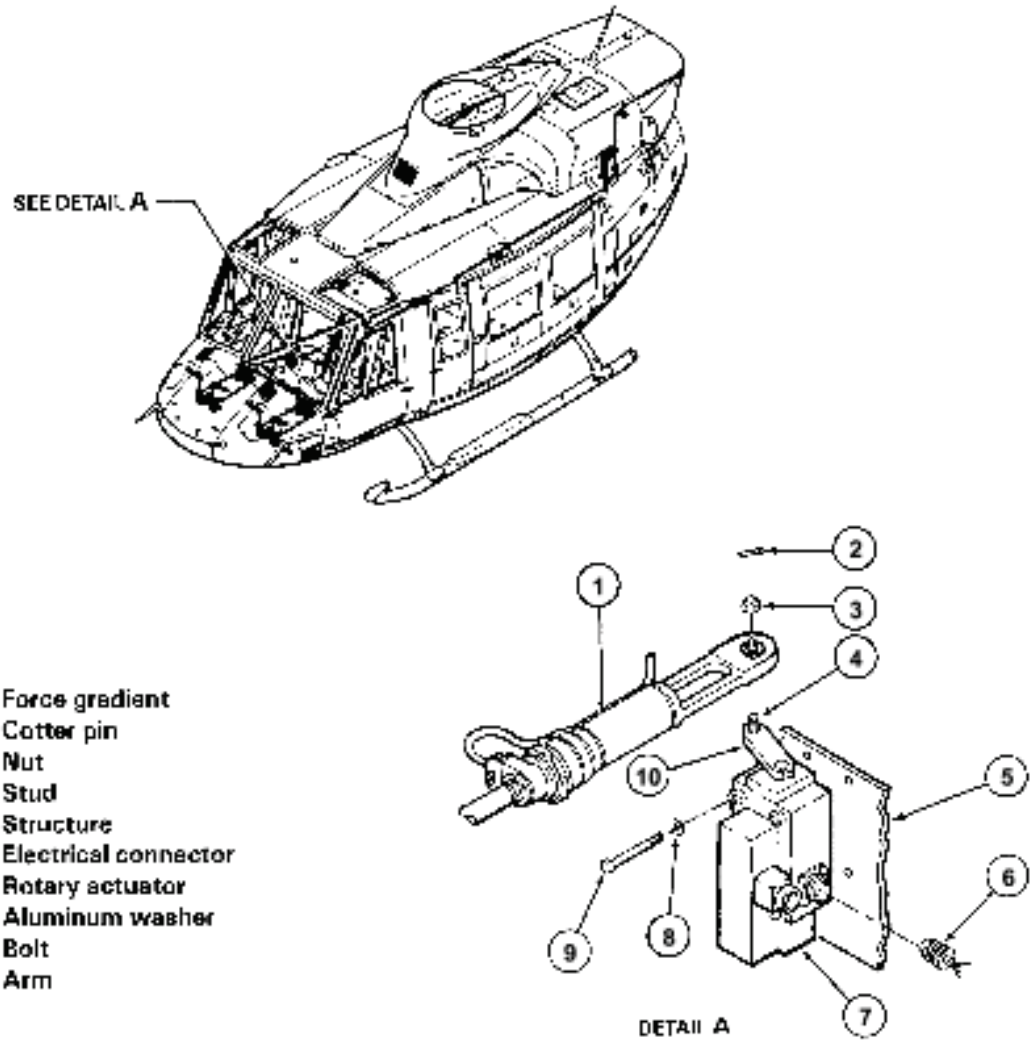
NOTE

The following inspection procedures are applicable to anti-torque rotary actuators used when the optional 4-axis AFCS/Flight Director kit is installed. For magnetic brakes used in the anti-torque system, refer to 412-A-67-20-00-08A-280A-A / 01521.

1. Inspect rotary actuator (7, Figure 1) for corrosion and mechanical damage. Superficial mechanical damage is acceptable without repair. Mechanical damage in excess of superficial damage and/or corrosion damage is not acceptable.

2. Inspect stud (4) for damaged threads. No damage that can be detected visually is acceptable.

3. Inspect electrical receptacle for bent pins and threads. Superficial damage is acceptable without repair.



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Figure 1. Anti-Torque Rotary Actuator (4-Axis Configuration) (Sheet 1 of 1)

Anti-Torque Rotary Actuator - Repair

Procedure

NOTE

The following repair procedure is applicable to anti-torque rotary actuators used when the optional 4-axis AFCS/Flight Director kit is installed. For magnetic brakes used in the anti-torque system, refer to 412-A-67-20-00-08A-685A-A / 01522.

1. Refer to vendor manual for magnetic brake repair instructions.

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Anti-Torque Rotary Actuator - Installation

Procedure

NOTE

The following installation procedures are applicable to anti-torque rotary actuators used when the optional 4-axis AFCS/Flight Director kit is installed. For magnetic brakes used in the anti-torque system, refer to 412-A-67-20-00-08A-720A-A / 01523.

NOTE

Ensure arm (10, Figure 1) is properly installed on rotary actuator (7) as outlined in 412-A-67-01-00-11A-276A-A / 01551 rigging instructions.

1. Position rotary actuator (7) on structure (5). Install four bolts (9) and four aluminum washers (8) as follows:

1.1. Install aluminum washers (8) with each bolt (9).

1.2. Install four bolts (9) of correct length to obtain proper thread engagement. Three different length bolts are required.

1.3. Install bolts (9) with drilled heads in two inboard holes in rotary actuator (7) and structure (5).

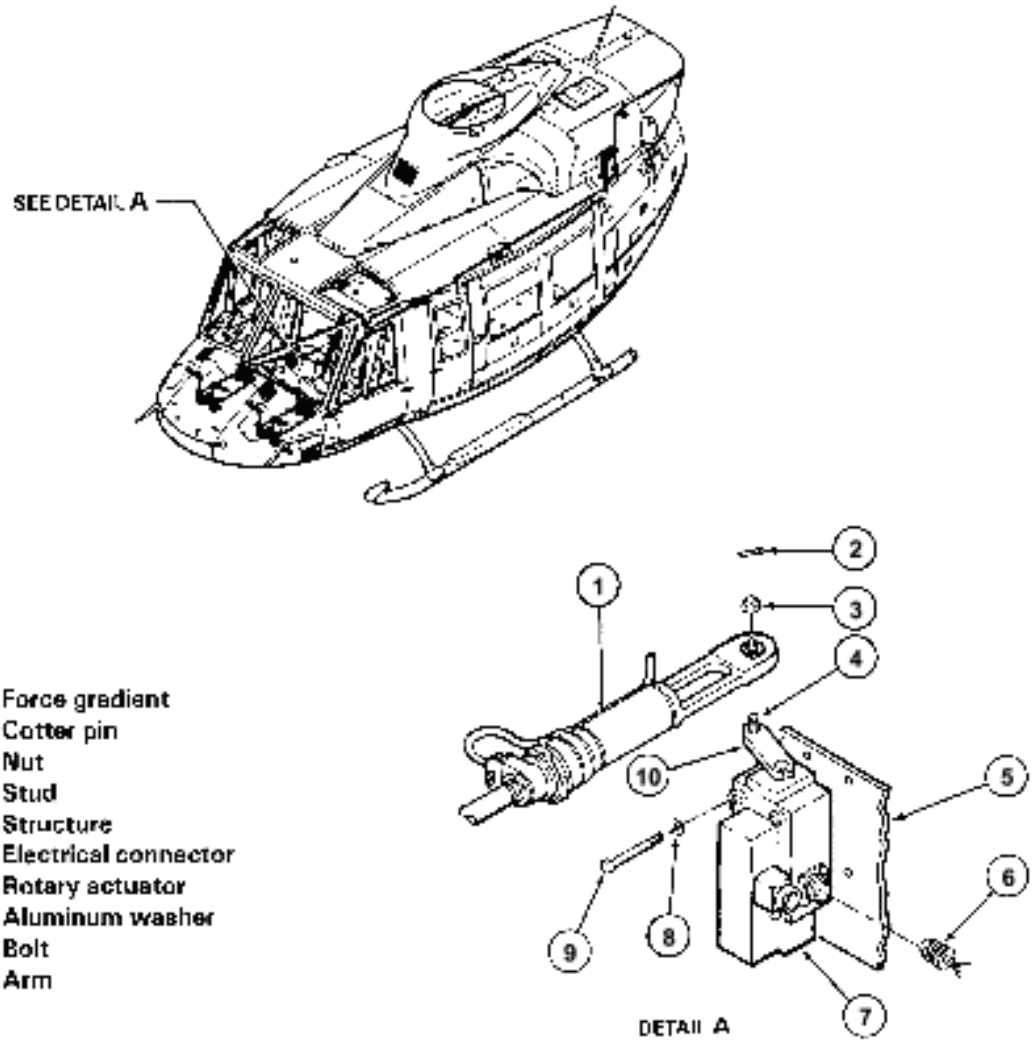
2. Position force gradient (1) on stud (4) of arm (10). Install nut (3) and cotter pin (2).

3. Install electrical connector (6).

4. Move anti-torque controls through full travel and check for binding and/or obstructions.

5. Confirm that anti-torque control rigging is within limits in 412-A-67-01-00-11A-276A-A / 01551.

6. Install access panel on cabin floor under pilot seat (412-A-52-40-00-02A-720A-A / 00840).



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Figure 1. Anti-Torque Rotary Actuator (4-Axis Configuration) (Sheet 1 of 1)

Anti-Torque Series Actuators - Removal

Procedure

1. Remove large round access panel from bottom of aft fuselage. Open access door on right lower side of aft fuselage (412-A-52-40-00-02A-520A-A / 00837).

2. Remove tailboom assembly (412-A-53-50-00-00A-520A-A / 00912).

3. Disconnect electrical connectors (8, Figure 1).

4. Remove cotter pin (1), nut (2), washer (3), bolt (6), and washer (5).

5. Remove cotter pin (14), nut (13), washers (10 and 12), and bolt (9).

NOTE

The removal of clamps that secure the electrical cable assemblies will allow sufficient deflection of cables for actuator removal.

6. Remove cotter pin (15), nut (16), washer (17), and bolt (18).

7. Remove cotter pin (19), nut (20), washer (21), spacer (22), and bolt (23). Remove bellcrank (24).

8. Disconnect clamps securing electrical cable assemblies, located aft of servo actuator.

9. Pull servo actuator aft as far as possible, then change angle and reposition actuator as necessary to allow for removal through round access panel at bottom of aft fuselage. Remove servo actuator (7).

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Figure 1. Anti-Torque Servo Actuator Removal/Installation (36087 and Subsequent) (Sheet 1 of 1)

Anti-Torque Series Actuators - Inspection

Procedure

1. Inspect servo actuators (7, Figure 1) in accordance with procedures in BHT-412-CR&O.
2. Inspect bellcrank sleeve (25) and bearing (26) for wear, damage, or corrosion.

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Figure 1. Anti-Torque Servo Actuator Removal/Installation (36087 and Subsequent) (Sheet 1 of 1)

Anti-Torque Series Actuators - Installation

Procedure



DISASSEMBLY OF SERVO ACTUATOR (7, Figure 1) IS NOT PERMITTED.

1. Establish overall dimensions of servo actuators (7, Figure 1) as shown in Figure 2.
2. Place servo actuator (7, Figure 1) into position with rod end bearing in bellcrank (4).
3. Install bolt (6), washer (5), washer (3), nut (2), and cotter pin (1).
4. Place forward end of servo actuator (7) on anti-torque control lever (11).
5. Install bolt (9), washer (10), washer (12), nut (13), and cotter pin (14).
6. Re-install clamps to secure cable assemblies.
7. Install bellcrank (24) and attach with bolt (23), spacer (22), washer (21), nut (20), and cotter pin (19).
8. Attach bolt (18), washer (17), nut (16), and cotter pin (15).
9. Move anti-torque controls through full travel and check for obstructions and/or binding.
10. Connect electrical connectors (8).
11. Install tailboom assembly (412-A-53-50-00-00A-720A-A / 00915).
12. Confirm that anti-torque rigging is within limits (412-A-67-02-00-02A-280A-A / 01545).
13. Install access panel on bottom aft fuselage (412-A-52-40-00-02A-720A-A / 00840).
14. Close access door on right side of aft fuselage (412-A-52-40-00-02A-720A-A / 00840).

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Figure 1. Anti-Torque Servo Actuator Removal/Installation (36087 and Subsequent) (Sheet 1 of 1)

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Figure 2. Anti-Torque Series Actuator (36087 and Subsequent) (Sheet 1 of 1)

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CHAPTER 71
POWER PLANT

MAINTENANCE MANUAL

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List of Effective Data Modules

The listed documents are included in issue 007, dated 2025-11-18, of this publication.

C = Changed data module, N = New data module

Document title	Data module code / Seq number		Issue date	Applicable to
Power Plant - Description	DMC-412-A-71-00-00-00A-028A-A / 01585		2023-05-31	ALL
Power Plant Maintenance - Description	DMC-412-A-71-00-00-01A-042A-A / 01586		2023-05-31	ALL
Power Plant Lubrication System - Priming	DMC-412-A-71-00-00-01A-232A-A / 01587		2023-05-31	ALL
Reduction (Combining) Gearbox - Pre-oil	DMC-412-A-71-00-00-01A-212A-A / 01588		2023-05-31	ALL
Power Plant Fuel System - Priming	DMC-412-A-71-00-00-01A-232B-A / 01589		2023-05-31	ALL
Power Plant - Wet Motoring Run	DMC-412-A-71-00-00-01A-913A-A / 01590		2023-05-31	ALL
Engine to Transmission - Alignment	DMC-412-A-71-00-00-01A-272A-A / 01591	C	2025-11-18	SN: 33001-Subsequent or 36001-36019 or 412-570-001-103: Not Installed or (BHT-412-SI-74: Not Installed BHT-412-SI-97: Not Installed)
Engine to Transmission (412Classic/SP to 412HP Upgrade) - Alignment	DMC-412-A-71-00-00-01A-272B-A / 01592	C	2025-11-18	SN: 36020-Subsequent or 412-570-001-103: Installed or BHT-412-SI-74: Installed or BHT-412-SI-97: Installed
Compressor - Wash	DMC-412-A-71-00-00-02A-042A-A / 01593		2023-05-31	ALL
Power Plant - Troubleshooting	DMC-412-A-71-00-00-02A-422A-A / 01594		2023-05-31	ALL
Power Plant - Operational Check	DMC-412-A-71-00-00-02A-320A-A / 01595		2023-05-31	ALL
Engine Assembly - Description	DMC-412-A-71-00-01-00A-028A-A / 01596		2023-05-31	ALL
Engine - Removal	DMC-412-A-71-00-01-01A-520A-A / 01597		2023-05-31	ALL
Engine - Installation	DMC-412-A-71-00-01-01A-720A-A / 01598		2024-12-19	ALL
Engine Power Section - Removal	DMC-412-A-71-00-01-02A-520A-A / 01599		2023-05-31	ALL

Document title	Data module code / Seq number	Issue date	Applicable to
Engine Power Section - Installation	DMC-412-A-71-00-01-02A-720A-A / 01600	2023-05-31	ALL
Reduction (Combining) Gearbox - Description	DMC-412-A-71-00-01-03A-042A-A / 01601	2023-05-31	ALL
Reduction (Combining) Gearbox - Removal	DMC-412-A-71-00-01-03A-520A-A / 01602	2023-05-31	ALL
Reduction (Combining) Gearbox - Installation	DMC-412-A-71-00-01-03A-720A-A / 01603	2024-12-19	ALL
Cowlings - Description	DMC-412-A-71-10-00-00A-028A-A / 01604	2023-05-31	ALL
Forward and Aft Pylon Fairings - Removal	DMC-412-A-71-10-00-01A-520A-A / 01605	2023-05-31	ALL
Forward and Aft Pylon Fairings - Inspection and Repair	DMC-412-A-71-10-00-01A-280A-A / 01606	2023-05-31	ALL
Forward and Aft Pylon Fairings - Installation	DMC-412-A-71-10-00-01A-720A-A / 01607	2023-05-31	ALL
Transmission Fairings and Engine Air Inlet Cowlings - Removal	DMC-412-A-71-10-00-02A-520A-A / 01608	2023-05-31	ALL
Transmission Fairings and Engine Air Inlet Cowlings - Inspection and Repair	DMC-412-A-71-10-00-02A-280A-A / 01609	2023-05-31	ALL
Engine Air Inlet Cowlings - Gap Inspection	DMC-412-A-71-10-00-02A-280B-A / 01610	2023-05-31	ALL
Transmission Fairing and Engine Air Inlet Cowlings - Installation	DMC-412-A-71-10-00-02A-720A-A / 01611	2023-05-31	ALL
Engine Air Inlet Cowlings - Adjustable Seal Replacement	DMC-412-A-71-10-00-02A-921A-A / 01612	2023-05-31	SN: 36521-Subsequent or BHT-412-SI-87: Installed
Engine Air Inlet Cowlings - Adjustable Seal Adjustment	DMC-412-A-71-10-00-02A-271A-A / 01613	2023-05-31	SN: 36521-Subsequent or BHT-412-SI-87: Installed
Engine Air Inlet Cowlings - Adjustable Seals - Repair	DMC-412-A-71-10-00-02A-685A-A / 01614	2023-05-31	SN: 36521-Subsequent or BHT-412-SI-87: Installed
Engine Cowlings - Removal	DMC-412-A-71-10-00-03A-520A-A / 01615	2023-05-31	ALL
Engine Cowlings - Installation	DMC-412-A-71-10-00-03A-720A-A / 01616	2023-05-31	ALL
Reduction (Combing) Gearbox Cowlings - Removal	DMC-412-A-71-10-00-04A-520A-A / 01617	2023-05-31	ALL
Reduction (Combing) Gearbox Cowlings - Inspection and Repair	DMC-412-A-71-10-00-04A-280A-A / 01618	2023-05-31	ALL
Reduction (Combing) Gearbox Cowlings - Installation	DMC-412-A-71-10-00-04A-720A-A / 01619	2023-05-31	ALL
Oil Cooler Airscoop Cowling - Removal	DMC-412-A-71-10-00-05A-520A-A / 01620	2023-05-31	ALL
Oil Cooler Airscoop Cowling - Inspection and Repair	DMC-412-A-71-10-00-05A-280A-A / 01621	2023-05-31	ALL

Document title	Data module code / Seq number	Issue date	Applicable to
Oil Cooler Airscoop Cowling - Installation	DMC-412-A-71-10-00-05A-720A-A / 01622	2023-05-31	ALL
Engine Mount - Description	DMC-412-A-71-20-00-00A-028A-A / 01623	2023-05-31	ALL
Forward Mount - Removal	DMC-412-A-71-20-00-01A-520A-A / 01624	2023-05-31	ALL
Forward Mount - Inspection and Repair	DMC-412-A-71-20-00-01A-280A-A / 01625	2023-05-31	ALL
Forward Mount - Installation	DMC-412-A-71-20-00-01A-720A-A / 01626	2023-05-31	ALL
Forward Mount Fittings - Removal	DMC-412-A-71-20-00-02A-520A-A / 01627	2023-05-31	ALL
Forward Mount Fittings - Inspection and Repair	DMC-412-A-71-20-00-02A-280A-A / 01628	2023-05-31	ALL
Forward Mount Fittings - Installation	DMC-412-A-71-20-00-02A-720A-A / 01629	2023-05-31	ALL
Forward Mount Deck Fittings - Removal	DMC-412-A-71-20-00-03A-520A-A / 01630	2023-05-31	ALL
Forward Mount Deck Fittings - Inspection and Repair	DMC-412-A-71-20-00-03A-280A-A / 01631	2023-05-31	ALL
Forward Mount Deck Fittings - Installation	DMC-412-A-71-20-00-03A-720A-A / 01632	2023-05-31	ALL
Aft Left Mount Tripod - Removal	DMC-412-A-71-20-00-04A-520A-A / 01633	2023-05-31	ALL
Aft Left Mount Tripod - Inspection and Repair	DMC-412-A-71-20-00-04A-280A-A / 01634	2023-05-31	ALL
Aft Left Mount Tripod - Installation	DMC-412-A-71-20-00-04A-720A-A / 01635	2023-05-31	ALL
Aft Left Mount Tripod Deck Fittings - Removal	DMC-412-A-71-20-00-05A-520A-A / 01636	2023-05-31	ALL
Aft Left Mount Tripod Deck Fittings - Inspection and Repair	DMC-412-A-71-20-00-05A-280A-A / 01637	2023-05-31	ALL
Aft Left Mount Tripod Deck Fittings - Installation	DMC-412-A-71-20-00-05A-720A-A / 01638	2023-05-31	ALL
Aft Left and Right Mount Engine Bearings - Removal	DMC-412-A-71-20-00-06A-520A-A / 01639	2023-05-31	ALL
Aft Left and Right Mount Engine Bearings - Inspection and Repair	DMC-412-A-71-20-00-06A-280A-A / 01640	2023-05-31	ALL
Aft Left and Right Mount Engine Bearings - Installation	DMC-412-A-71-20-00-06A-720A-A / 01641	2023-05-31	ALL
Aft Right Mount Bipod - Removal	DMC-412-A-71-20-00-07A-520A-A / 01642	2023-05-31	ALL
Aft Right Mount Bipod - Inspection and Repair	DMC-412-A-71-20-00-07A-280A-A / 01643	2023-05-31	ALL
Aft Right Mount Bipod - Installation	DMC-412-A-71-20-00-07A-720A-A / 01644	2023-05-31	ALL
Aft Right Mount Bipod Deck Fittings - Removal	DMC-412-A-71-20-00-08A-520A-A / 01645	2023-05-31	ALL
Aft Right Mount Bipod Deck Fittings - Inspection and Repair	DMC-412-A-71-20-00-08A-280A-A / 01646	2023-05-31	ALL

Document title	Data module code / Seq number	Issue date	Applicable to
Aft Right Mount Bipod Deck Fittings - Installation	DMC-412-A-71-20-00-08A-720A-A / 01647	2024-05-09	ALL
Firewalls - Description	DMC-412-A-71-30-00-00A-028A-A / 01648	2023-05-31	ALL
Air Inlet Cowling Firewall Assembly - Description	DMC-412-A-71-30-00-01A-042A-A / 01649	2023-05-31	ALL
Air Inlet Cowling Assembly - Removal	DMC-412-A-71-30-00-01A-520A-A / 01650	2023-05-31	ALL
Air Inlet Cowling Assembly - Inspection	DMC-412-A-71-30-00-01A-280A-A / 01651	2023-05-31	ALL
Air Inlet Cowling Assembly - Installation	DMC-412-A-71-30-00-01A-720A-A / 01652	2023-05-31	ALL
Air Inlet Cowling Assembly - Repair	DMC-412-A-71-30-00-01A-685A-A / 01653	2023-05-31	ALL
Middle Firewall Assembly - Removal	DMC-412-A-71-30-00-02A-520A-A / 01654	2023-05-31	ALL
Middle Firewall Assembly - Inspection	DMC-412-A-71-30-00-02A-280A-A / 01655	2023-05-31	ALL
Middle Firewall Assembly - Installation	DMC-412-A-71-30-00-02A-720A-A / 01656	2023-05-31	ALL
Centerline Firewall Assembly - Removal	DMC-412-A-71-30-00-03A-520A-A / 01657	2023-05-31	ALL
Centerline Firewall Assembly - Inspection	DMC-412-A-71-30-00-03A-280A-A / 01658	2023-05-31	ALL
Centerline Firewall Assembly - Installation	DMC-412-A-71-30-00-03A-720A-A / 01659	2023-05-31	ALL
Aft Firewall Assembly - Removal	DMC-412-A-71-30-00-04A-520A-A / 01660	2023-05-31	ALL
Aft Firewall Assembly - Inspection	DMC-412-A-71-30-00-04A-280A-A / 01661	2023-05-31	ALL
Aft Firewall Assembly - Installation	DMC-412-A-71-30-00-04A-720A-A / 01662	2023-05-31	ALL
Gas Producer Tachometer Generator - Description	DMC-412-A-71-01-00-01A-042A-A / 01663	2023-05-31	ALL
Gas Producer Tachometer Generator - Removal	DMC-412-A-71-01-00-01A-520A-A / 01664	2023-05-31	ALL
Gas Producer Tachometer Generator - Installation	DMC-412-A-71-01-00-01A-720A-A / 01665	2023-05-31	ALL
Power Turbine Tachometer Generator - Description	DMC-412-A-71-01-00-02A-042A-A / 01666	2023-05-31	ALL
Power Turbine Tachometer Generator - Removal	DMC-412-A-71-01-00-02A-520A-A / 01667	2023-05-31	ALL
Power Turbine Tachometer Generator - Installation	DMC-412-A-71-01-00-02A-720A-A / 01668	2023-05-31	ALL
Differential Pressure Indicator - Description	DMC-412-A-71-01-00-03A-042A-A / 01669	2023-05-31	ALL
Differential Pressure Indicator - Removal	DMC-412-A-71-01-00-03A-520A-A / 01670	2023-05-31	ALL
Differential Pressure Indicator - Installation	DMC-412-A-71-01-00-03A-720A-A / 01671	2023-05-31	ALL

Document title	Data module code / Seq number	Issue date	Applicable to
Torque Pressure Transmitters - Description	DMC-412-A-71-01-00-04A-042A-A / 01672	2023-05-31	ALL
Torque Pressure Transmitters - Removal	DMC-412-A-71-01-00-04A-520A-A / 01673	2023-05-31	ALL
Torque Pressure Transmitters - Installation	DMC-412-A-71-01-00-04A-720A-A / 01674	2023-05-31	ALL
Power Section Oil Pressure Switch - Description	DMC-412-A-71-01-00-05A-042A-A / 01675	2023-05-31	ALL
Power Section Oil Pressure Switch - Removal	DMC-412-A-71-01-00-05A-520A-A / 01676	2023-05-31	ALL
Power Section Oil Pressure Switch - Installation	DMC-412-A-71-01-00-05A-720A-A / 01677	2023-05-31	ALL
Reduction (Combining) Gearbox Oil Pressure Switch - Description	DMC-412-A-71-01-00-06A-042A-A / 01678	2023-05-31	ALL
Reduction (Combining) Gearbox Oil Pressure Switch - Removal	DMC-412-A-71-01-00-06A-520A-A / 01679	2023-05-31	ALL
Reduction (Combining) Gearbox Oil Pressure Switch - Installation	DMC-412-A-71-01-00-06A-720A-A / 01680	2023-05-31	ALL
Reduction (Combining) Gearbox Oil Temperature Bulb - Description	DMC-412-A-71-01-00-07A-042A-A / 01681	2023-05-31	ALL
Reduction (Combining) Gearbox Oil Temperature Bulb - Removal	DMC-412-A-71-01-00-07A-520A-A / 01682	2023-05-31	ALL
Reduction (Combining) Gearbox Oil Temperature Bulb - Installation	DMC-412-A-71-01-00-07A-720A-A / 01683	2023-05-31	ALL
Reduction (Combining) Gearbox Oil Pressure Transmitter - Description	DMC-412-A-71-01-00-08A-042A-A / 01684	2023-05-31	ALL
Reduction (Combining) Gearbox Oil Pressure Transmitter - Removal	DMC-412-A-71-01-00-08A-520A-A / 01685	2023-05-31	ALL
Reduction (Combining) Gearbox Oil Pressure Transmitter - Installation	DMC-412-A-71-01-00-08A-720A-A / 01686	2023-05-31	ALL
Power Section Oil Pressure Transmitter - Description	DMC-412-A-71-01-00-09A-042A-A / 01687	2023-05-31	ALL
Power Section Oil Pressure Transmitter - Removal	DMC-412-A-71-01-00-09A-520A-A / 01688	2023-05-31	ALL
Power Section Oil Pressure Transmitter - Installation	DMC-412-A-71-01-00-09A-720A-A / 01689	2023-05-31	ALL
Power Section Oil Temperature Bulb - Description	DMC-412-A-71-01-00-10A-042A-A / 01690	2023-05-31	ALL
Power Section Oil Temperature Bulb - Removal	DMC-412-A-71-01-00-10A-520A-A / 01691	2023-05-31	ALL
Power Section Oil Temperature Bulb - Installation	DMC-412-A-71-01-00-10A-720A-A / 01692	2023-05-31	ALL
Starter-Generator - Description	DMC-412-A-71-01-00-11A-042A-A / 01693	2023-05-31	ALL
Starter-Generator - Removal	DMC-412-A-71-01-00-11A-520A-A / 01694	2023-05-31	ALL
Starter-Generator - Inspection	DMC-412-A-71-01-00-11A-280A-A / 01695	2023-05-31	ALL
Starter Generator - Repair	DMC-412-A-71-01-00-11A-685A-A / 01696	2023-05-31	ALL

Document title	Data module code / Seq number	Issue date	Applicable to
Starter Generator - Cleaning	DMC-412-A-71-01-00-11A-251A-A / 01697	2023-05-31	ALL
Starter Generator - Installation	DMC-412-A-71-01-00-11A-720A-A / 01698	2023-05-31	ALL
Reduction (Combining) Gearbox Oil Temperature Switch - Description	DMC-412-A-71-01-00-12A-042A-A / 01699	2023-05-31	ALL
Air Management System - Description	DMC-412-A-71-61-00-00A-028A-A / 01700	2023-05-31	ALL
Air Management System Ejectors - Removal	DMC-412-A-71-61-00-01A-520A-A / 01701	2025-05-13	ALL
Air Management System Ejectors - Inspection	DMC-412-A-71-61-00-01A-280A-A / 01702	2025-05-13	ALL
Air Management System Ejectors - Repair	DMC-412-A-71-61-00-01A-685A-A / 01703	2023-05-31	ALL
Air Management System Ejectors - Installation	DMC-412-A-71-61-00-01A-720A-A / 01704	2025-05-13	ALL
Air Management System Transition Ducts - Removal	DMC-412-A-71-61-00-02A-520A-A / 01705	2023-05-31	ALL
Air Management System Transition Ducts - Inspection	DMC-412-A-71-61-00-02A-280A-A / 01706	2023-05-31	ALL
Air Management System Transition Ducts - Repair	DMC-412-A-71-61-00-02A-685A-A / 01707	2023-05-31	ALL
Air Management System Transition Ducts - Installation	DMC-412-A-71-61-00-02A-720A-A / 01708	2023-05-31	ALL
Air Management System Valve Assembly - Removal	DMC-412-A-71-61-00-03A-520A-A / 01709	2023-05-31	ALL
Air Management System Valve Assembly - Inspection	DMC-412-A-71-61-00-03A-280A-A / 01710	2023-05-31	ALL
Air Management System Valve Assembly - Repair	DMC-412-A-71-61-00-03A-685A-A / 01711	2023-05-31	ALL
Air Management System Valve Assembly - Installation	DMC-412-A-71-61-00-03A-720A-A / 01712	2023-05-31	ALL
Air Management System Forward Ducts - Removal	DMC-412-A-71-61-00-04A-520A-A / 01713	2023-05-31	ALL
Air Management System Forward Ducts - Inspection	DMC-412-A-71-61-00-04A-280A-A / 01714	2023-05-31	ALL
Air Management System Forward Ducts - Installation	DMC-412-A-71-61-00-04A-720A-A / 01715	2023-05-31	ALL
Air Management System Induction Baffles - Removal	DMC-412-A-71-61-00-05A-520A-A / 01716	2023-05-31	ALL
Air Management System Induction Baffles - Installation	DMC-412-A-71-61-00-05A-720A-A / 01717	2023-05-31	ALL
Air Management System Exhaust Ducts - Removal	DMC-412-A-71-61-00-06A-520A-A / 01718	2023-05-31	ALL
Air Management System Exhaust Dusts - Inspection	DMC-412-A-71-61-00-06A-280A-A / 01719	2023-05-31	ALL
Air Management System Exhaust Ducts - Repair	DMC-412-A-71-61-00-06A-685A-A / 01720	2023-05-31	ALL
Air Management System Exhaust Ducts - Installation	DMC-412-A-71-61-00-06A-720A-A / 01721	2023-05-31	ALL

Power Plant - Description

The power plant is the overall power package. Major components in this chapter are the engine assembly (consisting of two power sections), mounts, firewalls, accessories, and air management system.

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Power Plant Maintenance - Description

The fuel system maintenance data is contained in 412-A-28-10-00-00A-042B-A / 00437.

The engine oil system maintenance data is contained in 412-A-79-00-00-00A-422A-A / 01767.

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Power Plant Lubrication System - Priming

Procedure

NOTE

The following procedure shall be accomplished when a power section or reduction (combining) gearbox lubrication system or oil cooler has been drained.

1. Open access doors.
2. Fill oil reservoir to specified level (412-A-12-00-00-00A-040A-A / 00126).
3. With qualified personnel at controls, perform dry motoring run (BHT-412-FM, Flight Manual).
4. After completion of motoring run, refill oil reservoir to specified level.
5. Close access doors.

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Reduction (Combining) Gearbox - Pre-oil

Procedure

1. Open access doors.
2. Fill gearbox oil reservoir (412-A-12-00-00-00A-040A-A / 00126).
3. Turn main driveshaft a minimum of 20 revolutions in direction of normal rotation by using main rotor turned by hand.
4. Check gearbox oil level. Visually check oil level to full mark on indicator. Service gearbox as necessary.
5. Close access doors.

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Power Plant Fuel System - Priming

Procedure

1. Disconnect fuel inlet tube from fuel flow divider and connect suitable drainage tubes to inlet tube open ends. Provide a suitable container to catch displaced fuel and/or preservative.

2. Perform a wet motoring run on each power section (412-A-71-00-00-01A-913A-A / 01590).

3. Verify fuel flows freely from each drainage line.

NOTE

If power section fuel systems have been preserved, ensure all preservative fluid is cleared. Repeat wet motoring run if necessary.

4. Disconnect drainage tubes.

5. Connect fuel inlet tube to flow divider. Torque 90 to 100 inch-pounds (10.17 to 11.30 Nm).

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Power Plant - Wet Motoring Run

Procedure



RESIDUAL VOLTAGE IN IGNITION EXCITER MAY BE DANGEROUSLY HIGH. ALWAYS DISCONNECT CABLE COUPLING NUTS AT IGNITION EXCITER END FIRST.



USE INSULATED TOOLS TO REMOVE CABLE COUPLING NUTS. DO NOT TOUCH EXCITER OUTPUT CONNECTORS OR CABLE COUPLING NUTS WITH BARE HANDS.

NOTE

The procedures that follow must be accomplished as a functional check of the fuel system components as well as to depreserve and preserve fuel system.

1. Disconnect the input source of power to the igniter exciter box on power section No. 1, power section No. 2, or both, whichever is necessary.
2. With approved personnel at controls, fully close throttles.
3. Set BATTERY BUS 1 and BATTERY BUS 2 switches ON and necessary power section GOV switch to AUTO.



DO NOT EXCEED STARTER LIMITATIONS (BHT-412-FM).

4. Move the START switch to ENG 1 or ENG 2, whichever is necessary, and after the gas generator (GAS PROD) RPM attains 12%, put throttle to idle detent.
5. While the starter is engaged, close the throttle, put the GOV switch to MANUAL, and open the throttle to idle detent.
6. Return the throttle to OFF and release START switch.
7. Allow the power section to coast down and put BATTERY BUS 1 and BATTERY BUS 2 switches to OFF.

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Engine to Transmission - Alignment

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Clevis	204-001-178-1	1	
Transmission Leveling Jacks	T101440	1	
Alignment Tool	T101579	1	

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Adhesive	C-317	AR	
Lockwire	C-447	AR	
Anti-seize thread compound	C-452	AR	

Procedure

NOTE

When the engine is replaced, an alignment check is not necessary provided engine mount dimensions and/or the shim stack-up is not changed.

1. Engine to transmission alignment is necessary when any of the following conditions exist:

- Main driveshaft misalignment is suspected for any reason
- Excessive wear of driveshaft crowned-tooth coupling
- Indications of excessive driveshaft coupling heating
- Hard landings
- Major repair or replacement of components is necessary in center fuselage or pylon support structure

2. Do an alignment check with the helicopter resting on skids on a level surface, and with engine and the pylon assembly installed in a correct manner and with the main rotor untied.

3. Remove the main driveshaft, leaving the adapter installed in the end of the engine output shaft (412-A-63-10-01-01A-520A-A / 00994).

4. Remove the cotter pins, nuts, and the washers from both lift link bolts (Figure 1). Discard the cotter pins.

5. Set the transmission leveling jacks (T101440) as follows:

5.1. With the hoist and clevis (204-011-178-1), lift the transmission until the lift link bolts can be turned freely by hand (Figure 1). Replace bolt if binding occurs due to corrosion or galling. Apply anti-seize thread compound (C-452) to bore and outside diameter of bolt.

5.2. Install four transmission leveling jacks (T101440) between the transmission support case and the pylon structure. Use jacks to hold the transmission at a height where lift link bolts turn freely.

NOTE

Use shim plates on jacks if necessary to get correct height.

5.3. Use a depth gauge to measure distance at each corner mount between the top side of the support case and the pylon structure.

5.4. The four depth gauge measurements must be within 0.020 inch (0.508 mm) of each other. Adjust jacks as necessary to get this measurement.

5.5. If the four measurements cannot be brought within tolerance, average the two front corner measurements and adjust aft corners to be within 0.020 inch (0.508 mm) of the average. Check the lower lift link bolt for freedom of movement (Step 5.1.).

6. Do engine to transmission alignment check as follows:

6.1. With an alignment tool (T101579) (Figure 2), install the support assembly (3) to blower drive pads (4) (Detail A). Install the beam assembly (2) to the support assembly and oil pressure transmitter pad (5). Loosely install knurled bolt (1).

6.2. Use a length of lockwire (C-447) for alignment wire (7), sufficient to reach from installed target plate (8) to installed knurled bolt (1). Tie a knot in one end of wire and thread wire through the target plate with a knot on the inner (counterbored) side of the target plate.

NOTE

Target plate (8) must be installed in a vertical position, with stamp TOP at the top.

6.3. Install target plate (8) to the transmission input drive quill (9).

6.4. Guide alignment wire (7) through center hole of the beam assembly bracket (6) and attach to the knurled bolt (1) (Detail B).

6.5. Tighten knurled bolt (1) if necessary to remove all slack, kinks, or bends from alignment wire (7). Alignment is correct if wire does not touch the edge of hole in beam assembly bracket (6) (Detail C).

6.6. If alignment wire (7) touches the edge of hole in beam assembly bracket (6), alignment is not correct. Change alignment by adjusting thickness of laminated shim(s) below the work deck of engine mount fitting(s) if necessary. Refer to 412-A-71-20-00-03A-520A-A / 01630, 412-A-71-20-00-05A-520A-A / 01636, and 412-A-71-20-00-08A-520A-A / 01645 for removal of engine mount deck fitting.

NOTE

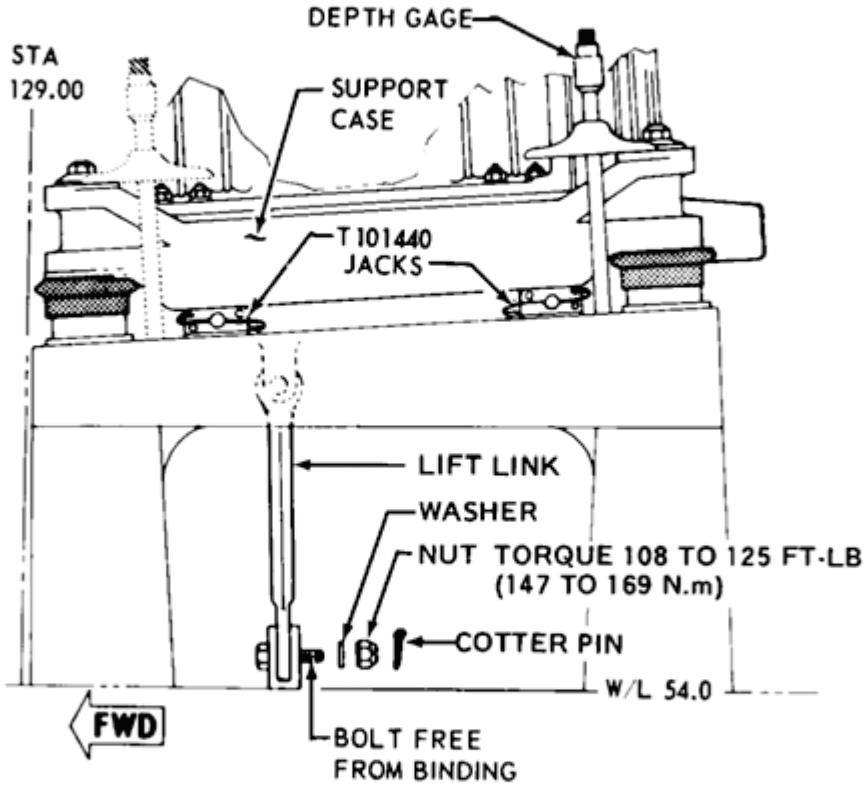
The maximum amount of shim allowed under fitting is 0.188 inch (4.775 mm). If more shim is necessary, fabricate a plate of 2024-T4 aluminum alloy 0.10 inch (2.54 mm) thick, same outside dimensions as shim stock. Structurally bond a plate to the engine service deck with adhesive (C-317). Total thickness of the shim and plate under engine mount fitting must not be more than 0.288 inch (7.315 mm).

7. Remove the transmission jacks, hoist, and the clevis.

8. Install the washer and nut on the upper lift link bolt. Torque nut 60 to 80 foot-pounds (81 to 108 Nm) and safety with a new cotter pin.

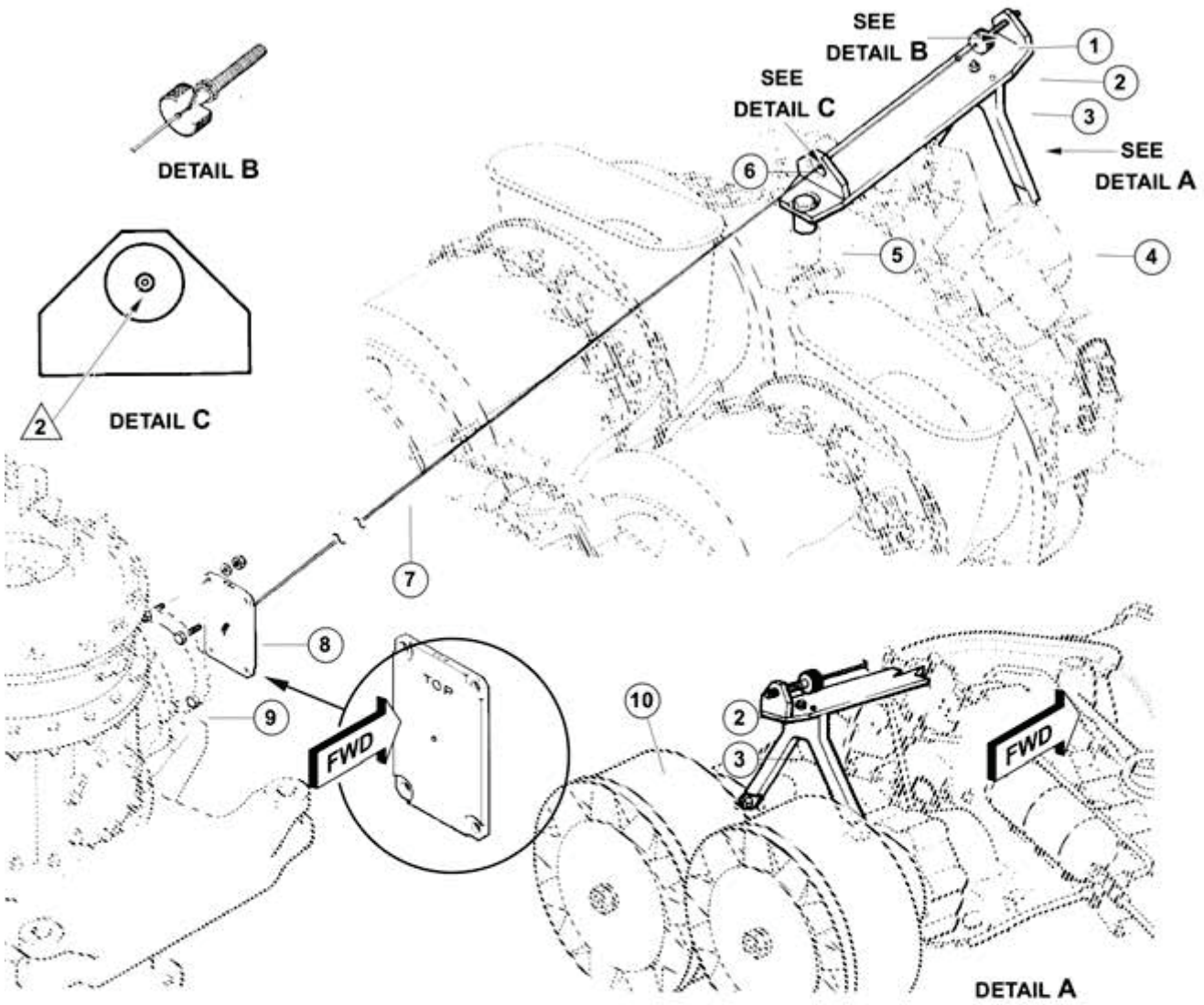
9. Install the washer and nut on the lower lift link bolt. Torque nut 108 to 125 foot-pounds (147 to 169 Nm) and safety with a new cotter pin.

10. Install the main driveshaft (412-A-63-10-01-01A-720A-A / 00999).



ICN-412-A-710000-B-97499-00013-A-001-01

Figure 1. Tool Application - Transmission Leveling Jacks (T101440) (Sheet 1 of 1)



1. Knurled bolt (T101579-17)
2. Beam assembly (T101579-3)
3. Support assembly (T101579-5)
4. Blower drive pad (2 Reqd)
5. Oil pressure transmitter pad
6. Bracket
7. Wire (MS20995C41)
8. Target plate (T101579-19)
9. Transmission input drive quill
10. Blower (installed)

NOTES

1. Applicable to S/N 33001 through 33213 and S/N 36001 through 36019 not modified by 412-570-001-103 or pre BHT-412-SI-74 or pre BHT-412-SI-97.

(2) Alignment wire (C-447) must not touch edge of hole.

ICN-412-A-710000-B-97499-00014-A-002-01

Figure 2. Tool Application - Engine to Transmission Alignment (T101579) (Sheet 1 of 1)

Engine to Transmission (412Classic/SP to 412HP Upgrade) - Alignment

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Lifting Clevis	204-011-178-001	1	
Transmission Leveling Jacks	T101440	1	
Tool Set, Driveshaft Alignment - Transmission to Engine	T101743-125	1	
Depth Micrometer	Commercial	1	Capable of measuring 5-6 and 6-7 inch depths.

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Cadmium plating solution	C-108	AR	
India stone	C-464	AR	
Sealant	C-559	AR	

Procedure

NOTE

The procedure that follows is for helicopters (412Classic/SP to 412HP Upgrade).

NOTE

When the engine is replaced, alignment check is not necessary provided engine mount dimensions or the shim stack-up is not changed.

1. Engine to transmission alignment is necessary when any of the following conditions exist:

- Main driveshaft misalignment is suspected for any reason
- Excessive wear of driveshaft crowned-tooth coupling
- Indications of excessive driveshaft coupling heating
- Hard landings
- Major repair or replacement of components is necessary in center fuselage or pylon support structure

2. During engine-to-transmission alignment procedure, obey the below conventions:

2.1. Clocking conventions applies to dial indicator sweeps on engine and the transmission adapters while the adapter is viewed looking into the face (i.e., transmission adapter viewed looking forward and the engine adapter viewed looking aft).

2.2. The dial indicator supplied with alignment tool (T101743) can be read in 0.001 inch (0.0254 mm) units. A fourth decimal place must be estimated as accurately as possible.

2.3. Movement of indicator probe into dial must be considered as positive (+) indicator deflection.

2.4. Movement of indicator probe away from dial must be considered as negative (-) indicator deflection.

3. Do the level transmission as follows:

3.1. Make alignment check with the helicopter resting on skids on a level surface, and with engine and the pylon assembly installed in a correct manner and with the main rotor untied.

3.2. Remove the transmission and the engine cowlings.

3.3. Remove the main driveshaft assembly (412-A-63-10-01-01A-520A-A / 00994).

3.4. Position the transmission with leveling jacks as follows:

3.4.1. Remove the four pylon isolation mount bolts (1, Figure 1) and washers (2 and 3).

3.4.2. Remove the cotter pin (6), nut (7), and the thin steel washer (8) from the lower lift link bolt (9). Do not remove the bolt (9). Discard the cotter pin (6).

3.4.3. With the hoist and lifting clevis (204-011-178-001), raise the transmission to point where lift link bolt can be turned freely by hand.

3.4.4. Install four transmission leveling jacks (T101440) between the transmission support case and the pylon structure. Use jacks to hold the transmission at a height where lift link bolts turn freely.

NOTE

The transmission leveling jacks will not lower simply by rotating the jack screw counterclockwise. To lower jack precise amount, rotate the jack screw counterclockwise 1/2 turn past the desired point, then tap lightly on end of the jack screw to drive in wedge within jack. Rotate the jack screw 1/2 turn clockwise to get elevation desired.

3.4.5. Carefully adjust four transmission leveling jacks equally so that the transmission is held at a height where the lift link bolt (9) turns freely.

3.4.6. Make sure that the transmission support points are at the same height above the support structure by measuring at each pylon mount with a depth micrometer (Figure 2).



THE transmission leveling jacks (T101440) ARE TO BE USED TO PITCH TRANSMISSION FORE OR AFT AND ROLL TRANSMISSION LEFT OR RIGHT TO PROPER LEVEL ASSEMBLY FOR DRIVESHAFT ALIGNMENT CHECKS. EXTREME CARE MUST BE TAKEN WHEN RAISING A JACK NOT TO DEFORM THE PYLON SUPPORT STRUCTURE. THIS CAN BE AVOIDED BY LOWERING THE JACK OR JACKS OF THE OPPOSITE SIDE OR END OF THE TRANSMISSION TO THE SAME AMOUNT WHILE A PARTICULAR JACK IS RAISED. IT IS ALSO IMPORTANT TO DETERMINE THE DEGREE OF LIFT LINK BOLT LOOSENESS ANY TIME A JACK IS ADJUSTED. IF DURING ROTATION OF LIFT LINK BOLT, THE BOLT APPEARS TO GET TIGHTER AFTER A JACK MOVE, STOP. MOVE WAS IN WRONG DIRECTION OR A FAILURE TO MAKE OPPOSITE MOVE ON OPPOSITE JACK OR JACKS HAS OCCURED.

3.4.7. Use a depth micrometer to measure from the top surface of each transmission support case mounting washer to the top of pylon support structure. Measure at each of the four mount locations; measurements must be within 0.010 inch (0.254 mm). Adjust the jacks to achieve necessary tolerance. After each adjustment, recheck lift link bolt for freedom of movement. If four measurements cannot be brought within the tolerance, average two forward corner measurements and adjust aft corners to be within 0.010 inch (0.254 mm) of this average.

3.4.8. Upon completion of the transmission leveling, check lift link bolt to make sure that it can be turned freely by hand.

4. Install the alignment tool for the transmission-end alignment check as follows:

4.1. The tool set, driveshaft alignment - transmission to engine (T101743-125) (Figure 3) is supplied with two adapters (T101743-127). Inspect the tool adapters and mating flanged adapters on the transmission and the engine for contamination and raised material on interfacing surfaces. Clean as necessary and remove any raised material, flush with surrounding surfaces using a fine india stone (C-464). Brush any damage to cadmium plated surfaces with brush cadmium plating solution (C-108).

4.2. Using a permanent ink marker, mark a 12 o'clock, 3 o'clock, 6 o'clock, and a 9 o'clock position on the face of each adapter (2 and 5). Label each position.

4.3. Install one adapter (5) on engine adapter (9). Install the other adapter (2) on transmission adapter (1). Secure using bolts (6), washers (7), and nuts (8). Torque nuts 50 to 70 inch-pounds (5.65 to 7.91 Nm).

4.4. Insert tooling ball on the adjustable end of the support assembly (4) in the pivot hole in adapter (5). Push aft against the adapter to compress plunger in the support assembly and insert tooling ball at opposite end in the adapter (2) pivot hole. Dial indicator mounting bracket must be at the transmission end of the support assembly.

4.5. Rotate the transmission end adapter (2) so that 12 o'clock mark is at 12 o'clock position.

4.6. Install the dial indicator (3) on the support assembly (4). Position so that indicator is at approximately mid range of travel and tighten thumb screw.

5. Take the indicator readings at the transmission end.

6. For a transmission end vertical alignment check, take indicator readings in the vertical plane 12 o'clock, and 6 o'clock positions, as follows:

NOTE

Because of the sensitivity of dial indicator measurements of extraneous pylon motion, only personnel actually involved in the measurement operations are to be allowed on the helicopter. Care must be taken not to lean or otherwise exert side loads on the engine, transmission, or support assembly (4) as measurements are being taken.

6.1. Zero the dial indicator at 12 o'clock position on the transmission end adapter (2) by rotating dial indicator bezel. Lock the bezel in place.

6.2. To help tracking direction of indicator needle travel, particularly when the indicator is viewed with a mirror, coat half the indicator lens (180°) in the area swept by initial negative indicator travel with the red permanent ink felt tipped marker.

6.3. While holding the support assembly (4) stationary with the dial indicator at 12 o'clock position, rotate the transmission adapter (1) until the 6 o'clock mark on the adapter (2) is in 12 o'clock position. Record the indicator reading as 6 o'clock transmission adapter runout and note whether reading is positive (+) or negative (-). The adapter runout reading is amount of runout inherent in the assembly and must be accounted for when determining actual amount of misalignment.

6.4. Rotate the adapter (2) to return the 12 o'clock mark to the 12 o'clock position. Rezero the indicator if necessary.

6.5. With the dial indicator zeroed at 12 o'clock position and holding the transmission adapter (1) stationary, rotate tool support until the dial indicator probe is at 6 o'clock position. Record the dial indicator reading as 6 o'clock transmission alignment, note whether reading is positive (+) or negative (-).

7. For a transmission end horizontal alignment check, take indicator readings in the horizontal plane (3 o'clock and 9 o'clock positions) as follows:

7.1. With the adapter (2) still positioned with 12 o'clock mark at 12 o'clock position, rotate the support assembly (4) and zero the indicator at 3 o'clock mark.

7.2. While holding the support assembly (4) stationary with the indicator at 3 o'clock position, rotate the transmission adapter (1) until the 9 o'clock mark on the adapter is in 3 o'clock position. Record the indicator reading as 9 o'clock transmission adapter runout and note whether reading is positive (+) or negative (-).

7.3. Return 3 o'clock mark on the adapter to 3 o'clock position. Rezero the indicator, if necessary.

7.4. With the indicator zeroed at 3 o'clock position and holding the transmission adapter (1) stationary, rotate the support assembly (4) until the indicator probe is at 9 o'clock position. Record the indicator reading as 9 o'clock transmission alignment, note whether reading is positive (+) or negative (-).

8. Remove and install alignment tool for engine end alignment check as follows:

8.1. Remove the dial indicator from the support tube. Disengage and remove the support tube by compressing engine end of the tube and lifting out transmission end from the transmission end adapter.

8.2. Install the support assembly (4) with the indicator end at engine end adapter. Insert tooling the ball of adjustable end of the support assembly in the pivot hole in the transmission end adapter (2). Push forward against the transmission end adapter to compress the plunger mechanism in the support assembly and insert tooling ball at the opposite end in engine and adapter pivot hole.

NOTE

Rotate engine end adapter counterclockwise (as viewed looking aft) requires less effort than rotating it clockwise. Freewheel clutches within the engine combining gearbox disengage much of the gearbox drive train as well as engine output shafts, when the adapter is rotated counterclockwise.

8.3. Rotate engine end adapter counterclockwise so that 12 o'clock mark on the adapter is at 12 o'clock position.

8.4. Install the dial indicator on the support assembly (4). Position the indicator so that the indicator is about mid range of its travel and tighten the thumb screw.

9. Take the indicator readings at engine end.

10. For engine end vertical alignment check, take the indicator readings in the vertical plane (12 o'clock and 6 o'clock positions) as follows:

10.1. Zero the dial indicator at 12 o'clock position on engine end adapter (2) by rotating the indicator bezel. Lock the bezel in a place.

10.2. While holding the support assembly (4) stationary with the indicator at 12 o'clock position, rotate engine adapter counterclockwise until 6 o'clock mark on the adapter is in 12 o'clock position. Record the indicator reading as 6 o'clock engine adapter runout and note whether reading is positive (+) or negative (-).

10.3. Rotate the adapter (5) to return the 12 o'clock mark on adapter to the 12 o'clock position. Rezero the indicator if necessary.

10.4. With the indicator zeroed at 12 o'clock position and holding engine adapter (9) stationary, rotate the support assembly (4) until the indicator probe is at 6 o'clock position. Record the indicator reading as 6 o'clock engine alignment, note whether reading is positive (+) or negative (-).

11. For engine end horizontal alignment check, take the indicator readings in the horizontal plane (3 o'clock and 9 o'clock positions) as follows:

11.1. With the adapter still positioned with 12 o'clock mark at 12 o'clock position, rotate the support assembly (4) and zero the indicator at 3 o'clock mark.

11.2. While holding the support assembly (4) stationary with the indicator at 3 o'clock position, rotate engine shaft adapter (5) counterclockwise until 9 o'clock mark on the adapter is in 3 o'clock position. Record the indicator reading as 9 o'clock engine adapter runout and note whether reading is positive (+) or negative (-).

11.3. Return 3 o'clock mark on the adapter (5) to 3 o'clock position. Rezero the indicator if necessary.

11.4. With the indicator zeroed at 3 o'clock position and holding engine adapter (2) stationary, rotate the support assembly (4) until the indicator probe is at 9 o'clock position. Record the indicator reading as 9 o'clock engine alignment, note whether reading is positive (+) or negative (-).

12. Adjust the indicator readings. All the indicator readings necessary to correctly evaluate magnitude of any misalignment between the transmission and the engine assembly are now available.

Indicator readings for the transmission and engine misalignment measured in Step 6.5., Step 7.4., Step 8.4., and Step 11.4. must be adjusted to cancel a portion of misalignment readings caused by the transmission and engine adapter runouts.

NOTE

In adjusting the various indicator readings, it is very important that algebraic signs of runout and misalignment readings be included.

Adjust the measured alignment indicators by subtracting algebraically the corresponding adapter runout readings.

13. Vertical alignment - If engine is within acceptable alignment with the transmission in the vertical plane, adjusted 6 o'clock indicator readings must be within (+) or (-) 0.012 inch (0.3048 mm) at the transmission end and within (+) or (-) 0.040 inch (1.0160 mm) at the engine end.

13.1. Example (Figure 4):

Suppose the 6 o'clock transmission alignment measured in Step 6.5. is (-) 0.015 inch (0.3810 mm) and the 6 o'clock adapter runout measured in Step 6.3. is (+) 0.002 inch (0.0050 mm).

Transmission alignment	(-) 0.015 inch (0.3810 mm)
Adapter runout	- (+) 0.002 inch (0.0508 mm)
	(-) 0.017 inch (0.4318 mm)

The magnitude of adjusted 6 o'clock transmission alignment reading in the vertical plane is (-) 0.017 inch (0.4318 mm).

13.2. Example (Figure 5):

Suppose the 6 o'clock transmission alignment measured in Step 6.5. is (+) 0.022 inch (0.5588 mm) and the 6 o'clock adapter runout measured in Step 6.3. is (-) 0.003 inch (0.0762 mm).

Transmission alignment	(+) 0.022 inch (0.5588 mm).
Adapter runout	- (-) 0.003 inch (0.0762 mm).
	(+) 0.025 inch (0.6350 mm).

Now, (+) 0.025 inch (0.6350 mm) is the magnitude of adjusted 6 o'clock transmission alignment reading in the vertical plane.

14. Horizontal alignment - If engine is within the acceptable alignment with the transmission in horizontal plane, adjusted 9 o'clock indicator reading must be between (+) 0.005 inch (0.13 mm) and (+) 0.035 inch (0.889 mm) at the transmission end and between (+) 0.044 inch (1.12 mm) and (-) 0.032 inch (0.81 mm) at the engine end.

The readings for acceptable alignment in the horizontal plane will not be evenly disposed about zero, as in the vertical plane, because the transmission assembly is rotated slightly to the right to compensate for torsional wind-up.

14.1. Example (Figure 6 and Figure 7):

Suppose the 9 o'clock engine alignment measured in Step 11.2. is (+) 0.012 inch (0.3048 mm) and the runout measured in Step 11.2. is (-) 0.0015 inch (0.0387 mm).

Engine alignment	(+) 0.0120 inch (3048 mm)
Adapter runout	- (-) 0.0015 inch (0.038 mm)
	(+) 0.0135 inch (0.3429 mm)

Now, (+) 0.0135 inch (0.3429 mm) is the magnitude of adjusted 9 o'clock engine alignment reading in the horizontal plane.

14.2. Example (Figure 8):

Suppose the 9 o'clock transmission alignment measured in Step 7.4. is (-) 0.002 inch (0.05 mm) and the 9 o'clock adapter runout measured in Step 7.2. is (+) 0.005 inch (0.13 mm).

Transmission alignment	(-) 0.002 inch (0.05 mm)
Adapter runout	- (+) 0.005 inch (0.13 mm)
	(-) 0.007 inch (0.18 mm)

Now, (-) 0.007 inch (0.18 mm) is the magnitude of adjusted 9 o'clock transmission alignment reading in the horizontal plane.

15. Adjust alignment - Review the adjusted alignment readings at the transmission and engine end together using Table 1 and Table 2 to determine which engine situation is causing misalignment. Refer Table 3 to estimate shim changes necessary to correct the vertical and horizontal misalignment.

15.1. Adjust the alignment by adding or removing laminated shims beneath engine deck fittings. Shim changes at mount locations shown in Table 3 will move pivot hole in engine adapter the showed amount and result in the tabulated change in indicator reading. All shim changes in Table 3 are shown in 0.010 inch (0.2540 mm) increments. Shim removal will result in motions exactly opposite in the direction but identical in magnitude.

15.2. Check the vertical and horizontal alignment after shim change.

15.3. After alignment is complete, remove the alignment tool, transmission jacks, hoist, and clevis.

15.4. Install the thin steel washer (8, Figure 1) and the nut (7) on the lower lift link bolt (9). Torque the nut 108 to 125 inch-pounds (146 to 169 Nm) and secure with a new cotter pin (6).

15.5. Install the four pylon isolation mount bolts (1) and washers (2 and 3).

15.6. Torque the four pylon isolation mount bolts (1) 90 to 105 inch-pounds (122 to 142 Nm).

16. Install the main driveshaft (412-A-63-10-01-01A-720A-A / 00999).

Table 1. Situation Matrix - Vertical Plane

CONDITION		VISUALIZATION	SITUATION
ADJUSTED 6 O'CLOCK INDICATOR READING		LOOKING INBOARD L/H SIDE	
TRANSMIS-SION END	ENGINE END		
OUT OF TOLERANCE (+)	WITHIN LIMITS		ENGINE IS LOW AND POSSIBLY PITCHED NOSE-UP
OUT OF TOLERANCE (-)	WITHIN LIMITS		ENGINE IS HIGH AND POSSIBLY PITCHED NOSE-DOWN
OUT OF TOLERANCE (+)	OUT OF TOLERANCE (+)		ENGINE IS LOW AND PITCHED NOSE-UP
OUT OF TOLERANCE (+)	OUT OF TOLERANCE (–)		ENGINE IS LOW AND POSSIBLY PITCHED NOSE-DOWN
OUT OF TOLERANCE (-)	OUT OF TOLERANCE (+)		ENGINE IS HIGH AND POSSIBLY PITCHED NOSE-UP
OUT OF TOLERANCE (-)	OUT OF TOLERANCE (-)		ENGINE IS HIGH AND PITCHED NOSE-DOWN
WITHIN LIMITS	OUT OF TOLERANCE (+)		ENGINE IS PITCHED NOSE-UP AND POSSIBLY IS LOW
WITHIN LIMITS	OUT OF TOLERANCE (-)		ENGINE IS PITCHED NOSE-DOWN AND POSSIBLY IS HIGH

Table 2. Situation Matrix - Horizontal Plane

CONDITION		VISUALIZATION	SITUATION
ADJUSTED 9 O'CLOCK INDICATOR READING		LOOKING DOWN FACING INBOARD FROM L/H SIDE	
TRANSMISSION END	ENGINE END		
GREATER THAN (+) .035 INCH (0.8890 MM)	WITHIN LIMITS		ENGINE IS TRANSLATED LEFT AND YAWED RIGHT
LESS THAN (+) .005 INCH (0.1270 MM) OR (-)	WITHIN LIMITS		ENGINE IS TRANSLATED RIGHT AND YAWED LEFT
GREATER THAN (+) .035 INCH (0.8890 MM)	OUT OF TOLERANCE (+)		ENGINE IS TRANSLATED LEFT OR YAWED LEFT OR BOTH
GREATER THAN (+) .035 INCH (0.8890 MM)	OUT OF TOLERANCE (-)		ENGINE IS TRANSLATED LEFT AND YAWED RIGHT
LESS THAN (+) .005 INCH (0.1270 MM) OR (-)	OUT OF TOLERANCE (+)		ENGINE IS TRANSLATED RIGHT AND YAWED LEFT

Table 2. Situation Matrix - Horizontal Plane (continued)

CONDITION		VISUALIZATION	SITUATION
ADJUSTED 9 O'CLOCK INDICATOR READING		LOOKING DOWN FACING INBOARD FROM L/H SIDE	
TRANSMISSION END	ENGINE END		
LESS THAN (+) .005 INCH (0.1270 MM) OR (-)	OUT OF TOLERANCE (-)		ENGINE IS TRANSLATED RIGHT OR YAWED RIGHT OR BOTH
WITHIN LIMITS	OUT OF TOLERANCE (+)		ENGINE IS YAWED LEFT AND POSSIBLY TRANSLATED RIGHT
WITHIN LIMITS	OUT OF TOLERANCE (-)		ENGINE IS YAWED RIGHT AND POSSIBLY TRANSLATED LEFT

Table 3. Situation Matrix

CHANGE IN INDICATOR
ADJUSTED READING
MOUNT LOCATION
MOVEMENT OF PIVOT HOLE IN ENGINE END ADAPTER REF

(1) 6 o'clock
(2) 9 o'clock

Fwd monopod mounts
UP 0.0023 (0.05842 mm)
XMSN END
-0.0002 inch (0.00508 mm)
0
AFT 0.0033 (0.08382 mm)
ENG END
+0.0014 inch (0.03556 mm)
0
Fwd leg, bipod mount
UP 0.0015 (0.03810 mm)
XMSN END
-0.0001 inch (0.00254 mm)
-0.0002 inch (0.00508 mm)
AFT 0.0058 (0.14732 mm)
ENG END
+0.0001 inch (0.00254 mm)
-0.0028 inch (0.07112 mm)
RT 0.0023 (0.05842 mm)

Aft leg, bipod mount
UP 0.0035 (0.08890 mm)
XMSN END
-0.0004 inch (0.01016 mm)
+0.0008 inch (0.002032 mm)
FWD 0.0058 (0.14732 mm)
ENG END
+0.0004 inch (0.01016 mm)
+0.003 inch (0.00762 mm)
LFT 0.0078 (0.19812 mm)

Fwd leg, tripod

Table 3. Situation Matrix (continued)

UP 0.0015 (0.03810 mm)
XMSN END
-0.0001 inch (0.00254 mm)
+0.0002 inch (0.00508 mm)
AFT 0.0058 (0.14732 mm)
ENG END
+0.0001 inch (0.00254 mm)
+0.0026 inch (0.06604 mm)
LFT 0.0023 (0.05842 mm)

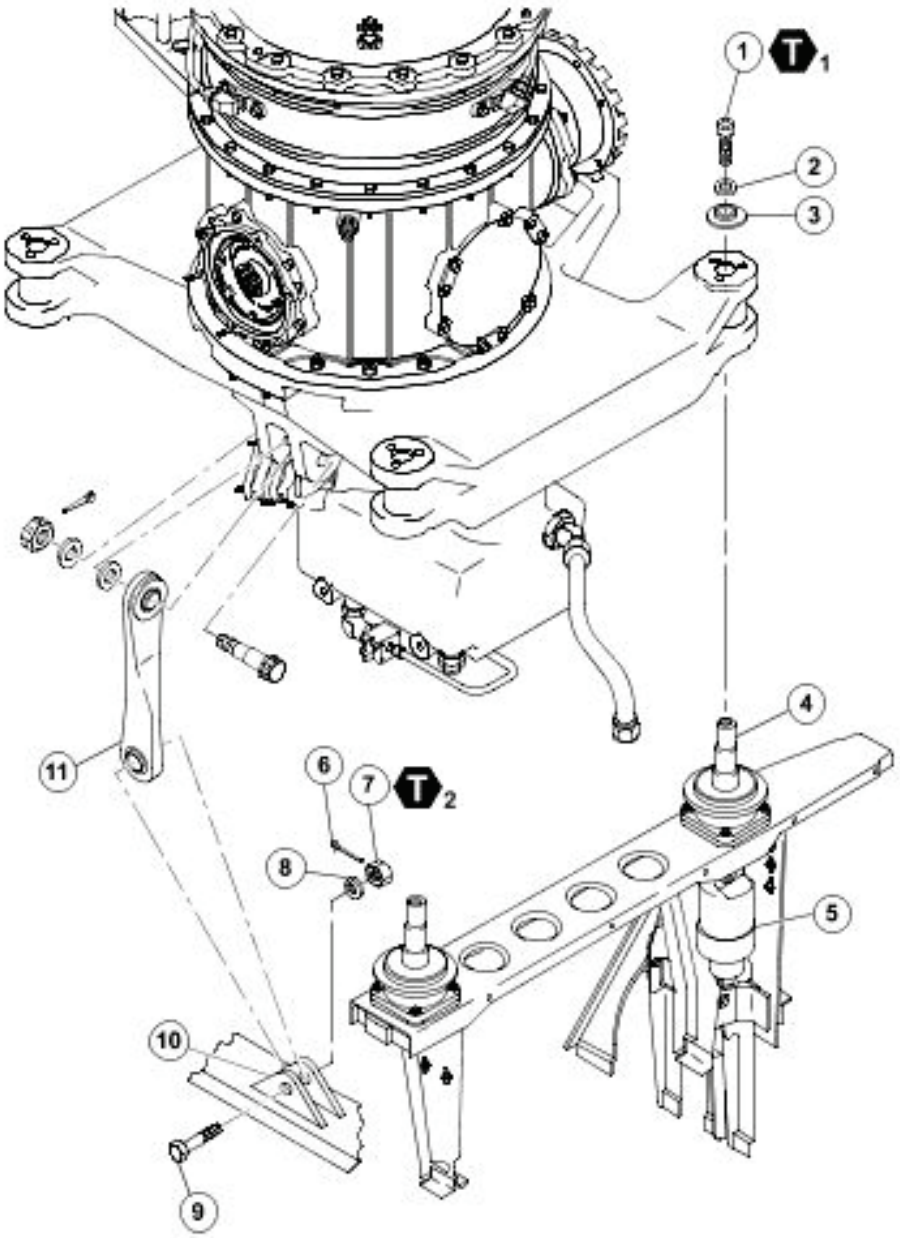
Aft, inbrd leg, tripod mount
UP 0.0003 (0.00762 mm)
XMSN END
0
+0.0012 inch (0.03048 mm)
LFT 0.0103 (0.261762 mm)
ENG END
0
+0.0012 inch (0.03048 mm)
Aft, outbrd leg, tripod mount
UP 0.0025 (0.06350 mm)
XMSN END
-0.0004 inch (0.01016 mm)
-0.0018 inch (0.04572 mm)
FWD 0.0061 (0.15494 mm)
ENG END
+0.0004 inch (0.01016 mm)
-0.0044 inch (0.11176 mm)
RT 0.0169 (0.42926 mm)

All
UP 0.0100 (0.2540 mm)
XMSN END
-0.0012 inch (0.03048 mm)
0

ENG END
+0.0012 inch (0.03048 mm)
0
NOTES:
1. Small pitch alignment errors can be cancelled by relevelling the transmission within the limits. The transmission cannot be repositioned again to meet vertical alignment requirements.
2. The maximum amount of shim allowed under fitting is 0.188 inch. (4.775 mm). If more shim is necessary, fabricate a plate of 2024-T4 aluminum alloy 0.100 inch. (2.540 mm) thick, same outside dimensions as shim stock. Seal the faying surfaces of the plate and engine deck with sealant (C-559). Total thickness of shim and plate under the engine mount fitting must not be exceed 0.288 inch. (7.320 mm).

1 The indicator is assumed to be zeroed at the 12 o'clock.

2 The indicator is assumed to be zeroed at the 3 o'clock.



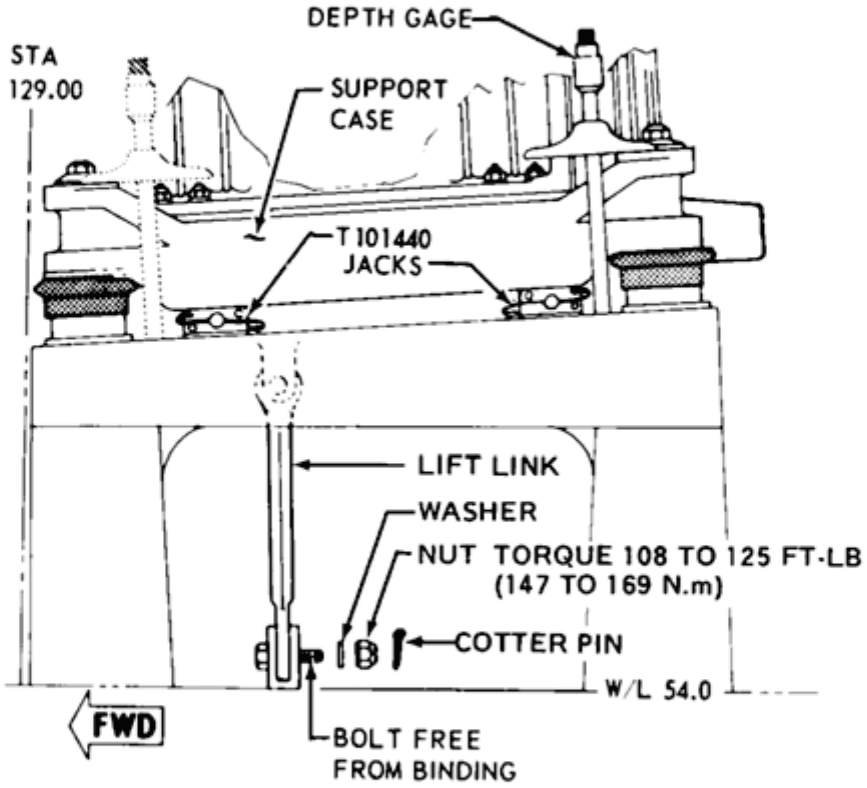
1. Bolt
2. Washer
3. Washer
4. Isolation mount
5. Damper
6. Cotter pin
7. Nut
8. Thin steel washer
9. Bolt
10. Lift beam lift link fitting
11. Lift link

T1 90 TO 105 FT-LBS (122 TO 142 Nm)

T2 108 TO 125 FT-LBS (147 TO 169 Nm)

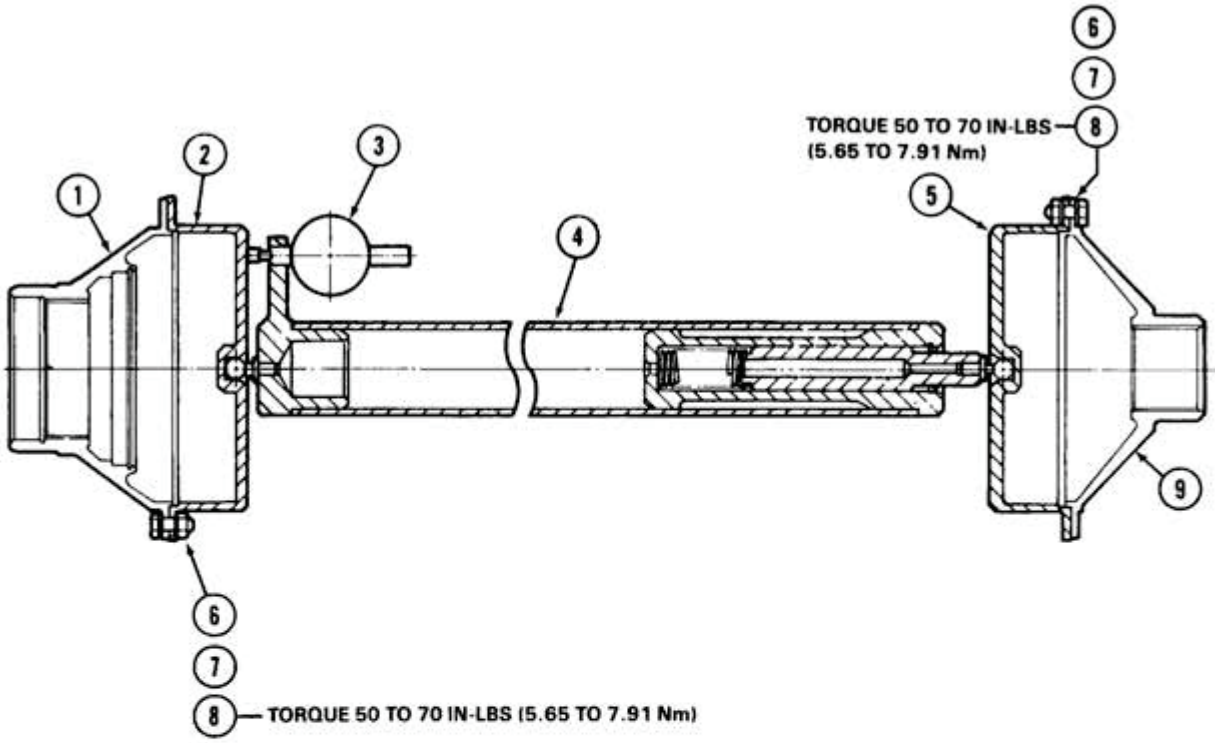
ICN-412-A-710000-B-97499-00015-A-001-01

Figure 1. Transmission Leveling (Sheet 1 of 1)



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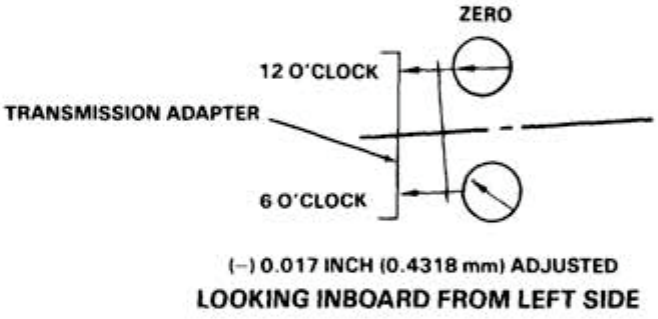
Figure 2. Tool Application - Transmission Leveling Jacks (T101440) (Sheet 1 of 1)



1. Transmission adapter
2. T101743-127 Adapter
3. Dial indicator
4. T101743-107 Support assembly
5. T101743-127 Adapter
6. Bolt (6 each)
7. Washer (12 each)
8. Nut (6 each)
9. Engine adapter

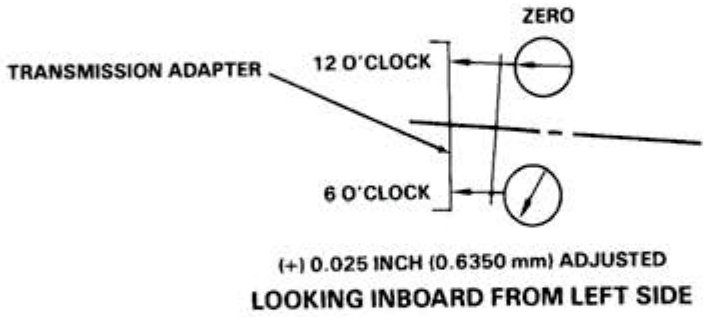
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Figure 3. Engine to Transmission Alignment (S/N 36020 and Subsequent and Helicopters Modified by 412-570-001-103 or Post BHT-412-SI-74 (412SP to 412HP Upgrade) or Post BHT-412-SI-97 (412Classis to 412HP upgrade) (Sheet 1 of 1)



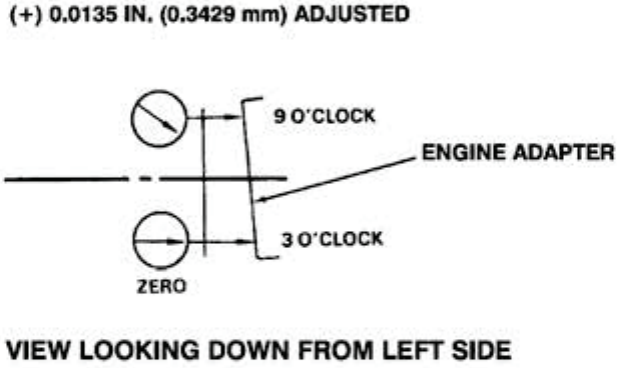
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Figure 4. Engine - High (Sheet 1 of 1)



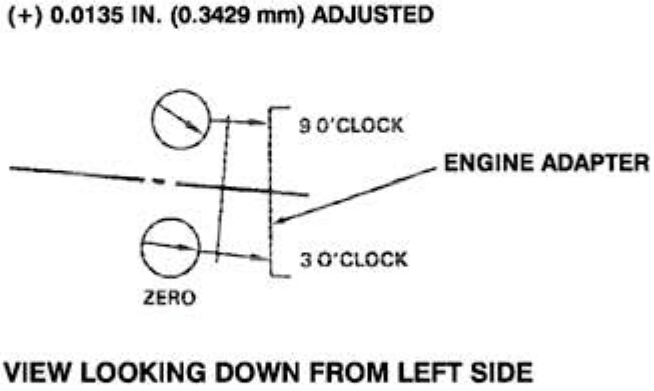
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Figure 5. Engine - Low (Sheet 1 of 1)



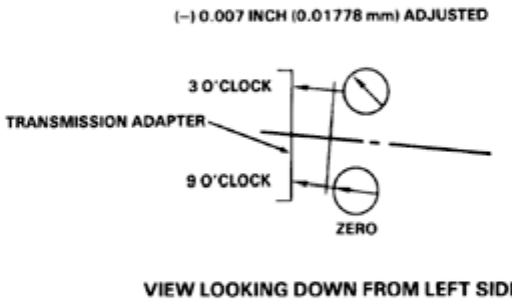
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Figure 6. Engine - Yawed Left (Sheet 1 of 1)



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Figure 7. Engine - Offset Left (Sheet 1 of 1)



ICN-412-A-710000-B-97499-00021-A-001-01

Figure 8. Engine - Offset Slightly Right (Sheet 1 of 1)

Compressor - Wash



CONNECT ENGINE MANIFOLD FUEL DRAIN HOSES AFTER FIRST DRYING RUN.

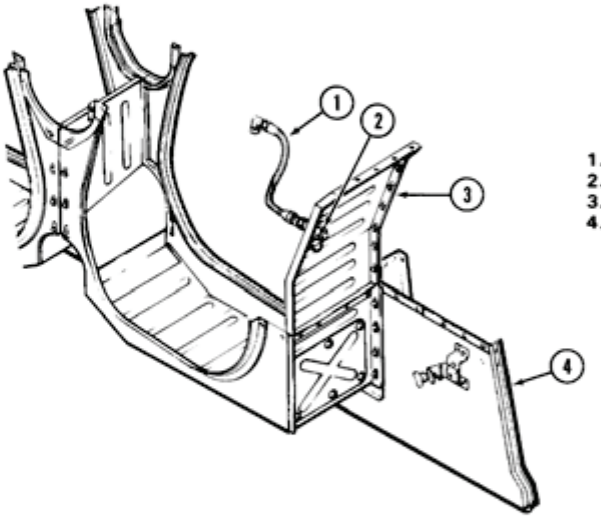
PLACE A SUITABLE CONTAINER BENEATH MANIFOLD VALVE ON EACH POWER SECTION TO CONTAIN FUEL WHICH WILL BE DUMPED UPON ENGINE STARTUP OR SHUTDOWN DURING DRYING RUN.



SN: 33168-Subsequent:

HELICOPTERS ANY MODEL 412 MODIFIED TO RETURN DUMPED FUEL BACK TO FUEL CELLS, REQUIRES DISCONNECTION OF FUEL MANIFOLD DRAIN HOSES FROM MANIFOLD VALVE ON EACH POWER SECTION WHEN ACCOMPLISHING AN ENGINE COMPRESSOR WASH OR MOTORING WASH.

There are two types of compressor wash procedures for cleaning the compressor section: desalination wash (to remove salt deposits) and performance recovery wash (removal of dirt and other deposits). Connections for the compressor wash are located in the upper forward induction baffles (Figure 1). Refer to the applicable Pratt & Whitney Canada Engine Maintenance Manual for compressor wash procedures.



1. Compressor wash hose
2. Compressor wash fitting
3. Induction baffle
4. Forward firewall

ICN-412-A-710000-B-97499-00022-A-001-01

Figure 1. Compressor Wash Fitting (Sheet 1 of 1)

Power Plant - Troubleshooting

Procedure

1. Refer to the applicable Pratt & Whitney Canada Engine Maintenance Manual for troubleshooting procedures.

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Power Plant - Operational Check

Procedure



AFTER POWER PLANT CHANGE AND/OR DISCONNECTING ENGINE OIL INLET HOSE, PRIME OIL PUMP PRIOR TO ENGINE START.

1. Prior to engine starting, motor the engine (BHT-412-FM) until the oil pressure is indicated. If an indication of oil pressure is not indicated in 30 seconds, stop the starter operation and determine the cause.

2. Do the engine check run (BHT-412-FM). Refer to the applicable Pratt & Whitney Canada Engine Maintenance Manual.

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Engine Assembly - Description

The engine assembly consists of a PT6T-3B or PT6T-3D engine and adapting parts to connect the engine to the related systems and helicopter airframe.

The PT6T-3 Series engine consists of two identical, free-turbine, twinned-turboshaft power sections driving a single output shaft through identical gear trains in a common reduction (combining) gearbox.

NOTE

Refer to the applicable Pratt & Whitney Canada Engine Maintenance Manual for engine maintenance data.

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Engine - Removal

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Engine Sling	SWE-13833	1	

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Lockwire	C-405	AR	

Procedure

1. Disconnect the electrical power.

2. Open the forward transmission fairing.

3. Remove the left and right engine air inlet cowling.

4. Remove the left and right engine lower cowling.

5. Remove the left and right engine upper cowl.

6. Remove the left and right twin gearbox cowl.

7. Remove the ejector on each side of the engine (412-A-71-61-00-01A-520A-A / 01701).

NOTE

The center section of the aft firewall will be removed with the engine.

8. Remove the upper aft firewall (412-A-71-30-00-04A-520A-A / 01660). Remove the screws to detach the center section of the aft firewall from the lower section and from the flange of the centerline firewall.

9. Remove the main driveshaft and adapter (412-A-63-10-01-02A-520A-A / 01001).

10. Remove the upper web (trough) of the centerline firewall (412-A-71-30-00-03A-520A-A / 01657, 412-A-71-30-00-03A-280A-A / 01658 and 412-A-71-30-00-03A-720A-A / 01659).

11. Remove the air management system, valve assembly, and transition duct from each side of the engine (412-A-71-61-00-02A-520A-A / 01705 and 412-A-71-61-00-03A-520A-A / 01709).

12. Remove the induction baffle from each side of the engine (412-A-71-61-00-05A-520A-A / 01716).

13. Remove the upper section of the middle firewall from each side of the engine (412-A-71-30-00-02A-520A-A / 01654).



DO NOT DEFLECT FIREWALL OR INDUCTION BAFFLE MORE THAN NECESSARY TO EFFECT REMOVAL.

14. Deflect the left and right inboard ears of the lower middle firewall and induction baffle forward, to allow for engine removal clearance. Secure with lockwire (C-405).

15. Remove the bulged access doors from the aft cabin bulkhead at each side of the pylon support.

16. Place suitable container under oil drain line and open oil drain valves (9 and 10, Figure 1) and valve on deck aft of reduction (combining) gearbox. Close oil drain valves.

17. Place suitable container under drain line and open fuel filter valves (6 and 13) to drain fuel filters. Close fuel filter valves.

18. Remove nuts and washers from tachometer generators and remove bonding straps (1 and 2).

19. Disconnect hoses (3, 4, 5, 7, 8, 11, 12, 14, 15, and 16).

20. Disconnect the following items at left side of engine; disconnect the same items at right side of engine unless otherwise noted:

20.1. Disconnect the bleed air tube (38, Figure 1) from the adapter (34) and the bleed air tube assembly (23).

20.2. On left side only, remove the clamp (24) with attaching hardware.

20.3. Remove the bleed air tube (38).

20.4. If necessary, remove the bleed air tube connector assembly (23) as follows:

20.4.1. Remove and discard the lockwire from the bolts (39).

20.4.2. Remove the bolts (39) and the washers (40).

20.4.3. Remove the bleed air tube connector assembly (23) and the gasket (41).

20.4.4. Put protective caps on all open ends.

20.5. Loosen screws (25) and slide oil cooler blower boot (26) aft from oil cooler blower.

20.6. On left side of engine only, remove cotter pin (31), nut (32), washers (33 and 28), and bolt (29). Separate actuator (30) from jackshaft assembly (27).

20.7. Disconnect hoses (35 and 36 or 37).

NOTE

On right side of engine, a control tube is installed instead of a link (22).

20.8. Remove cotter pin (17), nut (18), washer (19), and bolt (21). Separate link (22) from lever (20).

21. At rear of engine, disconnect hoses (52, 54, 46, 47, 48, 49, 50, 51, and 53).

22. Disconnect electrical connections at front of engine (Figure 2).

22.1. Disconnect electrical leads (9, Figure 2) from flight idle solenoid (7). Remove clamps (8 and 10) with attaching hardware.

22.2. Disconnect electrical connector (12) from fuel pressure transmitter (11).

22.3. Disconnect engine electrical disconnect cable (13).

22.4. Disconnect electrical leads (14) and remove clamp (16).

22.5. Disconnect engine electrical disconnect cable (1).

22.6. Disconnect electrical connector (2) from fuel pressure transmitter (3). Remove clamp (4) with attaching hardware.

22.7. Disconnect electrical leads (6) from flight idle solenoid (5).

23. Disconnect electrical connectors at left side of engine (Figure 2).

23.1. Disconnect starter generator cable (26). Remove clamp (27) with attaching hardware.

23.2. Disconnect electrical connectors (21, 22, and 22A).

23.3. Disconnect electrical connector (19). Remove clamp (20).

WARNING

RESIDUAL VOLTAGE IN IGNITION EXCITER MAY BE DANGEROUSLY HIGH. ALWAYS DISCONNECT CABLE COUPLING NUTS AT IGNITION EXCITER END FIRST. USE INSULATED TOOLS TO REMOVE CABLE COUPLING NUTS. DO NOT TOUCH EXCITER OUTPUT CONNECTORS OR CABLE COUPLING NUTS WITH BARE HANDS.

23.4. Disconnect electrical connector (25). Remove screws, washers, and nuts (23). Secure ignition exciter (24) to engine with tape or equivalent.

24. Disconnect electrical connectors at right side of engine (Figure 2, Sheet 3).

24.1. Disconnect starter generator cable (31). Remove clamp (30) with attaching hardware.

24.2. Disconnect fire detector connectors (32).



RESIDUAL VOLTAGE IN IGNITION EXCITER MAY BE DANGEROUSLY HIGH. ALWAYS DISCONNECT CABLE COUPLING NUTS AT IGNITION EXCITER END FIRST. USE INSULATED TOOLS TO REMOVE CABLE COUPLING NUTS. DO NOT TOUCH EXCITER OUTPUT CONNECTORS OR CABLE COUPLING NUTS WITH BARE HANDS.

24.3. Disconnect electrical connector (35). Remove screws, washers, and nuts (33). Secure ignition exciter (34) to engine with tape or equivalent.

24.4. Disconnect electrical connectors (36, 36A, and 37).

25. Inspect engine area. Ensure all hoses, electrical connectors, and controls are disconnected from engine.

26. Install engine sling (SWE-13833) (Figure 3).



LIFTING CAPACITY OF HOIST SHALL BE 2000 POUNDS (908 KG) OR GREATER.

27. Attach hoist (minimum 2000 pounds (908 kg) capacity) to sling and take up slack in sling cables without placing load on engine mount bolts.

28. Disconnect engine mounts as follows:

28.1. Remove cotter pin (24, Figure 4), nut (25), steel washer (26), and bolt (27), attaching the bipod fitting (23) or tripod fitting (28) to the bearing (2 or 3).

28.2. Remove cotter pin (11), nut (12), thin corrosion steel washers (13 and 15), and bolt (16). Separate tube (10) from bellcrank (14).

28.3. Remove cotter pin (21), nut (20), washers (19 and 18), and bolt (17).

29. Lift engine assembly clear of helicopter.

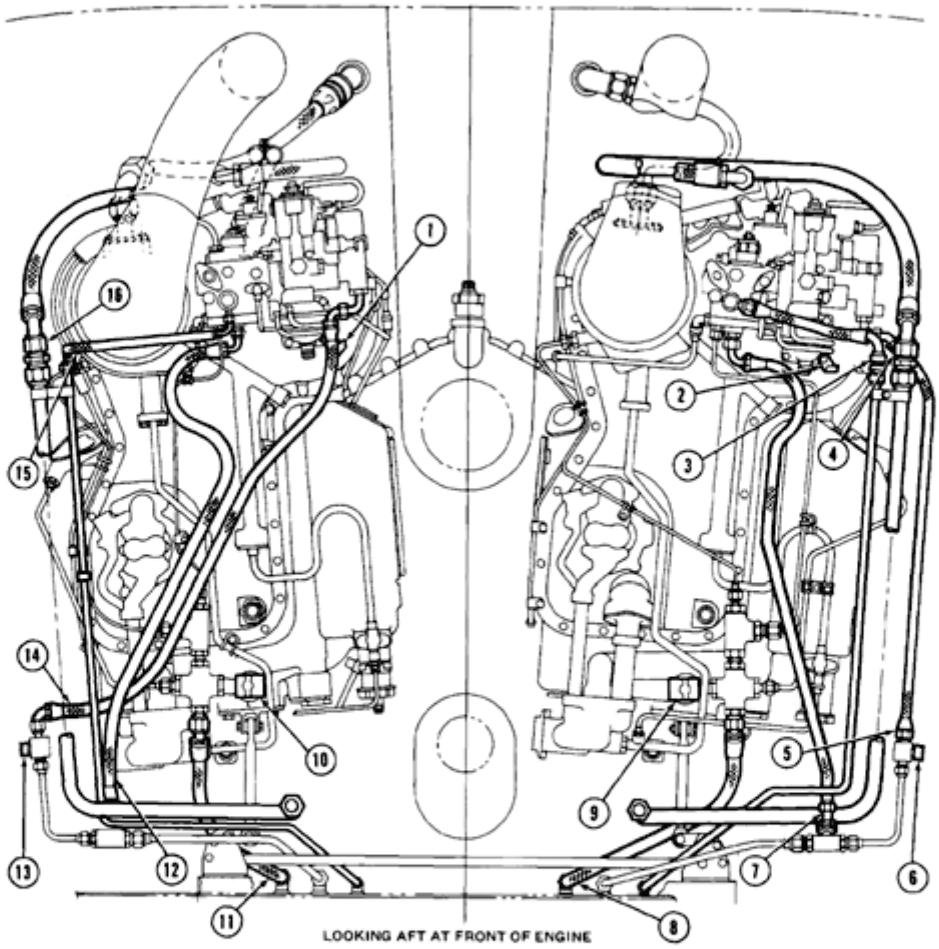
30. Remove cotter pin (9), nut (8), thin steel washers (7 and 6), bolt (5), and tube (10) from fitting (1).

31. Remove center section of aft firewall between power sections (412-A-71-30-00-04A-520A-A / 01660).

32. Install engine assembly in suitable stand and remove hoist.

NOTE

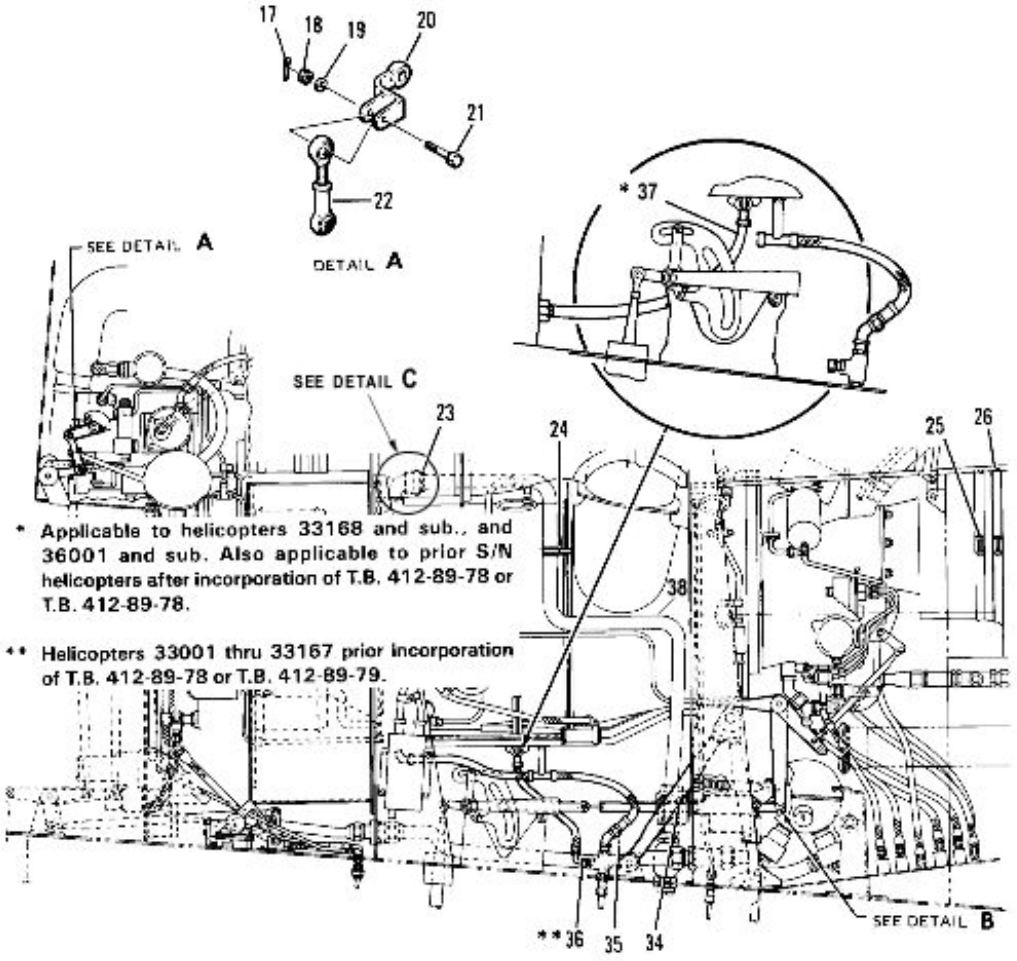
Refer to the BHT-412-CR&O for build-up of replacement engine.



1. Bonding strap
2. Bonding strap
3. Governor bleed hose
4. Fuel inlet hose
5. Fuel filter drain hose
6. Fuel filter drain valve
7. Fuel pump seal drain hose
8. Oil drain hose
9. Oil drain valve
10. Oil drain valve
11. Oil drain hose
12. Fuel pump seal drain hose
13. Fuel filter drain valve
14. Fuel filter drain hose
15. Governor bleed hose
16. Fuel inlet hose

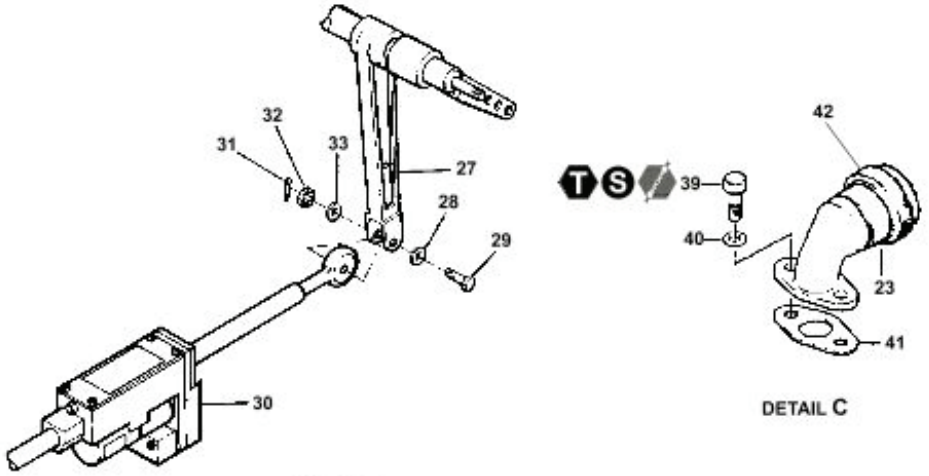
ICN-412-A-710001-B-97499-00001-A-001-01

Figure 1. Engine Installation Connections (Sheet 1 of 4)



ICN-412-A-710001-B-97499-00002-A-001-01

Figure 1. Engine Installation Connections (Sheet 2 of 4)



17. Cotter pin
18. Nut
19. Washer
20. Lever
21. Bolt
22. Link (control tube on right side)
23. Bleed air tube connector assembly
24. Clamp, screw, washer, and nut (left side only)
25. Screw
26. Oil cooler blower boot
27. Jackshaft assembly
28. Aluminum washer
29. Bolt
30. Actuator
31. Cotter pin
32. Nut
33. Thin aluminum washer
34. Adapter
35. Combustion chamber drain hose
36. Solenoid dump valve hose
37. Engine manifold drain hose
38. Bleed air tube
39. Bolt
40. Washer
41. Gasket
42. Nut

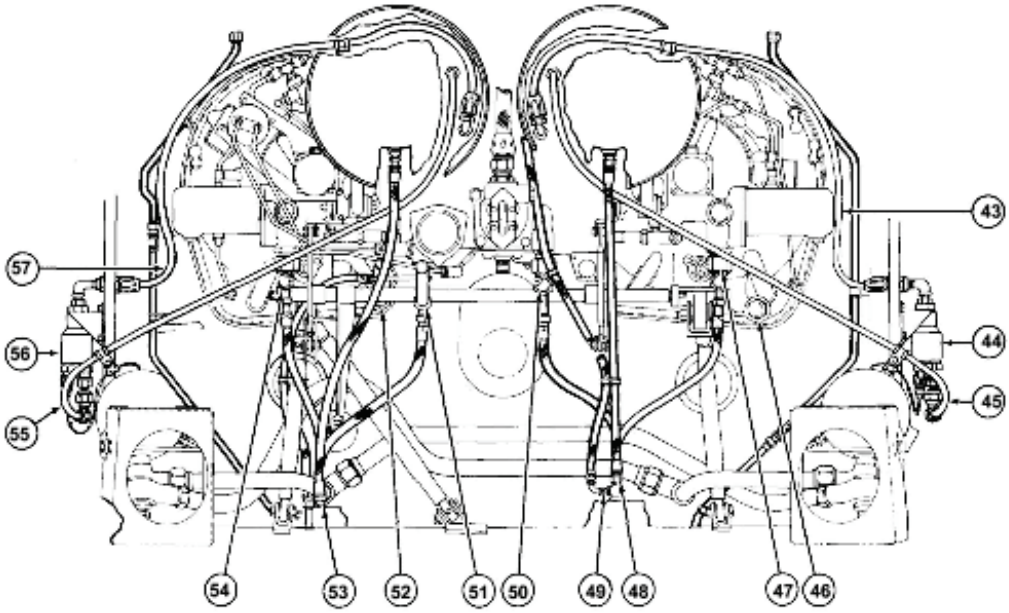
LOCKWIRE (C-405)

SEALANT (C-308)

85 TO 95 IN-LBS
(9.60 TO 10.73 Nm)

ICN-412-A-710001-B-97499-00003-A-001-01

Figure 1. Engine Installation Connections (Sheet 3 of 4)

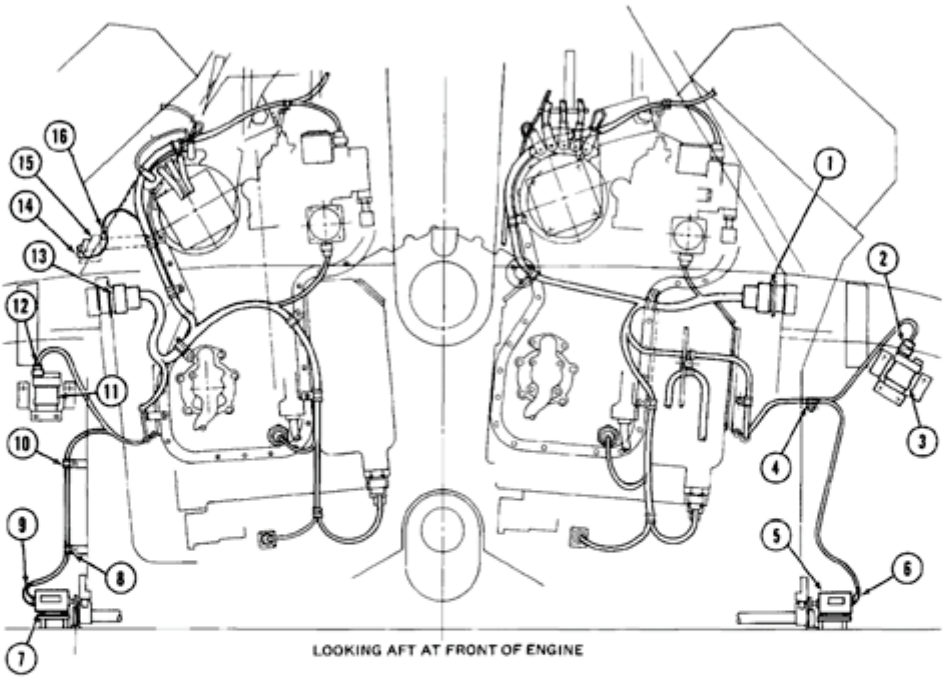


LOOKING FORWARD AT REAR OF ENGINE

43. Torque transmitter pressure hose (right)
44. Torque transmitter (right)
45. Torque transmitter vent hose (right)
46. Right power section oil outlet hose
47. Right power section oil inlet hose
48. Blower seal drain hose
49. Reduction gearbox oil filler adapter drain hose
50. Reduction gearbox oil outlet hose
51. Reduction gearbox oil inlet hose
52. Left power section oil outlet hose
53. Blower seal drain hose
54. Left power section oil inlet hose
55. Torque transmitter vent hose (left)
56. Torque transmitter (left)
57. Torque transmitter pressure hose (left)

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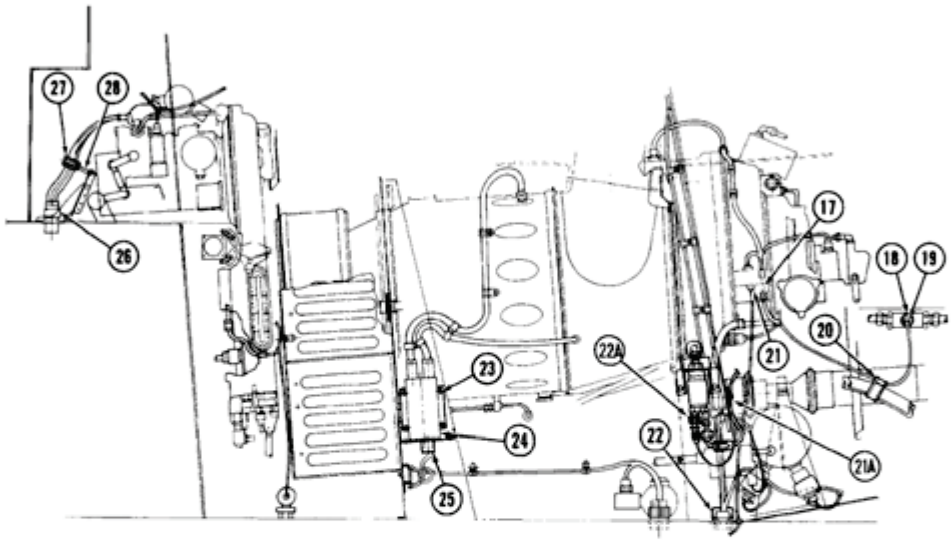
Figure 1. Engine Installation Connections (Sheet 4 of 4)



1. Engine electrical disconnect cable
2. Electrical connector
3. Fuel pressure transmitter
4. Clamp, screw, washer, and nut
5. Flight idle solenoid
6. Electrical leads
7. Flight idle solenoid
8. Clamp, screw, washer, and nut
9. Electrical leads
10. Clamp, screw, washer, and nut
11. Fuel pressure transmitter
12. Electrical connector
13. Engine electrical disconnect cable
14. Electrical leads
15. No. 2 generator overheat switch
16. Clamp

ICN-412-A-710001-B-97499-00005-A-001-01

Figure 2. Engine Electrical Connections (Sheet 1 of 4)

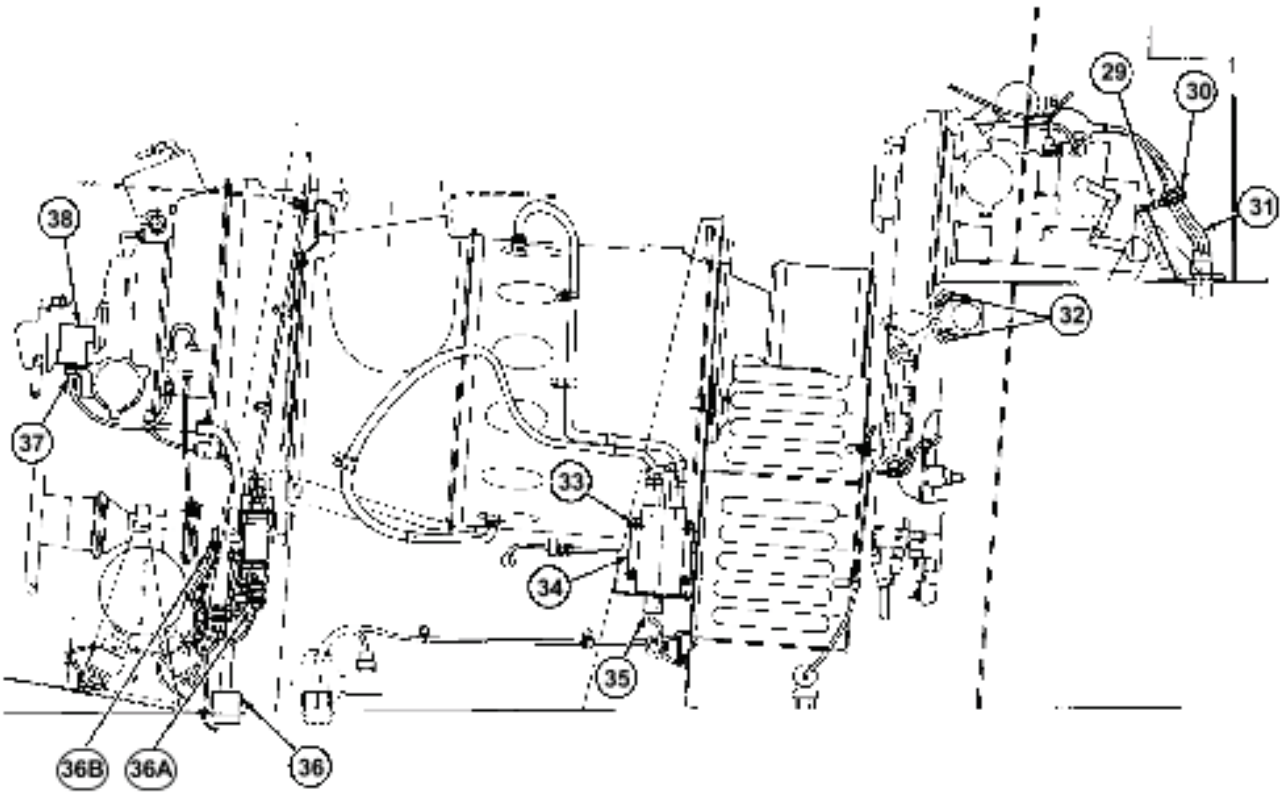


LOOKING INBOARD AT LEFT SIDE OF ENGINE

17. ITT compensator
18. Reduction gearbox oil temperature switch
19. Electrical connector
20. Clamp
21. Electrical connector
21A. Clamp
22. Engine aft electrical disconnect cable
22A. Electrical connector
23. Screw, washer, and nut
24. Ignition exciter
25. Electrical connector
26. Starter generator cable
27. Clamp
28. Screw, spacer, washer, and nut

ICN-412-A-710001-B-97499-00006-A-001-01

Figure 2. Engine Electrical Connections (Sheet 2 of 4)

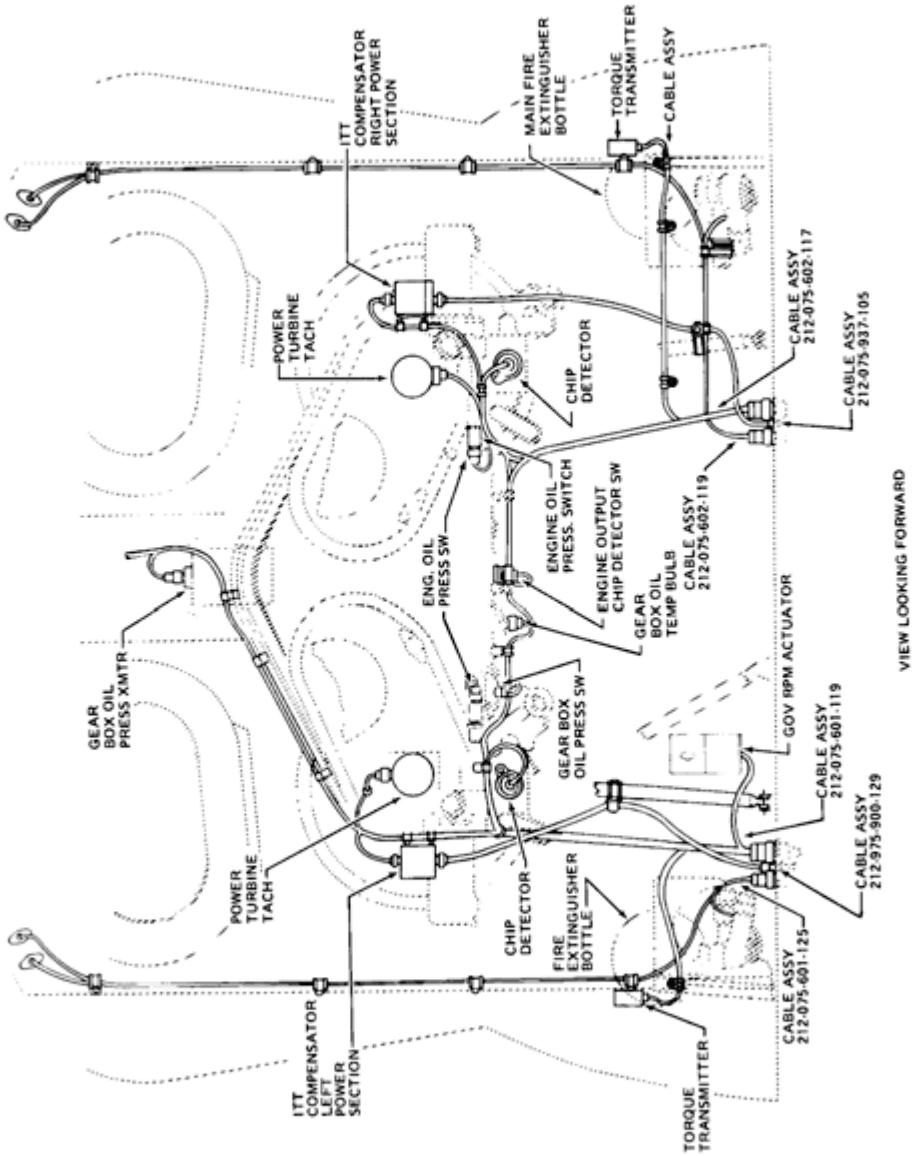


LOOKING INBOARD AT RIGHT-HAND SIDE OF ENGINE

29. Screw, spacer, washer, and nut
30. Clamp
31. Starter generator cable
32. Fire detector connector
33. Screw, washer, and nut
34. Ignition exciter
35. Electrical connector
36. Engine aft electrical disconnect cable
36A. Electrical connector
36B. Clamp
37. Electrical connector
38. Tachometer generator

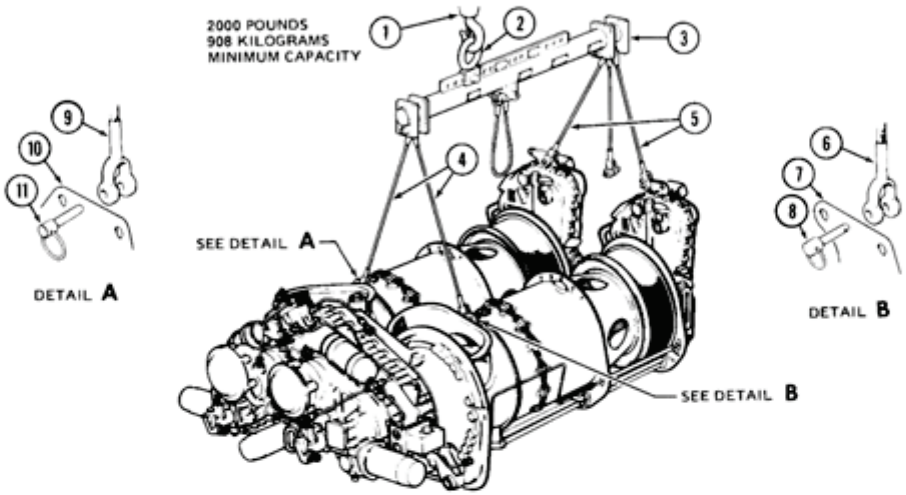
ICN-412-A-710001-B-97499-00007-A-001-01

Figure 2. Engine Electrical Connections (Sheet 3 of 4)



ICN-412-A-710001-B-97499-00008-A-001-01

Figure 2. Engine Electrical Connections (Sheet 4 of 4)



1. Hoist
2. Lifting eye
3. Engine sling, SWE-13833
4. Sling cables
5. Sling cables
6. Clevis
7. Lifting bracket (accessory gearbox)
8. Pin
9. Clevis
10. Lifting bracket (engine flange C)
11. Pin

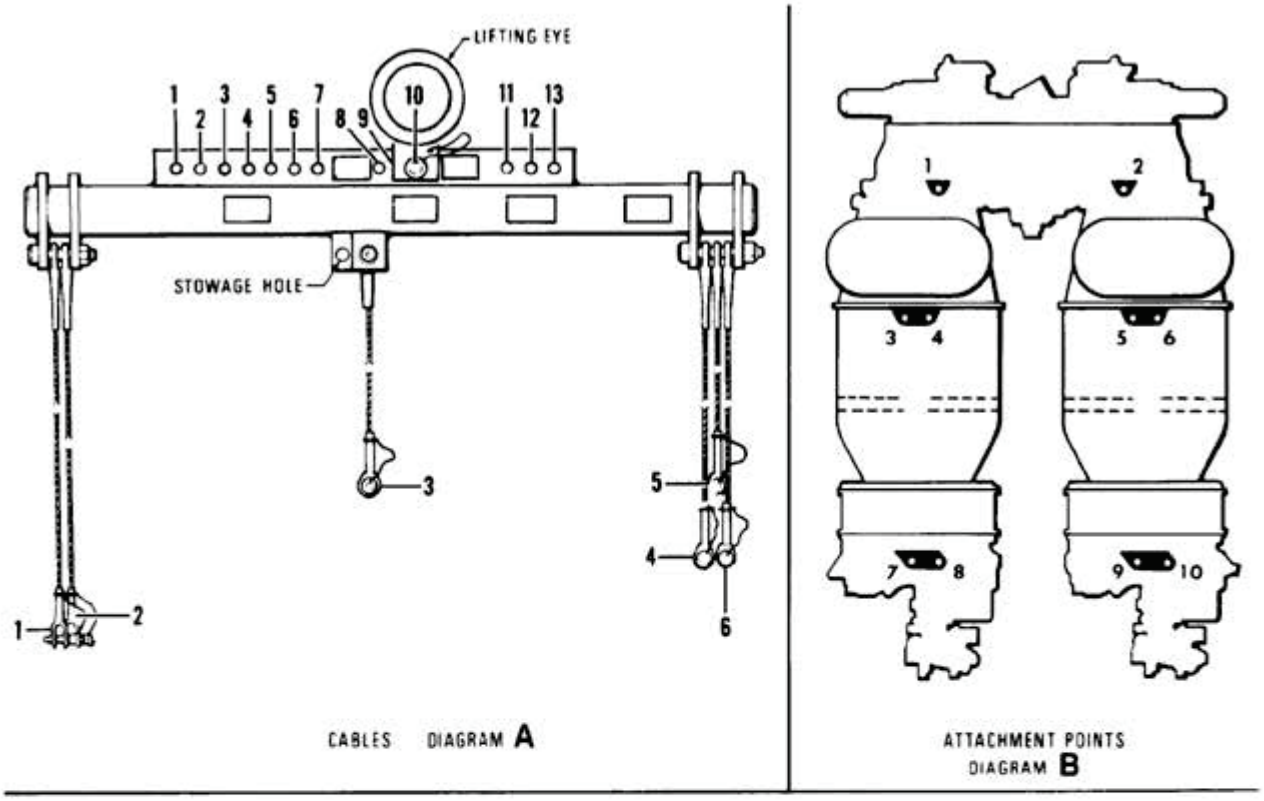
NOTES:

1. Determine which components listed in columns 1, 2 and 3 on sheet 2 are installed on engine to be hoisted.
2. Position lifting eye (2) in correct sling holes number for engine to be hoisted as shown in column 4 and diagram A.
3. Attach cables shown in column 5 to engine at points shown in column 6 and diagram B.

412-M-71-12-1

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Figure 3. Tool Application - Engine Sling (SWE-13833) (Sheet 1 of 2)

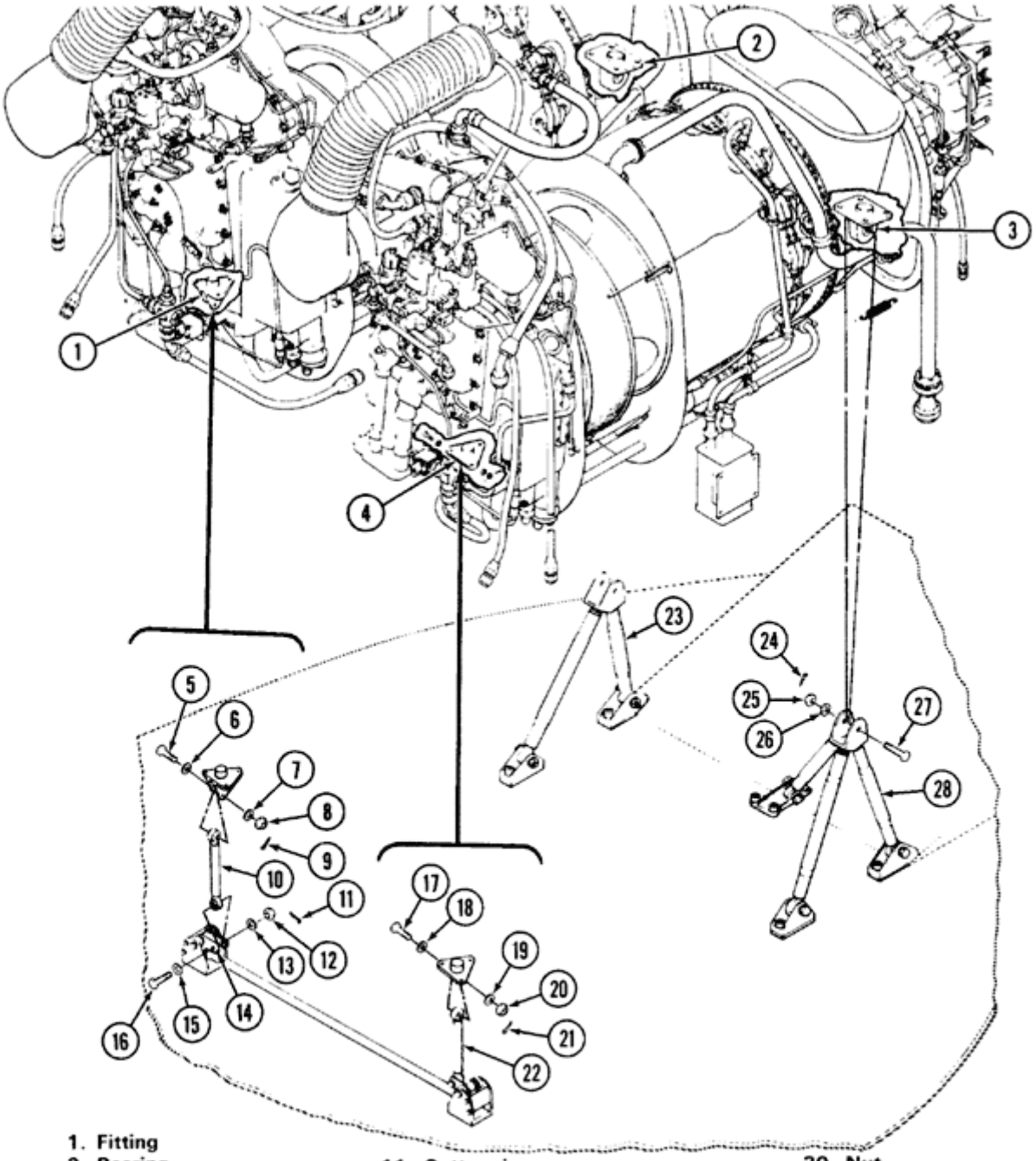


COLUMN

1	2	3	4	5	6
ITEM REMOVAL/INSTALLATION	Starter/ Generator Installed	Blower Installed	Sling Hole Number	Cables (See Diagram 'A')	Attachment Points (See Diagram 'B')
COMPLETE ENGINE	YES	YES	6	1, 2, 4, 6,	3, 6, 7, 9,
COMPLETE ENGINE	YES	NO	7	1, 2, 4, 6,	3, 6, 7, 9,
COMPLETE ENGINE	NO	YES	4	1, 2, 4, 6,	3, 6, 7, 9,
COMPLETE ENGINE	NO	NO	5	1, 2, 4, 6,	3, 6, 7, 9,
COMPLETE POWER SECTION LEFT	YES		10	1, 4,	5, 9,
COMPLETE POWER SECTION RIGHT	YES		10	1, 4,	4, 8,
COMPLETE POWER SECTION LEFT	NO		9	1, 4,	5, 9,
COMPLETE POWER SECTION RIGHT	NO		9	1, 4,	4, 8,
REDUCTION GEARBOX			11	3, 5,	1, 2,

ICN-412-A-710001-B-97499-00010-A-001-01

Figure 3. Tool Application - Engine Sling (SWE-13833) (Sheet 2 of 2)



1. Fitting
2. Bearing
3. Bearing
4. Fitting
5. Bolt
6. Thin steel washer
7. Thin steel washer
8. Nut
9. Cotter pin or safety pin
10. Tube
11. Cotter pin
12. Nut
13. Thin corrosion steel washer
14. Bellcrank
15. Thin corrosion steel washer
16. Bolt
17. Bolt
18. Thin steel washer
19. Thin steel washer
20. Nut
21. Cotter pin or safety pin
22. Tube
23. Bipod fitting
24. Cotter pin
25. Nut
26. Steel washer
27. Bolt
28. Tripod fitting

ICN-412-A-710001-B-97499-00011-A-001-01

Figure 4. Engine Mount Connections (Sheet 1 of 1)

Engine - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Solid film lubricant	C-021	AR	
Sealant	C-308	AR	
Lockwire	C-405	AR	

Procedure

1. Install SWE-13833 engine sling. (Refer to Figure 1, Sheet 2 for cable usage and engine attachment points.)

2. Position center section of aft firewall between power sections (412-A-71-30-00-04A-720A-A / 01662).



LIFTING CAPACITY OF HOIST SHALL BE 2000 POUNDS (908 KG) OR GREATER.

3. Attach hoist to sling and remove engine from stand.

4. Connect tube (10, Figure 2) to fitting (1) with bolt (5), thin steel washers (6 and 7), nut (8), and cotter pin or safety pin(9).

5. Lower engine carefully into place, checking position as follows:

5.1. Bearings (2 and 3) align with bipod fitting (23) and tripod fitting (28).

5.2. Tube (22) aligns with fitting (4) and tube (10) aligns with bellcrank (14).

5.3. Forward engine fireshield is aft of lower induction baffle (412-A-71-30-00-02A-720A-A / 01656).

5.4. Middle engine fireshield is aft of lower middle firewall.

5.5. Aft engine fireshield is forward of lower aft firewall.

6. Install bolt (27), steel washer (26), nut (25), and cotter pin (24) to connect bipod fitting (23) and tripod fitting (28) to bearings (2 and 3).

7. Install bolt (16), thin corrosion steel washers (15 and 13), nut (12), and cotter pin (11).

8. Install bolt (17), thin steel washers (18 and 19), nut (20), and cotter pin or safety pin(21).

9. Remove hoist and sling.

10. Connect electrical components at right side of engine (Figure 3, Sheet 3).

10.1. Connect electrical connectors (36, 36A, and 37, Figure 3).

10.2. Position ignition exciter (34) on firewall flange, install screws, washers, and nuts (33). Connect electrical connector (35).

10.3. Connect fire detector connectors (32).

10.4. Connect starter generator cable (31). Install clamp (30) using screw spacer, washer, and nut (29).

11. Connect electrical components at left side of engine (Figure 3, Sheet 2).

11.1. Connect electrical connectors (21, 22, and 22A).

11.2. Connect electrical connector (19). Install clamp (20) using existing hardware.

11.3. Position ignition exciter (24) on firewall flange. Install screws, washers, and nuts (23). Connect electrical connector (25).

11.4. Connect starter generator cable (26). Install clamp (27) using screw, spacer, washer, and nut (28).

12. Connect electrical components at front of engine (Figure 3, Sheet 1).

12.1. Connect electrical leads (9, Figure 3). Install clamps (8 and 10) with attaching hardware.

12.2. Connect electrical connectors (12 and 13).

12.3. Connect electrical leads (14). Install clamp (16) using existing hardware.

12.4. Connect electrical leads (6). Install clamp (4) with attaching hardware.

12.5. Connect electrical connector (2) and engine electrical disconnect cable (1).

13. Connect the following items at left side of engine, connect same items at right side of engine unless otherwise noted (Figure 4, Sheet 2).

13.1. If removed, install the bleed air tube connector assembly (23, Figure 4) as follows:

13.1.1. Remove the protective caps from all open ends.

13.1.2. Put the gasket (41) and the bleed air tube connector assembly (23) in the correct position.

Applicable: SN: 33001-Subsequent or 36001-36424:

13.1.3. Do as follows:

13.1.3.1. Install the washers (40) and the bolts (39).

13.1.3.2. Torque the bolts (39) ◍.

Applicable: SN: 36425-Subsequent:

13.1.4. Do as follows:

NOTE

Make sure that the washer's flat surface faces the elbow machined radius to prevent possible fouling condition.

13.1.4.1. Install washers (40).

NOTE

Do not apply sealant (C-308) to threads of the bolts.

13.1.4.2. Wet install bolts (39) with sealant (C-308).

13.1.4.3. Torque bolts (39) ◍.

13.2. Install lockwire (C-405) on the bolts (39).

13.3. Apply solid film lubricant (C-021) to the threads of bleed air tube connector assembly (23) and the nut (42).

13.4. Make sure that the orifice is in the correct position in the nut (42).

13.5. Connect the bleed air tube (38) to the bleed air tube connector assembly (23) and the adapter (34).

13.6. On the left side only, install the clamp (24) with attaching hardware.

13.7. Connect the electrical connector (19, Figure 3).

13.8. Position oil cooler blower boot (26, Figure 4) around oil cooler blower. Tighten screw (25) to secure boot.

13.9. On left side of engine only, connect actuator (30) to jackshaft (27) using bolt (29), washers (28 and 33), nut (32), and cotter pin (31).

13.10. Connect hoses (35 and 36 or 37).

NOTE

On right side of engine, a control tube is installed instead of a link (22).

13.11. Connect link (22) to lever (20) using bolt (21), washer (19), nut (18), and cotter pin (17).

14. At rear of engine, connect hoses (52, 54, 46, 47, 48, 49, 50, 51, and 53) shown in Figure 4, Sheet 3.

15. Connect the following items at front of engine (Figure 4, Sheet 1):

15.1. Connect hoses (3, 4, 5, 7, 8, 11, 12, 14, 15, and 16).

15.2. Remove nuts and washers from tachometer generator where bonding straps (1 and 2) mount. Clean area, if necessary, to ensure a good electrical ground. Install bonding straps (1 and 2), washers, and nuts.

16. Install upper section of middle firewall and induction baffle (412-A-71-20-00-04A-720A-A / 01635) on each side of engine.

17. Install air management system valve assembly and transition duct (412-A-71-61-00-03A-720A-A / 01712 and 412-A-71-61-00-02A-720A-A / 01708) on each side of engine.

18. Install upper web (trough) of centerline firewall (412-A-71-30-00-03A-720A-A / 01659).

19. Install main driveshaft adapter (412-A-63-10-01-02A-720A-A / 01004) and main driveshaft (412-A-63-10-01-01A-720A-A / 00999).

20. Install upper aft firewall (412-A-71-30-00-04A-720A-A / 01662).

21. Install ejector on each side of engine (412-A-71-61-00-01A-720A-A / 01704).

22. Install left and right twin gearbox cowl.

23. Service engine and reduction gearbox (412-A-12-00-00-00A-040A-A / 00126).

24. Install left and right engine top cowl.

25. Install the left and right engine lower cowling.

26. Install the left and right engine air inlet cowling.

27. Do an internal thermocouple resistance check on each power section (412-A-96-12-00-09A-340A-A / 02002).

28. Prime the power plant lubrication system (412-A-71-00-01-01A-520A-A / 01597 and 412-A-71-00-01-01A-720A-A / 01598).

29. Pre-oil the reduction gearbox (412-A-71-00-00-01A-212A-A / 01588).

30. Do a wet motoring run (412-A-71-00-00-01A-913A-A / 01590).

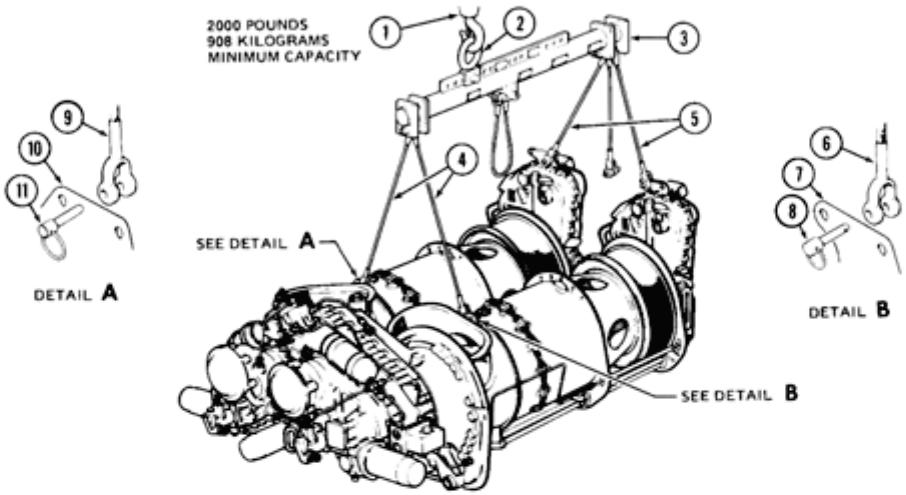
31. Prime the power plant fuel system (412-A-71-00-00-01A-232B-A / 01589).

32. Rig the power lever controls (412-A-76-10-00-00A-276A-A / 01725).

33. Rig the droop compensator controls (412-A-76-11-00-08A-280A-A / 01761).

34. *Check the torque transmitter adjustment (*412-A-76-10-00-00A-271A-A / 01727 *or* 412-A-76-10-00-00A-271B-A / 01728*).*

35. *Check the torque limiter adjustment (*412-A-76-10-00-00A-271C-A / 01730 *or* 412-A-76-10-00-00A-271D-A / 01731*).*



1. Hoist
2. Lifting eye
3. Engine sling, SWE-13833
4. Sling cables
5. Sling cables
6. Clevis
7. Lifting bracket (accessory gearbox)
8. Pin
9. Clevis
10. Lifting bracket (engine flange C)
11. Pin

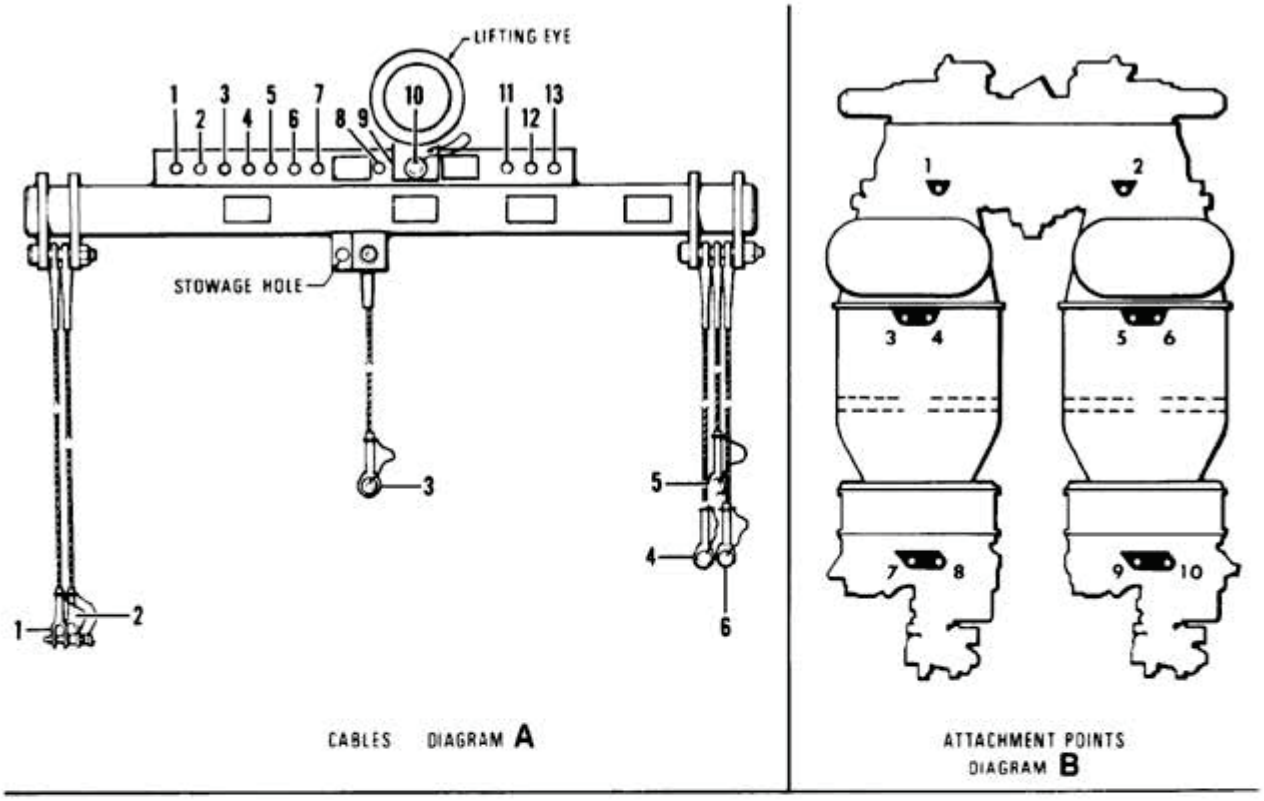
NOTES:

1. Determine which components listed in columns 1, 2 and 3 on sheet 2 are installed on engine to be hoisted.
2. Position lifting eye (2) in correct sling holes number for engine to be hoisted as shown in column 4 and diagram A.
3. Attach cables shown in column 5 to engine at points shown in column 6 and diagram B.

412-M-71-12-1

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Figure 1. Tool Application - Engine Sling (SWE-13833) (Sheet 1 of 2)

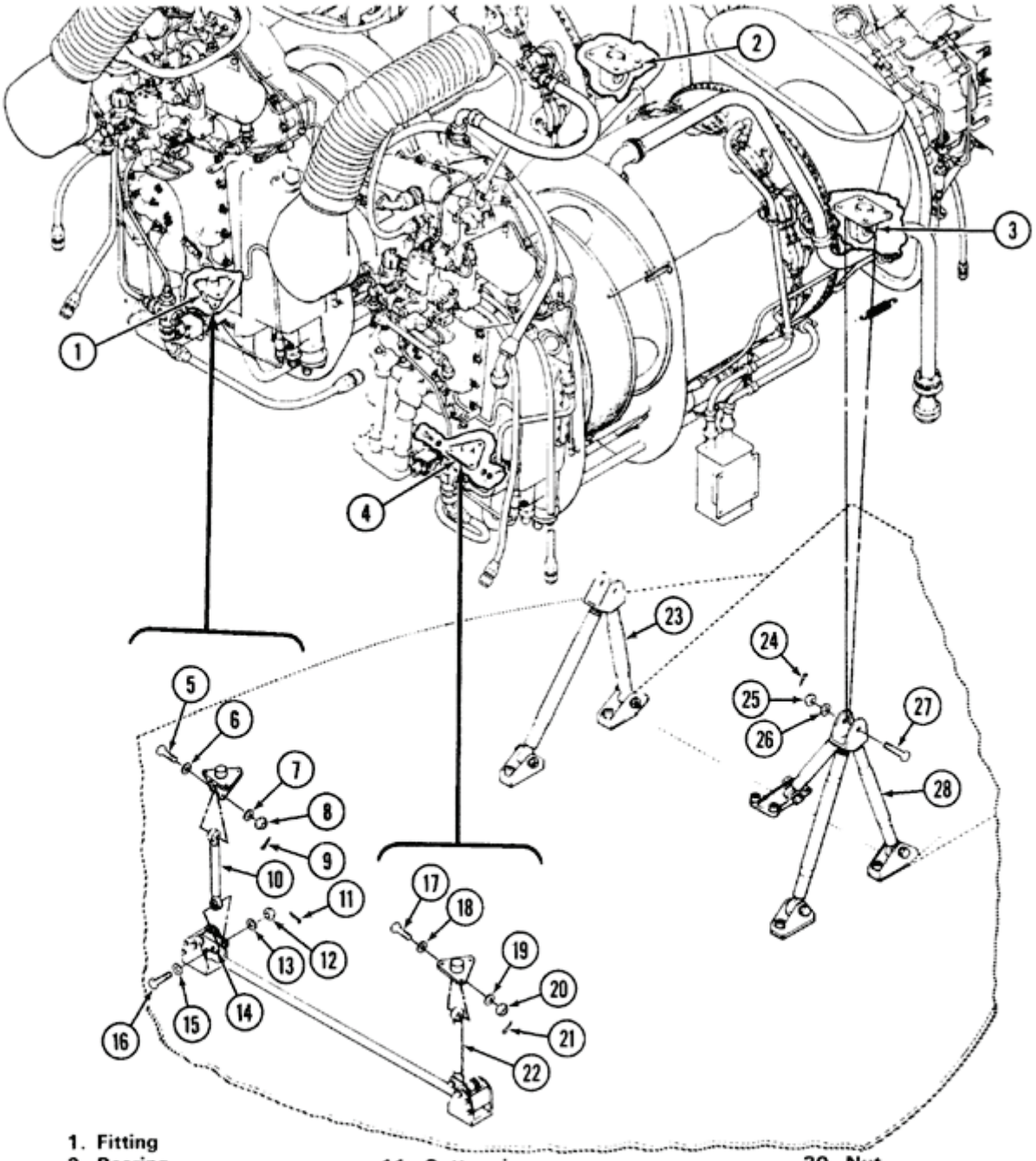


COLUMN

1	2	3	4	5	6
ITEM REMOVAL/INSTALLATION	Starter/ Generator Installed	Blower Installed	Sling Hole Number	Cables (See Diagram 'A')	Attachment Points (See Diagram 'B')
COMPLETE ENGINE	YES	YES	6	1, 2, 4, 6,	3, 6, 7, 9,
COMPLETE ENGINE	YES	NO	7	1, 2, 4, 6,	3, 6, 7, 9,
COMPLETE ENGINE	NO	YES	4	1, 2, 4, 6,	3, 6, 7, 9,
COMPLETE ENGINE	NO	NO	5	1, 2, 4, 6,	3, 6, 7, 9,
COMPLETE POWER SECTION LEFT	YES		10	1, 4,	5, 9,
COMPLETE POWER SECTION RIGHT	YES		10	1, 4,	4, 8,
COMPLETE POWER SECTION LEFT	NO		9	1, 4,	5, 9,
COMPLETE POWER SECTION RIGHT	NO		9	1, 4,	4, 8,
REDUCTION GEARBOX			11	3, 5,	1, 2,

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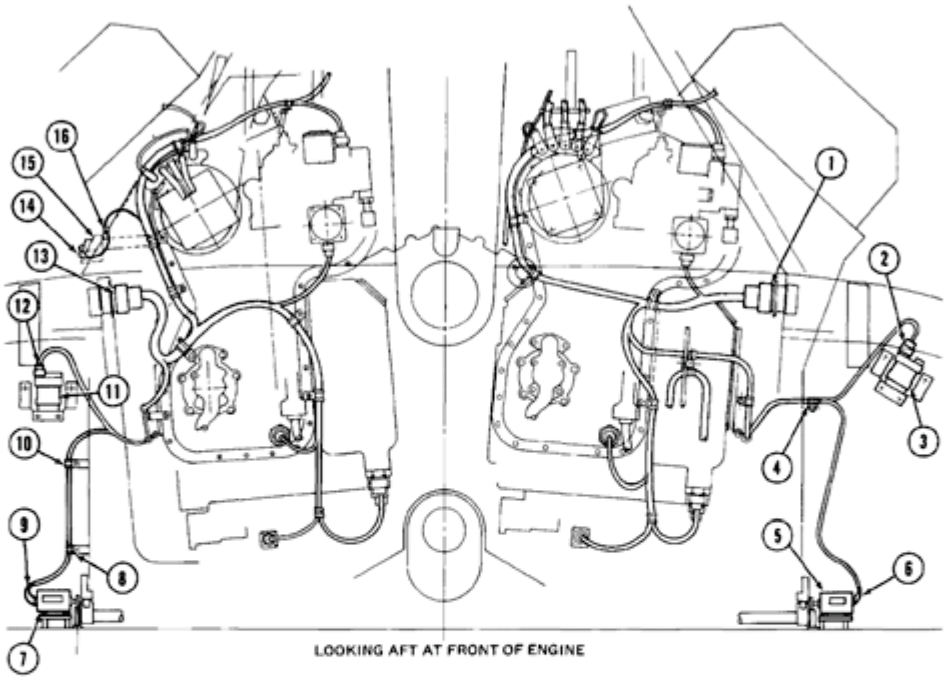
Figure 1. Tool Application - Engine Sling (SWE-13833) (Sheet 2 of 2)



1. Fitting
2. Bearing
3. Bearing
4. Fitting
5. Bolt
6. Thin steel washer
7. Thin steel washer
8. Nut
9. Cotter pin or safety pin
10. Tube
11. Cotter pin
12. Nut
13. Thin corrosion steel washer
14. Bellcrank
15. Thin corrosion steel washer
16. Bolt
17. Bolt
18. Thin steel washer
19. Thin steel washer
20. Nut
21. Cotter pin or safety pin
22. Tube
23. Bipod fitting
24. Cotter pin
25. Nut
26. Steel washer
27. Bolt
28. Tripod fitting

ICN-412-A-710001-B-97499-00011-A-001-01

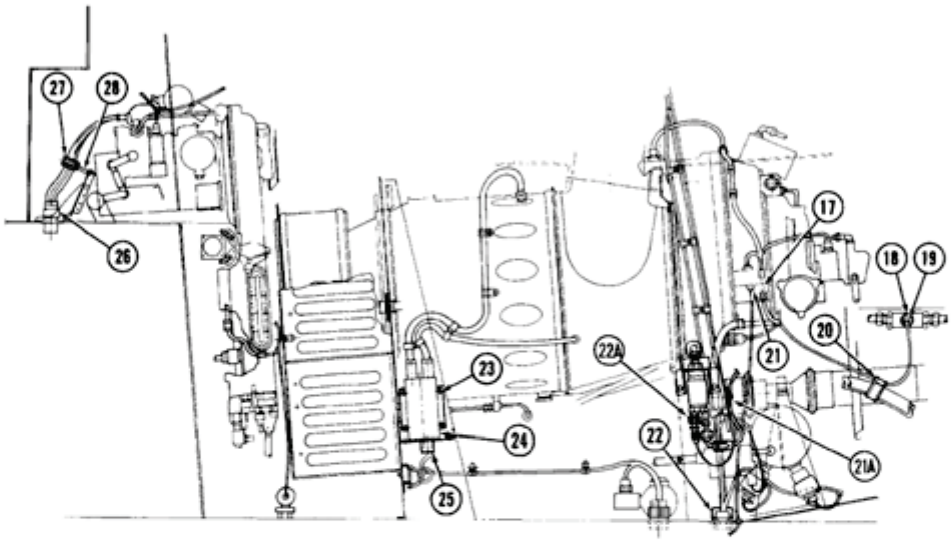
Figure 2. Engine Mount Connections (Sheet 1 of 1)



1. Engine electrical disconnect cable
2. Electrical connector
3. Fuel pressure transmitter
4. Clamp, screw, washer, and nut
5. Flight idle solenoid
6. Electrical leads
7. Flight idle solenoid
8. Clamp, screw, washer, and nut
9. Electrical leads
10. Clamp, screw, washer, and nut
11. Fuel pressure transmitter
12. Electrical connector
13. Engine electrical disconnect cable
14. Electrical leads
15. No. 2 generator overheat switch
16. Clamp

ICN-412-A-710001-B-97499-00005-A-001-01

Figure 3. Engine Electrical Connections (Sheet 1 of 4)

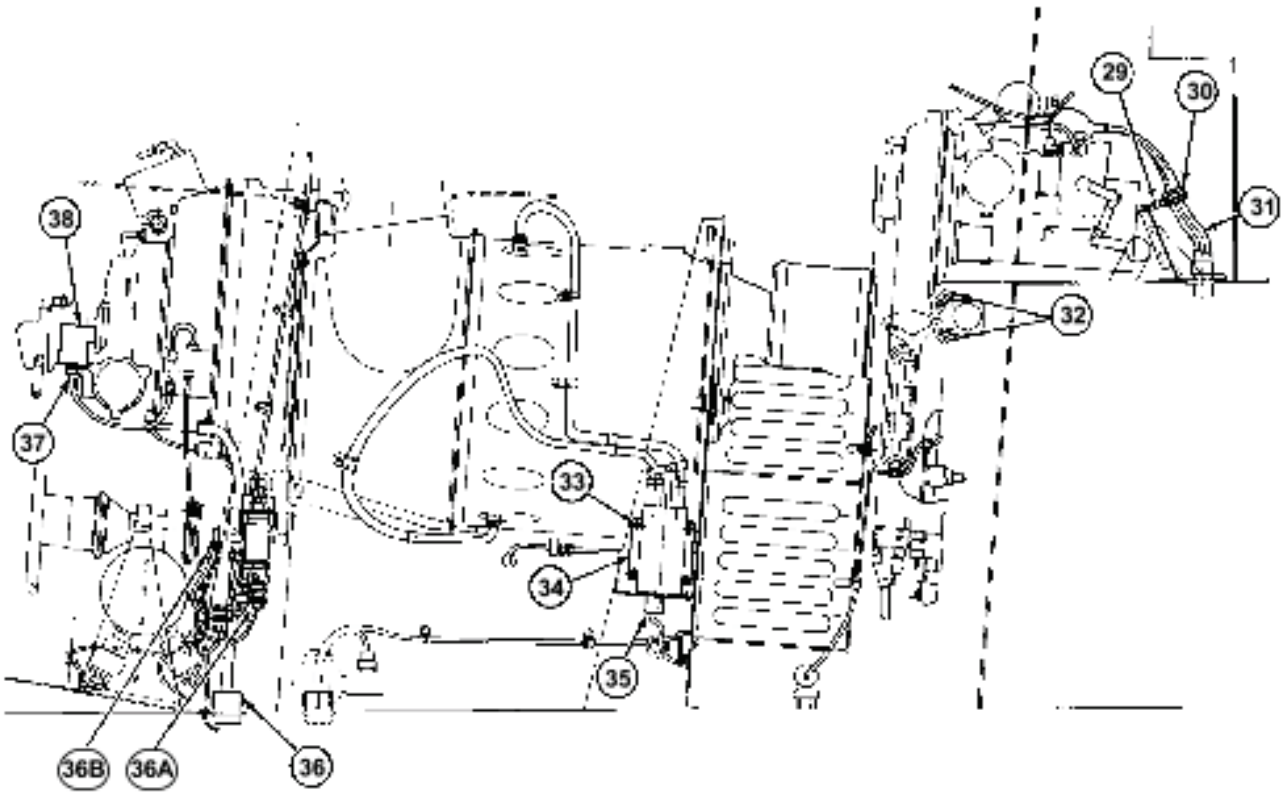


LOOKING INBOARD AT LEFT SIDE OF ENGINE

17. ITT compensator
18. Reduction gearbox oil temperature switch
19. Electrical connector
20. Clamp
21. Electrical connector
21A. Clamp
22. Engine aft electrical disconnect cable
22A. Electrical connector
23. Screw, washer, and nut
24. Ignition exciter
25. Electrical connector
26. Starter generator cable
27. Clamp
28. Screw, spacer, washer, and nut

ICN-412-A-710001-B-97499-00006-A-001-01

Figure 3. Engine Electrical Connections (Sheet 2 of 4)

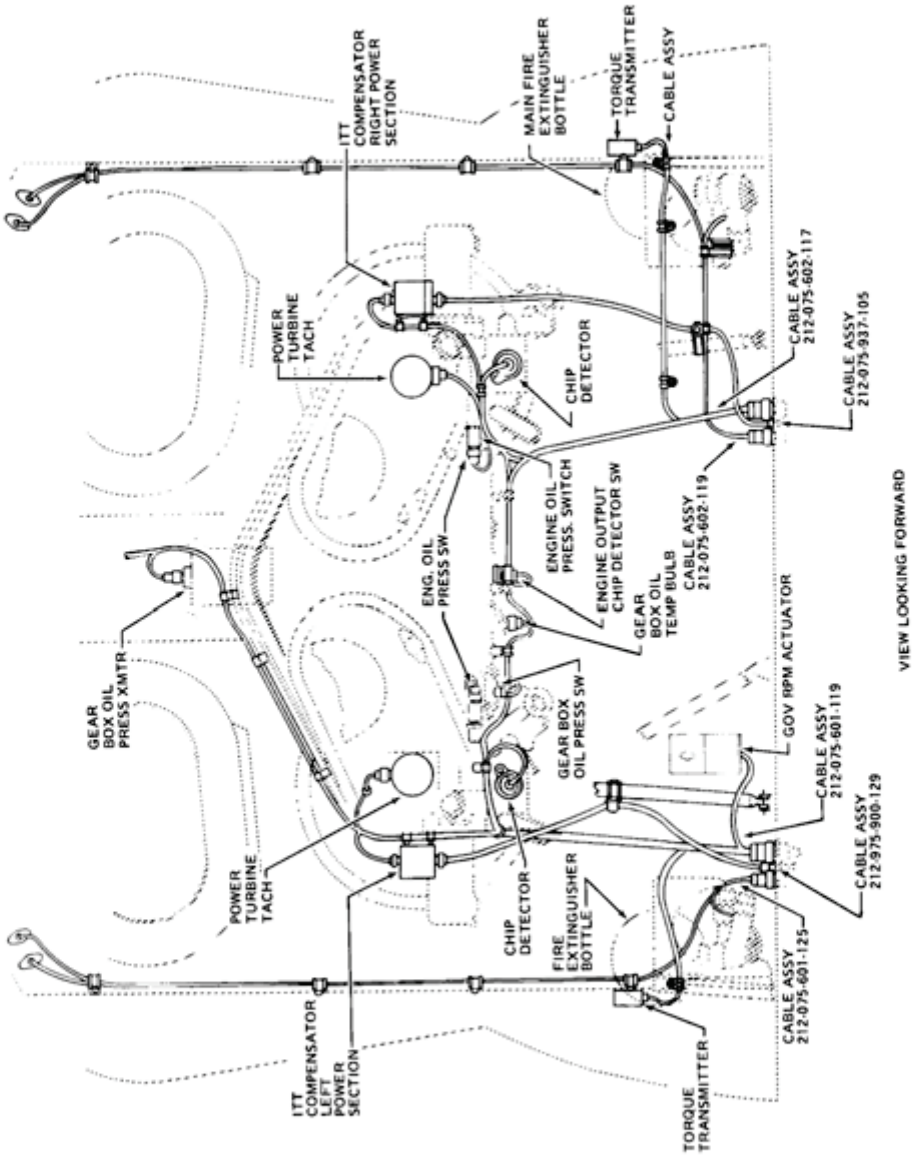


LOOKING INBOARD AT RIGHT-HAND SIDE OF ENGINE

29. Screw, spacer, washer, and nut
30. Clamp
31. Starter generator cable
32. Fire detector connector
33. Screw, washer, and nut
34. Ignition exciter
35. Electrical connector
36. Engine aft electrical disconnect cable
36A. Electrical connector
36B. Clamp
37. Electrical connector
38. Tachometer generator

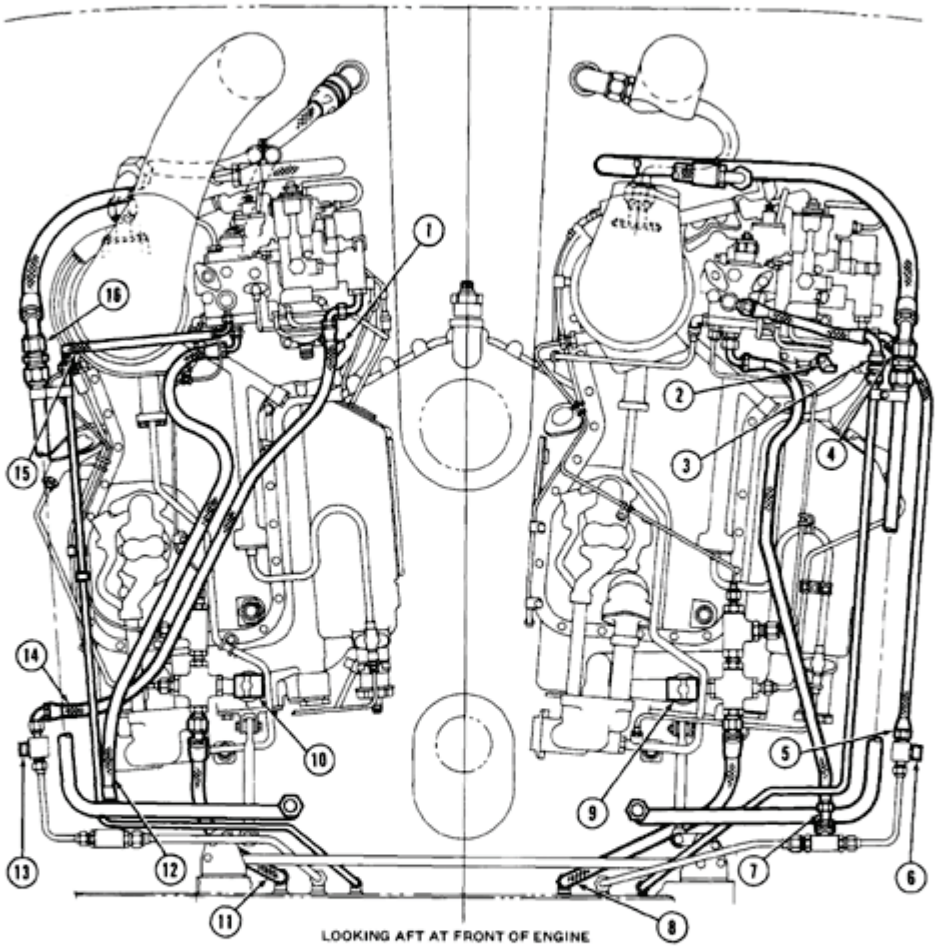
ICN-412-A-710001-B-97499-00007-A-001-01

Figure 3. Engine Electrical Connections (Sheet 3 of 4)



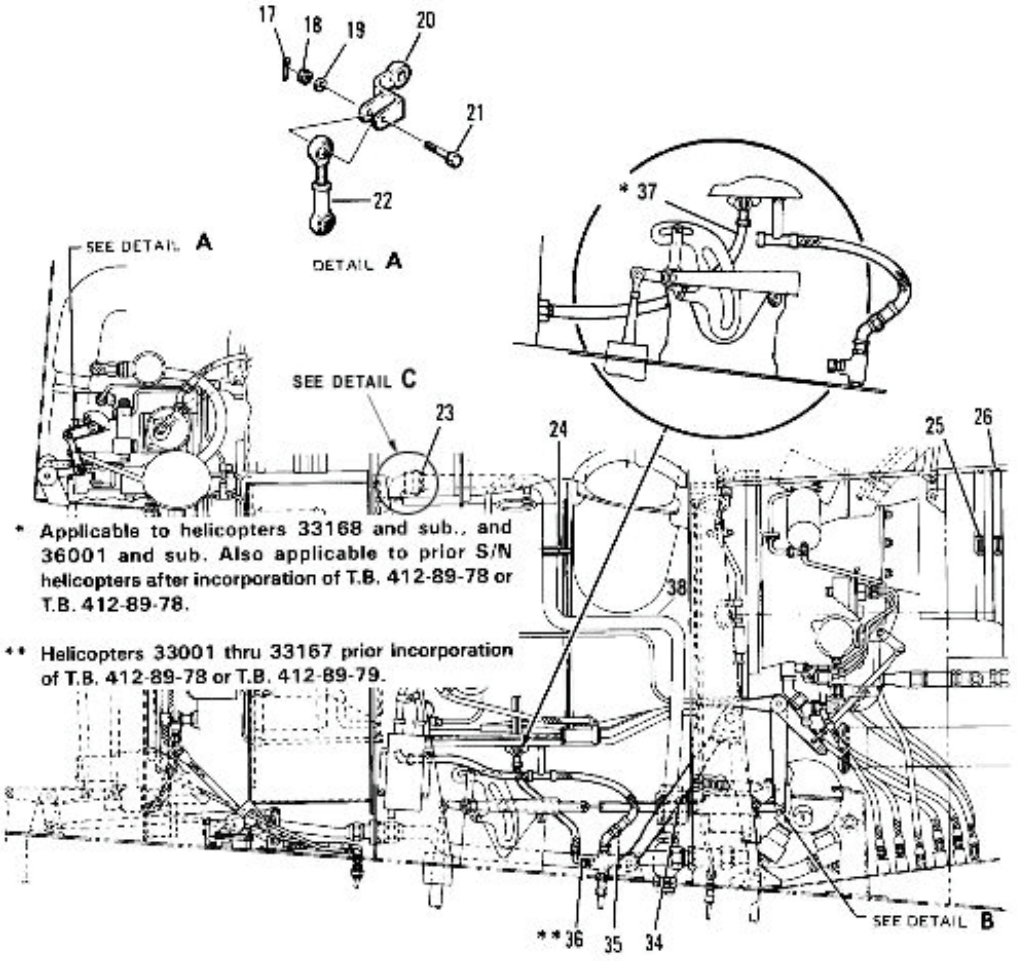
ICN-412-A-710001-B-97499-00008-A-001-01

Figure 3. Engine Electrical Connections (Sheet 4 of 4)



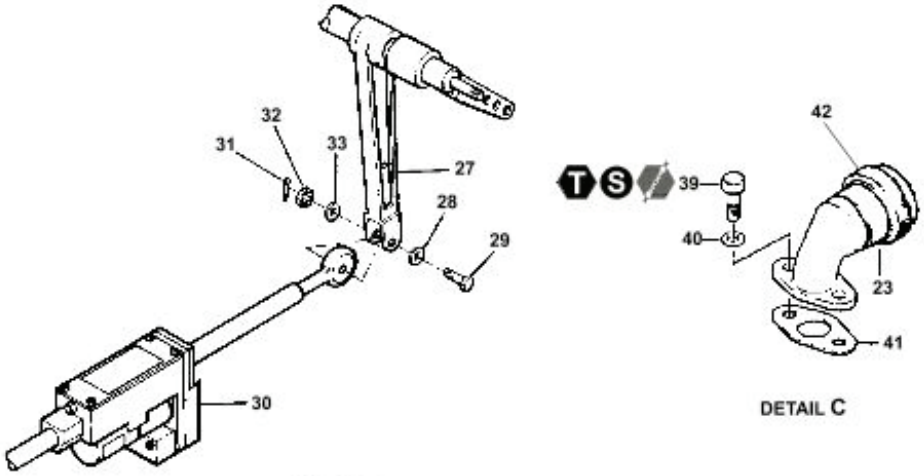
ICN-412-A-710001-B-97499-00001-A-001-01

Figure 4. Engine Installation Connections (Sheet 1 of 4)



ICN-412-A-710001-B-97499-00002-A-001-01

Figure 4. Engine Installation Connections (Sheet 2 of 4)



17. Cotter pin
18. Nut
19. Washer
20. Lever
21. Bolt
22. Link (control tube on right side)
23. Bleed air tube connector assembly
24. Clamp, screw, washer, and nut (left side only)
25. Screw
26. Oil cooler blower boot
27. Jackshaft assembly
28. Aluminum washer
29. Bolt
30. Actuator
31. Cotter pin
32. Nut
33. Thin aluminum washer
34. Adapter
35. Combustion chamber drain hose
36. Solenoid dump valve hose
37. Engine manifold drain hose
38. Bleed air tube
39. Bolt
40. Washer
41. Gasket
42. Nut

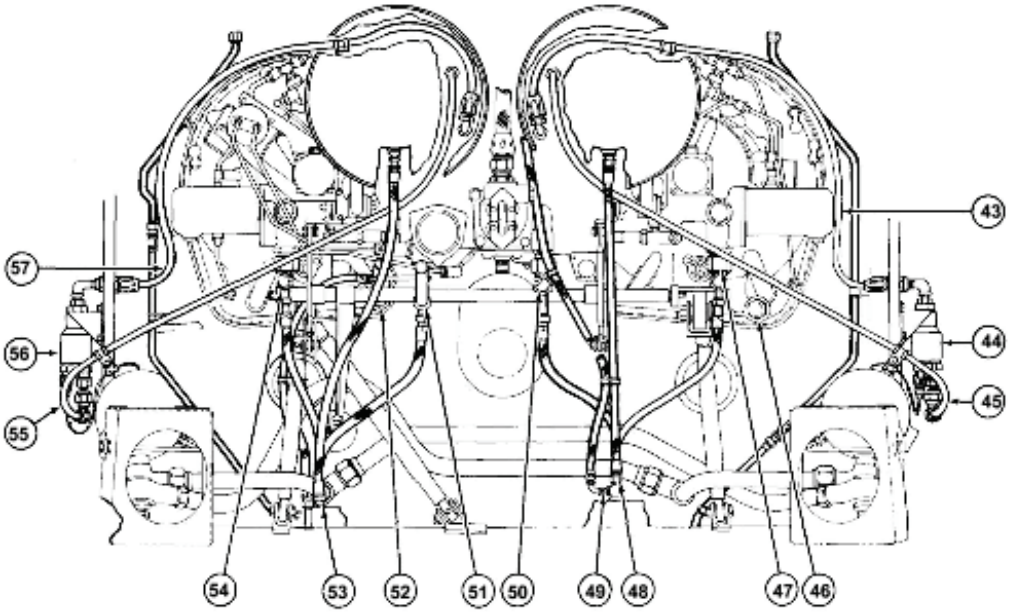
LOCKWIRE (C-405)

SEALANT (C-308)

85 TO 95 IN-LBS
(9.60 TO 10.73 Nm)

ICN-412-A-710001-B-97499-00003-A-001-01

Figure 4. Engine Installation Connections (Sheet 3 of 4)



LOOKING FORWARD AT REAR OF ENGINE

43. Torque transmitter pressure hose (right)
44. Torque transmitter (right)
45. Torque transmitter vent hose (right)
46. Right power section oil outlet hose
47. Right power section oil inlet hose
48. Blower seal drain hose
49. Reduction gearbox oil filler adapter drain hose
50. Reduction gearbox oil outlet hose
51. Reduction gearbox oil inlet hose
52. Left power section oil outlet hose
53. Blower seal drain hose
54. Left power section oil inlet hose
55. Torque transmitter vent hose (left)
56. Torque transmitter (left)
57. Torque transmitter pressure hose (left)

ICN-412-A-710001-B-97499-00004-A-001-01

Figure 4. Engine Installation Connections (Sheet 4 of 4)

Engine Power Section - Removal

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Engine Sling	SWE-13833	1	

Procedure

NOTE

Either the left or right power section may be removed and installed by the following procedures, with the opposite power section and reduction (combining) gearbox remaining in place.

1. Do step 1 through step 24 of 412-A-71-00-01-01A-520A-A / 01597 as applicable for the power section being removed.

2. Install the engine sling (SWE-13833). Refer to Figure 1 for cable usage and power section attachment points.



LIFTING CAPACITY OF HOIST SHALL BE 2000 POUNDS (908 KG) OR GREATER.

3. Attach the hoist (minimum 2000 pounds (908 kg) capacity) to the sling and take up slack without placing load on the engine mount bolts.

4. Disconnect the engine mount from the applicable power section (412-A-71-00-01-01A-520A-A / 01597, step 28).

5. Disconnect power section from engine. Refer to the applicable Pratt & Whitney Canada Engine Maintenance Manual.

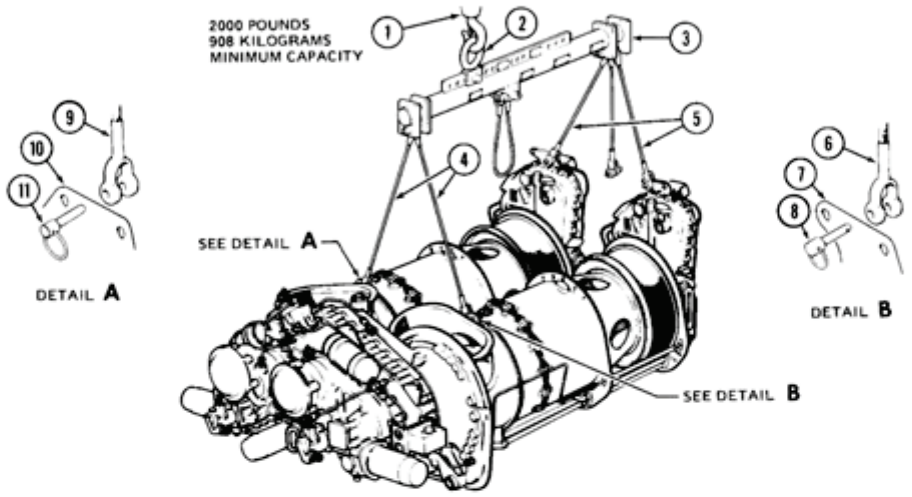
6. Inspect the power section to ensure all the connections are separated from the engine and helicopter airframe.

7. Move the power section approximately 1.0 inch (25.4 mm) forward until clear of the reduction (combining) gearbox.

8. Lift the power section vertically until clear of the helicopter. Watch all sides to ensure adequate clearance.

9. Place the power section on a suitable stand. Disconnect the hoist and sling.

10. Cover the openings on the power section and gearbox.



1. Hoist
2. Lifting eye
3. Engine sling, SWE-13833
4. Sling cables
5. Sling cables
6. Clevis
7. Lifting bracket (accessory gearbox)
8. Pin
9. Clevis
10. Lifting bracket (engine flange C)
11. Pin

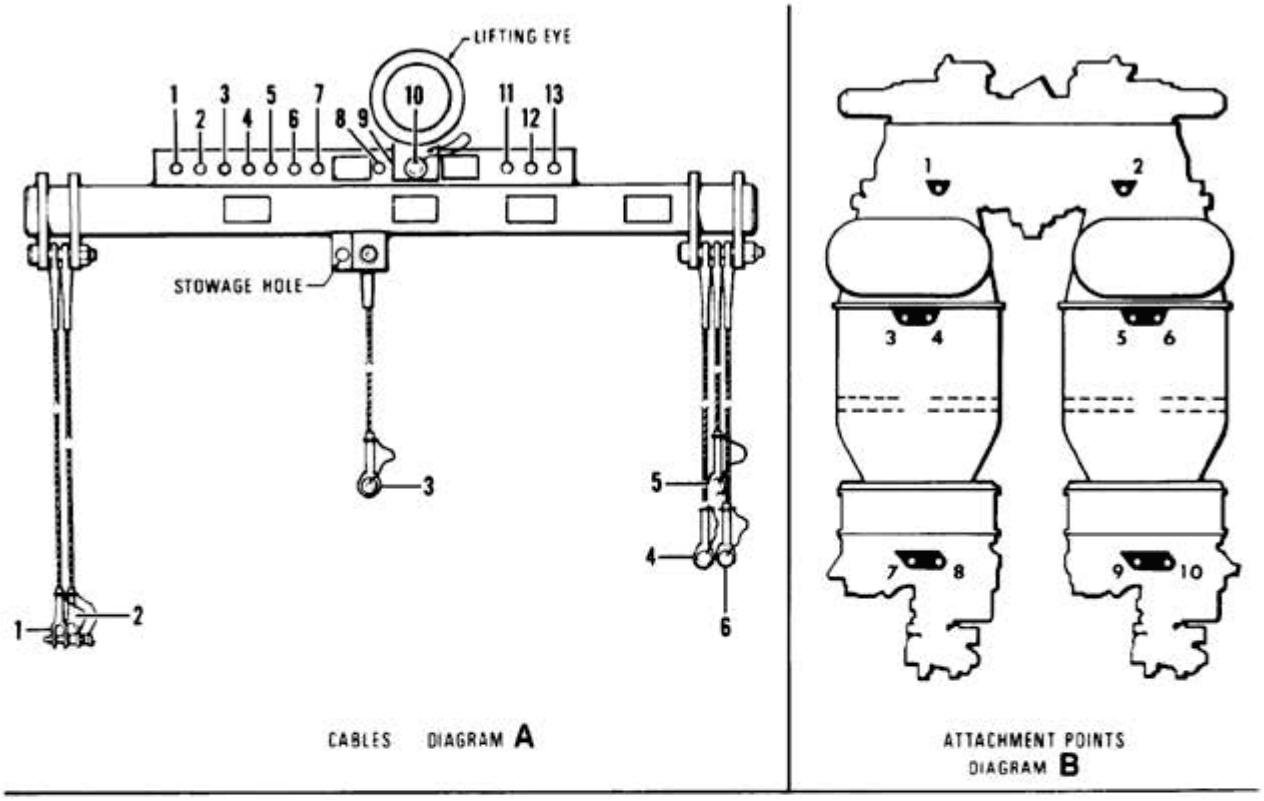
NOTES:

1. Determine which components listed in columns 1, 2 and 3 on sheet 2 are installed on engine to be hoisted.
2. Position lifting eye (2) in correct sling holes number for engine to be hoisted as shown in column 4 and diagram A.
3. Attach cables shown in column 5 to engine at points shown in column 6 and diagram B.

412-M-71-12-1

ICN-412-A-710001-B-97499-00009-A-001-01

Figure 1. Tool Application - Engine Sling (SWE-13833) (Sheet 1 of 2)



COLUMN

1	2	3	4	5	6
ITEM REMOVAL/INSTALLATION	Starter/ Generator Installed	Blower Installed	Sling Hole Number	Cables (See Diagram 'A')	Attachment Points (See Diagram 'B')
COMPLETE ENGINE	YES	YES	6	1, 2, 4, 6,	3, 6, 7, 9,
COMPLETE ENGINE	YES	NO	7	1, 2, 4, 6,	3, 6, 7, 9,
COMPLETE ENGINE	NO	YES	4	1, 2, 4, 6,	3, 6, 7, 9,
COMPLETE ENGINE	NO	NO	5	1, 2, 4, 6,	3, 6, 7, 9,
COMPLETE POWER SECTION LEFT	YES		10	1, 4,	5, 9,
COMPLETE POWER SECTION RIGHT	YES		10	1, 4,	4, 8,
COMPLETE POWER SECTION LEFT	NO		9	1, 4,	5, 9,
COMPLETE POWER SECTION RIGHT	NO		9	1, 4,	4, 8,
REDUCTION GEARBOX			11	3, 5,	1, 2,

ICN-412-A-710001-B-97499-00010-A-001-01

Figure 1. Tool Application - Engine Sling (SWE-13833) (Sheet 2 of 2)

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Engine Power Section - Installation

Procedure

NOTE

Either the left or right power section of the engine assembly can be installed by the following procedures, with the opposite power section and reduction gearbox remaining in place.

1. Install the engine sling (SWE-13833). Refer to Figure 1 for cable usage and power section attachment points.



LIFTING CAPACITY OF HOIST SHALL BE 2000 POUNDS (908 KG) OR GREATER.

2. Attach the hoist to the engine sling and move the power section near the helicopter.

3. If installing the right power section, attach the forward engine mount tube to the accessory gearbox mount fitting (412-A-71-00-01-01A-720A-A / 01598, step 4).

4. Make sure that the new seals are installed on the mating flange and oil tubes on the rear end of the power section. Lubricate the seals with engine oil (412-A-12-01-00-00A-040A-A / 00141).

5. Install the power section in the helicopter.

5.1. Position the power section above the engine compartment and lower until 6.0 inches (150 mm) clearance remains above the firewalls.

5.2. Route the electrical connector and wire connected at the lower left and right corner of the aft fire shield through the space below the reduction (combining) gearbox.

5.3. Push the middle panel of the lower middle firewall at least 1.4 inches (35.5 mm) forward.

5.4. Carefully lower the power section into position while observing the following:

5.4.1. Make sure that the seals on the mating flange and oil tubes on the rear of the power section are not damaged.

5.4.2. Position the mid-fireshield on the power section aft of the lower middle firewall.

5.4.3. Make sure that the seal on the air induction baffle is forward of the fireshield on the power section.

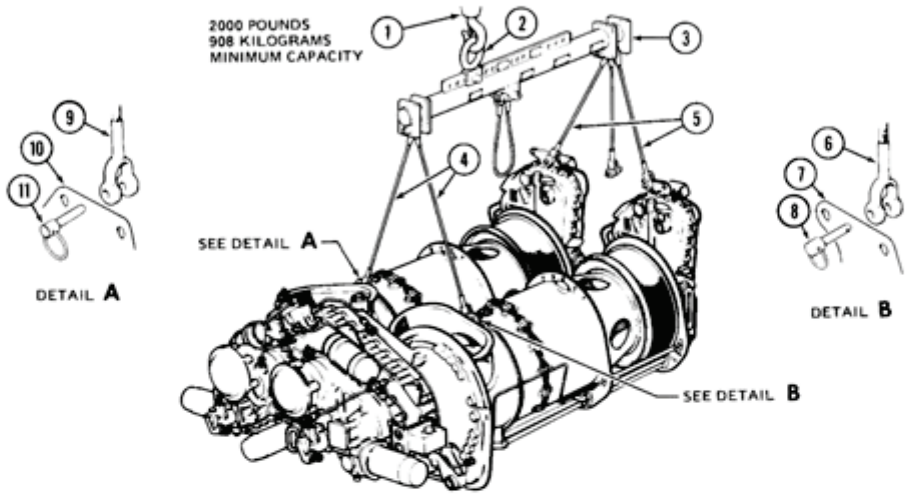
6. Mate the power section with the reduction gearbox and install the retaining nuts and washers. Refer to the applicable Pratt & Whitney Canada Engine Maintenance Manual.

7. Connect the engine mount (412-A-71-00-01-01A-720A-A / 01598).

8. Remove the hoist and sling.

9. Connect all the attachments between the power section and the engine. Refer to the applicable Pratt & Whitney Canada Engine Maintenance Manual.

10. Do step 10. through step 34 of 412-A-71-00-01-01A-720A-A / 01598, as necessary for the power section being installed.



1. Hoist
2. Lifting eye
3. Engine sling, SWE-13833
4. Sling cables
5. Sling cables
6. Clevis
7. Lifting bracket (accessory gearbox)
8. Pin
9. Clevis
10. Lifting bracket (engine flange C)
11. Pin

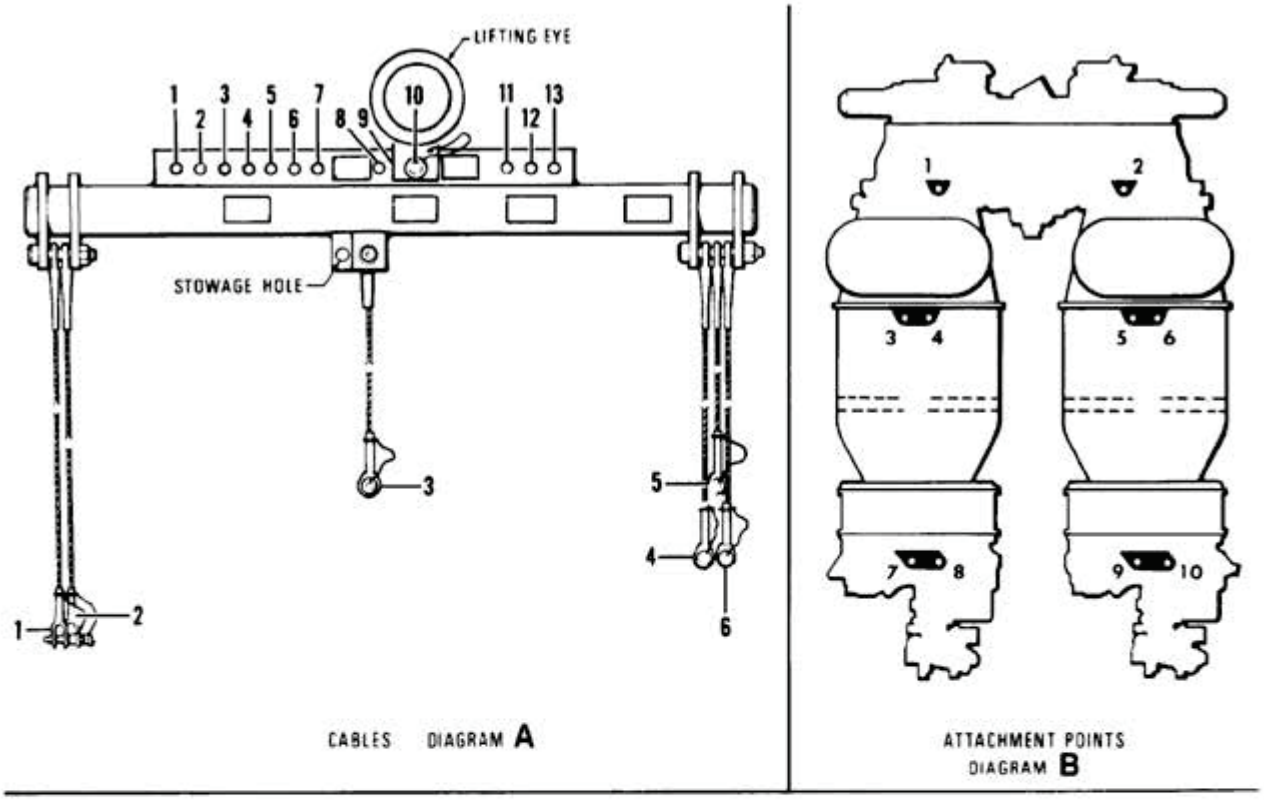
NOTES:

1. Determine which components listed in columns 1, 2 and 3 on sheet 2 are installed on engine to be hoisted.
2. Position lifting eye (2) in correct sling holes number for engine to be hoisted as shown in column 4 and diagram A.
3. Attach cables shown in column 5 to engine at points shown in column 6 and diagram B.

412-M-71-12-1

ICN-412-A-710001-B-97499-00009-A-001-01

Figure 1. Tool Application - Engine Sling (SWE-13833) (Sheet 1 of 2)



COLUMN

1	2	3	4	5	6
ITEM REMOVAL/INSTALLATION	Starter/ Generator Installed	Blower Installed	Sling Hole Number	Cables (See Diagram 'A')	Attachment Points (See Diagram 'B')
COMPLETE ENGINE	YES	YES	6	1, 2, 4, 6,	3, 6, 7, 9,
COMPLETE ENGINE	YES	NO	7	1, 2, 4, 6,	3, 6, 7, 9,
COMPLETE ENGINE	NO	YES	4	1, 2, 4, 6,	3, 6, 7, 9,
COMPLETE ENGINE	NO	NO	5	1, 2, 4, 6,	3, 6, 7, 9,
COMPLETE POWER SECTION LEFT	YES		10	1, 4,	5, 9,
COMPLETE POWER SECTION RIGHT	YES		10	1, 4,	4, 8,
COMPLETE POWER SECTION LEFT	NO		9	1, 4,	5, 9,
COMPLETE POWER SECTION RIGHT	NO		9	1, 4,	4, 8,
REDUCTION GEARBOX			11	3, 5,	1, 2,

ICN-412-A-710001-B-97499-00010-A-001-01

Figure 1. Tool Application - Engine Sling (SWE-13833) (Sheet 2 of 2)

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Reduction (Combining) Gearbox - Description

NOTE

The reduction (combining) gearbox can be replaced without removing either power section.

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Reduction (Combining) Gearbox - Removal

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Engine Sling	SWE-13833	1	

Procedure

1. Disconnect the electrical power.

2. Open the transmission cowling.

3. Remove the air inlet cowling.

4. Remove the engine cowling.

5. Remove the air scoop cowling.

6. Remove the gearbox cowling.

7. Remove the ejectors (412-A-71-61-00-01A-520A-A / 01701).

8. Remove the upper aft firewall (412-A-71-30-00-04A-520A-A / 01660).

9. Remove the main driveshaft and the adapter (412-A-63-10-01-01A-520A-A / 00994) and the adapter (412-A-63-10-01-02A-520A-A / 01001).

10. Remove the upper web (trough) of the centerline firewall (412-A-71-30-00-03A-520A-A / 01657).

11. Drain the oil from the gearbox.

12. Remove the oil cooler blowers (412-A-79-20-00-06A-520A-A / 01784).

13. Disconnect the hoses, electrical connectors, and engine controls that attach the gearbox to the helicopter (refer to engine removal, 412-A-71-00-01-01A-520A-A / 01597).

14. Remove the exhaust ducts and cover the openings.

15. Disconnect the engine connections between the gearbox and the power sections. Do not remove the nuts and washers around the mounting flange. Refer to the applicable Pratt & Whitney Canada Engine Maintenance Manual.

16. Support the power sections with a small hydraulic jack at flange C (exhaust duct to gas generator case) or a suitable sling and the hoist.

17. Install the engine sling (SWE-13833). Refer to Figure 1 for cable usage and gearbox attachment points.



LIFTING CAPACITY OF HOIST SHALL BE 2000 POUNDS (908 KG) OR GREATER.

18. Attach the hoist (minimum 2000 pounds (908 kg) capacity) to the sling and remove the tension. Do not put the load on the engine mount bolts.

19. Remove the nuts and washers that secure the gearbox to the power sections.

20. Disconnect the engine mounts from the gearbox (412-A-71-00-01-01A-520A-A / 01597, Step 28).

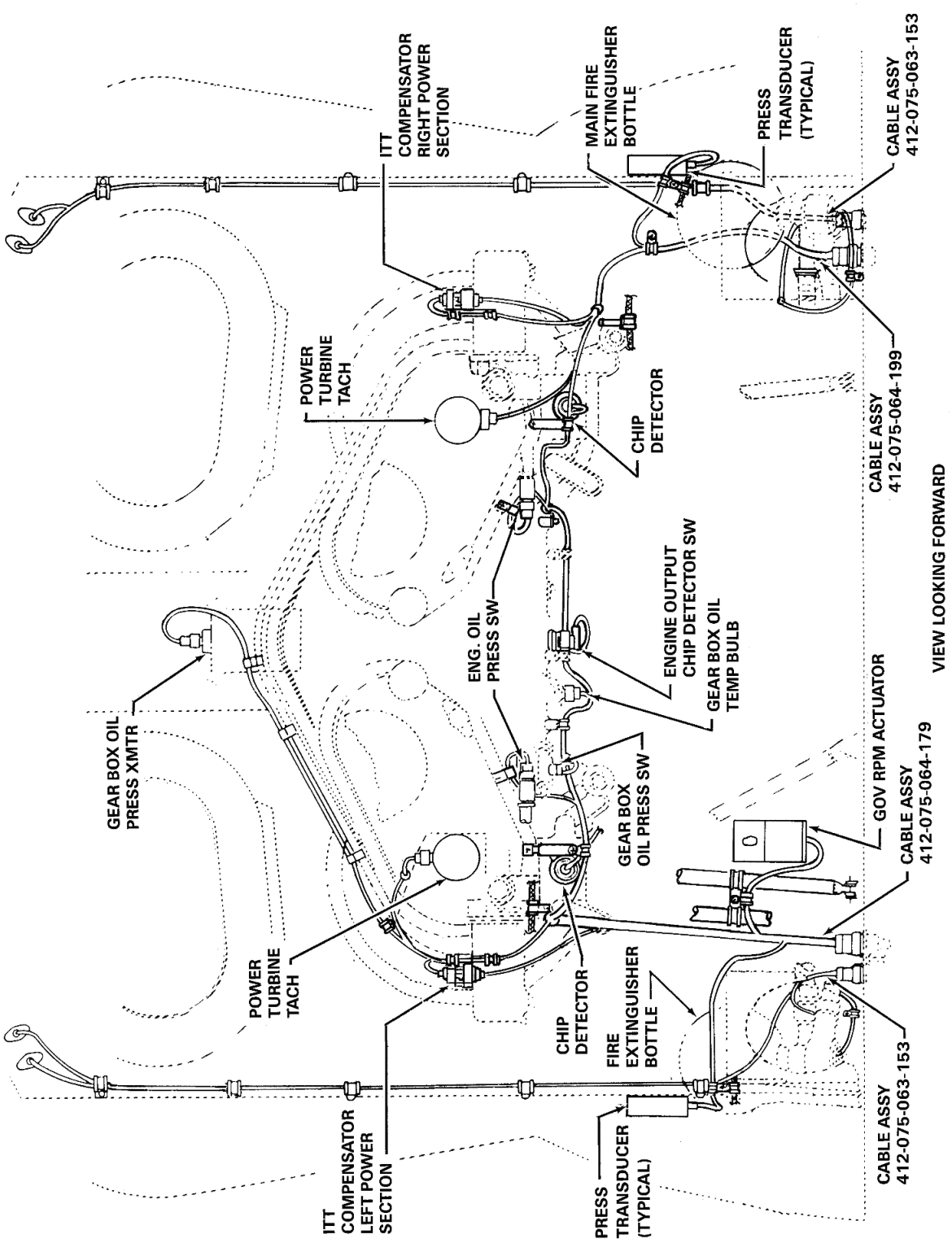
21. Inspect the gearbox to make sure that all the connections are separated from the power sections and the helicopter airframe.

22. Support the weight of the gearbox with the hoist. Slide the gearbox aft, approximately 1.0 inch (25.4 mm), to clear the power sections.

23. Lift the gearbox vertically until clear of the helicopter. Look all sides to make sure that there is sufficient clearance.

24. Place the gearbox on a suitable stand.

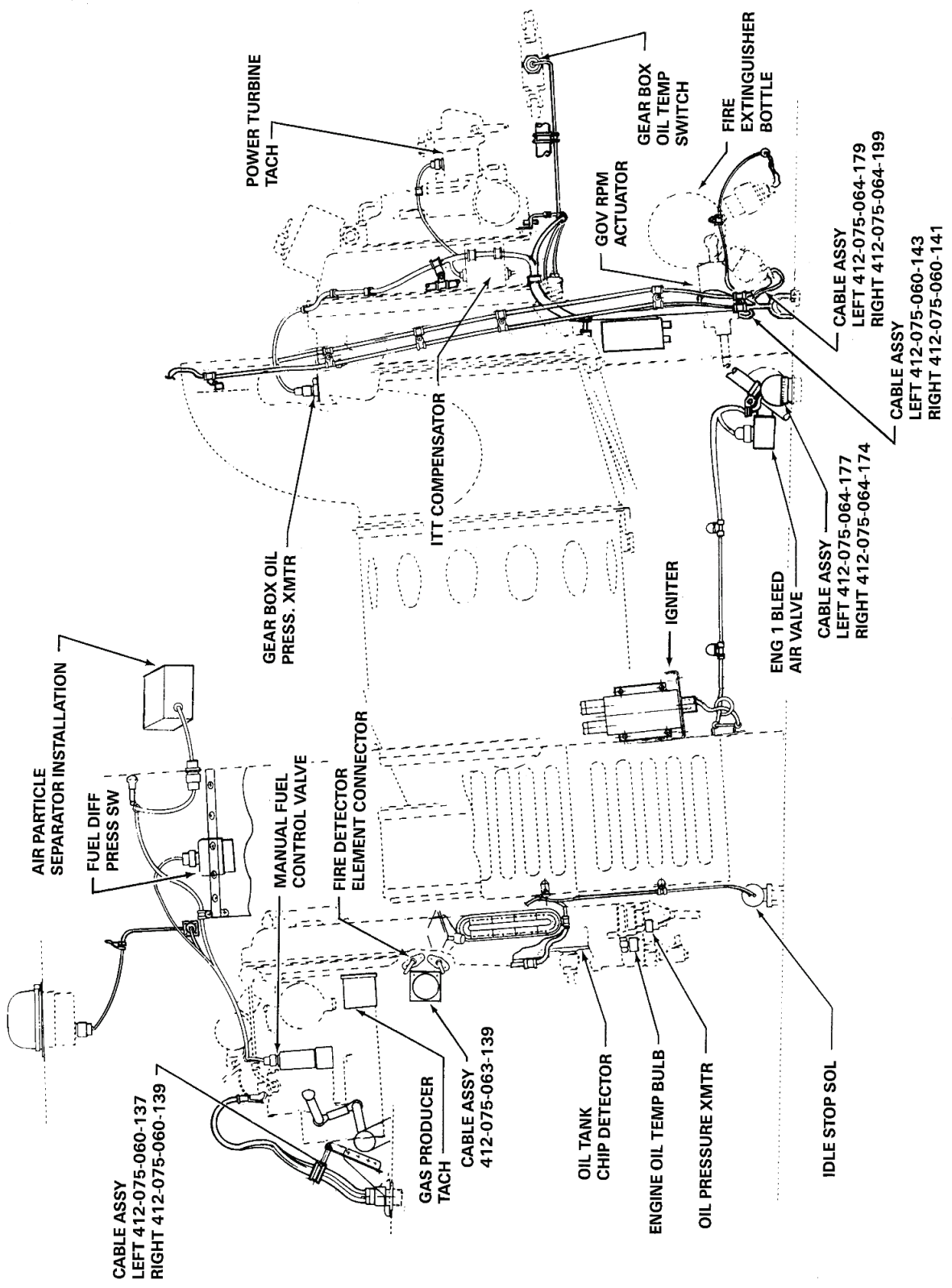
25. Cover the openings on the power sections and the reduction (combining) gearbox.



EFFECTIVITY: ALL

ICN-412-A-710000-B-97499-00009-A-001-01

Figure 1. Tool Application - Engine Sling (SWE-13833) (Sheet 1 of 2)



ICN-412-A-710000-B-97499-00010-A-001-01

Figure 1. Tool Application - Engine Sling (SWE-13833) (Sheet 2 of 2)

Reduction (Combining) Gearbox - Installation

Procedure

1. Prepare the reduction (combining) gearbox for installation.

1.1. Transfer the parts not provided on the replacement gearbox from the removed gearbox.

1.2. Install new packings on the mating faces and the oil transfer tubes.

2. Install the engine sling (SWE-13833). Refer to Figure 1 for cable usage and gearbox attachment points.



LIFTING CAPACITY OF HOIST SHALL BE 2000 POUNDS (908 KG) OR GREATER.

3. Attach the hoist (minimum 2000 pounds (908 kg) capacity) to the engine sling and move the gearbox into position aft of the power sections.

4. Mate the gearbox with the power sections and install the retaining nuts and washers. Refer to the applicable Pratt & Whitney Canada Engine Maintenance Manual.

5. Connect the engine mounts (412-A-71-00-01-01A-720A-A / 01598).

6. Remove the jacks or sling used to support the power sections. Remove the engine sling and hoist from the gearbox.

7. Connect all the attachments between the gearbox and power sections. Install the exhaust ducts. Refer to the applicable Pratt & Whitney Canada Engine Maintenance Manual.

8. Connect the hoses, electrical connectors and engine controls that attach the gearbox to the helicopter (refer to engine installation, 412-A-71-00-01-01A-720A-A / 01598).

9. Install the oil cooler blowers (412-A-79-20-00-06A-720A-A / 01786).

10. Install the upper web (trough) of the centerline firewall (412-A-71-20-00-02A-720A-A / 01629).

11. Install the main driveshaft adapter (412-A-63-10-01-02A-720A-A / 01004) and the driveshaft (412-A-63-10-01-01A-720A-A / 00999).

12. Install the upper aft firewall (412-A-71-20-00-03A-720A-A / 01632).

13. Install the ejectors (412-A-71-61-00-01A-720A-A / 01704).

14. Install the combining gearbox cowling.

15. Install the air scoop cowling.

16. Install the engine cowling.

17. Install the air inlet cowling.

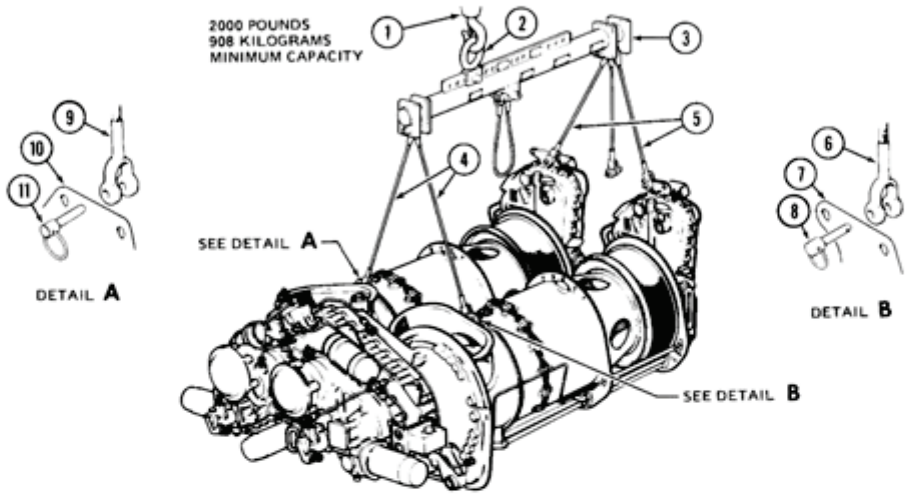
18. Close the transmission cowlings.

19. Service the gearbox (412-A-12-00-00-00A-040A-A / 00126).

20. Pre-oil the gearbox (412-A-71-00-00-01A-212A-A / 01588).

21. *Check the torque transmitter adjustment (*412-A-76-10-00-00A-271A-A / 01727 *or* 412-A-76-10-00-00A-271B-A / 01728*).*

22. *Check the torque limiter adjustment (*412-A-76-10-00-00A-271C-A / 01730 *or* 412-A-76-10-00-00A-271D-A / 01731*).*



1. Hoist
2. Lifting eye
3. Engine sling, SWE-13833
4. Sling cables
5. Sling cables
6. Clevis
7. Lifting bracket (accessory gearbox)
8. Pin
9. Clevis
10. Lifting bracket (engine flange C)
11. Pin

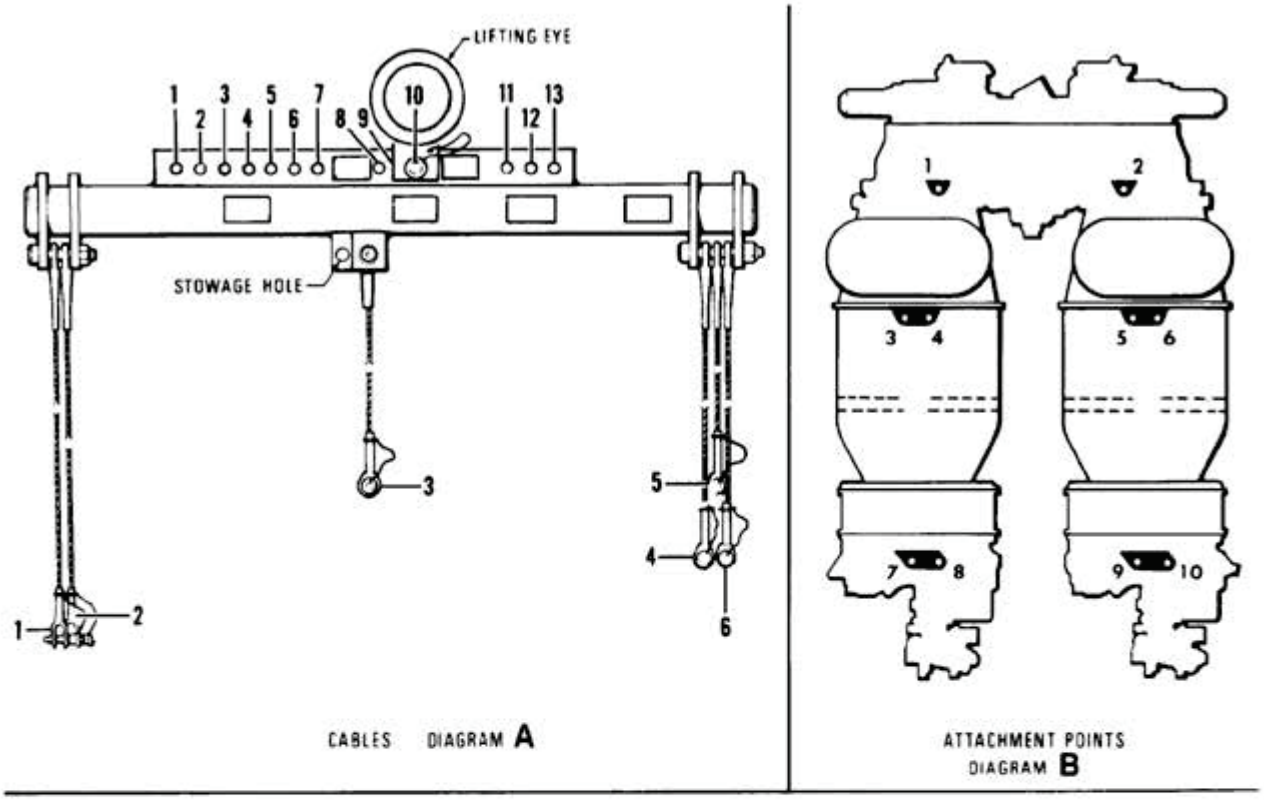
NOTES:

1. Determine which components listed in columns 1, 2 and 3 on sheet 2 are installed on engine to be hoisted.
2. Position lifting eye (2) in correct sling holes number for engine to be hoisted as shown in column 4 and diagram A.
3. Attach cables shown in column 5 to engine at points shown in column 6 and diagram B.

412-M-71-12-1

ICN-412-A-710001-B-97499-00009-A-001-01

Figure 1. Tool Application - Engine Sling (SWE-13833) (Sheet 1 of 2)



COLUMN

1	2	3	4	5	6
ITEM REMOVAL/INSTALLATION	Starter/ Generator Installed	Blower Installed	Sling Hole Number	Cables (See Diagram 'A')	Attachment Points (See Diagram 'B')
COMPLETE ENGINE	YES	YES	6	1, 2, 4, 6,	3, 6, 7, 9,
COMPLETE ENGINE	YES	NO	7	1, 2, 4, 6,	3, 6, 7, 9,
COMPLETE ENGINE	NO	YES	4	1, 2, 4, 6,	3, 6, 7, 9,
COMPLETE ENGINE	NO	NO	5	1, 2, 4, 6,	3, 6, 7, 9,
COMPLETE POWER SECTION LEFT	YES		10	1, 4,	5, 9,
COMPLETE POWER SECTION RIGHT	YES		10	1, 4,	4, 8,
COMPLETE POWER SECTION LEFT	NO		9	1, 4,	5, 9,
COMPLETE POWER SECTION RIGHT	NO		9	1, 4,	4, 8,
REDUCTION GEARBOX			11	3, 5,	1, 2,

ICN-412-A-710001-B-97499-00010-A-001-01

Figure 1. Tool Application - Engine Sling (SWE-13833) (Sheet 2 of 2)

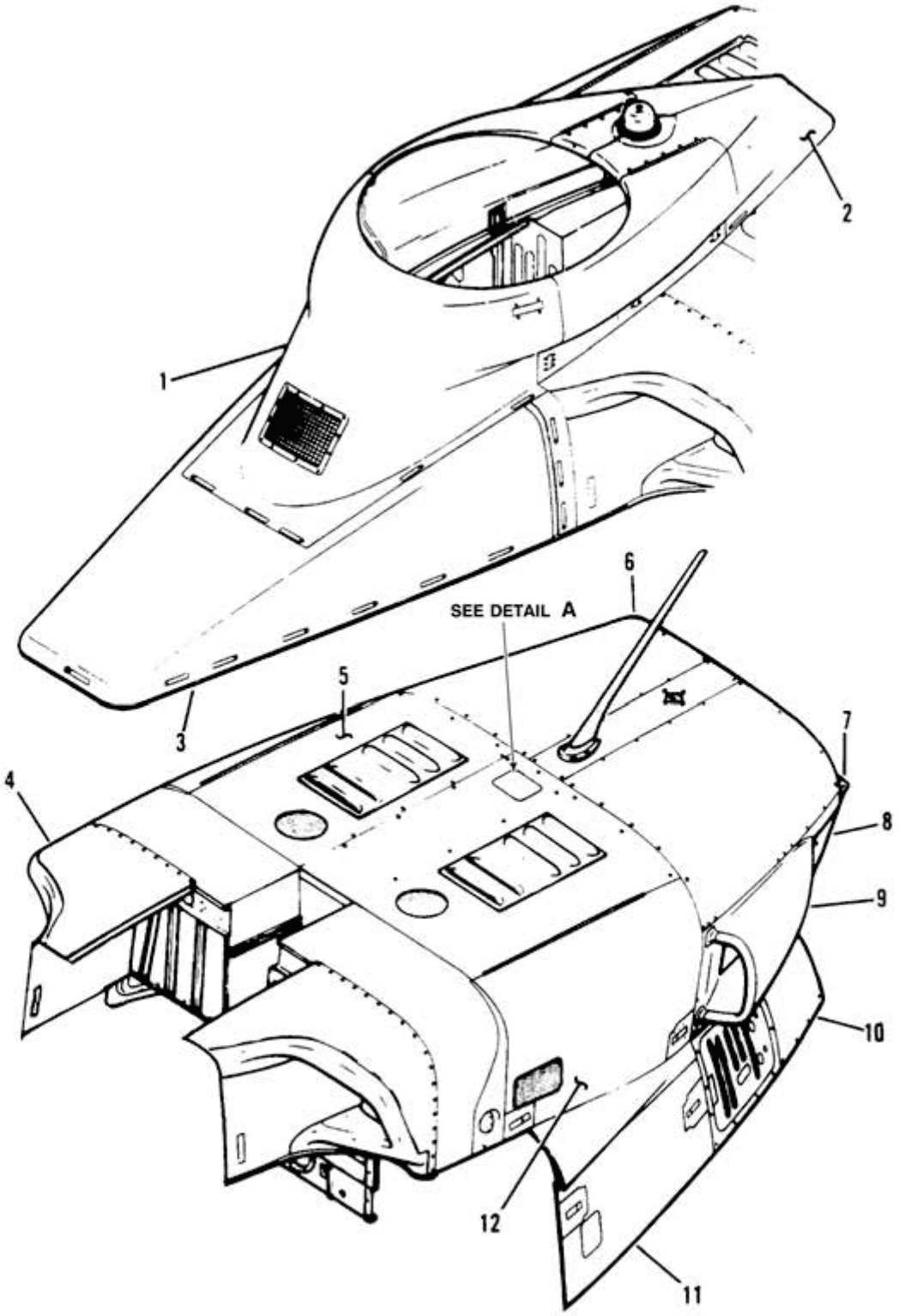
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Cowlings - Description

Cowlings are utilized to protect and provide easy access to engine, reduction (combining) gearbox air intake, exhaust ejectors, and top of main transmission.

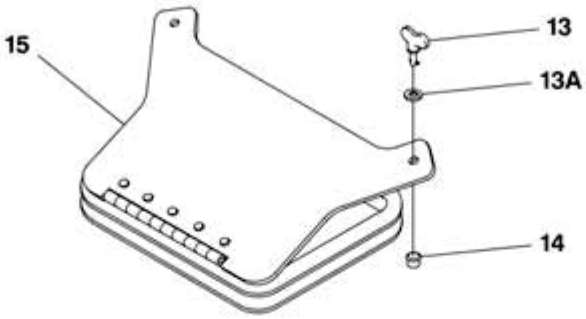
Applicable: SN: 36346-Subsequent:

The forward pylon fairing (1, Figure 1) is secured with two latches, seven fasteners, and one hinge assembly. If applicable, the aft pylon fairing (2) is secured with six latches, six fasteners, and four screws, and supports the anti-collision light. The transmission fairing (3) is secured with 19 studs. The engine air inlet cowlings (4) are secured with three fasteners, one latch, and screws. The lower and upper engine cowlings (11 and 12) may be constructed of sheet metal or composite materials. The composite engine cowlings were introduced in production. The lower engine cowling (11) is secured by two latches, a cowl lock, and a brace assembly that serves as the hinge. The upper engine cowling (12) is secured by two latches and hinges. The upper engine cowling is held open with a hold open brace. The following cowlings are secured by screws: engine top cowling (5), reduction gearbox cowling (6), oil cooler cowling (7), engine oil cooler support cowling (8), oil cooler side scoop cowling (9), and reduction gearbox side cowling (10). Reduction (combining) gearbox side cowling has an inspection door in the front half.



ICN-412-A-711000-B-97499-00001-A-001-01

Figure 1. Transmission and Engine Cowling (Sheet 1 of 2)



DETAIL A ⚠1

1. Forward pylon fairing
2. Aft pylon fairing
3. Transmission fairing
4. Engine air inlet cowling
5. Engine top cowling
6. Reduction gearbox cowling
7. Oil cooler cowling
8. Engine oil cooler support cowling
9. Oil cooler side scoop
10. Reduction gearbox side cowling
11. Lower engine cowling
12. Upper engine cowling
13. Stud panel fastener
13A. Washer
14. Receptacle panel fastener
15. Main drive aft inspection door

NOTE

⚠1 Applicable to S/N 36189, 36190, 36206 and subsequent and helicopters modified by TB 412-99-155.

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Figure 1. Transmission and Engine Cowling (Sheet 2 of 2)

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Forward and Aft Pylon Fairings - Removal

Procedure

Applicable: SN: 33001-33213 or 36001-36167 TB 412-00-169: Pre:

1. Remove the forward pylon fairing (1, Figure 1) as follows:

1.1. Press latches (5) open. Unfasten fasteners (6), and open.

NOTE

As an alternative, the forward pylon fairing (1) can be removed with the transmission fairing (412-A-71-10-00-02A-520A-A / 01608) without disconnecting the hinge assembly.

1.2. To remove the forward pylon fairing (1), remove cotter pins (9), washers (8), and pins (7), and remove fairing from the hinge assembly (10).

Applicable: SN: 36168-Subsequent or TB 412-00-169: Post:

2. Remove the forward pylon fairing (1) as follows:

2.1. Press latches (5) open and unfasten fasteners (6).

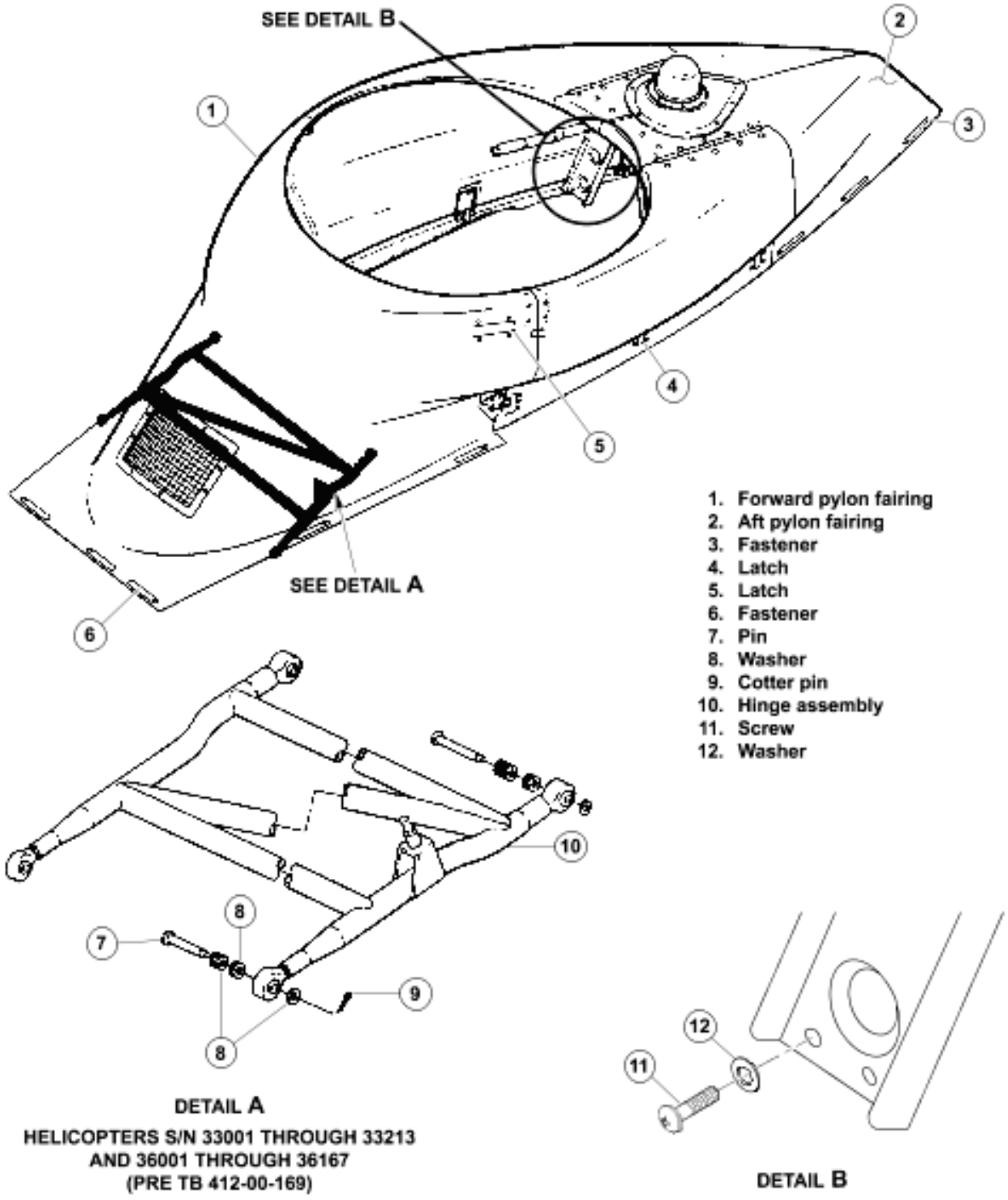
2.2. Remove the forward pylon fairing (1).

2.3. Remove the aft pylon fairing (2) as follows:

2.4. Disconnect the electrical connector P50 or 8DS8P1, as applicable, from the upper anticollision light.

2.5. Remove the screws (11) and washers (12).

2.6. Press latches (4) open and unfasten fasteners (3). Remove aft pylon fairing (2).



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Figure 1. Forward and Aft Pylon Fairings (Sheet 1 of 1)

Forward and Aft Pylon Fairings - Inspection and Repair

Procedure

1. Inspect latches, fittings, fasteners, and hinges for wear, damage, and serviceability.
2. Inspect seals for cracks, tears, deterioration, and security of bonding.
3. Replace latches, fittings, and seals if unserviceable.
4. Inspect fairings for dents, cracks, and damage. Repair fairings using maintenance practices in accordance with FAA Publication AC 43.13.1, Aircraft Inspection and Repair.

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Forward and Aft Pylon Fairings - Installation

Procedure

1. If removed, install the engine air inlet cowlings and the transmission fairing (412-A-71-10-00-02A-720A-A / 01611).

2. Install the aft pylon fairing (2, Figure 1) as follows:

2.1. Put the aft pylon fairing (2) in position and secure latches (4) and fasteners (3).

2.2. Install the four screws (11) and washers (12). Tighten the screws.

2.3. Connect the electrical connector P50 or 8DS8P1, as applicable, to the upper anticollision light.

3. Install the forward pylon fairing (1) as follows:

NOTE

Washers may be stacked in any combination required on inboard side of bearings.

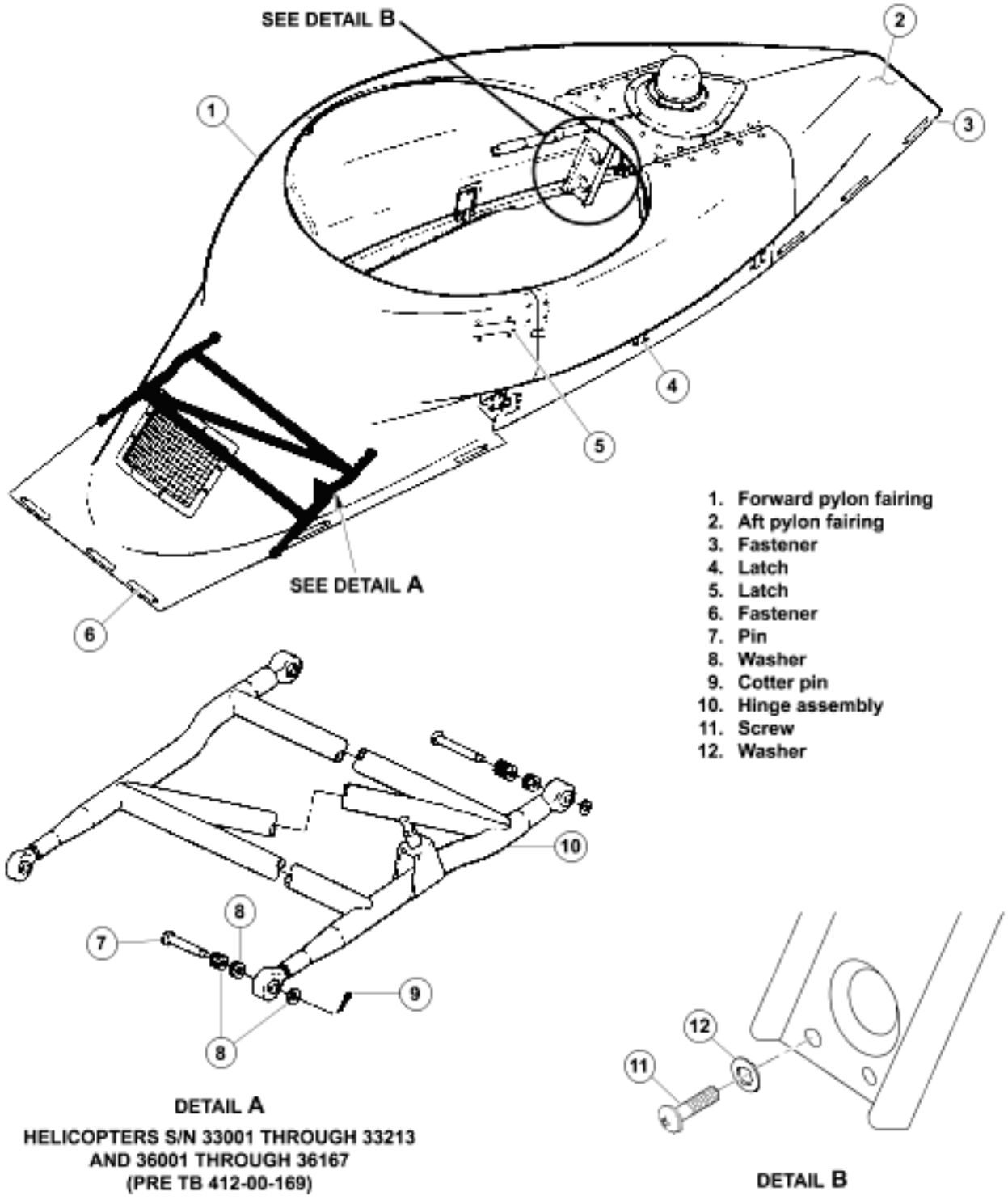
Applicable: SN: 33001-33213 or 36001-36167 TB 412-00-169: Pre:

3.1. Position the forward pylon fairing (1) in hinged open position. Install hinge with pins (7), washers (8), and new cotter pins (9).

Applicable: SN: 36168-Subsequent or TB 412-00-169: Post:

3.2. Put the forward pylon fairing (1) in position.

3.3. Close forward pylon fairing (1), if applicable, and secure latches (5) and fasteners (6).



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Figure 1. Forward and Aft Pylon Fairings (Sheet 1 of 1)

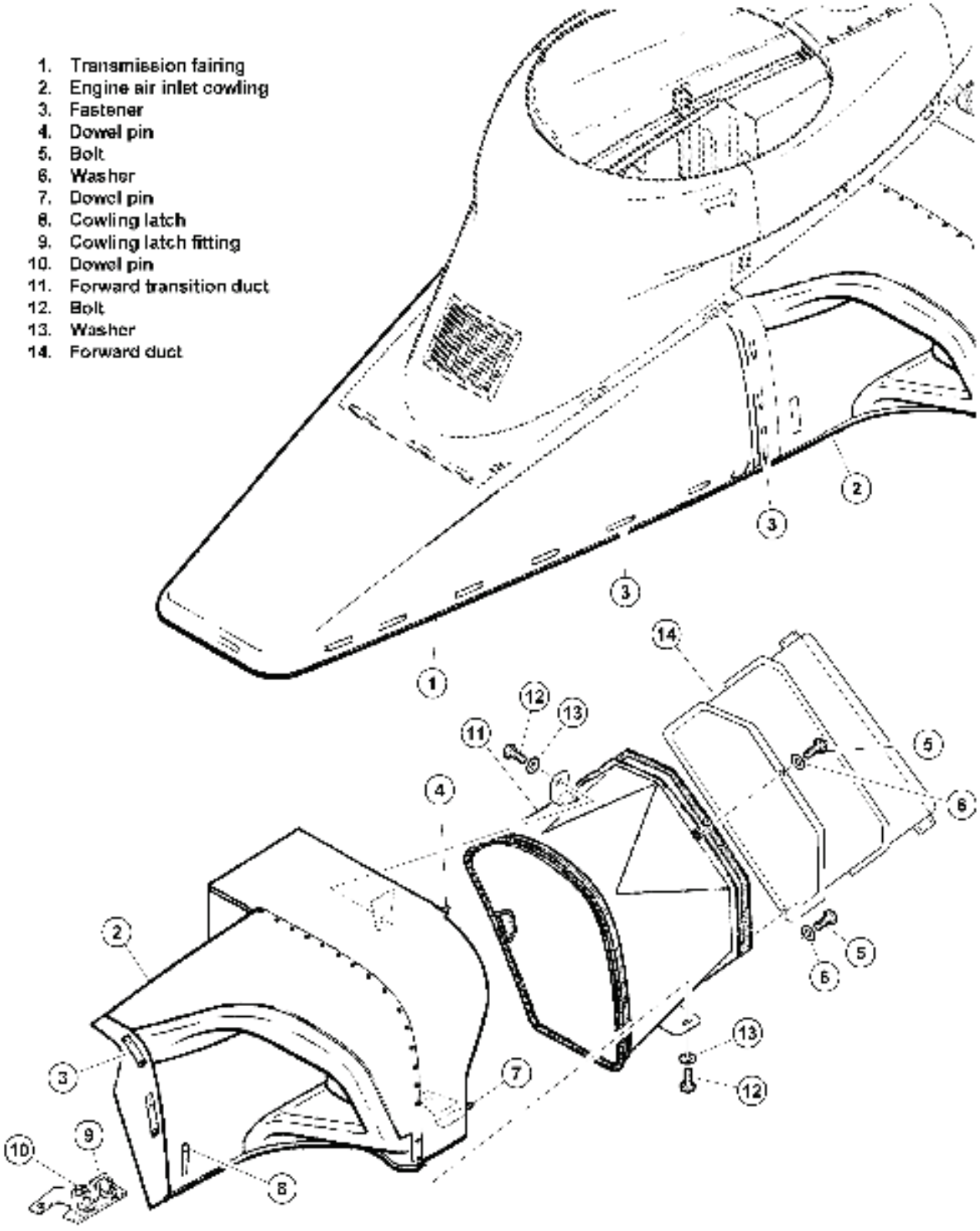
Transmission Fairings and Engine Air Inlet Cowlings - Removal

Procedure

1. Remove forward and aft pylon fairings (412-A-71-10-00-01A-520A-A / 01605).

2. Remove transmission fairing (1, Figure 1). Unfasten fasteners (3), and remove fairing.

3. Remove the left-hand or right-hand engine air inlet cowlings (2). Remove two bolts (5) and washers (6) from the forward duct (14) and forward transition duct (11). Press cowling latch (8), lift forward end of engine air inlet cowling to clear dowel pin (10), pull forward and remove cowling.



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Figure 1. Transmission Fairings and Engine Air Inlet Cowlings (Sheet 1 of 1)

Transmission Fairings and Engine Air Inlet Cowlings - Inspection and Repair

Procedure

1. Inspect latches, cowl locks, fittings, fasteners, and hinges for wear, damage, and serviceability.
2. Inspect seals for proper fit against mating surfaces as well as for cracks, tears, deterioration, and security of bonding.
3. Replace latches, cowl locks, fittings, and seals if unserviceable.
4. Inspect cowling, duct, and firewall for dents, cracks, and damage. Repair sheet metal cowlings, ducts, and firewalls using maintenance practices in accordance with FAA Publication AC 43.13.1B, Aircraft Inspection and Repair. For information on damage limits and repairs to composite cowlings, please contact Product Support Engineering for assistance.

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Engine Air Inlet Cowlings - Gap Inspection

Procedure

NOTE

This inspection will determine if gaps exist between the lower surface of the engine air inlet cowlings (4, Figure 1) and the associated mating surfaces of the horizontal engine firewall (2, Figure 2).

1. Remove the forward pylon fairing (1, Figure 1), aft pylon fairing (2), and transmission fairing (3) (412-A-71-10-00-01A-520A-A / 01605 and 412-A-71-10-00-02A-520A-A / 01608).

2. Open the upper and lower engine cowlings (12 and 11) on the left-hand and right-hand sides of the helicopter.

NOTE

Although a depiction of the engine air inlet cowlings with adjustable seals installed is shown in Figure 3, this figure is also provided to identify the areas to be inspected on engine air inlet cowlings without adjustable seals. Regardless of the seal configuration of the engine air inlet cowlings, inspect for gaps in the areas identified by adjustable seals (2, 3, and 4).

3. Using a bright light and a mirror, inspect for gaps between the lower seal surfaces of the left-hand and right-hand engine air inlet cowlings (1) and the associated mating surfaces of the horizontal engine firewall (2, Figure 2). Regardless of the seal configuration of the engine air inlet cowlings, inspect for gaps in the areas identified by adjustable seals (2, 3, and 4, Figure 3).

4. If a gap does not exist, proceed as follows:

4.1. Close the upper and lower engine cowlings (12 and 11, Figure 1) on the left-hand and right-hand sides of the helicopter.

4.2. Install the transmission fairing (3), aft pylon fairing (2), and forward pylon fairing (1) (412-A-71-10-00-02A-720A-A / 01611 and 412-A-71-10-00-01A-720A-A / 01607).

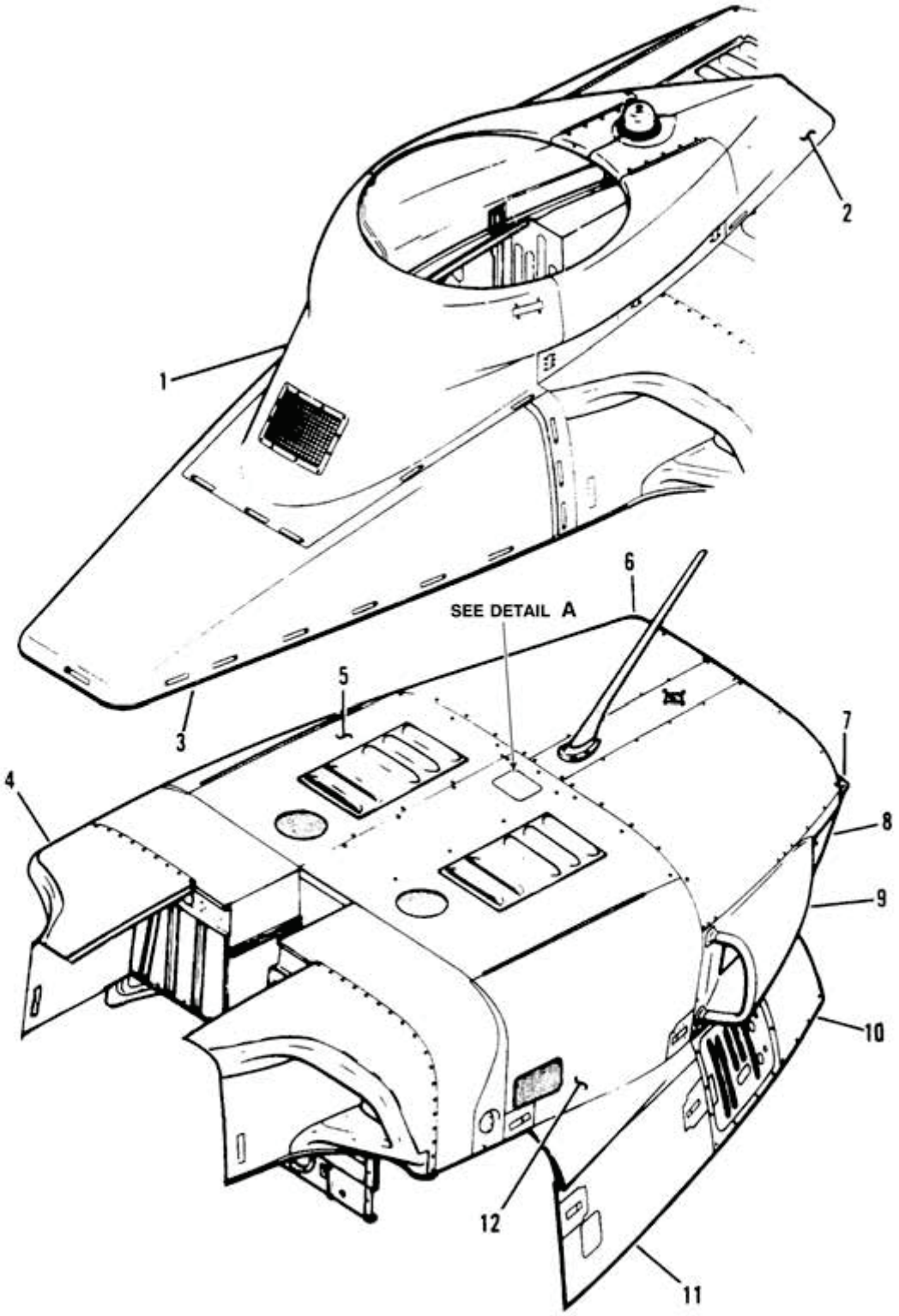
5. If a gap exists, remove the left-hand and right-hand engine air inlet cowlings (4) (412-A-71-10-00-02A-520A-A / 01608) and inspect the horizontal engine firewall (2, Figure 2) as follows:

5.1. Using a straight edge, inspect the forward portion of the horizontal engine firewall (2) on the left-hand and right-hand sides at W.L. 80.0 and F.S. 154.46 between B.L. 4.0 and B.L. 14.31 along the retainer (1). The surface of the retainer must be flat to within 0.063 inch (1.6 mm) over 10.0 inches (254 mm).

5.2. If the retainer (1) of the horizontal engine firewall (2) is not flat, investigate and repair to obtain the required flatness of 0.063 inch (1.6 mm) over 10.0 inches (254 mm). If required, contact Product Support Engineering for an approved repair.

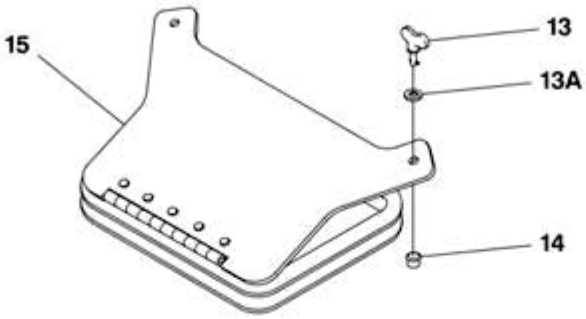
5.3. If the retainer (1) of the horizontal engine firewall (2) was repaired to obtain the required flatness of 0.063 inch (1.6 mm) over 10.0 inches (254 mm), install the left-hand and right-hand engine air inlet cowlings (4, Figure 1) (412-A-71-10-00-02A-720A-A / 01611) and inspect for gaps (Step 3.).

5.4. If a gap exists, carry out Part II of ASB 412-10-140 or adjust/replace the engine air inlet cowling adjustable seals (2, 3, and 4, Figure 3)(Post BHT-412-SI-87 and S/N 36521 and Subsequent), as applicable (412-A-71-10-00-02A-921A-A / 01612 and 412-A-71-10-00-02A-271A-A / 01613).



ICN-412-A-711000-B-97499-00001-A-001-01

Figure 1. Transmission and Engine Cowling (Sheet 1 of 2)



DETAIL A ⚠1

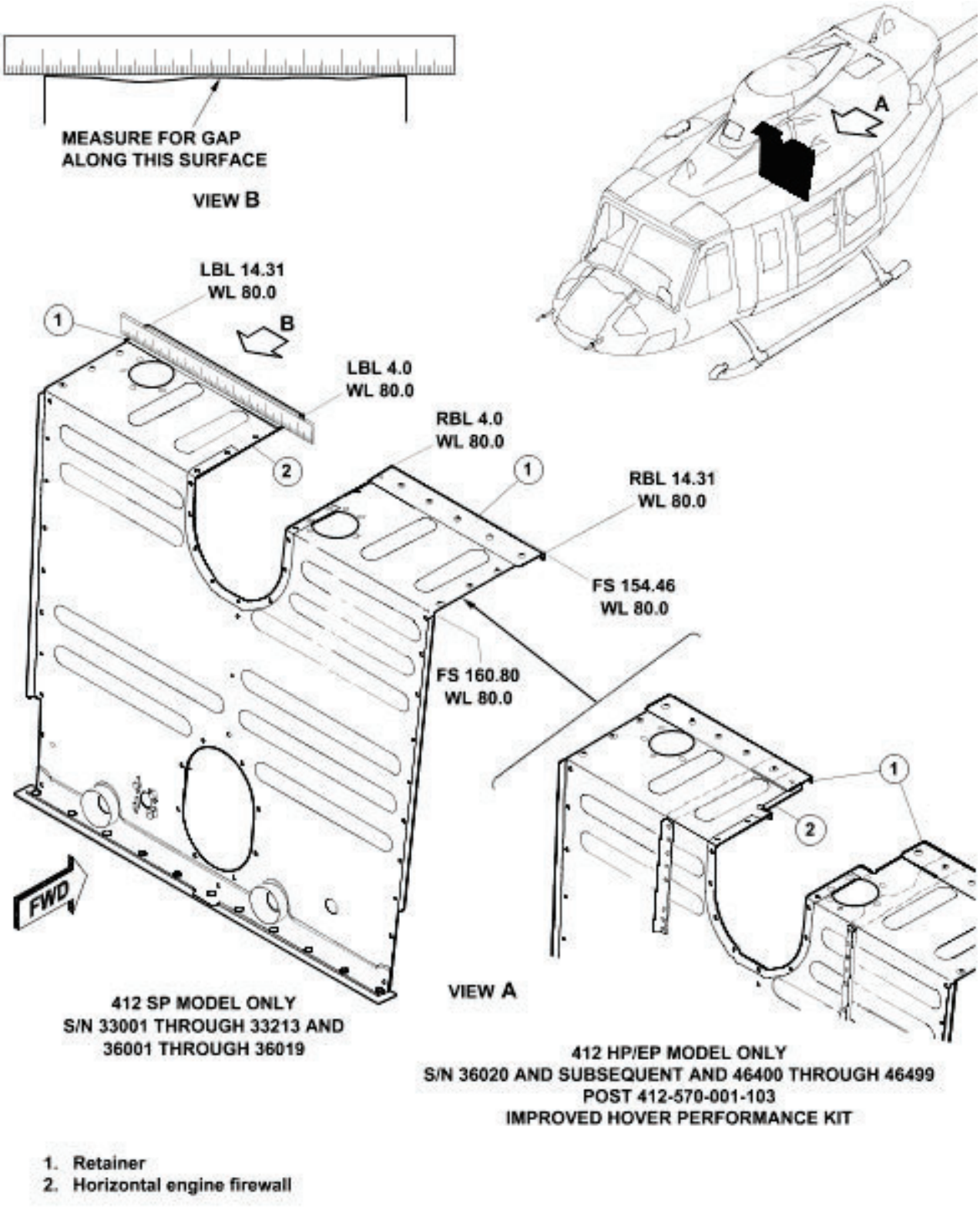
1. Forward pylon fairing
2. Aft pylon fairing
3. Transmission fairing
4. Engine air inlet cowling
5. Engine top cowling
6. Reduction gearbox cowling
7. Oil cooler cowling
8. Engine oil cooler support cowling
9. Oil cooler side scoop
10. Reduction gearbox side cowling
11. Lower engine cowling
12. Upper engine cowling
13. Stud panel fastener
13A. Washer
14. Receptacle panel fastener
15. Main drive aft inspection door

NOTE

⚠1 Applicable to S/N 36189, 36190, 36206 and subsequent and helicopters modified by TB 412-99-155.

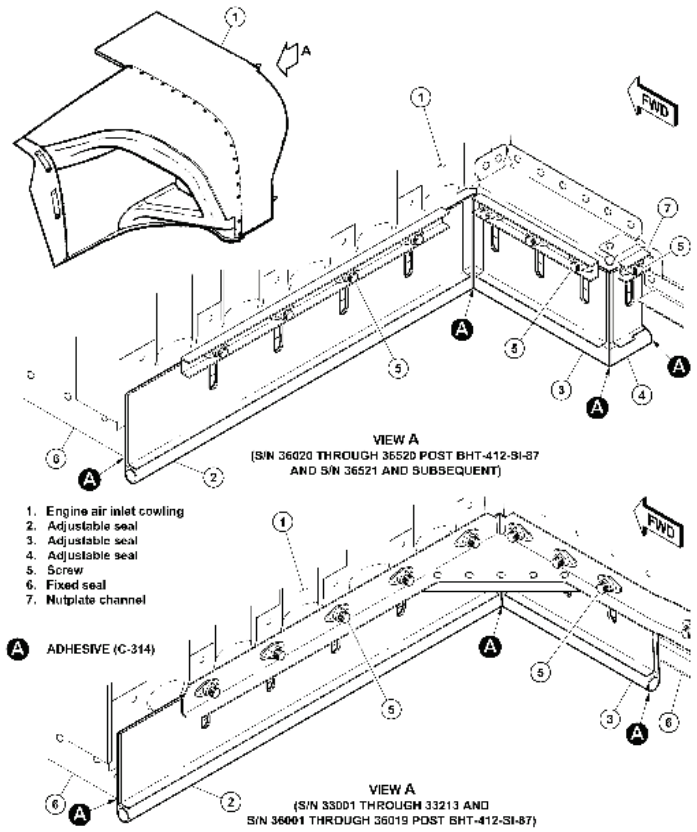
ICN-412-A-711000-B-97499-00002-A-001-01

Figure 1. Transmission and Engine Cowling (Sheet 2 of 2)



ICN-412-A-711000-B-97499-00005-A-001-01

Figure 2. Horizontal Engine Firewall - Straightness Check (Sheet 1 of 1)



ICN-412-A-711000-B-97499-00006-A-001-01

Figure 3. Engine Air Inlet Cowlings - Adjustable Seals (Sheet 1 of 1)

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Transmission Fairing and Engine Air Inlet Cowlings - Installation

Procedure

1. Install the left-hand or right-hand engine air inlet cowling (2, Figure 1), as applicable, as follows:

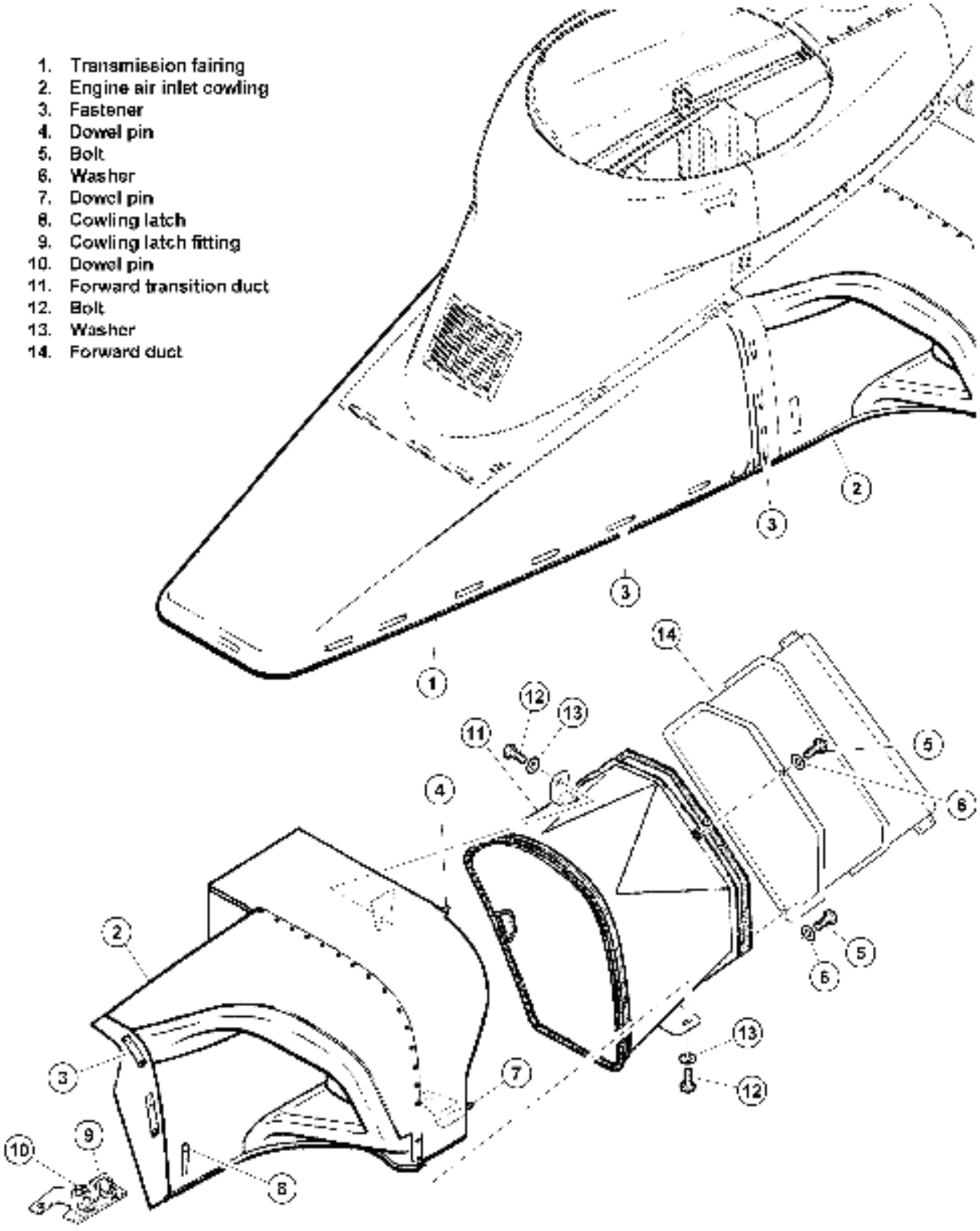
1.1. Put the engine air inlet cowling (2) into position with the forward end raised to clear dowel pin (10). Align and engage dowel pins (4 and 7), and lower on dowel pin (10).

1.2. Install two bolts (5) and washers (6) to secure the forward transition duct (11) to forward duct (14).

1.3. Hook cowling latch (8) to cowling latch fitting (9).

2. Position transmission fairing (1) in place and secure with fasteners (3).

3. Install forward and aft pylon fairings (412-A-71-10-00-01A-720A-A / 01607).



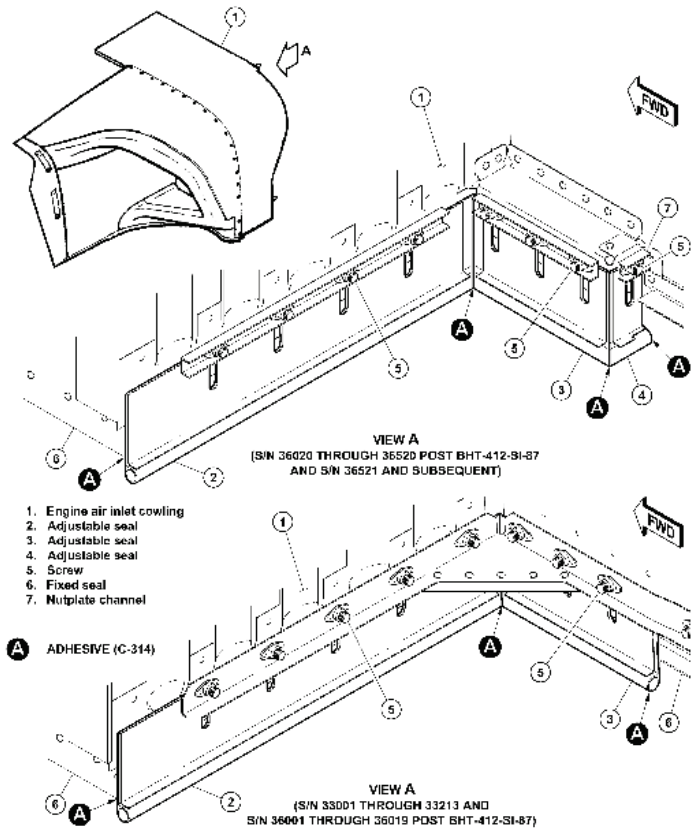
ICN-412-A-711000-B-97499-00004-A-001-01

Figure 1. Transmission Fairings and Engine Air Inlet Cowlings (Sheet 1 of 1)

Engine Air Inlet Cowlings - Adjustable Seal Replacement

Procedure

1. Remove the left-hand or right-hand engine air inlet cowlings (412-A-71-10-00-02A-520A-A / 01608).
2. Cut and remove the adhesive from the ends of the adjustable seals (2, 3, and 4, Figure 1), as applicable.
3. Loosen the screws (5) until the adjustable seals (2, 3, and 4) become loose.
4. Slide out the adjustable seals (2, 3, and 4), as applicable.
5. Slide in the new adjustable seals (2, 3, and 4), as applicable, and tighten screws (5).
6. Perform the adjustment of the adjustable seals (412-A-71-10-00-02A-271A-A / 01613).



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Figure 1. Engine Air Inlet Cowlings - Adjustable Seals (Sheet 1 of 1)

Engine Air Inlet Cowlings - Adjustable Seal Adjustment

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Adhesive	C-314	AR	

Procedure

1. Remove the left-hand or right-hand engine air inlet cowlings (412-A-71-10-00-02A-520A-A / 01608).

2. Cut and remove the adhesive from the ends of the adjustable seals (2, 3, and 4 Figure 1), as applicable.

3. Loosen the screws (5) until the adjustable seals (2, 3, and 4) can move with finger pressure, but not too loose that they can slide off the engine air inlet cowling (1).

4. Extend out the adjustable seals (2, 3, and 4) prior to installing the engine air inlet cowling (1).

NOTE

The adjustable seals (2, 3, and 4) will be pushed up when they make contact with the flanges of the lower forward firewall.

5. Carefully Install the applicable engine air inlet cowling (1) (412-A-71-10-00-02A-720A-A / 01611).

6. Tighten the screws (5) to secure the adjustable seals (2, 3, and 4), as applicable.

NOTE

Make sure the adjustable seals (2, 3, and 4) and fixed seals (6) make a smooth transition from one seal to the next.

7. Use an inspection mirror to check for gaps under the adjustable seals (2, 3, and 4), as applicable.

8. If a gap is still visible under the adjustable seals (2, 3, and 4), remove the engine air inlet cowling (1). Loosen the applicable screws (5). Slide out the adjustable seal as required and tighten the screws.

9. When the N1 throttle linkages are actuated they cause a distortion of the mating surfaces of the adjustable seals, which can cause a gap under the adjustable seals.

9.1. During the following verification, locate a small wooden block between the shoulder of the idle stop plunger and flange of the idle stop solenoid mounting bracket to assist in holding the idle stop plunger in the retracted position (412-A-76-10-00-00A-276A-A / 01725).

9.2. Cycle the applicable throttle through its entire range of motion several times and observe if a gap becomes visible under the adjustable seals (2, 3, and 4), as applicable.

9.3. If a gap becomes visible, remove the engine air inlet cowling (1). Loosen the applicable screws (5). Slide out the adjustable seal (2, 3, or 4) as required and tighten the screws. Reinstall the engine air inlet cowling and cycle the throttle again to check for gaps. Repeat as required.

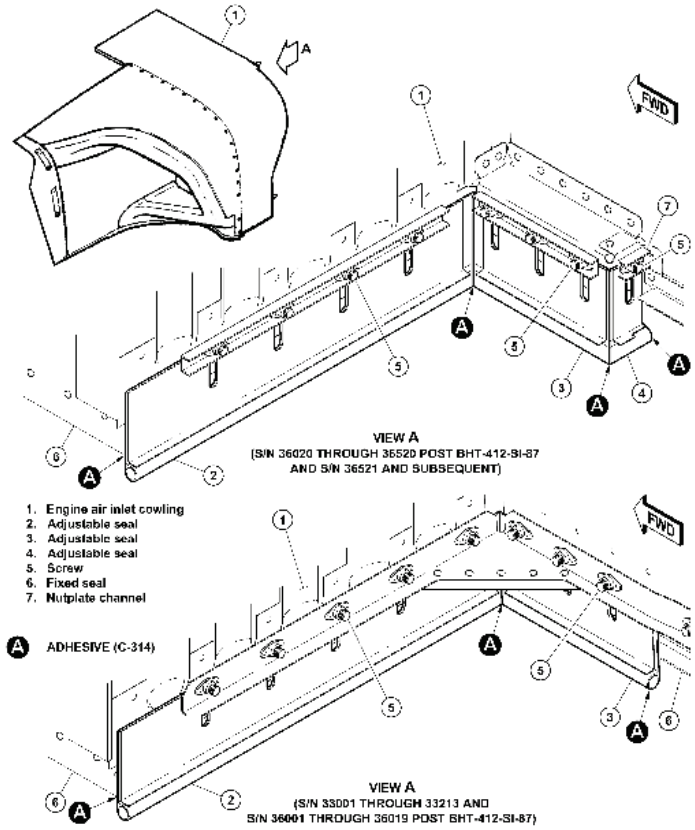
9.4. Remove the small wooden block from the idle stop solenoid mounting bracket.

10. Remove the left-hand or right-hand engine air inlet cowling as applicable (412-A-71-10-00-02A-520A-A / 01608).

11. Apply adhesive (C-314) at the intersection of the adjustable seals (2, 3, and 4) and fixed seals (6) to bond the ends together (BHT-ALL-SPM, Chapter 7, Standard Practices Manual).

12. Install the left-hand or right-hand engine air inlet cowlings, as applicable, and transmission fairing (412-A-71-10-00-02A-720A-A / 01611).

13. Install forward and aft pylon fairings (412-A-71-10-00-01A-720A-A / 01607).



ICN-412-A-711000-B-97499-00006-A-001-01

Figure 1. Engine Air Inlet Cowlings - Adjustable Seals (Sheet 1 of 1)

Engine Air Inlet Cowlings - Adjustable Seals - Repair

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Toluene	C-306	AR	
Adhesive	C-300	AR	
Abrasive cloth or paper	C-423	AR	
Clean cloth	C-516	AR	
Cheesecloth	C-486	AR	
Primer	C-337	AR	

Procedure

NOTE

The following procedure is applicable to helicopters S/N 36521 and subsequent and Post BHT-412-SI-87.

1. Remove the existing seal from the adjustable seal assemblies (2, 3, or 4, Figure 1) on the engine air inlet cowling (1).

2. Repair the adjustable seal assemblies (2, 3, or 4) on the engine air inlet cowling (1) as follows:

2.1. Prepare the mating surfaces of the rubber seal (P/N 2330-00542-00) before bonding as follows:

2.1.1. Abrade the rubber seal with 80 girt abrasive cloth or paper (C-423).

2.1.2. Clean the abraded area with a clean cloth (C-516) moist with toluene (C-306).

2.1.3. Wipe the area dry with a clean cheesecloth (C-486).

NOTE

If the titanium parts need to be replaced, they can be manufactured using MIL-T-9046, COMP, CP-2, 0.016 inch (0.41 mm) thick (backing plate) and 0.025 inch (0.64 mm) thick (retainer plate).

2.1.4. Abrade the metal surfaces with 240 grit abrasive cloth or paper (C-423).

2.1.5. Clean the abraded area with a clean cloth (C-516) moist with toluene (C-306).

2.1.6. Wipe the area dry with a clean cheesecloth (C-486).

2.2. Apply a layer of primer (C-337) by a brush to the metallic surface and allow to air dry for 30 minutes at room temperature.



DO NOT CAP ADHESIVE (C-300) AFTER MIXTURE. MIXTURE RELEASES HYDROGEN GAS WHICH CAN CAUSE HIGH PRESSURES. POT LIFE OF THIS ADHESIVE (C-300) IS 2-4 HOURS. IT IS RECOMMENDED TO APPLY IMMEDIATELY AFTER MIXING.

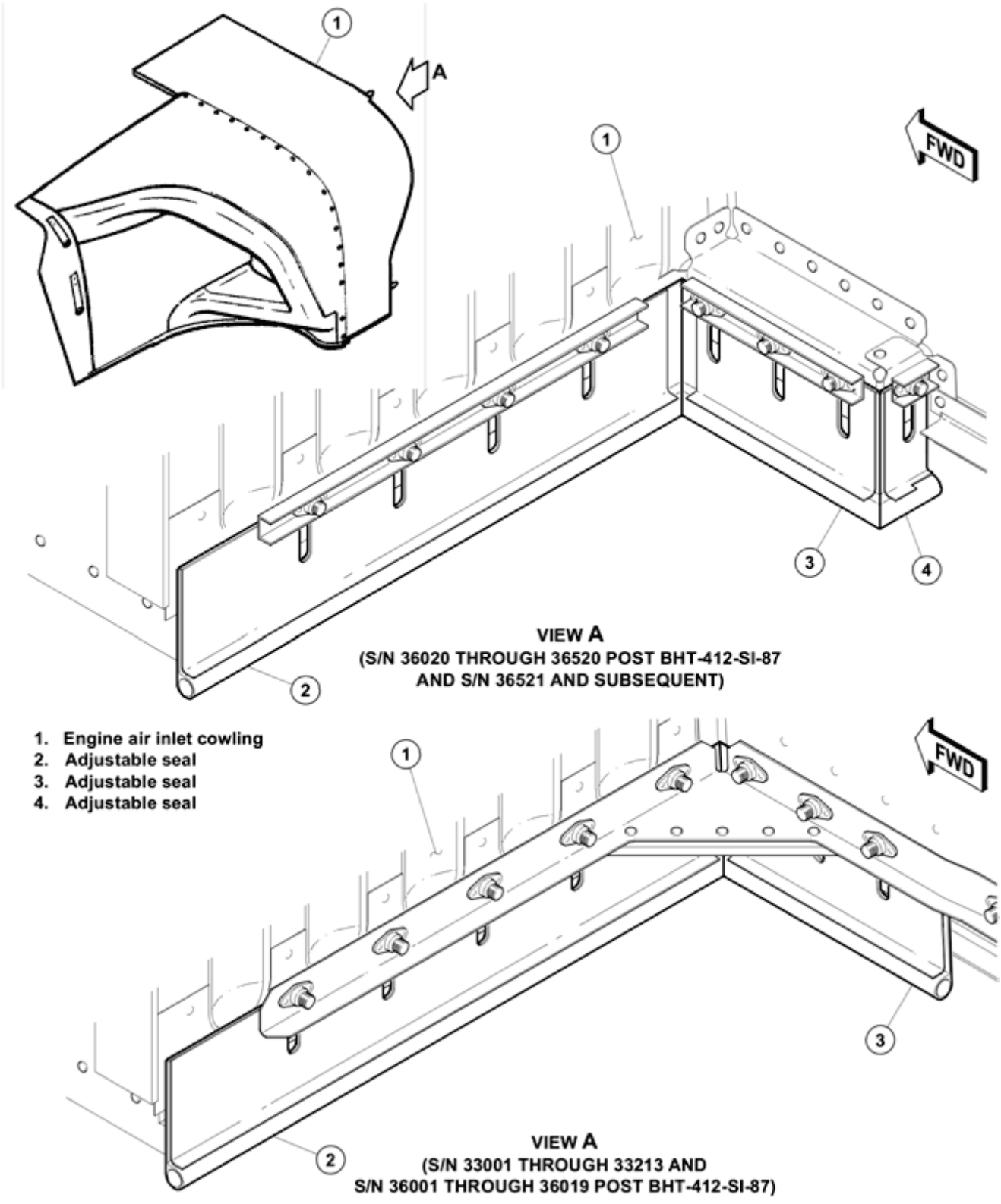
2.3. Mix the adhesive (C-300) 100 parts of Part A to 5 of Part B and apply a layer to bond surfaces of the seal and metal with a brush (BHT-ALL-SPM, Chapter 7, Standard Practices Manual).

2.4. Allow to air dry for 15 to 30 minutes at room temperature to let the solvents evaporate and get a strong tack.

NOTE

Handling strength is 50% of the maximum strength and is obtained in 24 hours. Maximum strength is obtained in five to seven days.

2.5. Press the two coated surfaces together and maintain firm contact pressure until handling strength (50% of maximum strength) has been obtained.



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Figure 1. Engine Air Inlet Cowlings - Adjustable Seals - Repair (Sheet 1 of 1)

Engine Cowlings - Removal

Procedure

1. Remove engine lower cowling (11, Figure 1) as follows:

1.1. Disengage two latches (2, Figure 2) and cowl lock (18). Open engine cowling (1).

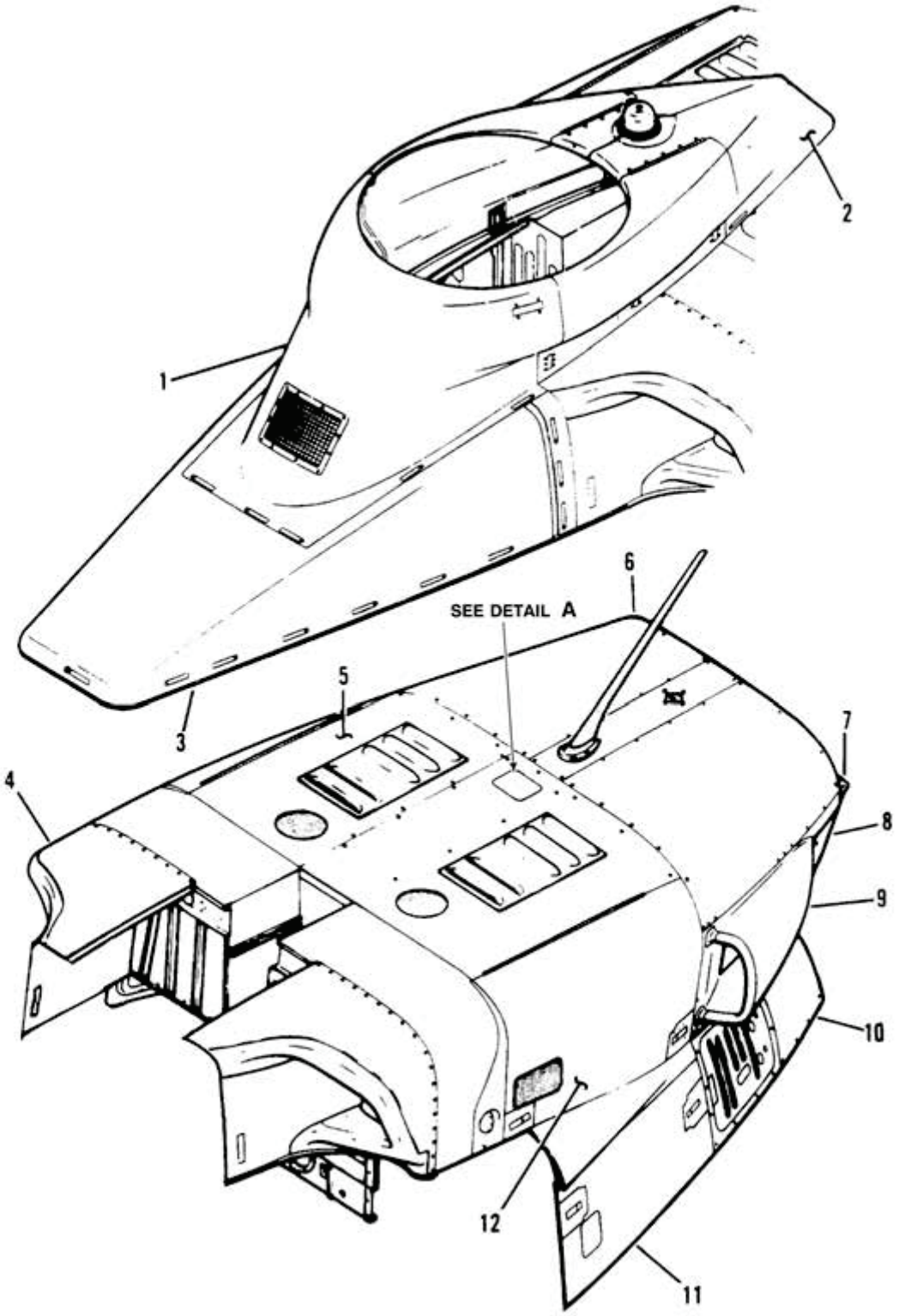
1.2. Pull quick release pin (8) from brace (6) and hinge fitting. Remove cowling.

2. Remove top and upper cowling as follows:

2.1. Disengage latches on engine upper cowling (12, Figure 1).

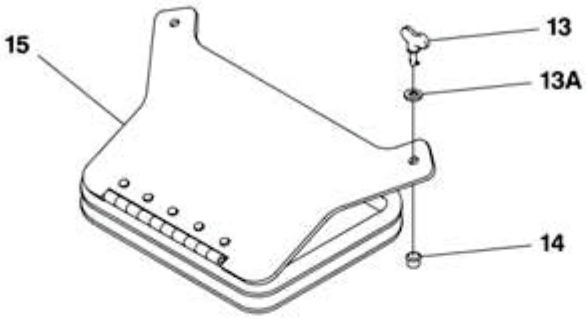
2.2. Remove screws securing engine top cowling (5) and remove engine top cowling and upper engine cowling (12) together.

3. Refer to 412-A-71-10-00-02A-280A-A / 01609 for inspection and repair.



ICN-412-A-711000-B-97499-00001-A-001-01

Figure 1. Transmission and Engine Cowling (Sheet 1 of 2)



DETAIL A ⚠1

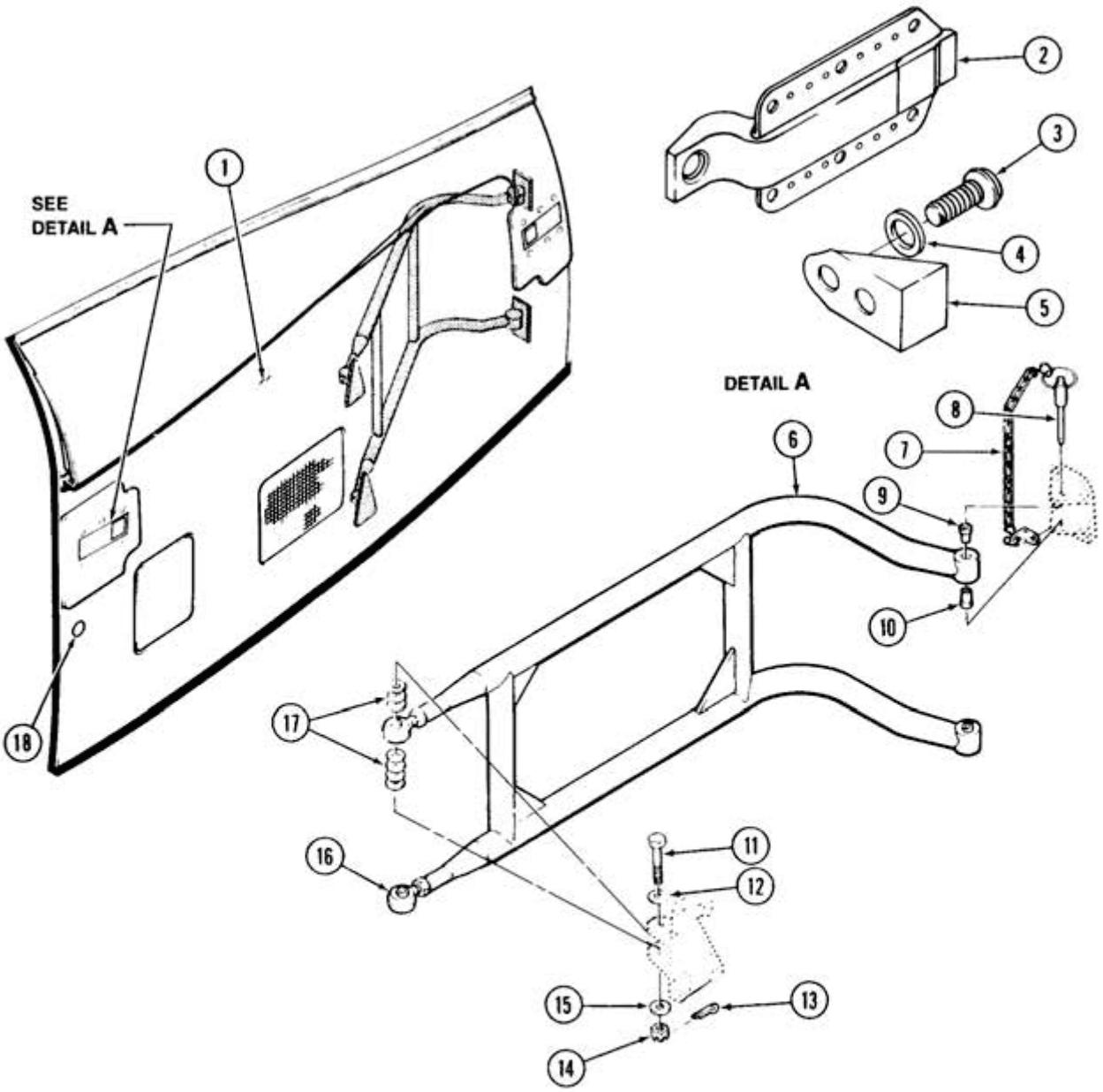
1. Forward pylon fairing
2. Aft pylon fairing
3. Transmission fairing
4. Engine air inlet cowling
5. Engine top cowling
6. Reduction gearbox cowling
7. Oil cooler cowling
8. Engine oil cooler support cowling
9. Oil cooler side scoop
10. Reduction gearbox side cowling
11. Lower engine cowling
12. Upper engine cowling
13. Stud panel fastener
13A. Washer
14. Receptacle panel fastener
15. Main drive aft inspection door

NOTE

⚠1 Applicable to S/N 36189, 36190, 36206 and subsequent and helicopters modified by TB 412-99-155.

ICN-412-A-711000-B-97499-00002-A-001-01

Figure 1. Transmission and Engine Cowling (Sheet 2 of 2)



1. Cowl assembly, engine lower
2. Latch
3. Bolt
4. Washer
5. Striker
6. Brace
7. Chain
8. Quick release pin
9. Spacer
10. Spacer
11. Bolt
12. Washer
13. Cotter pin
14. Nut
15. Washer
16. Rod end
17. Washer (7)
18. Cowl lock

ICN-412-A-711000-B-97499-00007-A-001-01

Figure 2. Lower Engine Cowling (Sheet 1 of 1)

Engine Cowlings - Installation

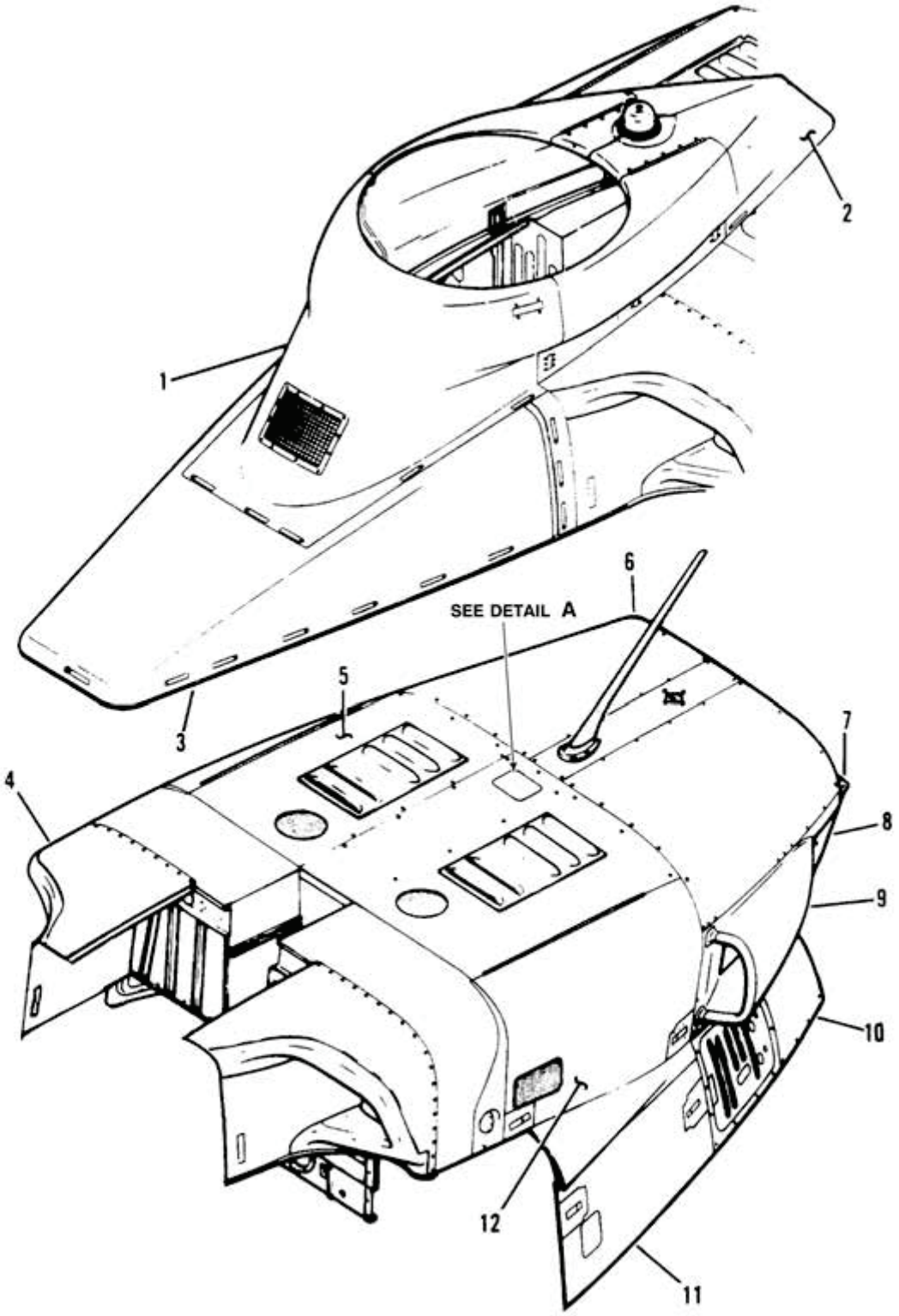
Procedure

NOTE

A combination of sheet metal and composite engine cowlings may be utilized. Weight differential between the sheet metal and composite cowlings is considered negligible.

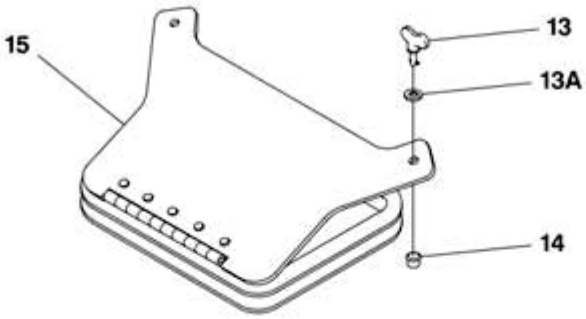
1. Position engine top cowling and upper engine cowling (5 and 12, Figure 1) on helicopter. Align screw holes and install (but do not tighten) screws. Engage latches on upper engine cowling and tighten screws.

2. Position lower engine cowling (11) on helicopter. Align brace (6, Figure 2) to hinge fitting. Install quick release pin (8), close cowling, and engage two latches (2) and cowl lock (18).



ICN-412-A-711000-B-97499-00001-A-001-01

Figure 1. Transmission and Engine Cowling (Sheet 1 of 2)



DETAIL A ⚠1

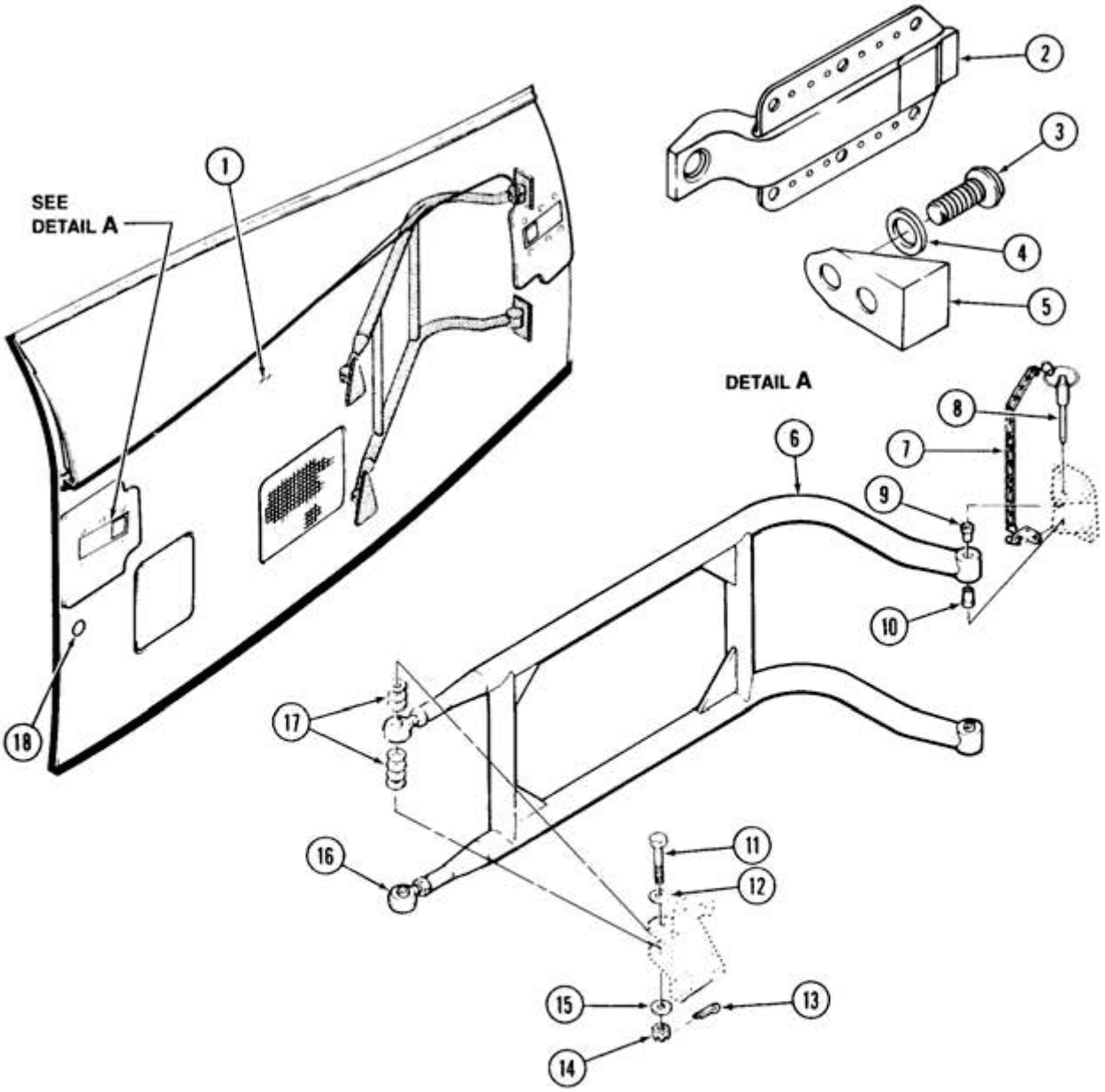
1. Forward pylon fairing
2. Aft pylon fairing
3. Transmission fairing
4. Engine air inlet cowling
5. Engine top cowling
6. Reduction gearbox cowling
7. Oil cooler cowling
8. Engine oil cooler support cowling
9. Oil cooler side scoop
10. Reduction gearbox side cowling
11. Lower engine cowling
12. Upper engine cowling
13. Stud panel fastener
13A. Washer
14. Receptacle panel fastener
15. Main drive aft inspection door

NOTE

⚠1 Applicable to S/N 36189, 36190, 36206 and subsequent and helicopters modified by TB 412-99-155.

ICN-412-A-711000-B-97499-00002-A-001-01

Figure 1. Transmission and Engine Cowling (Sheet 2 of 2)



1. Cowl assembly, engine lower
2. Latch
3. Bolt
4. Washer
5. Striker
6. Brace
7. Chain
8. Quick release pin
9. Spacer
10. Spacer
11. Bolt
12. Washer
13. Cotter pin
14. Nut
15. Washer
16. Rod end
17. Washer (7)
18. Cowl lock

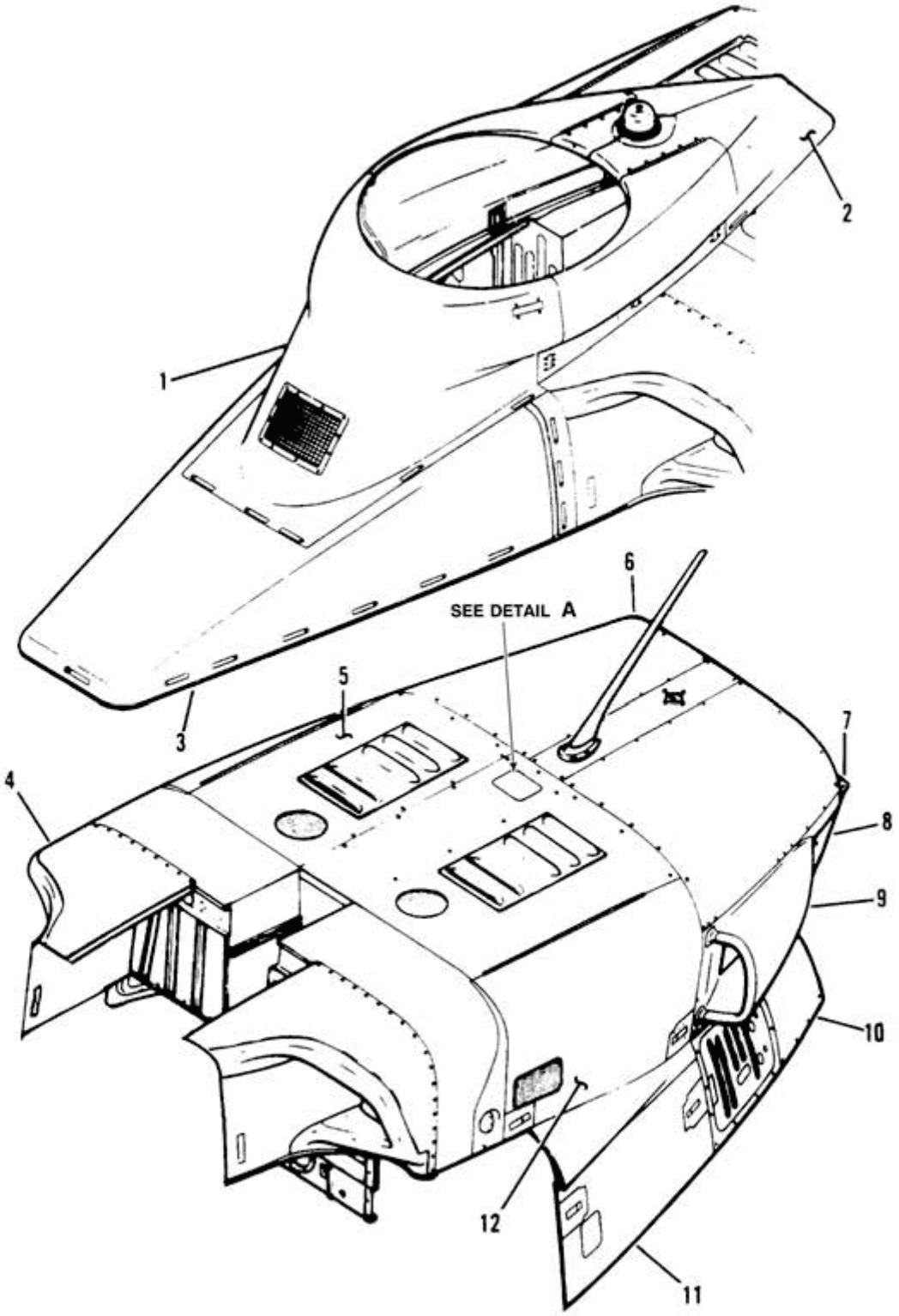
ICN-412-A-711000-B-97499-00007-A-001-01

Figure 2. Lower Engine Cowling (Sheet 1 of 1)

Reduction (Combing) Gearbox Cowlings - Removal

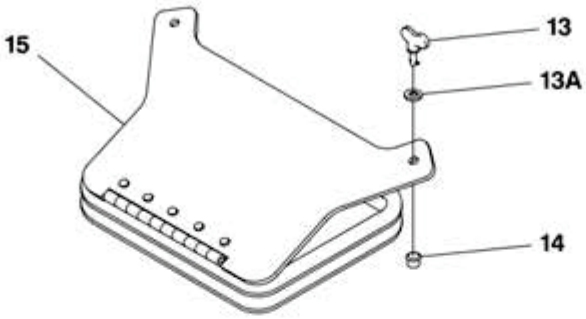
Procedure

1. Remove screws around reduction gearbox side cowling (10, Figure 1) and remove cowling.
2. Remove screws around reduction gearbox cowling (6) and remove cowling.
3. Remove four screws from oil filler adapter. Disconnect antenna, if installed. Remove screws from ends of center top cowling. Remove cowling.



ICN-412-A-711000-B-97499-00001-A-001-01

Figure 1. Transmission and Engine Cowling (Sheet 1 of 2)



DETAIL A ⚠1

1. Forward pylon fairing
2. Aft pylon fairing
3. Transmission fairing
4. Engine air inlet cowling
5. Engine top cowling
6. Reduction gearbox cowling
7. Oil cooler cowling
8. Engine oil cooler support cowling
9. Oil cooler side scoop
10. Reduction gearbox side cowling
11. Lower engine cowling
12. Upper engine cowling
13. Stud panel fastener
13A. Washer
14. Receptacle panel fastener
15. Main drive aft inspection door

NOTE

⚠1 Applicable to S/N 36189, 36190, 36206 and subsequent and helicopters modified by TB 412-99-155.

ICN-412-A-711000-B-97499-00002-A-001-01

Figure 1. Transmission and Engine Cowling (Sheet 2 of 2)

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Reduction (Combing) Gearbox Cowlings - Inspection and Repair

Procedure

1. Refer to 412-A-71-10-00-02A-280A-A / 01609 for inspection and repair.

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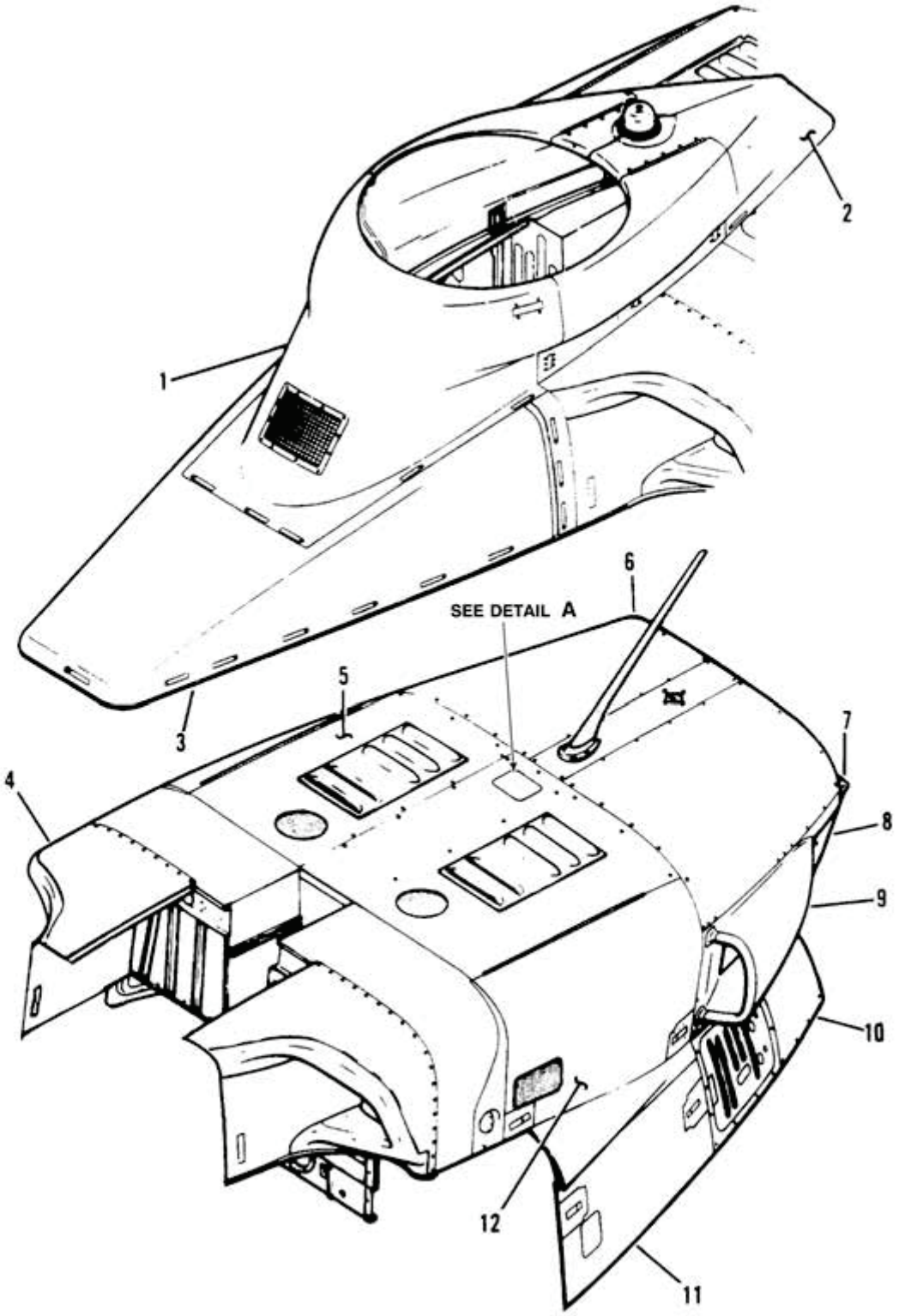
Reduction (Combing) Gearbox Cowlings - Installation

Procedure

1. Position center top cowling on helicopter, and install four screws in ends of cowling. If installed, connect rotating light and radio antenna. Install four screws in center top cowling and oil filler adapter.

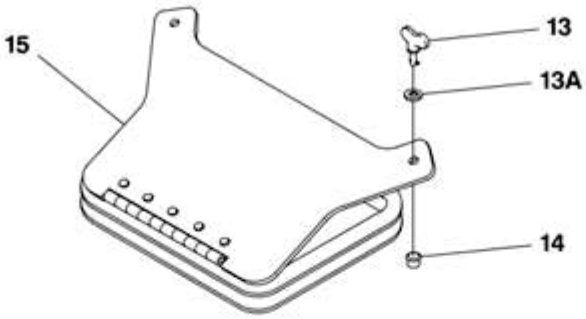
2. Position reduction gearbox cowling (6, Figure 1) on helicopter, align screw holes, and install and tighten screws.

3. Position reduction gearbox side cowling (10) on helicopter, align screw holes, and install and tighten screws.



ICN-412-A-711000-B-97499-00001-A-001-01

Figure 1. Transmission and Engine Cowling (Sheet 1 of 2)



DETAIL A ⚠1

1. Forward pylon fairing
2. Aft pylon fairing
3. Transmission fairing
4. Engine air inlet cowling
5. Engine top cowling
6. Reduction gearbox cowling
7. Oil cooler cowling
8. Engine oil cooler support cowling
9. Oil cooler side scoop
10. Reduction gearbox side cowling
11. Lower engine cowling
12. Upper engine cowling
13. Stud panel fastener
13A. Washer
14. Receptacle panel fastener
15. Main drive aft inspection door

NOTE

⚠1 Applicable to S/N 36189, 36190, 36206 and subsequent and helicopters modified by TB 412-99-155.

ICN-412-A-711000-B-97499-00002-A-001-01

Figure 1. Transmission and Engine Cowling (Sheet 2 of 2)

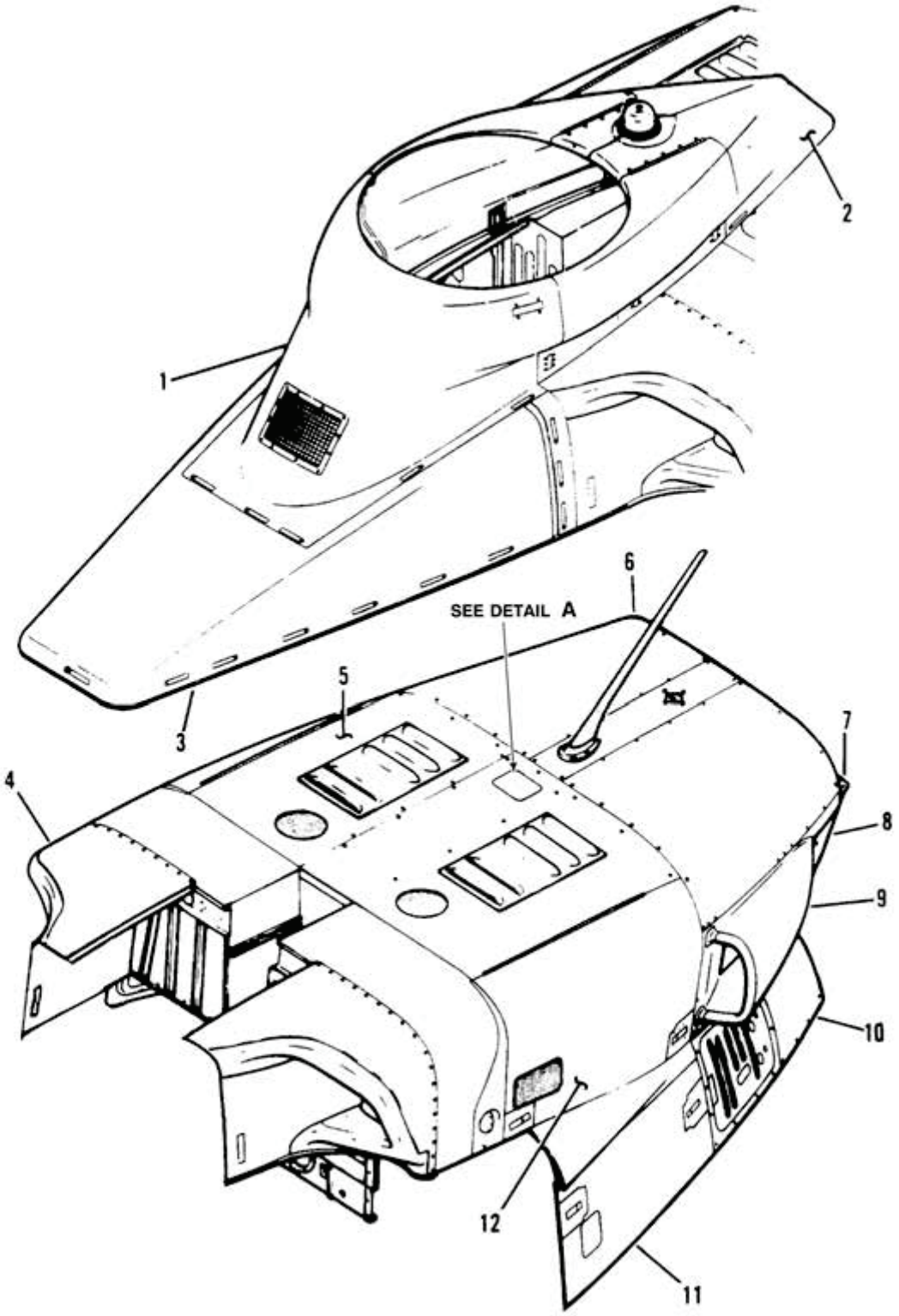
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Oil Cooler Airscoop Cowling - Removal

Procedure

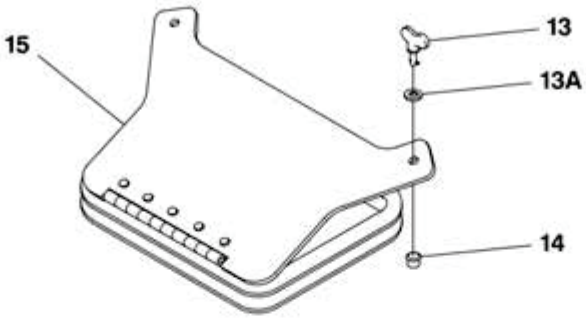
1. Remove oil cooler airscoop cowling (9, Figure 1). Disengage latches (7, 8, and 9, Figure 2). Install hold open strut.

2. Remove nut (5), screw (3), and washers (4) from hinge fitting (2 and 6). Disengage hold open brace, remove cowling (1).



ICN-412-A-711000-B-97499-00001-A-001-01

Figure 1. Transmission and Engine Cowling (Sheet 1 of 2)



DETAIL A ⚠1

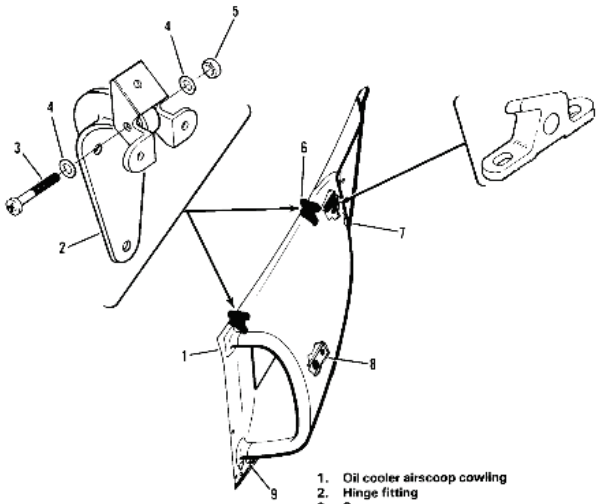
1. Forward pylon fairing
2. Aft pylon fairing
3. Transmission fairing
4. Engine air inlet cowling
5. Engine top cowling
6. Reduction gearbox cowling
7. Oil cooler cowling
8. Engine oil cooler support cowling
9. Oil cooler side scoop
10. Reduction gearbox side cowling
11. Lower engine cowling
12. Upper engine cowling
13. Stud panel fastener
13A. Washer
14. Receptacle panel fastener
15. Main drive aft inspection door

NOTE

⚠1 Applicable to S/N 36189, 36190, 36206 and subsequent and helicopters modified by TB 412-99-155.

ICN-412-A-711000-B-97499-00002-A-001-01

Figure 1. Transmission and Engine Cowling (Sheet 2 of 2)



ICN-412-A-711000-B-97499-00008-A-001-01

Figure 2. Oil Cooler Air Scoop Cowling (Sheet 1 of 1)

Oil Cooler Airscoop Cowling - Inspection and Repair

Procedure

1. Refer to 412-A-71-10-00-02A-280A-A / 01609 for inspection and repair.

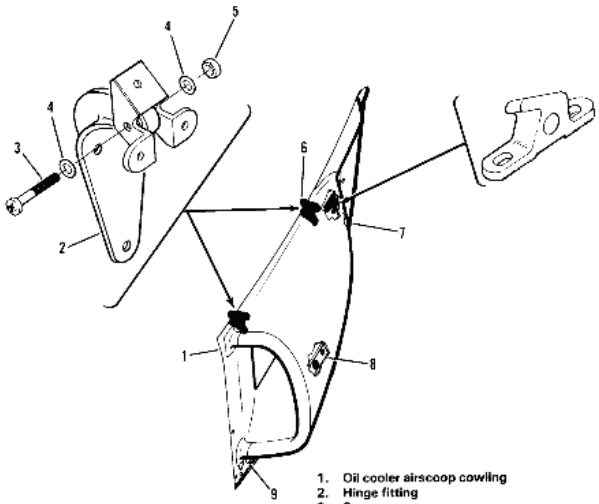
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Oil Cooler Airscoop Cowling - Installation

Procedure

1. Position oil cooler airscoop cowling (1, Figure 1) on helicopter and install hold open strut. Install three AN960JD10L washers each side of hinge (2 and 6), to equalize gap at fore and aft ends of oil cooler airscoop cowling (1). Install screw (3), washers (4), and nuts (5). Tighten nuts.

2. Disengage strut, close oil cooler airscoop cowling (1). Check gap between cowling ends. Engage latches (7, 8, and 9).

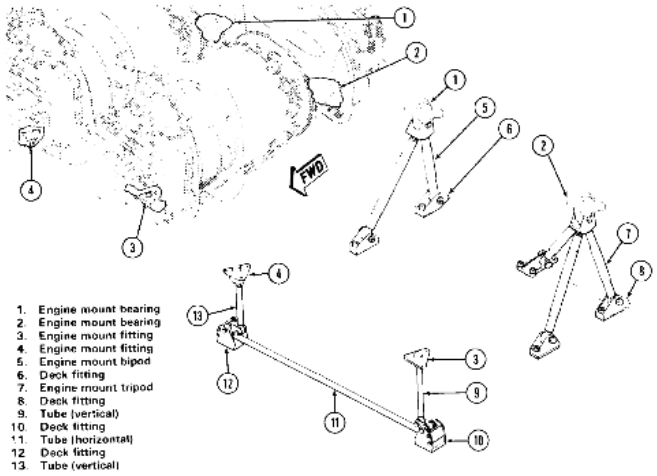


ICN-412-A-711000-B-97499-00008-A-001-01

Figure 1. Oil Cooler Air Scoop Cowling (Sheet 1 of 1)

Engine Mount - Description

The accessory gearbox on front end of each power section is supported by vertical steel tubes interconnected through bellcrank fittings to a horizontal steel tube (Figure 1). The bearing rod end at top of each vertical tube is attached to a fitting on accessory gearbox. The other end is attached to a bellcrank fitting, which is pivoted in a deck fitting with movement restricted by a bellcrank stop. The reduction (combining) gearbox on aft end of engine is supported by a tripod mount on left side and by a bipod mount on right side. The tripod and bipod have steel tube legs with bearing rod ends, which attach to fittings on engine deck. The other end has a fitting, which attaches to bearing fittings on reduction gearbox. Laminated shims under engine deck fittings are used to align engine to transmission.



ICN-412-A-712000-B-97499-00018-A-001-01

Figure 1. Engine Mounts (Sheet 1 of 1)

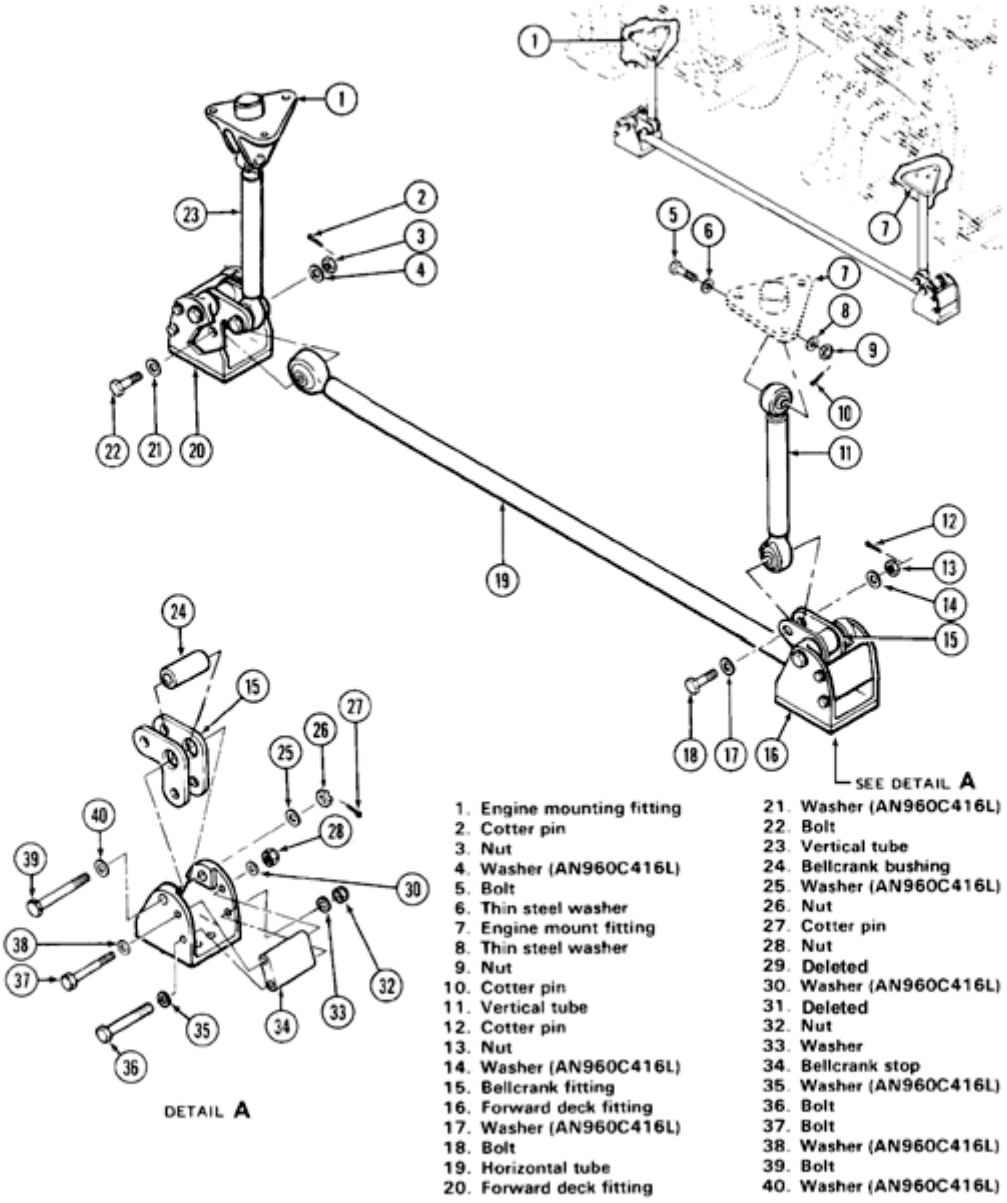
Forward Mount - Removal

Procedure



LIFTING CAPACITY OF HOIST SHALL BE 2,000 POUNDS (908 KG) OR GREATER.

1. If the engine mount is to be removed while engine is installed, support the engine with the hoist and the engine sling. Refer to 412-A-71-00-01-00A-028A-A / 01596 for the procedure to install sling on the engine and get access to forward engine mount tubes.
2. Remove bolt (5, Figure 1). Remove bolt (18). Remove the tube (11).
3. Remove the tube (23) in the same manner as given in preceding step.
4. Remove bolts (36 and 37). Remove the stop (34).
5. Remove the stop from fitting (20) in the same manner as given in the preceding step.
6. Remove bolt (22). Separate the tube (19) from the fitting (20).
7. Remove bolt and separate the tube (19) from the fitting (16) in the same manner as given in the preceding step. Remove the tube (19).
8. Remove bolt (39). Remove the fitting (15) and the bushing (24).
9. Remove the fitting (20) in the same manner as given in preceding step.



1. Engine mounting fitting
2. Cotter pin
3. Nut
4. Washer (AN960C416L)
5. Bolt
6. Thin steel washer
7. Engine mount fitting
8. Thin steel washer
9. Nut
10. Cotter pin
11. Vertical tube
12. Cotter pin
13. Nut
14. Washer (AN960C416L)
15. Bellcrank fitting
16. Forward deck fitting
17. Washer (AN960C416L)
18. Bolt
19. Horizontal tube
20. Forward deck fitting
21. Washer (AN960C416L)
22. Bolt
23. Vertical tube
24. Bellcrank bushing
25. Washer (AN960C416L)
26. Nut
27. Cotter pin
28. Nut
29. Deleted
30. Washer (AN960C416L)
31. Deleted
32. Nut
33. Washer
34. Bellcrank stop
35. Washer (AN960C416L)
36. Bolt
37. Bolt
38. Washer (AN960C416L)
39. Bolt
40. Washer (AN960C416L)

ICN-412-A-712000-B-97499-00019-A-001-01

Figure 1. Forward Mount Tube (Sheet 1 of 1)

Forward Mount - Inspection and Repair

Procedure



IF RECORDS OR PHYSICAL APPEARANCE INDICATE ENGINE MOUNT HAS BEEN INVOLVED IN AN ACCIDENT OR INCIDENT, REFER TO 412-A-05-50-00-00A-284A-A / 00063 FOR CONDITIONAL INSPECTION REQUIREMENTS.

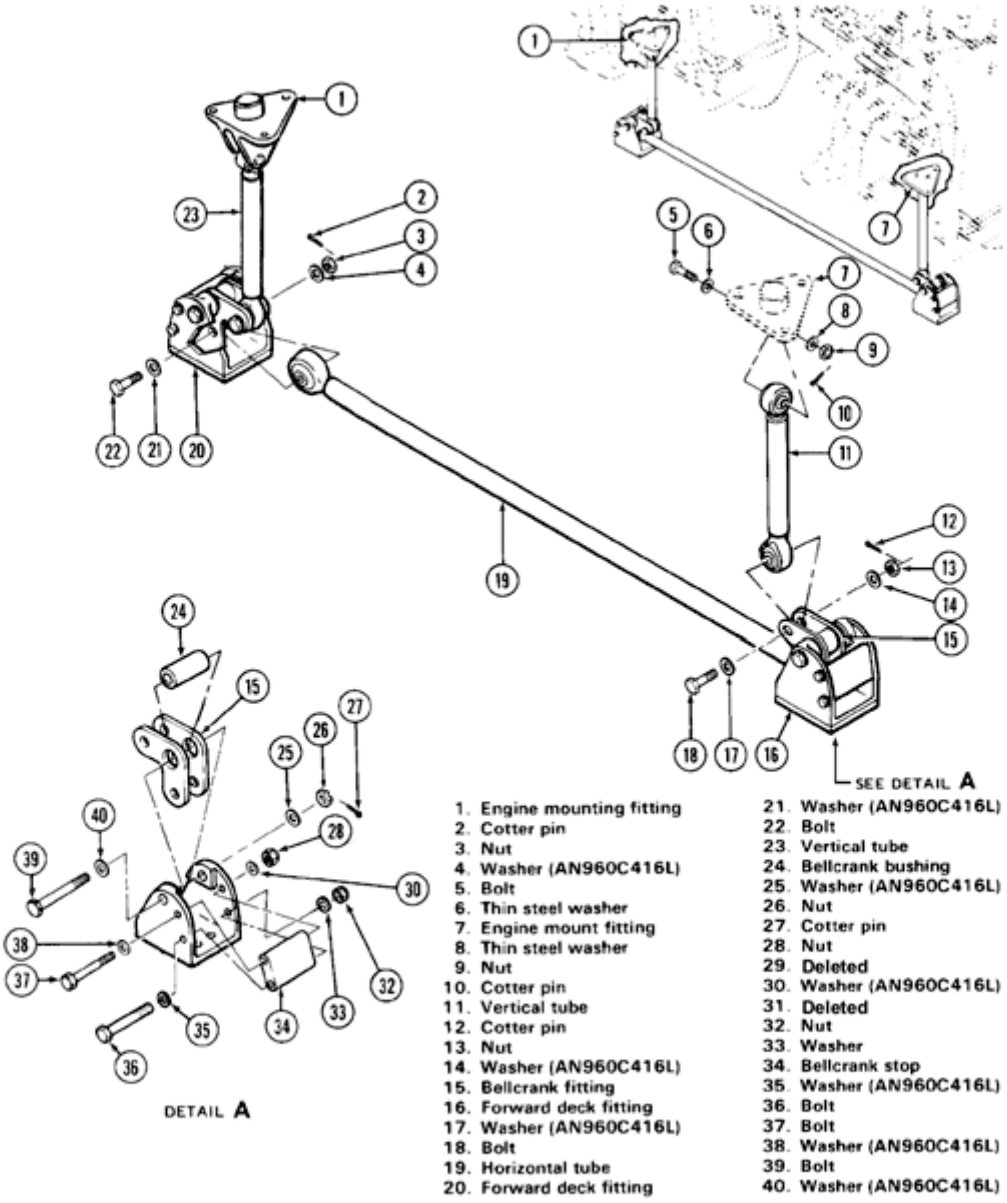
1. For inspection and repair procedure, refer to 412-CR&O, Chapter 71.

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Forward Mount - Installation

Procedure

1. Install the bushing (24, Figure 1).
2. Install the fitting (15) with the bolt (39), washers (40 and 25) and the nut (26).
3. Do Step 1. and Step 2. to install the fitting (20).
4. Install the tube (19) with the bolt (22), washers (21 and 4), nut (3), and cotter pin (2).
5. Install the opposite end of the tube (19) in similar manner.
6. Install the stop (34) with the bolts (36 and 37), washers (35, 33, 38, and 30) and the nuts (28 and 32).
7. Install the stop in the fitting (20) in the same manner as given in the preceding step.
8. Install the bolt (18), washers (17 and 14), nut (13), and cotter pin (12) into the bottom of the tube (11).
9. Install bottom of the tube (23) in the same manner as given in the preceding step.
10. Install the bolt (5), washers (6 and 8), nut (9), and cotter pin (10) into the top of the tube (11).
11. Install the top of tube (23) in the same manner as given in the preceding step.



ICN-412-A-712000-B-97499-00019-A-001-01

Figure 1. Forward Mount Tube (Sheet 1 of 1)

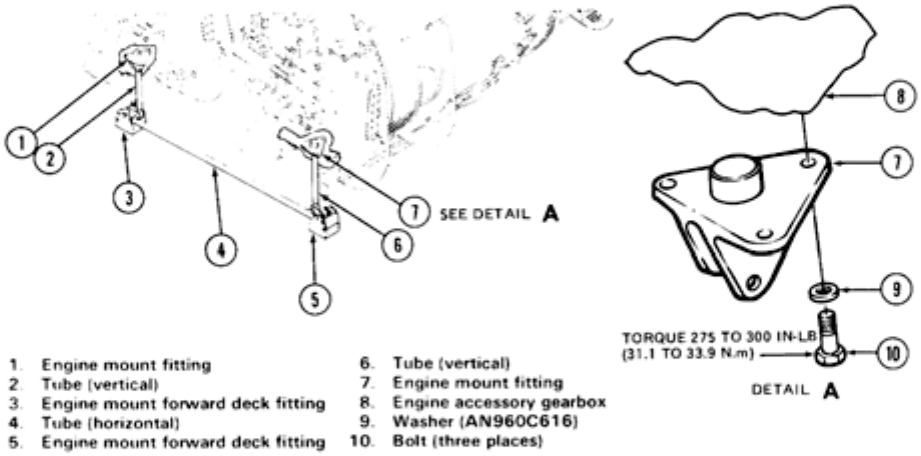
Forward Mount Fittings - Removal

Procedure

1. Remove tubes (2 and 6, Figure 1).

2. Remove bolts (10). Remove fitting (7).

3. Remove fitting (1) in the same manner as given in the preceding step.



1. Engine mount fitting
2. Tube (vertical)
3. Engine mount forward deck fitting
4. Tube (horizontal)
5. Engine mount forward deck fitting
6. Tube (vertical)
7. Engine mount fitting
8. Engine accessory gearbox
9. Washer (AN960C616)
10. Bolt (three places)

ICN-412-A-712000-B-97499-00020-A-001-01

Figure 1. Forward Mount Fitting (Sheet 1 of 1)

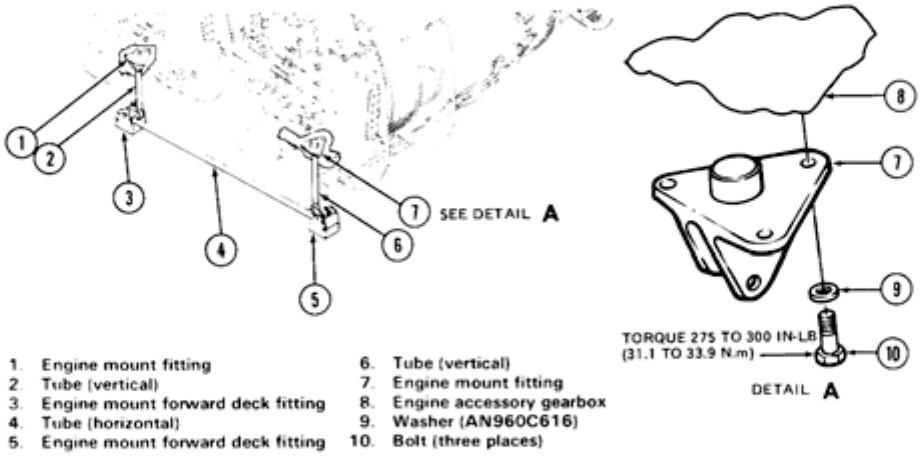
Forward Mount Fittings - Inspection and Repair

Procedure



IF RECORDS OR THE PHYSICAL APPEARANCE SHOW THAT THE ENGINE MOUNT IS INVOLVED IN AN ACCIDENT OR INCIDENT, REFER TO 412-A-05-50-00-00A-284A-A / 00063 FOR CONDITIONAL INSPECTION REQUIREMENTS.

1. Examine and repair the forward engine mount fittings (1 and 7, Figure 1) per 412-CR&O, Chapter 71.



1. Engine mount fitting
2. Tube (vertical)
3. Engine mount forward deck fitting
4. Tube (horizontal)
5. Engine mount forward deck fitting
6. Tube (vertical)
7. Engine mount fitting
8. Engine accessory gearbox
9. Washer (AN960C616)
10. Bolt (three places)

ICN-412-A-712000-B-97499-00020-A-001-01

Figure 1. Forward Mount Fitting (Sheet 1 of 1)

Forward Mount Fittings - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Lockwire	C-405	AR	

Procedure

NOTE

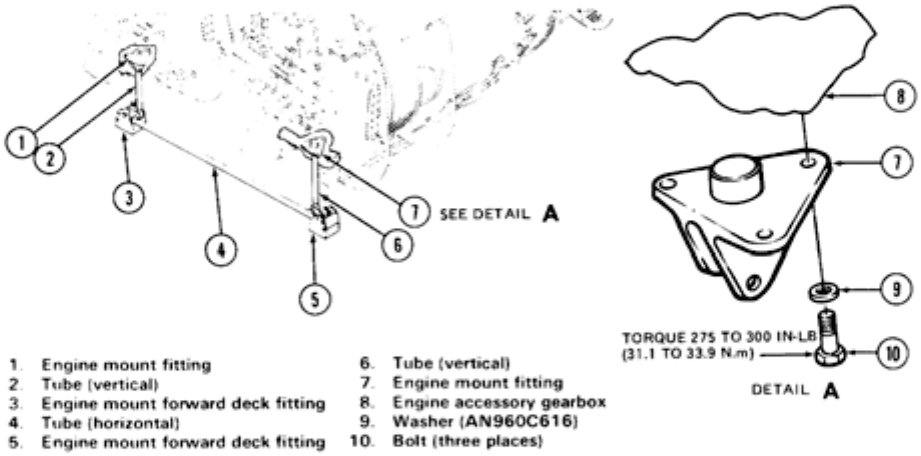
Fittings (1 and 7, Figure 1) are not interchangeable.

1. Install fitting (7) with the three bolts (10) and washers (9). Torque the bolts 275 to 300 inch-pounds (31.1 to 33.9 Nm).

2. Safety the bolts (10) with the lockwire (C-405).

3. Install the fitting (1) in same manner as given in Step 1. and Step 2..

4. Install tubes (2 and 6) (412-A-71-20-00-01A-720A-A / 01626).



1. Engine mount fitting
2. Tube (vertical)
3. Engine mount forward deck fitting
4. Tube (horizontal)
5. Engine mount forward deck fitting
6. Tube (vertical)
7. Engine mount fitting
8. Engine accessory gearbox
9. Washer (AN960C616)
10. Bolt (three places)

ICN-412-A-712000-B-97499-00020-A-001-01

Figure 1. Forward Mount Fitting (Sheet 1 of 1)

Forward Mount Deck Fittings - Removal

Procedure



WHEN THE FITTINGS (2 OR 4, Figure 1) ARE REMOVED, INDEX SHIM (7) OR SIMILAR SHIM UNDER THE FITTING (2), FOR REINSTALLATION IN SAME LOCATION. IF SHIM IS NOT REINSTALLED IN SAME LOCATION, ENGINE TO TRANSMISSION ALIGNMENT MAY BE AFFECTED. IF SHIMS ARE LOST OR DAMAGED, ENGINE TO TRANSMISSION ALIGNMENT CHECK IS NECESSARY (412-A-71-00-00-01A-272A-A / 01591).

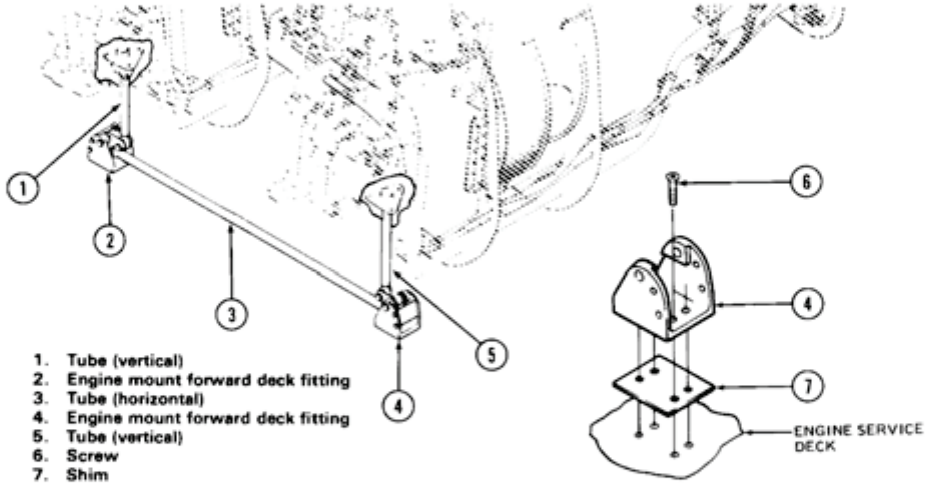
NOTE

Fittings (2 or 4) should be removed only if required for engine to transmission alignment, or to replace a damaged fitting.

1. Remove tubes (1, 3, and 5, Figure 1).

2. Remove four screws (6). Remove fitting (4) and shim (7). Index shim (7) for reinstallation in same location.

3. Remove fitting (2) and index shim in same manner outlined in preceding step.



ICN-412-A-712000-B-97499-00021-A-001-01

Figure 1. Forward Mount Deck Fitting (Sheet 1 of 1)

Forward Mount Deck Fittings - Inspection and Repair

Procedure



IF RECORDS OR PHYSICAL APPEARANCE INDICATES MOUNT HAS BEEN INVOLVED IN AN ACCIDENT OR INCIDENT, REFER TO 412-A-05-50-00-00A-284A-A / 00063 FOR CONDITIONAL INSPECTION REQUIREMENTS.

1. For inspection and repair, refer to 412-CR&O, Chapter 71.

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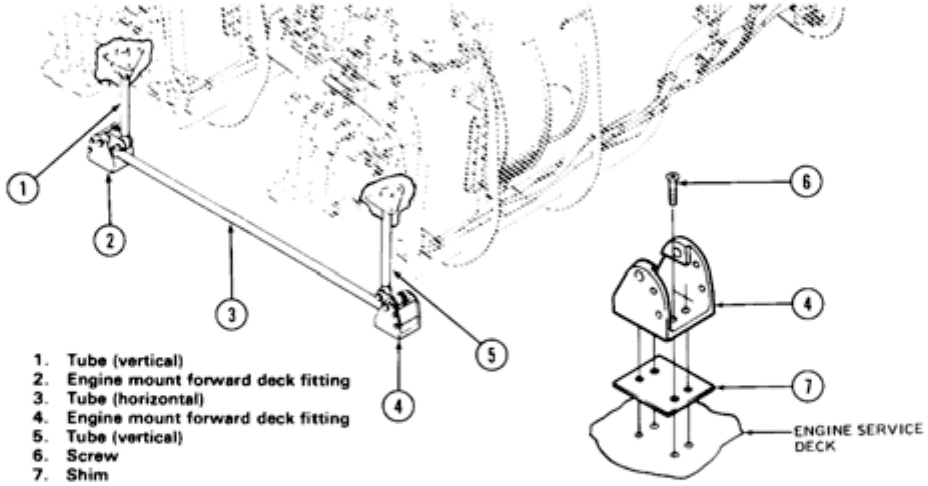
Forward Mount Deck Fittings - Installation

Procedure



SHIM (7, Figure 1) AND SIMILAR SHIM UNDER FITTING (2) SHOULD HAVE BEEN INDEXED FOR REINSTALLATION IN SAME LOCATION WHEN REMOVED. THE SAME SHIMS REMOVED, OR SHIMS OF IDENTICAL THICKNESS, SHALL BE INSTALLED. IF SHIMS HAVE BEEN LOST, DAMAGED, OR MIXED, PERFORM ENGINE TO TRANSMISSION ALIGNMENT CHECK (412-A-71-00-00-01A-272A-A / 01591 OR 412-A-71-00-00-01A-272B-A / 01592).

1. Install fitting (4, Figure 1) and shim (7) with screws (6).
2. Install fitting (2) in same manner outlined in Step 1..
3. Install tubes (1, 3, and 5) (412-A-71-20-00-01A-720A-A / 01626).



ICN-412-A-712000-B-97499-00021-A-001-01

Figure 1. Forward Mount Deck Fitting (Sheet 1 of 1)

Aft Left Mount Tripod - Removal

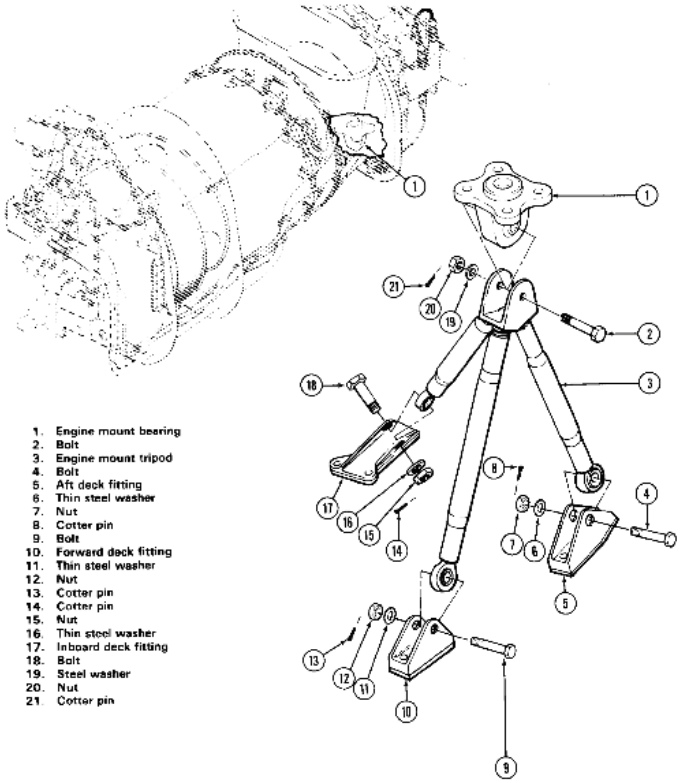
Procedure



LIFTING CAPACITY OF ENGINE HOIST SHALL BE 2,000 LBS (908 KG) OR GREATER.

1. If the engine mount tripod is to be removed while engine is installed, support engine with hoist and engine sling (412-A-71-00-01-01A-520A-A / 01597).

2. Remove bolt (2, Figure 1). Remove bolts (4, 9, and 18).



ICN-412-A-712000-B-97499-00022-A-001-01

Figure 1. Aft Left Mount Tripod (Sheet 1 of 1)

Aft Left Mount Tripod - Inspection and Repair

Procedure



IF RECORDS OR PHYSICAL APPEARANCE INDICATES MOUNT HAS BEEN INVOLVED IN AN ACCIDENT OR INCIDENT, REFER TO 412-A-05-50-00-00A-284A-A / 00063 FOR CONDITIONAL INSPECTION REQUIREMENTS.

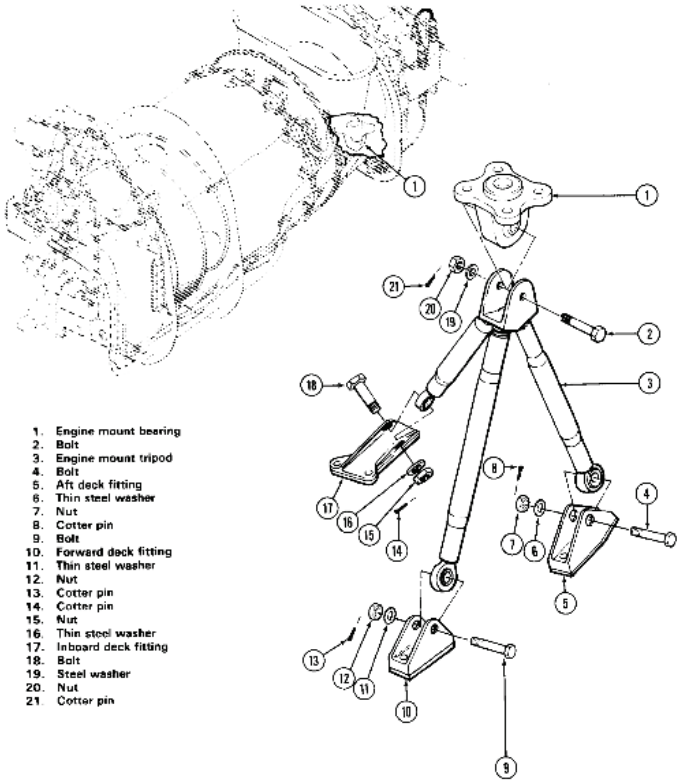
1. For inspection and repair of mount tripod, refer to 412-CR&O, Chapter 71.

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Aft Left Mount Tripod - Installation

Procedure

1. Install tripod (3, Figure 1), with bolt (9), washer (11), nut (12), and cotter pin (13).
2. Install bolt (4), washer (6), nut (7), and cotter pin (8).
3. Install bolt (18), washer (16), nut (15) and cotter pin (14).
4. Install bolt (2) through tripod mount (3) and engine mount bearing (1). Install washer (19), nut (20), and cotter pin (21).



ICN-412-A-712000-B-97499-00022-A-001-01

Figure 1. Aft Left Mount Tripod (Sheet 1 of 1)

Aft Left Mount Tripod Deck Fittings - Removal

Procedure

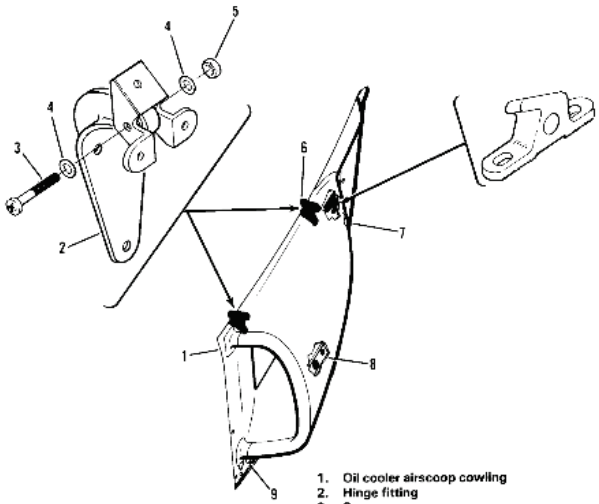


WHEN FITTINGS (3, 10, AND 16, Figure 1) ARE REMOVED, INDEX SHIMS (4, 5, 11, 17, AND 18) FOR REINSTALLATION IN SAME LOCATION. IF SHIMS ARE NOT REINSTALLED IN SAME LOCATION, ENGINE TO TRANSMISSION ALIGNMENT MAY BE AFFECTED. IF SHIMS ARE LOST OR DAMAGED, ENGINE TO TRANSMISSION ALIGNMENT PROCEDURE SHALL BE ACCOMPLISHED.

NOTE

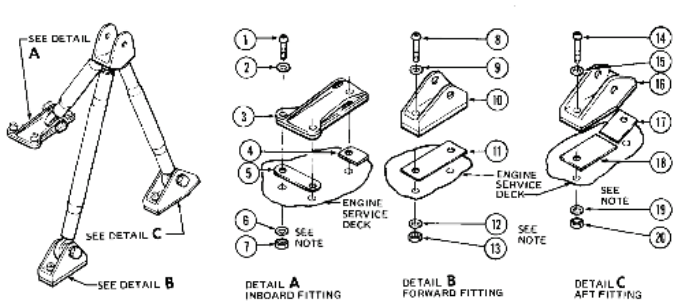
Fittings (3, 10, and 16) should be removed only if required to perform engine to transmission alignment, or to replace a damaged fitting.

1. Remove engine mount tripod (412-A-71-20-00-04A-520A-A / 01633).
2. Remove inboard fitting (3, Figure 2) (detail A).
3. Remove forward fitting (10) (detail B).
4. Remove aft fitting (16) (detail C).



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Figure 1. Oil Cooler Air Scoop Cowling (Sheet 1 of 1)



NOTE: Use maximum of three steel washers under each nut to obtain proper thread engagement.

1. Bolt (3 places)
2. Steel washer (MS20002C4)
3. Inboard deck fitting
4. Deck fitting shim
5. Deck fitting shim
6. Steel washer (see note)
7. Nut (3 places)
8. Bolt (2 places)
9. Steel washer (MS20002C4)
10. Forward deck fitting
11. Deck fitting shim
12. Steel washer (see note)
13. Nut (2 places)
14. Bolt (2 places)
15. Steel washer (MS20002C6)
16. Aft deck fitting
17. Deck fitting shim
18. Deck fitting shim
19. Steel washer (see note)
20. Nut (2 places)

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Figure 2. Aft Left Mount Tripod Deck Fitting (Sheet 1 of 1)

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Aft Left Mount Tripod Deck Fittings - Inspection and Repair

Procedure



IF RECORDS OR PHYSICAL APPEARANCE INDICATE DECK FITTINGS HAVE BEEN INVOLVED IN AN ACCIDENT OR INCIDENT, REFER TO 412-A-05-50-00-00A-284A-A / 00063 FOR CONDITIONAL INSPECTION REQUIREMENTS.

1. For inspection and repair of fittings, refer to 412-CR&O, Chapter 71.

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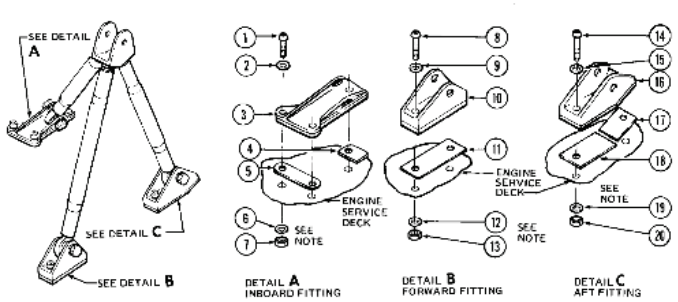
Aft Left Mount Tripod Deck Fittings - Installation

Procedure



SHIMS (4, 5, 11, 17, AND 18, Figure 1) SHALL BE SAME SHIMS AS REMOVED, OR SHIMS OF IDENTICAL THICKNESS. IF SHIMS HAVE BEEN LOST, DAMAGED OR MIXED, PERFORM ENGINE TO TRANSMISSION ALIGNMENT CHECK (412-A-71-00-00-01A-272A-A / 01591 OR 412-A-71-00-00-01A-272B-A / 01592).

1. Install inboard fitting (3, Figure 1) and shims (4 and 5) (detail A).
2. Install forward fitting (10) and shim (11) as shown (detail B).
3. Install aft fitting (16) and shims (17 and 18) (detail C).
4. Install engine mount tripod (412-A-71-20-00-04A-720A-A / 01635).



NOTE: Use maximum of three steel washers under each nut to obtain proper thread engagement.

1. Bolt (3 places)
2. Steel washer (MS20002C4)
3. Inboard deck fitting
4. Deck fitting shim
5. Deck fitting shim
6. Steel washer (see note)
7. Nut (3 places)
8. Bolt (2 places)
9. Steel washer (MS20002C4)
10. Forward deck fitting
11. Deck fitting shim
12. Steel washer (see note)
13. Nut (2 places)
14. Bolt (2 places)
15. Steel washer (MS20002C6)
16. Aft deck fitting
17. Deck fitting shim
18. Deck fitting shim
19. Steel washer (see note)
20. Nut (2 places)

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Figure 1. Aft Left Mount Tripod Deck Fitting (Sheet 1 of 1)

Aft Left and Right Mount Engine Bearings - Removal

Procedure

1. Remove engine mount tripod (412-A-71-20-00-04A-520A-A / 01633).
2. Remove engine mount bipod (412-A-71-20-00-07A-520A-A / 01642).

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Aft Left and Right Mount Engine Bearings - Inspection and Repair

Procedure



IF RECORDS OR PHYSICAL APPEARANCE INDICATE ENGINE MOUNT BEARINGS HAVE BEEN INVOLVED IN AN ACCIDENT OR INCIDENT, REFER TO 412-A-05-50-00-00A-284A-A / 00063 FOR CONDITIONAL INSPECTION REQUIREMENTS.

1. For inspection and repair of mount bearings, refer to 412-CR&O, Chapter 71.

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Aft Left and Right Mount Engine Bearings - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Lockwire	C-405	AR	

Procedure

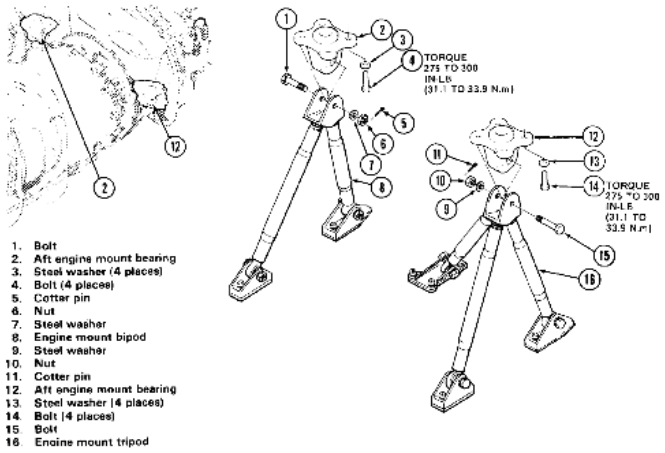
1. Install bearing (12, Figure 1) with bolts (14) and washers (13).

2. Torque bolts (14) 275 to 300 in.lbs. (31.1 to 33.9 N-m). Secure bolts in pairs with lockwire (C-405).

3. Install bearing (2) in same manner outlined in preceding Step 1.. and Step 2..

4. Install engine mount tripod (412-A-71-20-00-04A-720A-A / 01635).

5. Install engine mount bipod (412-A-71-20-00-07A-720A-A / 01644).



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Figure 1. Aft Left and Right Mount Engine Bearing (Sheet 1 of 1)

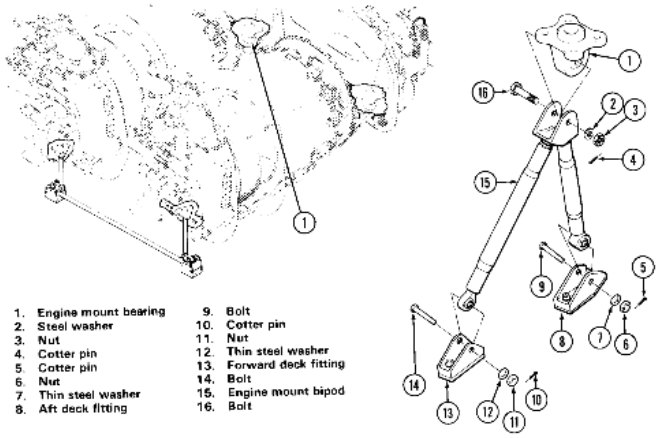
Aft Right Mount Bipod - Removal

Procedure



LIFTING CAPACITY OF ENGINE HOIST SHALL BE 2,000 POUNDS (908 KG) OR GREATER.

1. If engine mount bipod is to be removed while engine is installed, support engine with hoist and engine sling (412-A-71-00-01-01A-520A-A / 01597).
2. Remove bolt (16, Figure 1).
3. Remove bolts (9 and 14).
4. Remove bipod (15).



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Figure 1. Aft Right Mount Bipod (Sheet 1 of 1)

Aft Right Mount Bipod - Inspection and Repair

Procedure



IF RECORDS OR PHYSICAL APPEARANCE INDICATE BIPOD HAS BEEN INVOLVED IN AN ACCIDENT OR INCIDENT, REFER TO 412-A-05-50-00-00A-284A-A / 00063 FOR CONDITIONAL INSPECTION REQUIREMENTS.

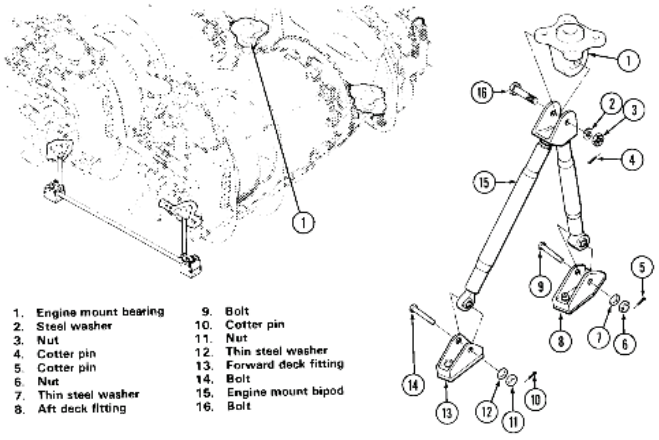
1. For inspection and repair of bipod, refer to 412-CR&O, Chapter 71.

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Aft Right Mount Bipod - Installation

Procedure

1. Install bipod (15, Figure 1).
2. Install bolt (16), washer (2), nut (3), and cotter pin (4).
3. Install bolt (9), washer (7), nut (6), and cotter pin (5).
4. Install bolt (14), washer (12), nut (11), and cotter pin (10).



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Figure 1. Aft Right Mount Bipod (Sheet 1 of 1)

Aft Right Mount Bipod Deck Fittings - Removal

Procedure

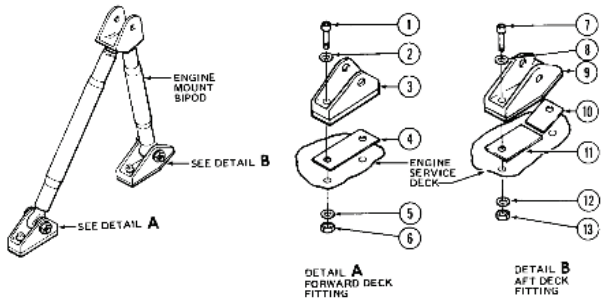


WHEN FITTINGS (3 AND 9, Figure 1) ARE REMOVED, INDEX SHIMS (4, 10, AND 11) UNDER FITTINGS FOR REINSTALLATION IN SAME LOCATION. IF SHIMS ARE NOT REINSTALLED IN SAME LOCATION, ENGINE TO TRANSMISSION ALIGNMENT MAY BE AFFECTED. IF SHIMS ARE LOST, DAMAGED, OR MIXED, ENGINE TO TRANSMISSION ALIGNMENT (412-A-71-00-00-01A-272A-A / 01591 OR 412-A-71-00-00-01A-272B-A / 01592) IS REQUIRED.

NOTE

Fittings (3 and 9) should be removed only to perform engine to transmission alignment or to replace a damaged fitting.

1. Remove engine mount bipod (412-A-71-20-00-07A-520A-A / 01642).
2. Remove forward fitting (3) as shown (detail A).
3. Remove aft fitting (9) as shown in (detail B).



NOTE: Use maximum of three steel washers under each nut to obtain proper thread engagement.

1. Bolt (2 places)
2. Steel washer (MS20002C4)
3. Forward deck fitting
4. Deck fitting shim
5. Steel washers (see NOTE)
6. Nut (2 places)
7. Bolt (2 places)
8. Steel washer (MS20002C6)
9. Aft deck fitting
10. Deck fitting shim
11. Deck fitting shim
12. Steel washer (see NOTE)
13. Nut (2 places)

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Figure 1. Aft Right Mount Bipod Deck Fitting (Sheet 1 of 1)

Aft Right Mount Bipod Deck Fittings - Inspection and Repair

Procedure



IF RECORDS OR PHYSICAL APPEARANCE INDICATE ENGINE MOUNT HAS BEEN INVOLVED IN AN ACCIDENT OR INCIDENT, REFER TO 412-A-05-50-00-00A-284A-A / 00063 FOR CONDITIONAL INSPECTION REQUIREMENTS.

1. For inspection and repair, refer to 412-CR&O, Chapter 71.

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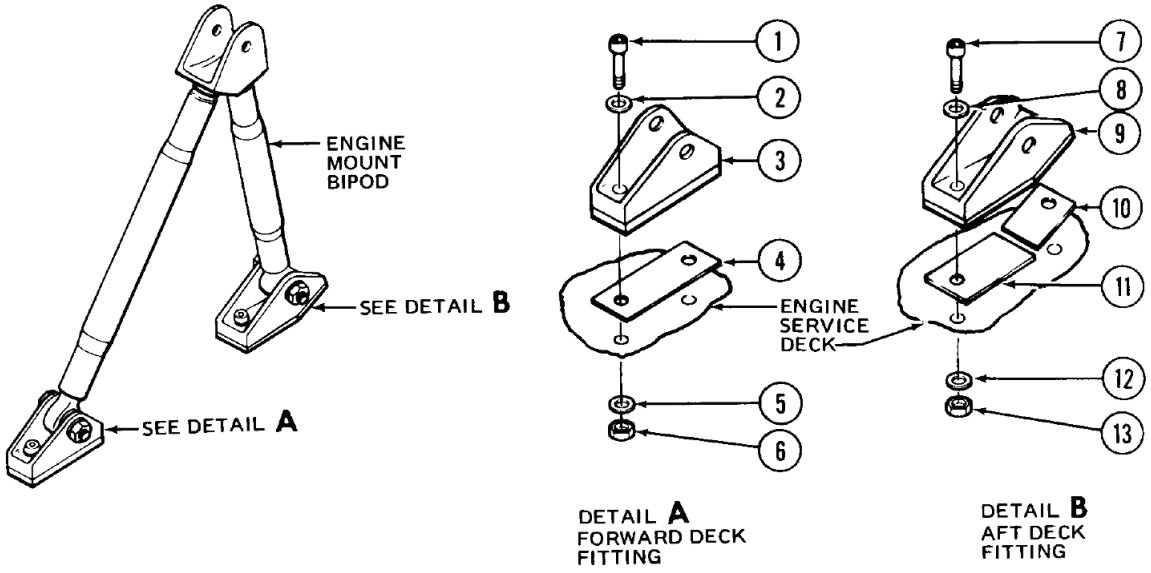
Aft Right Mount Bipod Deck Fittings - Installation

Procedure



SHIMS (4, 10 AND 11, Figure 1) MUST BE THE SAME SHIMS REMOVED, OR SHIMS OF IDENTICAL THICKNESS. IF SHIMS HAVE BEEN LOST, DAMAGED, OR MIXED, PERFORM ENGINE TO TRANSMISSION ALIGNMENT CHECK (412-A-71-00-00-01A-272A-A / 01591 OR 412-A-71-00-00-01A-272B-A / 01592).

1. Install forward fitting (3) and shim (4) (detail A).
2. Install aft fitting (9) and shims (10 and 11) (detail B).
3. Install engine mount bipod (412-A-71-20-00-07A-720A-A / 01644).



NOTE: Use maximum of three steel washers under each nut to obtain proper thread engagement.

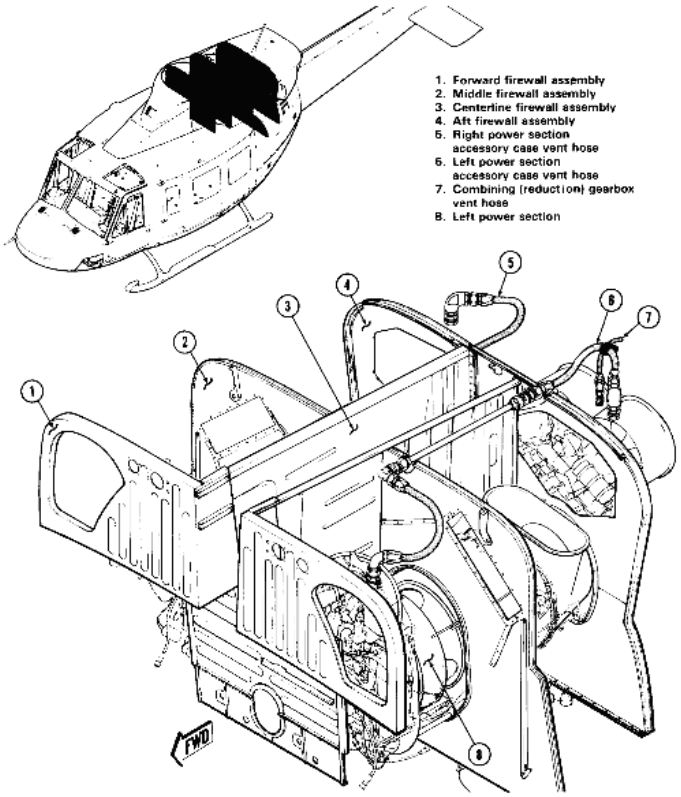
1. Bolt (2 places)
2. Steel washer (MS20002C4)
3. Forward deck fitting
4. Deck fitting shim
5. Steel washers (see NOTE)
6. Nut (2 places)
7. Bolt (2 places)
8. Steel washer (MS20002C6)
9. Aft deck fitting
10. Deck fitting shim
11. Deck fitting shim
12. Steel washer (see NOTE)
13. Nut (2 places)

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Figure 1. Aft Right Mount Bipod Deck Fitting (Sheet 1 of 1)

Firewalls - Description

The forward, middle, centerline, and aft firewall assemblies (1, 2, 3 and 4, Figure 1) are provided to isolate both power sections, combustion chambers, combining gearbox areas, and main driveshaft tunnel in case of fire in the engine compartment.

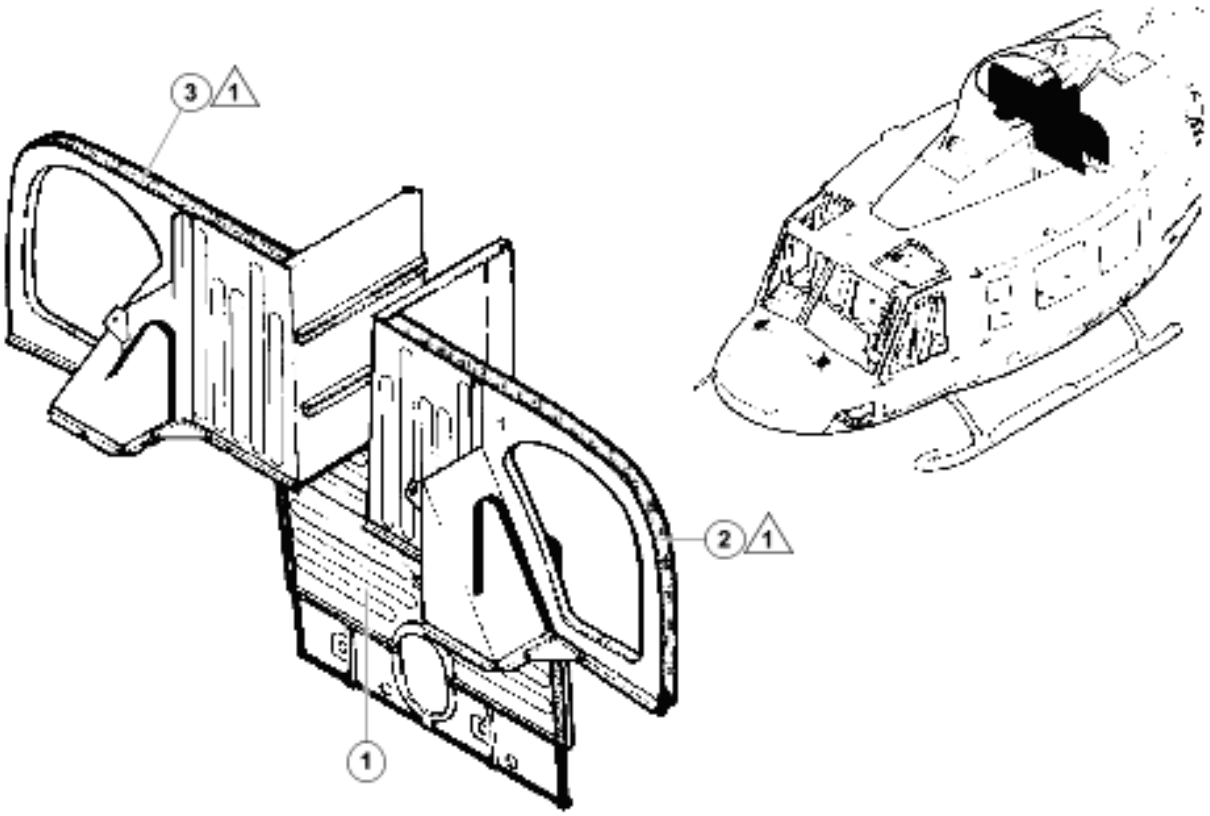


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Figure 1. Engine Firewalls (Sheet 1 of 1)

Air Inlet Cowling Firewall Assembly - Description

The left-hand and right-hand air inlet cowling firewalls (2 and 3, Figure 1) are integral parts of the left-hand and right-hand engine air inlet cowlings. The maintenance of the left-hand and right-hand air inlet cowling firewalls are covered in the cowlings section (412-A-71-10-00-02A-520A-A / 01608, 412-A-71-10-00-02A-280A-A / 01609, 412-A-71-10-00-02A-280B-A / 01610, 412-A-71-10-00-02A-720A-A / 01611, 412-A-71-10-00-02A-921A-A / 01612 and 412-A-71-10-00-02A-271A-A / 01613).



1. Lower forward firewall
2. Left air inlet cowling firewall
3. Right air inlet cowling firewall

NOTE

1 The left and right air inlet cowling firewalls (2 and 3) are an integral part of the engine air inlet cowlings.

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Figure 1. Forward Firewall (Sheet 1 of 1)

Air Inlet Cowling Assembly - Removal

Procedure

1. Remove the forward and aft pylon fairings (412-A-71-10-00-01A-520A-A / 01605).

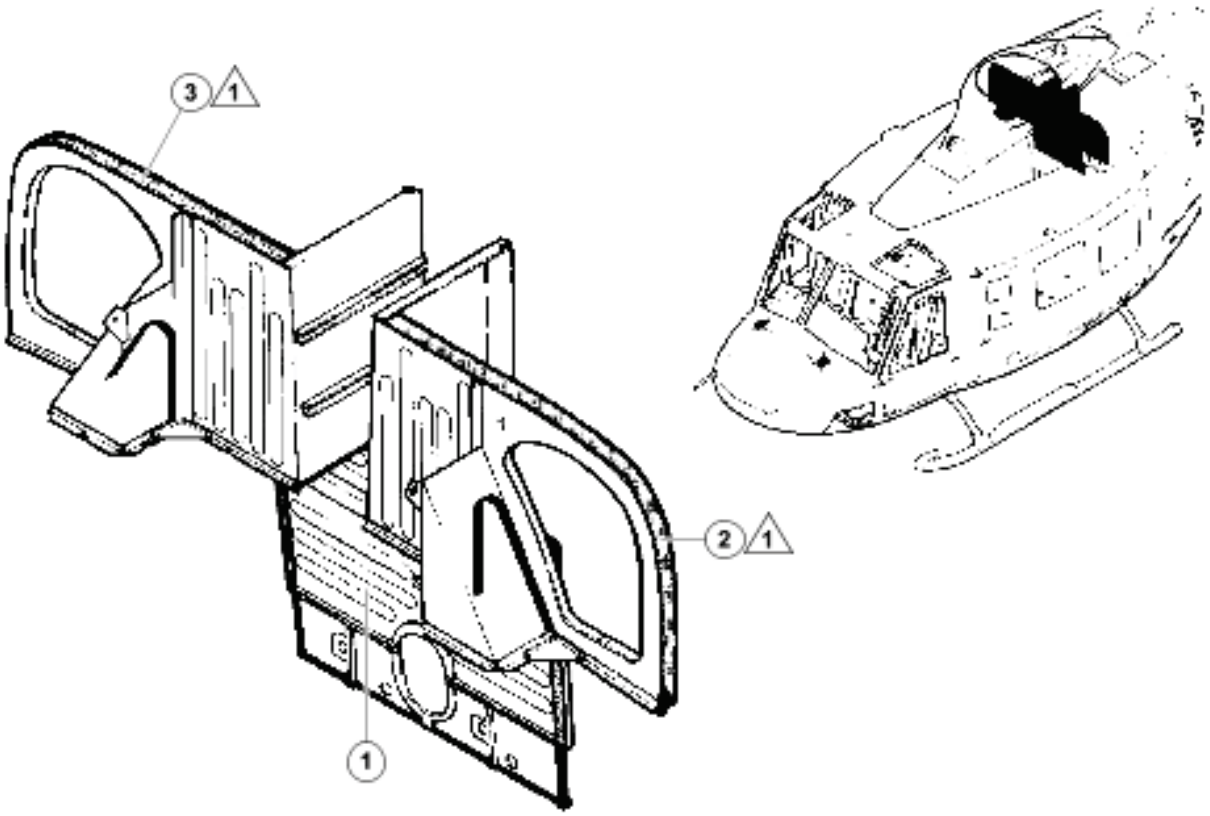
2. Remove the transmission fairing and the left-hand and right-hand engine air inlet cowlings (412-A-71-10-00-02A-520A-A / 01608).

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Air Inlet Cowling Assembly - Inspection

Procedure

1. Inspect lower forward firewall (1, Figure 1) for distortion, holes, and cracks.
2. Inspect for loose and/or missing attaching screws and nut plates. No missing attaching parts are acceptable.
3. Inspect seals for secure installation and deterioration.



1. Lower forward firewall
2. Left air inlet cowling firewall
3. Right air inlet cowling firewall

NOTE

1 The left and right air inlet cowling firewalls (2 and 3) are an integral part of the engine air inlet cowlings.

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Figure 1. Forward Firewall (Sheet 1 of 1)

Air Inlet Cowling Assembly - Installation

Procedure

1. Install the left-hand and right-hand engine air inlet cowlings and the transmission fairing (412-A-71-10-00-02A-720A-A / 01611).

2. Install the forward and aft pylon fairings (412-A-71-10-00-01A-720A-A / 01607).

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Air Inlet Cowling Assembly - Repair

Procedure

1. Minor damage is acceptable if repaired in accordance with FAA Publication AC 43.13-1B, Aircraft Inspection and Repair Manual.

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Middle Firewall Assembly - Removal

Procedure

NOTE

Instructions in this paragraph are for removal of upper left induction baffle and upper portion of middle firewall assemblies, which must be removed and installed when engine is removed and installed. Removal procedures for upper right middle firewall assembly and upper right induction baffle (not illustrated) are the same.

1. Remove forward and aft pylon fairings (412-A-71-10-00-01A-520A-A / 01605).

2. Remove transmission fairing and left-hand and right-hand engine air inlet cowlings (412-A-71-10-00-02A-520A-A / 01608).

3. Remove centerline firewall (412-A-71-30-00-03A-520A-A / 01657).

4. Remove transition duct (412-A-71-61-00-02A-520A-A / 01705).

5. Remove valve assembly (412-A-71-61-00-03A-520A-A / 01709 71-140).

6. Disconnect engine power section accessory case vent hose (not illustrated) from forward side of elbow (2, Figure 1). Remove vent line (not illustrated) from aft side of elbow (2).

7. Remove screws (33).

8. Disconnect hose (25) from engine compressor wash manifold.

9. Remove bolts (40).

10. Remove screws (37).

11. Remove screws (35).

12. Remove bolts (42).

13. Remove screw (32). Remove induction baffle (upper section) (43) and forward duct (24) as an assembly.

14. Remove right induction baffle and forward duct (not illustrated) in same manner outlined in Step 1. through Step 11..

15. Disconnect brace (cowl strut) (3) from fastener (12).

16. Remove screws (4, 6, 9, 18, 21, and 23).

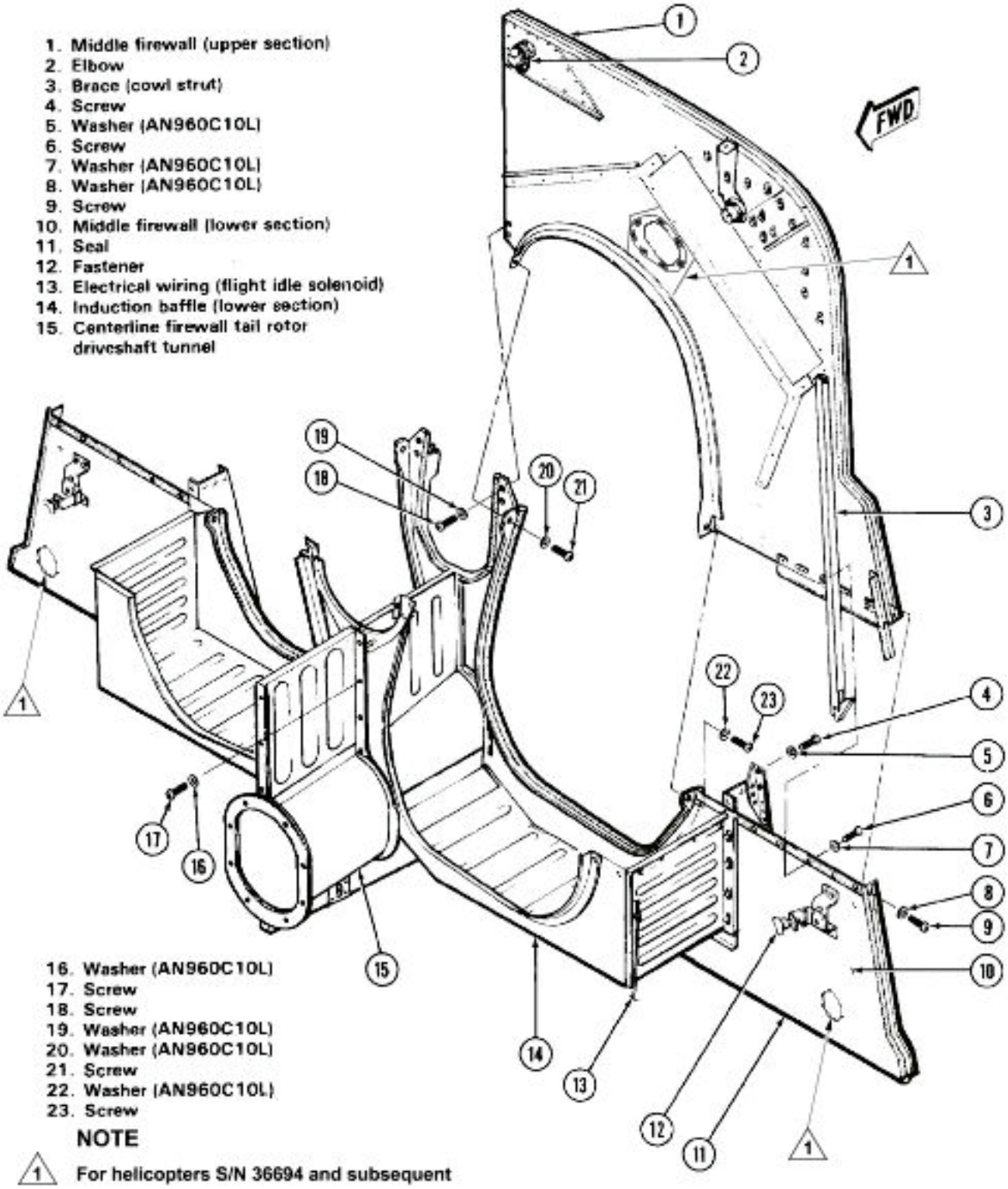
17. Remove middle upper firewall (1).



DO NOT DEFLECT FIREWALL OR INDUCTION BAFFLE MORE THAN NECESSARY TO ACCOMPLISH ENGINE REMOVAL.

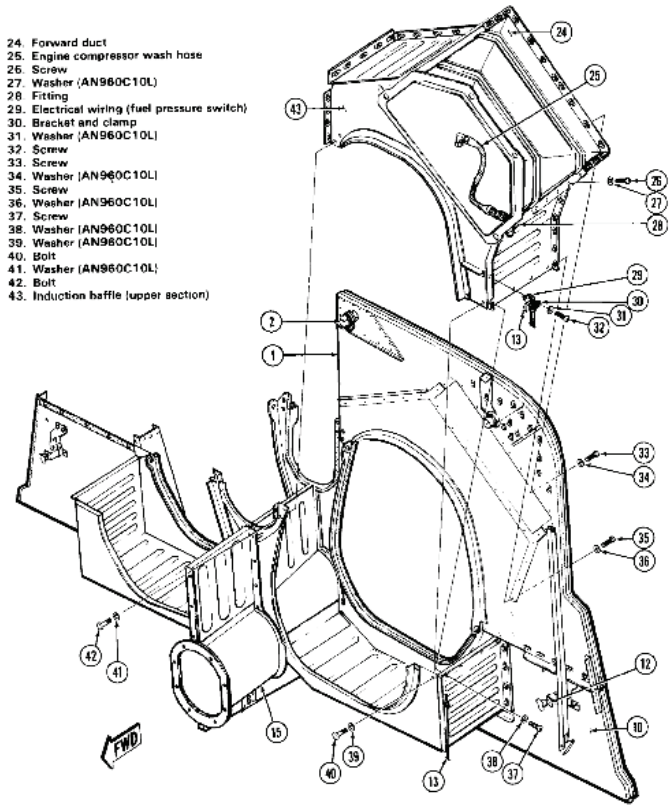
18. If engine is to be removed, deflect section of firewall adjacent to screw (21) forward. Deflect section of induction baffle adjacent to bolt (42) aft. Secure these parts together with wire.

19. Remove right upper firewall (not illustrated) in same manner.



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Figure 1. Middle Firewall (Sheet 1 of 2)



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Figure 1. Middle Firewall (Sheet 2 of 2)

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Middle Firewall Assembly - Inspection

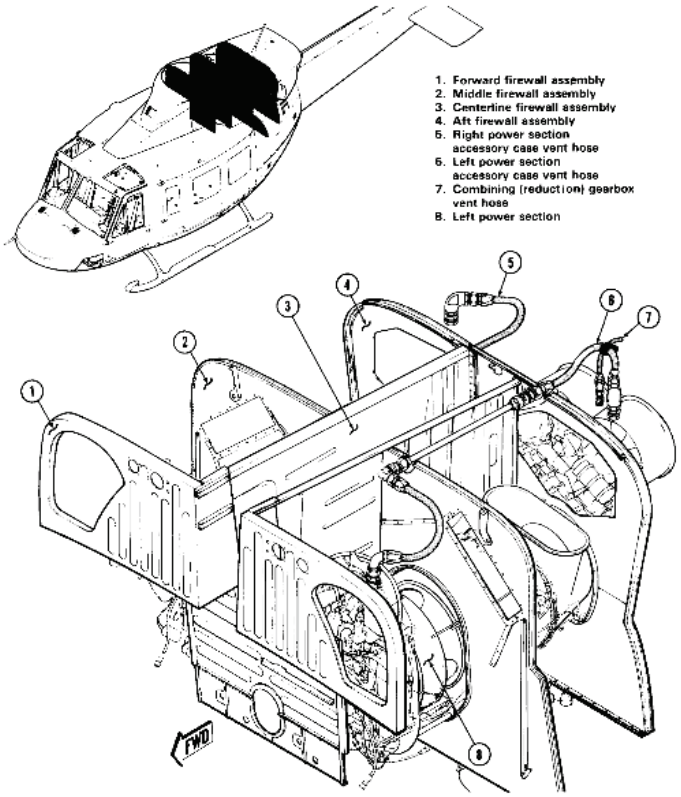
Procedure

1. Inspect middle firewall assembly (2, Figure 1) for distortion, holes, and cracks. Minor damage is acceptable if repaired by standard sheet metal repair procedures.

2. Inspect for loose and/or missing attaching screws and nutplates. No missing attaching parts are acceptable.

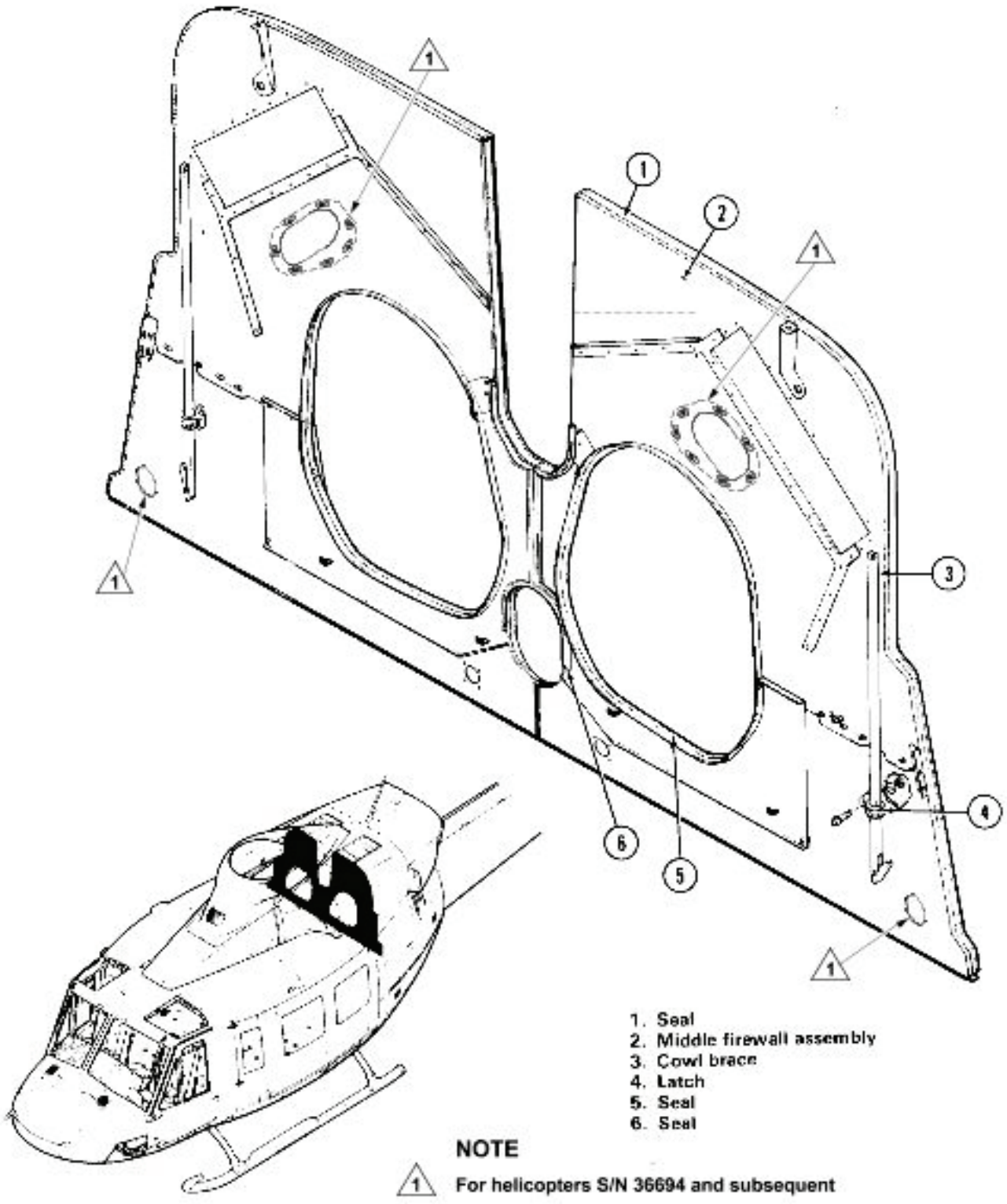
3. Inspect cowl brace (3, Figure 2) and latch (4) for secure installation, distortion, and cracks. Damage affecting function is not acceptable.

4. Inspect seals (1, 5, and 6) for condition and security.



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Figure 1. Engine Firewalls (Sheet 1 of 1)



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Figure 2. Middle Firewall Assembly (Sheet 1 of 1)

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Middle Firewall Assembly - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Sealing compound	C-353	AR	

Procedure

NOTE

Instructions in this paragraph are for installation of upper left induction baffle and upper portion of middle firewall assembly (2, Figure 1), which must be removed and installed when engine is removed and installed. Installation procedures for upper right middle firewall and upper right induction baffle (not illustrated) are the same.

NOTE

Use of beeswax, paste-type drill and tap lubricant, or bar soap on bolts and screws will reduce damage to nutplates and make installation easier.

1. If not previously accomplished, deflect section of firewall adjacent to screw (21, Figure 2) forward. Deflect section of induction baffle adjacent to bolt (42) aft. Secure these parts together with wire. This will allow engine to be lowered vertically into position unless aft firewall has been removed. Remove wire securing induction baffle and firewall after engine is installed.

2. If removed, install screws (17) and washers (16).

3. Position middle firewall (upper section) (1) on middle firewall (lower section) (10).

4. Install screw (21) and washer (20).

5. Install screws (6) and washers (7).

6. Install screw (23) and washer (22).

7. Install screws (9) and washers (8).

8. Install screws (4) and washers (5).

9. Secure brace (cowl strut) (3) with fastener (12).

10. Install right induction baffle (not illustrated) in same manner.

11. If not previously accomplished, install engine compressor wash hose (25) on fitting (28).

NOTE

Ensure correct part number forward duct (24) is installed in following step.

12. If not previously accomplished, install forward duct (24) on induction baffle (upper section) (43) and install screws (26) and washers (27).

13. Position induction baffle (upper section) (43) and forward duct (24) on forward side of middle firewall (upper section) (1).

14. Install screws (37) and washers (38).

15. Install bolts (42) and washers (41).

16. Install screws (35) and washers (36).

17. Install bolts (40) and washer (39).

18. Attach engine compressor wash hose (25) to engine compressor wash manifold (not illustrated).

19. Install screws (33) and washers (34).

20. Secure electrical wiring (29 and 13) with bracket and clamp (30), washer (31), and screw (32).

21. Install engine power section accessory case vent hose (not illustrated) on forward side of elbow (2). Install vent line (not illustrated) on aft side of elbow.

22. Install right induction baffle and forward duct (not illustrated) in same manner outlined in Step 11. through Step 21..

23. Install valve assembly (412-A-71-61-00-03A-720A-A / 01712).

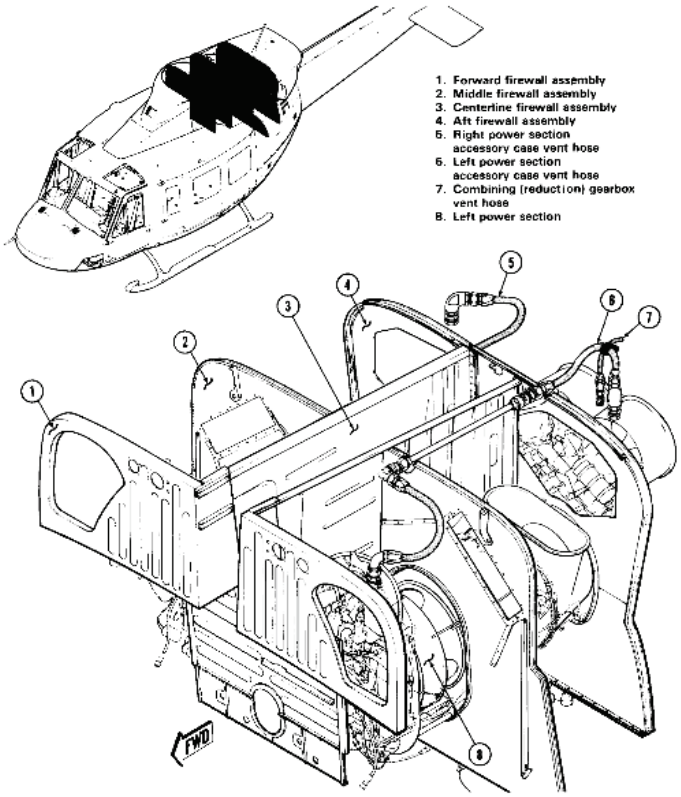
24. Install transition duct (412-A-71-61-00-02A-720A-A / 01708).

25. Install centerline firewall assembly (412-A-71-30-00-03A-720A-A / 01659).

26. Seal any small openings in firewalls with sealing compound (C-353).

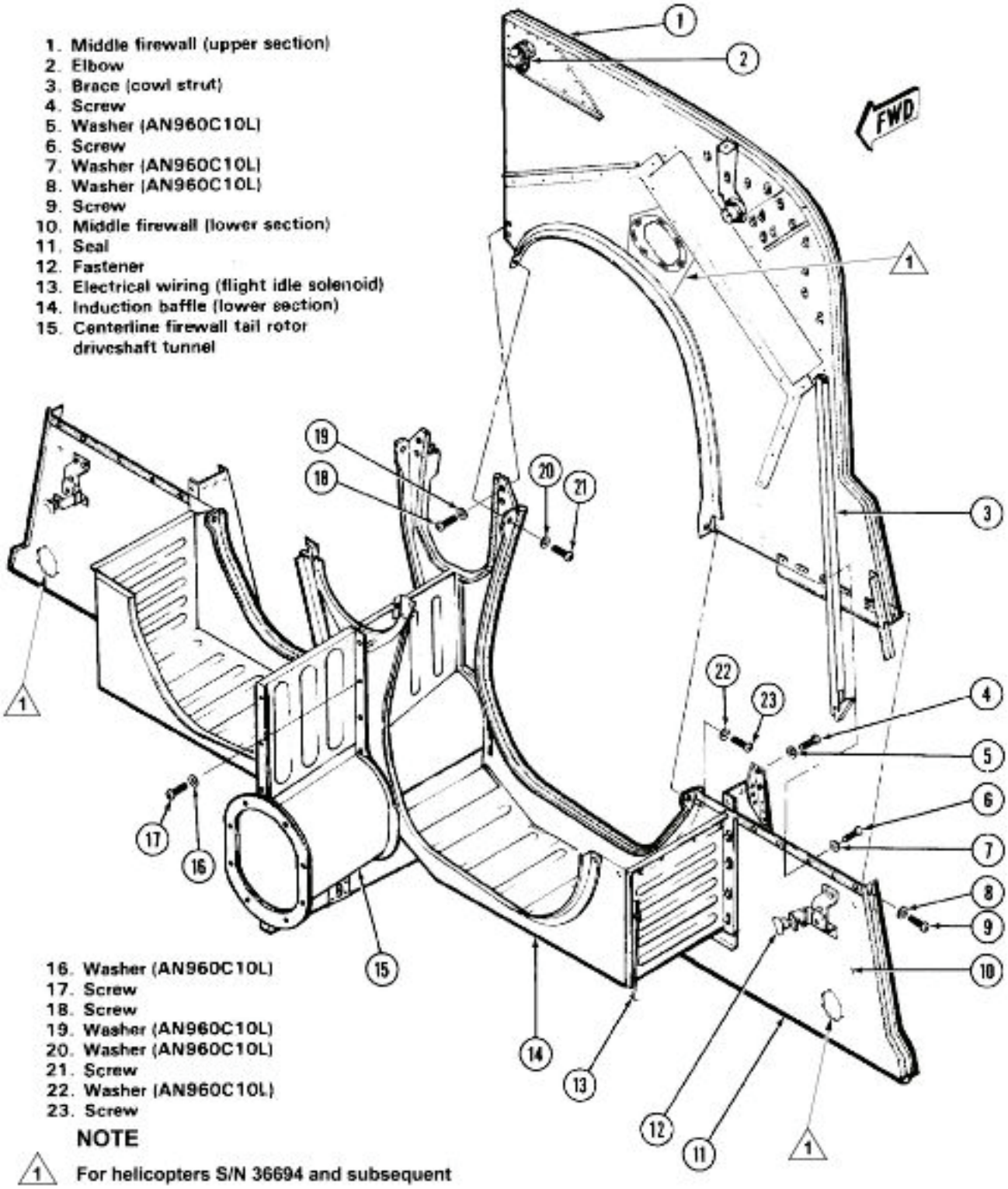
27. Install left-hand and right-hand engine air inlet cowlings and transmission fairing (412-A-71-10-00-02A-720A-A / 01611).

28. Install forward and aft pylon fairings (412-A-71-10-00-01A-720A-A / 01607).



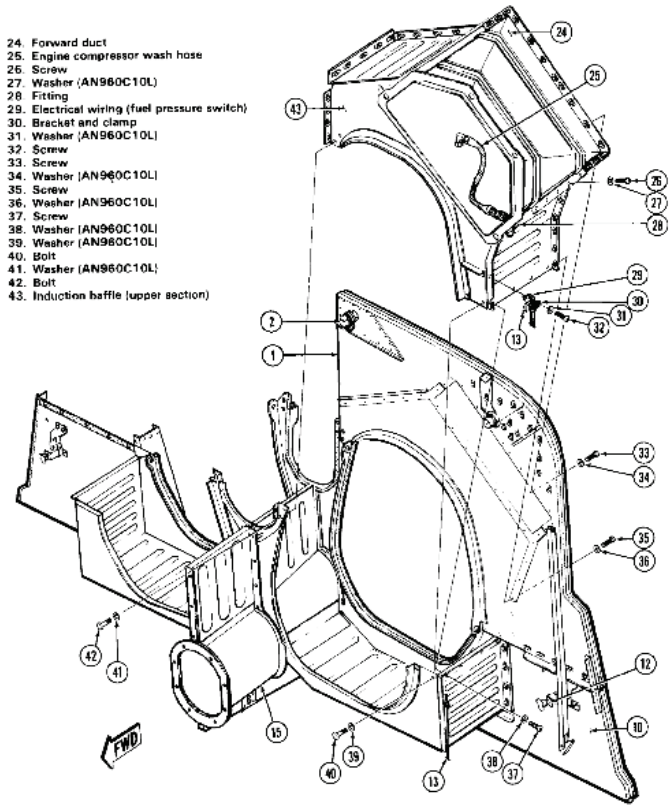
ICN-412-A-713000-B-97499-00001-A-001-01

Figure 1. Engine Firewalls (Sheet 1 of 1)



ICN-412-A-713000-B-97499-00003-A-001-01

Figure 2. Middle Firewall (Sheet 1 of 2)



ICN-412-A-713000-B-97499-00004-A-001-01

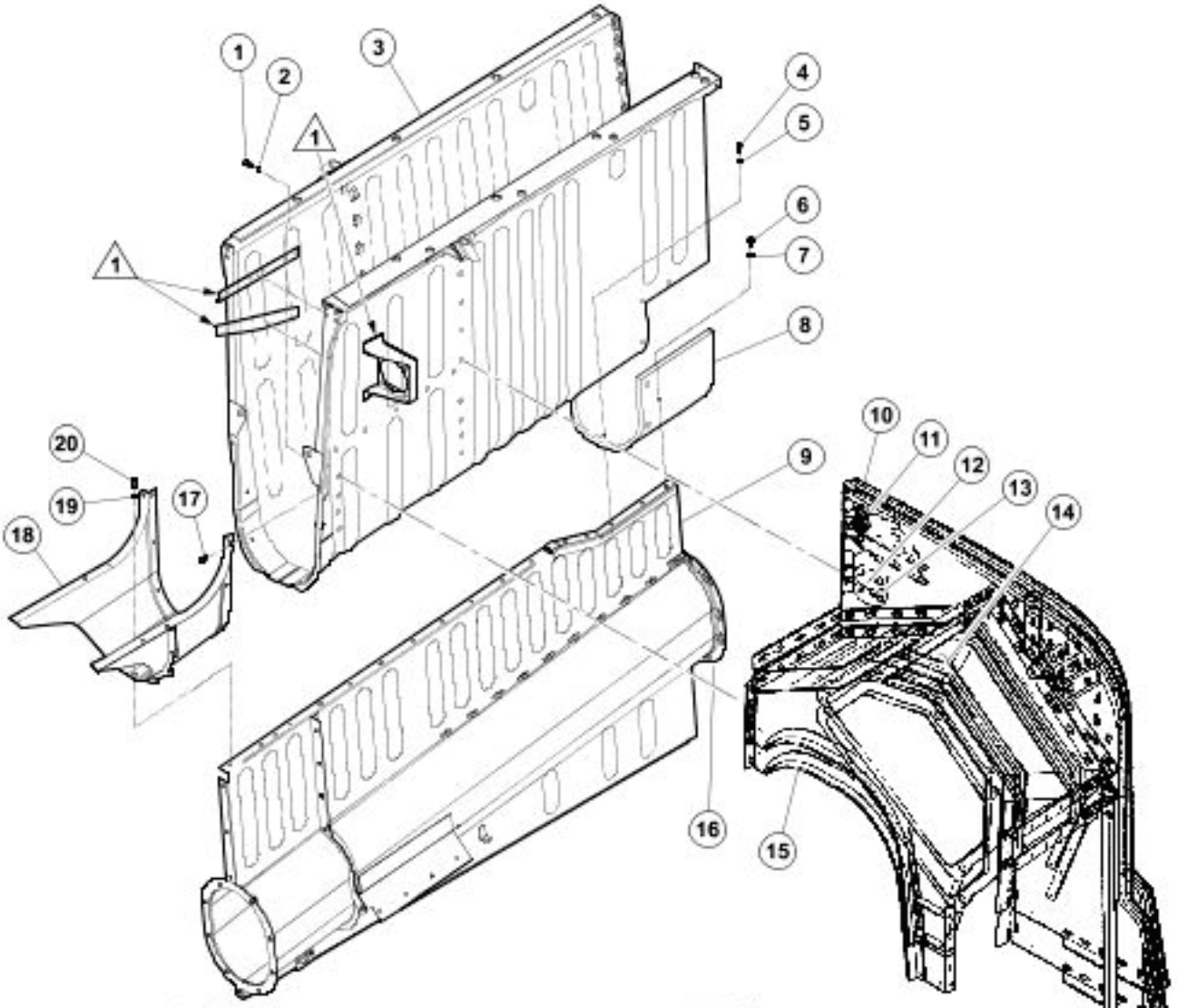
Figure 2. Middle Firewall (Sheet 2 of 2)

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Centerline Firewall Assembly - Removal

Procedure

1. Remove engine-to-transmission driveshaft (412-A-63-10-01-01A-520A-A / 00994).
2. Remove engine-to-driveshaft adapter (412-A-63-10-01-02A-520A-A / 01001).
3. Remove screws (17, Figure 1), screws (20), and washers (19). Remove forward cover (18).
4. Remove screws (not illustrated) on each side directly above aft cover (8).
5. Remove screws (13) and washer (12) on each side securing middle firewall (left) (10) to centerline firewall (3).
6. Remove screws and washers (not illustrated) on each side securing centerline firewall (3) to middle firewall (left) (10) and to right middle firewall assembly (not illustrated).
7. Remove screws (4) and washers (5) securing aft cover (8).
8. Remove screws (1) and washers (2) on each side securing centerline firewall (3) to induction baffle (15).
9. Remove screws (4) and washers (5) securing centerline firewall (3) to web (9).
10. Remove electrical connector (inside) at front of centerline firewall (3).
11. Remove two hoses on each side (outside) of centerline firewall (3).
12. Remove centerline firewall (3).
13. Remove screws (6) and washers (7).
14. Remove aft cover (8).



1. Screw
2. Washer (AN960C10L)
3. Centerline firewall
4. Screw
5. Washer (AN960C10L)
6. Screw
7. Washer (AN960C10L)
8. Aft cover
9. Web
10. Middle firewall (left)
11. Elbow
12. Washer (AN960C10L)
13. Screw
14. Forward duct
15. Induction baffle
16. Lower cover (tail rotor driveshaft)
17. Screw
18. Forward cover
19. Washer (AN960C10L)
20. Screw

NOTE

1 For helicopters S/N 36694 and subsequent

ICN-412-A-713000-B-97499-00006-A-001-01

Figure 1. Centerline Firewall (Sheet 1 of 1)

Centerline Firewall Assembly - Inspection

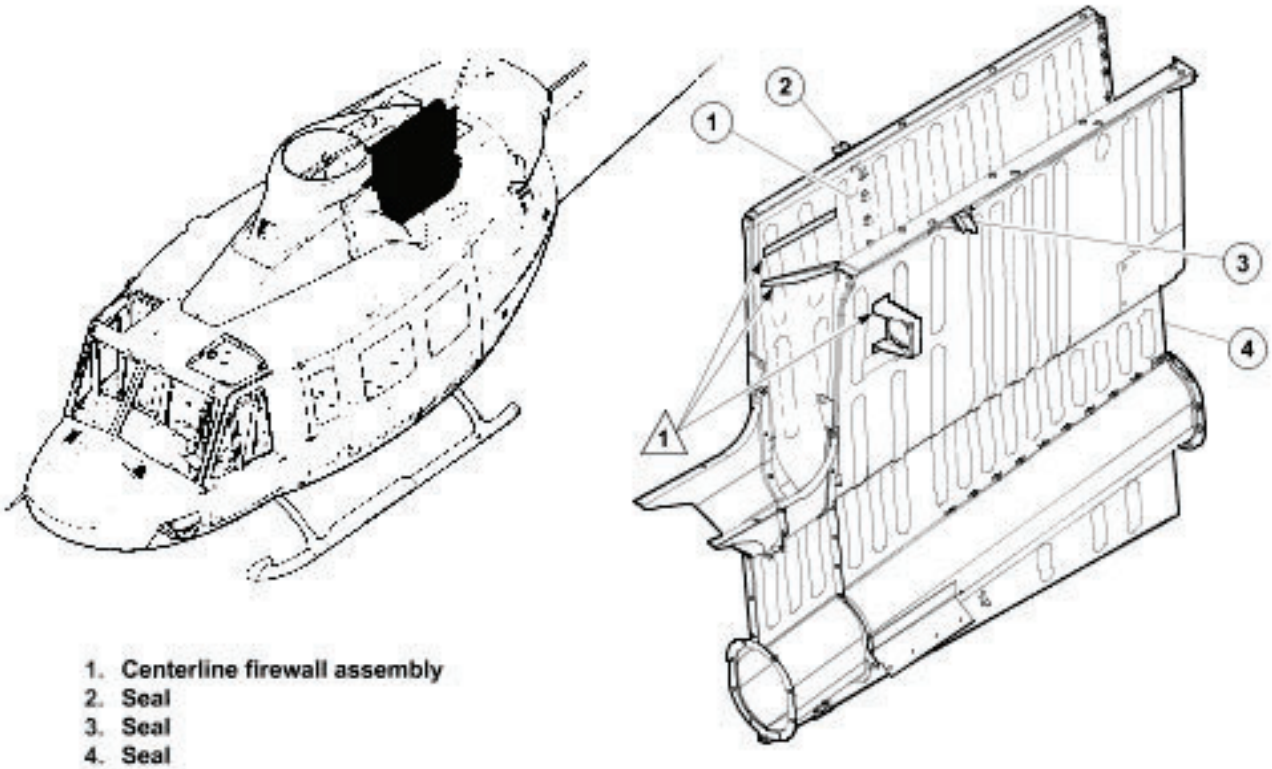
Procedure

1. Inspect centerline firewall assembly (1, Figure 1) for distortion, holes, and cracks.

   Minor damage is acceptable if repaired in accordance with FAA Publication AC 43.13-1B, Aircraft Inspection and Repair Manual.

2. Inspect for loose and/or missing attaching screws and nutplates. No missing attaching parts are acceptable.

3. Inspect seals (2, 3 and 4) for condition and security.



1. Centerline firewall assembly
2. Seal
3. Seal
4. Seal

NOTE

1 For helicopters S/N 36694 and subsequent

ICN-412-A-713000-B-97499-00007-A-001-01

Figure 1. Centerline Firewall (Sheet 1 of 1)

Centerline Firewall Assembly - Installation

Procedure

NOTE

Use of beeswax, paste-type drill and tap lubricant, or bar soap on bolts and screws will reduce damage to nutplates and make installation easier.

1. Position aft cover (8, Figure 1) on web (9). Install screws (6) and washers (7) at lower aft holes. Do not install screws at locations for screws (4) that also pass through centerline firewall assembly (3).

2. Position centerline firewall assembly (3) between middle firewall (10) and forward firewall (not illustrated).

3. Install screws (4) and washers (5) to secure centerline firewall to web (9).

4. Install screws (1) and washers (2) on each side to secure centerline firewall assembly (3) to induction baffle (15).

5. Install screws (4) and washers (5) to complete installation of aft cover (8).

6. Install screws (not illustrated) on each side to secure centerline firewall (3) to left middle firewall (10) and to right middle firewall (not illustrated).

7. Install screws (13) and washer (12) on each side to secure middle firewall (10) to centerline firewall assembly (3).

8. Install screws and washer (not illustrated) on each side directly above aft cover (8).

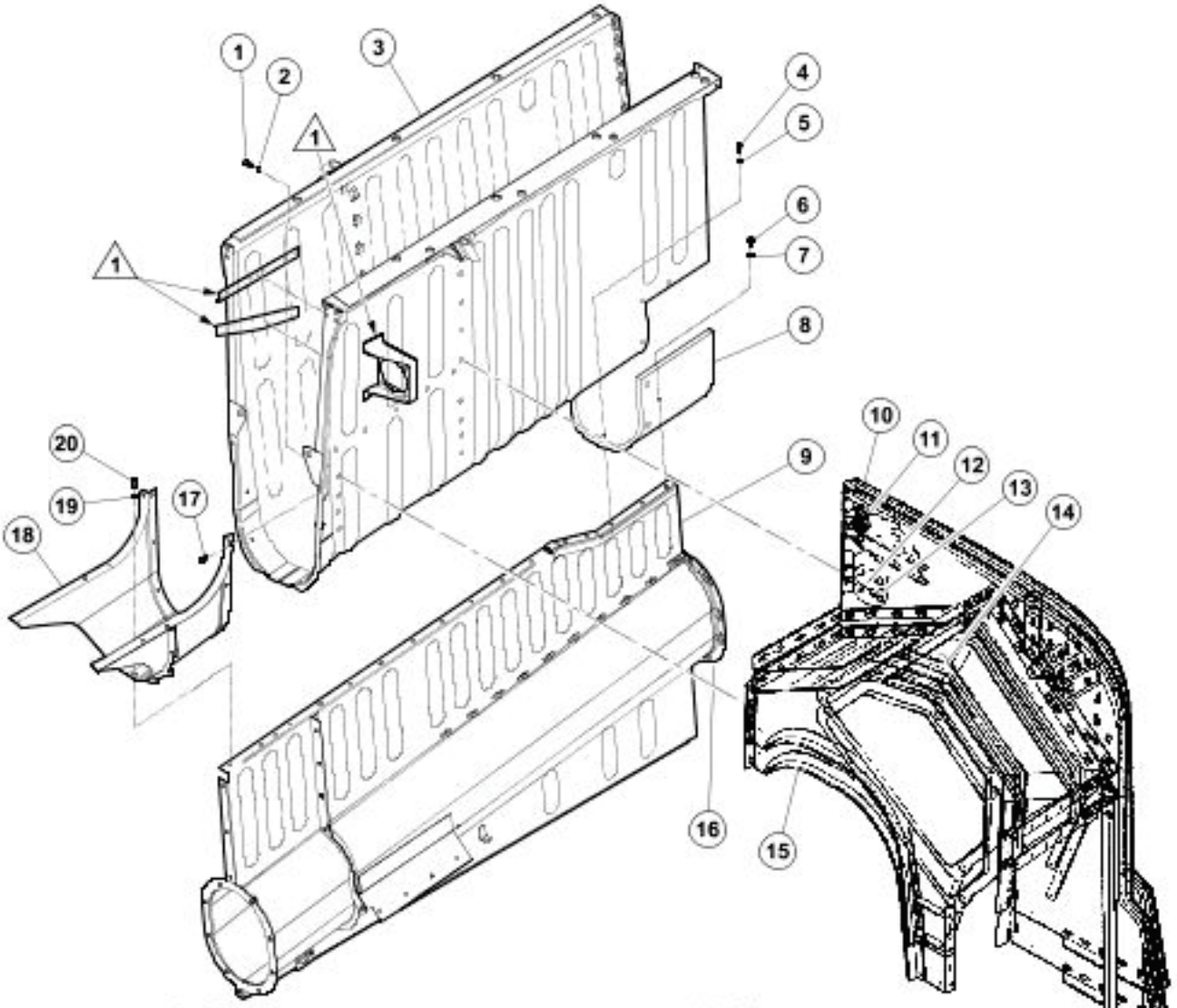
9. Position forward cover (18) on centerline firewall assembly (3). Install screws (20) and washers (19). Install countersunk screws (17).

10. Install electrical connector.

11. Install hoses on outside of each side of centerline firewall.

12. Install engine to driveshaft adapter (412-A-63-10-01-02A-720A-A / 01004).

13. Install engine to transmission driveshaft (412-A-63-10-01-01A-720A-A / 00999).



1. Screw
2. Washer (AN960C10L)
3. Centerline firewall
4. Screw
5. Washer (AN960C10L)
6. Screw
7. Washer (AN960C10L)
8. Aft cover
9. Web
10. Middle firewall (left)
11. Elbow
12. Washer (AN960C10L)
13. Screw
14. Forward duct
15. Induction baffle
16. Lower cover (tail rotor driveshaft)
17. Screw
18. Forward cover
19. Washer (AN960C10L)
20. Screw

NOTE

1 For helicopters S/N 36694 and subsequent

ICN-412-A-713000-B-97499-00006-A-001-01

Figure 1. Centerline Firewall (Sheet 1 of 1)

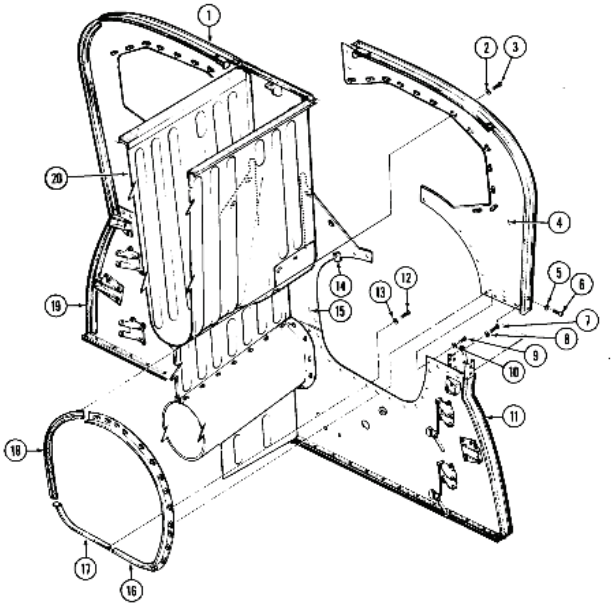
Aft Firewall Assembly - Removal

Procedure

NOTE

Instructions in this paragraph are for removal of left upper section (4, Figure 1) and center section (15) which must be removed and installed when engine is removed and installed. Removal procedure for the right upper section is the same.

1. Remove ejector (412-A-71-61-00-01A-520A-A / 01701).
2. Remove torque transmitter (412-A-71-01-00-04A-520A-A / 01673).
3. Remove screws (7, 6, and 4).
4. Remove screws (9) and stiffener (16).
5. Remove screws (3) and stiffener (18).
6. Remove screws (12) and stiffener (17).
7. Remove right upper section (not illustrated) in same manner outlined in Step 1.. through Step 8..
8. Remove engine (412-A-71-00-01-01A-520A-A / 01597), if desired. Firewall center section (15) will remain with engine.
9. Bend firewall center section (15) into a U shape and remove from engine.



1. Aft firewall (upper section right side)
2. Washer
3. Screw
4. Aft firewall (upper section left side)
5. Washer (AN960C10L)
6. Screw
7. Screw
8. Washer (AN960C10L)
9. Screw
10. Washer (AN960C10L)
11. Aft firewall (lower section left side)
12. Screw
13. Washer (AN960C10L)
14. Clip
15. Aft firewall (center section)
16. Stiffener
17. Stiffener
18. Stiffener
19. Aft firewall (lower section right side)
20. Centerline firewall

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Figure 1. Aft Firewall (Sheet 1 of 1)

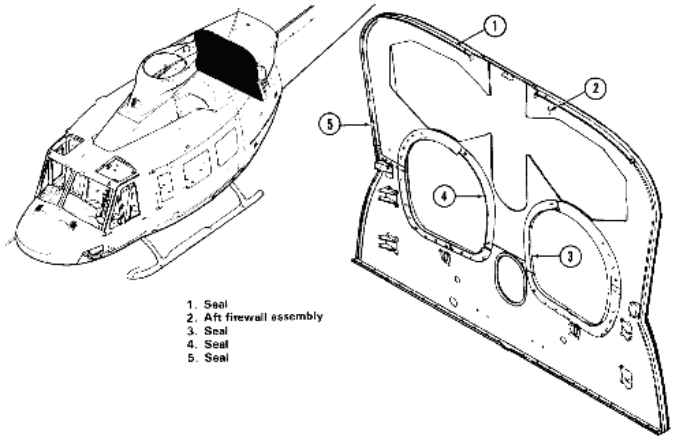
Aft Firewall Assembly - Inspection

Procedure

1. Inspect the aft firewall assembly (2, Figure 1) for distortion, holes, and cracks. Minor damage is acceptable if repaired in accordance with the FAA Publication AC 43.13-1B, Aircraft Inspection and Repair Manual.

2. Inspect for loose and/or missing attaching screws and nutplates. No missing attaching parts are acceptable.

3. Inspect the seals (1, 3, 4, and 5) for secure installation and for deterioration.



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Figure 1. Aft Firewall (Sheet 1 of 1)

Aft Firewall Assembly - Installation

Procedure

NOTE

The following instructions are for the installation of the left upper section (4, Figure 1) and center section (15), which must be removed and installed when engine is removed and installed.

NOTE

The use of beeswax, paste-type drill and tap lubricant, or bar soap on the bolts and screws will reduce damage to nutplates and make installation easier.

1. Install a piece of flexible wire (or string) approximately 18 inches (460 mm) long in the bottom center hole of the aft firewall center section (15, Figure 1).

NOTE

Do Step 2. with the engine installed on the stand.

2. Bend the firewall center section (15) into a U shape and position the firewall between the engine power sections immediately forward of the engine combining gearbox.

3. Secure the wire (or string) installed in Step 1. to the manual fuel control linkage jackshaft to hold the firewall center section (15) aft (preventing interference with lower firewall section (11) when engine is installed).

4. Carefully position the engine in the helicopter and make sure the firewall center section (15) engages the lower section (11). Secure the engine in the engine mounts (412-A-71-00-01-01A-720A-A / 01598).

5. Remove the wire (or string) installed in Step 3..

6. Position the stiffener (17) on the lower section (11) and forward of the engine fireshield (not illustrated). Install the screws (12) and washers (13).

7. Position the stiffener (18) under the clip (14) forward of the engine fireshield. Install the screws (3) and washers (2).

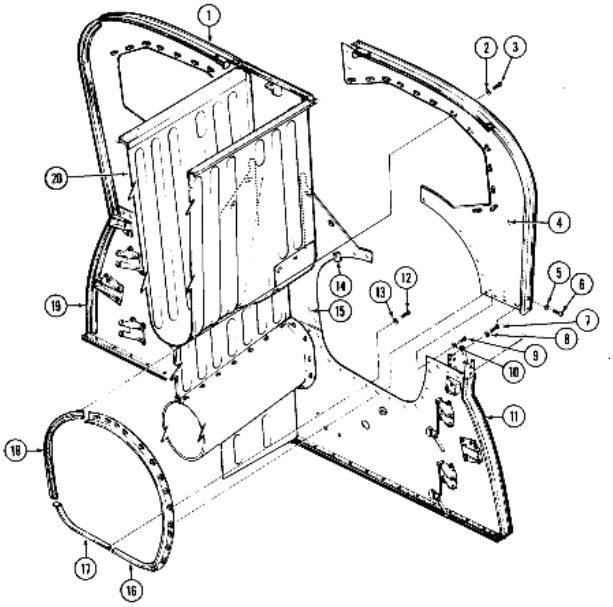
8. Position the stiffener (16) on the firewall. Install the screws (9) and washers (10) through the firewall lower section (11).

9. Position the firewall upper section (4) on the lower section (11). Install the screws (6) and washers (5). Install the screws (7) and washers (8).

10. Install the screws (9) and washers (10). Do not install the top screws at this time. The top screws will be installed with the ejector.

11. Install the ejector (412-A-71-61-00-01A-720A-A / 01704).

12. Install the torque transmitter (412-A-71-01-00-04A-720A-A / 01674).



1. Aft firewall (upper section right side)
2. Washer
3. Screw
4. Aft firewall (upper section left side)
5. Washer (AN960C10L)
6. Screw
7. Screw
8. Washer (AN960C10L)
9. Screw
10. Washer (AN960C10L)
11. Aft firewall (lower section left side)
12. Screw
13. Washer (AN960C10L)
14. Clip
15. Aft firewall (center section)
16. Stiffener
17. Stiffener
18. Stiffener
19. Aft firewall (lower section right side)
20. Centerline firewall

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Figure 1. Aft Firewall (Sheet 1 of 1)

Gas Producer Tachometer Generator - Description

A gas producer tachometer generator is mounted on the front of each power section accessory gearbox. The generators are connected by electrical harness to the respective gas producer (GAS PROD) indicator on the instrument panel.

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Gas Producer Tachometer Generator - Removal

Procedure

NOTE

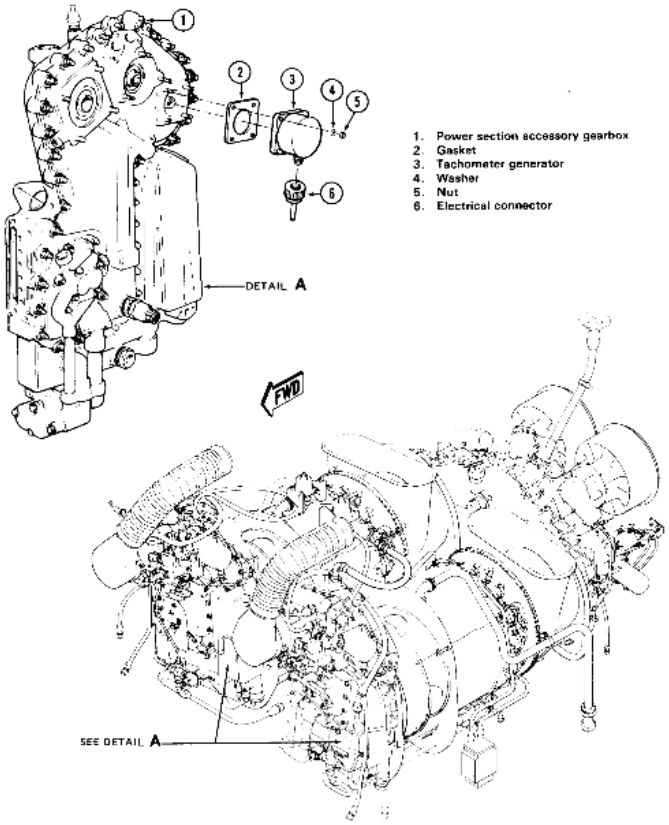
Maintenance procedures are the same for both gas producer tachometer generators.

1. Open engine cowling.

2. Disconnect electrical connector (6, Figure 1).

3. Remove nuts (5) and washers (4). Remove generator (3).

4. Remove and discard gasket (2), if unserviceable.



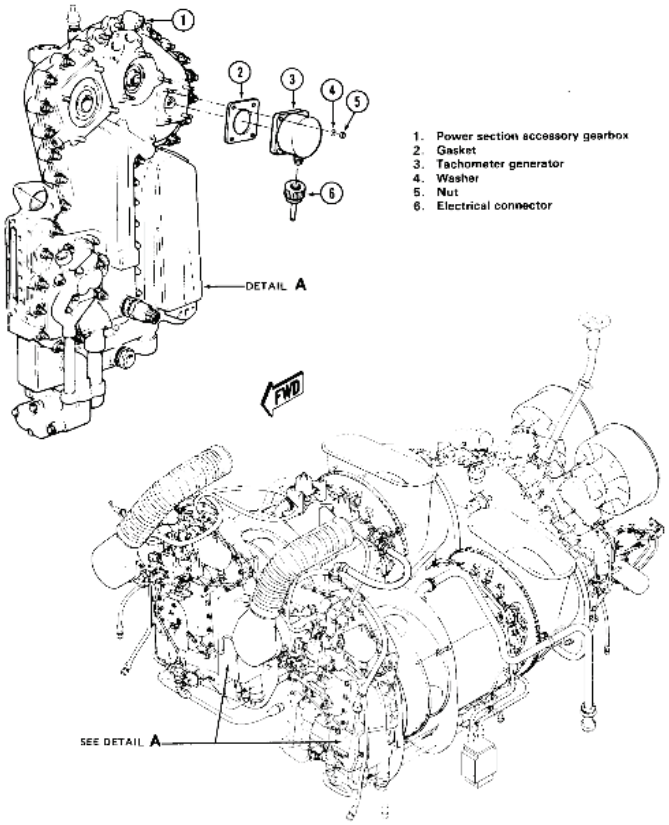
ICN-412-A-710100-B-97499-00001-A-001-01

Figure 1. Gas Producer Tachometer Generator (Sheet 1 of 1)

Gas Producer Tachometer Generator - Installation

Procedure

1. Install gasket (2, Figure 1) if removed.
2. Install generator on mounting studs with electrical receptacle pointing down. Install washers (4) and nuts (5).
3. Connect electrical connector (6).
4. Close engine cowling.



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Figure 1. Gas Producer Tachometer Generator (Sheet 1 of 1)

Power Turbine Tachometer Generator - Description

Two power turbine tachometer generators are mounted on the rear of the engine reduction gearbox. The generators are connected by electrical harness to the triple tachometer on the instrument panel.

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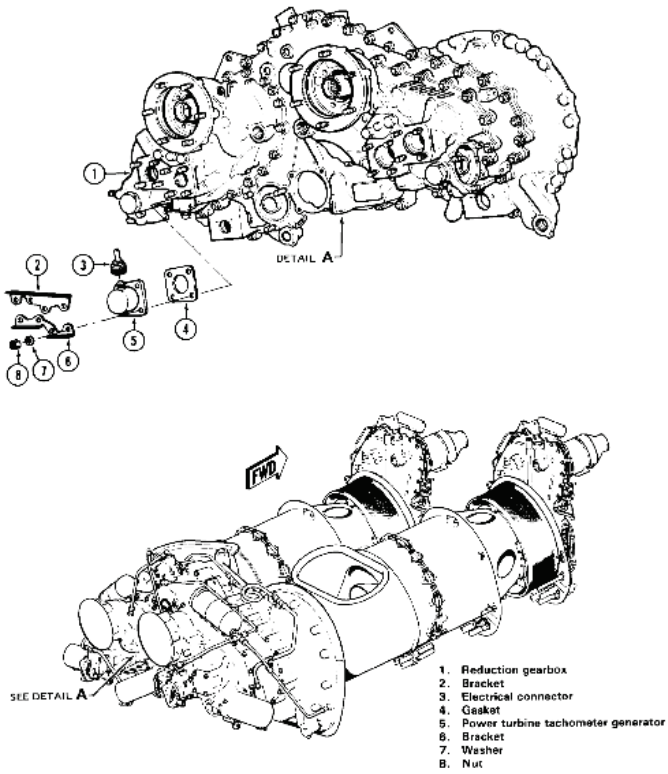
Power Turbine Tachometer Generator - Removal

Procedure

NOTE

Maintenance procedures are the same for both power turbine tachometer generator, except brackets (2 and 6, Figure 1) are mounted with the left tachometer generator.

1. Open engine cowling.
2. Disconnect electrical connector (3, Figure 1).
3. Remove nuts (8) and washers (7).
4. If removing left generator, remove brackets (2 and 6) from generator mounting studs.
5. Remove generator (5).
6. Remove and discard gasket (4), if unserviceable.



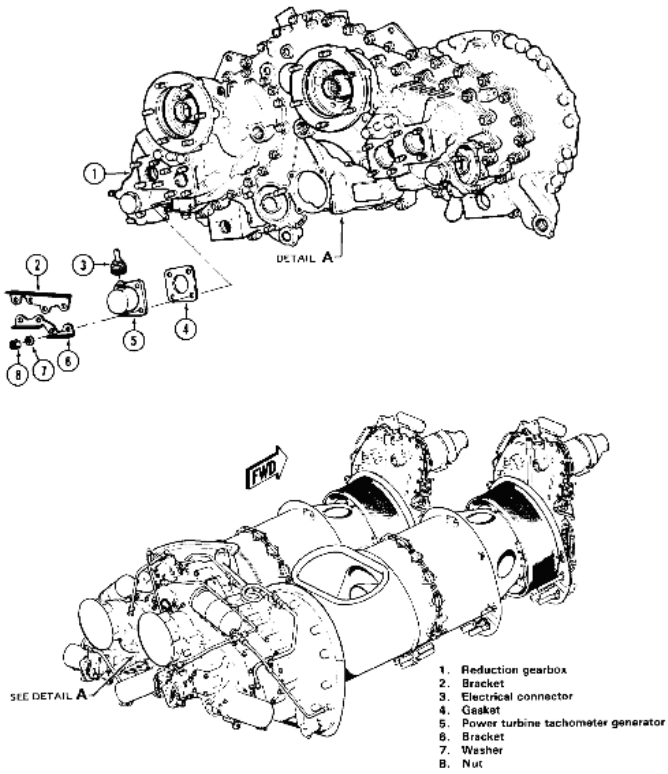
ICN-412-A-710100-B-97499-00002-A-001-01

Figure 1. Power Turbine Tachometer Generator (Sheet 1 of 1)

Power Turbine Tachometer Generator - Installation

Procedure

1. Install new gasket (4, Figure 1), if removed.
2. Install generator (5) on mounting studs with electrical receptacle pointing up. If installing left generator, place brackets (2 and 6) over mounting studs.
3. Install washers (7) and nuts (8).
4. Connect electrical connector (3).
5. Close engine cowling.



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Figure 1. Power Turbine Tachometer Generator (Sheet 1 of 1)

Differential Pressure Indicator - Description

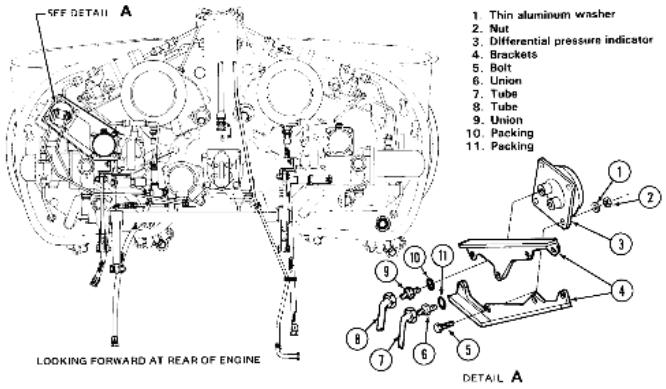
A differential pressure indicator is mounted on the left aft side of the reduction gearbox. The indicator is connected to inlet and outlet sides of gearbox oil filter. A button located on the side of the indicator will extend at approximately 27 psi (186 kPa) indicating an impending filter bypass condition.

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Differential Pressure Indicator - Removal

Procedure

1. Open left engine cowling.
2. Disconnect tubes (7 and 8, Figure 1).
3. Remove nuts (2), washers (1), and bolts (5).
4. Remove indicator (3).
5. Remove unions (6 and 9) with packings (10) and (11). Discard packings.



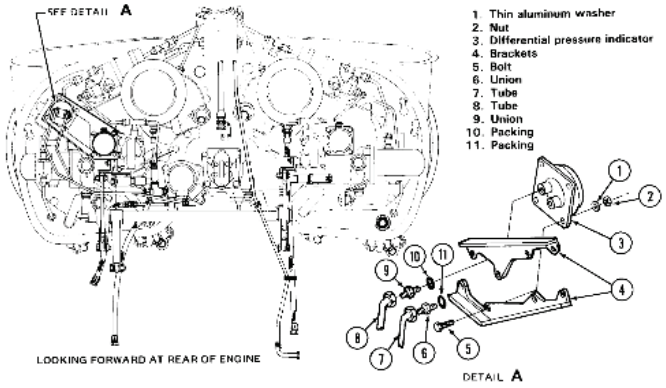
ICN-412-A-710100-B-97499-00003-A-001-01

Figure 1. Differential Pressure Indicator (Sheet 1 of 1)

Differential Pressure Indicator - Installation

Procedure

1. Install unions (6 and 9, Figure 1) with packings (10 and 11) in indicator (3).
2. Position indicator (3) on brackets (4). Install bolts (5), washers (1), and nuts (2).
3. Connect tubes (7 and 8).
4. Close engine cowling.



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Figure 1. Differential Pressure Indicator (Sheet 1 of 1)

Torque Pressure Transmitters - Description

Two torque transmitters are mounted on rear of firewall support. The transmitters are connected by electrical harness to the dual torque pressure indicator on instrument panel.

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Torque Pressure Transmitters - Removal

Procedure

NOTE

Removal procedures are identical for opposite torque pressure transmitter.

1. Open reduction (combining) gearbox access door.

2. Remove pressure hose (1, Figure 1) from elbow (2).

3. Remove electrical connector (9) from transmitter (8).

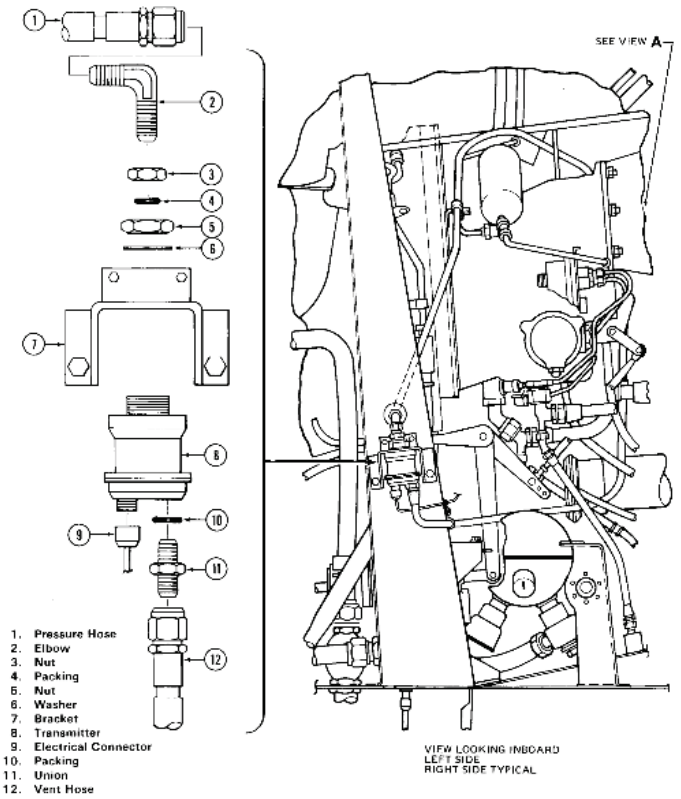
4. Remove vent hose (12) from union (11).

5. Remove nut (5) and washer (6). Remove transmitter from bracket (7). Cap open lines.

6. If replacing transmitter, accomplish the following:

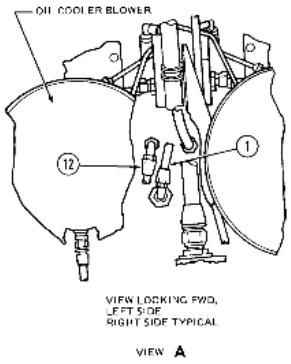
6.1. Remove nut (3) with elbow (2) and packing (4).

6.2. Remove union (11) and packing (10). Discard packings (4 and 10).



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Figure 1. Torque Pressure Transmitter (Sheet 1 of 2)



ICN-412-A-710100-B-97499-00005-A-001-01

Figure 1. Torque Pressure Transmitter (Sheet 2 of 2)

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Torque Pressure Transmitters - Installation

Procedure

1. If installing new transmitter, accomplish the following:

1.1. Install union (11) with packing (10) in vent port transmitter (8).

1.2. Install elbow (2) with nut (3) and packing (4) in pressure port of transmitter.

2. Position transmitter (8) in bracket (7) with pressure elbow directed through hole in bracket.

3. Install washer (6) and nut (5). Tighten nut (5).

4. Install vent hose (12) on union (11).

5. Install electrical connector (9) on transmitter (8).

6. Install pressure hose (1) on elbow (2).

7. Perform operational check for proper calibration.

NOTE

Installation and operational check procedures are identical for opposite torque pressure transmitter.

8. Close reduction gearbox access door.

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Power Section Oil Pressure Switch - Description

An oil pressure switch for each power section is installed on lower aft side of reduction (combining) gearbox. The pressure switches are electrically connected to segments (ENG 1 OIL PRESSURE and ENG 2 OIL PRESSURE) in caution panel located in pedestal. The pressure switch causes segment in caution panel to extinguish when oil pressure reaches 40 psi after engine start. If oil pressure in either power section decreases below approximately 31 psi, the appropriate segment in the caution panel illuminates.

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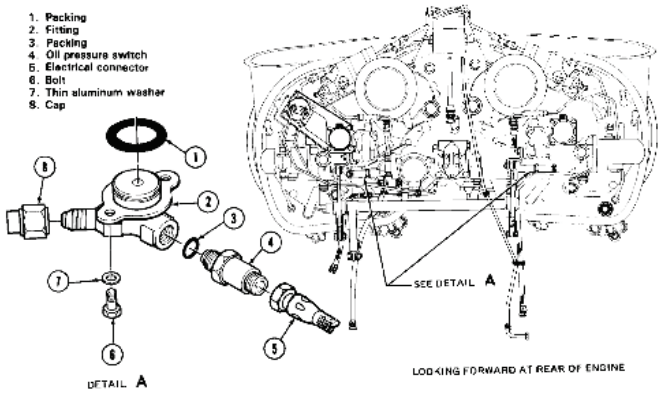
Power Section Oil Pressure Switch - Removal

Procedure

NOTE

Maintenance procedures are the same for each pressure switch.

1. Open engine cowling.
2. Disconnect electrical connector (5, Figure 1).
3. Remove switch (4) with packing (3). Discard packing.
4. Remove cap (8).
5. Remove bolts (6), washers (7), fitting (2), and packing (1). Discard packing.



ICN-412-A-710100-B-97499-00006-A-001-01

Figure 1. Power Section Oil Pressure Switch (Sheet 1 of 1)

Power Section Oil Pressure Switch - Installation

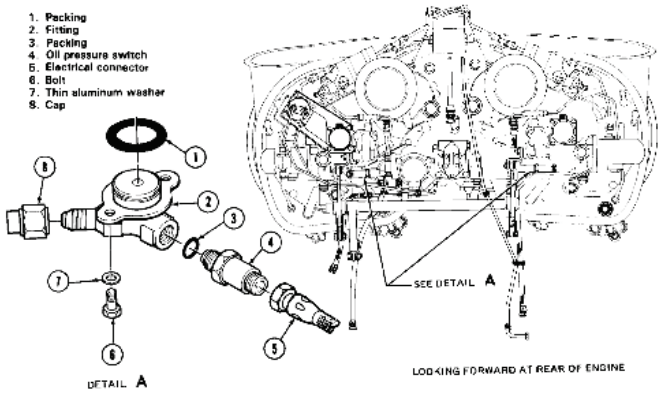
Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Lockwire	C-405	AR	
Lockwire	C-508	AR	

Procedure

1. Install fitting (2, Figure 1) with packing (1), bolts (6), and washers (7).
2. Install cap (8).
3. Install switch (4) with packing (3).
4. Secure bolts (6) to switch (4) with lockwire (C-405).
5. Connect electrical connector (5) and lockwire (C-508).
6. Check power section oil level, service if required (412-A-12-00-00-00A-040A-A / 00126).
7. Close engine cowling.



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Figure 1. Power Section Oil Pressure Switch (Sheet 1 of 1)

Reduction (Combining) Gearbox Oil Pressure Switch - Description

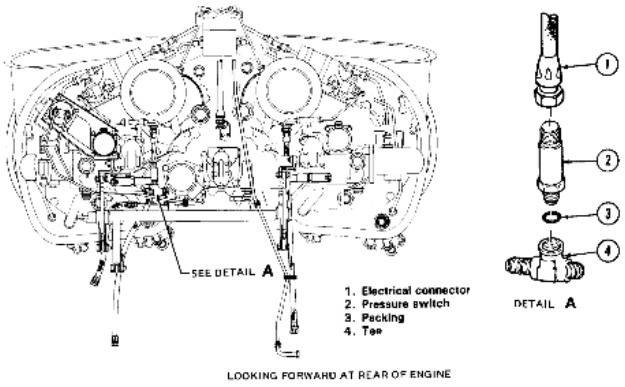
The oil pressure switch is installed on aft side of reduction (combining) gearbox, left of oil filter. The pressure switch is electrically connected to a warning segment (C BOX OIL PRESSURE) in caution panel located in pedestal. The pressure switch causes the segment to extinguish when oil pressure reaches 40 psi after engine start. If oil pressure in gearbox decreases below approximately 31 psi, the segment illuminates.

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Reduction (Combining) Gearbox Oil Pressure Switch - Removal

Procedure

1. Open engine cowling.
2. Disconnect electrical connector (1, Figure 1).
3. Remove switch (2) with packing (3). Discard packing.



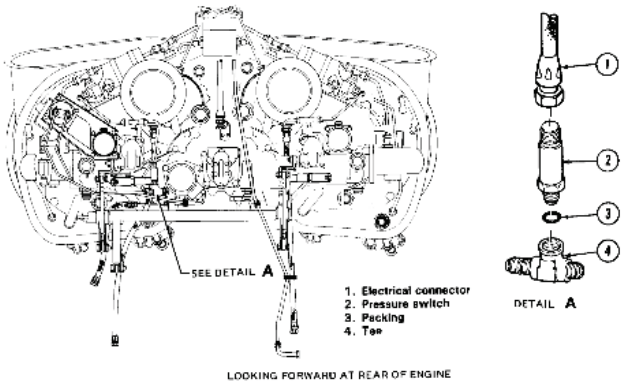
ICN-412-A-710100-B-97499-00007-A-001-01

Figure 1. Reduction (Combining) Gearbox Oil Pressure Switch (Sheet 1 of 1)

Reduction (Combining) Gearbox Oil Pressure Switch - Installation

Procedure

1. Install switch (2, Figure 1) with packing (3) in tee (4).
2. Connect electrical connector (1).
3. Check gearbox oil level, service if required (412-A-12-00-00-00A-040A-A / 00126).
4. Close engine cowling.



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Figure 1. Reduction (Combining) Gearbox Oil Pressure Switch (Sheet 1 of 1)

Reduction (Combining) Gearbox Oil Temperature Bulb - Description

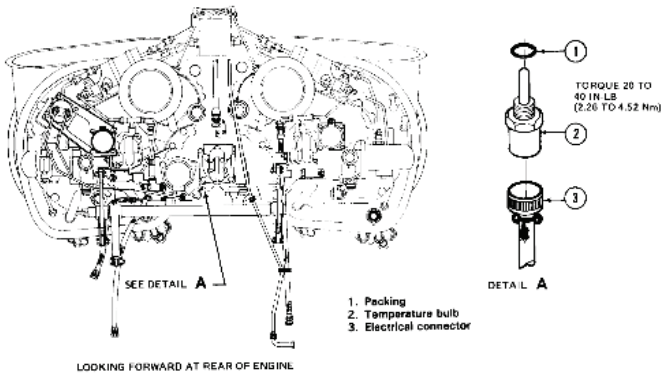
The temperature bulb is installed on lower side of reduction (combining) gearbox near the oil filter. The bulb is electrically connected to the C BOX TEMP warning segment in the caution panel.

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Reduction (Combining) Gearbox Oil Temperature Bulb - Removal

Procedure

1. Open engine cowling.
2. Disconnect electrical connector (3, Figure 1).
3. Remove bulb (2) and packing (1). Discard packing.



ICN-412-A-710100-B-97499-00008-A-001-01

Figure 1. Reduction (Combining) Gearbox Oil Temperature Bulb (Sheet 1 of 1)

Reduction (Combining) Gearbox Oil Temperature Bulb - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Lockwire	C-405	AR	

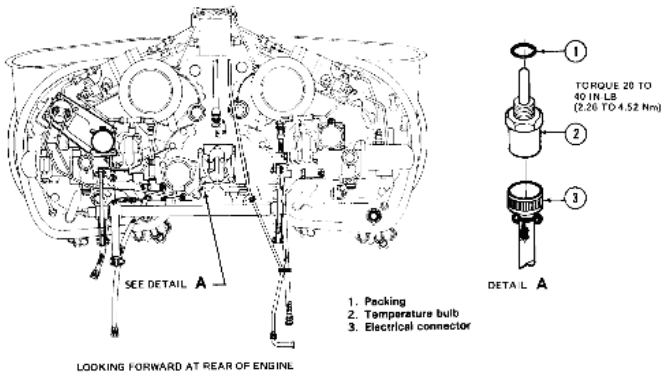
Procedure

1. Install bulb (2, Figure 1) with packing (1). Torque bulb (2) 20 to 40 inch-pounds (2.26 to 4.52 Nm) and lockwire (C-405).

2. Connect electrical connector (3).

3. Check reduction gearbox oil level, service if required (412-A-12-00-00-00A-040A-A / 00126).

4. Close engine cowling.



ICN-412-A-710100-B-97499-00008-A-001-01

Figure 1. Reduction (Combining) Gearbox Oil Temperature Bulb (Sheet 1 of 1)

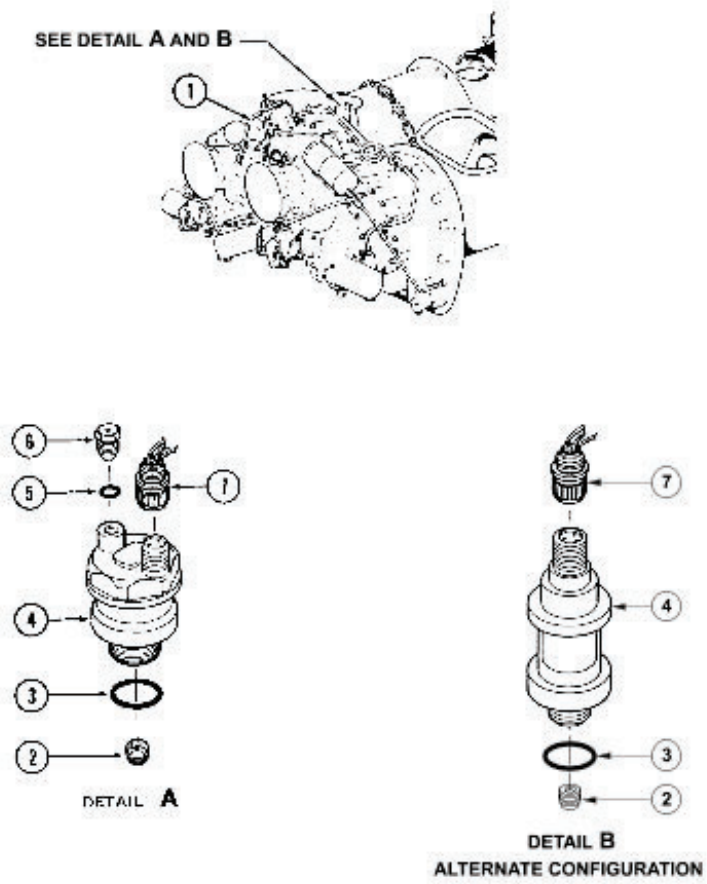
Reduction (Combining) Gearbox Oil Pressure Transmitter - Description

The oil pressure transmitter is installed on top of reduction (combining) gearbox. The transmitter is electrically connected to GEARBOX OIL temperature and pressure indicator mounted on instrument panel.

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Reduction (Combining) Gearbox Oil Pressure Transmitter - Removal Procedure

1. Open engine cowling.
2. Disconnect electrical connector (7, Figure 1).
3. Remove pressure transmitter (4) and packing (3). Discard packing.
4. Remove restrictor (2).
5. If applicable, remove plug (6) and packing (5). Discard packing.



1. Reduction gearbox
2. Restrictor
3. Packing
4. Pressure transmitter
5. Packing
6. Plug
7. Electrical connector

ICN-412-A-710100-B-97499-00009-A-001-01

Figure 1. Reduction (Combining) Gearbox Oil Pressure Transmitter (Sheet 1 of 1)

Reduction (Combining) Gearbox Oil Pressure Transmitter - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Sealant	C-320	AR	

Procedure

1. If applicable, install plug (6, Figure 1) with packing (5).

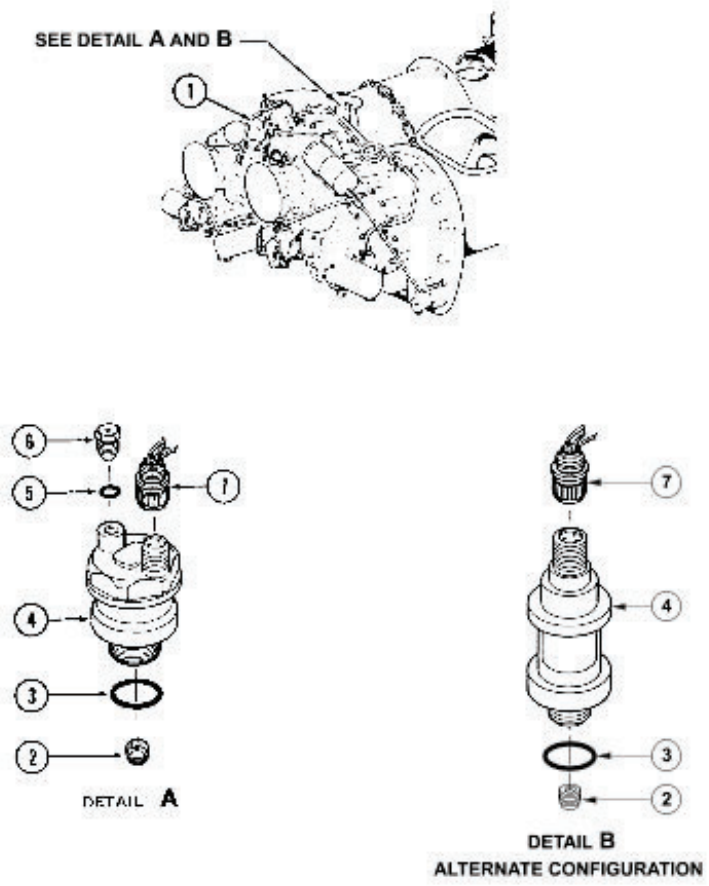
2. Apply sealant (C-320) on outer threads of restrictor (2). Install restrictor until it bottoms out in pressure transmitter (4).

3. Install the restrictor until it bottom out in the pressure transmitter (4).

4. Install the pressure transmitter (4) with new packing (3).

5. Connect the electrical connector (7).

6. Close the engine cowling.



1. Reduction gearbox
2. Restrictor
3. Packing
4. Pressure transmitter
5. Packing
6. Plug
7. Electrical connector

ICN-412-A-710100-B-97499-00009-A-001-01

Figure 1. Reduction (Combining) Gearbox Oil Pressure Transmitter (Sheet 1 of 1)

Power Section Oil Pressure Transmitter - Description

An oil pressure transmitter is installed on bottom of each power section accessory gearbox below power section oil tank. The transmitters are electrically connected to ENGINE OIL 1 and 2 temperature and pressure indicators located on instrument panel.

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Power Section Oil Pressure Transmitter - Removal

Procedure

NOTE

Maintenance procedures are the same for each pressure transmitter.

1. Open engine cowling.

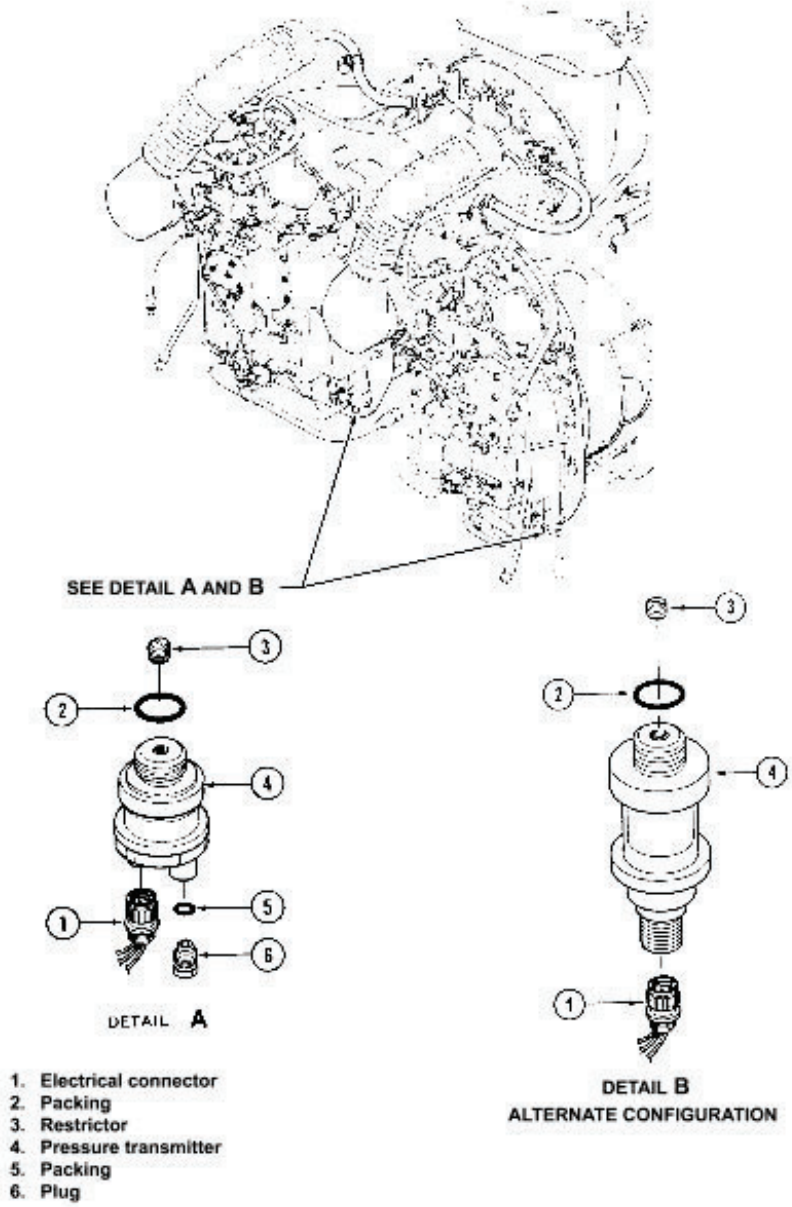
2. Disconnect electrical connector (1, Figure 1).

3. Remove transmitter (4) and packing (2). Discard packing.

4. Remove restrictor (3).

5. If applicable, remove plug (6) and packing (5). Discard packing.

6. For engine 2, right side, remove access panel inside helicopter on right side of transmission pylon.



ICN-412-A-710100-B-97499-00010-A-001-01

Figure 1. Engine Power Section Oil Pressure Transmitter (Sheet 1 of 1)

Power Section Oil Pressure Transmitter - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Sealant	C-320	AR	

Procedure

1. If applicable, install plug (6, Figure 1) with packing (5).

2. Apply sealant (C-320) on outer threads of restrictor (3). Install restrictor (3) until it bottoms out in pressure transmitter (4).

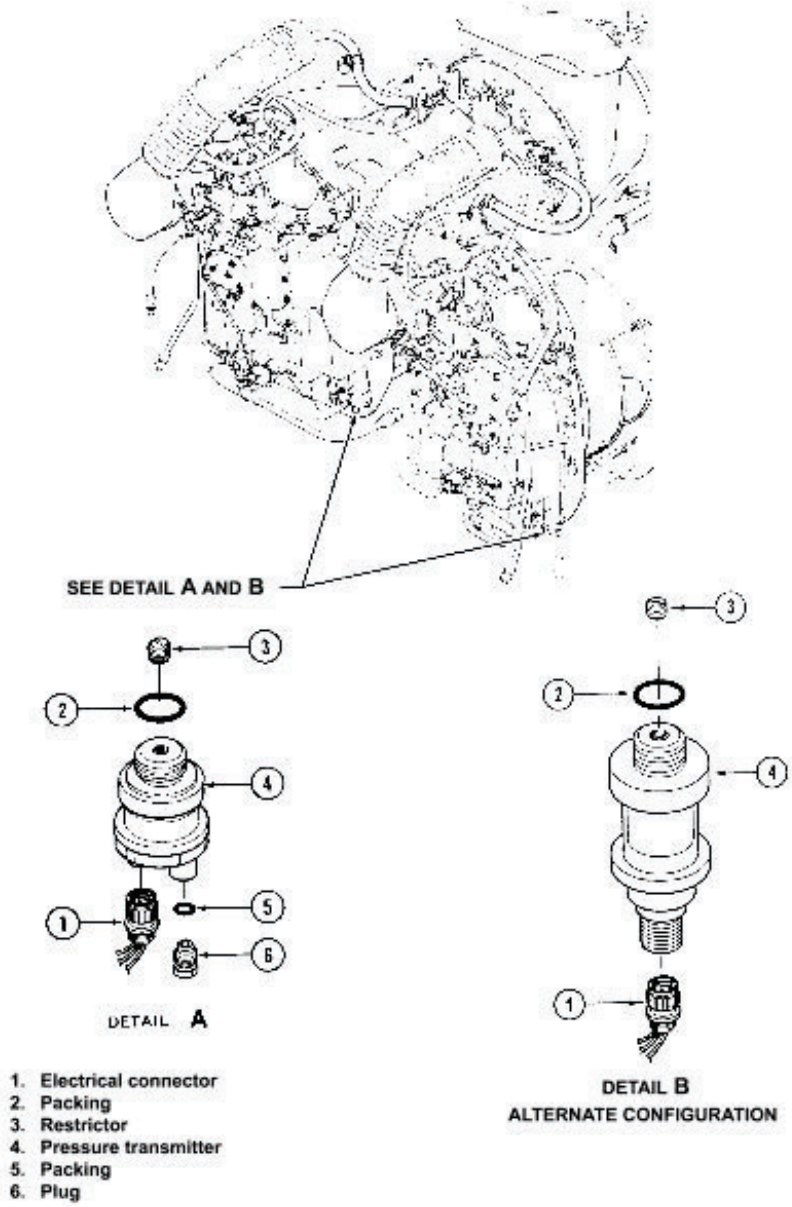
3. Install pressure transmitter (4) with packing (2).

4. Connect electrical connector (1).

5. Check power section oil level. Service if required (412-A-12-00-00-00A-040A-A / 00126).

6. Close engine cowling.

7. Secure access panel inside helicopter for engine No. 2, right side on transmission pylon.



ICN-412-A-710100-B-97499-00010-A-001-01

Figure 1. Engine Power Section Oil Pressure Transmitter (Sheet 1 of 1)

Power Section Oil Temperature Bulb - Description

An oil temperature bulb is installed on bottom of each power section accessory gearbox below power section oil tank. The temperature bulbs are electrically connected to ENGINE OIL 1 and 2 temperature and pressure indicators on instrument panel.

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Power Section Oil Temperature Bulb - Removal

Procedure

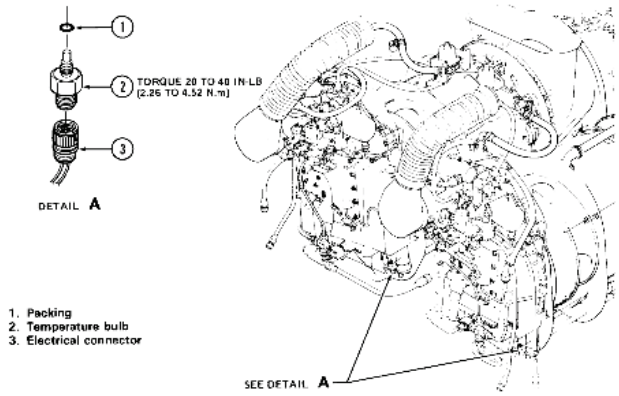
NOTE

Maintenance procedures are the same for each temperature bulb.

1. Open engine cowling.

2. Disconnect electrical connector (3, Figure 1).

3. Remove temperature bulb (2) and packing (1). Discard packing.



ICN-412-A-710100-B-97499-00011-A-001-01

Figure 1. Engine Power Section Oil Temperature Bulb (Sheet 1 of 1)

Power Section Oil Temperature Bulb - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Lockwire	C-405	AR	

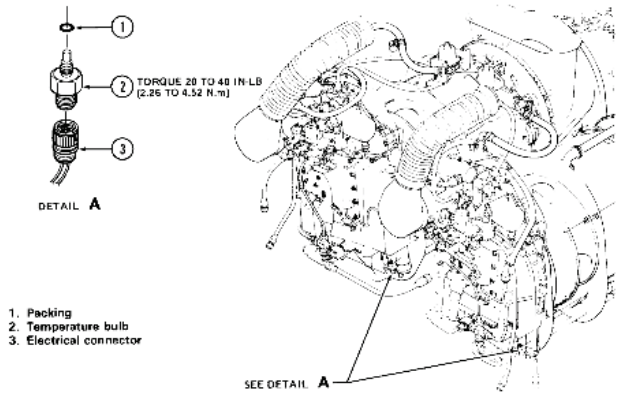
Procedure

1. Install temperature bulb (2, Figure 1) with packing (1). Torque temperature bulb (2) to 20 to 40 inch-pounds (2.26 to 4.52 Nm) and install lockwire (C-405).

2. Connect electrical connector (3).

3. Check power section oil level. Service if required (412-A-12-00-00-00A-040A-A / 00126).

4. Close engine cowling.



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Figure 1. Engine Power Section Oil Temperature Bulb (Sheet 1 of 1)

Starter-Generator - Description

The starter-generator is a 30 Vdc unit mounted on upper forward right of each power section accessory gearbox. The starter-generator drives the compressor input shaft through the gearing in the accessory gearbox during engine starting cycle and provides primary 28 Vdc power for the helicopter.

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Starter-Generator - Removal

Procedure

NOTE

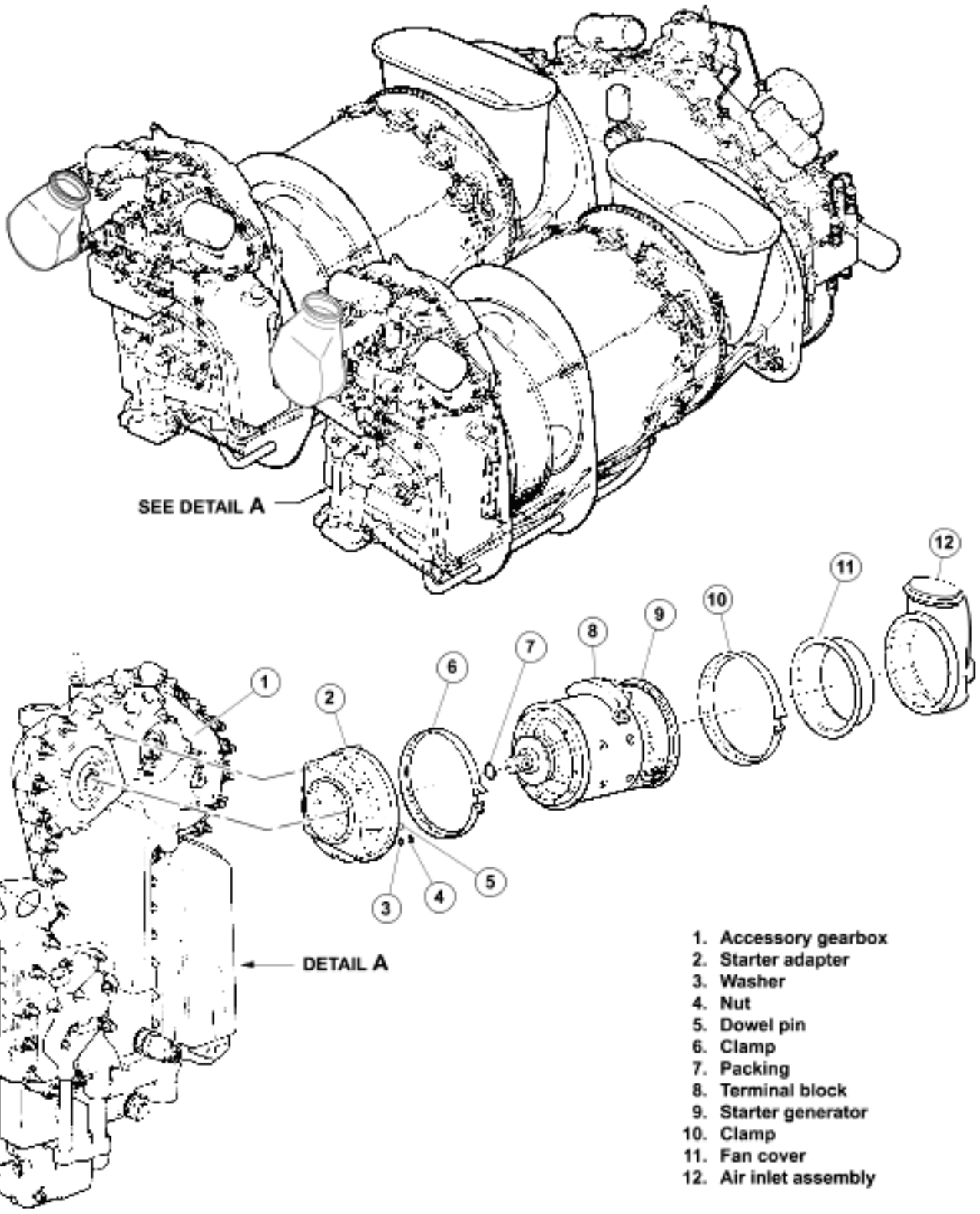
Maintenance procedures are the same for each starter generator except where noted.

1. Disconnect electrical power.
2. Open engine cowling.
3. Disconnect duct from air inlet assembly (12, Figure 1).
4. Remove clamp (10) attaching air inlet assembly (12) to starter generator (9). Remove air inlet assembly.
5. Remove fan cover (11) from starter generator (9).

NOTE

Identify location of electrical cables to aid during installation.

6. Disconnect electrical cables from terminal block (8).
7. While supporting starter generator (9), remove clamp (6). Remove starter generator.
8. Remove nuts (4), washers (3), and starter adapter (2).
9. Remove packing (7) from shaft of starter generator (9) and discard packing.



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Figure 1. Starter Generator (Sheet 1 of 1)

Starter-Generator - Inspection

Procedure

1. Inspect exterior for condition and damage.

2. Inspect the air inlet assembly (12, Figure 1) for security of attachment, cracks, and dents which might interfere with fan operation.

NOTE

To carry out the following steps, removal of the starter-generator will be required.

3. Inspect brushes, brush holders, commutator and bearings.

3.1. Remove the brush inspection cover.

NOTE

Prior to removing brushes, identify their location in the unit. Brushes must be reinstalled in their original location.

3.2. Remove screw securing brush leads to the brush holders.

3.3. Carefully lift the brush springs and remove the brushes.

NOTE

Brushes must be replaced as a set. If one brush is defective or has reached the wear limit, replace the entire set of brushes.

3.4. Inspect brushes for cracks, chips, frayed leads, loose rivets, and loose shunt connections.

3.5. Inspect brushes as per Figure 2, Detail A. It is recommended that brushes be replaced at or near the time the brush wear indicates 1/4 life remaining.

3.6. Inspect brush holders for security and condition of springs.

3.7. Inspect the commutator for a burnished appearance with light filming. A defective commutator will be black, burned, or pitted.

3.8. Inspect the mica undercut, it must be 0.02 inch (0.51 mm) minimum.

NOTE

Interpreting the feel of bearings requires experience. Some bearings use a high temperature lubricant that will cause a good bearing to feel rough when turned.

3.9. Inspect the bearings for roughness by turning the armature in the normal direction of rotation slowly by hand using your fingers to obtain the “feel” of the bearings. Bearing rotation should be relatively smooth, and the bearings should not sound dry.



RAPID RELEASE OF THE BRUSH SPRINGS CAN DAMAGE THE BRUSHES.

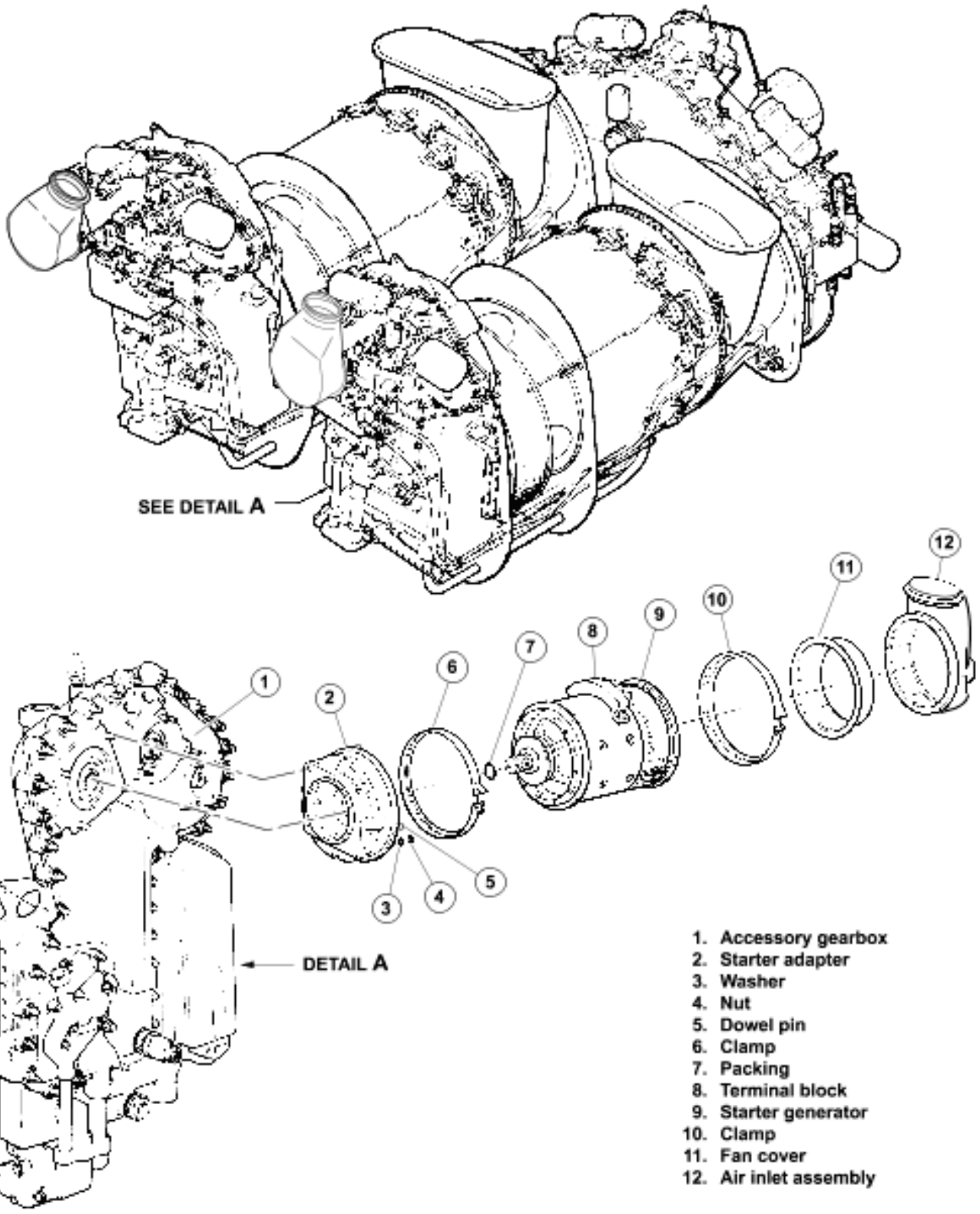
3.10. Carefully insert the brushes in their original location on the brush holder assembly. Referring to Figure 2, ensure the shorter side of the brush will be the first to contact the commutator when the armature is rotated in the normal direction.

3.11. Gently lower the brush springs onto the top of the brush clips.

3.12. Position the brush leads over the brush spring clips.

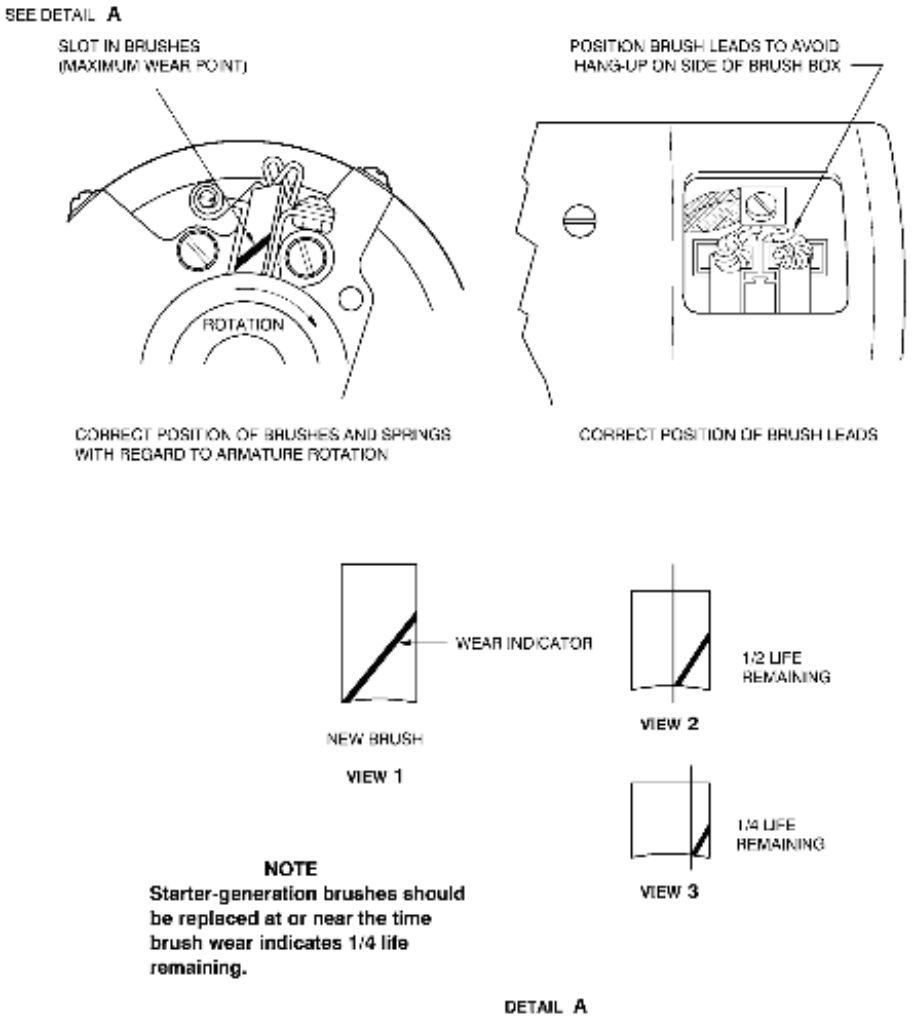
3.13. Install the brush inspection cover.

4. Visually inspect the drive shaft spline for excessive wear or damage.



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Figure 1. Starter Generator (Sheet 1 of 1)



ICN-412-A-710100-B-97499-00013-A-001-01

Figure 2. Starter-Generator Brush Installation (Sheet 1 of 1)

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Starter Generator - Repair

Procedure

1. For starter generator repair procedures, refer to applicable vendor maintenance publication. For operation check, refer to 412-A-96-61-00-04A-320C-A / 01925.

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Starter Generator - Cleaning

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Aliphatic naphtha	C-305	AR	

Procedure

1. Internal cleaning.

1.1. Remove the brush inspection cover.



WHEN YOU USE COMPRESSED AIR, ADJUST THE PRESSURE TO 29 PSI (200 KPA) MAXIMUM. WEAR GOGGLES OR A FACE-SHIELD TO PROTECT EYES. USE A DUST FILTER RESPIRATOR. DO NOT POINT THE AIR JET AT EXPOSED SKIN.



TO PREVENT DAMAGE TO THE BEARING SEALS AND TO AVOID FORCING CONTAMINANTS INTO BEARINGS, DO NOT DIRECT THE AIR JET AT THE BEARINGS.

1.2. Remove the carbon copper dust from the armature and stator windings using compressed air. Start at the drive end of the unit and work towards the brushes and commutator area.

1.3. Inspect the interior of the unit for oil and other contaminants. Clean with a cloth moistened with aliphatic naphtha (C-305) to remove the contaminants.

1.4. Inspect the air intake and exhaust screens for foreign material. Carefully remove the foreign material and check for possible internal damage to the unit.

2. External cleaning.

2.1. Fully clean the exterior of the unit with a soft cloth moistened with aliphatic naphtha (C-305). Wipe with a clean-lint-free cloth to dry.

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Starter Generator - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Zinc chromate primer	C-201	AR	

Procedure

NOTE

Installation procedures are the same for each starter generator except as described below.

1. Clean the accessory gearbox (1, Figure 1) and starter adapter (2) mating surfaces to ensure good electrical bonding.

2. Install the starter adapter (2) as follows:

2.1. Put the starter adapter (2) with one dowel pin (5) at the 12 o'clock position, if you install the left starter generator.

2.2. Put the starter adapter (2) with one dowel pin (5) at the 9 o'clock position, if you install the right starter generator.

2.3. Apply a coat to the washers (3) with zinc chromate primer (C-201). Install the washers (3) and nuts (4).

3. Install the starter generator (9) as follows:

3.1. Lubricate the packing (7) with engine oil (412-A-12-00-00-00A-040A-A / 00126) and install on shaft of the starter generator (9).

3.2. Put the clamp (6) on starter adapter (2).

3.3. Put the terminal block (8) at the 12 o'clock position, put the starter generator (9) on the starter adapter (2) and tighten the clamp (6), if you install the left starter generator.

3.4. Put the terminal block (8) at the 11 o'clock position, put the starter generator (9) on the starter adapter (2) and tighten the clamp (6), if you install the right starter generator.

3.5. On the left starter generator, put the fan cover (11) and air inlet assembly (12) at 12 o’clock and tighten the clamp (10).

3.6. On the right starter generator, put the fan cover (11) and air inlet assembly (12) at 1 o’clock and tighten the clamp (10).

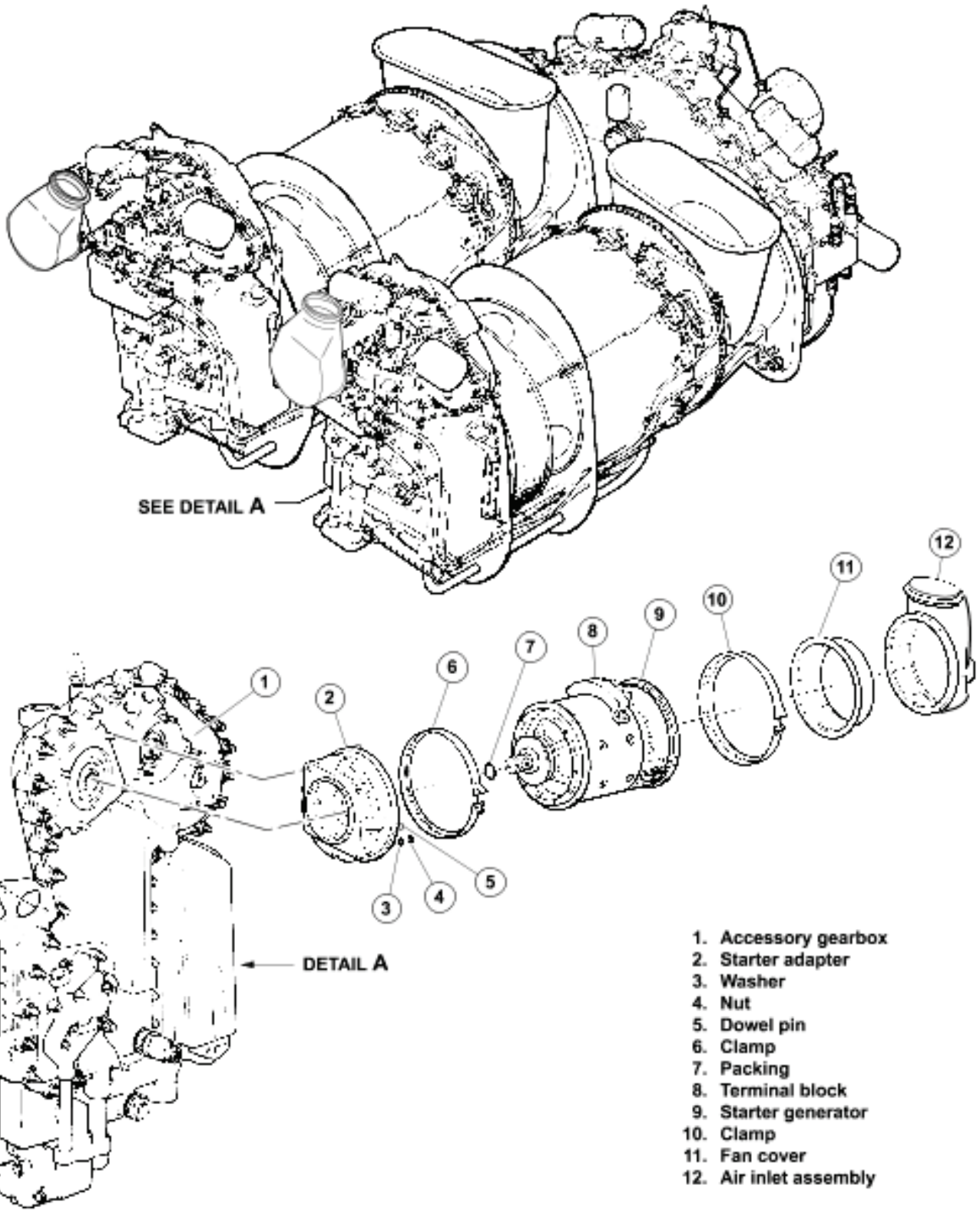
4. Make sure that the electrical terminals on starter generator and ends of electrical cables are clean.

5. Connect electrical cables to starter generator, positioning, as shown on Figure 2.

6. Place nipples over ends of electrical cables and secure cables using plate and straps.

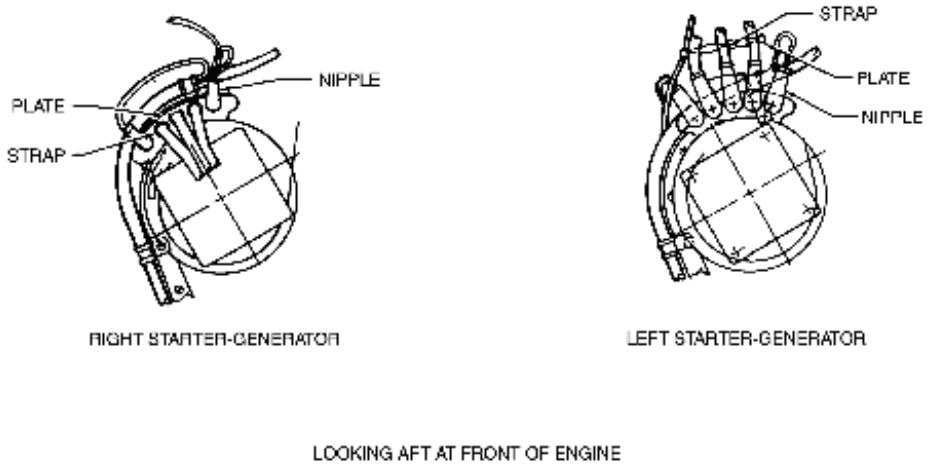
7. Ensure no interference exists between electrical cables and engine cowling.

8. Attach duct to air inlet assembly (12, Figure 1).



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Figure 1. Starter Generator (Sheet 1 of 1)



ICN-412-A-710100-B-97499-00014-A-001-01

Figure 2. Starter Generator Electrical Cable (Sheet 1 of 1)

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Reduction (Combining) Gearbox Oil Temperature Switch - Description

Refer to 412-A-79-20-00-01A-520A-A / 01769 and 412-A-79-20-00-01A-720A-A / 01770.

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Air Management System - Description

NOTE

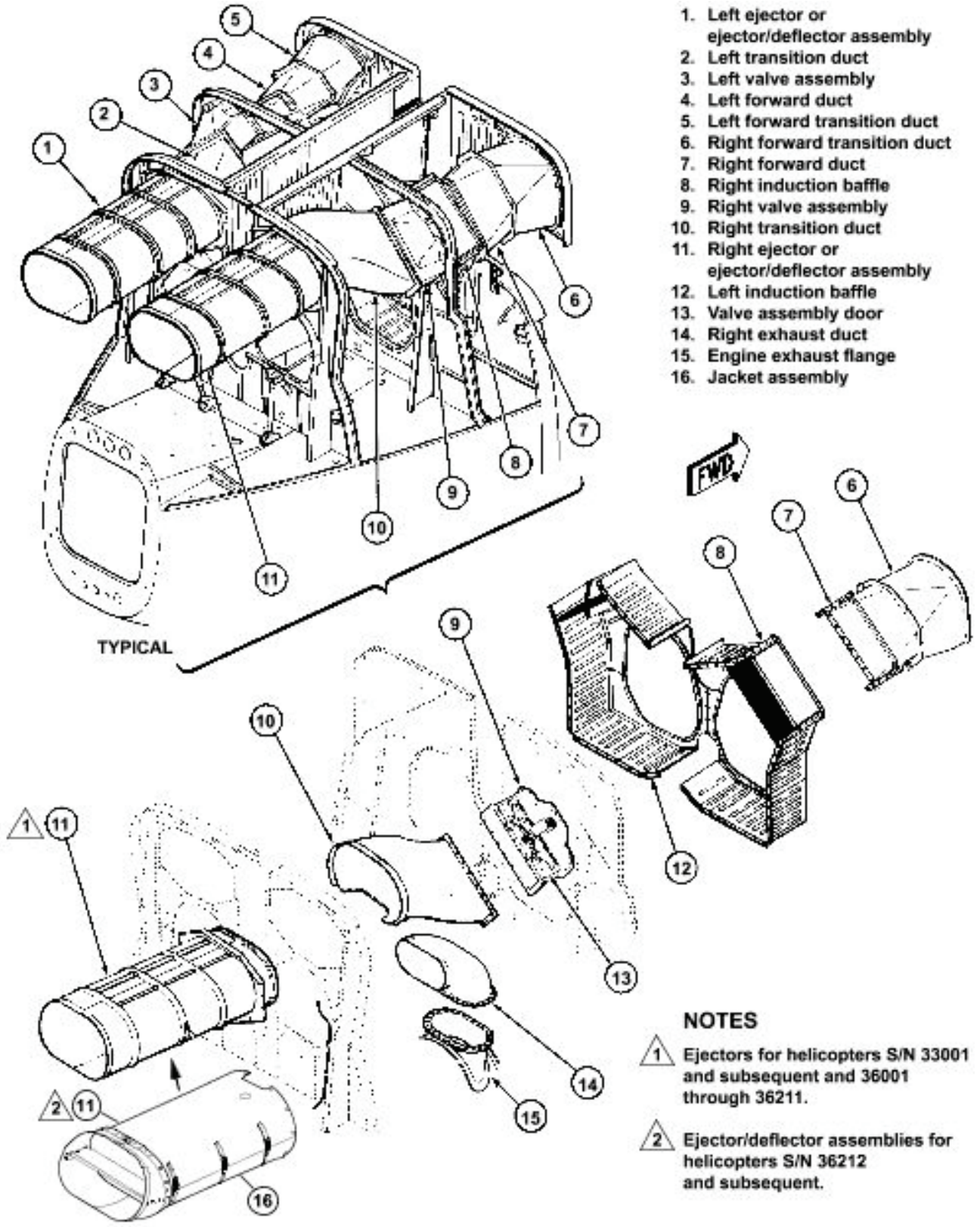
SN: 36212-Subsequent:

Helicopters have ejector/deflector assemblies installed in the place of ejectors.

The air management system (Figure 1) consists of a series of ducts located within the engine cowling. Inlet air flows through two separate duct systems (Figure 2) to each of the two engine power sections. During normal operation, air is drawn in through the forward ducts (4 and 7, Figure 1) and the forward transition ducts (5 and 6) by the combined action of the engine power section compressors, exhaust flow through the ejectors (1 and 11) or ejector/deflector assemblies, and the ram effect of forward movement of the helicopter. Doors (13) in the valve assemblies (3 and 9) are normally open. Part of the air, with any foreign particles present, passes through the valve assemblies, transition ducts (2 and 10), and out through the ejectors. The remaining air makes a sharp turn downward into induction baffles (8 and 12). Most of the airborne particles cannot make the sharp turn, but continue aft through the valve assemblies. The clean air passes from induction baffles into the engine power sections, then out through the exhaust ducts (14) and ejectors.

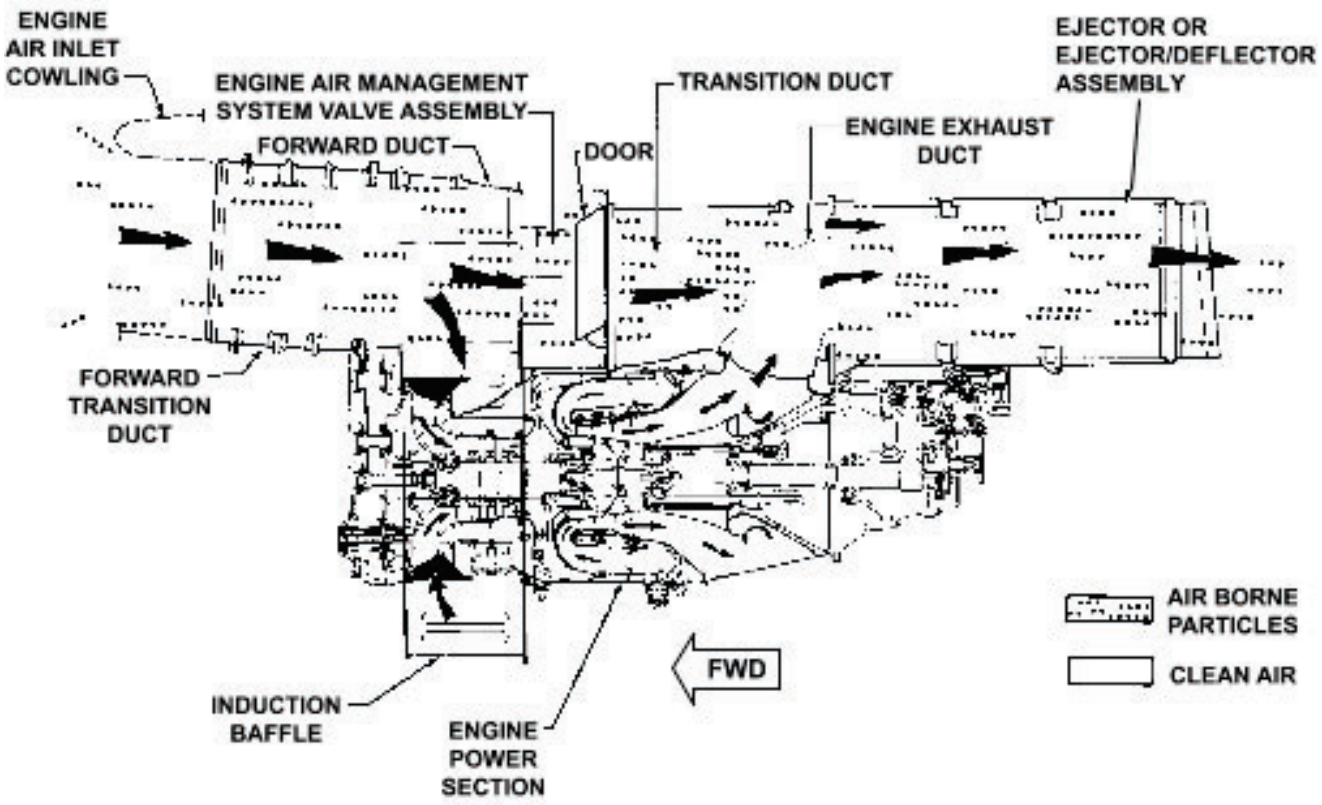
Doors (13) in the valve assemblies (3 and 9) are positioned either full open or full closed by actuators. Electrical power for the actuators is furnished by the 28 VDC essential bus. The electrical circuits are connected through circuit breakers ENG 1 PART SEP and ENG 2 PART SEP. The electrical actuators are operated either automatically (by the engine low RPM warning system) or manually (by pulling either FIRE PULL handle). When the actuators are operated to close the doors, either automatically or manually, the caution panel segment PART SEP OFF will illuminate and all intake air is directed to the power section.

The forward transition ducts (5 and 6, Figure 1) are integral parts of the engine air inlet cowlings (2, Figure 3).



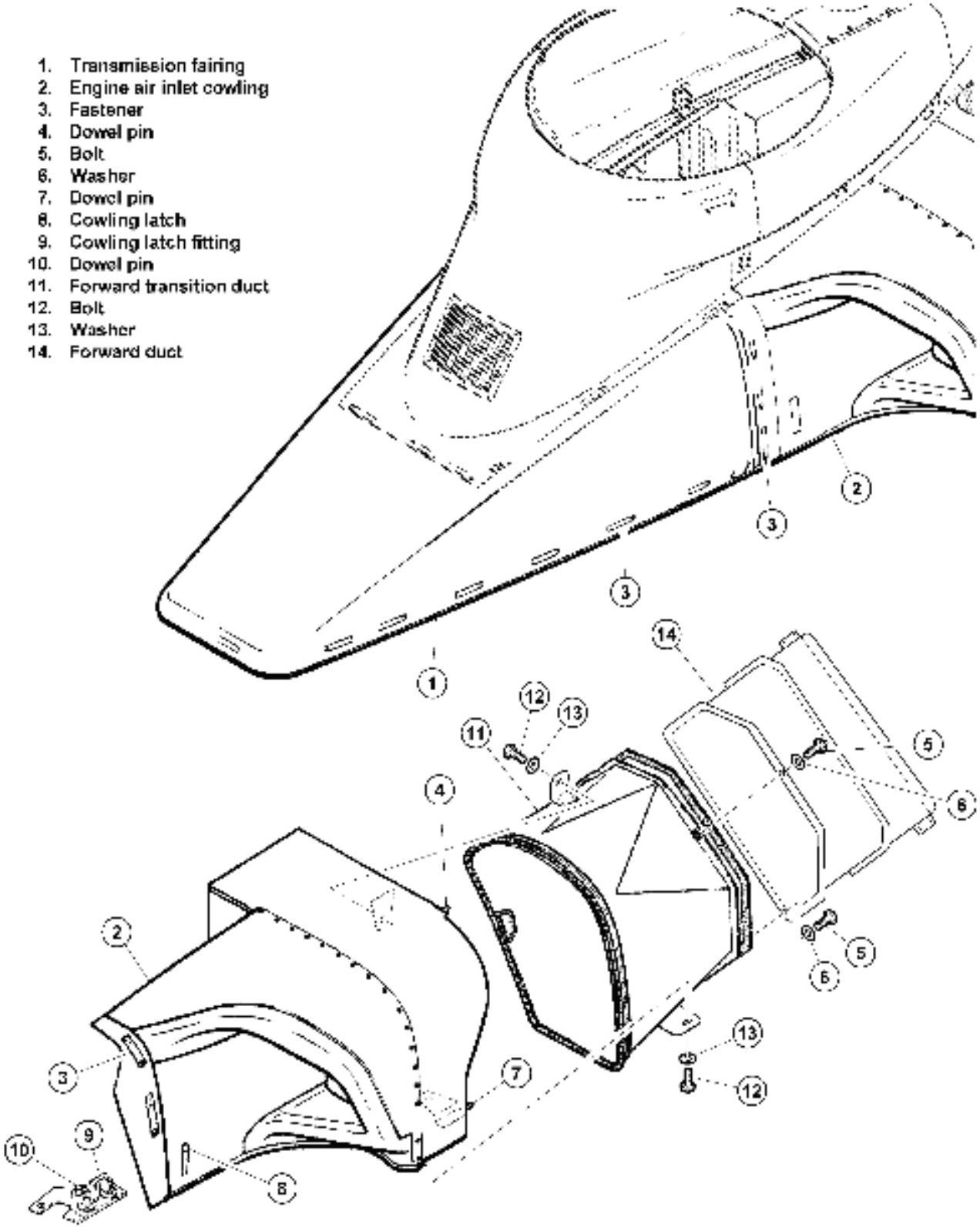
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Figure 1. Air Management System, Helicopters S/N 33001 and Subsequent and 36001 through 36211 and 36212 and Subsequent (Sheet 1 of 1)



ICN-412-A-716100-B-97499-00002-A-001-01

Figure 2. Air Management System Schematic (Sheet 1 of 1)



ICN-412-A-711000-B-97499-00004-A-001-01

Figure 3. Transmission Fairings and Engine Air Inlet Cowlings (Sheet 1 of 1)

Air Management System Ejectors - Removal

Procedure

NOTE

Maintenance procedures for the ejectors (1 and 5, Figure 1) are the same except as noted. Helicopter S/N 36212 through 36999, 37002 through 37999, 38001 through 38999 and 39101 through 39999 have ejector/deflector assemblies installed.

1. Remove the engine cowling, aft pylon fairing, and tailpipe fairing.

NOTE

Step 2. *is applicable to left ejector (1).* Step 3. *is applicable to right ejector (5).*

2. *Remove vent hoses (17 and 18) from fittings on left ejector (1).*

3. Remove vent hose (51) from fitting on right ejector (5).

4. Disconnect drain line (19) from drain line fitting (20).

NOTE

Step 5. *is applicable to left ejector (1).* Step 6. *is applicable to right ejector (5).*

5. If ejector will not be reinstalled, remove fittings (27 and 28) from left ejector (1).

6. If ejector will not be reinstalled, remove fitting from right ejector (5) that is similar to fitting (28) on left ejector (1).

Applicable: (S/N 36212 THRU 36999, 37002 THRU 37999, 38001 THRU 38999, 39101 THRU 39999) :

7. *Remove the jacket assembly (4, Figure 2) as follows:*

NOTE

This procedure is applicable to the left and the right side jacket assembly. The procedure for the left side jacket assembly is given. Differences for the right side jacket assembly are identified.

7.1. *Remove and discard the lockwire from the nuts (1).*

7.2. *Remove the nuts (1) from the jacket clamps.*

NOTE

Remove the screw opposite to the jacket assembly overlap last.

7.3. *Remove the screws (2) and the washers (3).*

7.4. *Remove the jacket assembly (4).*

8. *Remove screws (9, 11, and 13, Figure 1) and washers (8, 10 and 12). Remove oil cooler support frame (7).*

Applicable: (S/N 33001 THRU 33213, 36001 THRU 36629) :

9. *Remove the ejector (1) as follows:*

NOTE

This removal procedure is for the left and the right side ejectors (1 and 5). The procedure for the left side ejector (1) is given. Differences for the right side ejector (5) are identified.

9.1. *Remove screws (14) and washers (15) that attach the ejector (1) to the aft firewall (2).*

9.2. *Remove the ejector (1) from the aft firewall (2).*

Applicable: (S/N 36630 THRU 36999, 37002 THRU 37999, 38001 THRU 38999, 39101 THRU 39999) :

10. *Remove the ejectors (1) as follows:*

NOTE

This removal procedure is for the left and the right side ejectors (1 and 5). The procedure for the left side ejector (1) is given. Differences for the right side ejector (5) are identified.

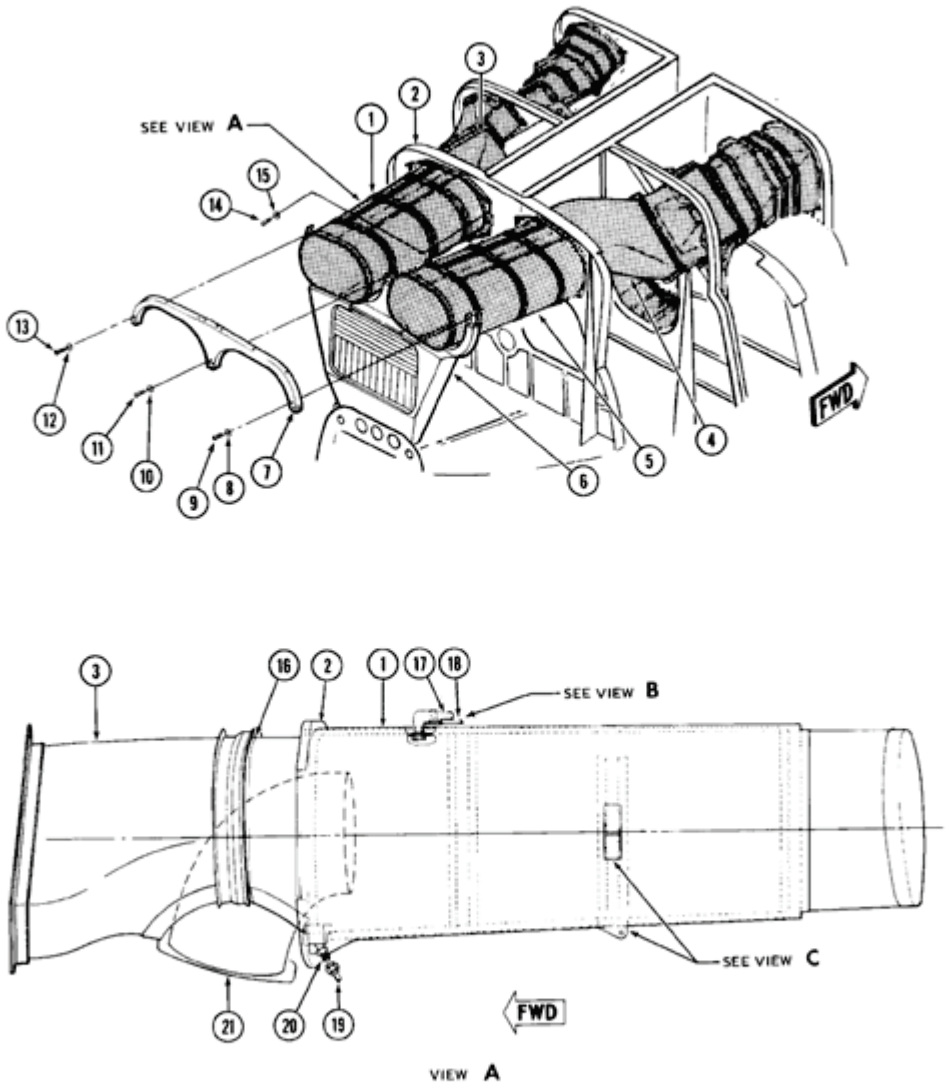
10.1. *Remove screws (64, 65, 66, 67 and 68) and washers (63) that attach the ejector (1) and shim (69) to the aft firewall (2).*

10.2. *Remove the nut (41), washers (34, 35, 39, 40), spacer (38) and bolt (33) to disconnect the link (36) from ejector (1).*

10.3. *Remove the nut (50), washers (49, 48, 44, 43), spacer (46) and bolt (42) to disconnect the link (47) from ejector (1).*

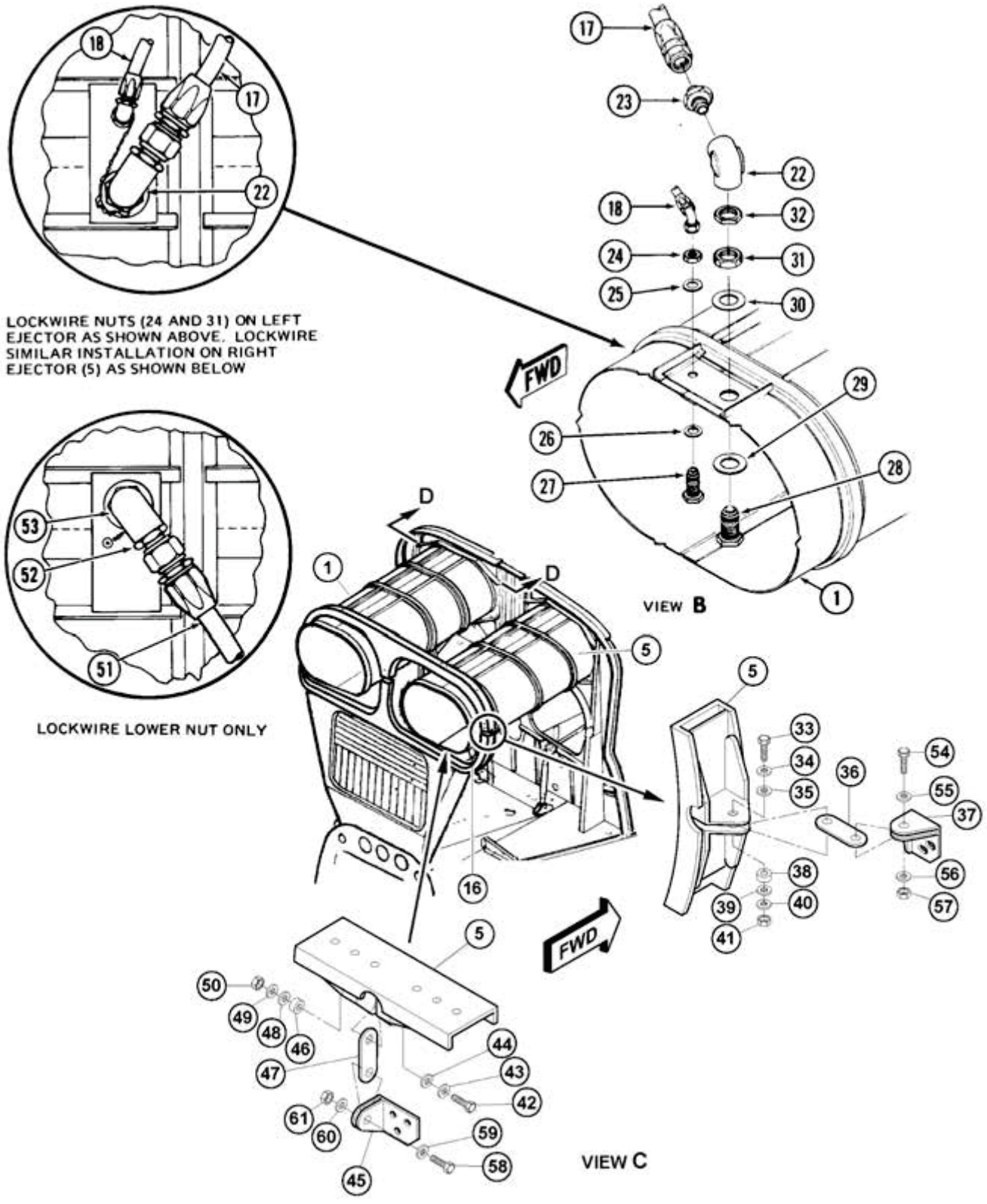
10.4. *Remove the ejector (1) from the aft firewall (2) and the shim (69).*

11. *DELETED*



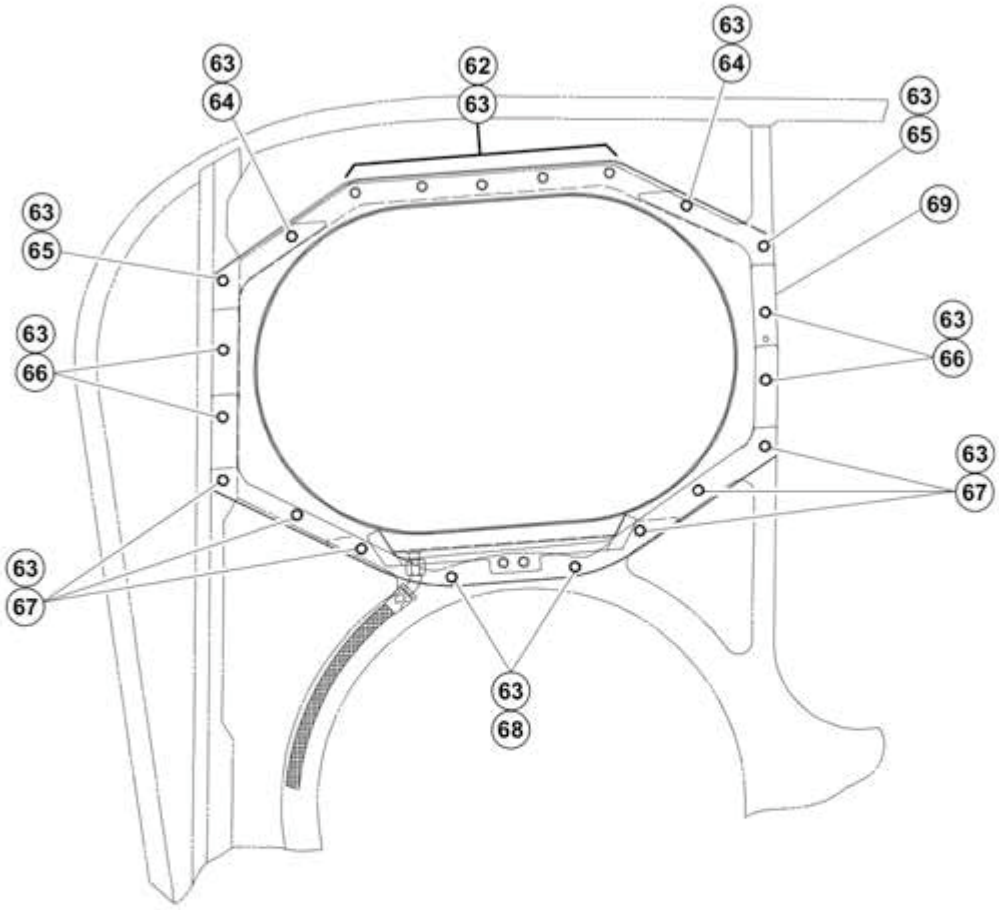
ICN-412-A-716100-B-97499-00003-A-001-01

Figure 1. Ejector - Removal/Installation (Sheet 1 of 4)



ICN-412-A-716100-B-97499-00004-A-002-01

Figure 1. Ejector - Removal/Installation (Sheet 2 of 4)



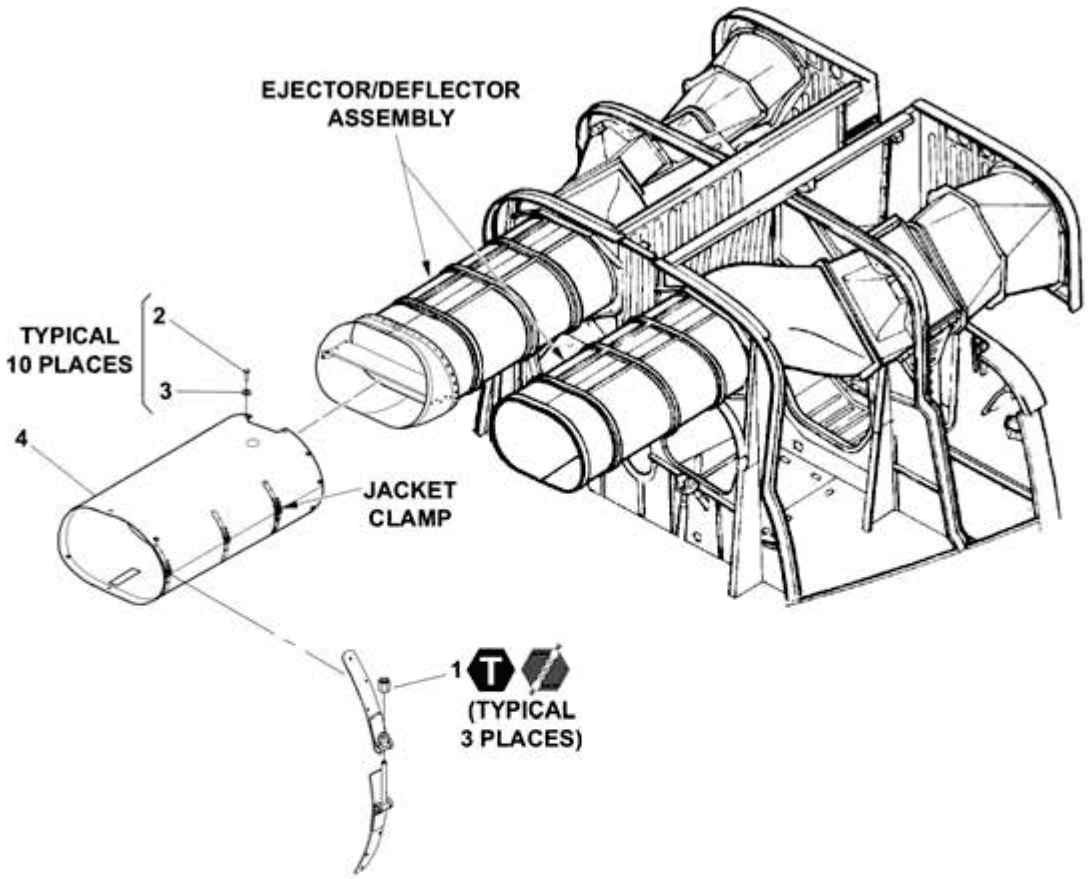
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Figure 1. Ejector - Removal/Installation (Sheet 3 of 4)

1. Left ejector
2. Aft firewall
3. Left transition duct
4. Right transition duct
5. Right ejector
6. Oil cooler cowling
7. Oil cooler support frame
8. Washer
9. Screw
10. Washer
11. Screw
12. Washer
13. Screw
14. Screw
15. Washer (AN960C10L)
16. Seal
17. Vent hose (left power section accessory gearbox)
18. Vent hose (combining gearbox)
19. Drain line
20. Drain line fitting
21. Exhaust duct
22. Elbow
23. Union
24. Nut
25. Thin steel washer
26. Thin steel washer
27. Fitting
28. Fitting
29. Thin steel washer
30. Thin steel washer
31. Nut
32. Nut
33. Bolt
34. Washer (AN960C10L)
35. Washer (AN960C10L)
36. Link
37. Bracket
38. Spacer
39. Washer (AN960C10L)
40. Washer (AN960C10L)
41. Nut
42. Bolt
43. Washer (AN960C10L)
44. Washer
45. Bracket
46. Spacer
47. Link
48. Washer
49. Washer
50. Nut
51. Vent hose (right power section accessory gearbox)
52. Union
53. Elbow
54. Bolt
55. Washer
56. Washer
57. Nut
58. Bolt
59. Washer
60. Washer
61. Nut
62. Bolt
63. Washer
64. Bolt
65. Bolt
66. Bolt
67. Bolt
68. Bolt
69. Shim

ICN-412-A-716100-B-97499-00005-A-002-01

Figure 1. Ejector - Removal/Installation (Sheet 4 of 4)



ICN-412-A-716100-B-97499-00006-A-001-01

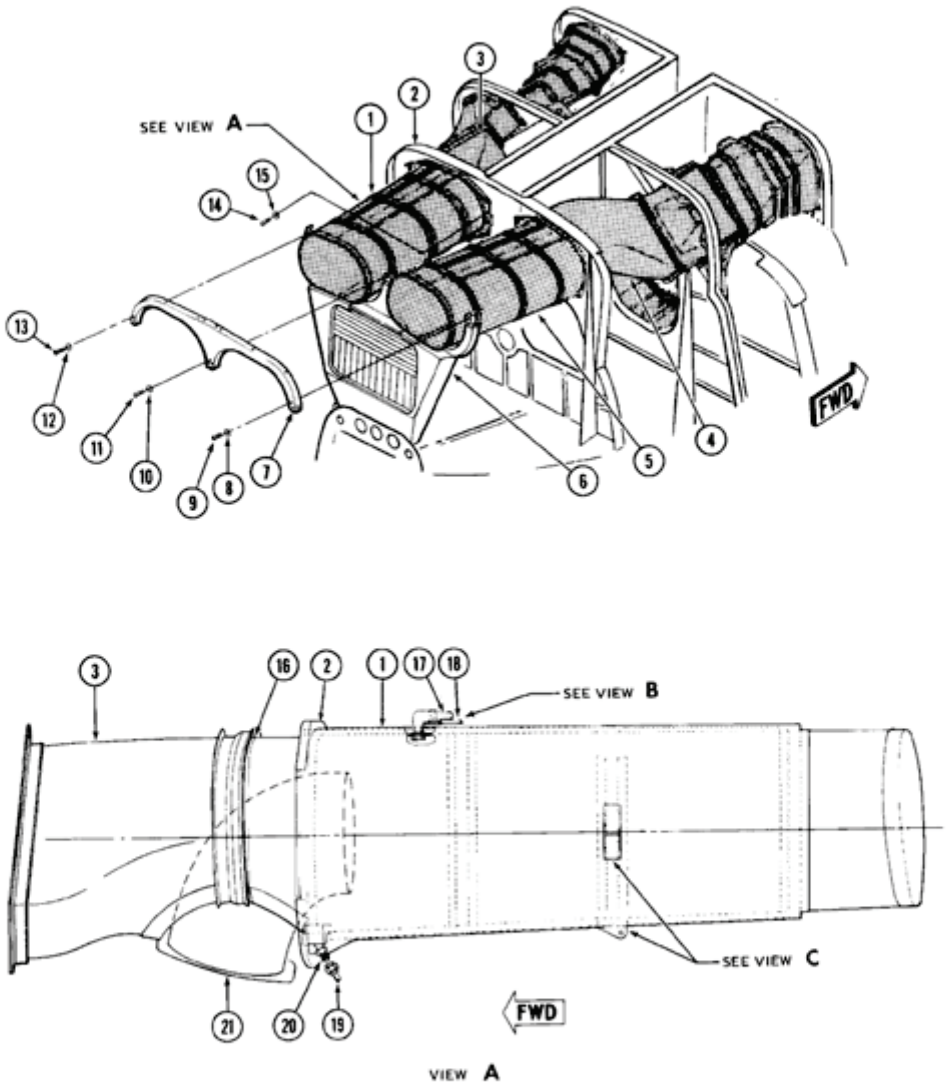
Figure 2. Jacket Assembly - Removal/Installation (Sheet 1 of 1)

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Air Management System Ejectors - Inspection

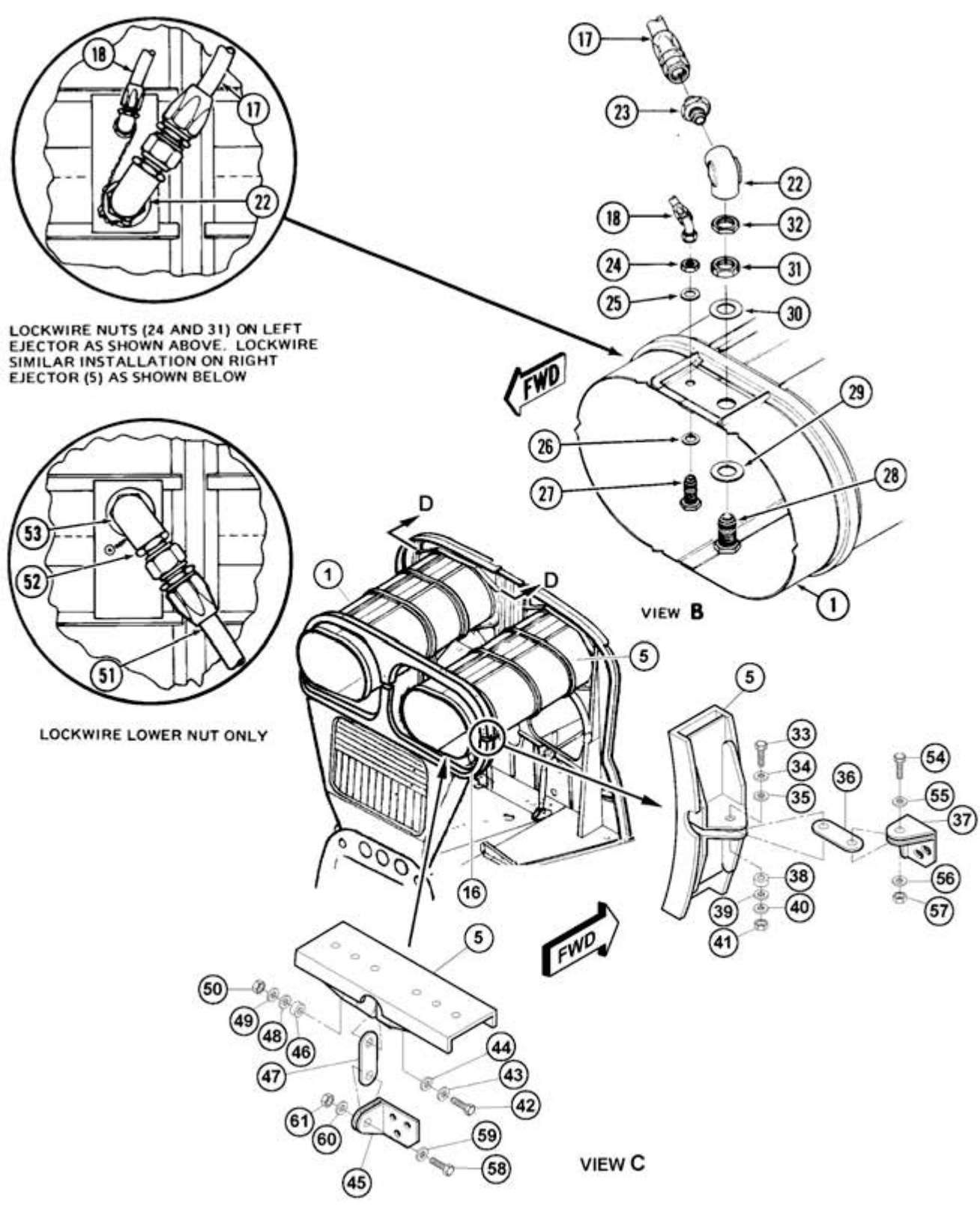
Procedure

1. Inspect ejectors (1 and 5, Figure 1) for evidence of overheating, distortion, cracks, and holes.
2. Inspect ejectors (1 and 5) for secure installation of attaching bolts (33 and 42) and screws (14).
3. Inspect drain line (19) on each ejector (1 and 5) for security and condition.
4. Inspect vent hoses (17, 18, and 51) for security and condition.
5. *Inspect for proper clearance between ejectors (1 and 5) and exhaust ducts (21) (*412-A-71-61-00-01A-720A-A / 01704*).*
6. *If installed, inspect the shims between the ejectors (1 and 5) and the firewall for damage, cracks, tears, delamination and debonding from the firewall.*



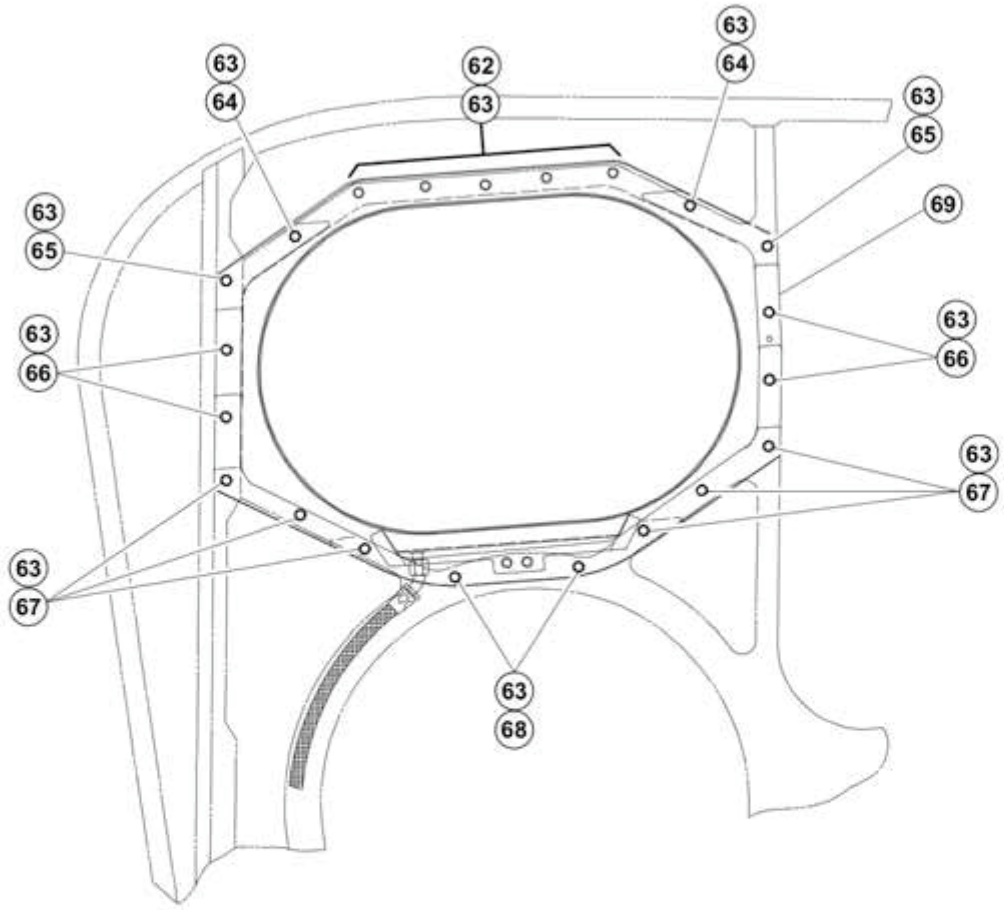
ICN-412-A-716100-B-97499-00003-A-001-01

Figure 1. Ejector (Sheet 1 of 4)



ICN-412-A-716100-B-97499-00004-A-002-01

Figure 1. Ejector (Sheet 2 of 4)



ICN-412-A-716100-B-97499-02958-A-001-01

Figure 1. Ejector (Sheet 3 of 4)

1. Left ejector
2. Aft firewall
3. Left transition duct
4. Right transition duct
5. Right ejector
6. Oil cooler cowling
7. Oil cooler support frame
8. Washer
9. Screw
10. Washer
11. Screw
12. Washer
13. Screw
14. Screw
15. Washer (AN960C10L)
16. Seal
17. Vent hose (left power section accessory gearbox)
18. Vent hose (combining gearbox)
19. Drain line
20. Drain line fitting
21. Exhaust duct
22. Elbow
23. Union
24. Nut
25. Thin steel washer
26. Thin steel washer
27. Fitting
28. Fitting
29. Thin steel washer
30. Thin steel washer
31. Nut
32. Nut
33. Bolt
34. Washer (AN960C10L)
35. Washer (AN960C10L)
36. Link
37. Bracket
38. Spacer
39. Washer (AN960C10L)
40. Washer (AN960C10L)
41. Nut
42. Bolt
43. Washer (AN960C10L)
44. Washer
45. Bracket
46. Spacer
47. Link
48. Washer
49. Washer
50. Nut
51. Vent hose (right power section accessory gearbox)
52. Union
53. Elbow
54. Bolt
55. Washer
56. Washer
57. Nut
58. Bolt
59. Washer
60. Washer
61. Nut
62. Bolt
63. Washer
64. Bolt
65. Bolt
66. Bolt
67. Bolt
68. Bolt
69. Shim

ICN-412-A-716100-B-97499-00005-A-002-01

Figure 1. Ejector (Sheet 4 of 4)

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Air Management System Ejectors - Repair

Procedure



ANY REPAIRS REQUIRING WELDING SHOULD ONLY BE CARRIED OUT BY COMPETENT CERTIFIED WELDERS USING PROPER EQUIPMENT.

NOTE

All repairs addressed are designed to be accomplished either with ejectors installed or on the bench without the aid of extensive equipment.

1. Dents and small cracks in inner and outer skins are repaired as follows (Figure 1):

1.1. Dents may be worked out using standard sheet metal methods; however, care must be exercised not to crack the metal, which is titanium.

1.2. Repair cracks in ejector skins that do not exceed 3.0 inches in length by stop drilling with 0.094 inch diameter drill and install a doubler patch 4.5 X 4.5 inches (114.3 X 114.3 mm) or 4.5 inches (114.3 mm) diameter X 0.025 inch (0.635 mm) thick titanium per MIL-T-9046-CP2 (formerly known as MIL-T-9046, TYP1, Comp C).

1.3. Install patch to outer surface of outer skin using NAS1738M4 monel rivets and/or to outer surface of inner skin using MS20615-3M monel rivets with rivet manufactured head on exhaust gas side of inner skin, rivet spacing to be 0.75 inch (19.05 mm) minimum to 1.0 inch (25.4 mm) maximum. TIG welding may be used in lieu of rivets.

1.4. Cracks and holes contained in an area of 0.50 inch (12.7 mm) diameter may be repaired (and). Patch to be fabricated from titanium per MIL-T-9046. Patch may be welded in place.

2. Cracks in intercostal members are repaired as follows:

2.1. Cracks in intercostal members that are confined to one flange only may be repaired as shown in . If crack exceeds 1.0 inch (25.4 mm) in length, member must be replaced or repaired in accordance with . Doubler used in or should be fabricated from titanium per MIL-T-9046.

2.2. Cracks in intercostal members affecting both flanges cannot be repaired with doublers defined in . Intercostal members shall be replaced.

3. Make repair to large areas as follows:

3.1. Outer skin - Repair of areas larger than 16 square inches (103.2 cm^2) and not greater than 8.0 x 12.0 inches (203.2 x 304.8 mm) may be repaired as shown in . Patch doubler may be fabricated from titanium per MIL-T-9046. Number of repairs of this size shall not exceed three per ejector. Consideration should be taken to combine adjacent small repairs into one large patch.

3.2. Inner skin - Repair of areas larger than 16 square inches (103.2 cm^2) and not greater than 8.0 x 12.0 inches (203.2 x 304.8 mm) may be repaired as shown in . Fabricate a patch from titanium per MIL-T-9046 using monel rivets (MS20615-3M) at each corner to locate patch. Number of repairs of this size shall not exceed two per ejector.

4. Frame repairs:

4.1. Cracks in frames may be repaired as shown in , , or . Cracks up to 1.5 inches (38.1 mm) (Dimension A, sheet 10) may be welded. Cracks up to 5.0 inches (127 mm) long (Dimension A) may be repaired by welding a 0.032 inch (0.813 mm) thick titanium per MIL-T-9046, patch over crack. Patch shall extend 0.50 inch (12.7 mm) beyond stop-drill holes and completely cover cracked area. Fillet weld around edges of frame.

4.2. When cracks extend into vertical flanges, repair as shown in using 0.025 inch (0.635 mm) thick titanium per MIL-T-9046. When cracks extend to attachment flanges, repair as shown in using 0.025 inch (0.635 mm) thick titanium per MIL-T-9046.

4.3. Rivet spacing (where used) shall be 0.75 inch (19.05 mm) minimum to 1.0 inch (25.4 mm) maximum. When crack extends through attachment flanges, frame may have only one repair in an 18.0 inches (457.2 mm) length of frame and no closer than 8.0 inches (203.2 mm) to any airframe attachment fitting.

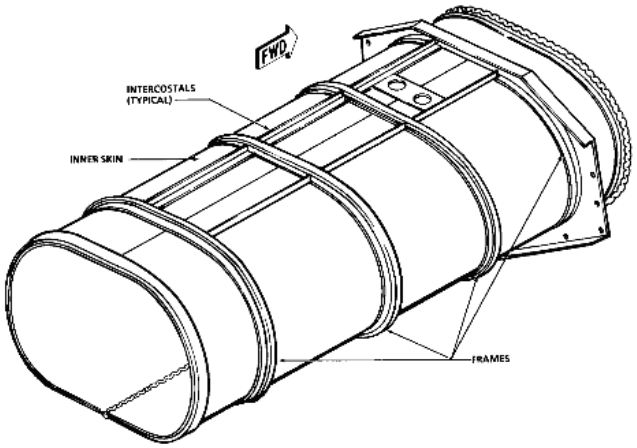
4.4. Support clips may have cracks in vertical flange weld repaired once. Subsequent repairs require replacement of clip which may be fabricated from same thickness titanium per MIL-T-9046.

4.5. Repair attachment flange as follows:

4.5.1. All loose nutplate attaching rivets are to be replaced using monel rivets (MS20427M3).

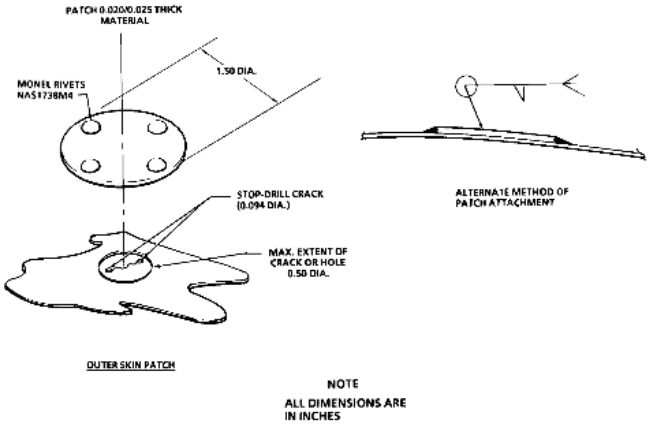
4.5.2. Cracks originating from rivet or bolt holes may be welded one weld per nutplate. Cracks which run from hole to hole within a nutplate pattern shall be reinforced by welding a doubler 0.025 inch (0.635 mm) thick titanium per MIL-T-9046 over hole pattern and redrilling holes.

4.5.3. Cracks in the attachment flange in between attachment holes may be welded and the weld finished smooth. When a crack runs across both flanges, a doubler 0.025 inch (0.635 mm) thick titanium per MIL-T-9046 should be used. A crack in the attachment flange is to be welded and finished smooth. Since the other the flange is attached to the ejector, weld as far as practical. Attach the doubler and weld. Repairs of this nature shall be restricted to a rivet per ejector and shall be no closer to other repairs by the length of the doubler.



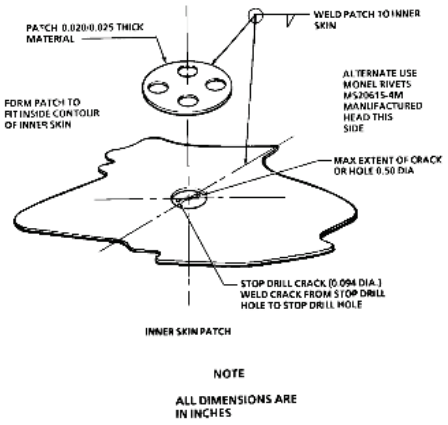
ICN-412-A-716100-B-97499-00007-A-001-01

Figure 1. Exhaust Ejector Repair (Sheet 1 of 11)



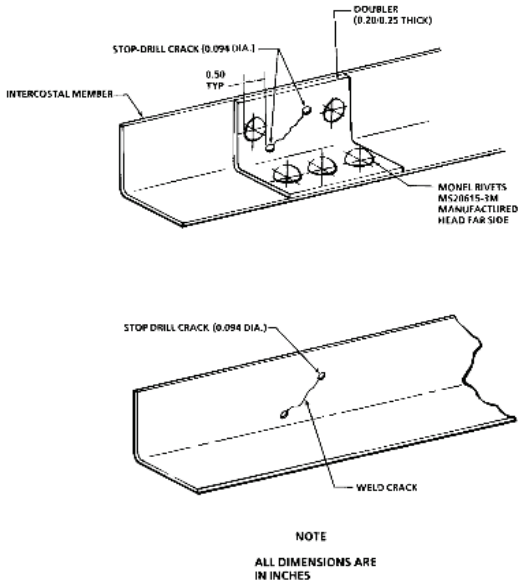
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Figure 1. Exhaust Ejector Repair (Sheet 2 of 11)



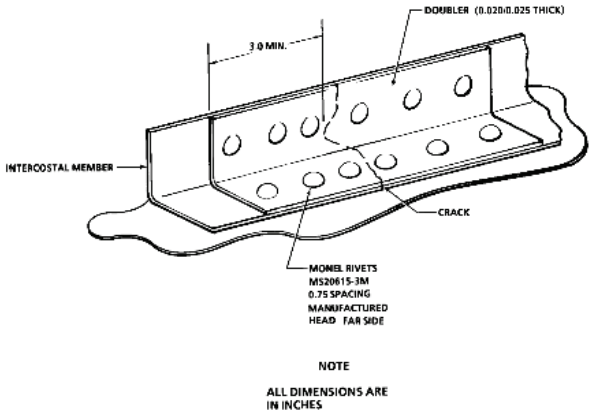
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Figure 1. Exhaust Ejector Repair (Sheet 3 of 11)



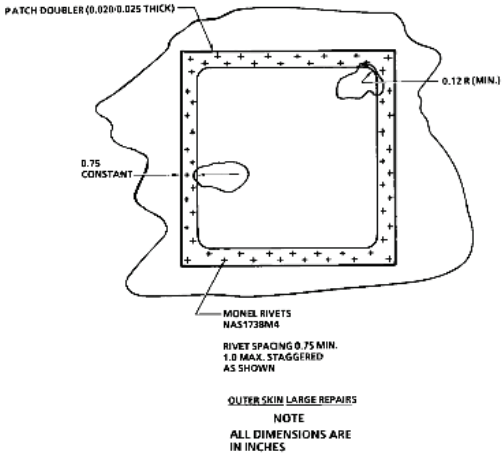
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Figure 1. Exhaust Ejector Repair (Sheet 4 of 11)



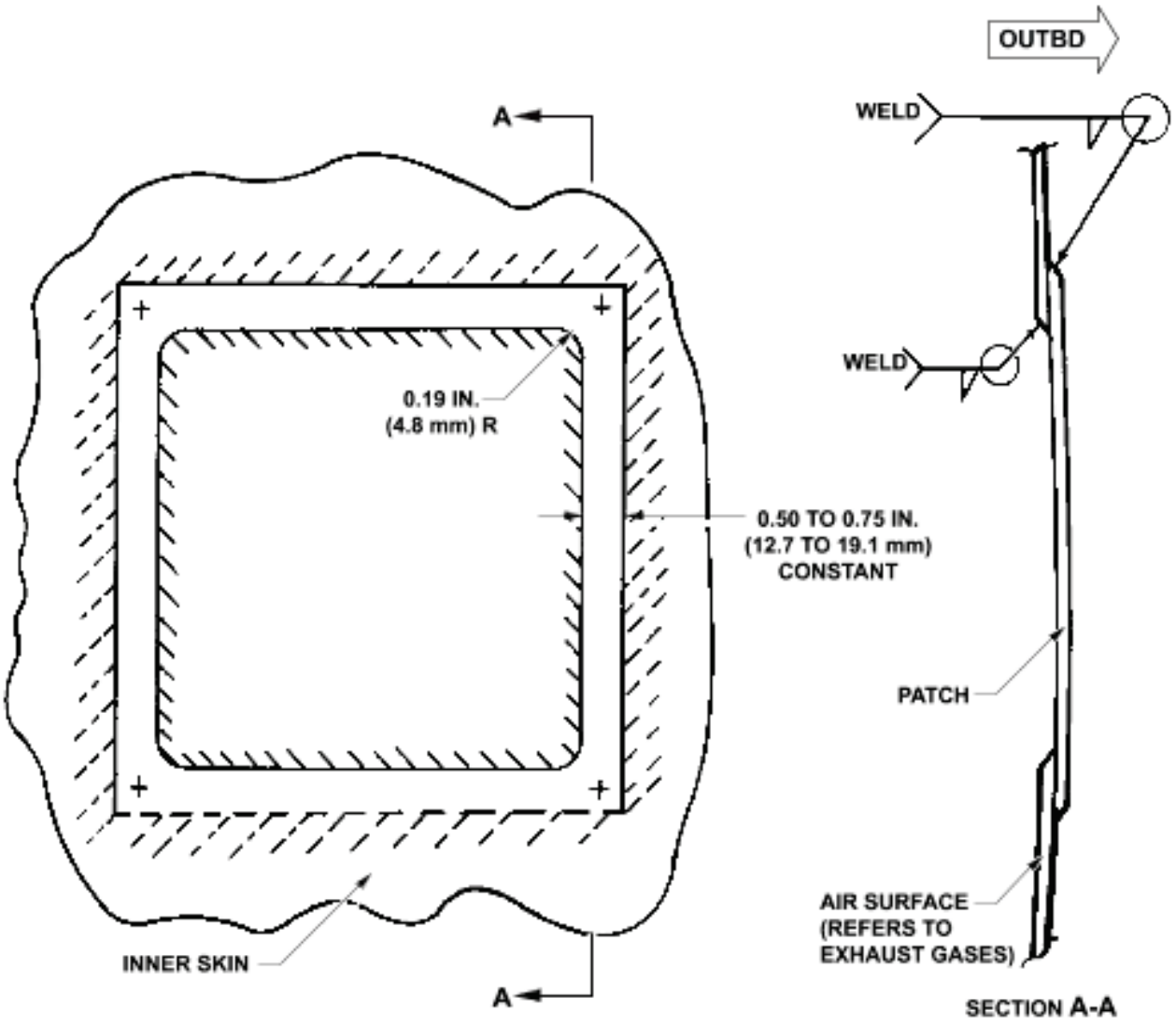
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Figure 1. Exhaust Ejector Repair (Sheet 5 of 11)



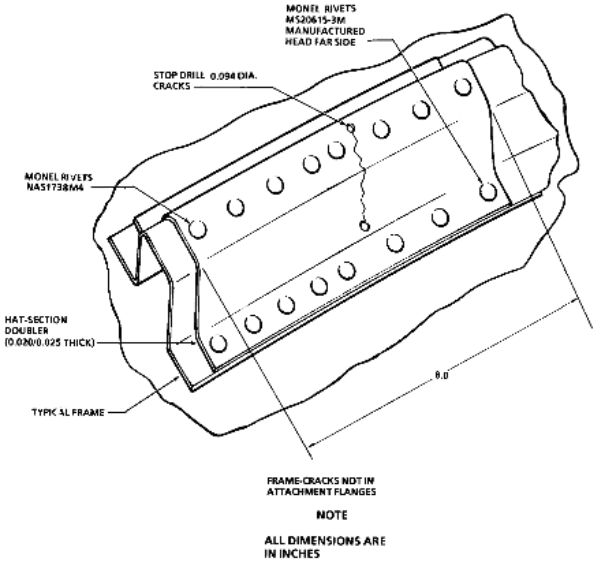
ICN-412-A-716100-B-97499-00012-A-001-01

Figure 1. Exhaust Ejector Repair (Sheet 6 of 11)



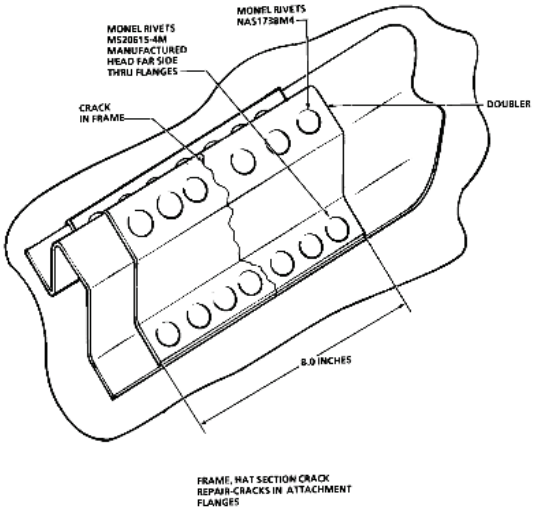
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Figure 1. Exhaust Ejector Repair (Sheet 7 of 11)



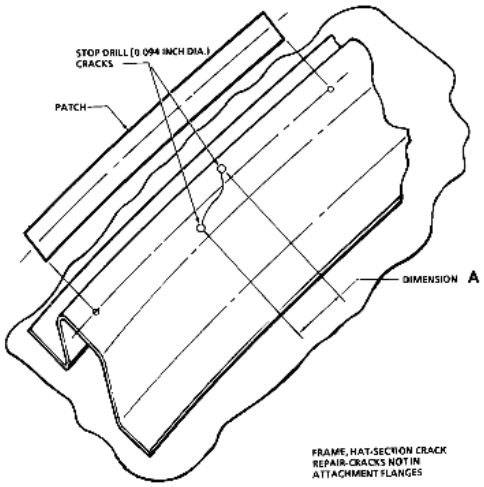
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Figure 1. Exhaust Ejector Repair (Sheet 8 of 11)



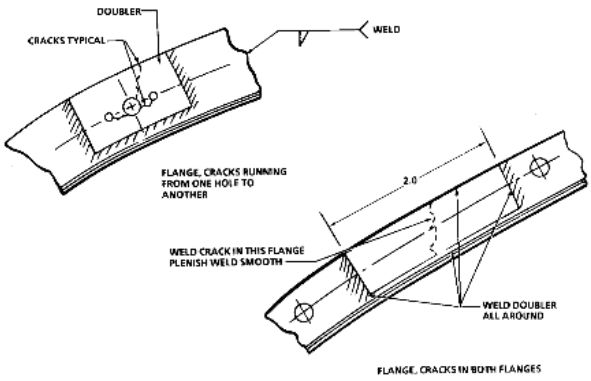
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Figure 1. Exhaust Ejector Repair (Sheet 9 of 11)



ICN-412-A-716100-B-97499-00016-A-001-01

Figure 1. Exhaust Ejector Repair (Sheet 10 of 11)



ICN-412-A-716100-B-97499-00017-A-001-01

Figure 1. Exhaust Ejector Repair (Sheet 11 of 11)

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Air Management System Ejectors - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Lockwire	C-405	AR	
Clean cloth	*C-516*	*AR*	
MEK	*C-309*	*AR*	
Sealing compound	*C-353*	*AR*	

Procedure

NOTE

Maintenance procedures for both ejectors (1 and 5, Figure 1) are the same except as noted in text.

Helicopter S/N 36212 through 36999, 37002 through 37999, 38001 through 38999 and 39101 through 39999 have ejector/deflector assemblies installed.

Step 1. *and* Step 2. *are for the left ejector (1).* Step 3. *is for the right ejector (5).*

NOTE

The use of beeswax, the paste-type drill and tap lubricant, or bar soap on bolts and screws will reduce damage to nutplates and make installation easier.

1. *If not done before, position the fitting (27) and thin steel washer (26) in the left ejector (1) from inside. Install the thin steel washer (25) and nut (24).*

2. If not done before, position the fitting (28) and thin steel washer (29) in the left ejector (1) from inside. Install the thin steel washer (30) and nut (31). Secure the nut (31) to the nut (24), as shown in View B, with lockwire (C-405). Install the nut (32), elbow (22), and the union (23).

3. If not done before, install the elbow (53) on the right ejector (5) in the same manner that the elbow (22) was installed in Step 2. Apply lockwire (C-405) as shown.

Applicable: (S/N 36630 THRU 36999, 37002 THRU 37999, 38001 THRU 38999, 39101 THRU 39999) :

4. *Install the ejector (1) with shim (69) as follows:*

NOTE

This installation procedure is for the left and the right side ejectors (1 and 5). The procedure for the left side ejector (1) is given. Differences for the right side ejector (5) are identified.

4.1. *Put the ejector (1) in its position on the aft firewall (2) and loosely install five bolts (62) with washers (63) on the upper horizontal row of holes in the upper flange of ejector (1). Do not tighten the five bolts (62) at this time.*

NOTE

Make sure to have a minimum clearance of 0.32 inch (8.13 mm) between the bottom of the ejector (1) and top surface of the oil cooler duct. A locally fabricated spacer can be temporarily installed to maintain this clearance.

NOTE

Use link AAL-061-210-001 and link AAL-061-210-003 only with ejector assembly AAL-061-204-001 and ejector assembly AAL-061-204-002.

NOTE

Make sure that there is a clearance of 0.010 to 0.012 inch (0.25 to 0.30 mm) below the bolt (54) head.

4.2. *If removed, attach the link (36) to the support bracket (37) with the bolt (54), washers (55 and 56), and the nut (57).*

4.3. *Put the spacer (38) and link (36) in the bracket on the side of the ejector (1). Install the bolt (33), washers (34, 35, 39, and 40), and the nut (41).*

NOTE

Make sure that there is a clearance of 0.010 to 0.012 inch (0.25 to 0.30 mm) below the bolt (58) head.

4.4. *If removed, attach the link (47) to the support bracket (45) with the bolt (58), washers (59 and 60), and the nut (61).*

4.5. *Put the spacer (46) and link (47) in the bracket on the bottom of the ejector (1). Install the bolt (42), washers (43, 44, 48, and 49), and nut (50).*

4.6. *Adjust the position of the support brackets (45 and 37) to make sure that the clearance between the side of ejector (1) and the oil cooler support are equal to within 0.063 inch (1.6 mm). Tighten the nuts (41, 57, 50 and 61).*

4.7. *Tighten the five bolts (62) and measure the clearance between the aft firewall (2) and ejector (1) at both end of the lower horizontal row of holes in the lower flange of ejector (1).*

4.8. *If the clearances are between 0.02 inch (0.5 mm) and 0.29 inch (7.37 mm), select the appropriate shim from Table 1.*

NOTE

If the clearance value is between thickness of two shim, then select the next higher one from Table 1. For example, if you get the clearance value 0.165 in. (4.20 mm), then select shim of thickness 0.193 inch (4.90 mm). If the clearance is greater than 0.290 inch (7.37 mm), contact Product Support Engineering for additional information.

Table 1. *SHIM THICKNESS*

Shim Part Number	*Shim Thickness*		*Nomenclature*
	(Inch)	*(mm)*	
AAL-238-011-001	*0.160*	*4.06*	*SHIM ASSY, LHS*
AAL-238-011-002	*0.160*	*4.06*	*SHIM ASSY, RHS*
AAL-238-011-003	*0.193*	*4.90*	*SHIM ASSY, LHS*
AAL-238-011-004	*0.193*	*4.90*	*SHIM ASSY, RHS*
AAL-238-011-005	*0.225*	*5.72*	*SHIM ASSY, LHS*
AAL-238-011-006	*0.225*	*5.72*	*SHIM ASSY, RHS*
AAL-238-011-007	*0.257*	*6.53*	*SHIM ASSY, LHS*
AAL-238-011-008	*0.257*	*6.53*	*SHIM ASSY, RHS*
AAL-238-011-009	*0.289*	*7.34*	*SHIM ASSY, LHS*
AAL-238-011-010	*0.289*	*7.34*	*SHIM ASSY, RHS*

4.9. *Remove the five bolts (62) from the upper flange of ejector (1) and disconnect the links (36 and 47) from the side and bottom of the ejector (1).*

4.10. *Remove the ejector (1). If installed, remove the locally fabricated spacer.*

4.11. *Clean the mating surfaces of the selected shim (69) and the aft firewall (2) with a clean cloth (C-516) moistened with MEK (C-309).*

4.12. *Apply sealing compound (C-353) to mating surfaces of the aft firewall (2) and tapered side the shim (69).*

4.13. *Put the shim (69) in its position on the aft firewall (2).*

4.14. *For the left ejector (1), put the ejector into the end of the transition duct (3).*

4.15. *For the right ejector (5), put the ejector into the end of the transition duct (4).*

4.16. *Perform Step 4.3. and Step 4.5. to connect the links (36 and 47) to the ejector (1) respectively.*

4.17. *Install and tighten the two bolts (64) with washers (63). Check the clearance between the exhaust duct (21) and ejector (1). The acceptable clearance is 0.10 to 0.40 inch (2.54 to 10.16 mm). Trim ejector (1) as necessary.*

4.18. *Install the remaining bolts (62, 65, 66, 67 and 68) with washers (63). Tighten the bolts (62, 65, 66, 67 and 68).*

Applicable: (S/N 33001 THRU 33213, 36001 THRU 36629) :

5. *Install the ejector (1) as follows:*

NOTE

This installation procedure is for the left and the right side ejectors (1 and 5). The procedure for the left side ejector (1) is given. Differences for the right side ejector (5) are identified.

5.1. *For the left ejector (1), put the ejector into the end of the transition duct (3).*

5.2. *For the right ejector (5), put the ejector into the end of the transition duct (4).*

NOTE

Use link AAL-061-210-001 and link AAL-061-210-003 only with ejector assembly AAL-061-204-001 and ejector assembly AAL-061-204-002.

5.3. If necessary, attach the link (47) to the bracket (45) with the bolt (58), washers (59 and 60), and the nut (61). Tighten the nut (61).

NOTE

Make sure that there is a clearance of 0.010 to 0.012 inch (0.25 to 0.30 mm) below the bolt (58) head.

5.4. Put the spacer (46) and link (47) in the bracket on the bottom of the ejector (1). Install the bolt (42), washers (43, 44, 48, and 49), and nut (50). Tighten the nut (50).

5.5. If necessary, attach the link (36) to the bracket (37) with the bolt (54), washers (55 and 56), and the nut (57). Tighten the nut (57).

NOTE

Make sure that there is a clearance of .010 to .012 inch (0.25 to 0.30 mm) below the bolt (54) head.

5.6. Put the spacer (38) and link (36) in the bracket on the side of the ejector (1). Install the bolt (33), washers (34, 35, 39, and 40), and the nut (41). Tighten the nut (41).

5.7. Install the two screws (14) and washers (15). Check clearance between the exhaust duct (21) and ejector (1). Acceptable clearance is 0.10 to 0.40 inch (2.54 to 10.16 mm). Trim the ejector (1) as necessary.

5.8. Install the remaining screws (14) and washers (15).

Applicable: (S/N 36212 THRU 36999, 37002 THRU 37999, 38001 THRU 38999, 39101 THRU 39999) :

6. *Install the jacket assembly (4,* Figure 2*) as follows:*

NOTE

This procedure is for the left and the right side jacket assembly. The procedure for the left side jacket assembly is given. Differences for the right side jacket assembly are identified.

6.1. Put the jacket assembly (4) in position on the left ejector (1, Figure 1).

NOTE

The jacket clamps are always on the inboard side, threaded portion up. The top of the jacket assembly must always overlap the bottom.

6.2. Install the nuts (1, Figure 2) on the jacket clamps.

6.3. Do not torque the nuts (1).

6.4. Install one screw (2) and one washer (3) at the opposite of the jacket assembly (4) overlap.

6.5. Torque the nuts (1) .

6.6. Install the rest of the screws (2) and the washers (3).

6.7. Install lockwire (C-405) on the nuts (1).

7. Connect drain line (19, Figure 1) to drain line fitting (20).

NOTE

Step 7. and Step 8. are for the left ejector. Step 9. is for the right ejector.

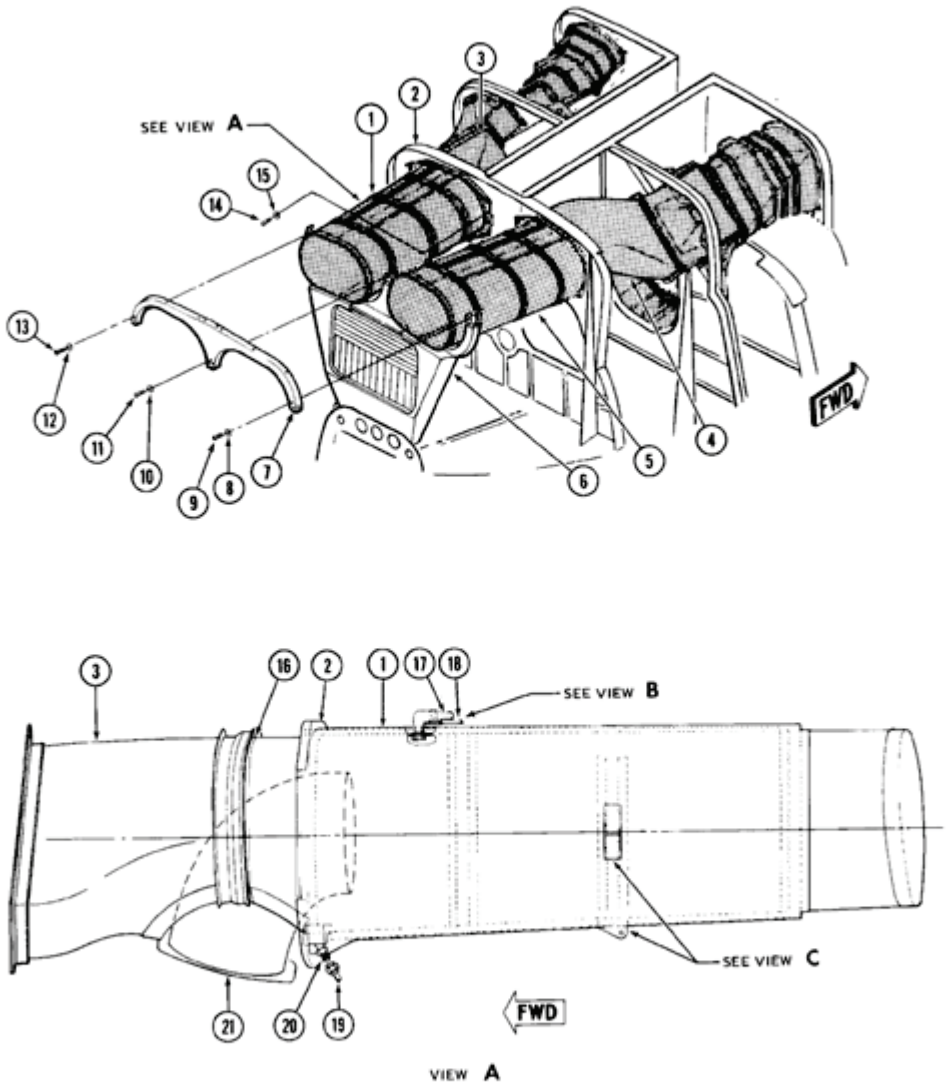
8. Install the vent hose (18) on the fitting (27).

9. Loosen the nut (32) and position the elbow (22) for installation of the vent hose (17). Tighten the nut (32) and install the vent hose (17).

10. Put the elbow (53) in the same manner as given in Step 9. and install vent hose (51).

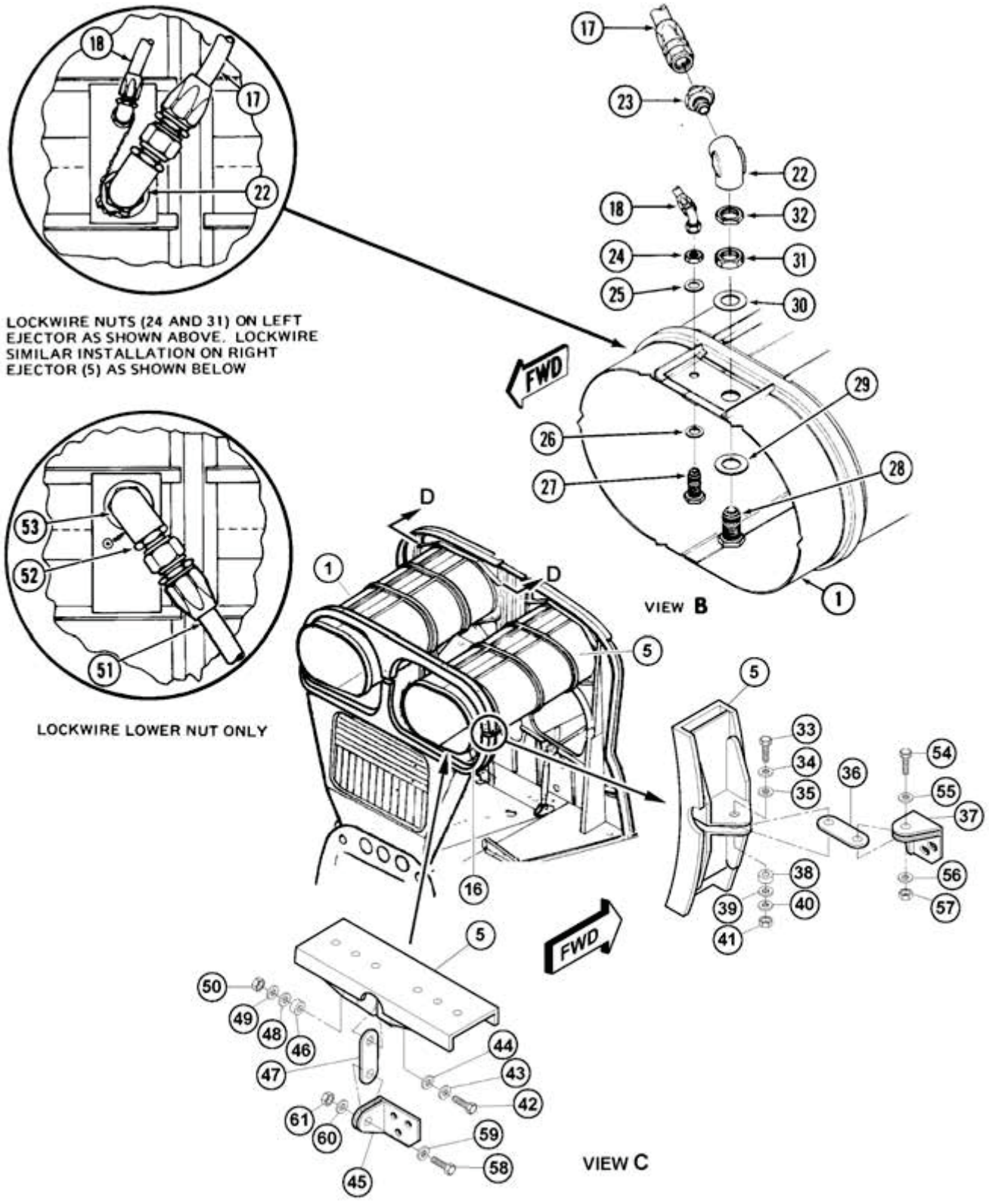
11. Put the oil cooler support frame (7) on the oil cooler cowling (6). Install the screws (9, 11, and 13) and washers (8, 10, and 12).

12. Install the engine cowling, aft pylon fairing, and the tailpipe fairing.



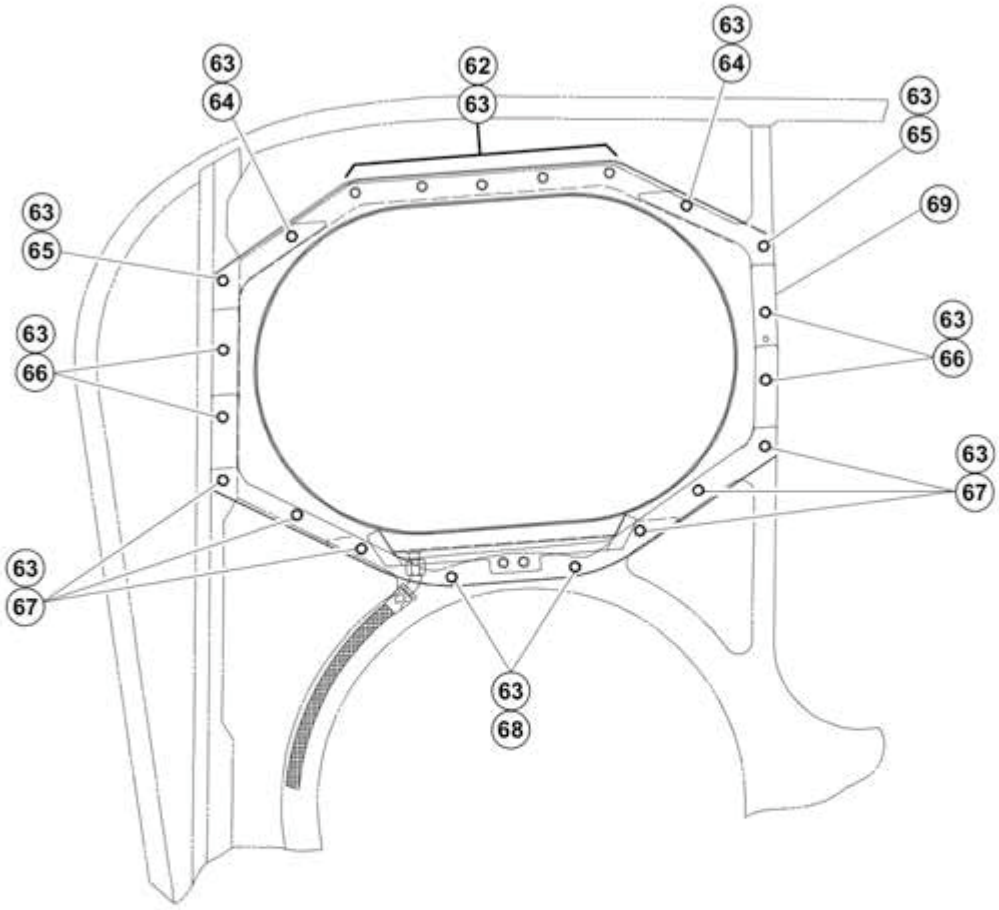
ICN-412-A-716100-B-97499-00003-A-001-01

Figure 1. Ejector -Removal/Installation (Sheet 1 of 4)



ICN-412-A-716100-B-97499-00004-A-002-01

Figure 1. Ejector -Removal/Installation (Sheet 2 of 4)



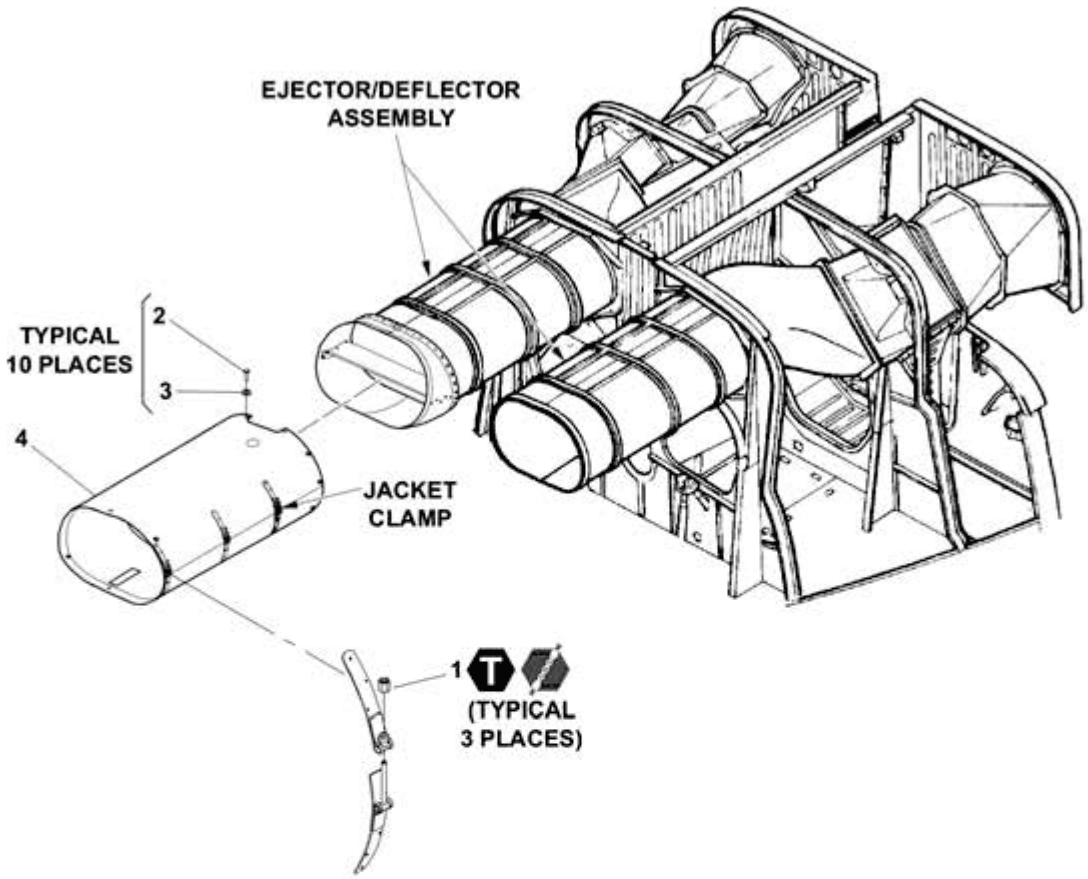
ICN-412-A-716100-B-97499-02958-A-001-01

Figure 1. Ejector -Removal/Installation (Sheet 3 of 4)

1. Left ejector
2. Aft firewall
3. Left transition duct
4. Right transition duct
5. Right ejector
6. Oil cooler cowling
7. Oil cooler support frame
8. Washer
9. Screw
10. Washer
11. Screw
12. Washer
13. Screw
14. Screw
15. Washer (AN960C10L)
16. Seal
17. Vent hose (left power section accessory gearbox)
18. Vent hose (combining gearbox)
19. Drain line
20. Drain line fitting
21. Exhaust duct
22. Elbow
23. Union
24. Nut
25. Thin steel washer
26. Thin steel washer
27. Fitting
28. Fitting
29. Thin steel washer
30. Thin steel washer
31. Nut
32. Nut
33. Bolt
34. Washer (AN960C10L)
35. Washer (AN960C10L)
36. Link
37. Bracket
38. Spacer
39. Washer (AN960C10L)
40. Washer (AN960C10L)
41. Nut
42. Bolt
43. Washer (AN960C10L)
44. Washer
45. Bracket
46. Spacer
47. Link
48. Washer
49. Washer
50. Nut
51. Vent hose (right power section accessory gearbox)
52. Union
53. Elbow
54. Bolt
55. Washer
56. Washer
57. Nut
58. Bolt
59. Washer
60. Washer
61. Nut
62. Bolt
63. Washer
64. Bolt
65. Bolt
66. Bolt
67. Bolt
68. Bolt
69. Shim

ICN-412-A-716100-B-97499-00005-A-002-01

Figure 1. Ejector -Removal/Installation (Sheet 4 of 4)



ICN-412-A-716100-B-97499-00006-A-001-01

Figure 2. Jacket Assembly - Removal/Installation (Sheet 1 of 1)

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Air Management System Transition Ducts - Removal

Procedure

NOTE

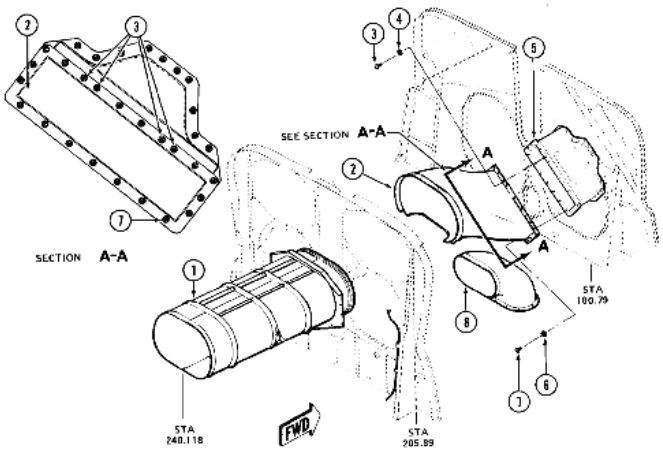
The removal procedure for right transition duct is given. The removal procedure for left transition duct is the same.

1. Remove engine cowling, aft pylon fairing, and tailpipe fairing.

2. Remove ejector (1, Figure 1) (412-A-71-61-00-01A-520A-A / 01701).

3. Remove screws (3 and 7).

4. Remove transition duct (2).



1. Ejector
2. Transition duct
3. Screw
4. Thin aluminum washer
5. Valve assembly
6. Thin aluminum washer
7. Screw
8. Exhaust duct

ICN-412-A-716100-B-97499-00018-A-001-01

Figure 1. Transition Duct (Sheet 1 of 1)

Air Management System Transition Ducts - Inspection

Procedure

NOTE

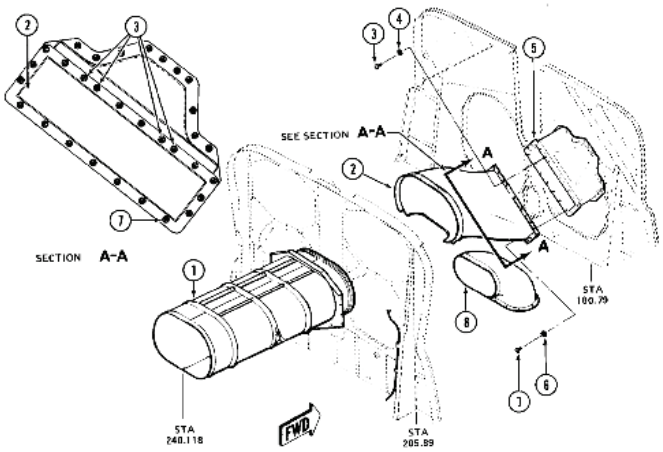
The inspection procedure for right transition duct is given. The inspection procedure for left transition duct is the same.

1. Inspect transition duct (2, Figure 1) for distortion, cracks, and holes.

2. Inspect transition duct (2) for secure installation of attaching screws.

3. Inspect transition duct (2) for proper clearance with exhaust duct (8). Clearance must be 0.10 to 0.70 inch (2.6 to 17.7 mm).

4. Inspect transition duct (2) for secure installation with ejector (1).



1. Ejector
2. Transition duct
3. Screw
4. Thin aluminum washer
5. Valve assembly
6. Thin aluminum washer
7. Screw
8. Exhaust duct

ICN-412-A-716100-B-97499-00018-A-001-01

Figure 1. Transition Duct (Sheet 1 of 1)

Air Management System Transition Ducts - Repair

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Epoxy polyamide primer	C-204	AR	
Sealant	C-308	AR	

Procedure

1. Repair transition duct (2, Figure 1) as follows:

1.1. Local dings, dents, or bumps not exceeding 0.25 inch (6.35 mm) in depth and 2.0 inches (50.8 mm) in diameter may be ignored or worked out at operator's discretion.

1.2. Dings, dents, and bumps exceeding 0.25 inch (6.35 mm) in diameter shall be worked out to original contour.

1.3. All dings, dents, and bumps penetrating the surface shall be worked out to original contour and treated as a crack.

1.4. All cracks up to 3.0 inches (76.2 mm) in length shall be stop-drilled (0.094 inch (2.387 mm) diameter), cracks cleaned out, and patch applied to outside surface (Figure 1, View A). Use sealant (C-308).

1.5. All holes up to 3.0 inches (76.2 mm) in diameter shall be cleaned and a patch applied to the outside surface (View B). Use sealant (C-308).

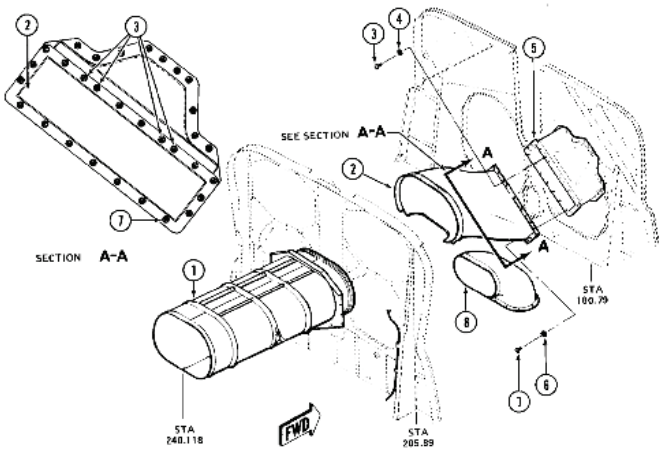
1.6. Repair cracks in flanges per View C. Only one repair on each side permitted. Use sealant (C-308).

1.7. Repairs to seal retainer shall be limited to stop-drilling cracks (0.094 inch (2.387 mm) diameter) and adding a doubler as shown in View D. Use sealant (C-308).

1.8. Repairs to duct requiring a patch shall have patch applied to outside surface except in area of seal retainer (aft end of duct), where internal surface is the only practical surface on which to apply patches.

1.9. All scratches, scores, and surface penetrations shall be buffed out and epoxy polyamide primer (C-204) applied.

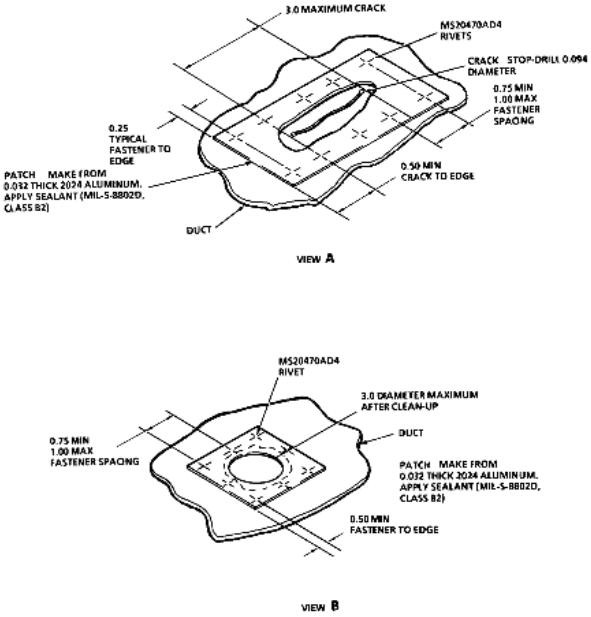
2. No more than four patches are allowed per duct.



1. Ejector
2. Transition duct
3. Screw
4. Thin aluminum washer
5. Valve assembly
6. Thin aluminum washer
7. Screw
8. Exhaust duct

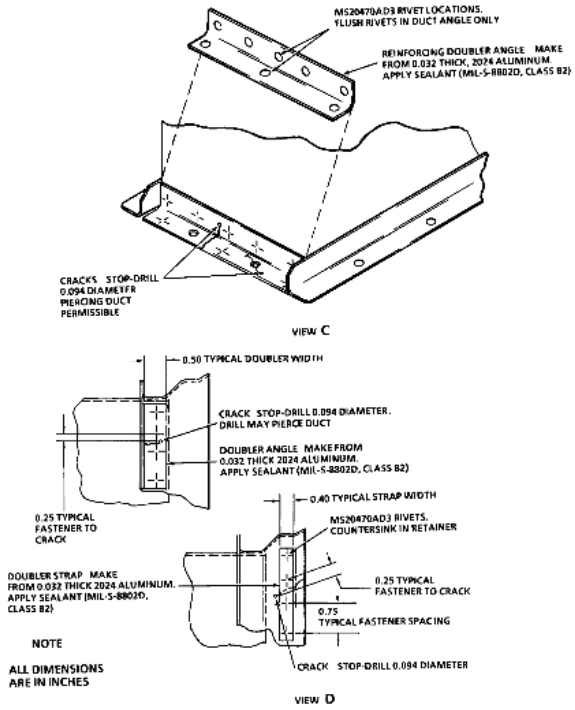
ICN-412-A-716100-B-97499-00018-A-001-01

Figure 1. Transition Duct (Sheet 1 of 1)



ICN-412-A-716100-B-97499-00019-A-001-01

Figure 2. Transition Duct Repair (Sheet 1 of 2)



ICN-412-A-716100-B-97499-00020-A-001-01

Figure 2. Transition Duct Repair (Sheet 2 of 2)

Air Management System Transition Ducts - Installation

Procedure

NOTE

The installation procedure for right transition duct is given. The installation procedure for left transition duct is the same.

NOTE

Use of beeswax, paste-type drill and tap lubricant, or bar soap on bolts and screws will reduce damage to nutplates and make installation easier.

1. Position transition duct (2, Figure 1) over exhaust duct (8).

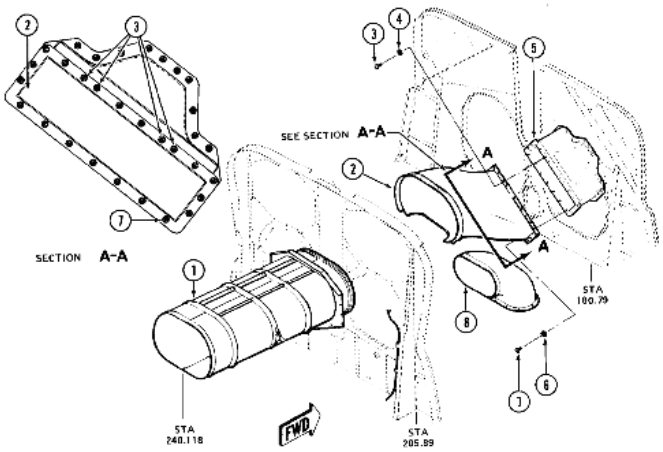
2. Align forward flanges of transition duct (2) with valve assembly (5).

3. Install screws (3 and 7) and thin aluminum washers (4 and 6) in locations shown in Section A-A.

4. Inspect clearance between transition duct (2) and exhaust duct (8). Clearance must be 0.10 to 0.70 inch (2.54 to 17.7 mm).

5. Install ejector (1) (412-A-71-61-00-01A-720A-A / 01704).

6. Install engine cowling, aft pylon fairing, and tailpipe fairing.



1. Ejector
2. Transition duct
3. Screw
4. Thin aluminum washer
5. Valve assembly
6. Thin aluminum washer
7. Screw
8. Exhaust duct

ICN-412-A-716100-B-97499-00018-A-001-01

Figure 1. Transition Duct (Sheet 1 of 1)

Air Management System Valve Assembly - Removal

Procedure

NOTE

Instructions in this paragraph are for removal of right valve assembly. Removal procedures for left valve assembly are the same.

1. Remove transition duct (2, Figure 1) (412-A-71-61-00-02A-520A-A / 01705).

2. Remove screws (6, Figure 2). Remove valve cover (38).

3. Remove bolt (36).

4. Remove screws (8). Remove valve body (37).

NOTE

Accomplish Step 5. through Step 6. only if it is necessary to remove actuator (9).

5. Disconnect electrical connector (11). Remove jamnut and separate electrical lead of actuator (9) from firewall.

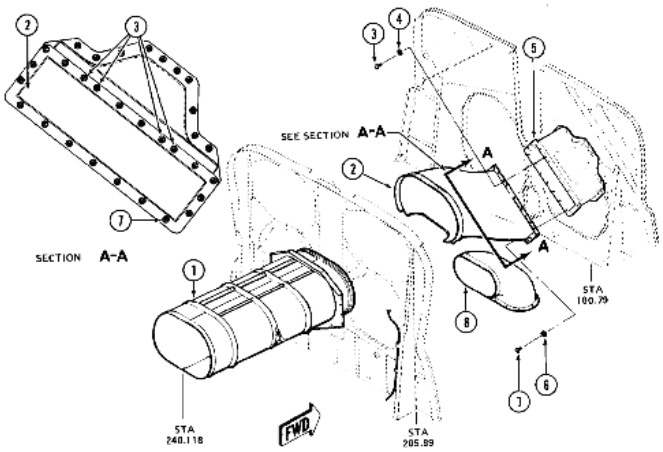
6. Remove bolt (18). Remove actuator (9).

NOTE

Accomplish Step 7. and Step 8. only if it is necessary to remove fitting (27).

7. Remove screws (26). Remove fitting (27).

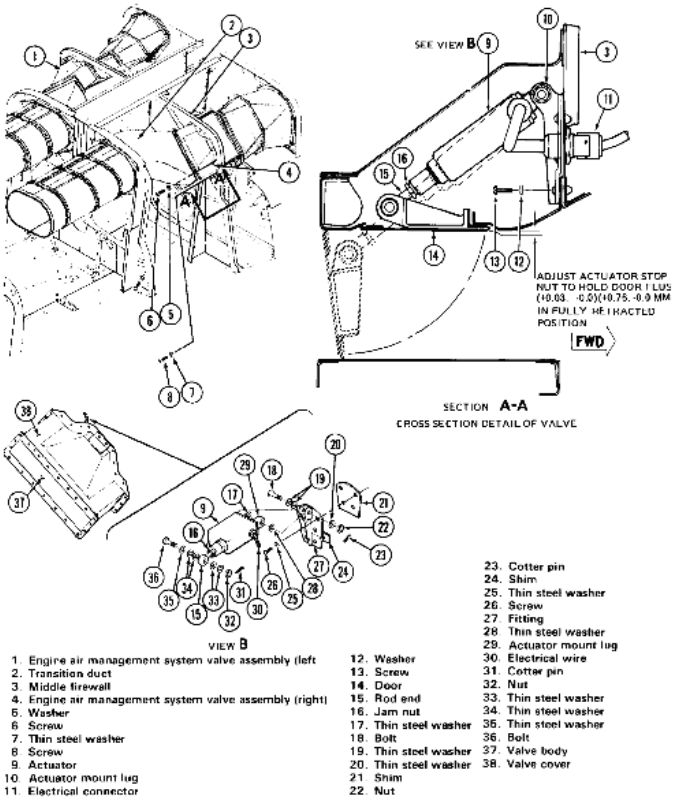
8. If shims (21 and 24) are present, index shims for reinstallation in same location.



1. Ejector
2. Transition duct
3. Screw
4. Thin aluminum washer
5. Valve assembly
6. Thin aluminum washer
7. Screw
8. Exhaust duct

ICN-412-A-716100-B-97499-00018-A-001-01

Figure 1. Transition Duct (Sheet 1 of 1)



ICN-412-A-716100-B-97499-00021-A-001-01

Figure 2. Air Management System Valve (Sheet 1 of 1)

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Air Management System Valve Assembly - Inspection

Procedure

NOTE

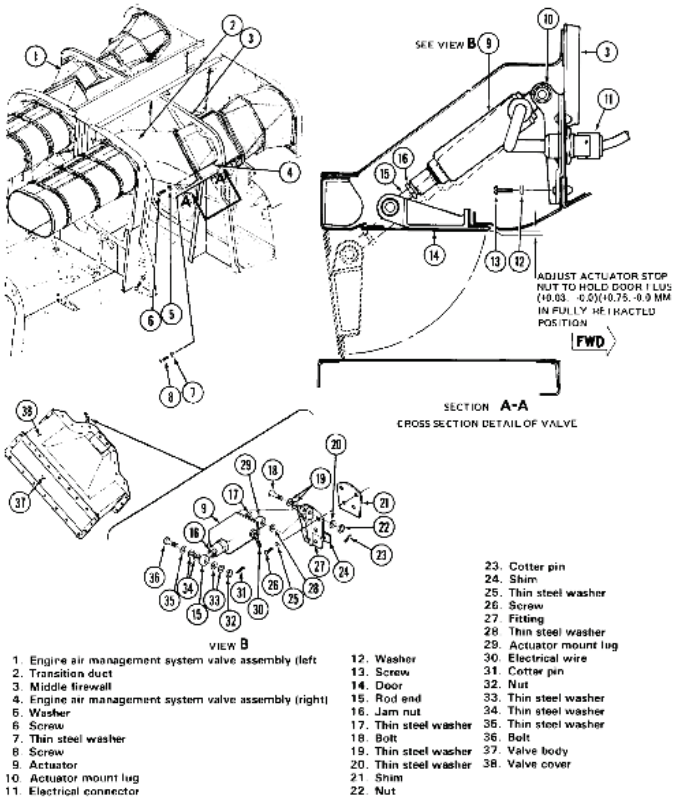
The inspection procedure for right valve assembly is given. Inspection procedure for left valve assembly is the same.

1. Inspect valve (4, Figure 1) for distortion, cracks, and holes.

2. Inspect door (14) for distortion, cracks, and proper adjustment in accordance with Section A-A.

3. Inspect actuator (9) for obvious damage and secure installation.

4. Inspect electrical connector (11) and electrical wiring (30) for obvious damage.



ICN-412-A-716100-B-97499-00021-A-001-01

Figure 1. Air Management System Valve (Sheet 1 of 1)

Air Management System Valve Assembly - Repair

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Sealing compound	C-353	AR	

Procedure

1. Local dings, dents, and bumps not exceeding 0.25 inch (6.35 mm) depth and 1.0 inch (25.4 mm) diameter may be ignored or repaired at operator discretion provided there is no interference with operation of valve box assembly.

2. Local dings, dents, or bumps exceeding 0.25 inch (6.35 mm) depth and greater than 1.0 inch (25.4 mm) diameter shall be repaired or replaced. Rework shall restore damaged area to original contour or limits stated in Step 1..

3. Damage in flat surfaces that can be cleaned up with a 2.0 inches (50.8 mm) diameter hole may be repaired using a doubler, as shown in Figure 1, View A. Apply a bead of sealing compound (C-353) or equivalent around edges of doubler.

4. Attachment flanges may be reworked to remove local wrinkles and surface deformities. Cracks in flange where crack does not originate from a hole may be stop-drilled (0.094 inch (2.387 mm) diameter) and an angle reinforcement added (Figure 1, View B). Where crack extends to spot-welded flange, stop-drill is permitted through flange and box. Where cracks are through nutplate hole pattern, remove nutplate, clean out flange area, and install nutplate on reinforcement angle (View B). Cracks through a single hole may be stop-drilled and a reinforcement angle installed.

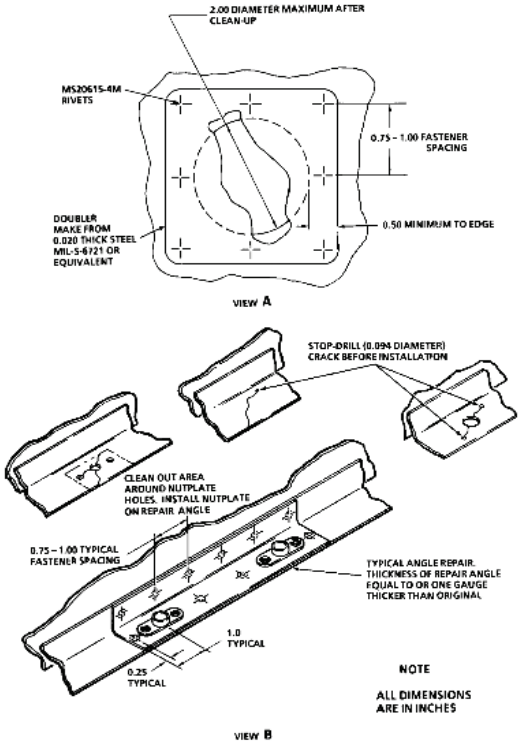
5. Only one repair per flange is allowed.

6. Damage on one surface which overlaps another shall be classified as being a repair on each of the surfaces. No more than two repairs per Figure 2, View A, are allowed per surface. Repairs on one surface that can be combined and repaired with same doubler shall be classified as one repair.

7. Repairs to door hinge fittings (other than buffing out minor surface damage) are not allowed. Repairs due to cracks in door shall consist of stop-drilling (0.094 inch (2.387 mm) diameter) all cracks and attaching a doubler made of 0.012 inch (0.305 mm) thick corrosion resistant steel, MIL-S-6721, or equivalent.

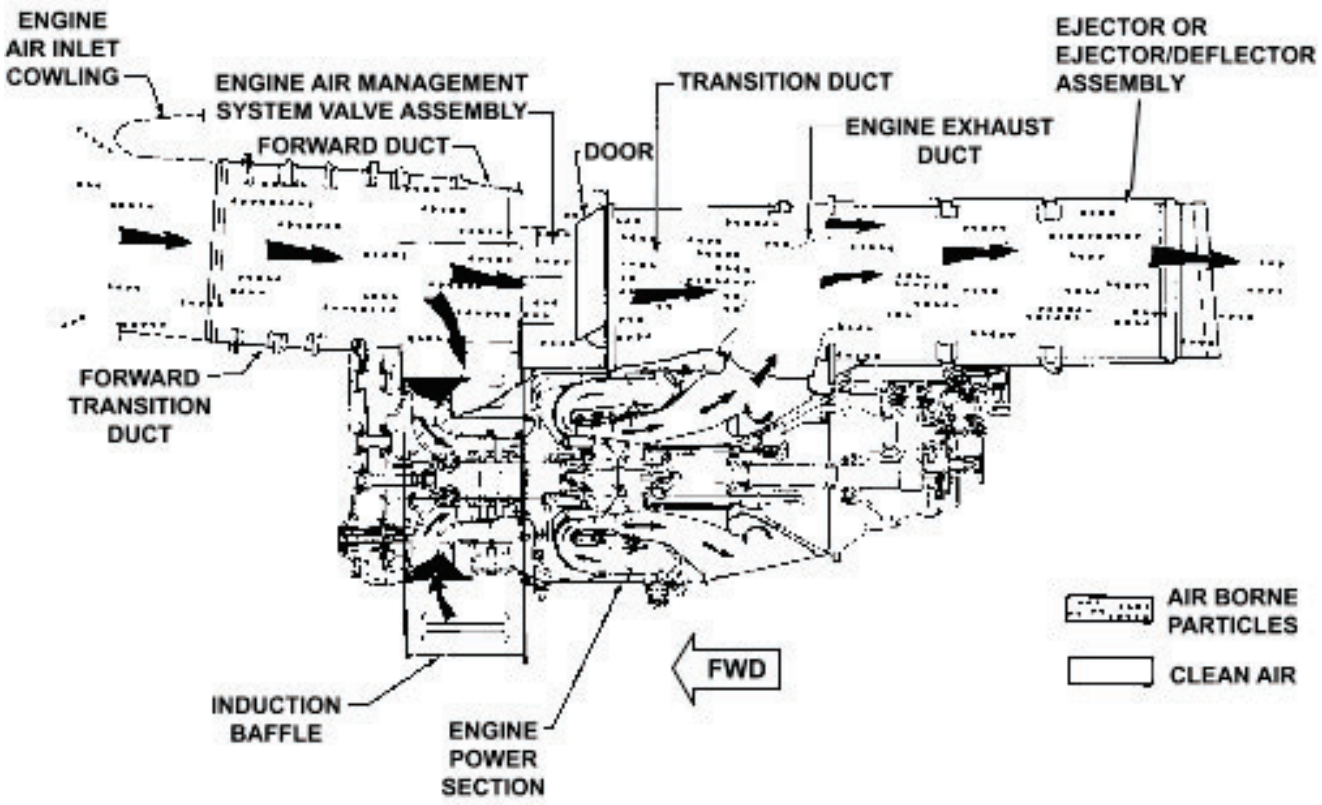
8. Cracks in support brackets are not repairable. Cracked support brackets shall be replaced.

9. Seals and gaskets are not repairable.



ICN-412-A-716100-B-97499-00022-A-001-01

Figure 1. Air Management Valve Assembly Repair (Sheet 1 of 1)



ICN-412-A-716100-B-97499-00002-A-001-01

Figure 2. Air Management System Schematic (Sheet 1 of 1)

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Air Management System Valve Assembly - Installation

Procedure

NOTE

The following instructions are for installation of right valve assembly. Installation procedures for left valve assembly are the same.

NOTE

Use of beeswax, paste type drill and tap lubricant, or bar soap on screws and bolts will reduce damage to nutplates and make installation easier.

1. Remove valve cover (38, Figure 1) from valve body (37).

2. Position valve body (37) on middle firewall (3). Install thin steel washers (7) and screws (8).

NOTE

Shims (21 and 24) are required only if interference occurs between actuator (9) and door (14).

3. Position fitting (27) on middle firewall (3). Install thin steel washers (25) and screws (26).

4. Position actuator (9) on fitting (27). Install bolt (18), thin steel washers (17, 19, 20, and 28), and nut (22) as illustrated. Install cotter pin (23).

5. Position actuator (9) on door (14). Install bolt (36), thin steel washers (33, 34, and 35), and nut (32). Install cotter pin (31).

6. Position electrical lead of actuator (9) through middle firewall (3) and install jam nut (16).

NOTE

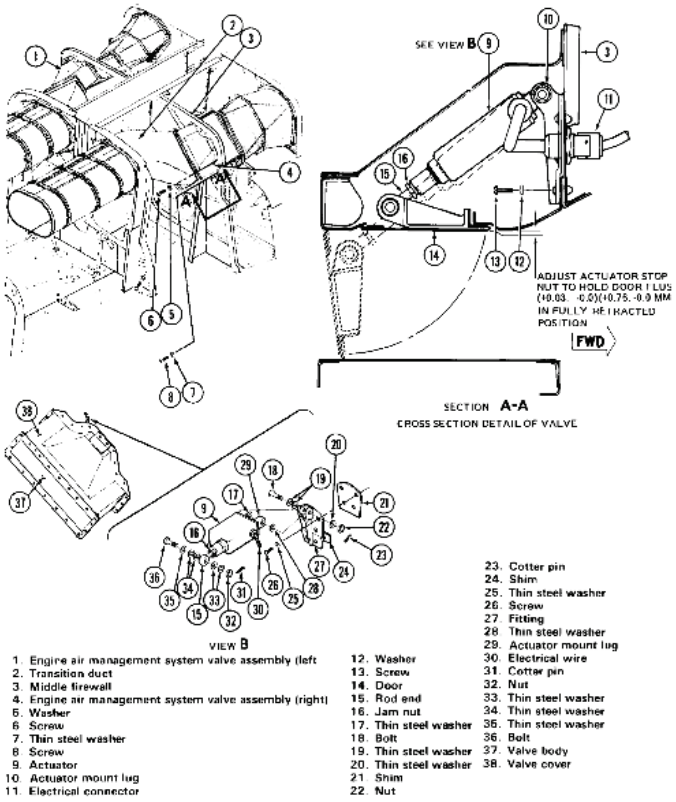
Use suitable electrical work aid to accomplish Step 7. and Step 8..

7. Operate actuator (9) to full retract position. Measure dimension between door (14) and valve body (37), as shown in Section A-A. If door is not flush within 0.03 inch (0.76mm), adjust rod end (15) to obtain proper door position.

8. Operate actuator (9) through full throw in both directions and check for binding and interference. If actuator touches edge of door (14), install shims (21 and 24), as required.

9. After actuator is adjusted, install electrical connector (11).

10. Position valve cover (38) on valve body (37) and install washers (5) and screws (6).



1. Engine air management system valve assembly (left
2. Transition duct
3. Middle firewall
4. Engine air management system valve assembly (right)
5. Washer
6. Screw
7. Thin steel washer
8. Screw
9. Actuator
10. Actuator mount lug
11. Electrical connector
12. Washer
13. Screw
14. Door
15. Rod end
16. Jam nut
17. Thin steel washer
18. Bolt
19. Thin steel washer
20. Thin steel washer
21. Shim
22. Nut
23. Cotter pin
24. Shim
25. Thin steel washer
26. Screw
27. Fitting
28. Thin steel washer
29. Actuator mount lug
30. Electrical wire
31. Cotter pin
32. Nut
33. Thin steel washer
34. Thin steel washer
35. Thin steel washer
36. Bolt
37. Valve body
38. Valve cover

ICN-412-A-716100-B-97499-00021-A-001-01

Figure 1. Air Management System Valve (Sheet 1 of 1)

Air Management System Forward Ducts - Removal

Procedure

NOTE

The removal procedure for right forward duct is given. The removal procedure for left forward duct is the same.

NOTE

The left and right forward transition ducts (5 and 6, Figure 1) are integral parts of the engine air inlet cowlings (2, Figure 2).

1. Remove the forward and aft pylon fairings (412-A-71-10-00-01A-520A-A / 01605).

2. Remove transmission fairing (412-A-71-10-00-02A-520A-A / 01608).

3. Remove the left-hand or right-hand engine air inlet cowling, as applicable (412-A-71-10-00-02A-520A-A / 01608).

4. Remove the forward transition duct (11) from the engine air inlet cowling (2) as follows:

4.1. Remove the two bolts (12) and washers (13) that attach the forward transition duct (11) to the engine air inlet cowling (2).

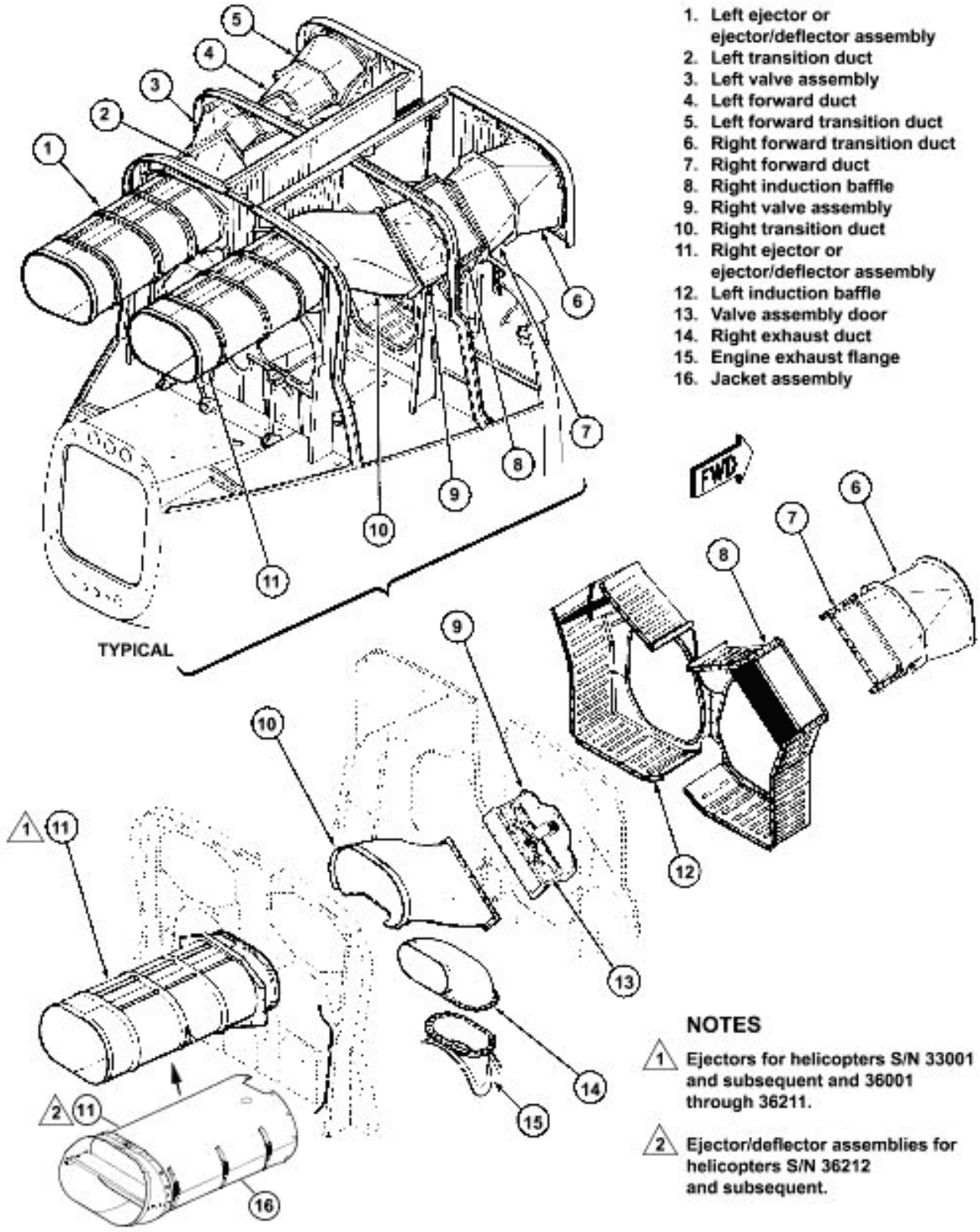
4.2. Remove the forward transition duct (11) from the engine air inlet cowlings (2).

5. Remove valve assembly (1, Figure 3) (412-A-71-61-00-03A-520A-A / 01709).

6. Remove the air management system forward duct (3, Figure 4) from the air management system induction baffle (2) as follows:

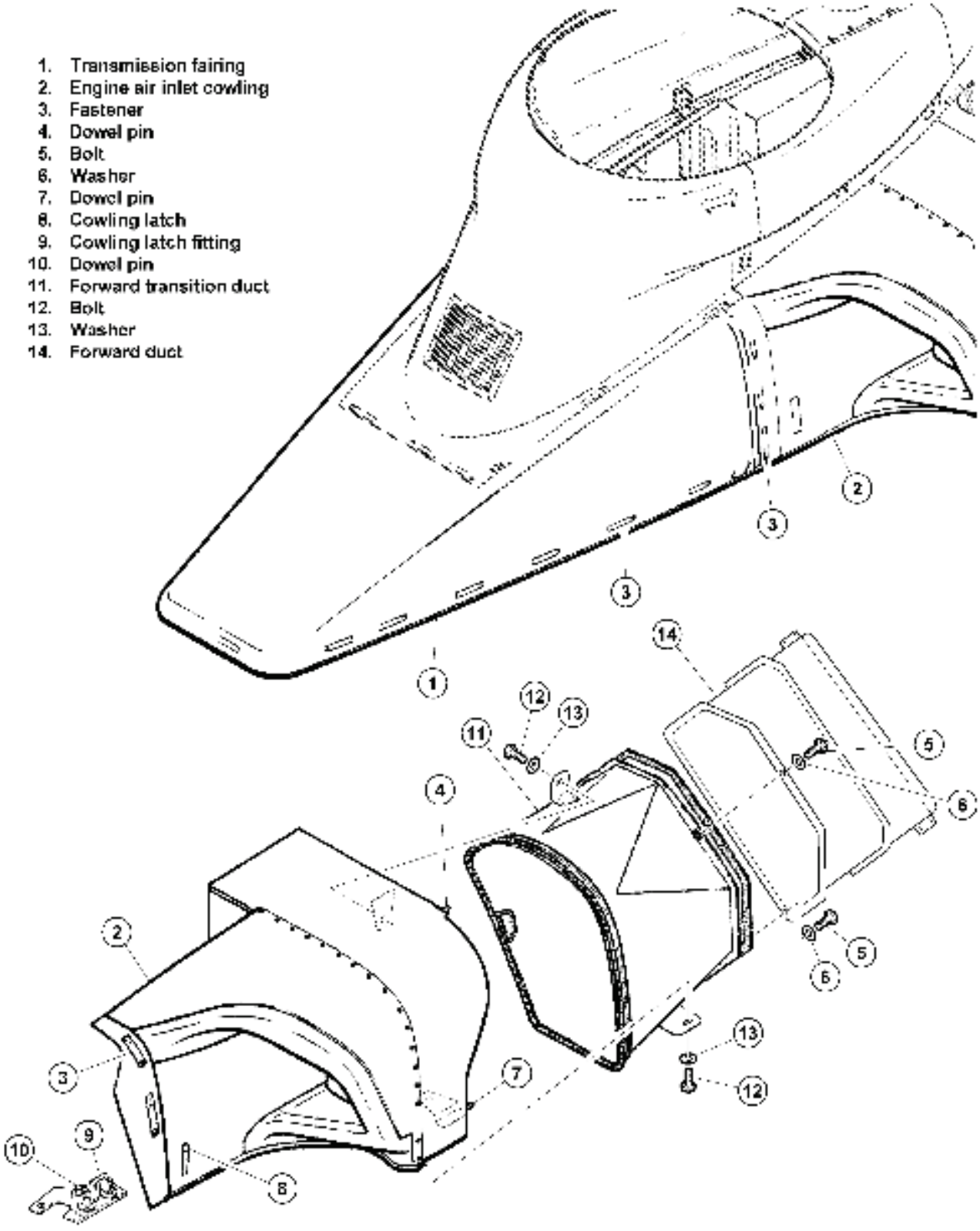
6.1. Remove screws (4) and thin aluminum washers (5), securing air management system forward duct (3) to air management system induction baffle (2).

6.2. Remove air management system forward duct (3).



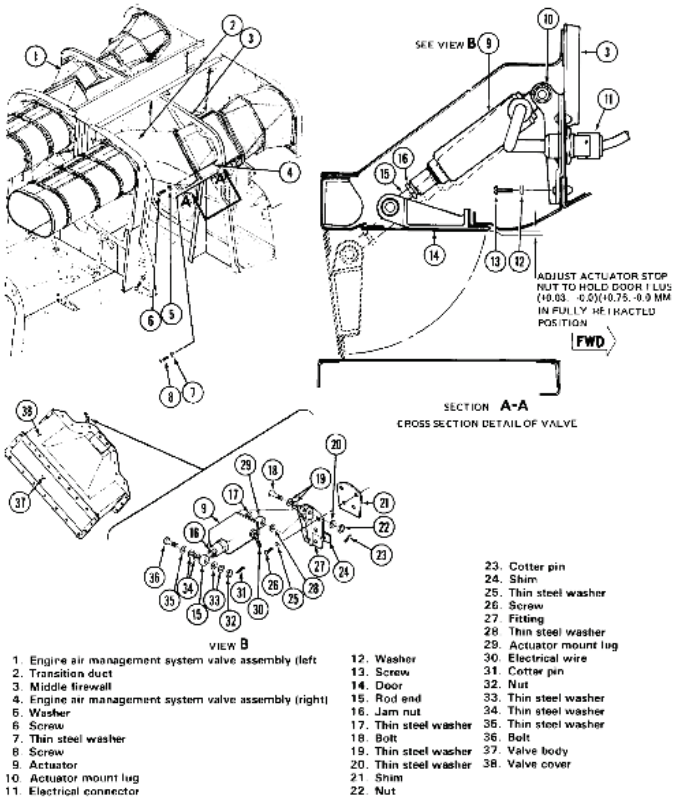
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Figure 1. Air Management System, Helicopters S/N 33001 and Subsequent and 36001 through 36211 and 36212 and Subsequent (Sheet 1 of 1)



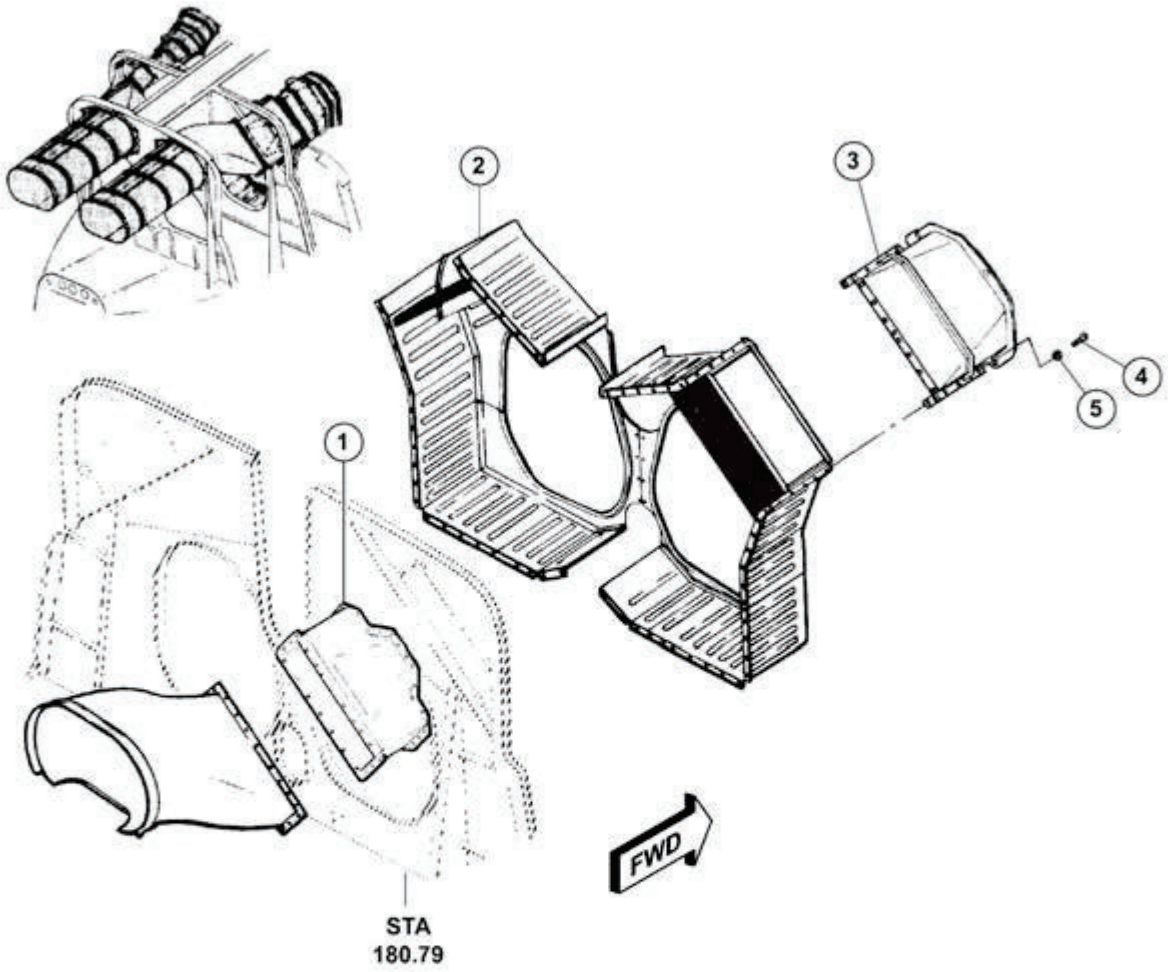
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Figure 2. Transmission Fairings and Engine Air Inlet Cowlings (Sheet 1 of 1)



ICN-412-A-716100-B-97499-00021-A-001-01

Figure 3. Air Management System Valve (Sheet 1 of 1)



1. Air management system valve assembly
2. Air management system induction baffle
3. Air management system forward duct
4. Screw
5. Thin aluminum washer

ICN-412-A-716100-B-97499-00023-A-001-01

Figure 4. Air Management System Forward Duct (Sheet 1 of 1)

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Air Management System Forward Ducts - Inspection

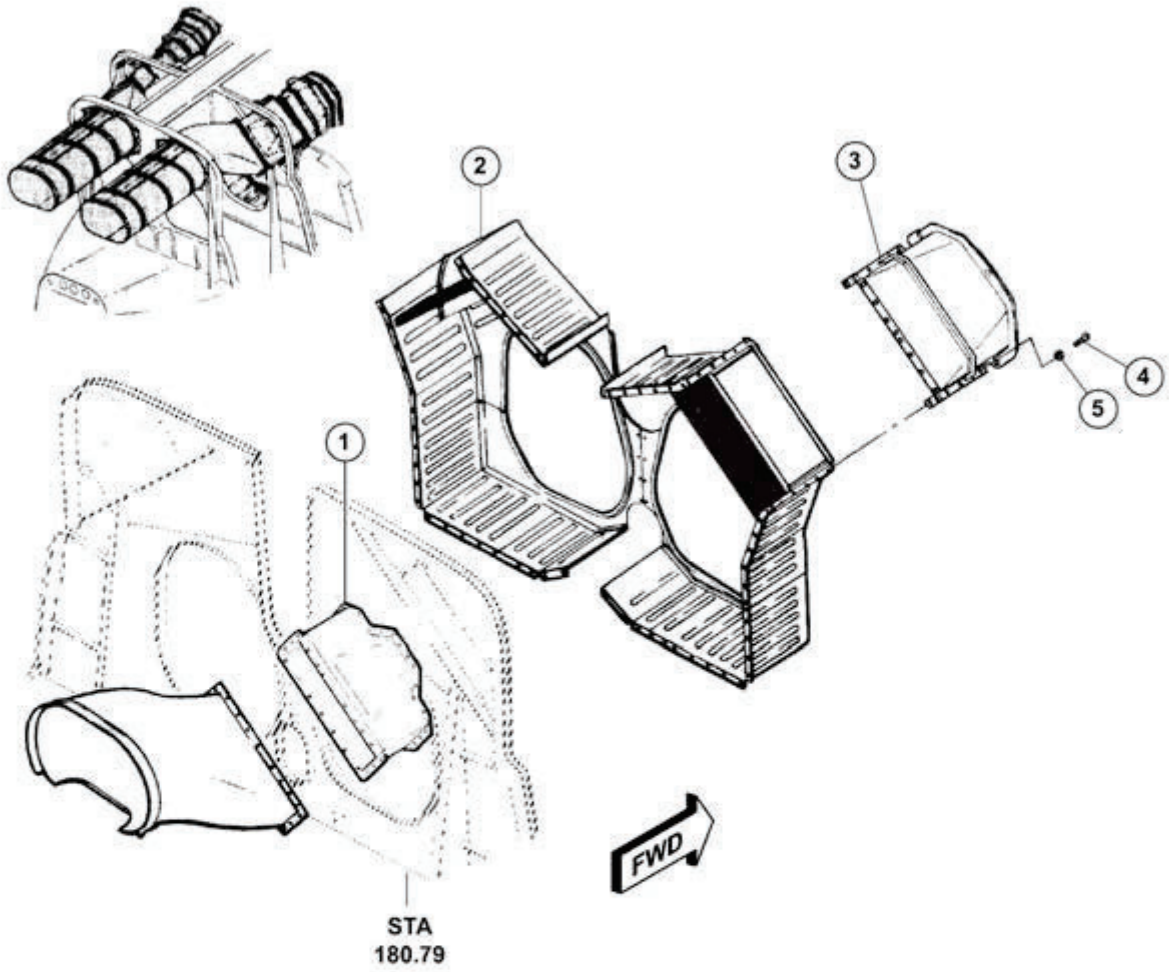
Procedure

NOTE

The inspection procedure for right forward duct is given. The inspection procedure for left forward duct is the same.

1. Inspect air management system duct (3, Figure 1) and forward transition duct (11, Figure 2) for evidence of distortion, cracks, and holes.

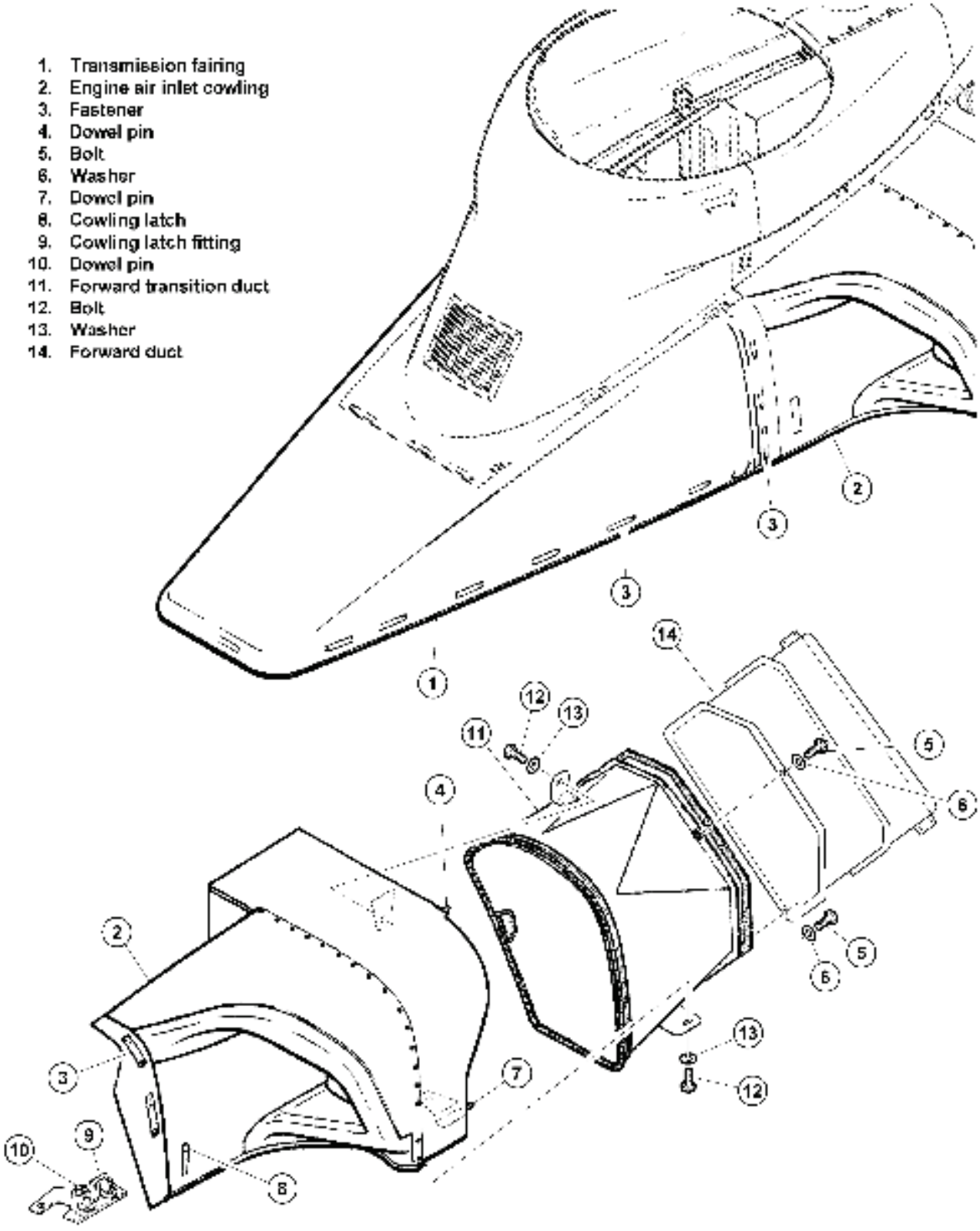
2. Inspect ducts for secure installation of attaching screws.



1. Air management system valve assembly
2. Air management system induction baffle
3. Air management system forward duct
4. Screw
5. Thin aluminum washer

ICN-412-A-716100-B-97499-00023-A-001-01

Figure 1. Air Management System Forward Duct (Sheet 1 of 1)



ICN-412-A-711000-B-97499-00004-A-001-01

Figure 2. Transmission Fairings and Engine Air Inlet Cowlings (Sheet 1 of 1)

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Air Management System Forward Ducts - Installation

Procedure

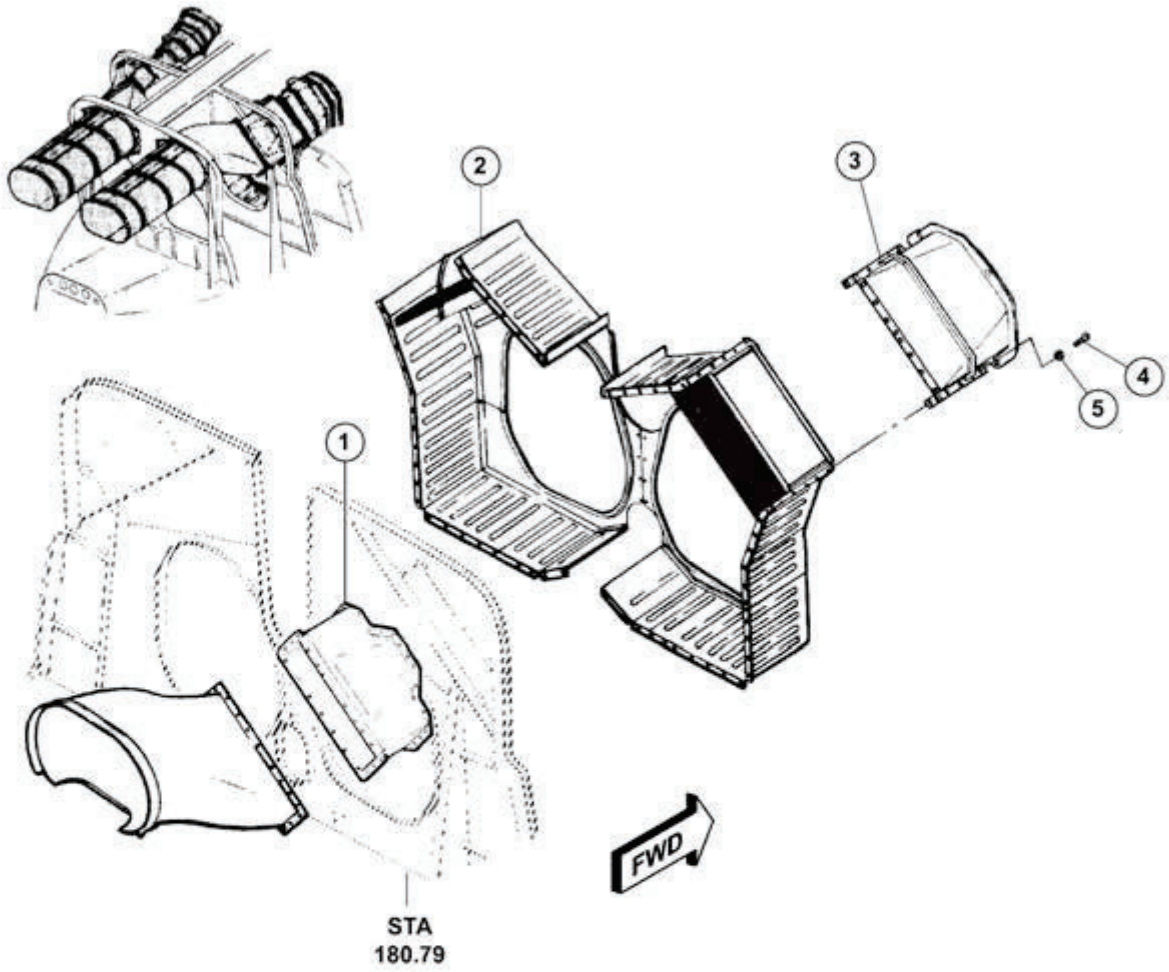
NOTE

The installation procedure for left forward duct is given. Installation procedure for right forward duct is the same.

NOTE

Use of beeswax, paste-type drill and tap lubricant, or bar soap on bolts and screws will reduce damage to nutplates and make installation easier.

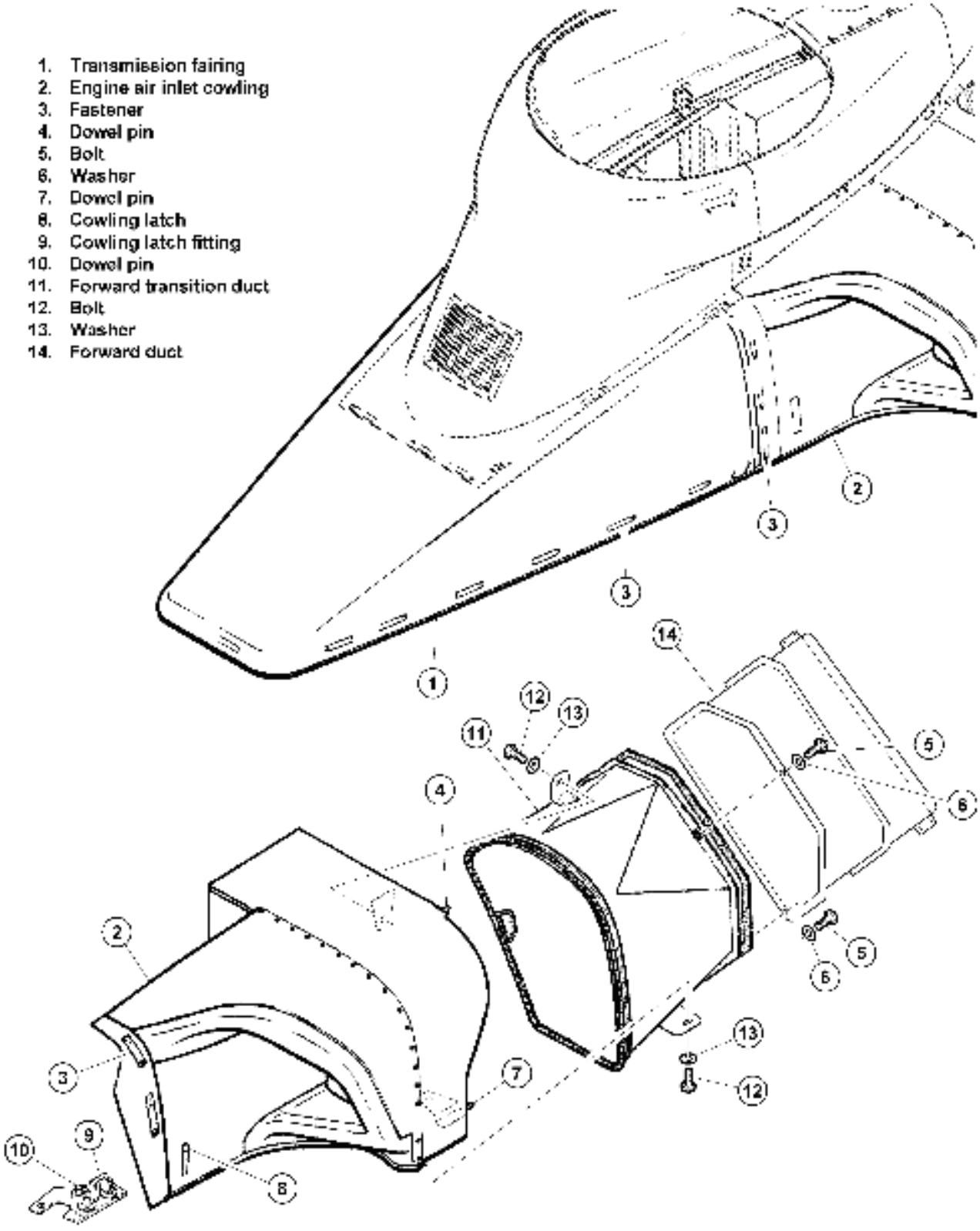
1. Put air management system forward duct (3, Figure 1) on induction baffle (2) and install screws (4) and thin aluminum washers (5).
2. Install valve assembly (1) (412-A-71-61-00-03A-720A-A / 01712).
3. Install the forward transition duct (11, Figure 2) into the engine air inlet cowling (2). Make sure the seal is correctly installed in the flange sealing ring.
4. Install the two bolts (12) and washers (13) to attach the forward transition duct (11) to the engine air inlet cowling (2).
5. Install the left-hand or right-hand engine air inlet cowling, as applicable (412-A-71-10-00-02A-720A-A / 01611).
6. Install transmission fairing (412-A-71-10-00-02A-720A-A / 01611).
7. Install the forward and aft pylon fairings (412-A-71-10-00-01A-720A-A / 01607).



1. Air management system valve assembly
2. Air management system induction baffle
3. Air management system forward duct
4. Screw
5. Thin aluminum washer

ICN-412-A-716100-B-97499-00023-A-001-01

Figure 1. Air Management System Forward Duct (Sheet 1 of 1)



ICN-412-A-711000-B-97499-00004-A-001-01

Figure 2. Transmission Fairings and Engine Air Inlet Cowlings (Sheet 1 of 1)

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Air Management System Induction Baffles - Removal

Procedure

1.

NOTE

Induction baffles are usually removed with the middle firewall and centerline firewall in conjunction with engine change.

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Air Management System Induction Baffles - Installation

Procedure

1.

NOTE

Induction baffles are usually installed with the centerline firewall and middle firewall in conjunction with engine change.

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Air Management System Exhaust Ducts - Removal

Procedure

NOTE

The removal procedure for the right exhaust duct (1, Figure 1) is given. The removal procedure for the left exhaust duct is the same.

1. Remove the right transition duct (2) (412-A-71-61-00-01A-520A-A / 01701).

2. Remove the nuts (3), bolts (4), and washers (5).

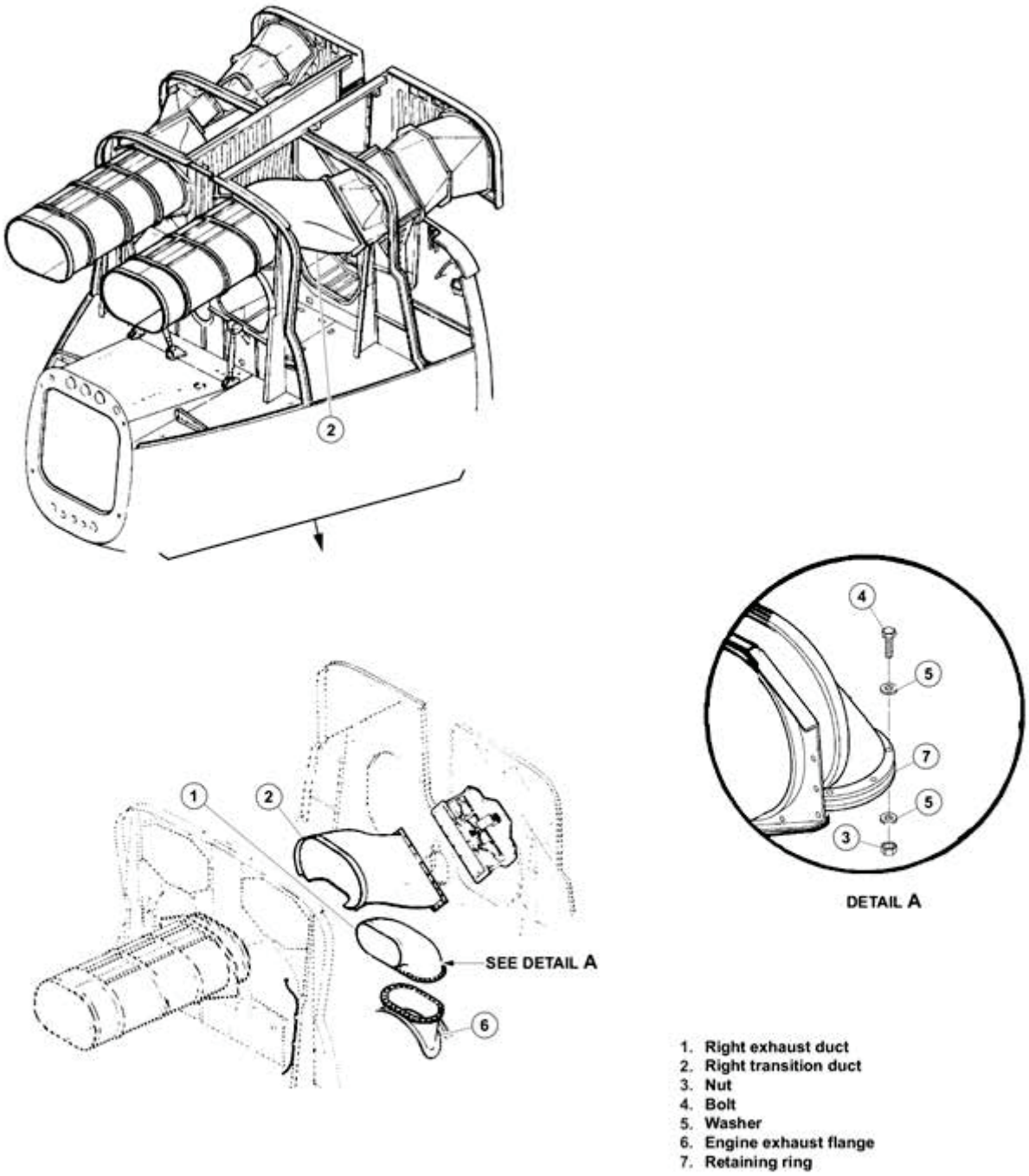
NOTE

Some exhaust ducts have 14 mounting holes.

3. Remove 14 nuts (3), 14 bolts (4), and 28 washers (5).

4. If the exhaust duct part number AAL-061-301-009 or -010 is installed, remove the retainer ring (7).

5. Remove the right exhaust duct (1).



ICN-412-A-716100-B-97499-00024-A-001-01

Figure 1. Air Management System Exhaust Duct (Sheet 1 of 1)

Air Management System Exhaust Dusts - Inspection

Procedure

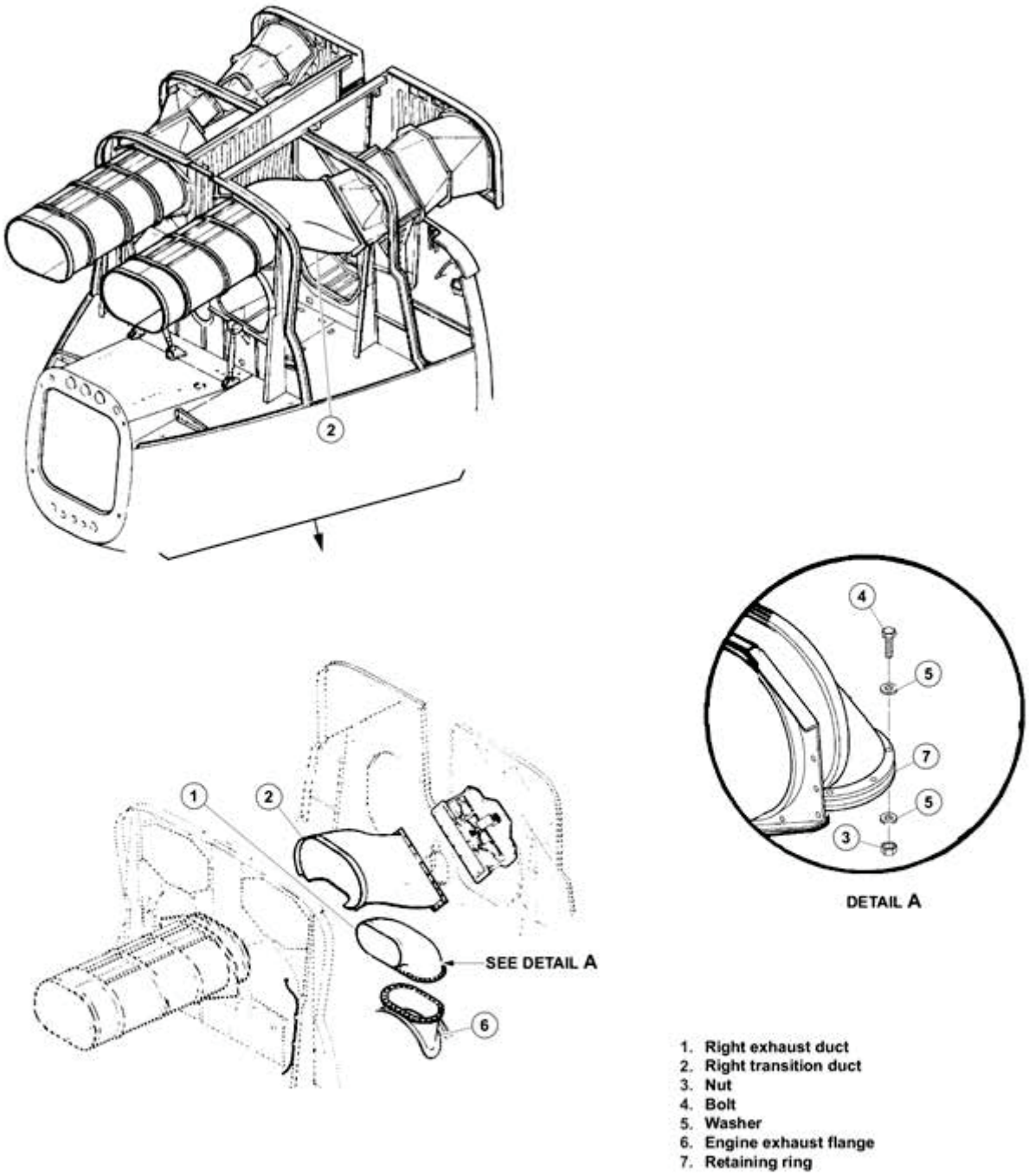
NOTE

The inspection procedure is for the right exhaust duct (1, Figure 1). Inspection procedure for left exhaust duct is the same.

1. Examine right exhaust duct (1) for distortion, cracks, and holes.

2. Examine right exhaust duct for secure installation of nuts (3) and bolts (4).

3. Examine duct for secure installation with engine exhaust flange (6).



ICN-412-A-716100-B-97499-00024-A-001-01

Figure 1. Air Management System Exhaust Duct (Sheet 1 of 1)

Air Management System Exhaust Ducts - Repair

Procedure

CAUTION

ANY REPAIRS REQUIRING WELDING SHALL BE CARRIED OUT BY COMPETENT CERTIFIED WELDERS USING PROPER EQUIPMENT.

1. If flange (1, Figure 1) is cracked, fabricate and install a doubler (2) from material described in Step 4., provided the following conditions are met:

1.1. If flange is cracked between bolt holes or from bolt holes to outboard edge of flange, cracks shall be welded and ground smooth. Use Tungsten Inert Gas (TIG) fusion method of welding with argon shielding.

1.2. Before installing doubler, check flange to ensure a flat, waveless, and clean surface.

1.3. After complying with Step 1.1. and Step 1.2., position doubler on flange and weld in place on three sides (Figure 2).

1.4. No more than four bolt holes are to be reinforced per single repair.

1.5. A total of two repairs may be made to a flange provided repairs are not closer than 3.0 inches (76.2 mm).

1.6. The engine exhaust flange (6, Figure 2) shall not be modified or redrilled to a larger size to accommodate additional bolts or larger diameter bolts.

CAUTION

A GAP IN EXCESS OF 0.030 INCH (0.762 MM) BETWEEN EXHAUST DUCT FLANGE AND ENGINE EXHAUST FLANGE MAY CAUSE NEW REPAIRS TO CRACK WHEN MOUNTING BOLTS ARE TIGHTENED.

1.7. After flange of right exhaust duct (1) is repaired per Step 1.1. through Step 1.6., make a careful check to ensure entire flange is flat and waveless. Ensure flange will fit engine exhaust flange within 0.030 inch (0.762 mm).

1.8. If a total of 25% or more of flange area requires repair, exhaust duct is not repairable. Replace with a serviceable exhaust duct.

2. If duct half (3 or 4, Figure 1) is cracked or damaged, repair is permitted provided the following conditions are met:

2.1. Not more than two repairs are permitted per duct half.

2.2. Use TIG fusion welding process with argon shielding to repair cracks 4.0 inches (101.6 mm) or less in length.

2.3. Grind and blend welded cracks to normal contour.

2.4. Fabricate a doubler (5) of material described in Step 4., formed to fit and overlap underlying weld area by 1.0 inch (25.4 mm) in all directions. Place doubler over weld area. Use TIG fusion welding method with argon shielding to attach doubler on all four sides to duct half.

2.5. Duct halves shall not be replaced. If exhaust duct is damaged to the extent replacement of duct half is indicated, exhaust duct is not repairable. Scrap damaged exhaust duct and replace with a serviceable exhaust duct.

2.6. Exhaust ducts with bulges, dents, or other damage (whether from repairs or abuse) are not airworthy. Scrap damaged exhaust duct and replace with a serviceable exhaust duct.

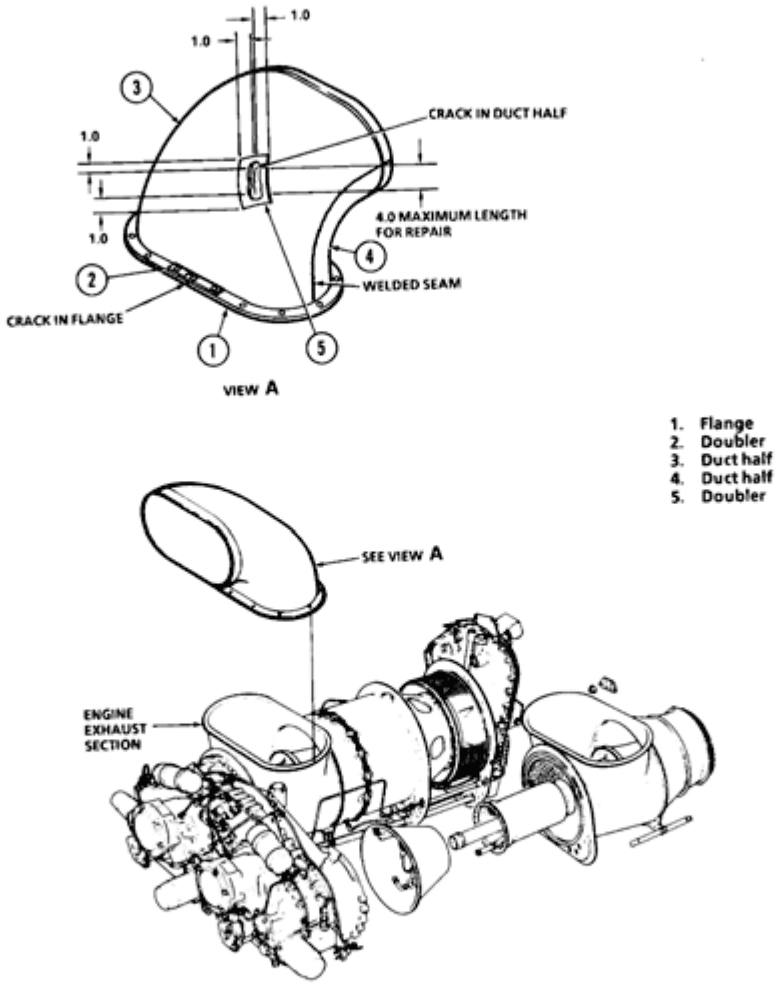
3. It is permissible to modify the 14-bolt-hole flange of exhaust duct to 28-bolt-hole flange by drilling fourteen 0.276 to 0.286 inch (7.01 to 7.26 mm) holes to match hole pattern, provided the following conditions are met:

3.1. No previous repair to flange and none required at time of modification.

3.2. Exhaust duct shall be free of cracks and repairs before and after modification. With any exception, exhaust duct is not airworthy. Scrap exhaust duct if cracked or if repair is required after modification.

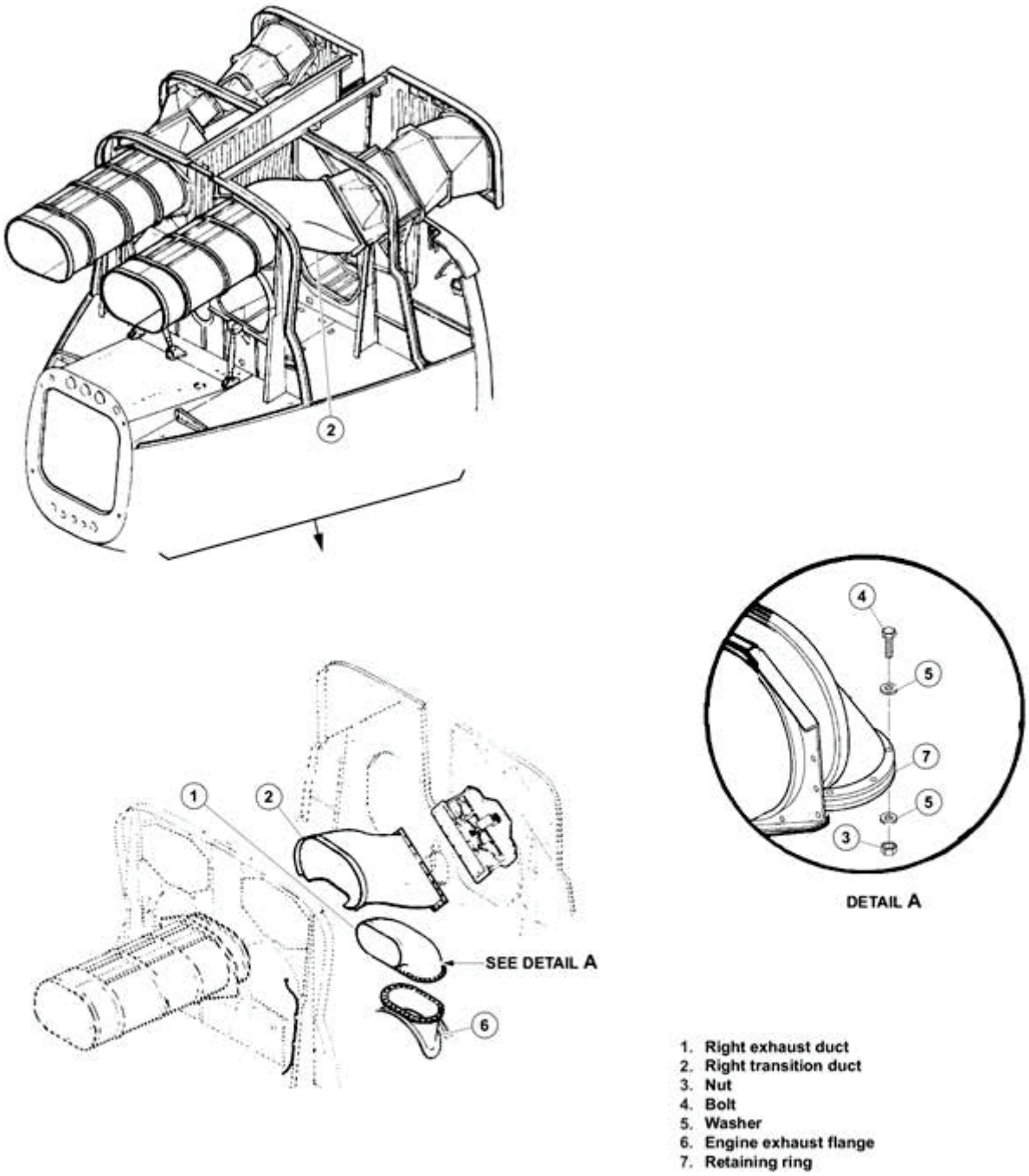
4. If bolt holes in flange of exhaust duct are worn, elongated, or distorted, inspect engine exhaust flange (6, Figure 2) for worn, elongated, or distorted bolt holes, cracks, and other damage. Refer to Pratt & Whitney Engine Customer Support Representative for evaluation of engine exhaust section.

5. Make doublers from corrosion resistant steel N-155 per AMS 5532. Hastalloy X material per AMS 5536 is an acceptable alternate. Material thickness varies from 0.025, 0.032, 0.040, to 0.050 inch. Use Hastalloy X filler rod per AMS 5798.



ICN-412-A-716100-B-97499-00025-A-001-01

Figure 1. Air Management System Exhaust Duct - Repair (Sheet 1 of 1)



ICN-412-A-716100-B-97499-00024-A-001-01

Figure 2. Air Management System Exhaust Duct (Sheet 1 of 1)

Air Management System Exhaust Ducts - Installation

Procedure

NOTE

The installation procedure for the right exhaust duct (1, Figure 1) is given. The installation procedure for left exhaust duct is the same.

1. Put the right exhaust duct (1) on the engine exhaust flange (6).

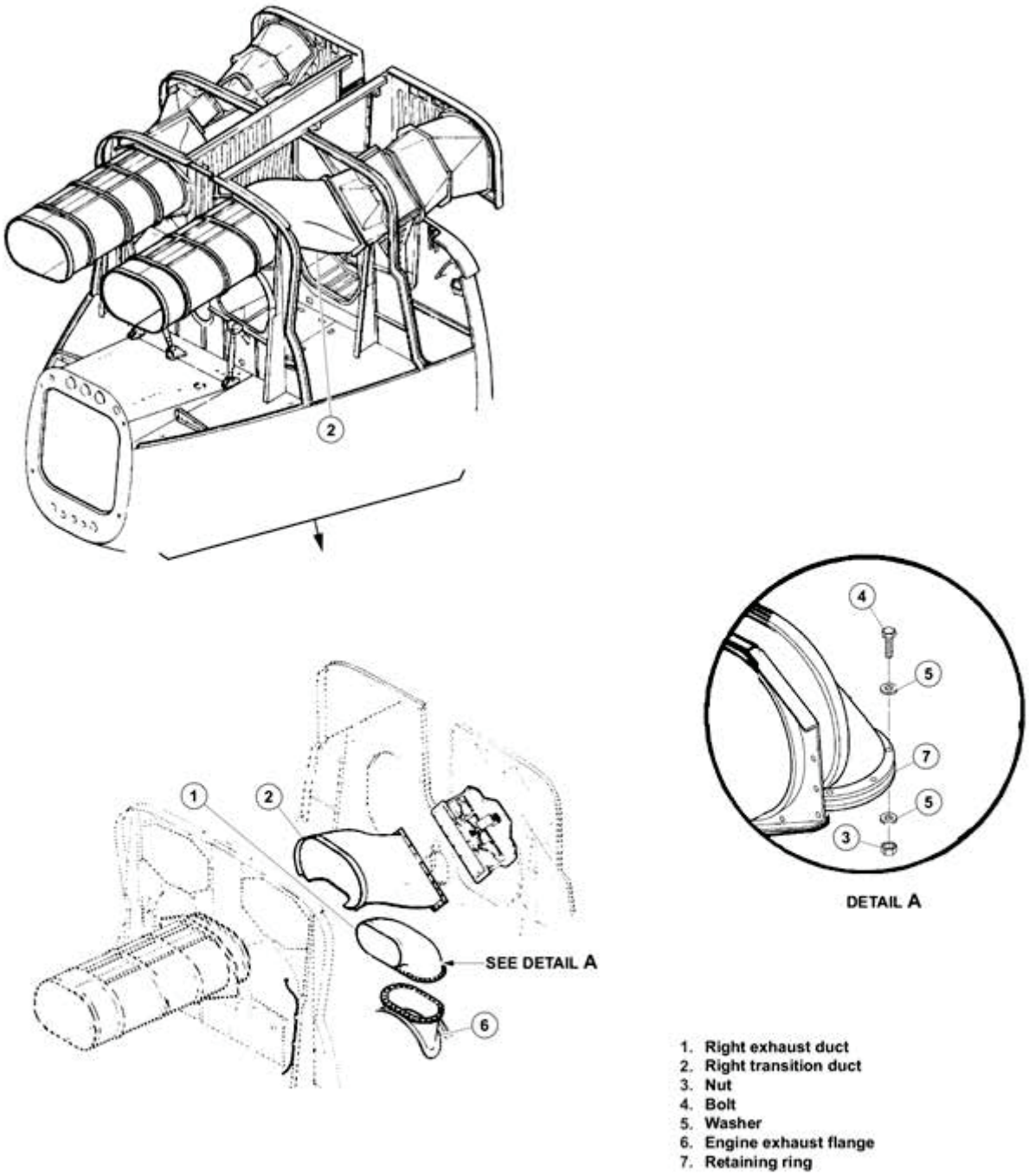
2. If the exhaust duct part number AAL-061-301-009 or -010 is installed, install the retainer ring (7).

3. Install the nuts (3), bolts (4), and washers (5).

NOTE

Some exhaust ducts have 14 mounting holes. Install 14 nuts (3), 14 bolts (4), and 28 washers (5).

4. Install the right transition duct (2) (412-A-71-61-00-02A-720A-A / 01708).



ICN-412-A-716100-B-97499-00024-A-001-01

Figure 1. Air Management System Exhaust Duct (Sheet 1 of 1)

CHAPTER 76
ENGINE CONTROLS

MAINTENANCE MANUAL

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List of Effective Data Modules

The listed documents are included in issue 007, dated 2025-11-18, of this publication.

C = Changed data module, N = New data module

Document title	Data module code / Seq number		Issue date	Applicable to
Engine Controls - Description	DMC-412-A-76-00-00-00A-028A-A / 01723	C	2025-11-18	ALL
Engine Power Controls - Description	DMC-412-A-76-10-00-00A-028A-A / 01724		2023-05-31	ALL
Power Lever (N1) Control - Rigging	DMC-412-A-76-10-00-00A-276A-A / 01725		2023-05-31	ALL
Engine Temperature Compensator (ITT Trim Values) - Verifications	DMC-412-A-76-10-00-00A-280A-A / 01726		2023-05-31	ALL
Engine Torque Transmitter - Adjustment and Functional Check	DMC-412-A-76-10-00-00A-271A-A / 01727	C	2025-11-18	SN: 33001-Subsequent or 36001-36019 or 412-570-001-103: Not Installed or (BHT-412-SI-74: Not Installed BHT-412-SI-97: Not Installed)
Engine Torque Transmitter - Adjustment and Functional Check	DMC-412-A-76-10-00-00A-271B-A / 01728	C	2025-11-18	SN: 36020-Subsequent or 412-570-001-103: Installed or BHT-412-SI-74: Installed or BHT-412-SI-97: Installed
Maximum Torque Check and Control Limit Setting - Checking	DMC-412-A-76-10-00-00A-280B-A / 01729		2023-05-31	ALL
Torque Limit - Adjustment	DMC-412-A-76-10-00-00A-271C-A / 01730	C	2025-11-18	ALL
Torque Limit - Adjustment - Alternate Method	DMC-412-A-76-10-00-00A-271D-A / 01731		2023-05-31	ALL
Maximum Torque Check and Control Limit Setting - Checking	DMC-412-A-76-10-00-00A-280C-A / 01732		2023-05-31	SN: 33001-33213 or 36001-36019
Automatic Fuel Control Idle Speed - Check	DMC-412-A-76-10-00-00A-280D-A / 01733		2023-05-31	ALL
Power Section Acceleration and Deceleration - Check	DMC-412-A-76-10-00-00A-280E-A / 01734		2023-05-31	ALL
PT6T-3B/BE Engine - Check and Adjustment	DMC-412-A-76-10-00-00A-280F-A / 01735		2023-05-31	ALL
PT6T-3B/BE Engine AFCU Maximum N1 Stop - Adjustment - Alternate Procedure	DMC-412-A-76-10-00-00A-271E-A / 01736		2023-05-31	ALL

Document title	Data module code / Seq number		Issue date	Applicable to
PT6T-3D/DE/DF Engine - Check and Adjustment	DMC-412-A-76-10-00-00A-280G-A / 01737		2023-05-31	ALL
Engine Idle Speed - Adjustment	DMC-412-A-76-10-00-00A-271F-A / 01738		2023-05-31	ALL
Droop Compensator - Description	DMC-412-A-76-11-00-00A-028A-A / 01740	C	2025-11-18	ALL
Linear Actuator - Removal	DMC-412-A-76-11-00-01A-520A-A / 01741		2023-05-31	ALL
Linear Actuator - Inspection	DMC-412-A-76-11-00-01A-280A-A / 01742		2023-05-31	ALL
Linear Actuator - Installation and Rigging	DMC-412-A-76-11-00-01A-720A-A / 01743		2023-05-31	ALL
ITT Actuator (ENG 2 Actuator) - Removal	DMC-412-A-76-11-00-02A-520A-A / 01744		2023-05-31	ALL
ITT Actuator (ENG 2 Actuator) - Inspection	DMC-412-A-76-11-00-02A-280A-A / 01745		2023-05-31	ALL
ITT Actuator (ENG 2 Actuator) - Installation and Rigging	DMC-412-A-76-11-00-02A-720A-A / 01746		2023-05-31	ALL
Droop Compensator Cambox - Removal	DMC-412-A-76-11-00-03A-520A-A / 01747		2023-05-31	SN: 33001-33107
Droop Compensator Cambox - Inspection	DMC-412-A-76-11-00-03A-280A-A / 01748		2023-05-31	SN: 33001-33107
Droop Compensator Cambox - Installation	DMC-412-A-76-11-00-03A-720A-A / 01749		2023-05-31	SN: 33001-33107
N2 Control - Rigging	DMC-412-A-76-11-00-04A-276A-A / 01750		2023-05-31	SN: 33001-33107
N2 Control - Rigging	DMC-412-A-76-11-00-04A-276B-A / 01751		2023-05-31	SN: 33001-33107 TB 412-93-118: Post
Droop Compensator and Cable - Removal	DMC-412-A-76-11-00-05A-520A-A / 01752		2023-05-31	SN: 33108-33213 or 36001-36096
Droop Compensator and Cable - Inspection	DMC-412-A-76-11-00-05A-280A-A / 01753		2023-05-31	SN: 33108-33213 or 36001-36096
Droop Compensator and Cable - Installation	DMC-412-A-76-11-00-05A-720A-A / 01754		2023-05-31	SN: 33108-33213 or 36001-36096
N2 Control - Rigging	DMC-412-A-76-11-00-06A-276A-A / 01755		2023-05-31	SN: 33108-33213 or 36001-36019

Document title	Data module code / Seq number		Issue date	Applicable to
N2 Control - Rigging	DMC-412-A-76-11-00-06A-276B-A / 01756	C	2025-11-18	SN: 36020-36096 or 412-570-001-103: Installed or BHT-412-SI-74: Installed or BHT-412-SI-97: Installed or (SN: 33108-33213 or 36001-36019 TB 412-93-118: Post)
Droop Compensator - Removal	DMC-412-A-76-11-00-07A-520A-A / 01757		2023-05-31	SN: 36097-Subsequent TB 412-94-122: Post
Droop Compensator - Inspection	DMC-412-A-76-11-00-07A-280A-A / 01758		2023-05-31	SN: 36097-Subsequent TB 412-94-122: Post
Droop Compensator - Installation	DMC-412-A-76-11-00-07A-720A-A / 01759	C	2025-11-18	SN: 36097-Subsequent or TB 412-94-122: Post
Droop Compensator Controls - Removal	DMC-412-A-76-11-00-08A-520A-A / 01760		2023-05-31	ALL
Droop Compensator Controls - Inspection	DMC-412-A-76-11-00-08A-280A-A / 01761		2023-05-31	ALL
Droop Compensator Controls - Installation	DMC-412-A-76-11-00-08A-720A-A / 01762	C	2025-11-18	ALL
N2 Control - Rigging	DMC-412-A-76-11-00-09A-276A-A / 01763		2023-05-31	SN: 33108-33213 or 36001-36019 TB 412-94-122: Post
N2 Control - Rigging	DMC-412-A-76-11-00-09A-276B-A / 01764	C	2025-11-18	SN: 36097-Subsequent or (SN: 36020-36096 TB 412-94-122: Post) or (TB 412-94-122: Post (412-570-001-103: Installed or BHT-412-SI-74: Installed or BHT-412-SI-97:

Document title	Data module code / Seq number	Issue date	Applicable to
			Installed)) or (SN: 33108-33213 or 36001-36019 TB 412-93-118: Post TB 412-94-122: Post)

Engine Controls - Description

Major components of the system are the Automatic Fuel Control Unit (AFCU) s, Manual Fuel Control Unit (MFCU) s, power turbine (N_2) governors, Torque Control Unit (TCU) , droop compensator control linkage and power lever control linkage (Figure 1 through Figure 4).

The engine control system permits the pilot to obtain maximum engine performance with minimum attention. Under normal flight conditions, the power turbine (N_2) speed is controlled by automatic fuel control units (2 and 7, Figure 1), power turbine governors (3 and 7, Figure 2 through Figure 4), and droop compensator (45).

The automatic fuel control units (2 and 7, Figure 1) safeguard engine power sections against overloading. On acceleration and deceleration, the automatic fuel control units avoid engine damage and/or flameout due to sudden power changes.

The manual fuel control units (3 and 8) can be operated in either automatic or manual mode. The pilot can select AUTO or MAN operation by means of engine GOV switches. When operating in MANUAL mode, the pilot positions lever (4 and/or 9) on manual fuel controls by means of twist grip throttles (29 and/or 30).

Depending on the helicopter serial number there are a number of different droop compensator control configurations.

Applicable: SN: 33001-33107:

(Figure 2) use control tubes to transmit collective changes to the droop compensator.

Applicable: SN: 33108-33213 or 36001-36096:

(Figure 3) use a combination control tube and control cable arrangement.

Applicable: SN: 36097-Subsequent or 33108-33213 or (SN: 36001-36096 TB 412-94-122: Post):

(Figure 4) use a control tube system similar to the earlier serial number helicopters.

Applicable: SN: 33001-33213 or 36001-36278 or 36279-Subsequent or (SN: 36020-36080 BHT-412-SI-54: Installed) or (SN: 36081-36278 TB 412-01-176: Post):

Power turbine (N_2) speed and ROTOR RPM (N_R) are adjustable by means of a linear actuator connected to the engine No. 1 (ENG 1) and engine No. 2 (ENG 2) power turbine governors through a jackshaft. The adjustment (beep) range is 97 to 101.5%. Helicopters have an expanded beep range of 97 to 103.5% for category A operation.

Applicable: SN: 33001-33213 or 36001-36019 or 412-570-001-103: Not Installed or (BHT-412-SI-74: Not Installed BHT-412-SI-97: Not Installed):

The TCU provides torque limiting and torque matching between engine 1 and engine 2.

Applicable: SN: 36020-Subsequent or 412-570-001-103: Installed or BHT-412-SI-74: Installed or BHT-412-SI-97: Installed:

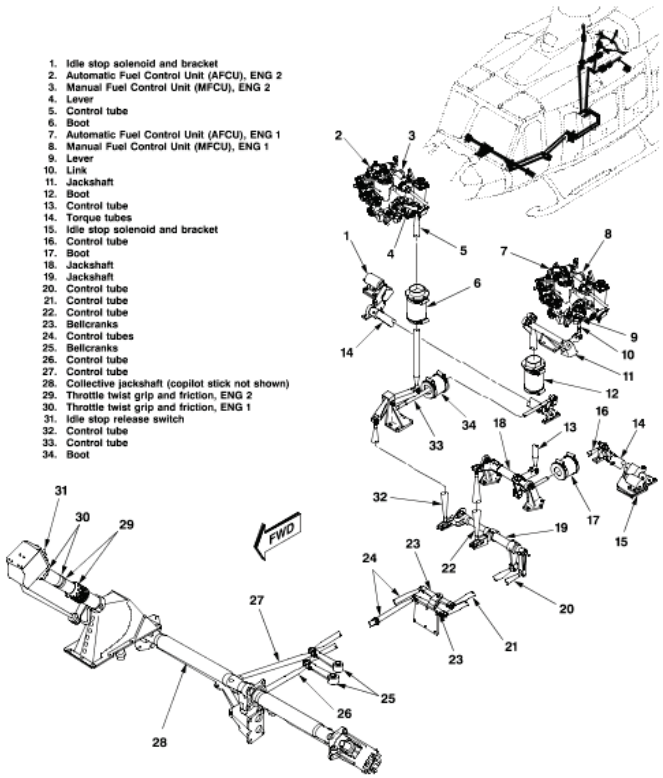
The TCU provides torque limiting function only.

An additional actuator connected to the engine No. 2 power turbine governor is used to provide either torque matching or Inter-Turbine Temperature (ITT) matching of ENG 1 and ENG 2.

Applicable: SN: 33001-33213 or (SN: 36001-36019 TB 412-93-118: Post):

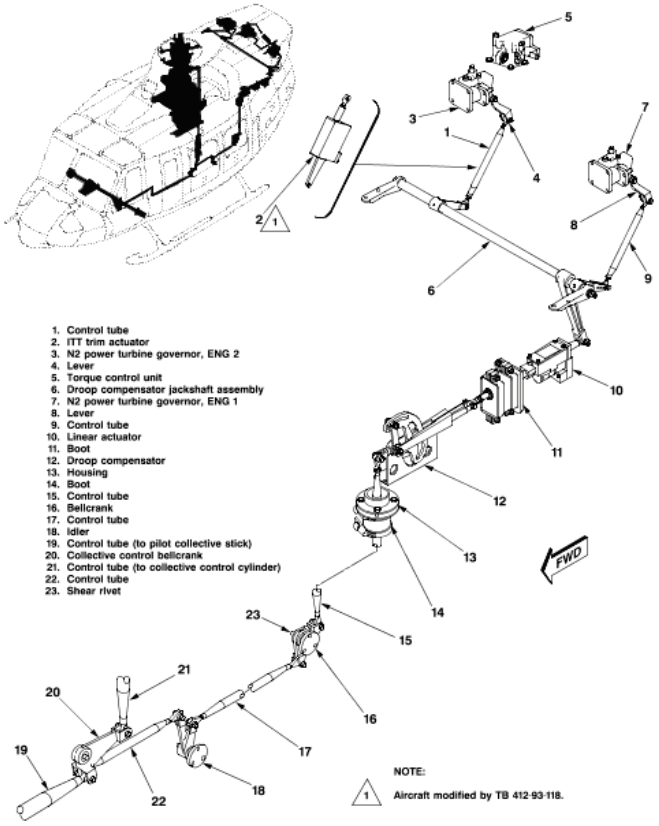
Helicopters incorporate the engine No. 2 ITT actuator.

Refer to Pratt & Whitney Canada PT6T-3 Series Maintenance Manual for detailed description of engine mounted components.



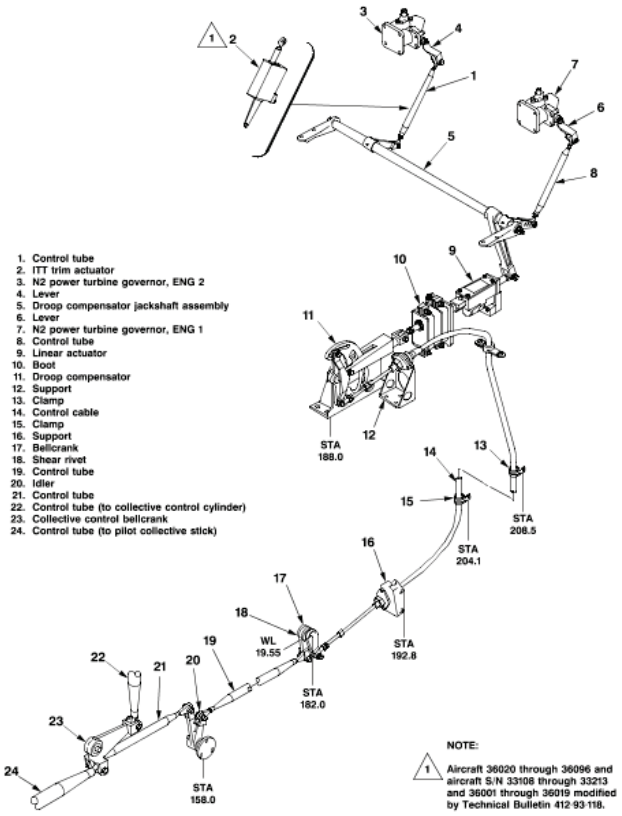
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Figure 1. Engine Power Lever Controls (Sheet 1 of 1)



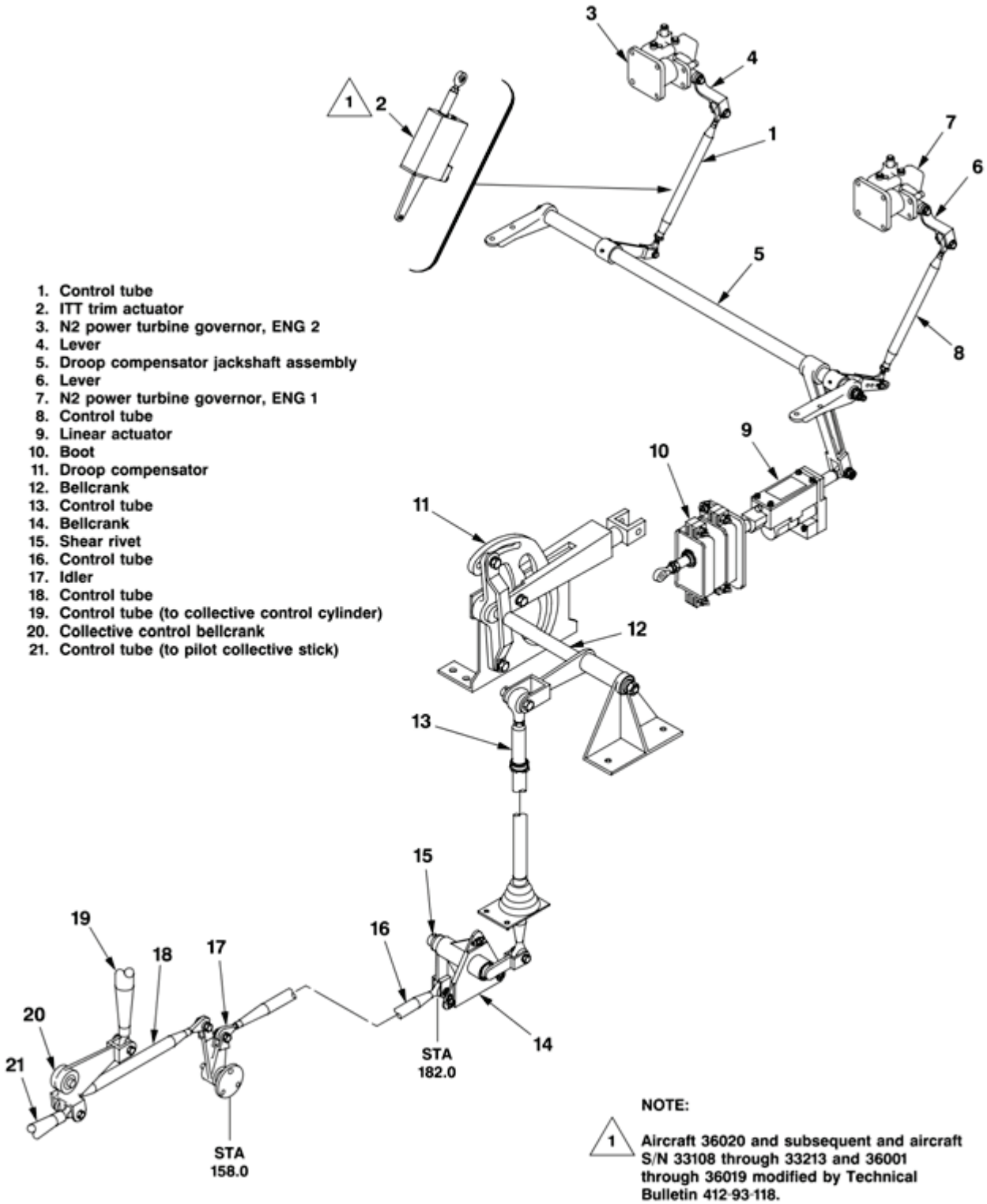
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Figure 2. Droop Compensator Linkage (S/N 33001 through 33107) (Sheet 1 of 1)



ICN-412-A-760000-B-97499-00003-A-001-01

Figure 3. Droop Compensator Linkage (S/N 33108 through 33213 and S/N 36001 through 36096) (Sheet 1 of 1)



ICN-412-A-760000-B-97499-00004-A-001-01

Figure 4. Droop Compensator Linkage (S/N 36097 and Subsequent and S/N 33108 through 33213 and 36001 through 36096 Modified by Technical Bulletin 412-94-122) (Sheet 1 of 1)

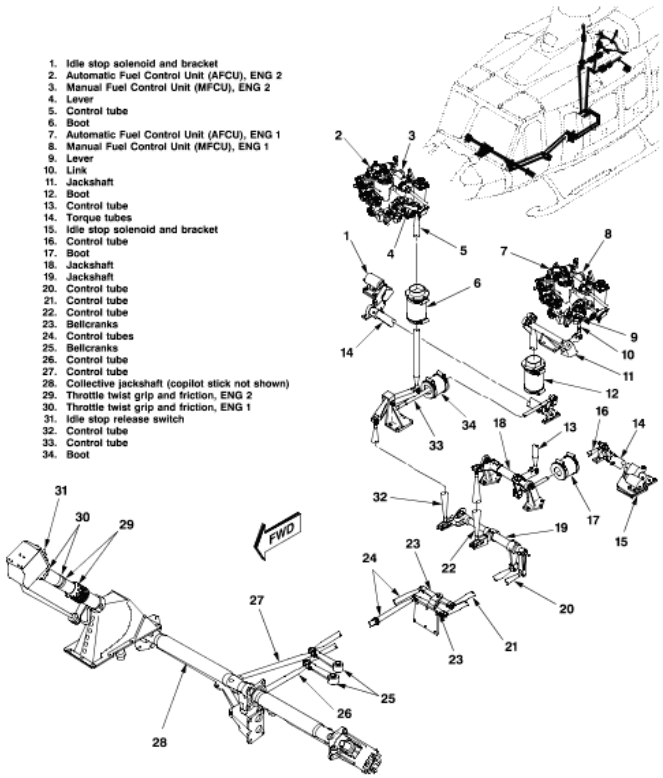
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Engine Power Controls - Description

The (N1) power lever control system consists of two parallel mechanical linkage subsystems, connecting dual throttle control twist grip (29 and 30, Figure 1) to Manual Fuel Control Units (3 and 8), on engine power sections. From the collective jackshaft, the power lever control linkage extends aft under the cabin floor between the main beams, then up inside the pylon, and through the cabin roof into the inlet fairing forward of the engine accessory gearboxes.

The power lever control linkages are actuated by throttle twist grips (29 and 30). A friction control is provided on pilots throttle only. The upper twist grip (30) marked ENG 1, controls engine 1 gas producer (GAS PROD) rpm for left (No. 1) power section. The lower twist grip (29), marked ENG 2, controls engine 2 gas producer (GAS PROD) rpm for right (No. 2) power section. Both twist grips are marked with arrows showing direction of movement to increase (INCR) rpm. When twist grip is turned right, rpm will be reduced until linkage contacts idle stop at approximately 61% GAS PROD rpm.

Idle stop solenoids (1 and 15), are operated by IDLE STOP ENG 1 and ENG 2 release switch on pilot collective switch box. When either idle stop release switch is activated, the corresponding idle stop solenoid plunger will retract allowing MFCU to be placed in cut-off position. The idle stop plunger will not retract if excessive pressure is applied by the twist grip toward the closed position.



ICN-412-A-760000-B-97499-00001-A-001-01

Figure 1. Engine Power Lever Controls (Sheet 1 of 1)

Power Lever (N1) Control - Rigging

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Corrosion preventive compound	C-101	AR	

Procedure

NOTE

If the dual controls kit is installed, refer to applicable Service Instruction (BHT-412-SI-13 or BHT-412-SI-44) for installation and removal instructions.

1. Obey the general procedures that follow when rigging power lever controls.

1.1. When a rigging procedure requires a specific position to do steps, manually hold controls in position or secure controls in position with pilot friction adjustment.

1.2. Tolerance for rigging dimensions is ±0.030 inch (0.76 mm) unless specified.

1.3. All adjustable control tubes must have a maximum of 1.00 inch (25.4 mm) of exposed threads after adjustment. If control tubes have inspection holes, threads must engage sufficiently to cover holes.

1.4. Apply corrosion preventive compound corrosion preventive compound (C-101) to exposed threads of all adjustable rod bearings and clevises after adjustment.

1.5. Torque control tube jamnuts 80 to 100 inch-pounds (9.0 to 11.3 N-m) after adjustment unless specified.

1.6. Control tubes must be free to rotate several degrees about longitudinal axis. If necessary, loosen jamnuts, adjust end fittings, and torque jamnuts.

2. Rig the collective control system (412-A-67-01-00-09A-276A-A / 01549) if not done.

3. Make sure that the engagement of the throttle flex shaft to the gear sector as follows:

3.1. Disconnect the tubes (25 and 11, Figure 1).

3.2. Disconnect the control tube (16) from the rod end bearing (17).

3.3. Disconnect the control tube (21) from the rod end bearing (22).

3.4. Inspect the marked tooth (41) on pilot engine 1 flex shaft (40) to make sure that the engagement of tenth tooth space on pilot engine 1 gear sector.

3.5. Inspect marked tooth on pilot engine 2 flex shaft (39) to make sure that the engagement of tenth tooth space on pilot engine 2 gear sector (42).

3.6. Inspect two gear sectors and flex shafts at base of copilot collective stick in same manner outlined in Step 3.4. and Step 3.5.

NOTE

If dual controls are installed, do Step 3.7., do not do Step 3.8. If dual controls are not installed, do Step 3.8., do not do Step 3.7.

3.7. If dual controls are installed, position the pilot ENG 1 (19) and ENG 2 (20) throttles to full open position against stops. Position the copilot ENG 1 and ENG 2 throttles to full open. Hold twist grips in position against the stops. Do not do Step 3.8. Go to Step 3.9.

3.8. If dual controls are not installed, position the pilot ENG 1 (19) and ENG 2 (20) throttles to full open position against stops. Hold twist grips in position against stops. Position the copilot engine 1 and engine 2 gear sectors (44 and 45) as shown in View D. Hold gear sectors in position while doing Step 3.9. and Step 3.10.

NOTE

Controls must be positioned as described in Step 3.7. or Step 3.8. as necessary, before doing Step 3.9. and Step 3.10.

3.9. Adjust control tube (21) to install double rod end bearing (22). Torque the jamnut on control tube (21) clevis. Install control tube (21) on the double rod end bearing (22).

3.10. Adjust control tube (16) to install the double rod end bearing (17). Torque the jamnut on control tube (16) clevis. Install control tube (16) on the double rod end bearing (17).

3.11. Move pilot throttle controls through the full range and make sure for binding and/or an obstruction of control movement.

3.12. Check pilot engine 2 gear sector (42) at the base of pilot collective and pilot engine 2 flex shaft (39), when engine 2 throttle (20) is moved through full range to make sure that the gear sector (42) is a minimum of one tooth of rolling off flex shaft (39) pinion.

3.13. Check pilot engine 1 flex shaft (40) pinion and the mating gear sector at the base of pilot collective is in the same manner as given in preceding step.

3.14. Check gear sectors (44 and 45) at the base of copilot collective in the same manner as given in Step 3.11. and Step 3.12.

4. Do the rigging procedure for engine 2 (ENG 2) as follows:

4.1. For ENG 2, examine the lever (3, Figure 1) for correct position in relation to the stop arm (32, view A). If lever is not aligned within 15 degrees of center line, loosen the nut (33). Position the lever (3) on the shaft of MFCU (2) within 15 degrees of center line, and tighten the nut (33). Make sure that serrations on the lever are correctly engaged with serrations on the shaft. Install a new cotter pin to secure the nut (33).

4.2. Disconnect tubes (28, 27 and 11).

4.3. Loosen clamps holding the boot (47) on the tube (28) and allow the boot to float on the tube.

4.4. Adjust tube (28) to a nominal dimension of 28.63 inches (727.20 mm). The measurement is from the center of the bearing bolt hole to the center of the other bearing bolt hole.

4.5. Adjust tube (11) to a nominal dimension to 26.94 inches (684.28 mm). The measurement is from the center of the bearing bolt hole to the center of the other bearing bolt hole.

NOTE

Refer to applicable Pratt & Whitney Aircraft of Canada Ltd. PT6T-3 Series Maintenance Manual for the MFCU clearance adjustment procedure.

4.6. With part power trim stop (12, Figure 2) in stowed position rotate lever (3, Figure 1) until the Automatic Fuel Control Unit (AFCU) (1) max stop is contacted. Check for 0.015 to 0.020 inch (0.38 to 0.51 mm) clearance between the Manual Fuel Control Unit (MFCU) stop (8, Figure 2) and the maximum stop screw (9). If necessary, adjust the AFCU/MFCU interconnect linkage as per applicable Pratt & Whitney Aircraft of Canada Ltd. PT6T-3 Series Maintenance Manual to get necessary clearance.

NOTE

During following steps a small wooden block can be used as a work aid to hold the idle stop solenoid plunger in the retracted position.

4.7. Connect tube (28, Figure 1) to the lever (3). With the idle stop solenoid (30) plunger retracted, check for the power lever travel from FULL OFF to MAX POWER. Make sure that maximum stop screw (9, Figure 2) and the minimum stop screw (6) contacts the stop (8).

4.8. Connect tube (27, Figure 1) in the center hole of the jackshaft (10) arm lever and extend the plunger of the idle stop solenoid (30) (remove wooden block).

4.9. Rotate jackshaft (10) to contact the idle solenoid stop. Make sure that the MFCU (2) pointer is aligned at approximately 37 degrees. If not, adjust tube (28) for correct alignment. Check for smooth rotation.

4.10. Connect tube (11) and hold the idle stop solenoid (30) plunger in retracted position (install wooden block).

4.11. Rotate ENG 2 throttle twist grip (20) to full open and check the overtravel (cushion). Twist grip must rotate an additional 4 to 6 degrees after maximum stop screw (9, Figure 2) contacts the stop (8). Rotate the throttle twist grip (20, Figure 1) to full closed. The twist grip must rotate an additional 4 to 6 degrees after the minimum stop screw (6, Figure 2) contacts the stop (8).

4.12. If overtravel is not equal between the full open and full closed positions, adjust the tube (11, Figure 1). Make sure that the rod end of tube (11) is visible in the witness hole. If twist grip (20) does not have sufficient travel, move the tube

(27) to next hole towards end of the arm on the jackshaft (10). If twist grip (20) has too much travel, move the tube (27) to the next hole in the arm nearer the jackshaft (10) (refer toFigure 1).

4.13. Rotate ENG 2 throttle twist grip (20) to full open and hold it with friction. Secure the boot (47) on the tube (28) with clamps.

4.14. Extend the idle stop solenoid (30) plunger (remove the wooden block). With idle stop lever (29) against the plunger, make sure that MFCU (2) pointer is aligned to approximately 37 degrees. If not, adjust the idle stop adjustment screw (1, Figure 3). If available adjustment of the idle stop adjustment screw (1) is not adequate (insufficient threads), replace the four NAS43DD3 spacers (5) under the idle stop solenoid (3) with longer or shorter lengths as necessary.

NOTE

Idle stop solenoid (30, Figure 1) is operated by IDLE STOP release switch (18) on the pilot collective stick. When IDLE STOP release switch is activated, the corresponding idle stop solenoid plunger is retracted for 5 seconds.

4.15. Move IDLE STOP release switch (18) to ENG 2 position and check clearance between the plunger (4, Figure 3) and the idle stop lever (2). Acceptable clearance is 0.003 to 0.015 inch (0.08 to 0.38 mm). If clearance is not within tolerance, adjust the position of the idle stop solenoid (3) on serrated bracket assembly (10) and base (11) to get necessary clearance.

4.16. Inspect all linkage shown on Figure 1 for correct installation of bolts, nuts, and cotter pins.

5. Do the following rigging procedure for engine 1 (ENG 1) as follows:

5.1. For ENG 1, examine the lever (6, Figure 1) for correct position in relation to the stop arm (36, view B). If the lever is not aligned within 15 degrees of centerline, loosen the nut (37). Position the lever (6) on the shaft of MFCU (5) within 15 degrees of centerline, and tighten the nut (37). Make sure that serrations on the lever are correctly engaged with serrations on the shaft. Install the cotter pin to secure nut (37).

5.2. Disconnect tubes (7, 26 and 25) and link (49).

5.3. Loosen clamps holding the boot on the tube and allow the boot to float on the tube.

5.4. Adjust tube (7) to a nominal dimension of 25.03 inch (635.76 mm). The measurement is from the center of the bearing bolt hole to the center of the other bearing bolt hole.

5.5. Adjust tube (25) to a nominal dimension of 25.68 inch (654.27 mm). The measurement is from the center of the bearing bolt hole to the center of the other bearing bolt hole.

NOTE

Refer to applicable Pratt & Whitney Aircraft of Canada Ltd. PT6T-3 Series Maintenance Manual for the MFCU clearance adjustment procedure.

5.6. With part power trim stop (12, Figure 2) in stowed position rotate the lever (6, Figure 1) until the AFCU (4) max stop is contacted. Check for 0.015 to 0.020 inch (0.38 to 0.51 mm) clearance between the MFCU stop (8, Figure 2) and the maximum stop screw (9). If necessary, adjust the AFCU/MFCU interconnect linkage as per applicable Pratt & Whitney Aircraft of Canada Ltd. PT6T-3 Series Maintenance Manual to get necessary clearance.

NOTE

During following steps a small wooden block can be used as a work aid to hold the idle stop solenoid plunger in the retracted position.

5.7. Connect tube (7, Figure 1) to the jackshaft (50) and link (49) to the lever (6). With the idle stop solenoid (9) plunger retracted, check for the power lever travel from FULL OFF to MAX POWER without fouling between the jackshaft (50) and tube (7). Make sure that the maximum stop screw (9, Figure 2) and the minimum stop screw (6) contact the stop (8). In case of fouling between the jackshaft (50, Figure 1) and tube (7), offset the lever (6) by one serration (±15 degrees) and repeat step.

5.8. Connect tube (26, Figure 1) in the center hole of the jackshaft (10) arm lever and extend the plunger of the idle stop solenoid (9) (remove wooden block).

5.9. Rotate the jackshaft (10) to contact the idle solenoid stop. Make sure that the MFCU (5) pointer is aligned at approximately 37 degrees. If not, adjust the tube (7) for correct alignment. Check for smooth rotation.

5.10. Connect tube (25) and hold the idle stop solenoid (9) plunger in retracted position (install the wooden block).

5.11. Rotate ENG 1 throttle twist grip (19) to full open and check the overtravel (cushion). Twist grip must rotate an additional 4 to 6 degrees after maximum stop screw (9, Figure 2) contacts the stop (8). Rotate the throttle twist grip (19, Figure 1) to full closed. The twist grip must rotate an additional 4 to 6 degrees after the minimum stop screw (6, Figure 2) contacts the stop (8).

5.12. If overtravel is not equal between the full open and full closed positions, adjust the tube (25, Figure 1). Make sure that the rod end of the tube (25) is visible in the witness hole. If twist grip (19) does not have sufficient travel, move the tube (26) to next hole towards end of the arm on the jackshaft (10). If twist grip (19) has too much travel, move the tube (26) to the next hole in the arm nearer the jackshaft (10) (refer to Figure 1).

5.13. Rotate ENG 1 throttle twist grip (19) to full open and hold it with friction. Secure the boot (48) on the tube (7) with clamps.

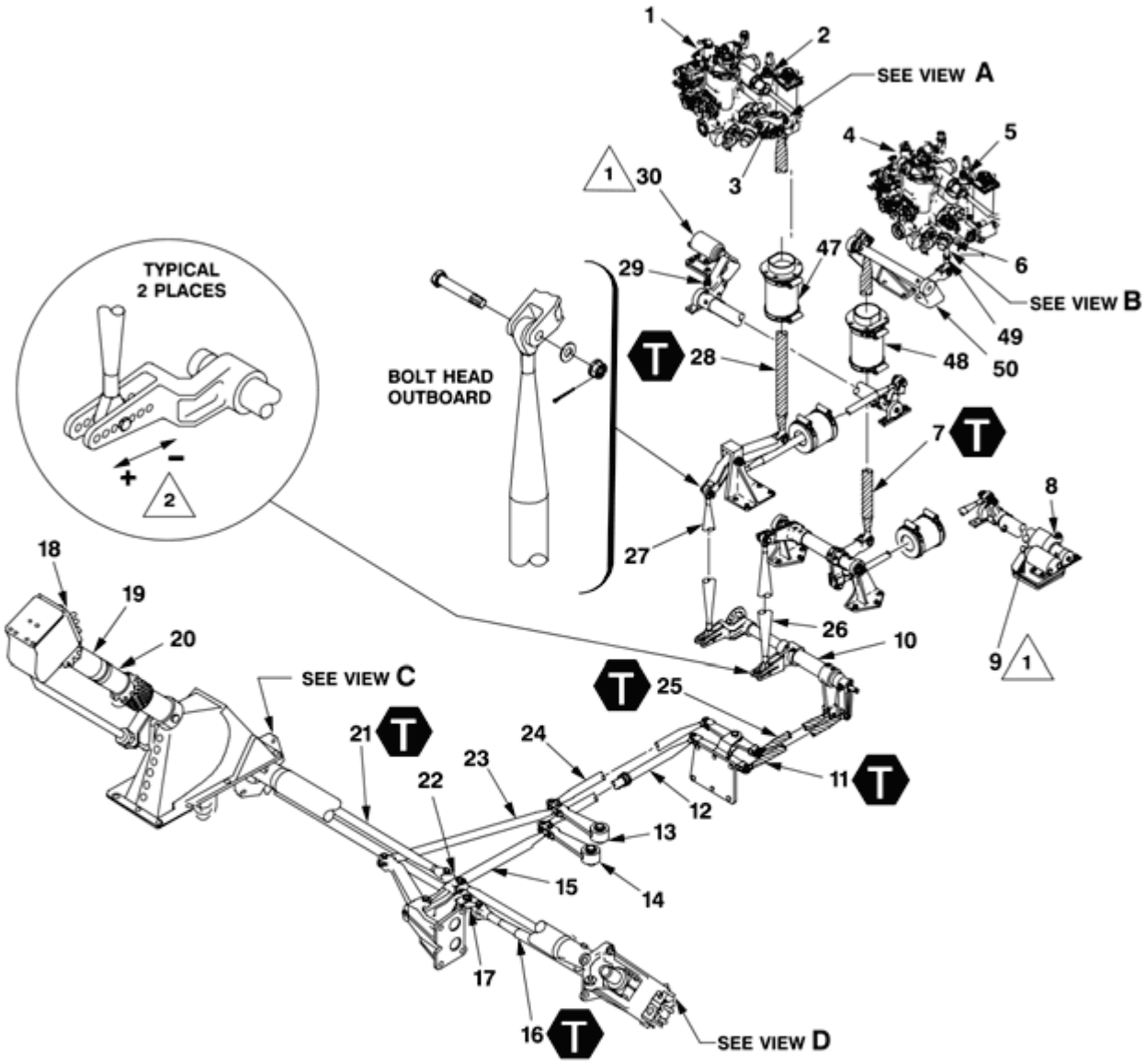
5.14. Extend the idle stop solenoid (9) plunger (remove the wooden block). With idle stop lever (8) against the plunger, make sure that MFCU (5) pointer is aligned to approximately 37 degrees. If not, adjust the idle stop adjustment screw (9, Figure 3). If available adjustment of the idle stop adjustment screw (9) is not adequate (insufficient threads), replace the four NAS43DD3 spacers (5) under the idle stop solenoid (7) with longer or shorter lengths as necessary.

NOTE

Idle stop solenoid (9, Figure 1) is operated by IDLE STOP release switch (18) on the pilot collective stick. When IDLE STOP release switch is activated, the corresponding idle stop solenoid plunger is retracted for 5 seconds.

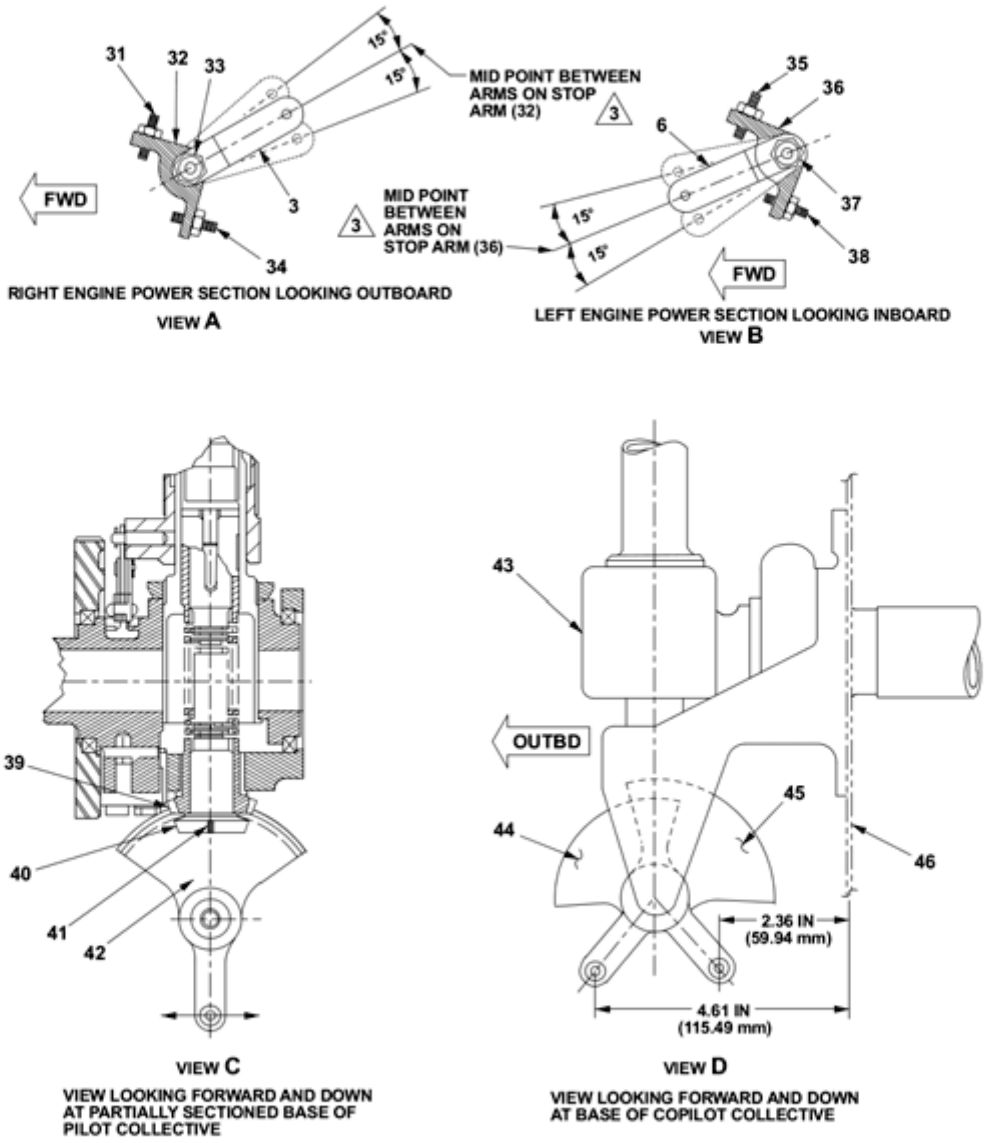
5.15. Move IDLE STOP release switch (18) to ENG 1 position and check clearance between the plunger (6, Figure 3) and the idle stop lever (8). Acceptable clearance is 0.003 to 0.015 inch (0.08 to 0.38 mm). If clearance is not within tolerance, adjust the position of the idle stop solenoid (7) on serrated bracket assembly (10) and base (11) to get necessary clearance.

5.16. Inspect all linkages shown on Figure 1 for correct installation of bolts, nuts, and cotter pins.



ICN-412-A-761000-B-97499-00001-A-001-01

Figure 1. Power Lever (N1) Controls Rigging (Sheet 1 of 3)



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Figure 1. Power Lever (N1) Controls Rigging (Sheet 2 of 3)

1. Automatic Fuel Control Unit (AFCU), ENG 2
2. Manual Fuel Control Unit (MFCU), ENG 2
3. Lever
4. Automatic Fuel Control Unit (AFCU), ENG 1
5. Manual Fuel Control Unit (MFCU), ENG 1
6. Lever
7. Control tube
8. Idle stop lever
9. Idle stop solenoid
10. Jackshaft
11. Control tube
12. Control tube
13. Bellcrank
14. Bellcrank
15. Control tube
16. Control tube
17. Double rod end bearing
18. IDLE STOP release switch
19. Throttle twist grip, ENG 1
20. Throttle twist grip, ENG 2
21. Control tube
22. Double rod end bearing
23. Control tube
24. Control tube
25. Control tube
26. Control tube
27. Control tube
28. Control tube
29. Idle stop lever
30. Idle stop solenoid
31. Minimum stop screw
32. Stop arm, manual fuel control
33. Nut
34. Maximum stop screw
35. Minimum stop screw
36. Stop arm, manual fuel control
37. Nut
38. Maximum stop screw
39. Pilot engine 2 flex shaft
40. Pilot engine 1 flex shaft
41. Marked tooth on flex shaft (40)
42. Pilot engine 2 gear sector
43. Copilot collective elbow and support
44. Copilot engine 1 gear sector
45. Copilot engine 2 gear sector
46. Structure
47. Boot
48. Boot
49. Link
50. Jackshaft

(T) 80 TO 100 IN-LBS
(9.0 TO 11.3 Nm)

NOTES

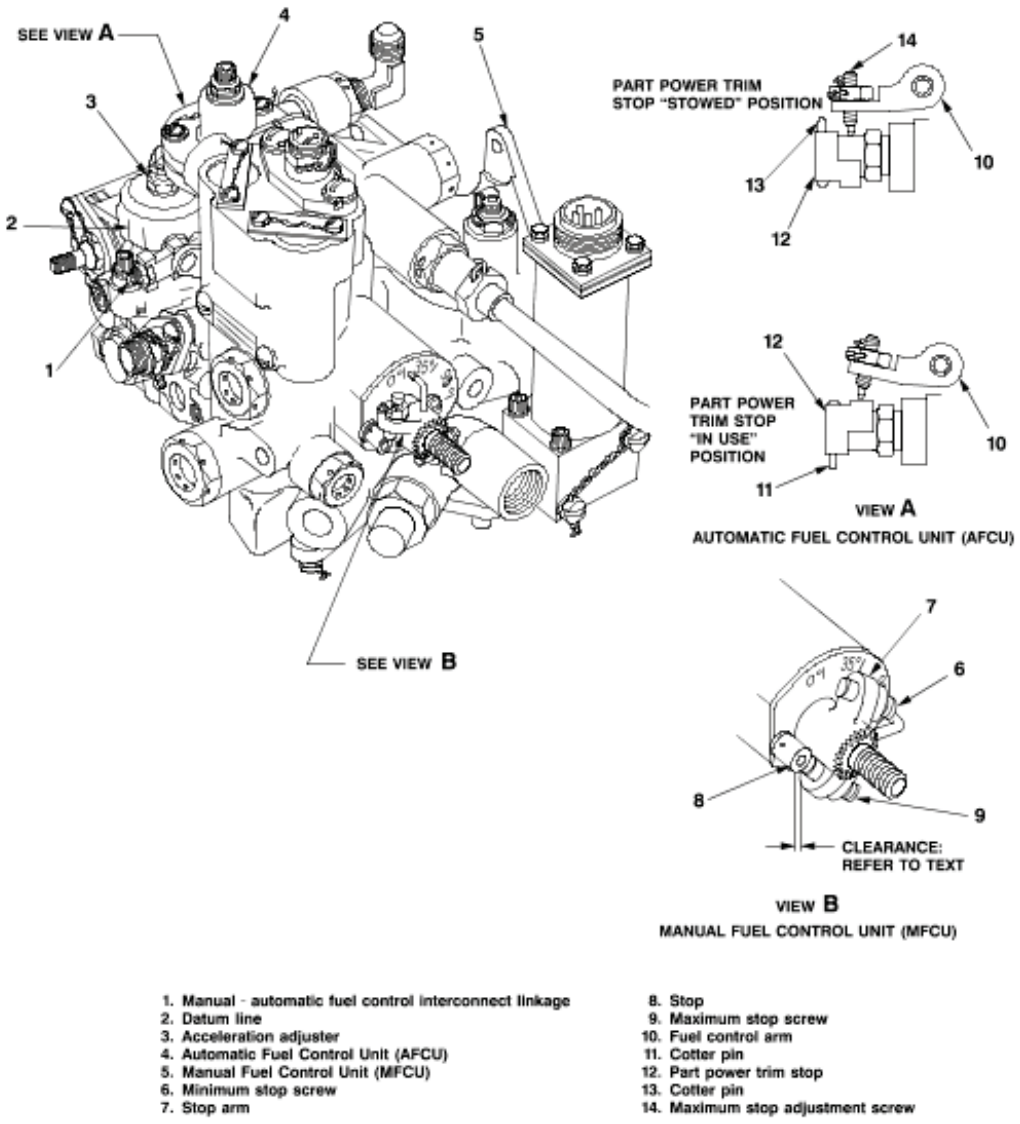
1. During rigging procedure, a small wooden block may be placed between the shoulder of the idle stop plunger and flange of the idle stop solenoid mounting bracket to assist in holding the idle stop plunger in the retract position.

2. To increase travel, reposition rod end in (+) direction, to decrease travel, reposition rod end in (-) direction.

3. Mid point shown as reference only. Mid position of lever (3 and 6) may not be achievable in reference to stop arms (32 and 36).

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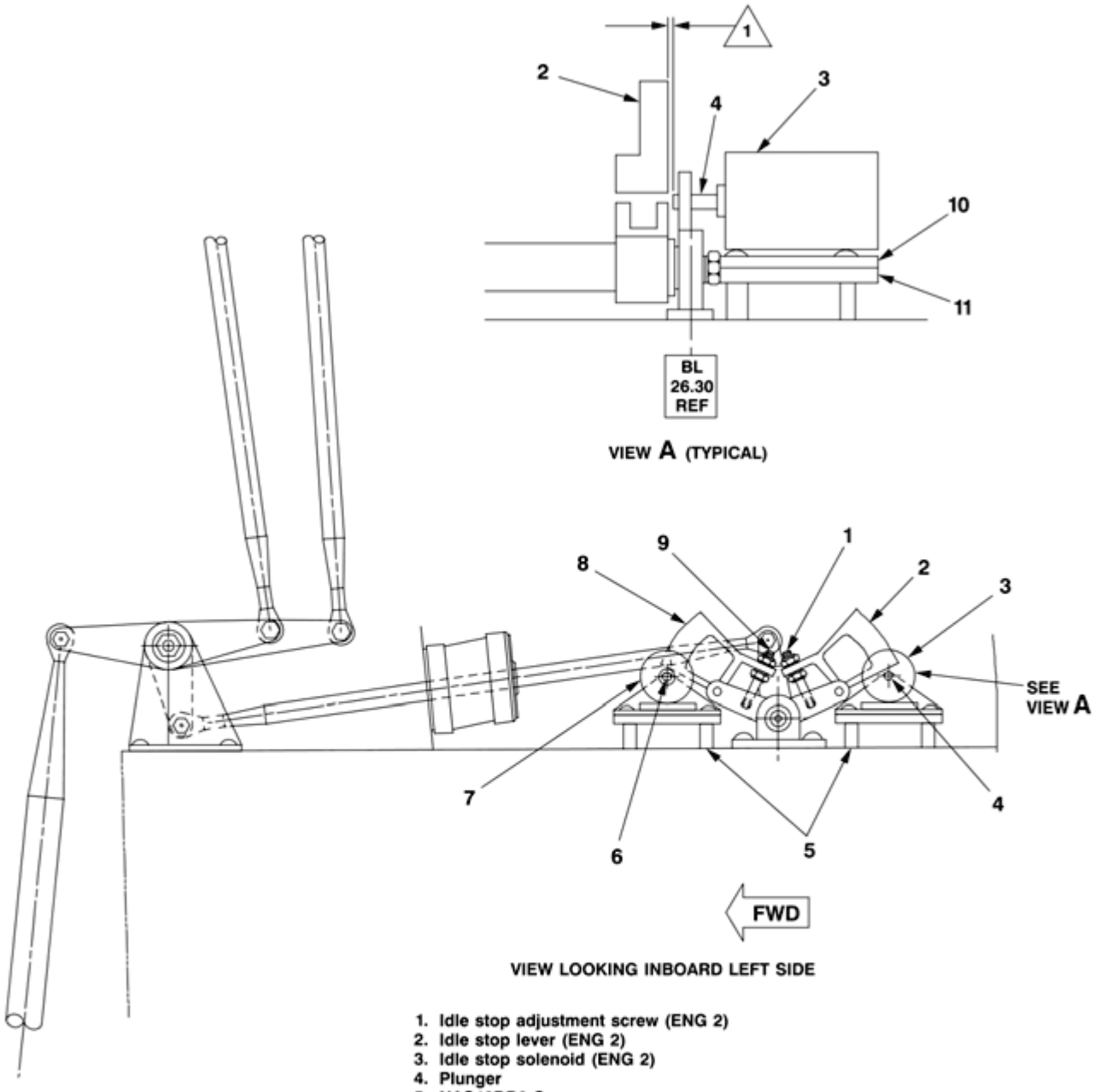
Figure 1. Power Lever (N1) Controls Rigging (Sheet 3 of 3)



1. Manual - automatic fuel control interconnect linkage
2. Datum line
3. Acceleration adjuster
4. Automatic Fuel Control Unit (AFCU)
5. Manual Fuel Control Unit (MFCU)
6. Minimum stop screw
7. Stop arm
8. Stop
9. Maximum stop screw
10. Fuel control arm
11. Cotter pin
12. Part power trim stop
13. Cotter pin
14. Maximum stop adjustment screw

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Figure 2. Manual and Automatic Fuel Control Adjustment (Sheet 1 of 1)



1. Idle stop adjustment screw (ENG 2)
2. Idle stop lever (ENG 2)
3. Idle stop solenoid (ENG 2)
4. Plunger
5. NAS43DD3 Spacers
6. Plunger
7. Idle stop solenoid (ENG 1)
8. Idle stop lever (ENG 1)
9. Idle stop adjustment screw (ENG 1)
10. Bracket assembly
11. Base

NOTE:

1 0.003 to 0.015 in. (0.08 to 0.38 mm) gap typical with plunger retracted.

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Figure 3. Engine Idle Adjustment (Sheet 1 of 1)

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Engine Temperature Compensator (ITT Trim Values) - Verifications

Procedure

NOTE

Reference in the Pratt & Whitney Aircraft of Canada Ltd. PT6T-3 Series Maintenance Manual to indicated temperature (T5) refers to ITT indicators on helicopter instrument panel.

1. Start engines and operate for 5 minutes to stabilize operating temperatures.

2. Remove top power connection from trim compensator and note indicated ITT rise by amount equal to trim value on power section data plate.

3. If not, remove lockwire and protective cap. Adjust trim compensator adjustment screw as necessary.

4. Connect top power connection to trim compensator. Indicated temperature (ITT) should drop by an amount equal to the trim value.

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Engine Torque Transmitter - Adjustment and Functional Check

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Deadweight Tester	2311F	1	or Equivalent
Pressure Gauge	302-00102	1	

Procedure

NOTE

The following procedure is applicable to the 412-075-205-101 torque transmitter with 412-075-008-101, 412-075-008-111, or 412-075-008-113 torque indicators. Refer to Figure 1 for calibration of 412-075-008-101 indicators and Figure 2 for 412-075-008-111/-113 indicators.

The following torque adjustment procedure is required when engine (Twin-Pac) or reduction gearbox is changed and/or when either torque transmitter is changed.

For additional information on engine torque indicating systems, refer to 412-A-95-61-02-00A-042A-A / 01845 and 412-A-96-12-00-02A-042A-A / 01986.

Torque transmitter adjustment procedures for left power section are given. Adjustment procedures for right power section are similar.

1. Locate data plate (1, Figure 3) on engine reduction gearbox and record torque transmitter settings for left power section (Figure 3, View B). The settings for left torque transmitter on sample data plate are:

LH valves	ABD
LH α°	87.7

2. Remove cover from left torque transmitter (5). Refer to Detail A for view of valves A, B, C, and D, with cover removed. Set valves to ON or OFF positions recorded at Step 1.. Note valves A, B, and D, are ON and valve C is OFF on sample shown.

3. Locate α° value (recorded at Step 1.) in Figure 1 for 412-075-008-101 torque indicators and Figure 2 for 412-075-008-111/-113 torque indicators. Record corresponding torque indicating reading.

4. Remove hose (2, Figure 3) and elbow from torque transmitter (5).

5. If torque transmitter (5) does not contain oil, fill transmitter with engine lubricating oil at port where hose (2) was removed.



ENSURE PRESSURE TESTER IS SERVICED WITH SAME OIL AS USED IN COMBINING GEARBOX.

PRESSURE TESTER AND GAUGE SHALL BE AT THE SAME HEIGHT AS REDUCTION GEARBOX TORQUE METER (APPROXIMATE LEVEL OF OIL COOLER BLOWER CENTERLINE).

6. Attach a deadweight tester (2311F) (or equivalent liquid pressure source) (Figure 4) to the torque transmitter (5, Figure 3) port where hose (2) was removed. Apply 35.4 ±0.25 PSIG to the transmitter with the tester. Make sure that no air is present in the line.

7. Position BATTERY BUS 1 and 2 switches ON. Put INV 1 AND INV 2 switches to ON. Record pilot (and copilot, if installed) ENG 1 TORQUE indication. Compare the indication with the value recorded at the Step 3.. If pilot TORQUE indicators are not the same as value recorded at Step 3., adjust the zero adjust the screw on the transmitter (5, Figure 3). Counterclockwise turn of the screw increases shown value. Vibrate transmitter during the calibration to make sure that the indication remains the same. After zero adjust the screw is set, apply the torque lacquer dot to screw.

8. Put INV 1 and 2 and BATTERY BUS 1 and 2 switches to OFF.

9. Disconnect and remove the tester.

10. Install the elbow and hose (2) on the transmitter (5).

11. Install the cover on transmitter (5) valves (Detail A).

12. Start the left power section (BHT-412-FM) and monitor the ENG 1 TORQUE indicator. If the indicator needle fluctuates, bleed air from hose (2) at the transmitter (5).

α° Value from torque transmitter setting instruction plate.	Corresponding torque indicator reading (in percent) for α° value at left.	α° Value from torque transmitter setting instruction plate.	Corresponding torque indicator reading (in percent) for α° value at left.
84.0	43.6%	88.0	45.6
.1	43.6	.1	45.7
.2	43.7	.2	45.7
.3	43.7	.3	45.8
.4	43.8	.4	45.9
.5	43.8	.5	45.9
.6	43.9	.6	46.0
.7	43.9	.7	46.0
.8	44.0	.8	46.1
.9	44.0	.9	46.1
85.0	44.1	89.0	46.2
.1	44.1	.1	46.2
.2	44.2	.2	46.3
.3	44.2	.3	46.3
.4	44.3	.4	46.4
.5	44.3	.5	46.4
.6	44.4	.6	46.5
.7	44.4	.7	46.5
.8	44.5	.8	46.6
.9	44.5	.9	46.6
86.0	44.6	90.0	46.7
.1	44.7	.1	46.7
.2	44.7	.2	46.8
.3	44.8	.3	46.8
.4	44.8	.4	46.9
.5	44.9	.5	46.9
.6	44.9	.6	47.0
.7	45.0	.7	47.0
.8	45.0	.8	47.1
.9	45.1	.9	47.1
87.0	45.1	91.0	47.2
.1	45.2		
.2	45.2		
.3	45.3		
.4	45.3		
.5	45.4		
.6	45.4		
.7 ◄—— SAMPLE ——►	45.5		
.8	45.5		
.9	45.6		

α° Value to Corresponding Torque Indicator Reading

ICN-412-A-761000-B-97499-00006-A-001-01

Figure 1. Torque Transmitter Calibration (412-075-008-101 Torque Indicators) (S/N 33001 through 33107) (Sheet 1 of 1)

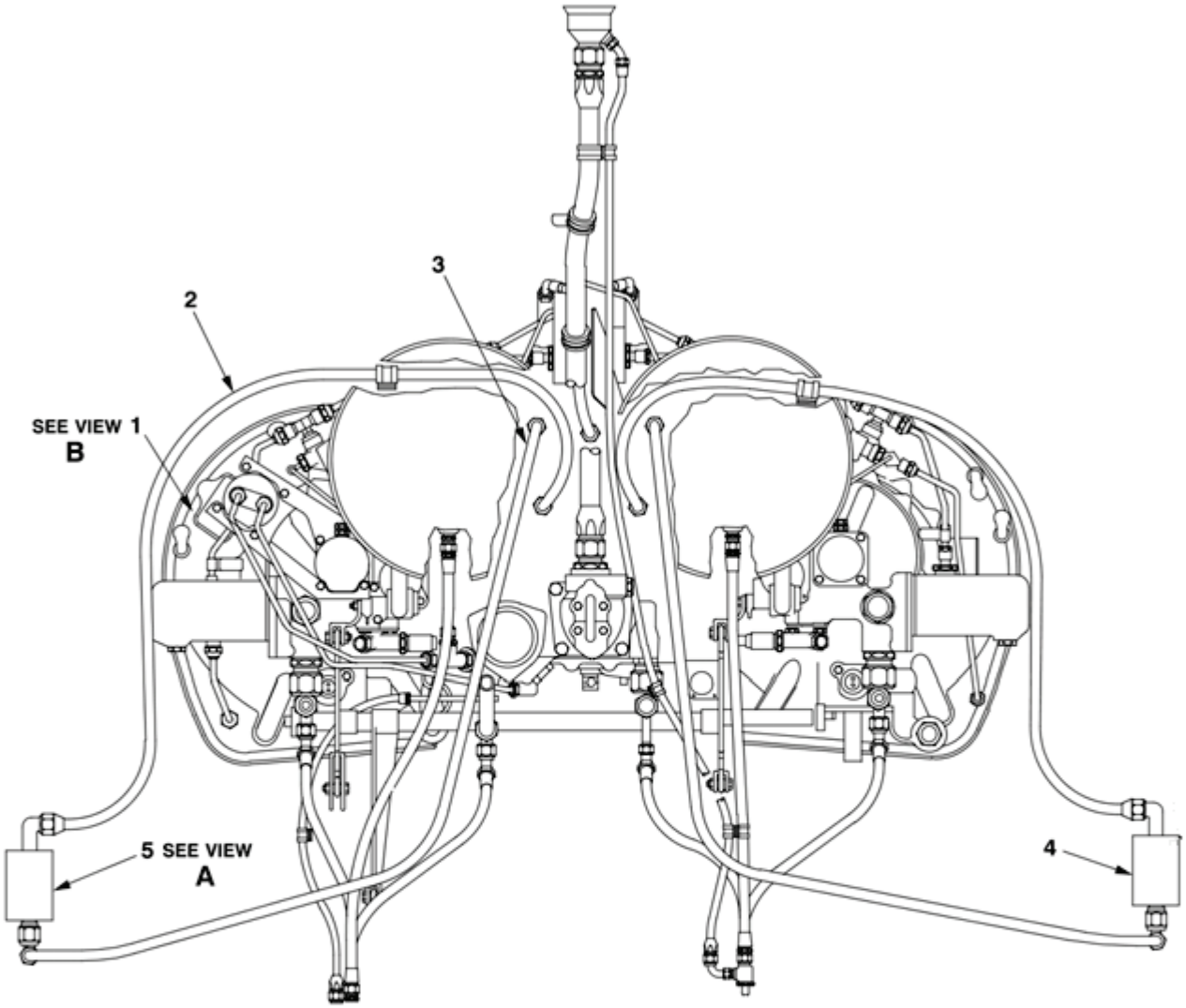
α° VALUE FROM TORQUE TRANSMITTER SETTING INSTRUCTION PLATE.	CORRESPONDING TORQUE INDICATOR READING (IN PERCENT) FOR α° VALUE AT LEFT.	α° VALUE FROM TORQUE TRANSMITTER SETTING INSTRUCTION PLATE.	CORRESPONDING TORQUE INDICATOR READING (IN PERCENT) FOR α° VALUE AT LEFT.
84.0	41.94	88.0	43.93
84.1	41.99	88.1	43.98
84.2	42.04	88.2	44.03
84.3	42.09	88.3	44.08
84.4	42.14	88.4	44.13
84.5	42.19	88.5	44.18
84.6	42.24	88.6	44.23
84.7	42.29	88.7	44.28
84.8	42.34	88.8	44.33
84.9	42.39	88.9	44.38
85.0	42.44	89.0	44.43
85.1	42.49	89.1	44.48
85.2	42.54	89.2	44.53
85.3	42.59	89.3	44.58
85.4	42.64	89.4	44.63
85.5	42.69	89.5	44.68
85.6	42.74	89.6	44.73
85.7	42.79	89.7	44.78
85.8	42.84	89.8	44.83
85.9	42.89	89.9	44.88
86.0	42.94	90.0	44.93
86.1	42.99	90.1	44.98
86.2	43.04	90.2	45.03
86.3	43.09	90.3	45.08
86.4	43.14	90.4	45.13
86.5	43.19	90.5	45.18
86.6	43.24	90.6	45.23
86.7	43.29	90.7	45.28
86.8	43.33	90.8	45.33
86.9	43.38	90.9	45.38
87.0	43.43	91.0	45.43
87.1	43.48	91.1	45.48
87.2	43.53	91.2	45.53
87.3	43.58	91.3	45.58
87.4	43.63	91.4	45.63
87.5	43.68	91.5	45.68
87.6	43.73	91.6	45.73
87.7	43.78	91.7	45.78
87.8	43.83	91.8	45.83
87.9	43.88	91.9	45.88
		92.0	45.93

K = 2.003

α° Value to Corresponding Torque Indicator Reading

ICN-412-A-761000-B-97499-00007-A-001-01

Figure 2. Torque Transmitter Calibration (412-075-008-111/-113 Torque Indicators) (Sheet 1 of 1)

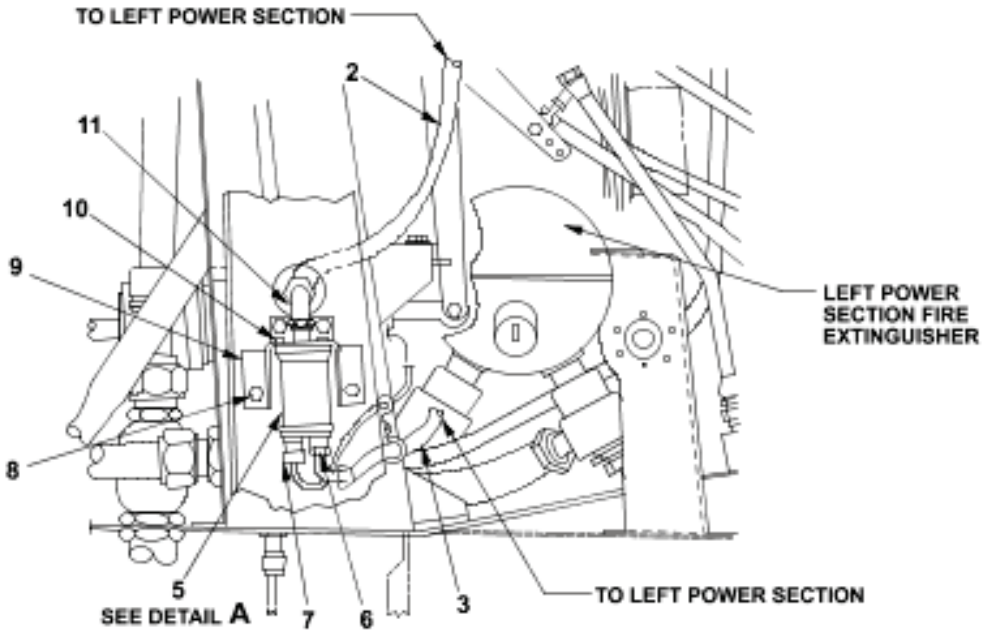


VIEW LOOKING FORWARD AT ENGINE

1. Data plate (Torque transmitter setting instruction plate)
2. Hose assembly
3. Hose assembly
4. Torque transmitter (right)
5. Torque transmitter (left)
6. Union and packing
7. Electrical connector
8. Bolts, nuts and washers (four required)
9. Bracket
10. Nut and washer (AN960C1716)
11. Elbow and nut

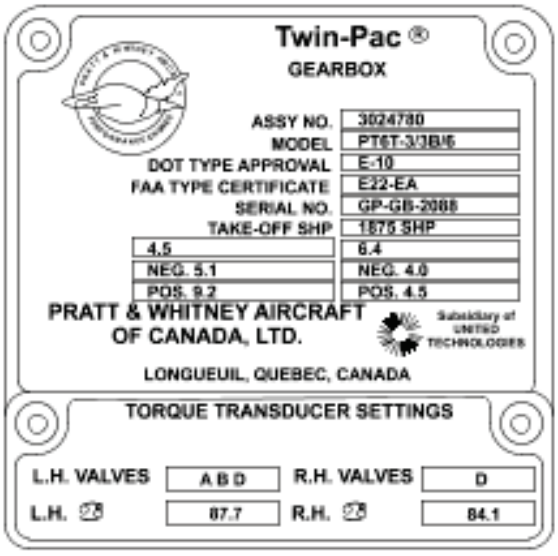
ICN-412-A-761000-B-97499-00008-A-001-01

Figure 3. Torque Transmitter (Typical) (Sheet 1 of 2)



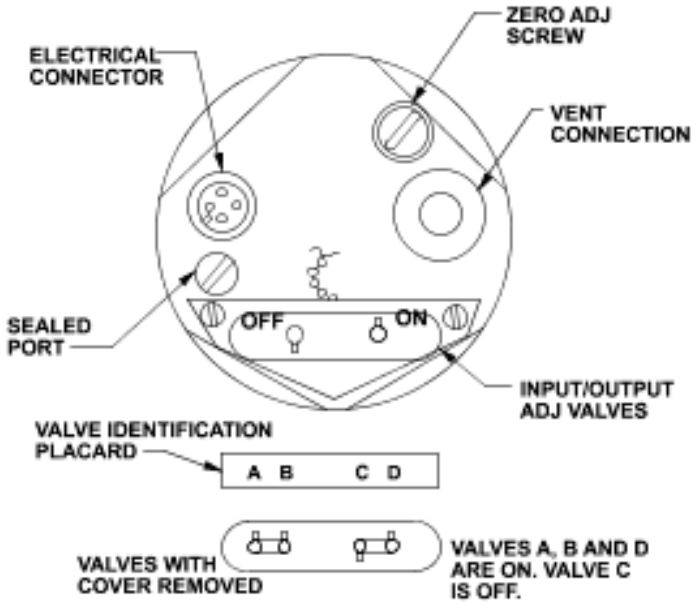
VIEW LOOKING INBOARD AT LEFT POWER SECTION TORQUE TRANSMITTER (5). RIGHT POWER SECTION TORQUE TRANSMITTER (4) IS OPPOSITE.

VIEW A



DATA PLATE (TORQUE TRANSMITTER SETTING INSTRUCTION PLATE)

VIEW B

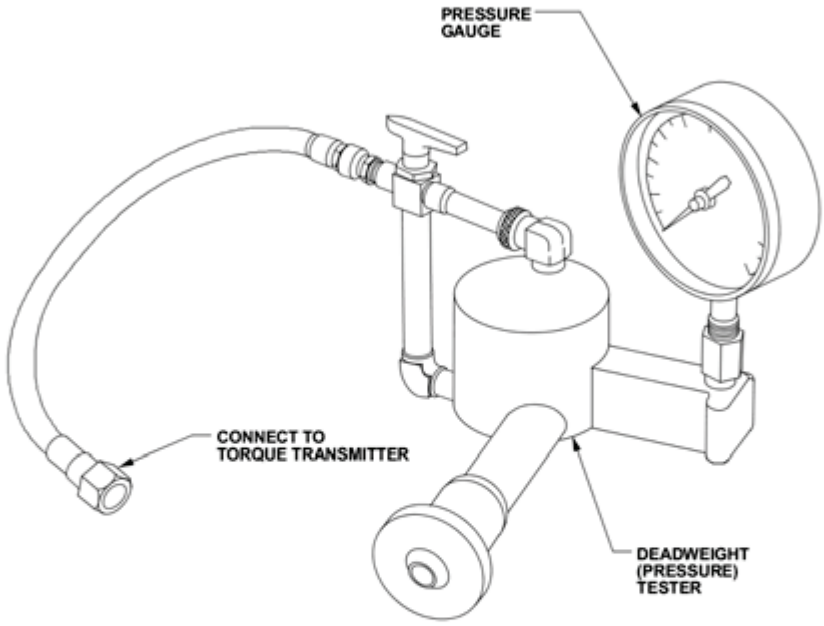


BOTTOM VIEW OF TORQUE TRANSMITTER SAMPLE TORQUE TRANSMITTER SETTING (S/N 33001 THROUGH 33213 AND 36001 THROUGH 36019)

DETAIL A

ICN-412-A-761000-B-97499-00009-A-001-01

Figure 3. Torque Transmitter (Typical) (Sheet 2 of 2)



*** CAUTION ***

ENSURE PRESSURE TESTER IS SERVICED WITH SAME OIL AS USED IN COMBINING GEARBOX

ICN-412-A-761000-B-97499-00011-A-001-01

Figure 4. Calibration Tool - Hand Pressure Tester (Sheet 1 of 1)

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Engine Torque Transmitter - Adjustment and Functional Check

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Deadweight Tester	2311F	1	or Equivalent
Pressure Gauge	302-00102	1	

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Lockwire	C-405	AR	

Procedure

NOTE

The following procedure is applicable to the torque transmitter 412-375-006-101/-103.

Do the following torque adjustment procedure when the engine (Twin-Pac) or the reduction gearbox is changed and/or when either the torque transmitter is changed.

For additional information on the engine torque indicating systems, refer to 412-A-95-61-02-00A-042A-A / 01845 and 412-A-96-12-00-02A-042A-A / 01986.

Torque transmitter adjustment procedures for the left power section are given. Adjustment procedures for the right power section are similar.

1. Locate the data plate on the engine reduction gearbox and record torque transducer settings for the left power section (Figure 1, Detail B). The torque transducer settings for the left torque transmitter on the sample data plate are:

LH valves	ABD
LH α°	87.7

2. Remove the slope adjustment cover from the left torque transmitter (Detail C). Using table (Detail A), make sure that the rotary switch is set to a position associated with the code letter on the data plate. Install the cover.



MAKE SURE THAT THE PRESSURE TESTER IS SERVICED WITH SAME OIL AS USED IN THE COMBINING GEARBOX.

THE PRESSURE TESTER AND GAUGE SHALL BE AT THE SAME HEIGHT AS REDUCTION GEARBOX TORQUE METER (APPROXIMATE LEVEL OF OIL COOLER BLOWER CENTERLINE).

3. Disconnect hose at the torque transmitter pressure port and attach a deadweight tester (2311F) pressure source to the torque transmitter pressure port where hose was removed. Apply a reference pressure of 35.4 ± 0.25 psig to the pressure port of torque transmitter. Make sure that no air is present in the line.

4. Energize essential busses by turning on either the BATTERY BUS 1 or 2 switch. Record pilot ENG 1 TORQUE indication.

5. Calculate the percentage torque corresponding to the referenced test pressure (35.4 psig) according to the following equation:

% Engine torque = $\propto \div 2$, $\propto$ previously recorded in Step 1.

EXAMPLE CALCULATION:

% Eng 1 torque = 87.7 ÷ 2 = 43.85% which rounds off to 43.8%

6. The pilot ENG 1 TORQUE pointer reading must match the engine torque percentage calculated in Step 5. If not, remove the zero adjustment cover (Detail C) on the pressure transducer and rotate the zero adjustment screw until the pilot engine 1 pointer shows the calculated torque value. The copilot indicator (if installed) must read within 1.0% of the pilot indicator. The ENG 2 TORQUE pointer on both the indicators must show zero torque.

7. Turn the BATTERY BUS 1 and 2 switches OFF.

8. Disconnect and remove the tester installed in Step 3.

9. Connect the hose on the pressure port of the torque transmitter.

10. Install the two covers and secure together with lockwire (C-405).

11. Start the left power section (BHT-412-FM) and monitor the ENG 1 TORQUE indicator. If the indicator needle fluctuates, bleed the air from the hose at the pressure port of the torque transmitter.

Gearbox Data Plate Slope Setting	Transducer Slope Switch Setting
NIL	1
A	2
B	3
AB	4
D	5
AD	6
BD	7
ABD	8
BCD	9
ALL	10

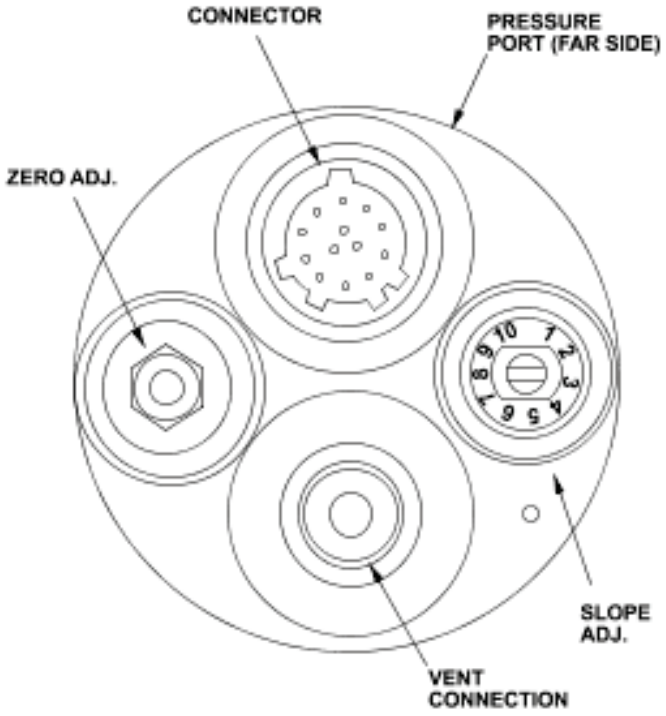
TRANSMITTER SETTING CHART

DETAIL A



DATA PLATE (TORQUE TRANSMITTER AND TCU SETTING INSTRUCTION PLATE)

DETAIL B



TORQUE TRANSMITTER

DETAIL C

ICN-412-A-761000-B-97499-00010-A-001-01

Figure 1. Torque Transmitter Setting (Typical) (Sheet 1 of 1)

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Maximum Torque Check and Control Limit Setting - Checking

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Deadweight Tester	2311F	1	or Equivalent
Pressure Gauge	302-00102	1	

Procedure



DO A TORQUE LIMITER CHECK AND ADJUSTMENT PROCEDURE AFTER EACH CHANGE OF REDUCTION GEARBOX, TORQUE CONTROL UNIT (TCU) , AND/OR INSTALLATION OF A COMPLETE ENGINE.

NOTE

Procedures outlined in 412-A-76-10-00-00A-271C-A / 01730 or 412-A-76-10-00-00A-271D-A / 01731 are applicable to all 412/412EP helicopters.

Applicable: SN: 33001-33213 or 36001-36019:

1. There are two preferred methods to adjust and check the Torque Control Unit (TCU) maximum torque limit setting. The primary method (412-A-76-10-00-00A-271C-A / 01730) requires the operation of the engines. The alternate method (412-A-76-10-00-00A-271D-A / 01731) is done without operating the engines. Both of these procedures are done on the ground. Refer to 412-A-76-10-00-00A-280C-A / 01732 for an additional method applicable.

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Torque Limit - Adjustment

Procedure

1. For helicopters equipped with the Torque Control Unit (TCU) per Figure 1, do as follows:

1.1. Remove the lockwire and disconnect the tube assemblies (6 and 7) from the fittings (5 and 8).

1.2. Plug the open ends of the tubes (6 and 7).

1.3. Use test equipment (Figure 2 or equivalent) with a tee fitting and two lines. Connect one line of the pressure tester to the fitting (5, Figure 1) and one line to the fitting (8).

2. For helicopters equipped with the Torque Control Unit (TCU) , as shown in Figure 3 or the universal Torque Control Unit (TCU) , as shown in Figure 4, do as follows:

2.1. Remove the lockwire and disconnect the tube assemblies (1 and 2, Figure 3 or Figure 4) from the tee fittings (3 and 4).

2.2. Plug the open ends of the tube assemblies (1 and 2) and cap Port D on the tee fitting (4).

2.3. Use the test equipment hoses, and connect the pressure tester to the Port A on the tee fitting (3). Connect the test hose between Port B and Port C.



MAKE SURE PRESSURE TESTER IS SERVICED WITH SAME OIL AS USED IN COMBINING GEARBOX.

PRESSURE TESTER AND GAUGE SHALL BE AT SAME HEIGHT AS REDUCTION GEARBOX TORQUE METER (APPROXIMATE LEVEL OF OIL COOLER BLOWER CENTER LINE).

3. Fill and bleed pressure tester, fitting and hoses.

4. Calculate torque limiter setting as follows:

NOTE

In the following steps Table 3 is used as an example. Use the following chart to determine the correct table for a particular helicopter configuration.

4.1.

SN: 33001-33107:	Table 1
SN: 33108-33213 or 36001-36019 or (SN: 33001-33107 TB 412-84-44: Post):	Table 2
SN: 36020-Subsequent or 412-570-001-103: Installed or BHT-412-SI-74: Installed or BHT-412-SI-97: Installed:	Table 3

Locate data plate on reduction (combining) gearbox and record SPAN and ZERO setting for left and right power section. Refer to Figure 5, Detail B for sample date plate. The span and zero setting for left and right power sections on the sample data plate are:

LH ZERO NEG 5.1 RH ZERO NEG 4.0

LH SPAN POS 9.2 RH SPAN POS 4.5

4.2. For left and right sides combining gearbox determine torque pressure equivalent to limiting torque as follows:

NOTE

If SPAN number falls between two ranges (e.g., is 5.0 or 8.0 etc.), use row at next higher range (e.g., if SPAN number is 5.0, follow row corresponding to 5 to 6 range).

4.3. Enter in Table 3 on left side row corresponding to range in which SPAN number falls.

NOTE

If ZERO number falls between two ranges (e.g., is 7 or 9 etc.), use row at the next lower (less negative) range (e.g., if ZERO number is -7.0, use column corresponding to -6 to -7.0 range).

4.4. Follow row from left to right until under column corresponding to range in which ZERO number falls. Record number at this location.

4.5. The pressure at which torque limiting should occur is the average left and right pressures found in Step 4.3. and Step 4.4.. Calculate this as follows:

$$\text{Average pressure} = \frac{\text{(LH press + RH press)}}{2}$$

EXAMPLE:

LH ZERO = NEG 5.1 RH ZERO = NEG 4.0

LH SPAN = POS 9.2 RH SPAN = POS 4.5

For LH pressure, enter left side of Table 3 at row corresponding to range 9 to 10. Follow this row until under column corresponding to range -5 to -6. The pressure at this location is 50.28 PSIG.

For RH pressure, enter left side of Table 3 at row corresponding to range 4 to 5. Follow this row until under column corresponding to range -3 to -4 (the lower is used as directed by NOTE above). The pressure at this location is 50.89 PSIG.

Average pressure for LH and RH sides:

$$\frac{50.28 + 50.89}{2} = 50.585 \text{ PSIG rounded to } 50.6 \text{ PSIG.}$$

5. Helicopters equipped with TCU (11, Figure 4) adjust as follows prior to ground run.



ON A NEW TCU, THREE LOCKWIRE/SEALS ARE INSTALLED FOR QUALITY CONTROL. THE SEAL BETWEEN THE COARSE ADJUSTMENT DOME AND THE COARSE ADJUSTMENT LOCKNUT MUST NOT BE REMOVED OR WARRANTY WILL BE VOIDED.

NOTE

Seals between the coarse adjustment dome and TCU and between the fine adjustment locknut can be removed and lockwire as required for maintenance.

5.1. Remove the lockwire/seal between the coarse adjustment dome (7) and TCU (11) body.

5.2. Remove the lockwire/seal between the fine adjustment screw (5) and fine adjustment locknut (6).

NOTE

On fine adjustment screw, internal stops allow nearly half a turn in each direction. Full travel (stop to stop) is equivalent to one coarse adjustment position change. The fine adjustment screw should be returned to the nominal (center) position prior to any coarse adjustment change (marks on inner and outer screw aligned).

5.3. Verify the fine adjustment screw (5) is in the nominal position (center); if necessary reset the fine adjustment screw before adjusting the coarse adjustment dome (7).

5.4. Set the coarse adjustment dome to the position (1 to 9) matching coarse adjustment range as per Table 4. The coarse adjustment range corresponds to the addition of LH and RH pressure as determined from Table 3. From the example, 50.28 to 50.89 = 101.17. This corresponds to dial position 4 from Table 4.

6. Check torque limiter adjustment as follows:

6.1. Start both power sections and stabilize at idle for 5 minutes. Verify if ENG 1 and ENG 2 switches are in AUTO position.

6.2. Increase twin-engine power to achieve 100% ROTOR (Nr) rpm, while maintaining flat pitch.



TO AVOID DAMAGE TO TCU, DO NOT EXCEED 57 PSIG FROM PRESSURE TESTER.

6.3. Increase pressure (57 psig maximum) from hand pressure calibrator and record what pressure rotor (Nr) rpm begins to droop.

NOTE

A 1/8 of a turn of screw (2, Figure 1) or (7, Figure 3) or (5, Figure 4) will change pressure at which droop occurs by approximately 3.0 psig.

6.4. If droop does not occur at calculated pressure ±0.35 psig (as read on tester), decrease applied pressure and adjust TCU adjustment screw clockwise to increase pressure or counterclockwise to decrease pressure.

6.5. Repeat Step 6.3. and Step 6.4. until limiter setting is adjusted within ±0.35 psig of pressure calculated.

6.6. Shut down engine power sections.

7. Remove pressure test equipment.

8. For aircraft equipped with Torque Control Unit (TCU) per Figure 1 proceed as follows:

8.1. Remove plugs from end of tubes (6 and 7).

8.2. Connect tubes assemblies (6 and 7) to fittings (5 and 8) and lockwire.

9. For aircraft equipped with Torque Control Unit (TCU) per Figure 3 and Figure 4 proceed as follows:

9.1. Remove test hose between ports “B” and “C”.

9.2. Remove plugs from tube assemblies (1 and 2) and from port “D” on tee fitting (4).

9.3. Connect tubes (1 and 2) to tee fittings (3 and 4) and lockwire.

Table 1. Model 412 Pratt and Whitney PT6T-3B Torque Meter Pressure at Torque Limit (S/N 33001 through 33107)

Zero Inc.	-12	-11	-10	-9	-8	-7	-6	-5	-4	-3	-2	-1	0	1	2
Span Ratio 22															
21	42.29	41.93	41.57	41.22	40.86	40.50	40.14	39.78	39.42	39.06	38.70	38.34	37.98	37.62	
20	42.51	42.15	41.79	41.43	41.07	40.70	40.34	39.98	39.62	39.26	38.90	38.54	38.18	37.82	
19	42.73	42.37	42.00	41.64	41.28	40.91	40.55	40.19	39.82	39.46	39.10	38.74	38.37	38.01	
18	42.95	42.58	42.22	41.85	41.49	41.13	40.76	40.40	40.03	39.67	39.30	38.94	38.57	38.21	
17	43.17	42.81	42.44	42.07	41.71	41.34	40.97	40.61	40.24	39.87	39.51	39.14	38.77	38.41	
16	43.40	43.03	42.66	42.29	41.92	41.55	41.19	40.82	40.45	40.08	39.71	39.34	38.97	38.61	
15	43.63	43.25	42.88	42.51	42.14	41.77	41.40	41.03	40.66	40.29	39.92	39.55	39.18	38.81	
14	43.86	43.48	43.11	42.74	42.37	41.99	41.62	41.25	40.88	40.50	40.13	39.76	39.39	39.01	
13	44.09	43.71	43.34	42.96	42.59	42.22	41.84	41.47	41.09	40.72	40.34	39.97	39.59	39.22	
12	44.32	43.95	43.57	43.19	42.82	42.44	42.06	41.69	41.31	40.94	40.56	40.18	39.81	39.43	
11	44.56	44.18	43.80	43.43	43.05	42.67	42.29	41.91	41.53	41.15	40.78	40.40	40.02	39.64	
10	44.80	44.42	44.04	43.66	43.28	42.90	42.52	42.14	41.76	41.38	41.00	40.62	40.24	39.85	
9	45.04	44.66	44.28	43.90	43.51	43.13	42.75	42.37	41.98	41.60	41.22	40.84	40.45	40.07	
8	45.29	44.90	44.52	44.14	43.75	43.37	42.98	42.60	42.21	41.83	41.44	41.06	40.67	40.29	
7	45.54	45.15	44.76	44.38	43.99	43.60	43.22	42.83	42.44	42.06	41.67	41.28	40.90	40.51	
6	45.79	45.40	45.01	44.62	44.23	43.84	43.46	43.07	42.68	42.29	41.90	41.51	41.12	40.73	
5	46.04	45.65	45.26	44.87	44.48	44.09	43.70	43.30	42.91	42.52	42.13	41.74	41.35	40.96	
4	46.30	45.91	45.51	45.12	44.73	44.33	43.94	43.55	43.15	42.76	42.37	41.97	41.58	41.19	
3	46.56	46.16	45.77	45.37	44.98	44.58	44.19	43.79	43.39	43.00	42.60	42.21	41.81	41.42	
2	46.82	46.42	46.02	45.63	45.23	44.83	44.43	44.04	43.64	43.24	42.84	42.45	42.05	41.65	
1	47.09	46.69	46.29	45.89	45.49	45.09	44.69	44.29	43.89	43.49	43.09	42.69	42.29	41.89	
0	47.35	46.95	46.55	46.15	45.75	45.34	44.94	44.54	44.14	43.73	43.33	42.93	42.53	42.13	
-1	47.63	47.22	46.82	46.41	46.01	45.60	45.20	44.79	44.39	43.98	43.58	43.18	42.77	42.37	
-2	47.90	47.49	47.09	46.68	46.27	45.87	45.46	45.05	44.65	44.24	43.83	43.42	43.02	42.61	
-3	48.18	47.77	47.36	46.95	46.54	46.13	45.72	45.31	44.90	44.49	44.09	43.68	43.27	42.86	
-4	48.46	48.05	47.64	47.22	46.81	46.40	45.99	45.58	45.17	44.75	44.34	43.93	43.52	43.11	

Table 2. Model 412 Pratt and Whitney PT6T-3B Torque Meter Pressure at Torque Limit (S/N 33108 through 33213, 36001 through 36019 and 33001 through 33107 Modified by TB 412-84-44)

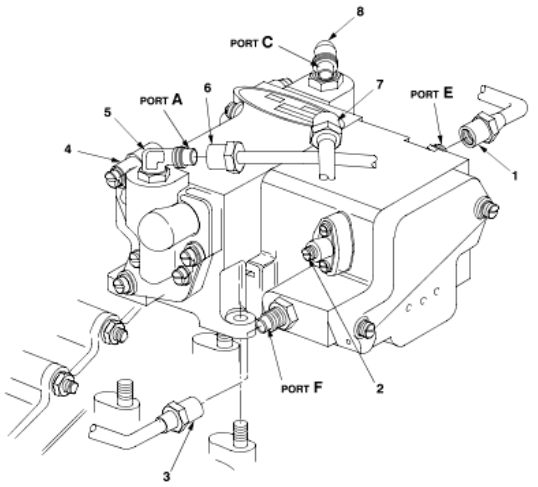
Zero Inc.	-12	-11	-10	-9	-8	-7	-6	-5	-4	-3	-2	-1	0	1	2
Span Ratio	22														
	21	43.71	43.35	43.00	42.64	42.28	41.92	41.56	41.20	40.84	40.48	40.12	39.76	39.40	39.04
	20	43.94	43.58	43.22	42.85	42.49	42.13	41.77	41.41	41.05	40.69	40.33	39.97	39.61	39.24
	19	44.16	43.80	43.44	43.07	42.71	42.35	41.99	41.62	41.26	40.90	40.53	40.17	39.81	39.45
	18	44.39	44.03	43.66	43.30	42.93	42.57	42.20	41.84	41.47	41.11	40.74	40.38	40.01	39.65
	17	44.62	44.26	43.89	43.52	43.16	42.79	42.42	42.06	41.69	41.32	40.96	40.59	40.22	39.86
	16	44.86	44.49	44.12	43.75	43.38	43.01	42.64	42.28	41.91	41.54	41.17	40.80	40.43	40.06
	15	45.09	44.72	44.35	43.98	43.61	43.24	42.87	42.50	42.13	41.76	41.39	41.02	40.64	40.27
	14	45.33	44.96	44.58	44.21	43.84	43.47	43.09	42.72	42.35	41.98	41.60	41.23	40.86	40.49
	13	45.57	45.19	44.82	44.45	44.07	43.70	43.32	42.95	42.57	42.20	41.82	41.45	41.08	40.70
	12	45.81	45.44	45.06	44.68	44.31	43.93	43.55	43.18	42.80	42.42	42.05	41.67	41.29	40.92
	11	46.06	45.68	45.30	44.92	44.54	44.17	43.79	43.41	43.03	42.65	42.27	41.89	41.52	41.14
	10	46.31	45.93	45.54	45.16	44.78	44.40	44.02	43.64	43.26	42.88	42.50	42.12	41.74	41.36
	9	46.56	46.17	45.79	45.41	45.03	44.64	44.26	43.88	43.50	43.11	42.73	42.35	41.97	41.48
	8	46.81	46.43	46.04	45.66	45.27	44.89	44.50	44.12	43.73	43.35	42.96	42.58	42.20	41.81
	7	47.07	46.68	46.29	45.91	45.52	45.13	44.75	44.36	43.97	43.59	43.20	42.81	42.43	42.04
	6	47.33	46.94	46.55	46.16	45.77	45.38	44.99	44.60	44.22	43.83	43.44	43.05	42.66	42.27
	5	47.59	47.20	46.81	46.42	46.02	45.63	45.24	44.85	44.46	44.07	43.68	43.29	42.90	42.51
	4	47.85	47.46	47.07	46.67	46.28	45.89	45.49	45.10	44.71	44.31	43.92	43.53	43.14	42.74
	3	48.12	47.73	47.33	46.94	46.54	46.14	45.75	45.35	44.96	44.56	44.17	43.77	43.38	42.98
	2	48.39	48.00	47.60	47.20	46.80	46.40	46.01	45.61	45.21	44.81	44.42	44.02	43.62	43.22
	1	48.67	48.27	47.87	47.47	47.07	46.67	46.27	45.87	45.47	45.07	44.67	44.27	43.87	43.47
	0	48.94	48.54	48.14	47.74	47.34	46.93	46.53	46.13	45.73	45.32	44.92	44.52	44.12	43.72
	-1	49.22	48.82	48.42	48.01	47.61	47.20	46.80	46.39	45.99	45.58	45.18	44.78	44.37	43.97
	-2	49.51	49.10	48.70	48.29	47.88	47.47	47.07	46.66	46.25	45.85	45.44	45.03	44.63	44.22
	-3	49.80	49.39	48.98	48.57	48.16	47.75	47.34	46.93	46.52	46.11	45.70	45.29	44.89	44.48
	-4	50.09	49.67	49.26	48.85	48.44	48.03	47.62	47.21	46.79	46.38	45.97	45.56	45.15	44.74

Table 3. ***Model 412 HP Pratt and Whitney PT6T-3BE/-3D/-3DE/-3DF Torquemeter Pressure at Torque Limit (S/N 36020 and Subsequent and Helicopters Modified by 412-570-001-101 or Post BHT-412-SI-74 (412SP to 412HP Upgrade) or Post BHT-412-SI-97 (412Classic to 412HP Upgrade))***

Zero Inc.	-12	-11	-10	-9	-8	-7	-6	-5	-4	-3	-2	-1	0	1	2
Span Ratio															
22															
21	49.63	49.00	48.65	48.29	47.93	47.57	47.21	46.85	46.49	46.13	45.77	45.41	45.05	44.69	
20	49.62	49.26	48.89	48.53	48.17	47.81	47.45	47.09	46.73	46.37	46.01	45.65	45.28	44.92	
19	49.87	49.51	49.15	48.78	48.42	48.06	47.69	47.33	46.97	46.61	46.24	45.88	45.52	45.15	
18	50.13	49.76	49.40	49.03	48.67	48.31	47.94	47.58	47.21	46.85	46.48	46.12	45.75	45.39	
17	50.39	50.02	49.66	49.29	48.92	48.56	48.19	47.82	47.46	47.09	46.72	46.36	45.99	45.62	
16	50.65	50.28	49.92	49.55	49.18	48.81	48.44	48.07	47.70	47.34	46.97	46.60	46.23	45.86	
15	50.92	50.55	50.18	49.81	49.44	49.07	48.70	48.33	47.95	47.58	47.21	46.84	46.47	46.10	
14	51.19	50.81	50.44	50.07	49.70	49.33	48.95	48.58	48.21	47.84	47.46	47.09	46.72	46.35	
13	51.46	51.08	50.71	50.34	49.96	49.59	49.21	48.84	48.46	48.09	47.71	47.34	46.97	46.59	
12	51.73	51.36	50.98	50.60	50.23	49.85	49.47	49.10	48.72	48.35	47.97	47.59	47.22	46.84	
11	52.01	51.63	51.25	50.88	50.50	50.12	49.74	49.36	48.98	48.60	48.23	47.85	47.47	47.09	
10	52.29	51.91	51.53	51.15	50.77	50.39	50.01	49.63	49.25	48.87	48.49	48.11	47.73	47.34	
9	52.57	52.19	51.81	51.43	51.04	50.66	50.28	49.90	49.51	59.13	48.75	48.37	47.98	47.60	
8	52.86	52.48	52.09	51.71	51.32	50.94	50.55	50.17	49.78	49.40	49.01	48.63	48.25	47.86	
7	53.15	52.76	52.38	51.99	51.60	51.22	50.83	50.44	50.06	49.67	49.28	48.90	48.51	48.12	
6	53.44	53.05	52.67	52.28	51.89	51.50	51.11	50.72	50.33	49.94	49.55	49.17	48.78	48.39	
5	53.74	53.35	52.96	52.57	52.18	51.78	51.39	51.00	50.61	50.22	49.83	49.44	49.05	48.66	
4	54.04	53.65	53.25	52.86	52.47	52.07	51.68	51.29	50.89	50.50	50.11	49.71	49.32	48.93	
3	54.34	53.95	53.55	53.16	52.76	52.36	51.97	51.57	51.18	50.78	50.39	49.99	49.60	49.20	
2	54.65	54.25	53.85	53.45	53.06	52.66	52.26	51.86	51.47	51.07	50.67	50.27	49.88	49.48	
1	54.96	54.56	54.16	53.76	53.36	52.96	52.56	52.16	51.76	51.36	50.96	50.56	50.16	49.76	
0	55.27	54.87	54.47	54.06	53.66	53.26	52.86	52.46	52.05	51.65	51.25	50.85	50.44	50.04	
-1	55.59	55.18	54.78	54.37	53.97	53.56	53.16	52.76	52.35	51.95	51.54	51.14	50.73	50.33	
-2	55.91	55.50	55.09	54.69	54.28	53.87	53.47	53.06	52.65	52.25	51.84	51.43	51.03	50.62	
-3	56.23	55.82	55.41	55.00	54.60	54.19	53.78	53.37	52.96	52.55	52.14	51.73	51.32	50.91	
-4	56.56	56.15	55.74	55.33	54.91	54.50	54.09	53.68	53.27	52.86	52.44	52.03	51.62	51.21	

Table 4. Coarse Adjustment Dome Range

DIAL POSITION	COARSE ADJUSTMENT RANGE
1	56.6 TO 68.4
2	68.5 TO 80.3
3	80.4 TO 92.2
4	92.3 TO 104.1
5	104.2 TO 116.0
6	116.1 TO 127.9
7	128.0 TO 140.8
8	140.9 TO 152.7
9	152.8 TO 164.6



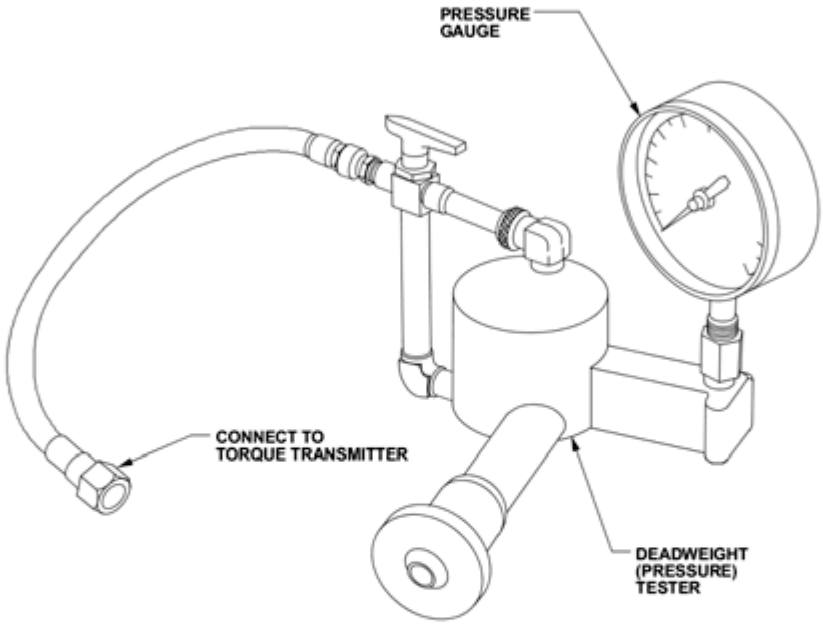
1. Tube
2. Torque limiter adjustment screw
3. Tube
4. Torque control unit
5. Fitting
6. Tube
7. Tube
8. Fitting

NOTE:

1. Port B and Port D not applicable to this installation.

ICN-412-A-761000-B-97499-00012-A-001-01

Figure 1. Torque Control Unit (TCU) Adjustment (Sheet 1 of 1)

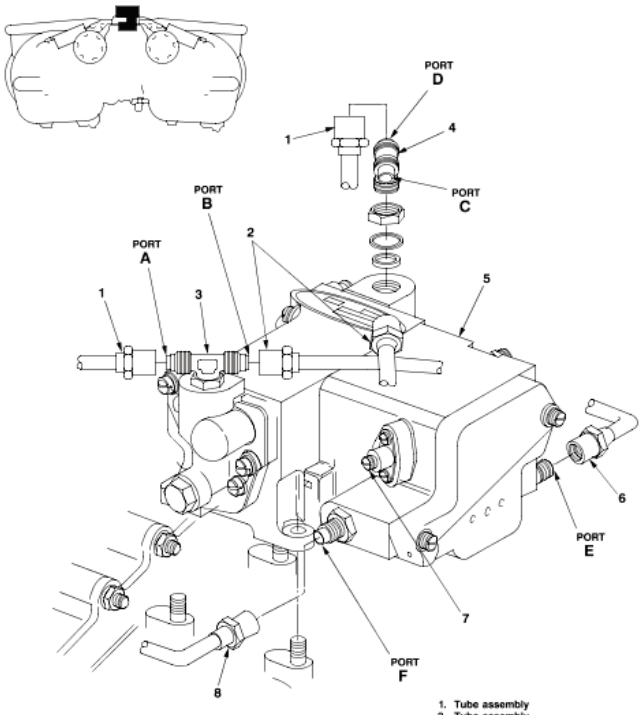


*** CAUTION ***

ENSURE PRESSURE TESTER IS SERVICED WITH SAME OIL AS USED IN COMBINING GEARBOX

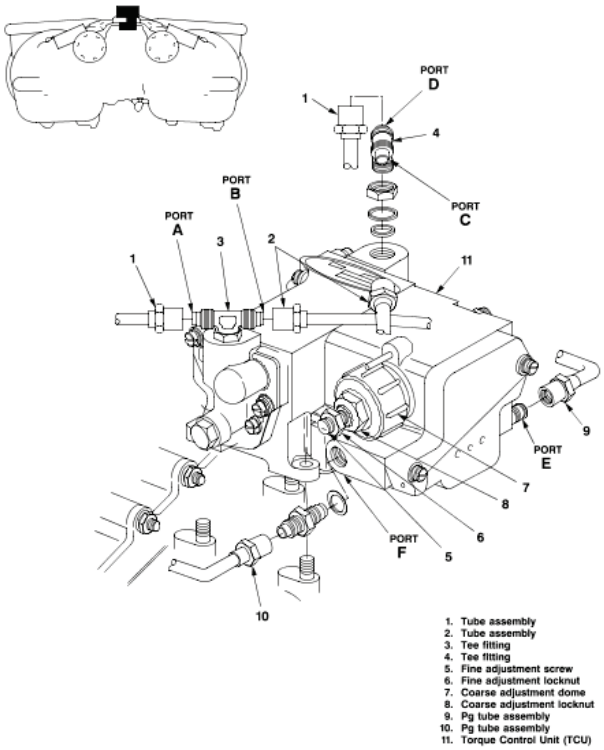
ICN-412-A-761000-B-97499-00011-A-001-01

Figure 2. Calibration Tool - Hand Pressure Tester (Sheet 1 of 1)



ICN-412-A-761000-B-97499-00013-A-001-01

Figure 3. Torque Control Unit (TCU) Torque Limit Adjustment (Sheet 1 of 1)



ICN-412-A-761000-B-97499-00014-A-001-01

Figure 4. Universal Torque Control Unit (TCU) Torque Limit Adjustment (Sheet 1 of 1)

Gearbox Data Plate Slope Setting	Transducer Slope Switch Setting
NIL	1
A	2
B	3
AB	4
D	5
AD	6
BD	7
ABD	8
BCD	9
ALL	10

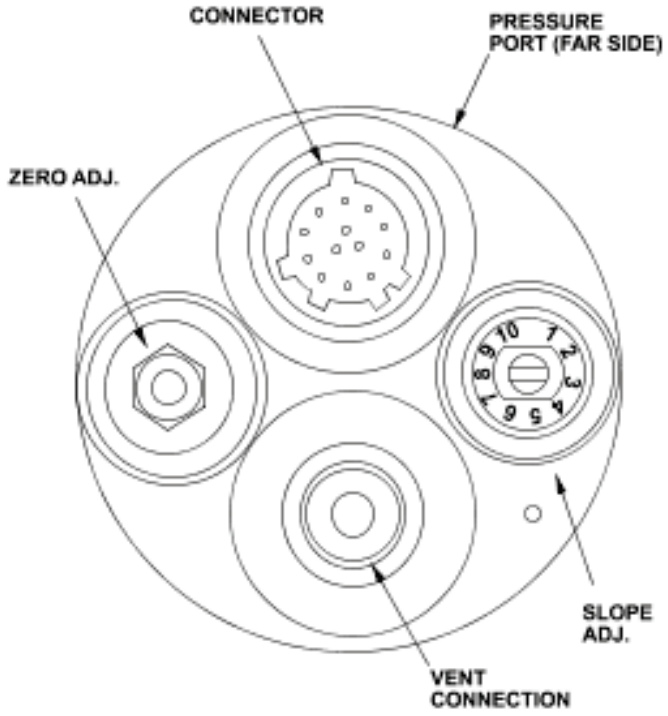
TRANSMITTER SETTING CHART

DETAIL A



DATA PLATE (TORQUE TRANSMITTER AND TCU SETTING INSTRUCTION PLATE)

DETAIL B



TORQUE TRANSMITTER

DETAIL C

ICN-412-A-761000-B-97499-00010-A-001-01

Figure 5. Torque Transmitter Setting (Typical) (S/N 36020 and Subsequent and Helicopters Modified by 412-570-001-103 or Post BHT-412-SI-74 (412SP to 412HP Upgrade) or Post BHT-412-SI-97 (412Classic to 412HP Upgrade)) (Sheet 1 of 1)

Torque Limit - Adjustment - Alternate Method

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Filtered, regulated air pressure source	Commercial	1	

Procedure

NOTE

This alternate procedure can be used when it is desirable to adjust the torque limiter without operating the engines.

1. Install equipment and calculate the torque limiter setting as per 412-A-76-10-00-00A-271C-A / 01730, Step 1. through Step 5..

2. Connect filtered, regulated air pressure source with in-line regulator to one or both Pg air ports (E and F, Figure 1, Figure 2 or Figure 3) on right and /or left side of TCU.

3. Adjust air pressure (Pg) to approximately 15 psig.



TO AVOID DAMAGE TO TCU, DO NOT EXCEED 57 PSIG FROM PRESSURE TESTER.

4. Increase pressure from pressure tester and record at what oil pressure the air pressure begins to drop.

NOTE

A 1/8 of a turn of screw (2, Figure 1) or (7, Figure 2) or (5, Figure 3) will change pressure at which droop occurs by approximately 3.0 psig.

5. If air pressure drop does not occur at calculated pressure ±0.35 psig (as read on tester), decrease applied pressure and adjust TCU adjustment screw clockwise to increase pressure or counterclockwise to decrease pressure.

6. Repeat Step 4. and Step 5. until limiter setting is adjusted within ±0.35 psig of pressure calculated.

7. Remove test equipment.

8. For aircraft equipped with Torque Control Unit (TCU) per Figure 1 proceed as follows:

8.1. Remove plugs from end of tubes (6 and 7).

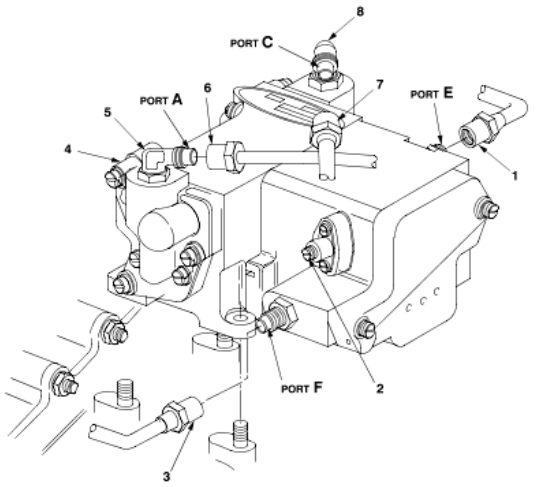
8.2. Connect tube assemblies (6 and 7) to fittings (5 and 8) and lockwire.

9. For aircraft equipped with Torque Control Unit (TCU) per Figure 2 and Figure 3 proceed as follows:

9.1. Remove test hose between ports “B” and “C”.

9.2. Remove the plugs from the tube assemblies (1 and 2) and from Port D on the tee fitting (4).

9.3. Connect the tubes (1 and 2) to the tee fittings (3 and 4) and lockwire.



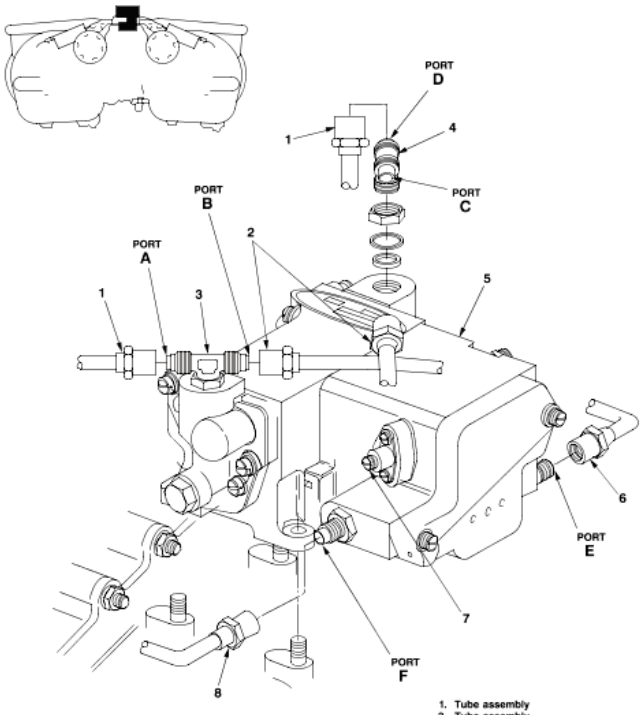
1. Tube
2. Torque limiter adjustment screw
3. Tube
4. Torque control unit
5. Fitting
6. Tube
7. Tube
8. Fitting

NOTE:

1. Port B and Port D not applicable to this installation.

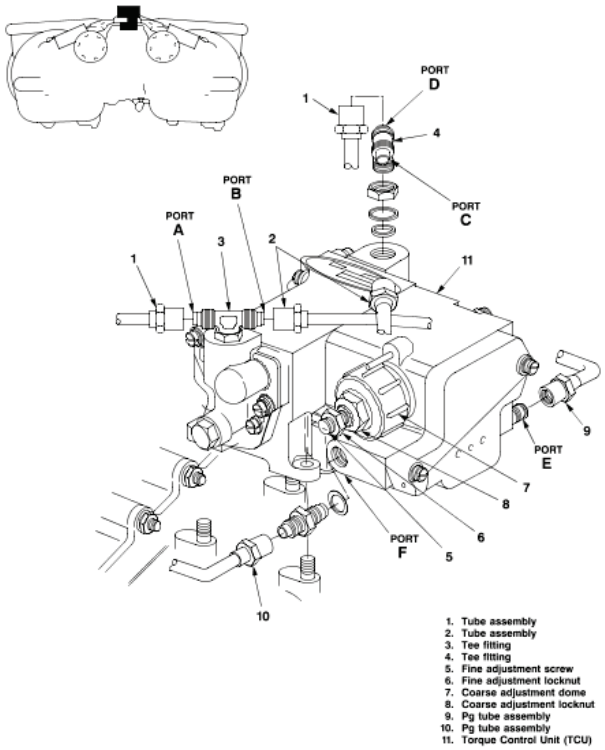
ICN-412-A-761000-B-97499-00012-A-001-01

Figure 1. Torque Control Unit (TCU) Adjustment (Sheet 1 of 1)



ICN-412-A-761000-B-97499-00013-A-001-01

Figure 2. Torque Control Unit (TCU) Torque Limit Adjustment (Sheet 1 of 1)



ICN-412-A-761000-B-97499-00014-A-001-01

Figure 3. Universal Torque Control Unit (TCU) Torque Limit Adjustment (Sheet 1 of 1)

Maximum Torque Check and Control Limit Setting - Checking

Procedure



DO A TORQUE LIMITER CHECK AND ADJUSTMENT PROCEDURE AFTER EACH CHANGE OF REDUCTION GEARBOX, TORQUE CONTROL UNIT (TCU) , AND/OR INSTALLATION OF A COMPLETE ENGINE.

WHEN YOU DO THE FOLLOWING STEPS, DO NOT EXCEED OPERATING LIMITATIONS.

NOTE

SN: 33001-33213 or 36001-36019:

It should only be used if test equipment listed in 412-A-76-10-00-00A-271C-A / 01730 is not available.

1. Start both power sections and let idle for 5 minutes.

NOTE

On a hot day or at high altitude conditions, it may not be possible to reach the torque limit required by the following test.

2. Do a maximum power climb and record the point at which torque limiting begins. This is indicated by the ROTOR speed decaying with further power application and the GAS PROD RPM appearing to be too low to have reached engine topping.

Applicable: SN: 33001-33107:

2.1. Torque limiting should start limiting at 103%.

Applicable: SN: 33108-33213 or 36001-36019 or (SN: 33001-33107 TB 412-84-44: Post):

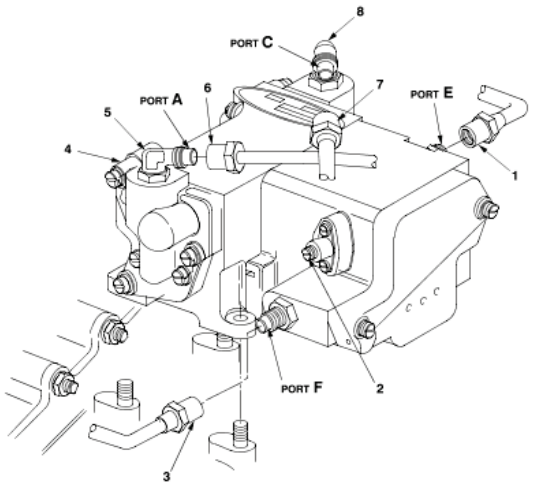
2.2. Torque limiting should start limiting at 100%.



IF ITT IS REACHED BEFORE ATTAINING DESIRED INITIAL TORQUE, DO NOT MAKE ANY ADJUSTMENT TO TORQUE CONTROL UNIT.

3. If the torque limit does not occur before reaching either the maximum initial value (100 or 103% as applicable), turn the adjustment screw (2, Figure 1 or 7, Figure 2) counterclockwise in 1/8 turn increments and repeat Step 2. after each adjustment until torque limiting occurs within the desired initial range.

4. Insert a wrench in the adjustment screw (2, Figure 1 or 7, Figure 2) and mark the starting location with a soft lead pencil or other nonpermanent marker. Rotate the adjustment screw 3/8 turn clockwise to establish the new limiting setting.



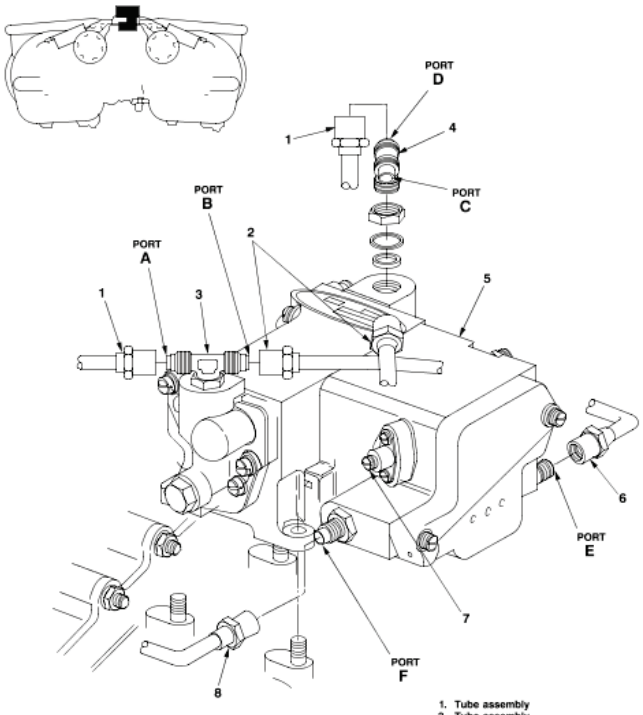
1. Tube
2. Torque limiter adjustment screw
3. Tube
4. Torque control unit
5. Fitting
6. Tube
7. Tube
8. Fitting

NOTE:

1. Port B and Port D not applicable to this installation.

ICN-412-A-761000-B-97499-00012-A-001-01

Figure 1. Torque Control Unit (TCU) Adjustment (Sheet 1 of 1)



ICN-412-A-761000-B-97499-00013-A-001-01

Figure 2. Torque Control Unit (TCU) Torque Limit Adjustment (Sheet 1 of 1)

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Automatic Fuel Control Idle Speed - Check

Procedure

NOTE

Refer to the applicable Pratt & Whitney Canada Engine Maintenance Manual for more details on the adjustment procedure.

1. Start both engines and let idle for 5 minutes.

2. Activate the ENG 1 IDLE STOP release switch (18, Figure 1).

3. Rotate the ENG 1 throttle twist grip (19) below the idle stop so that the Manual Fuel Control Unit (MFCU) (5, Figure 2) pointer (15) is positioned between 30 and 35° on the quadrant (16).

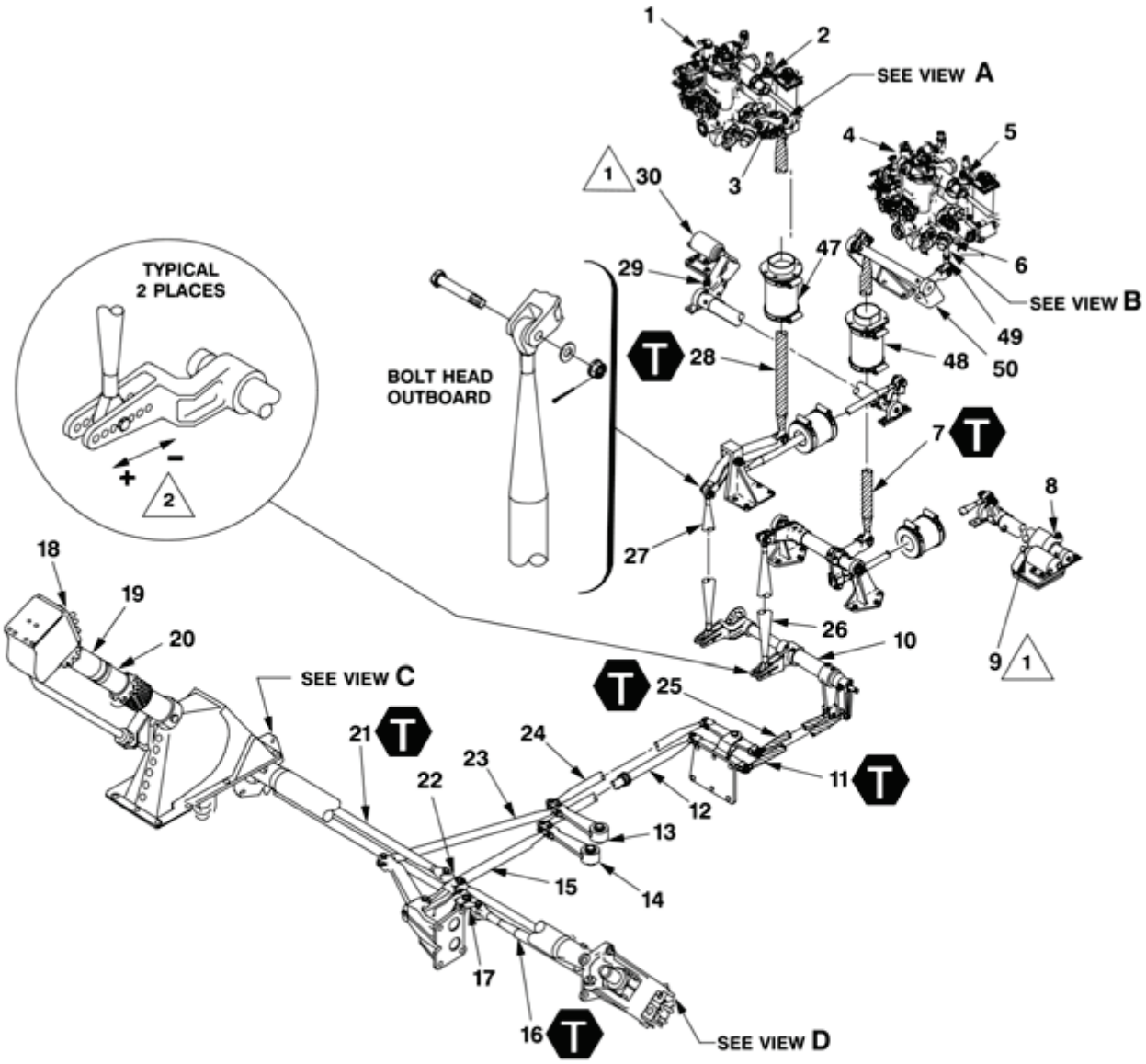
4. For helicopters with PT6T-3B/BE engine, make sure that the gas producer speed (N_1) is between 51 and 52.5%. For helicopters with PT6T-3D/DE/DF engine, make sure that the N_1 speed is between 53.8 and 55%. If necessary, adjust the Automatic Fuel Control Unit (AFCU) (4) idle to bring N_1 within the required range.

NOTE

The difference between N_1 IDLE speeds of both engines must not exceed 1%.

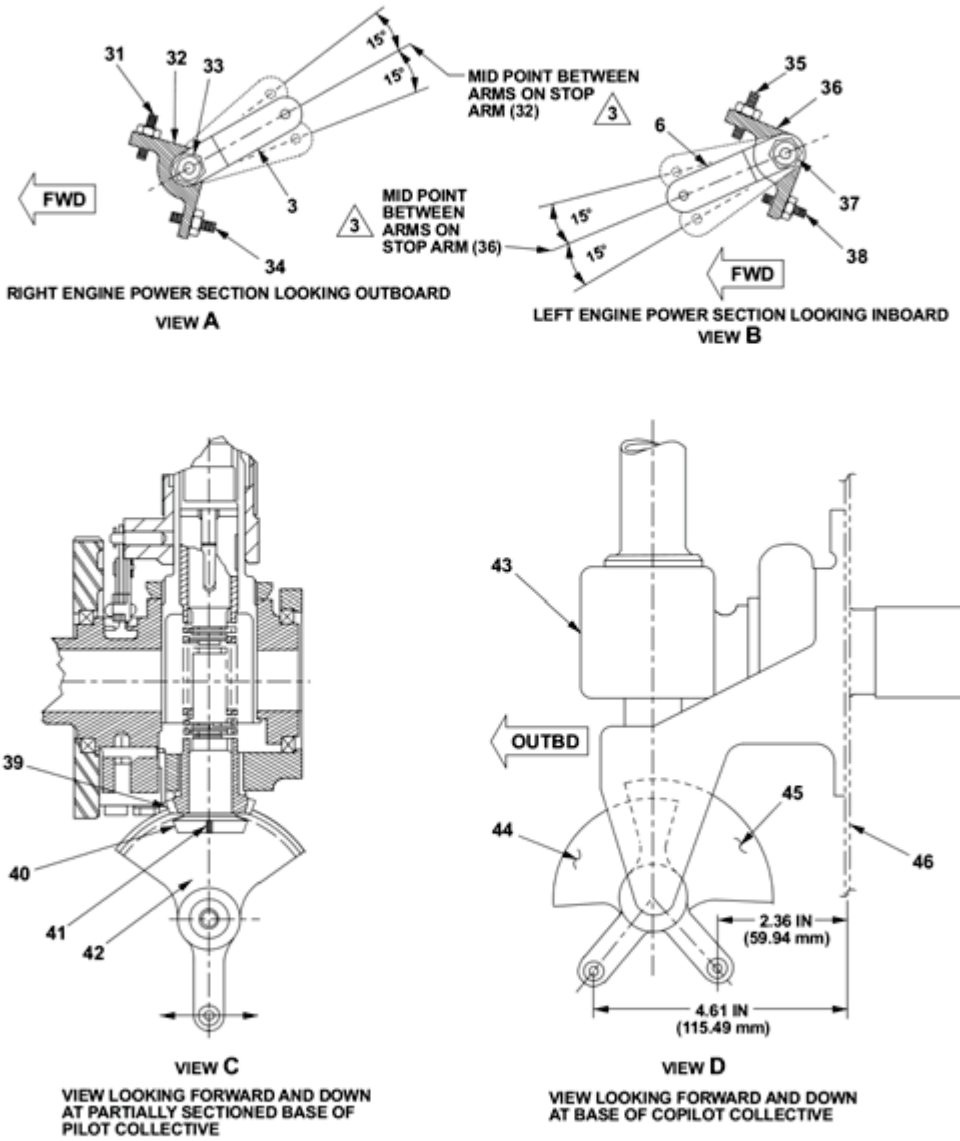
5. Do Step 2. and Step 4. with ENG 2.

6. Torque the idle stop locknut and lockwire (refer to Pratt & Whitney Maintenance Manual).



ICN-412-A-761000-B-97499-00001-A-001-01

Figure 1. Power Lever (N1) Controls Rigging (Sheet 1 of 3)



ICN-412-A-761000-B-97499-00002-A-001-01

Figure 1. Power Lever (N1) Controls Rigging (Sheet 2 of 3)

1. Automatic Fuel Control Unit (AFCU), ENG 2
2. Manual Fuel Control Unit (MFCU), ENG 2
3. Lever
4. Automatic Fuel Control Unit (AFCU), ENG 1
5. Manual Fuel Control Unit (MFCU), ENG 1
6. Lever
7. Control tube
8. Idle stop lever
9. Idle stop solenoid
10. Jackshaft
11. Control tube
12. Control tube
13. Bellcrank
14. Bellcrank
15. Control tube
16. Control tube
17. Double rod end bearing
18. IDLE STOP release switch
19. Throttle twist grip, ENG 1
20. Throttle twist grip, ENG 2
21. Control tube
22. Double rod end bearing
23. Control tube
24. Control tube
25. Control tube
26. Control tube
27. Control tube
28. Control tube
29. Idle stop lever
30. Idle stop solenoid
31. Minimum stop screw
32. Stop arm, manual fuel control
33. Nut
34. Maximum stop screw
35. Minimum stop screw
36. Stop arm, manual fuel control
37. Nut
38. Maximum stop screw
39. Pilot engine 2 flex shaft
40. Pilot engine 1 flex shaft
41. Marked tooth on flex shaft (40)
42. Pilot engine 2 gear sector
43. Copilot collective elbow and support
44. Copilot engine 1 gear sector
45. Copilot engine 2 gear sector
46. Structure
47. Boot
48. Boot
49. Link
50. Jackshaft

T 80 TO 100 IN-LBS
(9.0 TO 11.3 Nm)

NOTES

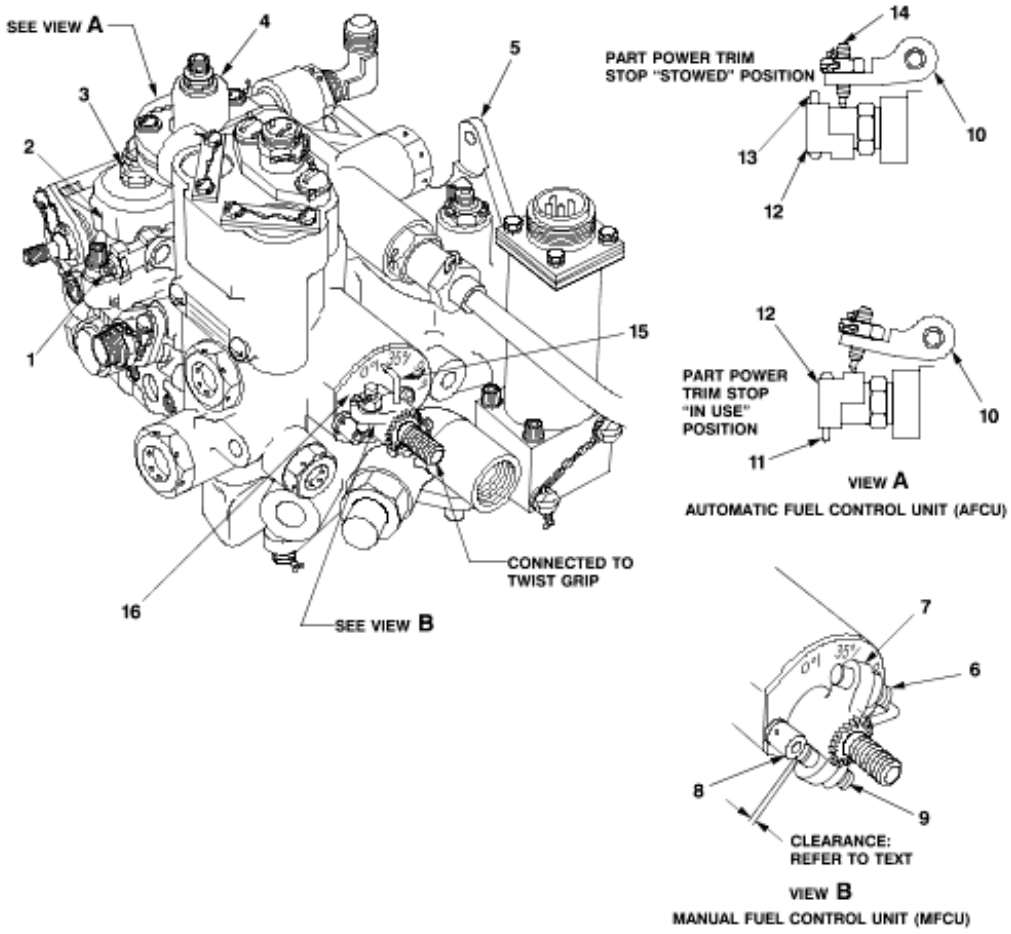
1. During rigging procedure, a small wooden block may be placed between the shoulder of the idle stop plunger and flange of the idle stop solenoid mounting bracket to assist in holding the idle stop plunger in the retract position.

2. To increase travel, reposition rod end in (+) direction, to decrease travel, reposition rod end in (-) direction.

3. Mid point shown as reference only. Mid position of lever (3 and 6) may not be achievable in reference to stop arms (32 and 36).

ICN-412-A-761000-B-97499-00003-A-001-01

Figure 1. Power Lever (N1) Controls Rigging (Sheet 3 of 3)



1. Manual - automatic fuel control interconnect linkage
2. Datum line
3. Acceleration Adjuster (dome)
4. Automatic Fuel Control Unit (AFCU)
5. Manual Fuel Control Unit (MFCU)
6. Minimum stop screw
7. Stop arm
8. Stop
9. Maximum stop screw
10. Fuel control arm
11. Cotter pin
12. Part power trim stop
13. Cotter pin
14. Maximum stop adjustment screw
15. Pointer
16. Quadrant

ICN-412-A-761000-B-97499-00015-A-001-01

Figure 2. Manual and Automatic Fuel Control Adjustment Provisions (Sheet 1 of 1)

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Power Section Acceleration and Deceleration - Check

Procedure

NOTE

Refer to the applicable Pratt & Whitney Canada Engine Series Maintenance Manual for greater detail on adjustment procedure.

1. Do an automatic fuel control idle speed check per 412-A-76-10-00-00A-280D-A / 01733 before the acceleration and deceleration check.

2. Start both engines and let idle for 5 minutes.

3. With the collective at flat pitch and both engines at IDLE, increase N_2 to maximum beep. Set the N_1 IDLE speed to 61% on each power section, using the throttles.

4. Check the acceleration time from 61 to 90% N_1 for ENG 1 as follows:



MONITOR ENGINE TORQUE AND ITT DURING THIS PROCEDURE.

NOTE

Measurement of acceleration time begins when ENG throttle is snapped to FULL OPEN and stops when N_1 reaches 90%.

4.1. Set the ENG 2 throttle to IDLE and apply friction to hold it in position.

4.2. Snap open the ENG 1 throttle to FULL OPEN. Start timing.

4.3. When 90% N_1 is reached, snap the ENG 1 throttle back to IDLE. Stop timing.

NOTE

Acceleration time from 61 to 90% N_1 should be 4 seconds maximum. Acceleration times less than 3 seconds are acceptable if outside air temperature is below 32°F (0°C), provided there is no compressor surge.

Match both power sections acceleration times as close as possible. The maximum acceptable difference in acceleration time is 0.5 second.

4.4. If the acceleration time is too slow or too fast (compressor surge), adjust the acceleration dome (3, Figure 1) (refer to Pratt & Whitney Maintenance Manual).

5. Repeat Step 4. with ENG 2.

6. Check the deceleration time from 90 to 61% N_1 for ENG 1 as follows:

NOTE

Measurement of deceleration time begins when ENG throttle is snapped to IDLE and stops when N_1 reaches 61%.

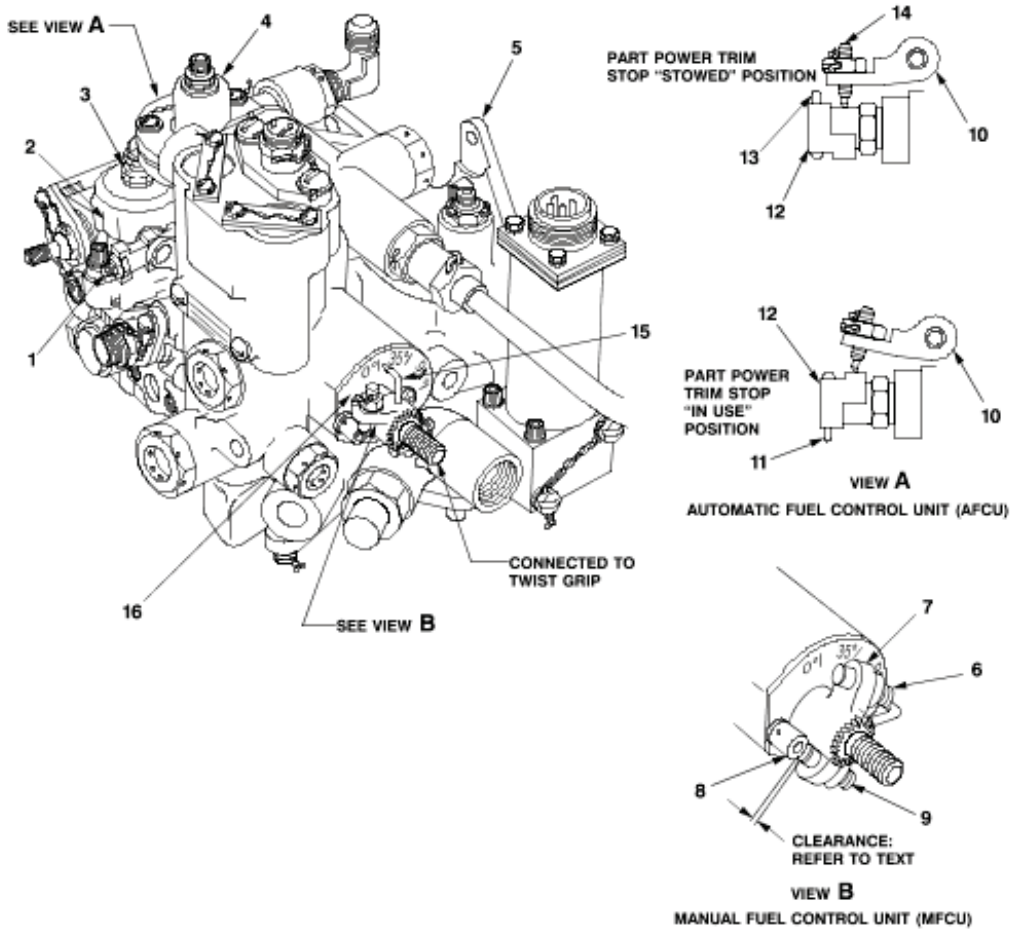
6.1. Setthe ENG 2 throttle to IDLE and apply friction to hold it in position.

6.2. Gradually open the ENG 1 throttle to FULL OPEN.

6.3. Increase power to 90% N_1 and stabilize, then snap the ENG 1 throttle back to IDLE.

6.4. Confirm a deceleration time from 90 to 61% N_1 of 5 seconds maximum below 6000 feet pressure altitude, and smooth deceleration to a stabilized IDLE of 61% N_1.

7. Repeat Step 6. with ENG 2.



1. Manual - automatic fuel control interconnect linkage
2. Datum line
3. Acceleration Adjuster (dome)
4. Automatic Fuel Control Unit (AFCU)
5. Manual Fuel Control Unit (MFCU)
6. Minimum stop screw
7. Stop arm
8. Stop
9. Maximum stop screw
10. Fuel control arm
11. Cotter pin
12. Part power trim stop
13. Cotter pin
14. Maximum stop adjustment screw
15. Pointer
16. Quadrant

ICN-412-A-761000-B-97499-00015-A-001-01

Figure 1. Manual and Automatic Fuel Control Adjustment Provisions (Sheet 1 of 1)

PT6T-3B/BE Engine - Check and Adjustment

Procedure

NOTE

Refer to Pratt & Whitney Aircraft of Canada Ltd. PT6T-3 Series Maintenance Manual No. 3017042 for detailed procedure.

This single power section check to be performed with helicopter heavily loaded and light on skids. However, pilot must be prepared for the possibility helicopter may become airborne.

1. Rotate in place and secure part power trim stop (12, Figure 1).

2. Start both engines and let idle for 5 minutes.

3. With collective at flat pitch and both engines at IDLE, increase ENG 1 throttle to FULL OPEN. Using INCR/DECR switch increase ROTOR rpm (NR) to full INCR.

NOTE

Maximum gas producer (N1) speed (topping) is achieved when N1 ceases to increase and N2/NR droops as collective is further applied. Do not droop below 90% N2. The observed reading at this point should be 99.8 to 100.8% N1. A minimum value of 99.8% N1 is required to assure in flight topping at 103.4% N1.

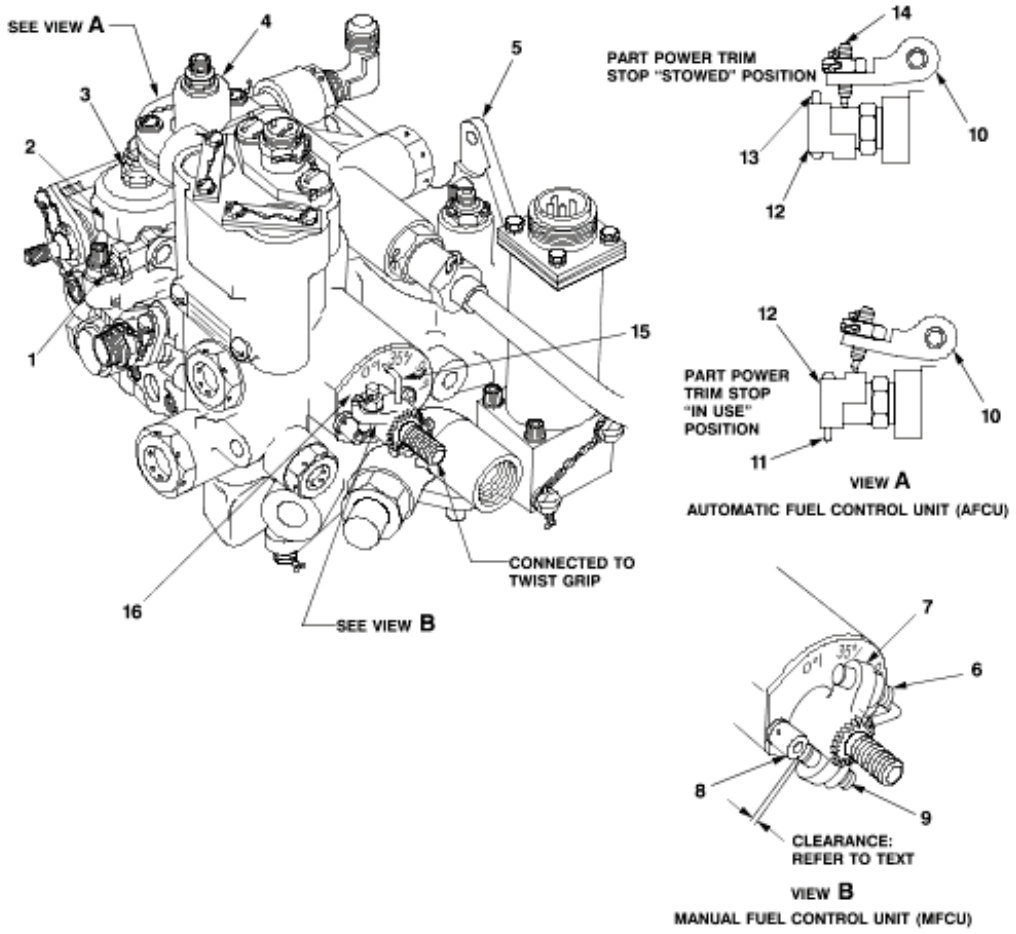
4. Gradually increase collective until N1 topping is achieved and note N1 value.

NOTE

If maximum stop screw is adjusted, readjust the interconnect linkage for proper AUTO/MANUAL gap and throttle cushions per 412-A-76-10-00-00A-276A-A / 01725 and Pratt & Whitney Aircraft of Canada Ltd. PT6T-3 Series Maintenance Manual. This is accomplished once maximum gas producer (N1) speed is attained and part power trim stop is retracted.

5. If required, adjust the topping adjustment screw and repeat Step 4..

6. Repeat Step 3., Step 4., and Step 5. with ENG 2.



1. Manual - automatic fuel control interconnect linkage
2. Datum line
3. Acceleration Adjuster (dome)
4. Automatic Fuel Control Unit (AFCU)
5. Manual Fuel Control Unit (MFCU)
6. Minimum stop screw
7. Stop arm
8. Stop
9. Maximum stop screw
10. Fuel control arm
11. Cotter pin
12. Part power trim stop
13. Cotter pin
14. Maximum stop adjustment screw
15. Pointer
16. Quadrant

ICN-412-A-761000-B-97499-00015-A-001-01

Figure 1. Manual and Automatic Fuel Control Adjustment Provisions (Sheet 1 of 1)

PT6T-3B/BE Engine AFCU Maximum N1 Stop - Adjustment - Alternate Procedure

Procedure



DO NOT EXCEED TORQUE LIMIT OF 73.2% OR ITT LIMIT OF 850 DEGREES C.

SN: 33001-33107 TB 412-84-44: Pre: DO NOT EXCEED TORQUE LIMIT OF 76.0% OR ITT LIMIT OF 850 DEGREES C.

MAXIMUM N1 TOPPING CHECKS SHOULD ONLY BE PERFORMED AFTER ENGINE CONTROL RIGGING IS ADJUSTED OR WHEN ANY OF THE FOLLOWING COMPONENTS ARE REPLACED: (TWO POWER SECTIONS AND REDUCTION GEARBOX), POWER SECTION, AFCU, MFCU AND FUEL PUMP.

NOTE

Field experience particularly in warm climates, has revealed it is not possible to operate both power sections at 102.4% N1 rpm with indicator 212-075-037-101 or 103.4% N1 rpm with indicator 212-075-037-113 at any altitude without exceeding maximum ITT limit of 850°C. This condition does not prevent engine from achieving required performance, but does present a functional test problem.

This single power section check to be performed in flight or with helicopter heavily loaded and light on skids. However, pilot must be prepared for the possibility helicopter may become airborne.

1. Start both engines and let idle for 5 minutes.

2. Increase both throttles to FULL OPEN one at a time and adjust ENG N2 rpm to INCR with RPM INCR/DECR switch.

3. Decrease throttle of one engine to flight IDLE leaving the other at FULL OPEN with ENG N2 rpm to full INCR.

NOTE

N1 values with indicator 212-075-037-101 installed is 102.4% and 103.4% for indicator 212-075-037-113. For maintenance purposes only adjust maximum N1 to 103.4% for both indicators.

4. Increase power by raising the collective, drooping ENG rpm to approximately 97% to obtain 103.4% N1 rpm.

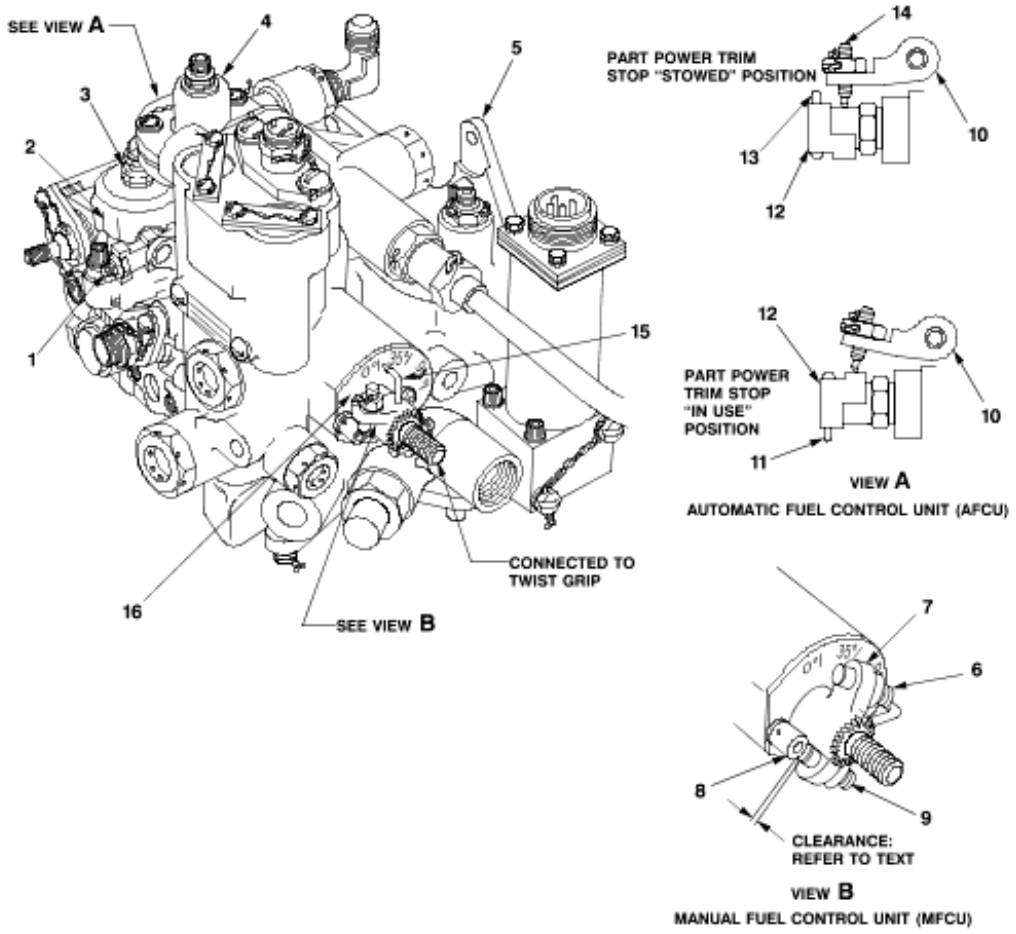
NOTE

If maximum stop screw is adjusted, readjust the interconnect linkage for proper AUTO/MANUAL gap and throttle cushions per 412-A-76-10-00-00A-276A-A / 01725 and Pratt & Whitney Aircraft of Canada Ltd. PT6T-3 Series Maintenance Manual. This is accomplished once maximum gas producer (N1) speed is attained and part power trim stop is retracted.

4.1. If ENG rpm droops below 97% prior to obtaining 103.4% N1 rpm, increase topping. Turn stop screw (14, Figure 1) counterclockwise to increase.

4.2. If 103.4% N1 rpm is reached without ENG rpm droop, apply power to a maximum of 103.6% N1 rpm. If droop does not occur between 103.4% and 103.6% N1 rpm, decrease topping. Turn stop screw (14) clockwise to decrease.

5. If stop screw (14) was adjusted, tighten jamnuts and lockwire.



1. Manual - automatic fuel control interconnect linkage
2. Datum line
3. Acceleration Adjuster (dome)
4. Automatic Fuel Control Unit (AFCU)
5. Manual Fuel Control Unit (MFCU)
6. Minimum stop screw
7. Stop arm
8. Stop
9. Maximum stop screw
10. Fuel control arm
11. Cotter pin
12. Part power trim stop
13. Cotter pin
14. Maximum stop adjustment screw
15. Pointer
16. Quadrant

ICN-412-A-761000-B-97499-00015-A-001-01

Figure 1. Manual and Automatic Fuel Control Adjustment Provisions (Sheet 1 of 1)

PT6T-3D/DE/DF Engine - Check and Adjustment

Procedure

NOTE

Refer to Pratt & Whitney Aircraft of Canada Ltd. PT6T-3 Series Maintenance Manual No. 3040592 for detailed procedure.

This single power section check to be performed with helicopter heavily loaded and light on skids. However, pilot must be prepared for the possibility helicopter may become airborne.

1. Rotate in place and secure part power trim stop (12, Figure 1).

2. Start both engines and let idle for 5 minutes.

3. With collective at flat pitch and both engines at IDLE, increase ENG 1 throttle to FULL OPEN. Using INCR/DECR switch increase ROTOR rpm (NR) to full INCR.

NOTE

Maximum gas producer (N1) speed (topping) is achieved when N1 ceases to increase and N2/NR droops as collective is further applied. Do not droop below 90% N2. The observed reading at this point should be 92.0 to 92.6% N1. A minimum value of 92.0% N1 is required to assure in flight topping at 109.2% N1.

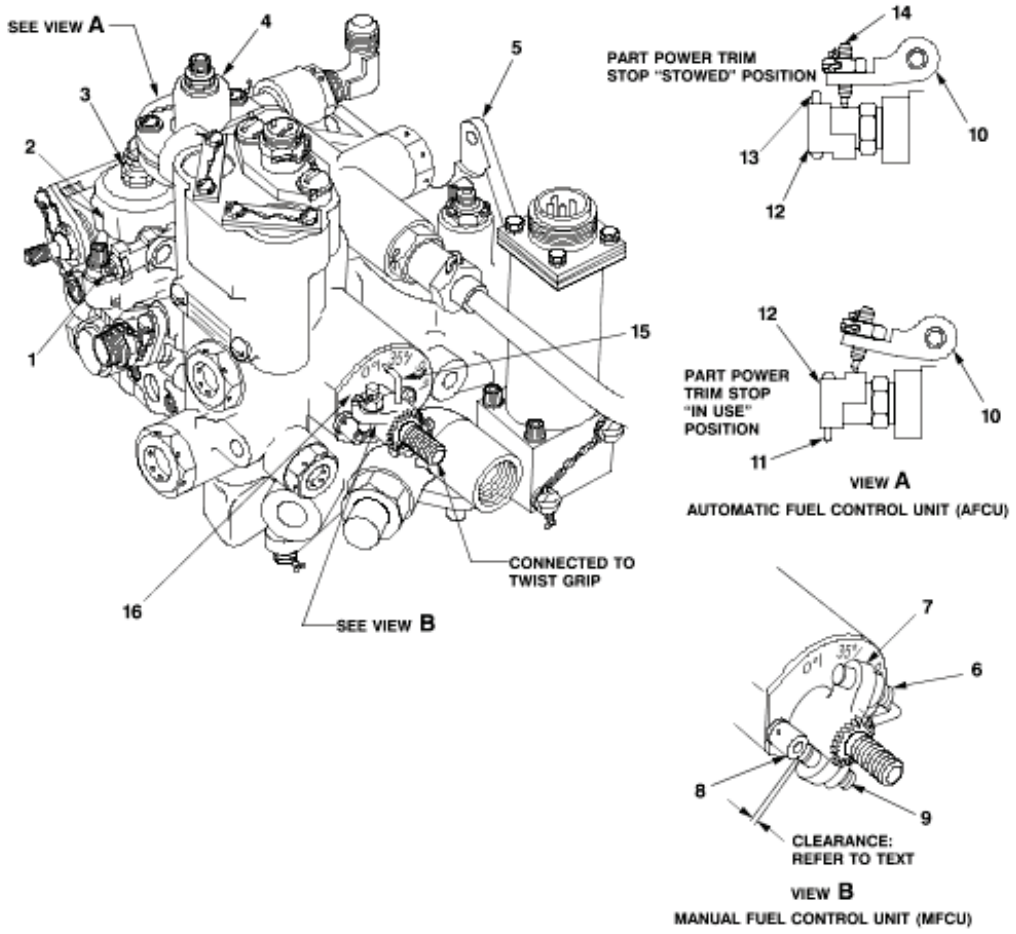
4. Gradually increase collective until N1 topping is achieved and note N1 value.

NOTE

If maximum stop screw is adjusted, readjust the interconnect linkage for proper AUTO/MANUAL gap and throttle cushions per 412-A-76-10-00-00A-276A-A / 01725 and Pratt & Whitney Aircraft of Canada Ltd. PT6T-3 Series Maintenance Manual. This is accomplished once maximum gas generator (N1) speed is attained and part power trim stop is retracted.

5. If required, adjust the topping adjustment screw and repeat Step 4..

6. Repeat Step 3., Step 4., and Step 5. with ENG 2.



1. Manual - automatic fuel control interconnect linkage
2. Datum line
3. Acceleration Adjuster (dome)
4. Automatic Fuel Control Unit (AFCU)
5. Manual Fuel Control Unit (MFCU)
6. Minimum stop screw
7. Stop arm
8. Stop
9. Maximum stop screw
10. Fuel control arm
11. Cotter pin
12. Part power trim stop
13. Cotter pin
14. Maximum stop adjustment screw
15. Pointer
16. Quadrant

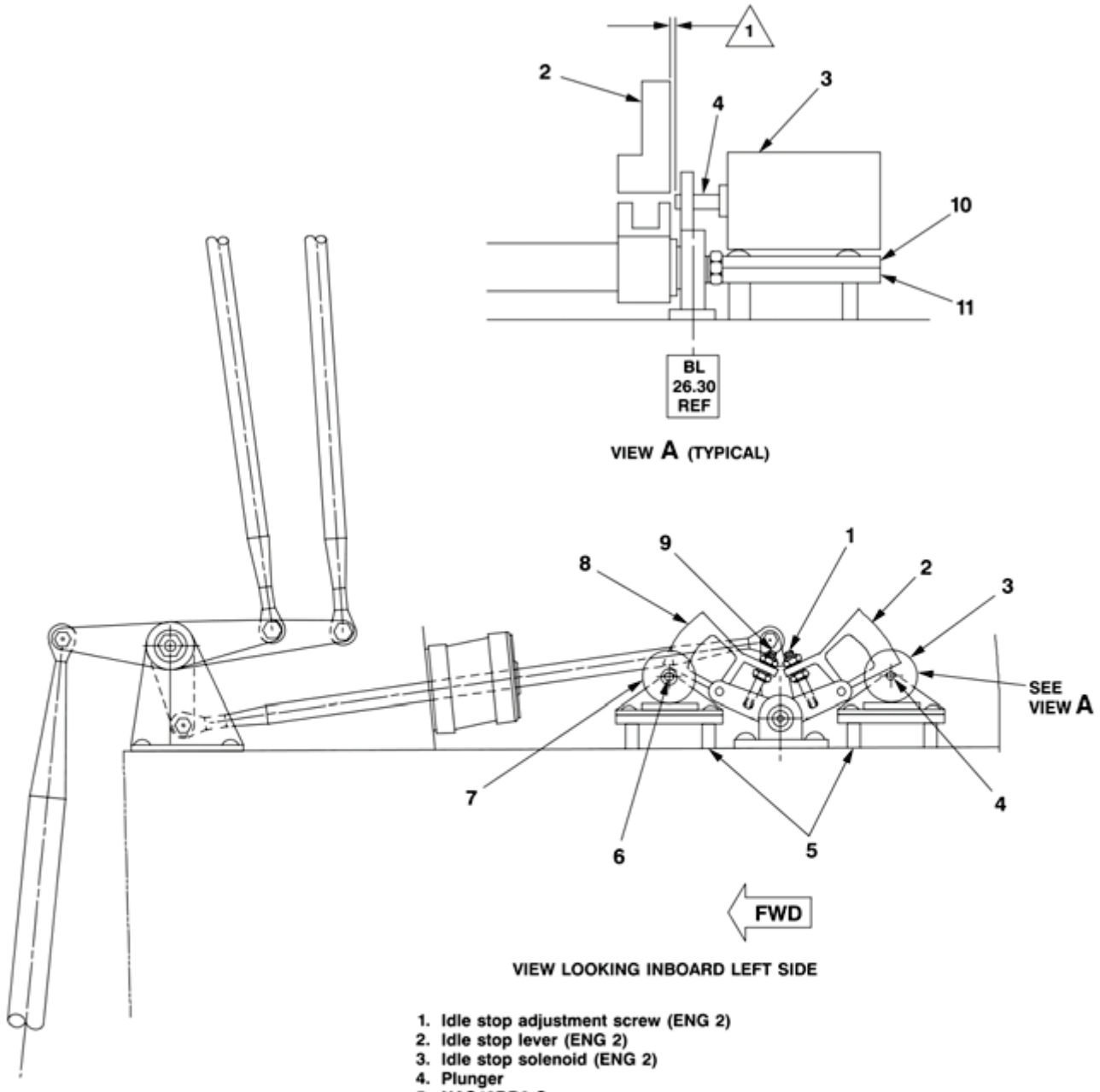
ICN-412-A-761000-B-97499-00015-A-001-01

Figure 1. Manual and Automatic Fuel Control Adjustment Provisions (Sheet 1 of 1)

Engine Idle Speed - Adjustment

Procedure

1. Start both engines and let idle for 5 minutes.
2. With collective at flat pitch and both engines at IDLE, set ENG 1 N1 at 61% using throttle.
3. Gradually increase ENG 1 throttle to FULL OPEN, then retard throttle to IDLE.
4. Verify ENG 1 idle speed is 61 ± 1% N1. If required, adjust idle stop adjustment screw (9, Figure 1). If available adjustment of idle stop adjustment screw (9) is not adequate (insufficient threads), replace four NAS43DD3 spacers (5) under the idle stop solenoid (7) with longer or shorter lengths as required (refer to 412-A-76-10-00-00A-276A-A / 01725).
5. Repeat Step 3. and Step 4. above with ENG 2.



1. Idle stop adjustment screw (ENG 2)
2. Idle stop lever (ENG 2)
3. Idle stop solenoid (ENG 2)
4. Plunger
5. NAS43DD3 Spacers
6. Plunger
7. Idle stop solenoid (ENG 1)
8. Idle stop lever (ENG 1)
9. Idle stop adjustment screw (ENG 1)
10. Bracket assembly
11. Base

NOTE:

1 0.003 to 0.015 in. (0.08 to 0.38 mm) gap typical with plunger retracted.

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Figure 1. Engine Idle Adjustment (Sheet 1 of 1)

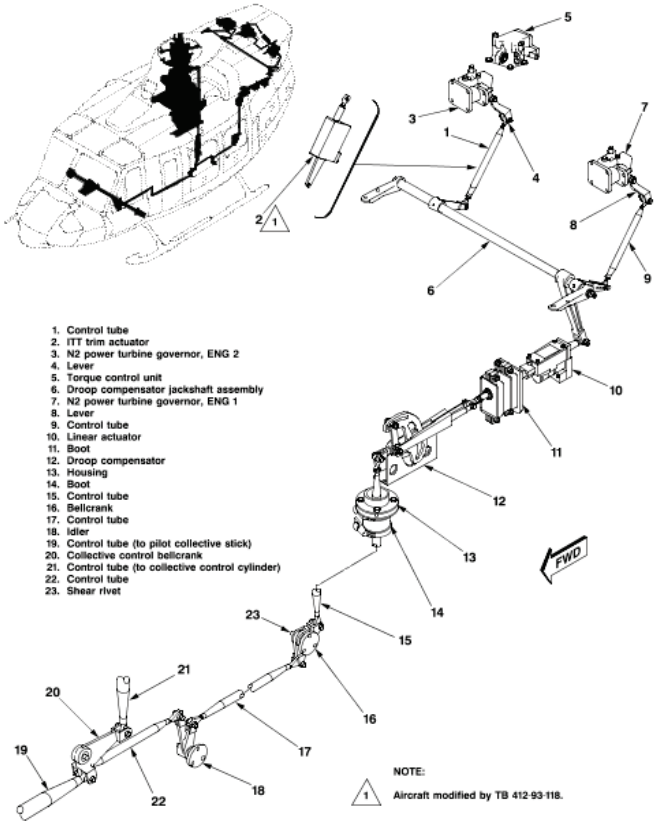
Droop Compensator - Description

The droop compensator linkage extends through droop compensator (12, Figure 1 or 11, Figure 2 and Figure 3) and linear actuator (10, Figure 1 or 9, Figure 2 and Figure 3) to N_2 power turbine governors (3 and 7, Figure 1 through Figure 3). The linear actuator (10, Figure 1 or 9, Figure 2 and Figure 3) can be retracted or extended by use of the RPM INCR/DECR (beep) switch on pilot or copilot collective switchbox to set engine power turbine ENG (N_2) RPM range.

Applicable: SN: 36020-Subsequent or 412-570-001-103: Installed or BHT-412-SI-74: Installed or BHT-412-SI-97: Installed or (SN: 33001-33213 or 36001-36019 TB 412-93-118: Post):

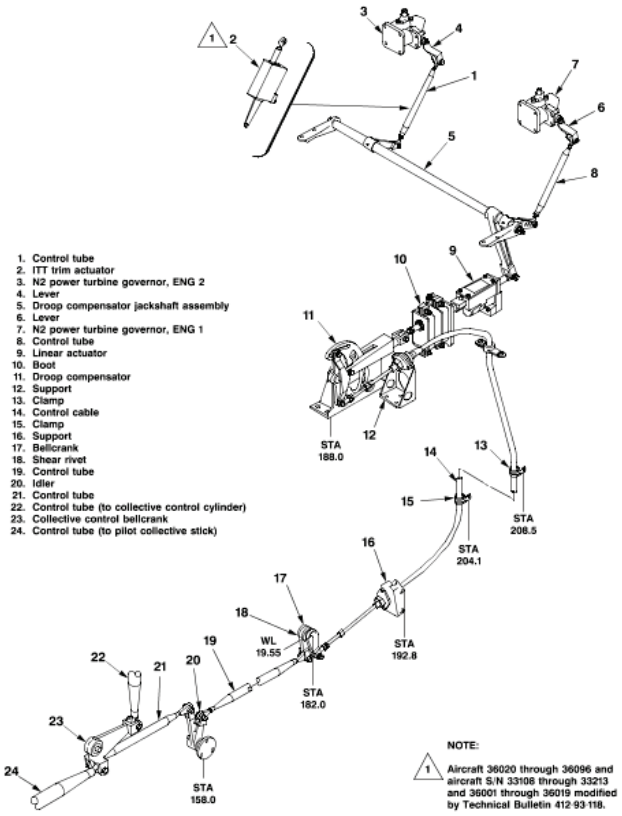
*Helicopters have a +2/-2 switch included in RPM INCR/DECR switch located on pilot collective switchbox (*Figure 4*). The +2/-2 switch controls the ITT actuator (2,* Figure 1 *through* Figure 3*), connected to ENG 2 power turbine governor (3,* Figure 1 *through* Figure 3*), can be extended or retracted to provide torque or ITT matching.*

When pilot/copilot collective controls are moved, droop compensator linkage control tubes (and control cable) are also moved. This repositions cam in droop compensator (12, Figure 1 or 11, Figure 2 and Figure 3) and positions power turbine governors (3 and 7, Figure 1 through Figure 3) to accommodate the need for increased or reduced power as main rotor blades change pitch.



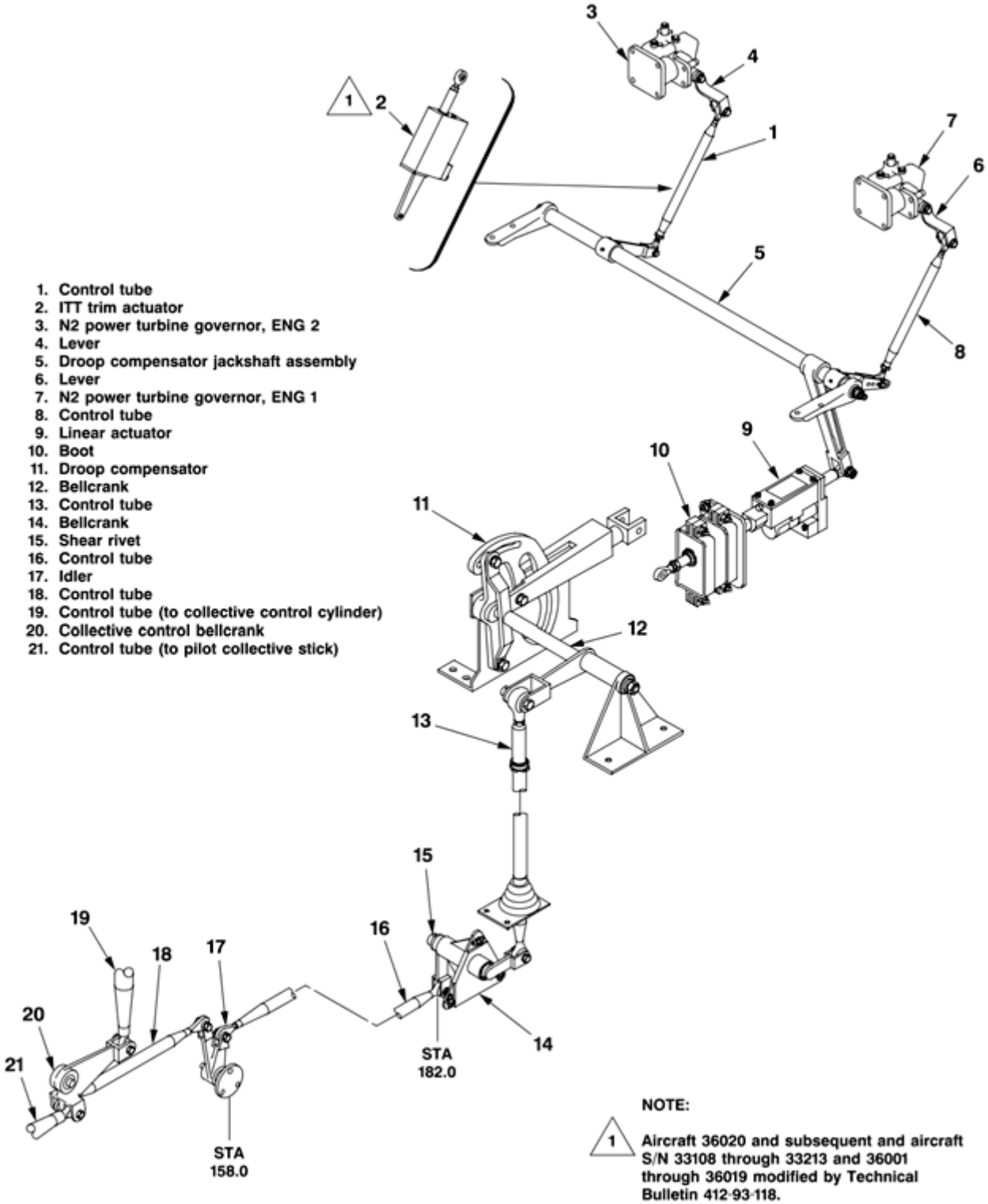
ICN-412-A-760000-B-97499-00002-A-001-01

Figure 1. Droop Compensator Linkage (S/N 33001 through 33107) (Sheet 1 of 1)



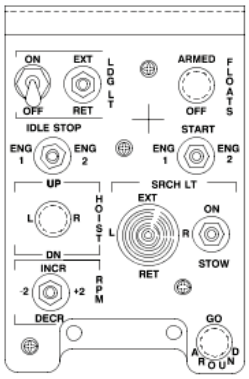
ICN-412-A-760000-B-97499-00003-A-001-01

Figure 2. Droop Compensator Linkage (S/N 33108 through 33213 and S/N 36001 through 36096) (Sheet 1 of 1)



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Figure 3. Droop Compensator Linkage (S/N 36097 and Subsequent and S/N 33108 through 33213 and 36001 through 36096 Modified by Technical Bulletin 412-94-122) (Sheet 1 of 1)



ICN-412-A-761100-B-97499-00001-A-001-01

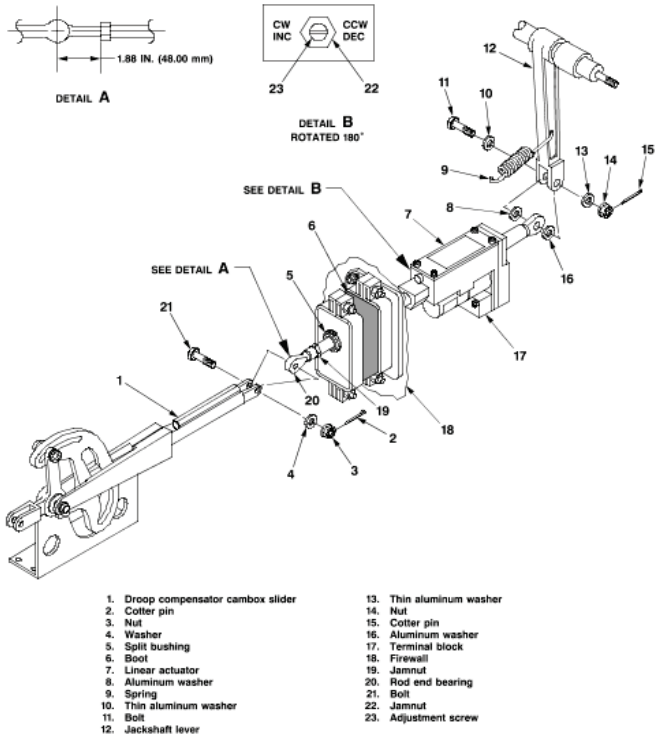
Figure 4. Pilot Collective Switchbox (S/N 36020 and Subsequent and Helicopters Modified by 412-570-001-103 or Post BHT-412-SI-74 (412SP to 412HP Upgrade) or Post BHT-412-SI-97 (412Classic to 412HP Upgrade) and S/N 33001 through 33213, S/N 36001 through 36019 Modified by TB 412-93-118) (Sheet 1 of 1)

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Linear Actuator - Removal

Procedure

1. Remove cotter pin (2, Figure 1), nut (3), washer (4), and bolt (21).
2. Loosen jamnut (19). Remove rod end bearing (20).
3. Remove spring (9).
4. Remove cotter pin (15), nut (14), washers (13, 16, 8, and 10), and bolt (11).
5. Index three wires on terminal block (17) for reinstallation in same location. Disconnect wires from terminal block.
6. Move actuator (7) aft out of the split bushing (5) and remove from helicopter.
7. Install jamnut (19) and rod end bearing (20) in actuator (7) tube.



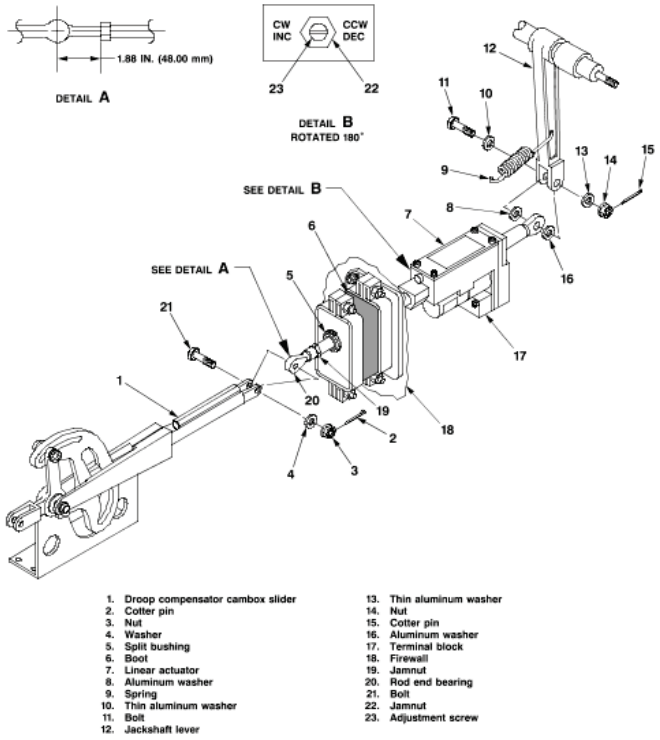
ICN-412-A-761100-B-97499-00002-A-001-01

Figure 1. Linear Actuator (Sheet 1 of 1)

Linear Actuator - Inspection

Procedure

1. Inspect linear actuator (7, Figure 1) for secure installation and damage.
2. Inspect electrical terminal block (17) and associated electrical wiring for damage.
3. Inspect split bushing (5) for damage.



1. Droop compensator cambox slider
2. Cotter pin
3. Nut
4. Washer
5. Split bushing
6. Boot
7. Linear actuator
8. Aluminum washer
9. Spring
10. Thin aluminum washer
11. Bolt
12. Jackshaft lever
13. Thin aluminum washer
14. Nut
15. Cotter pin
16. Aluminum washer
17. Terminal block
18. Firewall
19. Jamnut
20. Rod end bearing
21. Bolt
22. Jamnut
23. Adjustment screw

ICN-412-A-761100-B-97499-00002-A-001-01

Figure 1. Linear Actuator (Sheet 1 of 1)

Linear Actuator - Installation and Rigging

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Corrosion preventive compound	C-101	AR	

Procedure

1. Remove rod end bearing (20, Figure 1) and jamnut (19) from linear actuator (7).

2. Position linear actuator (7) tube through split bushing (5). Install jamnut (19) and rod end bearing (20).

NOTE

Make sure electrical wires are connected to correct terminals.

3. Connect three wires to terminal block (17) (412-A-98-00-00-29A-051A-A / 02489, or 412-A-98-00-00-30A-051A-A / 02490, or 412-A-98-00-00-31A-051A-A / 02491, or 412-A-98-00-00-32A-051A-A / 02492).

4. Position aft end of linear actuator (7) in jackshaft lever (12). Install bolt (11), washers (10, 8, 16, and 13), nut (14), and cotter pin (15).

5. If not previously accomplished, adjust total stroke (from full retract to full extend) of linear actuator (7) as follows:

NOTE

SN: 33001-33213 or 36001-36278:

Linear actuator (7) stroke is 0.62 inch (15.7 mm).

NOTE

(SN: 36279-Subsequent or 36020-36080 BHT-412-SI-54: Installed TB 412-01-176: Post) or (SN: 36081-36278 TB 412-01-176: Post):

Linear actuator (7) stroke is 0.850 inch (21.6 mm).

5.1. While measuring linear actuator (7) stroke, position power turbine governor lever against maximum power stop by holding lever (12) full down (forward).

5.2. Position RPM switch on collective switch box to DECR. Hold until linear actuator (7) reaches full extend position.

NOTE

In the following steps a locally fabricated metal block or similar device may be used as a workaid to assist measuring the actuator stroke.

5.3. Place a metal block on service deck in contact with forward end of linear actuator (7).

5.4. Place RPM INCR/DECR switch to INCR. Hold until linear actuator (7) reaches full retract position.

5.5. Measure and record distance between block positioned in Step 5.3., and forward end of linear actuator (7) rod. If distance is 0.62 or 0.850 inch (15.7 or (21.6 mm), proceed to Step 5.7.. If distance is not 0.62 or 0.850 inch (15.7 mm or 21.6 mm), accomplish Step 5.6. and Step 5.7..



SET LINEAR ACTUATOR (7) TO MIDPOINT TRAVEL WITH RPM SWITCH PRIOR TO MAKING ADJUSTMENT ON SCREW (23) OR DAMAGE TO LINEAR ACTUATOR (7) MAY RESULT.

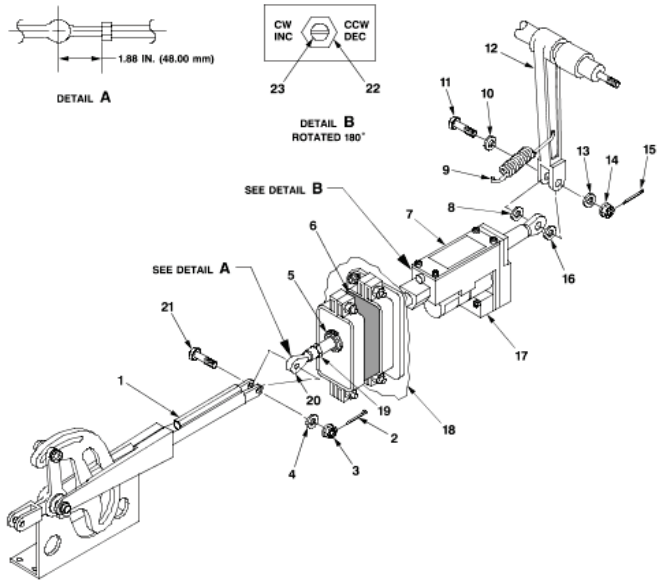
5.6. Set linear actuator (7) to midpoint of travel. Loosen jamnut (22) and rotate screw (23) slightly in INCR direction or DECR direction in accordance with arrows on decal. Repeat making small adjustments until total stroke is set to the proper dimension. Tighten jamnut.

5.7. Remove metal block (installed in Step 5.3.) from service deck.

6. Coat threads of rod end bearing (20) with corrosion preventive compound (C-101). Install rod end bearing and jamnut (19) on linear actuator (7). Dimension from center of rod end bearing bolt hole to edge of jamnut (19) to be 1.88 inches (47.8 mm).

7. Install bolt (21), nut (3), washer (4), and cotter pin (2).

8. Install spring (9).



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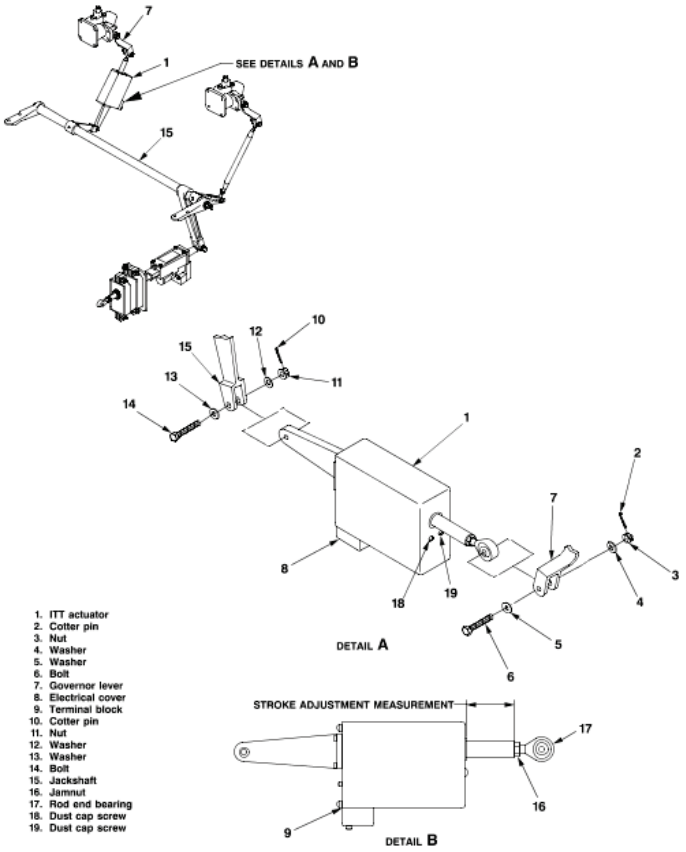
Figure 1. Linear Actuator (Sheet 1 of 1)

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ITT Actuator (ENG 2 Actuator) - Removal

Procedure

1. Remove electrical cover (8, Figure 1) to gain access to the three wires.
2. Index three wires on terminal block (9) for reinstallation in same location. Disconnect wires from terminal block.
3. Remove cotter pin (2), nut (3), washers (4 and 5), and bolt (6) from governor lever (7).
4. Remove cotter pin (10), nut (11), washers (12 and 13), and bolt (14) from jackshaft (15).
5. Remove ITT actuator (1) from helicopter.



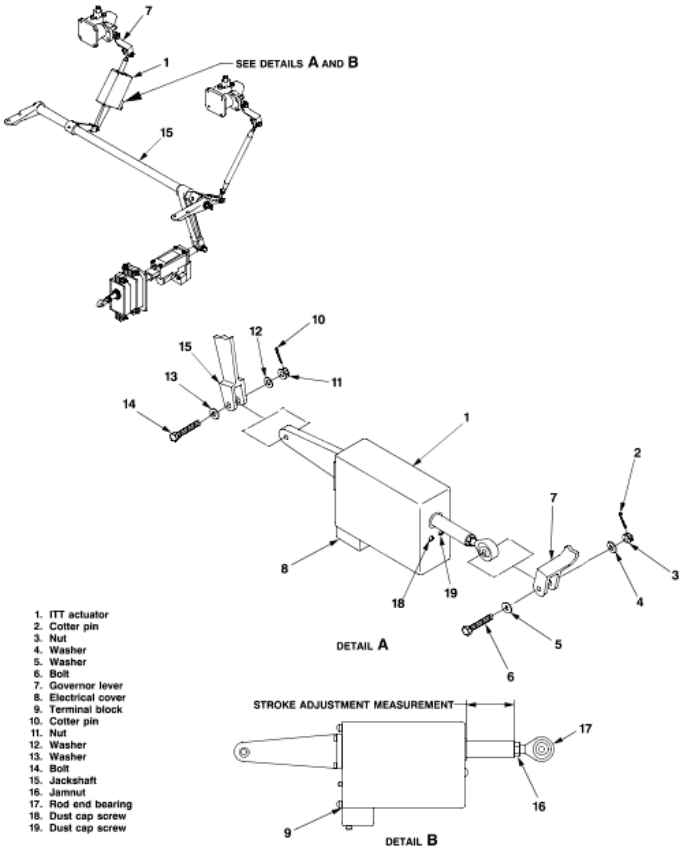
ICN-412-A-761100-B-97499-00003-A-001-01

Figure 1. ITT Actuator Installation (Sheet 1 of 1)

ITT Actuator (ENG 2 Actuator) - Inspection

Procedure

1. Inspect ITT actuator (1, Figure 1) for secure installation and damage.
2. Inspect terminal block (9) and associated electrical wiring for damage.



ICN-412-A-761100-B-97499-00003-A-001-01

Figure 1. ITT Actuator Installation (Sheet 1 of 1)

ITT Actuator (ENG 2 Actuator) - Installation and Rigging

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Corrosion preventive compound	C-101	AR	

Procedure

NOTE

ITT actuator (1, Figure 1) is installed with fixed end connected to jackshaft (15), terminal block (9) facing down, and adjustable rod end (17) connected to lever (7).

1. Position fixed end of actuator (1) on jackshaft (15).

2. Install bolt (14), washers (12 and 13), nut (11), and cotter pin (10).

NOTE

Ensure wires are connected to correct terminals.

3. Connect three wires to terminal block (9) (412-A-98-00-00-29A-051A-A / 02489, or 412-A-98-00-00-30A-051A-A / 02490, or 412-A-98-00-00-31A-051A-A / 02491, or 412-A-98-00-00-32A-051A-A / 02492).

4. Install actuator electrical cover (8).

5. If not previous accomplished, adjust total stroke on actuator (1) to 0.150 in. (3.81 mm) as follows:

5.1. Gain access to adjustment screws on actuator (1) by remove dust cap screws (18 and 19).

5.2. Turn both adjustment screws in direction of arrows till the screws bottom out.

5.3. Using RPM (INCR/DECREASE+2/-2) switch on pilot collective stick box, activate the +2 so that the actuator is fully retracted. Using a fine marker, draw a line on the shaft at the entrance of the actuator body.

5.4. Using RPM (INCR/DECREASE+2/-2) switch on pilot collective stick box, activate the -2 so that the actuator is fully extended. Measure the distance from marker line on the shaft to the entrance of the actuator body and record.

5.5. The recorded value is the total stroke dimension of the actuator. Divide the total stroke by 2 and record.

5.6. The recorded value is the mid stroke of the actuator. Using mid stroke dimension, measure from the marker line on shaft performed in Step 5.3. towards the actuator body. Using a fine marker, draw a line on the shaft; this will be the mid stroke of the actuator.

5.7. Remove marker line on shaft that was installed in Step 5.3..

5.8. Using RPM (INCR/DECREASE+2/-2) switch on pilot collective stick box, activate the +2 so that the actuator retracts to the mid stoke mark on the shaft.

5.9. From the mid position using RPM (INCR/DECREASE+2/-2) switch on pilot collective stick box, activate the –2 so that the actuator extends to 0.075 in. (1.90 mm).

5.10. Turn the EXT (extend) screw in the opposite direction of the arrow (Figure 1) until the EXT (extend) electrical stop switch is engaged. This will be observed by a clicking sound and the adjustment screw will no longer turn.

5.11. From the extend position using RPM (INCR/DECREASE+2/-2) switch on pilot collective stick box; activate the +2 so that the actuator retracts to 0.150 in. (3.81 mm).

5.12. Turn the RET (retract) screw in the opposite direction of the arrow (Figure 1) until the RET (retract) electrical stop switch is engaged. This will be observed by a clicking sound and the adjustment screw will no longer turn.

5.13. Verify actuator total stroke is 0.150 in. (3.81 mm).

5.14. Install dust cap screws (18 and 19).

NOTE

SN: 33001-33213 or 36001-36278:

Helicopters actuator (1) total length is 9.415 in. (239.140 mm).

NOTE

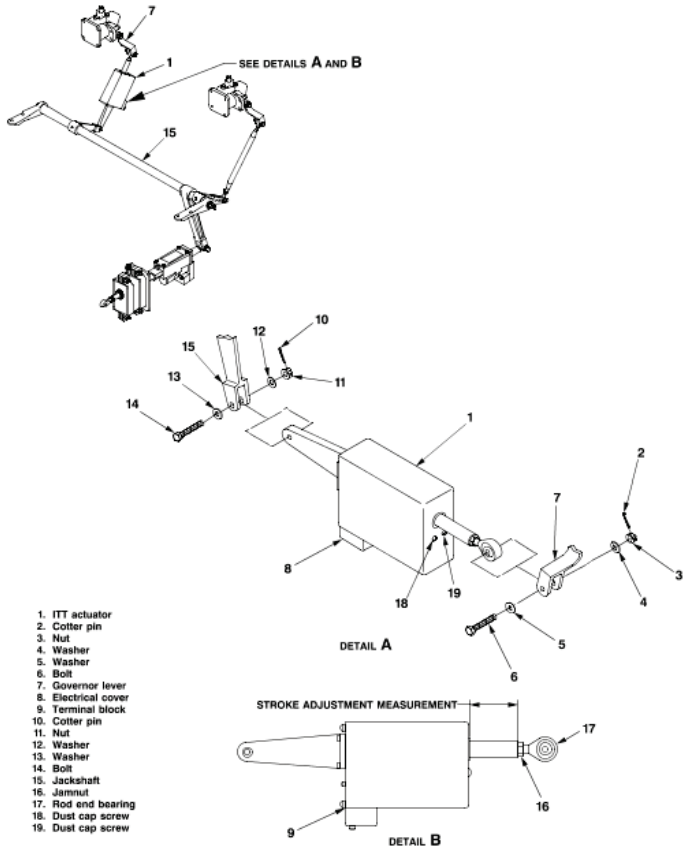
(SN: 36279-Subsequent or 36020-36080 BHT-412-SI-54: Installed TB 412-01-176: Post) or (SN: 36081-36278 TB 412-01-176: Post):

Actuator (1) total length is 9.455 in. (240.157 mm).

6. With actuator (1) fully extended, loosen jamnut (16) and turn rod end bearing (17) to adjust the total length of the actuator to 9.415 in. (239.140 mm) or 9.455 in. (240.157 mm). Tighten jamnut (16). The measurement is taken between attachment points (center of rod end bearing (17) to the center of the actuator attachment hole).

7. Install actuator rod end bearing (17) into governor lever (7).

8. Install bolt (6), washers (5 and 4), nut (3) and cotter pin (2).



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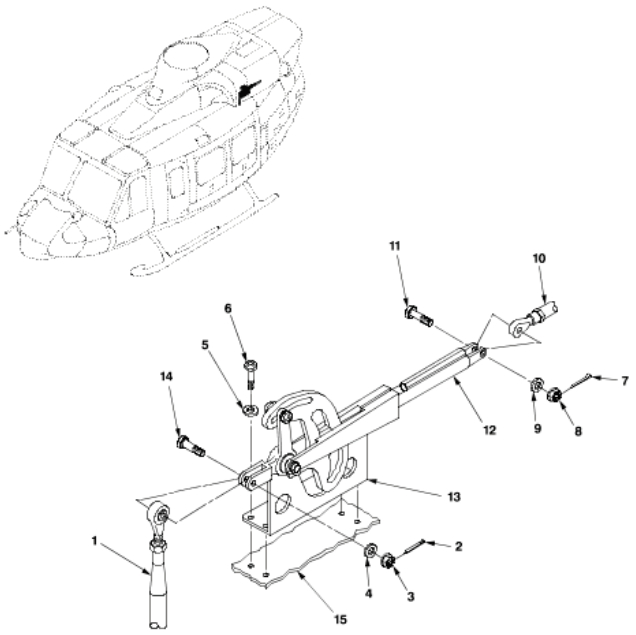
Figure 1. ITT Actuator Installation (Sheet 1 of 1)

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Droop Compensator Cambox - Removal

Procedure

1. Remove cotter pin (7, Figure 1), nut (8), washer (9), and bolt (11).
2. Remove cotter pin (2), nut (3), washer (4), and bolt (14).
3. Remove four screws (6) and washers (5).
4. Remove cambox (13) from service deck (15).



1. Fuel and governor engine control tube
2. Cotter pin
3. Nut
4. Washer (AN960C10)
5. Thin steel washer
6. Screw
7. Cotter pin
8. Nut
9. Washer (AN960C10)
10. Linear actuator tube
11. Bolt
12. Slider assembly
13. Droop compensator cambox assembly
14. Bolt
15. Service deck

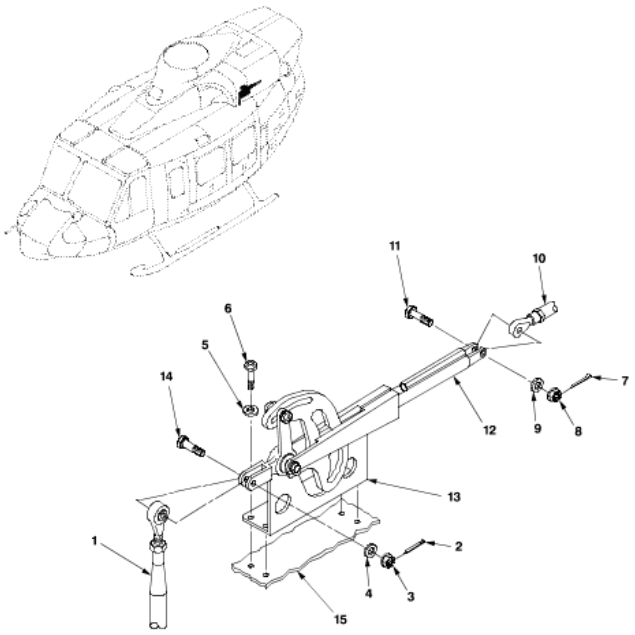
ICN-412-A-761100-B-97499-00004-A-001-01

Figure 1. Droop Compensator Cambox (Aircraft S/N 33001 through 33107) (Sheet 1 of 1)

Droop Compensator Cambox - Inspection

Procedure

1. Inspect cambox (13, Figure 1) for secure installation.
2. Inspect cambox for binding when collective controls are moved through full range.



1. Fuel and governor engine control tube
2. Cotter pin
3. Nut
4. Washer (AN960C10)
5. Thin steel washer
6. Screw
7. Cotter pin
8. Nut
9. Washer (AN960C10)
10. Linear actuator tube
11. Bolt
12. Slider assembly
13. Droop compensator cambox assembly
14. Bolt
15. Service deck

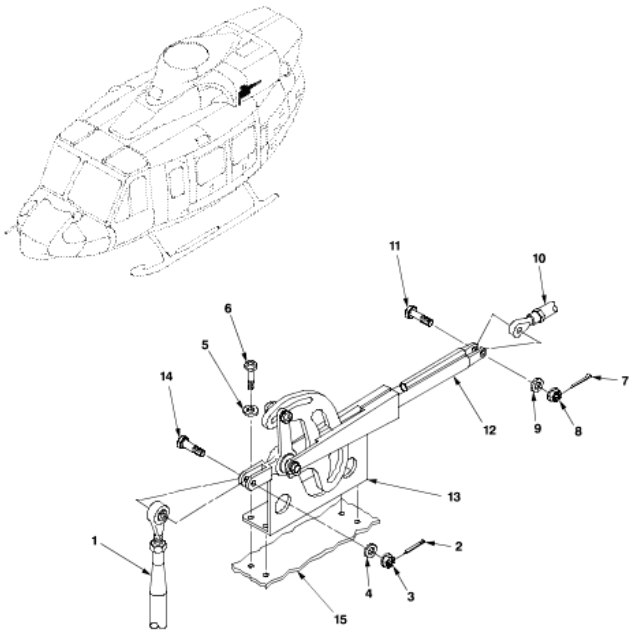
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Figure 1. Droop Compensator Cambox (Aircraft S/N 33001 through 33107) (Sheet 1 of 1)

Droop Compensator Cambox - Installation

Procedure

1. Position cambox (13, Figure 1) on service deck (15).
2. Install screws (6) and washers (5).
3. Position tube (10) on slider (12). Install bolt (11), washer (9), nut (8), and cotter pin (7).
4. Position tube (1) on forward end of cambox (13). Install bolt (14), washer (4), nut (3), and cotter pin (2).
5. Move collective controls through full range and inspect for binding and interference in droop compensator control installation.
6. Rig droop compensator controls per 412-A-76-11-00-04A-276A-A / 01750 or 412-A-76-11-00-04A-276B-A / 01751 depending on aircraft configuration.



1. Fuel and governor engine control tube
2. Cotter pin
3. Nut
4. Washer (AN960C10)
5. Thin steel washer
6. Screw
7. Cotter pin
8. Nut
9. Washer (AN960C10)
10. Linear actuator tube
11. Bolt
12. Slider assembly
13. Droop compensator cambox assembly
14. Bolt
15. Service deck

ICN-412-A-761100-B-97499-00004-A-001-01

Figure 1. Droop Compensator Cambox (Aircraft S/N 33001 through 33107) (Sheet 1 of 1)

N2 Control - Rigging

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Corrosion preventive compound	C-101	AR	

Procedure

NOTE

Refer to 412-A-76-10-00-00A-276A-A / 01725 for N1 (power lever control) rigging.

1. Comply with the following general procedures when rigging droop compensator controls.

1.1. When a rigging procedure requires a specific position to accomplish following steps, manually hold controls in position or secure controls in position with pilot friction adjustment.

1.2. Tolerance for rigging dimensions is ±0.030 in. (0.76 mm) unless otherwise stated.

1.3. All adjustable control tubes must have a maximum of 1.00 in. (25.4 mm) of exposed threads after adjustment. If control tubes have inspection holes, threads must be engaged sufficiently to cover holes.

1.4. Apply corrosion preventive compound (C-101) to threads of all adjustable rod end bearings and clevises after adjustment.

1.5. Torque control jamnuts 80 to 100 in-lbs (9.0 to 11.3 N-m) after adjustment unless otherwise specified.

1.6. Control tubes must be free to rotate several degrees about longitudinal axis. If necessary, loosen jamnuts, adjust end fitting and re-torque jamnuts.

2. Make sure that collective pitch control system rigging has been accomplished (412-A-67-01-00-01A-276A-A / 01323).

3. make sure that levers (2 and 6, Figure 1) are correctly installed on control shafts of power turbine governors (1 and 5). Nominal setting positions the centerline of levers (2 and 6) 15 degrees (one serration) clockwise from centerline of governor stop arm (21), viewed from left looking inboard (detail A). In subsequent steps levers (2 and 6) may be repositioned ±15 degrees (one serration) from the nominal setting if required to achieve proper rigging and eliminate any fouling conditions.

4. Connect control tube (3) to governor lever (2) and jackshaft (4).

NOTE

The dimension given in Step 5. is approximate for initial rigging. Minor adjustments may be necessary during ground and flight test.

5. Make sure that adjustable tube (7) is approximately 9.34 in. (237.23 mm) measured between attachment points (center of rod end bearings).

6. Connect tube (7) to governor lever (6) and to the middle hole of jackshaft lever (28) (detail C).

7. Disconnect actuator (8) from lever of jackshaft (4).

8. Disconnect actuator (8) rod end bearing from cambox slider (25).

9. Disconnect tube (18) from cambox (10).

NOTE

The dimension given in Step 10. is approximate for initial rigging. Minor adjustments may be necessary during ground and flight test.

10. Make sure that long arm of cambox bellcrank (26) is attached approximately in the middle of adjustment slot of cam (23) (detail B).

11. Use RPM INCR/DECR (beep) switch on collective switchbox to operate actuator. Make sure that actuator (8) is adjusted to a total stroke of approximately 0.62 in. (15.7 mm) from full extend (DECR) to full retract (INCR) positions. If not, refer to 412-A-76-11-00-01A-720A-A / 01743 for actuator rigging procedure.

12. Adjust rod end on actuator (8) to approximately 1.88 in. (47.8 mm) from center of bearing bolt hole to face of jamnut (detail D).

13. Connect actuator (8) rod end bearing to cambox slider (25).

14. Connect actuator (8) to lever of jackshaft (4).

15. Connect tube (18) to cambox (10).

16. Disconnect tube (16) from bellcrank (17) and rotate bellcrank (17) to make sure that there is no fouling. Rotate cambox bellcrank (26) (detail B) and make sure that there is no fouling.

17. Connect tube (16) to bellcrank (17).

NOTE

If hydraulic test unit is not available disconnect control tube (15).

18. Connect hydraulic test unit (412-A-29-00-00-00A-320A-A / 00616).

19. With hydraulic boost ON, place collective stick full down. Adjust the length of tube (18) to hold cam (23) so lower end of cam slot is approximately 0.280 in. (7.11 mm) below slider housing (24).

20. With hydraulic boost ON, place collective stick full up. Verify cam slot is approximately 0.280 in. (7.11 mm) above slider housing (24).

21. With collective stick full up, hold RPM INCR/DECR switch to INCR until actuator (8) is fully retracted. Make sure that minimum clearance of 0.010 in. (0.25 mm) (detail A) between stop arm (21) and maximum power stop screw (22) on both governors (1 and 5).

22. With hydraulic boost ON, place collective stick full down. Hold RPM INCR/DECR switch to DECR until actuator (8) is fully extended. Make sure tha minimum clearance of 0.010 in. (0.25 mm) between stop arm (21) and minimum power stop screw (20) on both governors (1 and 5).

23. Disconnect hydraulic test unit or connect control tube (15) as applicable.



MAKE SURE THAT ALL CONTROLS ARE PROPERLY CONNECTED PRIOR TO GROUND RUN AND FLIGHT TEST.

NOTE

Prior to accomplishing ground run make sure that Automatic Fuel Control Unit (AFCU) and Manual Fuel Control Unit (MFCU) maximum stops screws are being contacted with throttles full open (refer to 412-A-76-10-00-00A-276A-A / 01725). During ground run make sure that acceleration time difference between power sections is within 0.5 seconds (refer to 412-A-76-10-00-00A-280F-A / 01735, 412-A-76-10-00-00A-271E-A / 01736 and 412-A-76-10-00-00A-280G-A / 01737).

24. Start both power sections per Flight Manual and idle for 5 minutes.

25. Actuate RPM INCR/DECR switch to full DECR.

26. Place collective stick full down. Rotate ENG 2 throttle to full INCR. Record ROTOR rpm (NR). The recorded value should be 95% ROTOR rpm. If ROTOR rpm is not 95% adjust as follows.



RETURN ENGINES TO IDLE FOR ANY ADJUSTMENTS TO CONTROLS WITH ENGINE RUNNING.

NOTE

Three turns of actuator (8) rod end is equal to 1% ROTOR rpm change.

26.1. If ROTOR rpm is below 95%, shorten rod end on actuator (8).

26.2. If ROTOR rpm is above 95%, lengthen rod end on actuator (8).

NOTE

The single engine beep range should be 95 to 99.5% ROTOR rpm.

27. Slowly actuate RPM INCR/DECR switch to full INCR and make sure that the ROTOR rpm is 99 to 99.5%. If the range is not within limits adjust actuator (8) stroke per 412-A-76-11-00-01A-720A-A / 01743.

28. Actuate RPM INCR/DECR switch to full DECR.

29. Return ENG 2 to IDLE, rotate ENG 1 throttle to full INCR. Record ROTOR rpm. The recorded value should be 95% ROTOR rpm. If ROTOR rpm is not 95% adjust as follows.



RETURN ENGINES TO IDLE FOR ANY ADJUSTMENTS TO CONTROLS WITH ENGINE RUNNING.

NOTE

Two turns of tube (7) rod end is equal to 1% ROTOR rpm change.

29.1. If ROTOR rpm is below 95%, shorten rod end on tube (7).

29.2. If ROTOR rpm is above 95%, lengthen rod end on tube (7).

NOTE

The single engine beep range should be 95 to 99.5% ROTOR rpm.

30. Slowly actuate RPM INCR/DECR switch to full INCR and make sure that the ROTOR rpm is 99 to 99.5%.

NOTE

When performing the following steps helicopter will become airborne.

31. Return ENG 1 to IDLE, rotate ENG 2 to full INCR. Slowly actuate RPM INCR/DECR switch to 98% ROTOR rpm. Gradually increase collective from flat pitch to 60% engine torque and not static ROTOR rpm droop.

32. Return ENG 2 to IDLE, rotate ENG 1 to full INCR. Slowly actuate RPM INCR/DECR switch to 98% ROTOR rpm. Gradually increase collective from flat pitch to 60% engine torque and not static ROTOR rpm droop.

33. If more than 2% ROTOR rpm droop is encountered perform troubleshooting of engine pneumatic lines and N2 power turbine governor per Pratt & Whitney Maintenance Manual before proceeding with further adjustments.

34. Actuate RPM INCR/DECR switch to full DECR.

35. With collective stick full down, rotate ENG 1 and ENG 2 throttles to FULL INCR. Actuate RPM INCR/DECR switch to 100% ROTOR rpm.

NOTE

Engine torque difference (split) between ENG 1 and ENG 2 shall not exceed 4% from flat pitch to maximum power.

36. Gradually increase collective from flat pitch to full power (either transmission torque, ITT or N1 limit) and note the difference in engine torque readings. If not within limits, adjust as follows:

36.1. If ENG 1 torque reading is greater than ENG 2 by more than 4% throughout the collective range, lengthen tube (7).

36.2. If ENG 1 torque reading is less than ENG 2 by more than 4% throughout the collective range, shorten tube (7).

36.3. If ENG 1 torque reading is greater than ENG 2 by more than 4% at high power range and too low at low power range, move tube (7) rod end forward (DECR) on lever (28) (detail C).

36.4. If ENG 1 torque reading is less than ENG 2 by more than 4% at high power range and/or too low at low power range, move tube (7) rod end AFT (INCR) on lever (28) (detail C).

37. Place collective stick full down; rotate both engine throttles to full INCR. Use RPM INCR/DECR switch to obtain 100% ROTOR rpm. Increase collective pitch slowly in a series of equal steps from flat pitch to full power (either transmission torque, ITT or N1 limit). ROTOR rpm should remain at 100 ± 1% throughout power sweep.

38. If ROTOR rpm droops (decays below tolerance) or overspeeds (increase above tolerance) during power applications, make a cam rate adjustment.

NOTE

Make adjustments in small increments of approximately 0.125 in. (3.18 mm) measured on cam slot.

38.1. For increased cam compensation to correct droop, adjust cam (23) counterclockwise relative to bellcrank (26) (detail B).

38.2. For reduced cam compensation to correct overspeed, adjust cam (23) clockwise relative to bellcrank (26).

38.3. After each cam adjustment, move the collective to “full down” and then “full up” positions to make sure that there is no fouling between slider housing (24) and cambox slot. make sure that a minimum of 0.010 in. (0.25 mm) cam slot shows below and above slider housing (24). If necessary adjust tube (18) for proper clearance.



RETURN ENGINES TO IDLE FOR ANY ADJUSTMENTS TO CONTROLS WITH ENGINE RUNNING.

NOTE

Adjustments to achieve proper twin-engine rpm range and adjustments to avoid torque needle split, will supersede the initial single engine beep ranges.

Three turns of actuator (8) rod end is equal to 1% ROTOR rpm change.

39. Actuate RPM INCR/DECR switch to full decrease with collective in full down position. Rotate both engine throttles to full INCR and record ROTOR rpm. The recorded ROTOR rpm should be 97%. If not adjust actuator (8) rod end.

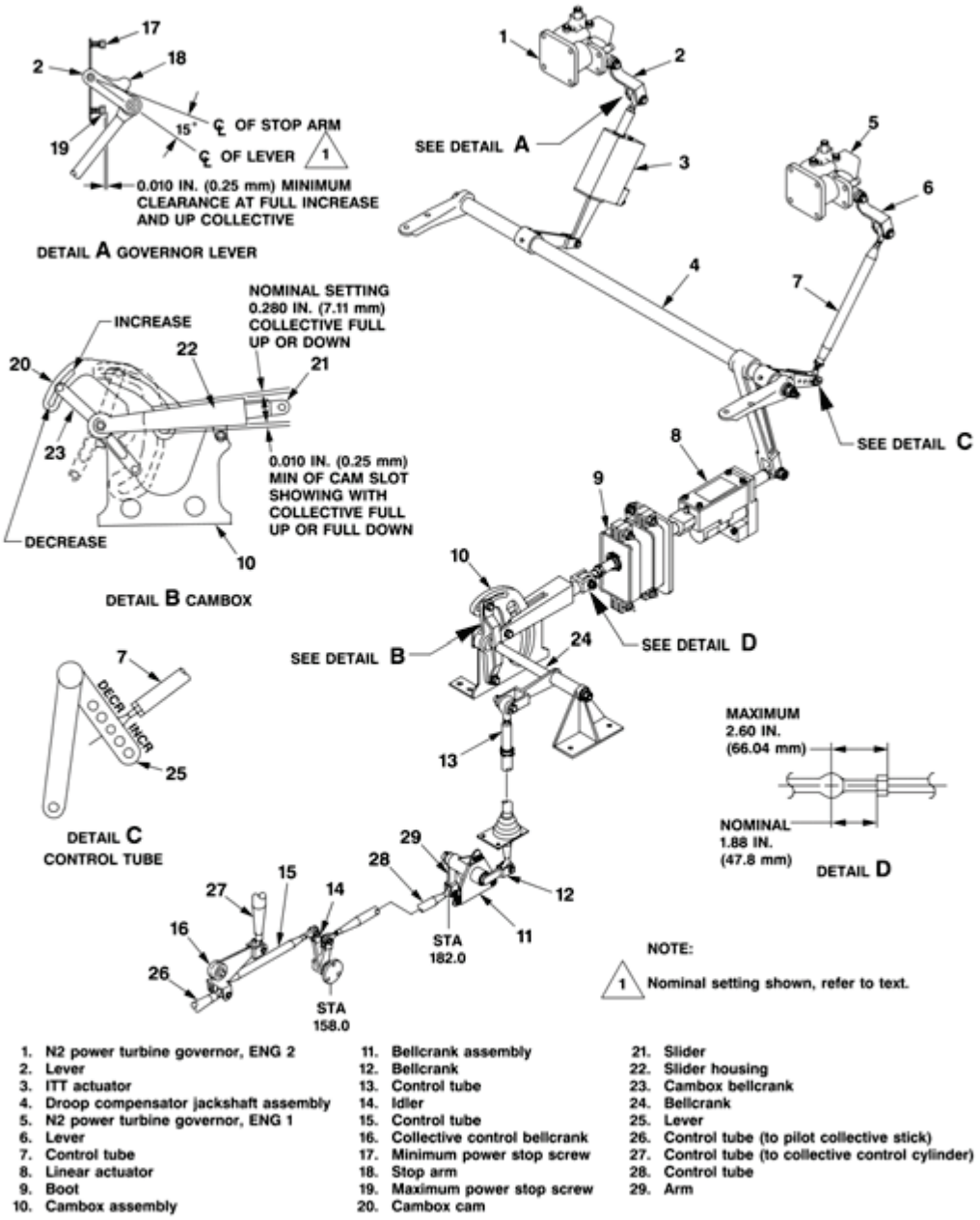
40. Actuate RPM INCR/DECR switch to full INCR and verify ROTOR rpm is 101.5%. The required range for twin-engine beep is 97 to 101.5% ROTOR rpm. If not, adjust actuator (8) stroke per 412-A-76-11-00-01A-720A-A / 01743.

NOTE

If hydraulic test unit is not available disconnect control tube (15).

41. With collective stick full up, hold RPM INCR/DECR switch to INCR until actuator (8) is fully retracted. Make sure that minimum clearance of 0.010 in. (0.25 mm) (detail A) between stop arm (21) and maximum power stop screw (22) on both governors (1 and 5).

42. Check complete system for security of all parts and make sure that no interference exists. Apply corrosion preventive compound (C-101) to all exposed threads.



1. N2 power turbine governor, ENG 2
2. Lever
3. ITT actuator
4. Droop compensator jackshaft assembly
5. N2 power turbine governor, ENG 1
6. Lever
7. Control tube
8. Linear actuator
9. Boot
10. Cambox assembly
11. Bellcrank assembly
12. Bellcrank
13. Control tube
14. Idler
15. Control tube
16. Collective control bellcrank
17. Minimum power stop screw
18. Stop arm
19. Maximum power stop screw
20. Cambox cam
21. Slider
22. Slider housing
23. Cambox bellcrank
24. Bellcrank
25. Lever
26. Control tube (to pilot collective stick)
27. Control tube (to collective control cylinder)
28. Control tube
29. Arm

ICN-412-A-761100-B-97499-00005-A-001-01

Figure 1. N2 Controls Rigging (S/N 33001 through 33107) (Sheet 1 of 1)

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N2 Control - Rigging

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Corrosion preventive compound	C-101	AR	

Procedure

NOTE

Refer to 412-A-76-10-00-00A-276A-A / 01725 for N1 (power lever control) rigging.

1. Comply with the following general procedures when rigging droop compensator controls.

1.1. When a rigging procedure requires a specific position to accomplish following steps, manually hold controls in position or secure controls in position with pilot friction adjustment.

1.2. Tolerance for rigging dimensions is ±0.030 in. (0.76 mm) unless otherwise stated.

1.3. All adjustable controls tubes must have a maximum of 1.00 in. (25.4 mm) of exposed threads after adjustment. If control tubes have inspection holes, threads must be engaged sufficiently to cover holes.

1.4. Apply corrosion preventive compound (C-101) to threads of all adjustable rod end bearings and clevises after adjustment.

1.5. Torque control jamnuts 80 to 100 in-lbs (9.0 to 11.3 N-m) after adjustment unless otherwise specified.

1.6. Control tubes must be free to rotate several degrees about longitudinal axis. If necessary, loosen jamnuts, adjust end fitting and re-torque jamnuts.

2. Make sure that collective pitch control system rigging has been accomplished (412-A-67-01-00-01A-276A-A / 01323).

3. Make sure that levers (2 and 6, Figure 1) are correctly installed on control shafts of power turbine governors (1 and 5). Nominal setting positions the centerline of levers (2 and 6) 15 degrees (one serration) clockwise from centerline of governor stop arm (21), viewed from left looking inboard (detail A). In subsequent steps levers (2 and 6) may be repositioned ±15 degrees (one serration) from the nominal setting if required to achieve proper rigging and eliminate any fouling conditions.

4. Connect ENG 2 ITT actuator (27) to governor lever (2) and jackshaft (4). Make sure that total stroke of actuator (27) is 0.150 in. (3.81 mm) and total length is 9.415 in. (239.14 mm) with actuator fully extended. (Refer to 412-A-76-11-00-02A-720A-A / 01746 for installation and rigging procedure.)

NOTE

The dimension given in Step 5. is approximate for initial rigging. Minor adjustments may be necessary during ground and flight test.

5. Make sure that adjustable tube (7) is approximately 9.34 in. (237.23 mm) measured between attachment points (center of rod end bearings).

6. Connect tube (7) to governor lever (6) and to the middle hole of jackshaft lever (28) (detail C).

7. Disconnect actuator (8) from lever of jackshaft (4).

8. Disconnect actuator (8) rod end bearing from cambox slider (25).

9. Disconnect tube (18) from cambox (10).

NOTE

The dimension given in Step 10. is approximate for initial rigging. Minor adjustments may be necessary during ground and flight test.

10. Make sure that long arm of cambox bellcrank (26) is attached approximately in the middle of adjustment slot of cam (23) (detail B).

11. Use RPM INCR/DECR (beep) switch on collective switchbox to operate actuator. Make sure that actuator (8) is adjusted to a total stroke of approximately 0.62 in. (15.7 mm) from full extend (DECR) to full retract (INCR) positions. If not, refer to 412-A-76-11-00-01A-720A-A / 01743 for actuator rigging procedure.

12. Adjust rod end on actuator (8) to approximately 1.88 in. (48 mm) from center of bearing bolt hole to face of jamnut (detail D).

13. Connect actuator (8) rod end bearing to cambox slider (25).

14. Connect actuator (8) to lever of jackshaft (4).

15. Connect tube (18) to cambox (10).

16. Disconnect tube (16) from bellcrank (17) and rotate bellcrank (17) to make sure that there is no fouling. Rotate cambox bellcrank (26) (detail B) and make sure that there is no fouling.

17. Connect tube (16) to bellcrank (17).

NOTE

If hydraulic test unit is not available disconnect control tube (15).

18. Connect hydraulic test unit (412-A-29-00-00-00A-320A-A / 00616).

19. With hydraulic boost ON, place collective stick full down. Adjust the length of tube (18) to hold cam (23) so lower end of cam slot is approximately 0.280 in. (7.11 mm) below slider housing (24).

20. With hydraulic boost ON, place collective stick full up. Make sure that cam slot is approximately 0.280 in. (7.11 mm) above slider housing (24).

21. With collective stick full up, hold RPM INCR/DECR switch to INCR until actuator (8) is fully retracted. Hold ENG 2 (+2/-2) switch to -2 until ITT actuator (27) is fully retracted. Make sure that minimum clearance of 0.010 in. (0.25 mm) (detail A) between stop arm (21) and maximum power stop screw (22) on both governors (1 and 5).

22. With hydraulic boost ON, place collective stick full down. Hold RPM INCR/DECR switch to DECR until actuator (8) is fully extended. Hold ENG 2 (+2/-2) switch to +2 until ITT actuator (27) is fully extended. Make sure that minimum clearance of 0.010 in. (0.25 mm) between stop arm (21) and minimum power stop screw (20) on both governors (1 and 5).

23. Disconnect hydraulic test unit or connect control tube (15) as applicable.



MAKE SURE THAT ALL CONTROLS ARE PROPERLY CONNECTED PRIOR TO GROUND RUN AND FLIGHT TEST.

NOTE

Prior to accomplishing ground run make sure that Automatic Fuel Control Unit (AFCU) and Manual Fuel Control Unit (MFCU) maximum stops screws are being contacted with throttles full open (refer to 412-A-76-10-00-00A-276A-A / 01725). During ground run make sure that acceleration time difference between power sections is within 0.5 seconds (refer to 412-A-76-10-00-00A-280F-A / 01735, 412-A-76-10-00-00A-271E-A / 01736 and 412-A-76-10-00-00A-280G-A / 01737).

24. Start both power sections per Flight Manual and idle for 5 minutes.

25. Actuate RPM INCR/DECR switch to full DECR.

NOTE

Positioning of ENG 2 ITT actuator (27) to mid stroke can be done during ground run by actuating ENG 2 (+2/-2) switch to full +2, record ROTOR rpm. Actuate ENG 2 (+2/-2) switch to full -2, record ROTOR rpm. Actuate ENG 2 (+2/-2) switch to set ROTOR rpm to middle of recorded range.

26. Ensure ENG 2 ITT actuator (27) is at mid stroke position, 9.34 in. (237.2 mm) before proceeding to next step.

27. Place the collective stick full down. Rotate the ENG 2 throttle to full INCR. Record the rotor RPM (NR). Slowly actuate the RPM INCR/DECR switch to full INCR until 101% rotor RPM is reached. Record the rotor RPM. Verify that full INCR produces a 99% rotor RPM minimum.



RETURN ENGINES TO IDLE FOR ANY ADJUSTMENTS TO CONTROLS WITH ENGINE RUNNING.

NOTE

Three turns of actuator (8) rod end is equal to 1% ROTOR rpm change.

28. If minimum ROTOR rpm cannot be obtained, make an interim adjustment to the rod end of actuator (8).

29. With the collective full down, rotate ENG 2 throttle to full INCR. Actuate RPM INCR/DECR switch to full DECR and ENG 2 (+2/-2) switch to full +2. Record ROTOR rpm. Actuate ENG 2 (+2/-2) switch to full -2 and record ROTOR rpm. The total spread between ROTOR rpm should be 2 1/4 ± 1/4%. If spread is not within limits, adjust ITT actuator (27) stroke per 412-A-76-11-00-02A-720A-A / 01746.

30. Return ENG 2 to IDLE, rotate ENG 1 throttle to full INCR. Record ROTOR rpm. Slowly actuate RPM INCR/DECR switch to full INCR but not to exceed 101% ROTOR rpm. Record ROTOR rpm. The minimum full INCR should be 99% ROTOR rpm.



RETURN ENGINES TO IDLE FOR ANY ADJUSTMENTS TO CONTROLS WITH ENGINE RUNNING.

NOTE

Two turns of tube (7) rod end is equal to 1% ROTOR rpm change.

31. If the minimum ROTOR rpm cannot be obtained, make an interim adjustment to the rod end of tube (7).

NOTE

When performing the following steps helicopter will become airborne.

32. Return ENG 1 to IDLE, rotate ENG 2 to full INCR. Slowly actuate RPM INCR/DECR switch to 98% ROTOR rpm. Gradually increase collective from flat pitch to 60% engine torque and note static ROTOR rpm droop.

33. Return ENG 2 to IDLE, rotate ENG 1 to full INCR. Slowly actuate RPM INCR/DECR switch to 98% ROTOR rpm. Gradually increase collective from flat pitch to 60% engine torque and note static ROTOR rpm droop.

34. If more than 2% ROTOR rpm droop is encountered perform troubleshooting of engine pneumatic lines and N2 power turbine governor per Pratt & Whitney Maintenance Manual before proceeding with further adjustments.

35. With rotor at flat pitch, increase both throttles to full INCR and adjust ROTOR rpm to 100% using RPM INCR/DECR switch.

36. Gradually increase collective from flat pitch to approximately 60% transmission torque. Using the +2/-2 switch, trim ENG 2 to minimum, and then to maximum and note the torque split. The torque split should be approximately equal above and below ENG 1 torque indication. If the torque split difference is greater than 5% above or below ENG 1 torque, adjust length of tube (7) as follows.

NOTE

Half turn of rod end on tube (7) is equal to approximately 5% engine torque change.

36.1. To increase torque on ENG 1, shorten rod end of tube (7).

36.2. To decrease torque on ENG 1, lengthen rod end of tube (7).

NOTE

Engine torque difference (split) between ENG 1 and ENG 2 shall not exceed 4% during power application throughout this maintenance procedure. However, during normal operation torque split may exceed 4%. +2/-2 switch may be used to trim engines for either torque matching or ITT matching.

37. At 100% ROTOR rpm, gradually increase collective to 60% transmission torque and match engines torques using +2/–2 switch. Gradually decrease collective to flat pitch and note the torque split. Gradually increase collective to full power (either transmission torque, ITT, or N1) and note the torque split throughout the power range, torque matching should remain constant within 4%. If torque does not remain constant throughout the power range, adjust as follows.

37.1. If ENG 1 torque reading is greater than ENG 2 by more than 4%, move tube (7) rod end forward (DECR) on lever (28) (detail C).

37.2. If ENG 1 torque reading is less than ENG 2 by more than 4%, move tube (7) rod end AFT (INCR) on lever (28) (detail C).

38. Place collective stick full down, rotate both engine throttles to full INCR. Use RPM INCR/DECR switch to obtain 100% ROTOR rpm and match engine torques using +2/-2 switch. Increase collective pitch slowly in a series of equal steps from flat pitch to full power (either transmission torque, engine ITT or N1 limit). ROTOR rpm should remain at 100 ± 1% throughout power sweep.

39. If ROTOR rpm droops (decays below tolerance) or overspeeds (increases above tolerance) during power applications, make a cam rate adjustment.

NOTE

Make adjustments in small increments of approximately 0.125 in. (3.18 mm) measured on cam slot.

39.1. For increased cam compensation to correct droop, adjust cam (23) counterclockwise relative to bellcrank (26) (detail B).

39.2. For reduced cam compensation to correct overspeed, adjust cam (23) clockwise relative to bellcrank (26).

39.3. After each cam adjustment, move the collective to “full down” and then “full up” positions to make sure that there is no fouling between slider housing (24) and cambox slot. Make sure that a minimum of 0.010 in. (0.25 mm) cam slot shows below and above slider housing (24). If necessary adjust tube (18) for proper clearance.



RETURN ENGINES TO IDLE FOR ANY ADJUSTMENTS TO CONTROLS WITH ENGINE RUNNING.

NOTE

Three turns of actuator (8) rod end is equal to 1% ROTOR rpm change.

40. Actuate RPM INCR/DECR switch to full decrease with collective in full down position. Rotate both engine throttles to full INCR and record ROTOR rpm. The recorded ROTOR rpm should be 97%. If not adjust actuator (8) rod end.

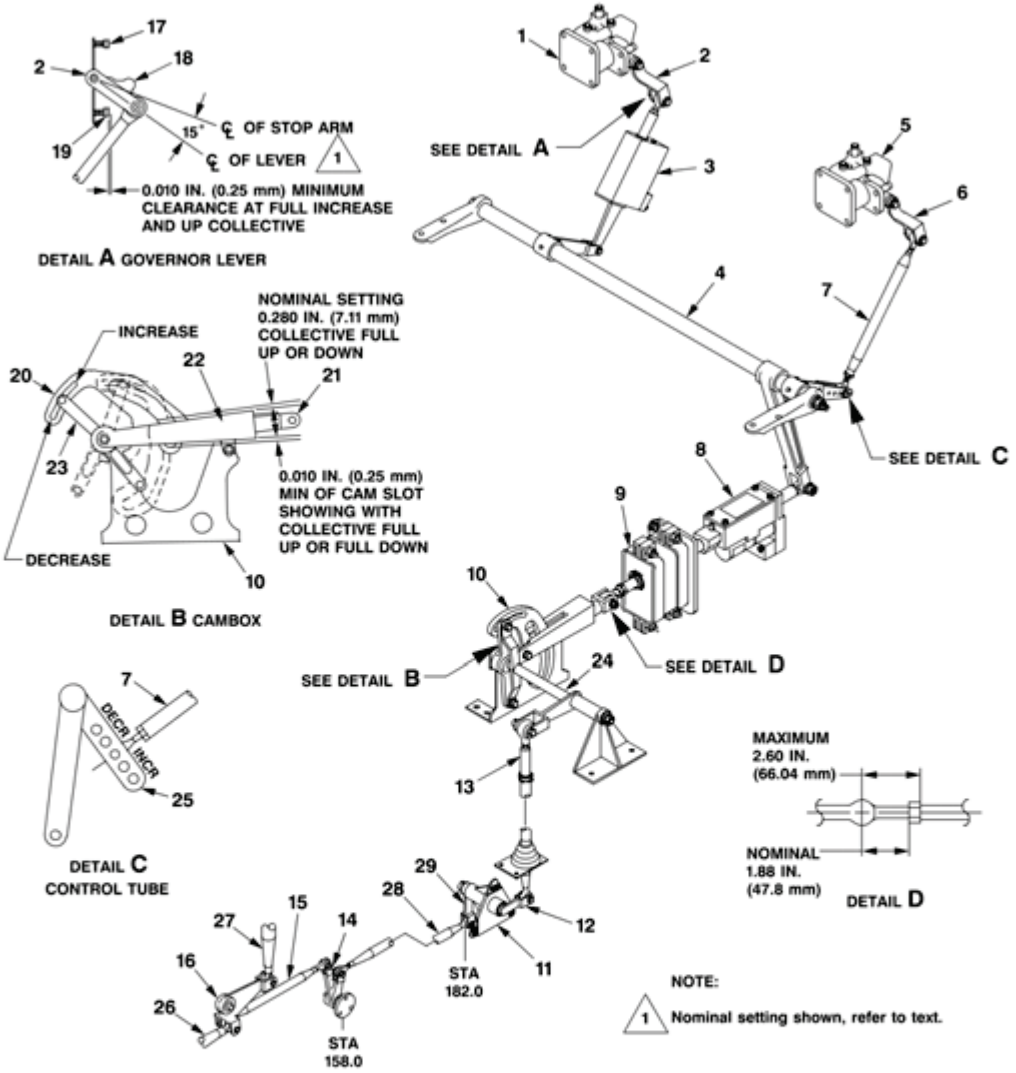
41. Actuate RPM INCR/DECR switch to full INCR and make sure that ROTOR rpm is 101.5%. The required range for twin-engine beep is 97 to 101.5% ROTOR rpm.

NOTE

If hydraulic test unit is not available disconnect control tube (15).

42. With collective stick full up, hold RPM INCR/DECR switch to INCR until actuator (8) is fully retracted. Hold +2/-2 switch to +2 until ITT actuator (27) is fully retracted. Make sure that minimum clearance of 0.010 in. (0.25 mm) (detail A) between stop arm (21) and maximum power stop screw (22) on both governors (1 and 5).

43. Check complete system for security of all parts and make sure that no interference exists. Apply corrosion preventive compound (C-101) to all exposed threads.



1. N2 power turbine governor, ENG 2
2. Lever
3. ITT actuator
4. Droop compensator jackshaft assembly
5. N2 power turbine governor, ENG 1
6. Lever
7. Control tube
8. Linear actuator
9. Boot
10. Cambox assembly
11. Bellcrank assembly
12. Bellcrank
13. Control tube
14. Idler
15. Control tube
16. Collective control bellcrank
17. Minimum power stop screw
18. Stop arm
19. Maximum power stop screw
20. Cambox cam
21. Slider
22. Slider housing
23. Cambox bellcrank
24. Bellcrank
25. Lever
26. Control tube (to pilot collective stick)
27. Control tube (to collective control cylinder)
28. Control tube
29. Arm

ICN-412-A-761100-B-97499-00005-A-001-01

Figure 1. N2 Controls Rigging (S/N 33001 through 33107) (Sheet 1 of 1)

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Droop Compensator and Cable - Removal

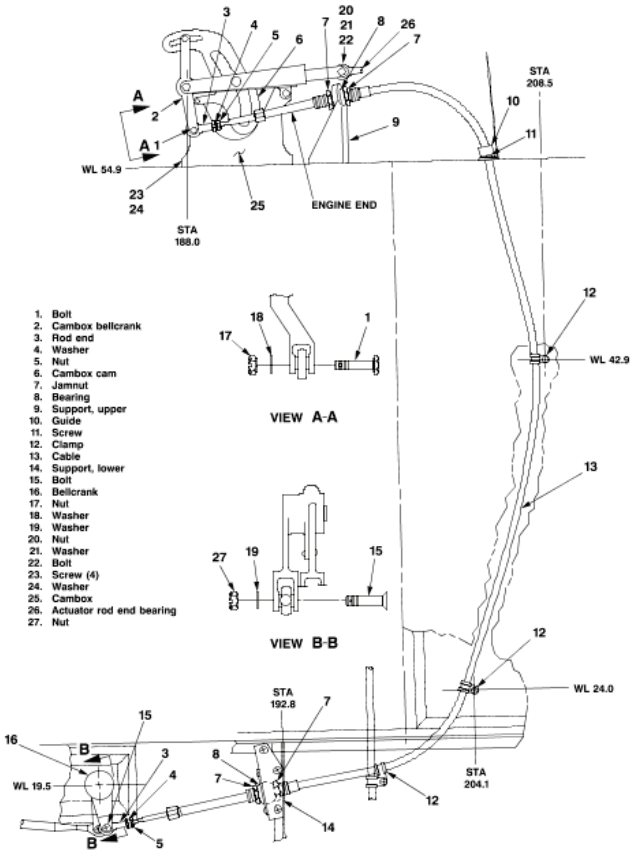
Procedure

1. Remove upper aft center fuel cell (412-A-28-00-00-00A-009A-A / 00323).
2. Remove nut (17, Figure 1), washer (19), and bolt (1) (view A-A), from cambox bellcrank (2).
3. Remove nut (27), washer (19), and bolt (15) (view B-B) from cambox bellcrank (16).
4. Remove rod ends (3), washer (4), and nuts (5) from both ends of cable (13).
5. Loosen jamnuts (7). Hold cable with wrench while loosening jamnuts.
6. Remove bearing (8) from lower support (14).
7. Remove clamps (12) securing cable. To gain access to upper most clamp, remove panel on aft wall of fuel cell cavity.
8. Remove screws (11) retaining guide (10).
9. Remove bearing (8) from upper support (9).
10. Remove both upper and lower bearings (8) from cable (13).
11. Remove jamnuts (7) from cable.



TO PREVENT DAMAGE TO CABLE DO NOT COIL IN A TIGHT LOOP.

12. Remove cable (13) through large access opening in bottom of helicopter.
13. Remove nut (20), washer, (21), and bolt (22).
14. Remove four screws (23) and washers (24).
15. Remove cambox (25) from service deck.



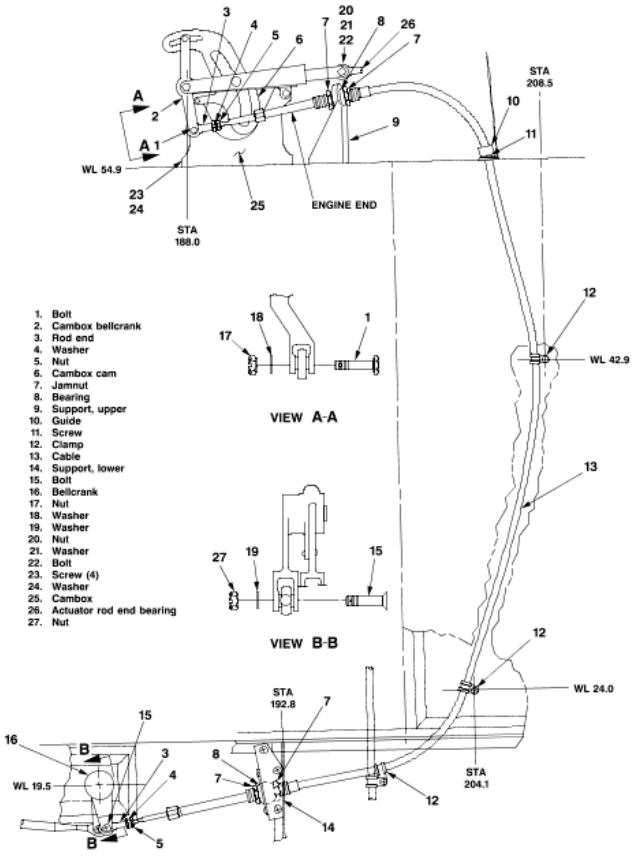
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Figure 1. Droop Compensator Cambox Control Cable (S/N 33108 through 33213 and S/N 36001 through 36096) (Sheet 1 of 1)

Droop Compensator and Cable - Inspection

Procedure

1. Inspect cambox (25, Figure 1) for secure installation.
2. Inspect cambox for binding when collective controls are moved through full range.
3. Inspect cable (13) for damage and corrosion. Inspect cable rod end bearing for damage.
4. Inspect cable to ensuring cable is free of twists, rod end should be stroked by hand to ensure cable is not binding or has excessively high operating loads and motion is smooth. Erratic or increasing load as cable is stroked is reason to replace cable.
5. Inspect bearing (8) for wear and damage.



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Figure 1. Droop Compensator Cambox Control Cable (S/N 33108 through 33213 and S/N 36001 through 36096) (Sheet 1 of 1)

Droop Compensator and Cable - Installation

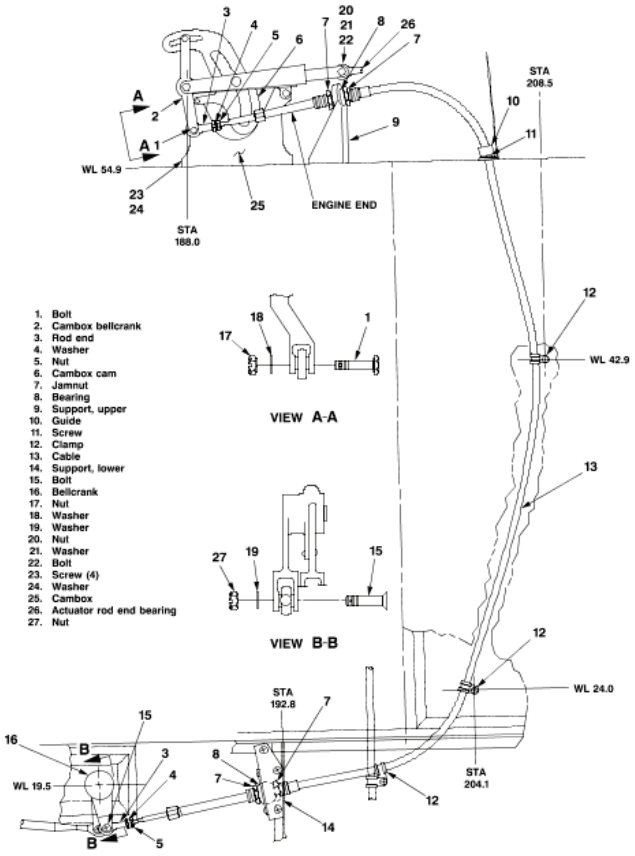
Procedure

1. Position cambox (25, Figure 1) on service deck.
2. Install four screws (23) and washers (24).
3. Position actuator rod end bearing (26) on cambox (25) slider and install bolt (22), washer (21) and nut (20).



TO PREVENT DAMAGE TO CABLE DO NOT COIL IN A TIGHT LOOP.

4. Upon removal from packing, lay cable (13) straight on a flat surface. Ensure cable is not twisted.
5. After ensuring cable is free of twists, rod end should be stroked by hand to ensure cable is not binding or has excessively high operating loads and motion is smooth. Erratic or increasing load as cable is stroked is reason to replace cable.
6. Stroke cable by hand to check for operation. No increase in load indicates cable has not been twisted and positioning is satisfactory.
7. Install guide (10) with screws (11).
8. Install jamnuts (7) on cable at both ends.
9. Install bearing (8) and jamnut (7) on upper portion of cable. Do not tighten at this time.
10. Install bearing (8) on upper support (9).
11. Install bearing (8) and jamnut (7) on lower portion of cable. Do not tighten at this time.
12. Install bearing (8) onto lower support (14).
13. Center bearings on both cable ends using jamnuts. Adjust jamnuts to prevent cable inner member from bottoming out in cable outer casing when collective is full down.
14. Do not apply torque to cable housing as this will introduce twist against cable housing while holding cable housing. Stroke cable by hand to check for freedom of operation.
15. Install nuts (5), washer (4), and rod ends (3) on both ends of cable. Install rod end at full thread engagement then back off six turns.
16. Connect rod end (3) by installing bolts (1), washer (18), and nut (17) on cambox bellcrank (2).
17. Connect rod end (3) by installing bolt (15), washer (19), and nut (17) on bellcrank (16).
18. Ensure both upper and lower rod ends of cable engage without inducing any cable twisting. Operate system and check for freedom of operation.
19. If proceeding check shows no appreciable increase in load or affect on smoothness of operation, control cable has been properly installed. Install clamps (12). Do not force cable to take an unusual configuration ensure clamps do not pinch or axially restrain conduit.
20. Install upper aft cell (412-A-28-10-00-70A-720A-A / 00555).
21. Rig droop compensator controls per 412-A-76-11-00-06A-276A-A / 01755, 412-A-76-11-00-06A-276B-A / 01756, or 412-A-76-11-00-07A-520A-A / 01757, 412-A-76-11-00-07A-280A-A / 01758 and 412-A-76-11-00-07A-720A-A / 01759 depending on aircraft configuration.



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Figure 1. Droop Compensator Cambox Control Cable (S/N 33108 through 33213 and S/N 36001 through 36096) (Sheet 1 of 1)

N2 Control - Rigging

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Corrosion preventive compound	C-101	AR	

Procedure

NOTE

Refer to 412-A-76-10-00-00A-276A-A / 01725 for N1 (power lever control) rigging.

1. Comply with the following general procedures when rigging droop compensator controls.

1.1. When a rigging procedure requires a specific position to accomplish following steps, manually hold controls in position or secure controls in position with pilot friction adjustment.

1.2. Tolerance for rigging dimensions is ±0.030 in. (0.76 mm) unless otherwise stated.

1.3. All adjustable control tubes must have a maximum of 1.00 in. (25.4 mm) of exposed threads after adjustment. If control tubes have inspection holes, threads must be engaged sufficiently to cover holes.

1.4. Apply corrosion preventive compound (C-101) to threads of all adjustable rod end bearings and clevises after adjustment.

1.5. Torque control jamnuts 80 to 100 in-lbs (9.0 to 11.3 N-m) after adjustment unless otherwise specified.

1.6. Control tubes must be free to rotate several degrees about longitudinal axis. If necessary, loosen jamnuts, adjust end fitting and re-torque jamnuts.

2. Ensure collective pitch control system rigging has been accomplished (412-A-67-01-00-01A-276A-A / 01323).

3. Verify levers (2 and 6, Figure 1) are correctly installed on control shafts of power turbine governors (1 and 5). Nominal setting positions the centerline of levers (2 and 6) 15 degrees (one serration) clockwise from centerline of governor stop arm (23), viewed from left looking inboard (detail A). In subsequent steps levers (2 and 6) may be repositioned ±15 degrees (one serration) from the nominal setting if required to achieve proper rigging and eliminate any fouling conditions.

4. Connect control tube (3) to governor lever (2) and jackshaft (4).

NOTE

The dimension given in Step 5. is approximate for initial rigging. Minor adjustments may be necessary during ground and flight test.

5. Verify adjustable tube (7) is approximately 9.34 in. (237.23 mm) measured between attachment points (center of rod end bearings).

6. Connect tube (7) to governor lever (6) and to the middle hole of jackshaft lever (30) (detail C).

7. Disconnect actuator (8) from lever of jackshaft (4).

8. Disconnect actuator (8) rod end bearing from cambox slider (26).

9. Disconnect control cable (13) from cambox (10).

NOTE

The dimension given in Step 10. is approximate for initial rigging. Minor adjustments may be necessary during ground and flight test.

10. Verify long arm of cambox bellcrank (28) is attached approximately in the middle of adjustment slot of cam (25) (detail B).

11. Use RPM INCR/DECR (beep) switch on collective switchbox to operate actuator. Verify actuator (8) is adjusted to a total stroke of approximately 0.62 in. (15.7 mm) from full extend (DECR) to full retract (INCR) positions. If not, refer to 412-A-76-11-00-01A-720A-A / 01743 for actuator rigging procedure.

12. Adjust rod end on actuator (8) to approximately 1.88 in. (47.8 mm) from center of bearing bolt hole to face of jamnut (detail D).

13. Connect actuator (8) rod end bearing to cambox slider (26).

14. Connect actuator (8) to lever of jackshaft (4).

15. Connect control cable (13) to cambox (10).

16. Disconnect tube (18) from bellcrank (16) and rotate bellcrank (16) to ensure there is no fouling. Rotate cambox bellcrank (28) (detail B) and ensure there is no fouling.

17. Connect tube (18) to bellcrank (16).

NOTE

If hydraulic test unit is not available, do not connect control tube (18) at this time.

18. Connect hydraulic test unit (412-A-29-00-00-00A-320A-A / 00616).

19. With hydraulic boost ON, place collective stick full down. Adjust the length or position of control cable (13) to hold cam (25) so lower end of cam slot is approximately 0.280 in. (7.11 mm) below slider housing (27).

20. With hydraulic boost ON, place collective stick full up. Verify cam slot is approximately 0.280 in. (7.11 mm) above slider housing (27).

21. With collective stick full up, hold RPM INCR/DECR switch to INCR until actuator (8) is fully retracted. Verify minimum clearance of 0.010 in. (0.25 mm) (detail A) between stop arm (23) and maximum power stop screw (24) on both governors (1 and 5).

22. With hydraulic boost ON, place collective stick full down. Hold RPM INCR/DECR switch to DECR until actuator (8) is fully extended. Verify minimum clearance of 0.010 in. (0.25 mm) between stop arm (23) and minimum power stop screw (22) on both governors (1 and 5).

23. Disconnect hydraulic test unit or connect control tube (18) as applicable.



VERIFY ALL CONTROLS ARE PROPERLY CONNECTED PRIOR TO GROUND RUN AND FLIGHT TEST.

NOTE

Prior to accomplishing ground run verify Automatic Fuel Control Unit (AFCU) and Manual Fuel Control Unit (MFCU) maximum stops screws are being contacted with throttles full open (refer to 412-A-76-10-00-00A-276A-A / 01725). During ground run verify acceleration time difference between power sections is within 0.5 seconds (refer to 412-A-76-10-00-00A-280F-A / 01735, 412-A-76-10-00-00A-271E-A / 01736 and 412-A-76-10-00-00A-280G-A / 01737).

24. Start both power sections per Flight Manual and idle for 5 minutes.

25. Actuate RPM INCR/DECR switch to full DECR.

26. Place collective stick full down. Rotate ENG 2 throttle to full INCR. Record ROTOR rpm (NR). The recorded value should be 95% ROTOR rpm. If ROTOR rpm is not 95% adjust as follows.



RETURN ENGINES TO IDLE FOR ANY ADJUSTMENTS TO CONTROLS WITH ENGINE RUNNING.

NOTE

Three turns of actuator (8) rod end is equal to 1% ROTOR rpm change.

26.1. If ROTOR rpm is below 95%, shorten rod end on actuator (8).

26.2. If ROTOR rpm is above 95%, lengthen rod end on actuator (8).

NOTE

The single engine beep range (spread) should be 95 to 99.5% rotor RPM.

27. Slowly actuate RPM INCR/DECR switch to full INCR and verify the rotor RPM is 99 to 99.5%. If the spread is not within limits, adjust actuator (8) stroke per 412-A-76-11-00-01A-720A-A / 01743.

28. Actuate RPM INCR/DECR switch to full DECR.

29. Return ENG 2 to IDLE, rotate ENG 1 throttle to full INCR. Record rotor RPM. The recorded value should be 95% rotor RPM. If rotor RPM is not 95%, adjust as follows.



RETURN ENGINES TO IDLE FOR ANY ADJUSTMENTS TO CONTROLS WITH ENGINE RUNNING.

NOTE

Two turns of control tube (7) rod end is equal to 1% rotor RPM change.

29.1. If rotor RPM is below 95%, shorten rod end on control tube (7).

29.2. If rotor RPM is above 95%, lengthen rod end on control tube (7).

NOTE

The single engine beep range should be 95 to 99.5% rotor RPM.

30. Slowly actuate RPM INCR/DECR switch to full INCR and verify the rotor RPM is 99 to 99.5%.

NOTE

When performing the following steps, helicopter will become airborne.

31. Return ENG 1 to IDLE, rotate ENG 2 to full INCR. Slowly actuate RPM INCR/DECR switch to 98% rotor RPM. Gradually increase collective from flat pitch to 60% engine torque and note static rotor RPM droop.

32. Return ENG 2 to IDLE, rotate ENG 1 to full INCR. Slowly actuate RPM INCR/DECR switch to 98% ROTOR RPM. Gradually increase collective from flat pitch to 60% engine torque and note static rotor RPM droop.

33. If more than 2% rotor RPM droop is encountered perform troubleshooting of engine pneumatic lines and N2 power turbine governor per Pratt & Whitney Maintenance Manual before proceeding with further adjustments.

34. Actuate RPM INCR/DECR switch to full DECR.

35. With collective stick full down, rotate ENG 1 and ENG 2 throttles to FULL INCR. Actuate RPM INCR/DECR switch to 100% rotor RPM.

NOTE

Engine torque difference (split) between ENG 1 and ENG 2 shall not exceed 4% from flat pitch to maximum power.

36. Gradually increase collective from flat pitch to full power (either transmission torque, ITT or N1 limit) and note the difference in engine torque readings. If not within limits, adjust as follows:

36.1. If ENG 1 torque reading is greater than ENG 2 by more than 4% throughout the collective range, lengthen control tube (7).

36.2. If ENG 1 torque reading is less than ENG 2 by more than 4% throughout the collective range, shorten control tube (7).

36.3. If ENG 1 torque reading is greater than ENG 2 by more than 4% at high power range and too low at low power range, move control tube (7) rod end forward (DECR) on lever (30) (Detail C).

36.4. If ENG 1 torque reading is less than ENG 2 by more than 4% at high power range and/or too low at low power range, move control tube (7) rod end AFT (INCR) on lever (30) (Detail C).

37. Place collective stick full down; rotate both engine throttles to full INCR. Use RPM INCR/DECR switch to obtain 100% rotor RPM. Increase collective pitch slowly in a series of equal steps from flat pitch to full power (either transmission torque, ITT or N1 limit). rotor RPM should remain at 100 ±1% throughout power sweep.

38. If rotor RPM droops (decays below tolerance) or overspeeds (increases above tolerance) during power applications, make a cam rate adjustment.

NOTE

Make adjustments in small increments of approximately 0.125 inch (3.18 mm) measured on cam slot.

38.1. For increased cam compensation to correct droop, adjust cambox cam (25) counterclockwise relative to cambox bellcrank (28) (Detail B).

38.2. For reduced cam compensation to correct overspeed, adjust cambox cam (25) clockwise relative to cambox bellcrank (28).

38.3. After each cam adjustment, move the collective to full down and then full up positions to ensure there is no fouling between slider housing (27) and cambox slot. Verify that a minimum of 0.010 inch (0.25 mm) cam slot shows below and above slider housing. If required, adjust control cable (13) for proper clearance.



RETURN ENGINES TO IDLE FOR ADJUSTMENTS TO CONTROLS WITH ENGINE RUNNING.

NOTE

Adjustments to achieve proper twin engine RPM range and adjustments to avoid torque needle split, will supersede the initial single engine beep ranges.

Three turns of linear actuator (8) rod end is equal to 1% rotor RPM change.

39. Actuate RPM INCR/DECR switch to full decrease with collective in full down position. Rotate both engine throttles to full INCR and record rotor RPM. The recorded rotor RPM should be 97%. If not, adjust linear actuator (8) rod end.

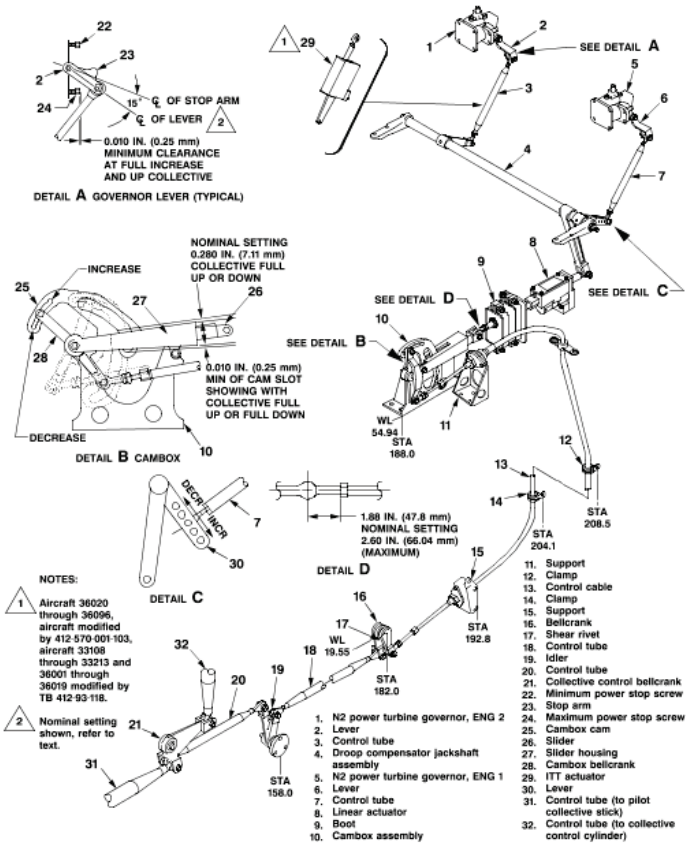
40. Actuate RPM INCR/DECR switch to full INCR and verify rotor RPM is 101.5%. The required range for twin-engine beep is 97 to 101.5% rotor RPM. If not, adjust linear actuator (8) stroke per 412-A-76-11-00-01A-720A-A / 01743.

NOTE

If hydraulic test unit is not available, disconnect control tube (32).

41. With collective stick full up, hold RPM INCR/DECR switch to INCR until linear actuator (8) is fully retracted. Verify minimum clearance of 0.010 inch (0.25 mm) (Detail A) between stop arm (23) and maximum power stop screw (24) on both power turbine governors (1 and 5).

42. Check complete system for security of all parts and ensure no interference exists. Apply corrosion preventive compound (C-101) to all exposed threads.



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Figure 1. N2 Controls Rigging (S/N 33108 through 33213 and S/N 36001 through 36096) (Sheet 1 of 1)

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N2 Control - Rigging

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Corrosion preventive compound	C-101	AR	

Procedure

NOTE

Refer to 412-A-76-10-00-00A-276A-A / 01725 for N1 (power lever control) rigging.

1. Comply with the following general procedures when rigging droop compensator controls.

1.1. When a rigging procedure requires a specific position to accomplish following steps, manually hold controls in position or secure controls in position with pilot friction adjustment.

1.2. Tolerance for rigging dimensions is ±0.030 inch (0.76 mm) unless otherwise stated.

1.3. All adjustable controls tubes must have a maximum of 1.00 inch (25.4 mm) of exposed threads after adjustment. If control tubes have inspection holes, threads must be engaged sufficiently to cover holes.

1.4. Apply corrosion preventive compound (C-101) to threads of all adjustable rod end bearings and clevises after adjustment.

1.5. Torque control jamnuts 80 to 100 inch-pounds (9.0 to 11.3 Nm) after adjustment unless otherwise specified.

1.6. Control tubes must be free to rotate several degrees about longitudinal axis. If necessary, loosen jamnuts, adjust end fitting and retorque jamnuts.

2. Make sure that collective pitch control system rigging has been accomplished (412-A-67-01-00-01A-276A-A / 01323).

3. Make sure that levers (2 and 6, Figure 1) are correctly installed on control shafts of power turbine governors (1 and 5). Nominal setting positions the centerline of levers 15° (one serration) clockwise from centerline of governor stop arm (23), viewed from left looking inboard (Detail A). In subsequent steps levers may be repositioned ±15° (one serration) from the nominal setting if required to achieve proper rigging and eliminate any fouling conditions.

NOTE

SN: 36020-Subsequent or (TB 412-01-176: Post (412-570-001-103: Installed or BHT-412-SI-74: Installed or BHT-412-SI-97: Installed)):

Total length of ITT actuator (29) is 9.455 inches (240.157 mm).

4. Connect ENG 2 ITT actuator (29) to governor lever (2) and droop compensator jackshaft assembly (4). Make sure that total stroke of actuator (29) is 0.150 inch (3.81 mm) and total length is 9.415 inches (239.14 mm) or 9.455 inches (240.157 mm) with actuator fully extended. Refer to 412-A-76-11-00-02A-720A-A / 01746 for installation and rigging procedure.

NOTE

The dimension given in Step 5. is approximate for initial rigging. Minor adjustments may be necessary during ground and flight test.

5. Make sure that adjustable control tube (7) is approximately 9.34 inches (237.23 mm) measured between attachment points (center of rod end bearings).

6. Connect control tube (7) to governor lever (6) and to the middle hole of jackshaft lever (30) (Detail C).

7. Disconnect linear actuator (8) from lever of droop compensator jackshaft assembly (4).

8. Disconnect linear actuator (8) rod end bearing from cambox slider (26).

9. Disconnect control cable (13) from cambox assembly (10).

NOTE

The dimension given in Step 10. is approximate for initial rigging. Minor adjustments may be necessary during ground and flight test.

10. Make sure that long arm of cambox bellcrank (28) is attached approximately in the middle of adjustment slot of cambox cam (25) (Detail B).

NOTE

SN: 36020-Subsequent or (TB 412-01-176: Post (412-570-001-103: Installed or BHT-412-SI-74: Installed or BHT-412-SI-97: Installed)):

Linear actuator (8) stroke is 0.850 inch (21.6 mm).

11. Use RPM INCR/DECR (beep) switch on collective switchbox to operate actuator. Make sure that linear actuator (8) is adjusted to a total stroke of approximately 0.62 inch (15.7 mm) or 0.850 inch (21.6 mm) from full extend (DECR) to full retract (INCR) positions. If not, refer to 412-A-76-11-00-01A-720A-A / 01743 for actuator rigging procedure.

12. Adjust rod end on linear actuator (8) to approximately 1.88 inches (47.8 mm) from center of bearing bolt hole to face of jamnut (Detail D).

13. Connect linear actuator (8) rod end bearing to cambox slider (26).

14. Connect linear actuator (8) to lever of droop compensator jackshaft assembly (4).

15. Connect control cable (13) to cambox assembly (10).

16. Disconnect control tube (18) from bellcrank (16) and rotate bellcrank (16) to make sure that there is no fouling. Rotate cambox bellcrank (28) (Detail B) and make sure that there is no fouling.

17. Connect control tube (18) to bellcrank (16).

NOTE

If hydraulic test unit is not available disconnect control tube (32).

18. Connect hydraulic test unit (412-A-29-00-00-00A-320A-A / 00616).

19. With hydraulic boost ON, place collective stick full down. Adjust the length or position of control cable (13) to hold cambox cam (25) so lower end of cam slot is approximately 0.280 inch (7.11 mm) below slider housing (27).

20. With hydraulic boost ON, place collective stick full up. Make sure that cam slot is approximately 0.280 inch (7.11 mm) above slider housing (27).

21. With collective stick full up, hold RPM INCR/DECR switch to INCR until linear actuator (8) is fully retracted. Hold ENG 2 (+2/-2) switch to -2 until ITT actuator (29) is fully retracted. Make sure that minimum clearance of 0.010 inch (0.25 mm) (Detail A) between stop arm (23) and maximum power stop screw (24) on both power turbine governors (1 and 5).

22. With hydraulic boost ON, place collective stick full down. Hold RPM INCR/DECR switch to DECR until linear actuator (8) is fully extended. Hold ENG 2 (+2/-2) switch to +2 until ITT actuator (29) is fully extended. Make sure that minimum clearance of 0.010 inch (0.25 mm) between stop arm (23) and minimum power stop screw (22) on both power turbine governors (1 and 5).

23. Disconnect hydraulic test unit or connect control tube (18) as applicable.



MAKE SURE THAT ALL CONTROLS ARE PROPERLY CONNECTED PRIOR TO GROUND RUN AND FLIGHT TEST.

NOTE

Prior to accomplishing ground run make sure that Automatic Fuel Control Unit (AFCU) and Manual Fuel Control Unit (MFCU) maximum stops screws are being contacted with throttles full open (412-A-76-10-00-00A-276A-A / 01725). During ground run make sure that acceleration time difference between power sections is within 0.5 seconds (412-A-76-10-00-00A-280F-A / 01735, 412-A-76-10-00-00A-271E-A / 01736 and 412-A-76-10-00-00A-280G-A / 01737).

24. Start both power sections per BHT-412-FM and idle for 5 minutes.

25. Actuate RPM INCR/DECR switch to full DECR.

NOTE

Positioning of ENG 2 ITT actuator (29) to mid stroke can be done during ground run by actuating ENG 2 (+2/-2) switch to full +2, record rotor RPM. Actuate ENG 2 (+2/-2) switch to full -2, record rotor RPM. Actuate ENG 2 (+2/-2) switch to set rotor RPM to middle of recorded range.

26. Make sure ENG 2 ITT actuator (29) is in mid stroke position, 9.34 inches (237.2 mm) before proceeding to next step.

27. Place the collective stick full down. Rotate the ENG 2 throttle to full INCR. Record the rotor RPM (N_R). Slowly actuate the RPM INCR/DECR switch to full INCR until 101% rotor RPM is reached. Record the rotor RPM. Verify that full INCR produces a 99% rotor RPM minimum.



RETURN ENGINES TO IDLE FOR ANY ADJUSTMENTS TO CONTROLS WITH ENGINE RUNNING.

NOTE

Three turns of actuator (8) rod end is equal to 1% rotor RPM change.

28. If minimum rotor RPM cannot be obtained, make an interim adjustment to the rod end of linear actuator (8).

29. With the collective full down, rotate ENG 2 throttle to full INCR. Actuate RPM INCR/DECR switch to full DECR and ENG 2 (+2/–2) switch to full +2. Record rotor RPM. Actuate ENG 2 (+2/–2) switch to full –2 and record rotor RPM. The total spread between rotor RPM should be 2 1/4 ±1/4%. If spread is not within limits, adjust ITT actuator (29) stroke per 412-A-76-11-00-02A-720A-A / 01746.

30. Return ENG 2 to IDLE, rotate ENG 1 throttle to full INCR. Record rotor RPM. Slowly actuate RPM INCR/DECR switch to full INCR but not to exceed 101% rotor RPM. Record rotor RPM. The minimum full INCR should be 99% rotor RPM.



RETURN ENGINES TO IDLE FOR ANY ADJUSTMENTS TO CONTROLS WITH ENGINE RUNNING.

NOTE

Two turns of control tube (7) rod end is equal to 1% rotor RPM change.

31. If the minimum rotor RPM cannot be obtained, make an interim adjustment to the rod end of control tube (7).

NOTE

When performing the following steps helicopter will become airborne.

32. Return ENG 1 to IDLE, rotate ENG 2 to full INCR. Slowly actuate RPM INCR/DECR switch to 98% rotor RPM. Gradually increase collective from flat pitch to 60% engine torque and note static rotor RPM droop.

33. Return ENG 2 to IDLE, rotate ENG 1 to full INCR. Slowly actuate RPM INCR/DECR switch to 98% rotor RPM. Gradually increase collective from flat pitch to 60% engine torque and note static rotor RPM droop.

34. If more than 2% rotor RPM droop is encountered perform troubleshooting of engine pneumatic lines and N_2 power turbine governor per Pratt & Whitney Maintenance Manual before proceeding with further adjustments.

35. With rotor at flat pitch, increase both throttles to full INCR and adjust rotor RPM to 100% using RPM INCR/DECR switch.

36. Gradually increase collective from flat pitch to approximately 60% transmission torque or mast torque (as applicable). Using the +2/–2 switch, trim ENG 2 to minimum, then to maximum and note the torque split. The torque split should be approximately equal above and below ENG 1 torque indication. If the torque split difference is greater than 5% above or below ENG 1 torque, adjust length of tube (7) as follows.

NOTE

Half turn of rod end on control tube (7) is equal to approximately 5% engine torque change.

36.1. To increase torque on ENG 1, shorten rod end of control tube (7).

36.2. To decrease torque on ENG 1, lengthen rod end of control tube (7).

NOTE

Engine torque difference (split) between ENG 1 and ENG 2 shall not exceed 4% during power application throughout this maintenance procedure. However, during normal operation torque split may exceed 4% +2/-2 switch may be used to trim engines for either torque matching or ITT matching.

37. At 100% rotor RPM, gradually increase collective to 60% transmission torque or mast torque (as applicable) and match engine torques using +2/-2 switch. Gradually decrease collective to flat pitch and note the torque split. Gradually increase collective to full power (either transmission torque or mast torque (as applicable), ITT, or N_1 limit) and note the torque split throughout the power range, torque matching should remain constant within 4%. If torque does not remain constant throughout the power range, adjust as follows.

37.1. If ENG 1 torque reading is greater than ENG 2 by more than 4%, move tube (7) rod end forward (DECR) on lever (30) (Detail C).

37.2. If ENG 1 torque reading is less than ENG 2 by more than 4%, move control tube (7) rod end AFT (INCR) on lever (30) (Detail C).

38. Place collective stick full down, rotate both engine throttles to full INCR. Use RPM INCR/DECR switch to obtain 100% rotor RPM and match engine torques using +2/-2 switch. Increase collective pitch slowly in a series of equal steps from flat pitch to full power (either transmission torque or mast torque (as applicable), ITT or N_1 limit). Rotor RPM should remain at 100 ±1% throughout power sweep.

39. If rotor RPM droops (decays below tolerance) or overspeeds (increases above tolerance) during power applications, make a cam rate adjustment.

NOTE

Make adjustments in small increments of approximately 0.125 inch (3.18 mm) measured on cam slot.

39.1. For increased cam compensation to correct droop, adjust cambox cam (25) counterclockwise relative to cambox bellcrank (28) (Detail B).

39.2. For reduced cam compensation to correct overspeed, adjust cambox cam (25) clockwise relative to cambox bellcrank (28).

39.3. After each cam adjustment, move the collective to full down and then full up positions to make sure that there is no fouling between slider housing (27) and cambox slot. Make sure that a minimum of 0.010 inch (0.25 mm) cam slot shows below and above slider housing. If necessary adjust or position control cable (13) for proper clearance.



RETURN ENGINES TO IDLE FOR ANY ADJUSTMENTS TO CONTROLS WITH ENGINE RUNNING.

NOTE

Three turns of linear actuator (8) rod end is equal to 1% rotor RPM change.

40. Actuate RPM INCR/DECR switch to full decrease with collective in full down position. Rotate both engine throttles to full INCR and record rotor RPM. The recorded rotor RPM should be 97%. If not adjust linear actuator (8) rod end.



DO NOT EXCEED 104% ROTOR RPM.

NOTE

TB 412-01-176: Pre: Twin-engine beep range is 97 to 101.5% rotor RPM

SN: 36020-36096 TB 412-01-176: Post: Twin-engine beep range is 97 to 103.5% rotor RPM for PT6T-3D/DE/DF engine equipped helicopters.

41. **TB 412-01-176: Pre:** Actuate RPM INCR/DECR switch to full INCR and make sure that rotor RPM is 101.5%. The required range for twin-engine beep is 97 to 101.5% rotor RPM.

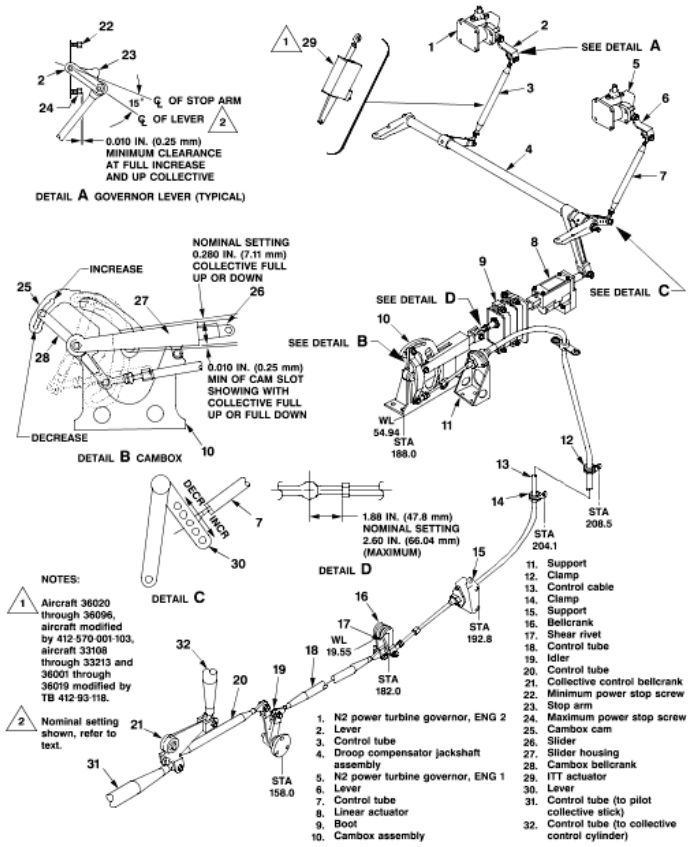
SN: 36020-36096 TB 412-01-176: Post: Actuate RPM INCR/DECR switch to full INCR and verify rotor RPM is 103.5%. The required range for twin-engine beep is 97 to 103.5% rotor RPM.

NOTE

If hydraulic test unit is not available disconnect control tube (32).

42. With collective stick full up, hold RPM INCR/DECR switch to INCR until actuator (8) is fully retracted. Hold +2/-2 switch to +2 until ITT actuator (29) is fully retracted. Make sure that minimum clearance of 0.010 inch (0.25 mm) (Detail A) between stop arm (23) and maximum power stop screw (24) on both governors (1 and 5).

43. Check complete system for security of all parts and make sure that no interference exists. Apply corrosion preventive compound (C-101) to all exposed threads.



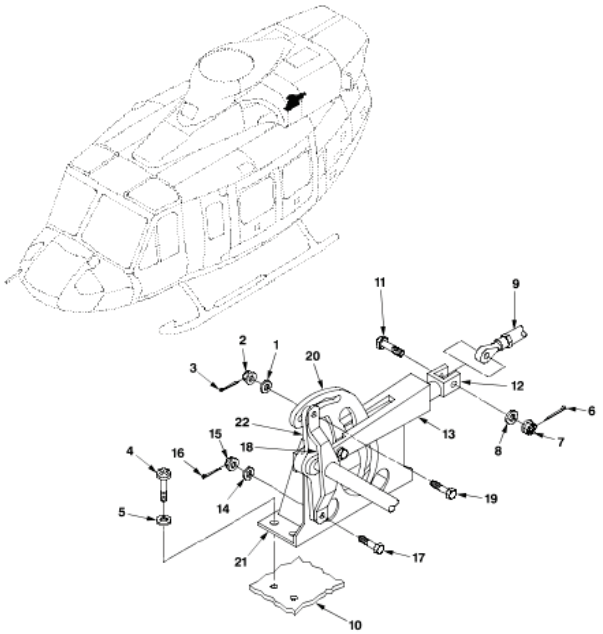
ICN-412-A-761100-B-97499-00007-A-001-01

Figure 1. N2 Controls Rigging (S/N 33108 through 33213 and S/N 36001 through 36096) (Sheet 1 of 1)

Droop Compensator - Removal

Procedure

1. Remove cotter pin (3, Figure 1), nut (2), washer (1), and bolt (19).
2. Remove cotter pin (16), nut (15), washer (14), and bolt (17), and disconnect bellcrank (18) from cambox bellcrank (22) and cambox cam (20).
3. Remove cotter pin (6), nut (7), washer (8), and bolt (11) from slider (12) and linear actuator tube (9).
4. Remove four screws (4) and four washers (5).
5. Remove cambox assembly (21) from service deck (10).



1. Washer
2. Nut
3. Cotter pin
4. Screw
5. Washer
6. Cotter pin
7. Nut
8. Washer
9. Linear actuator tube
10. Service deck
11. Bolt
12. Slider
13. Slider housing
14. Washer
15. Nut
16. Cotter pin
17. Bolt
18. Bellcrank
19. Bolt
20. Cambox cam
21. Cambox assembly
22. Cambox bellcrank

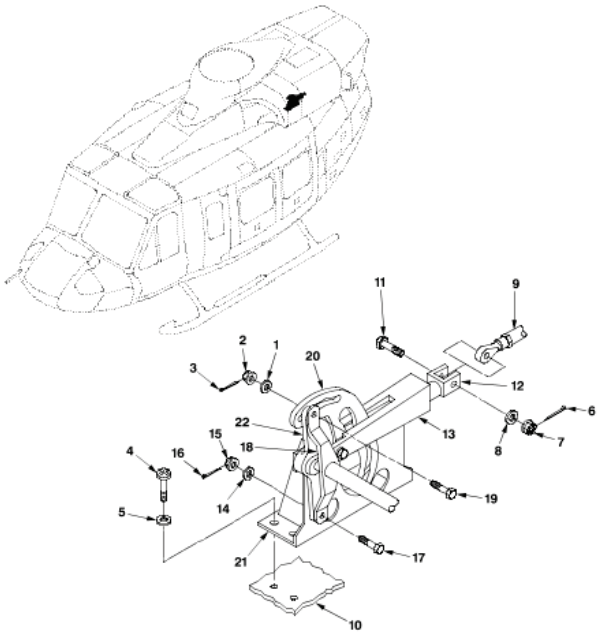
ICN-412-A-761100-B-97499-00008-A-001-01

Figure 1. Droop Compensator Cambox (S/N 36097 and Subsequent and S/N 33108 through 33213, 36019 through 36096 Modified by TB 412-94-122) (Sheet 1 of 1)

Droop Compensator - Inspection

Procedure

1. Inspect cambox assembly (21, Figure 1) for secure installation.
2. Inspect cambox assembly (21) for binding when collective controls are moved through full travel.



1. Washer
2. Nut
3. Cotter pin
4. Screw
5. Washer
6. Cotter pin
7. Nut
8. Washer
9. Linear actuator tube
10. Service deck
11. Bolt
12. Slider
13. Slider housing
14. Washer
15. Nut
16. Cotter pin
17. Bolt
18. Bellcrank
19. Bolt
20. Cambox cam
21. Cambox assembly
22. Cambox bellcrank

ICN-412-A-761100-B-97499-00008-A-001-01

Figure 1. Droop Compensator Cambox (S/N 36097 and Subsequent and S/N 33108 through 33213, 36019 through 36096 Modified by TB 412-94-122) (Sheet 1 of 1)

Droop Compensator - Installation

Procedure

1. Position cambox assembly (21, Figure 1) on service deck (10).
2. Install four screws (4) and four washers (5).
3. Position linear actuator tube (9) on slider (12). Install bolt (11), washer (8), nut (7), and cotter pin (6).
4. Position bellcrank (18) on cambox bellcrank (22) and cambox cam (20). Install bolt (19), washer (1), nut (2), and cotter pin (3).
5. Install bolt (17), washer (14), nut (15), and cotter pin (16).
6. Move collective controls through full travel and inspect for binding and interference in droop compensator control installation.

Applicable: SN: 33001-33213 or 36001-36019 or 412-570-001-103: Not Installed or (BHT-412-SI-74: Not Installed BHT-412-SI-97: Not Installed):

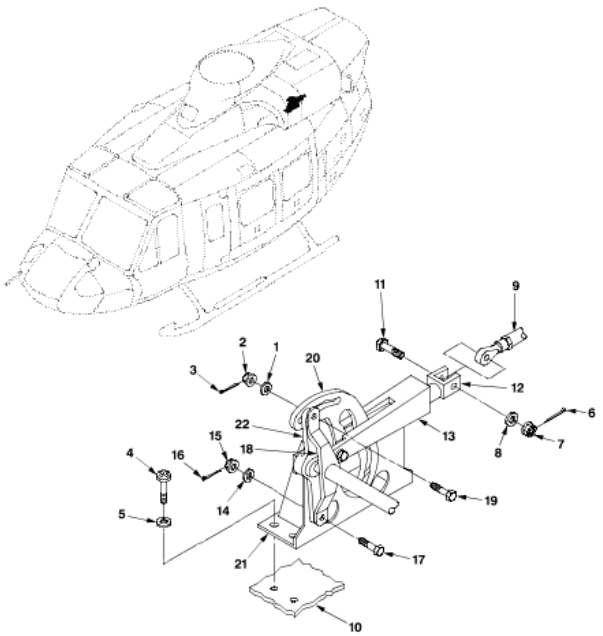
7. *Rig modified droop compensator controls (*412-A-76-11-00-09A-276A-A / 01763*).*

Applicable: SN: 36020-36096 or 412-570-001-103: Installed or BHT-412-SI-74: Installed or BHT-412-SI-97: Installed or (SN: 33108-33213 or 36001-36019 TB 412-94-122: Post):

8. *Rig modified droop compensator controls (*412-A-76-11-00-09A-276B-A / 01764*).*

Applicable: SN: 36097-Subsequent:

9. Rig droop compensator controls (412-A-76-11-00-09A-276B-A / 01764).
10. Perform maintenance test flight.



1. Washer
2. Nut
3. Cotter pin
4. Screw
5. Washer
6. Cotter pin
7. Nut
8. Washer
9. Linear actuator tube
10. Service deck
11. Bolt
12. Slider
13. Slider housing
14. Washer
15. Nut
16. Cotter pin
17. Bolt
18. Bellcrank
19. Bolt
20. Cambox cam
21. Cambox assembly
22. Cambox bellcrank

ICN-412-A-761100-B-97499-00008-A-001-01

Figure 1. Droop Compensator Cambox (S/N 36097 and Subsequent and S/N 33108 through 33213, 36019 through 36096 Modified by TB 412-94-122) (Sheet 1 of 1)

Droop Compensator Controls - Removal

Procedure

1. Remove the cotter pin (38, Figure 1), nut (37), washer (36), and bolt (35) connecting control tube (15) to the bellcrank assembly (10).

2. Remove the cotter pin (21), nut (22), washer (23), and bolt (24) connecting the control tube (15) to the bellcrank assembly (20).

3. Remove the screws (12), washers (11), retaining ring (32), split bushing (33), and seal assembly (34) from the control tube (15) and engine service deck.

4. Remove the screws (18) and washers (27) connecting the boot assembly to the shelf.

5. Remove the clamp (16), retaining ring (28), and boot assembly (17) from the control tube (15). Remove the control tube (15).

6. Remove the cotter pin (4), nut (3), washer (2), and bolt (1) connecting the bellcrank assembly (10) to the cambox cam (39).

7. Remove the cotter pin (6), nut (7), washer (8), and bolt (9) connecting the bellcrank assembly (10) to the cambox cam (39).

8. Remove the cotter pin (40), nut (41), and washer (42) connecting the bellcrank assembly (10) to the fitting assembly (13).

9. Remove the two screws (14) and two washers (29) connecting the fitting assembly (13) to the engine service deck.

10. Remove the fitting assembly (13) and the bellcrank assembly (10).

11. Remove the cotter pin (49), nut (48), washers (46 and 47), and bolt (45). Disconnect the control tube (50) from the bellcrank assembly (20).

12. Remove the fitting assembly (19) and bellcrank assembly (20).

13. If damaged, remove the bearings (55) from the bellcrank assembly (20) as follows:

13.1. Remove the pin (56).

13.2. Remove the washer (57).

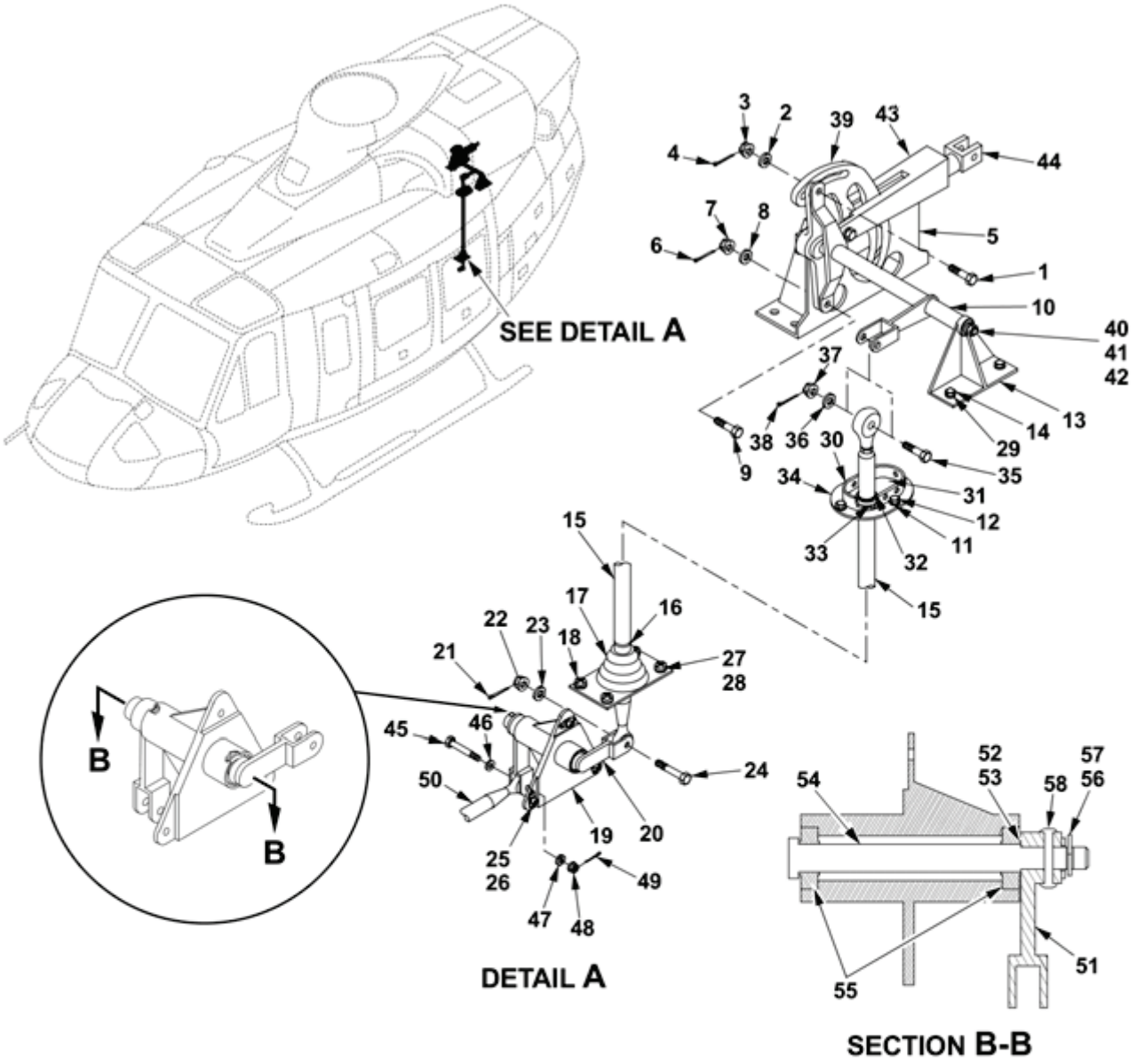
13.3. Remove and discard the rivet (58).

13.4. Remove the arm (51).

13.5. Remove the washers (52 and 53).

13.6. Remove the bellcrank (54).

13.7. Remove the bearings (55).



1. Bolt
2. Washer
3. Nut
4. Cotter pin
5. Cambox assembly
6. Cotter pin
7. Nut
8. Washer
9. Bolt
10. Bellcrank assembly
11. Washer (4 ea)
12. Screw (4 ea)
13. Fitting assembly
14. Screw (2 ea)
15. Control tube
16. Clamp
17. Boot assembly
18. Screw (4 ea)
19. Fitting assembly
20. Bellcrank assembly
21. Cotter pin
22. Nut
23. Washer
24. Bolt
25. Washer (3 ea)
26. Bolt (3 ea)
27. Washer (2 ea)
28. Retaining ring
29. Washer (2 ea)
30. Shield
31. Slider plate
32. Retaining ring
33. Split bushing
34. Seal assembly
35. Bolt
36. Washer
37. Nut
38. Cotter pin
39. Cambox cam
40. Cotter pin
41. Nut
42. Washer
43. Slider housing
44. Slider
45. Bolt
46. Washer
47. Washer
48. Nut
49. Cotter pin
50. Control tube
51. Arm
52. Washer
53. Washer
54. Bell crank
55. Bearing
56. Pin
57. Washer
58. Rivet

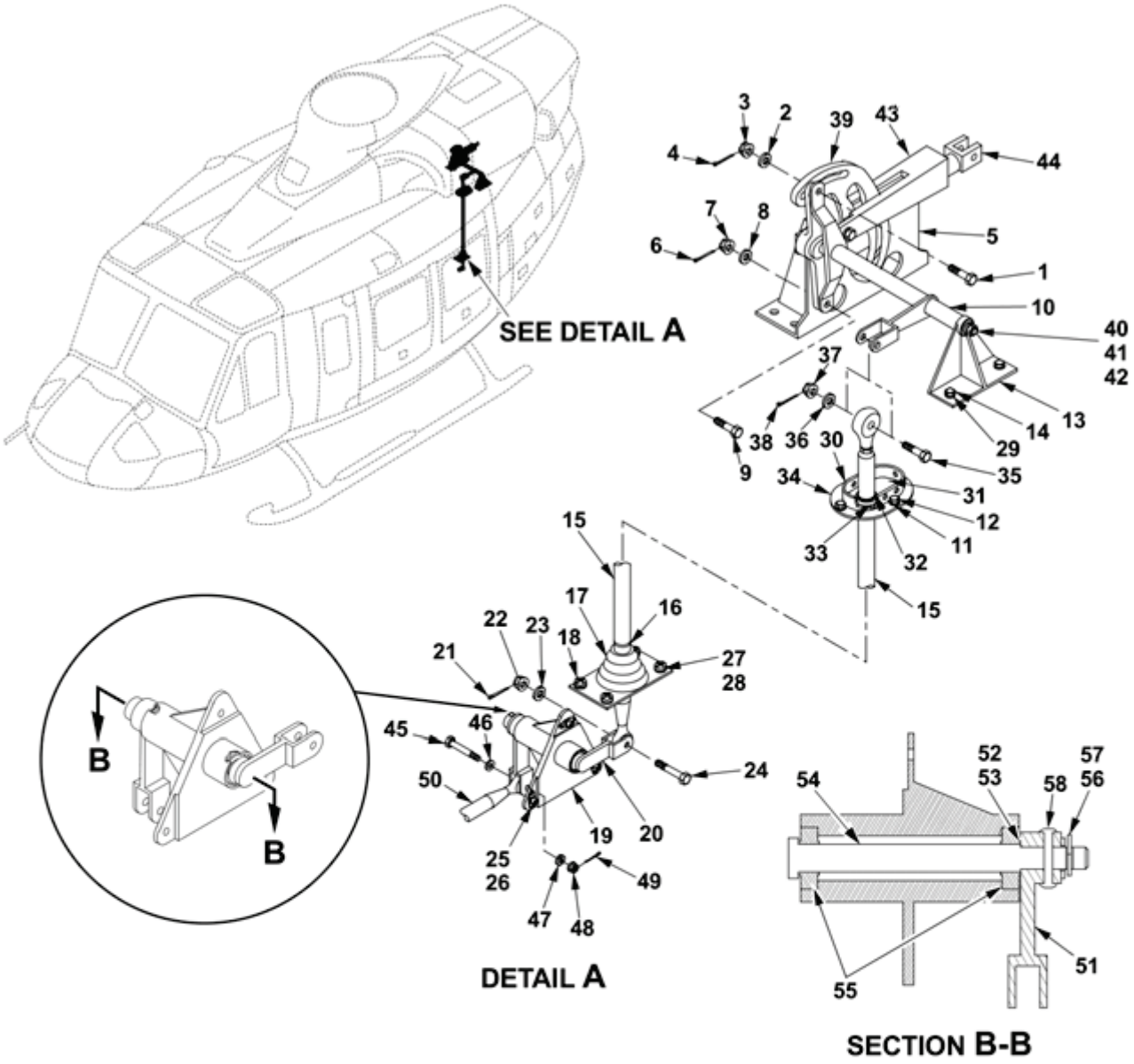
ICN-412-A-761100-B-97499-00009-A-001-01

Figure 1. Droop Compensator Controls (S/N 36097 and Subsequent and S/N 33108 through 33213, S/N 36019 through 36096 Modified by TB 412-94-122) (Sheet 1 of 1)

Droop Compensator Controls - Inspection

Procedure

1. Inspect the droop compensator controls (Figure 1) for secure installation.
2. Inspect the droop compensator controls for binding when the collective controls are moved through full travel.



1. Bolt
2. Washer
3. Nut
4. Cotter pin
5. Cambox assembly
6. Cotter pin
7. Nut
8. Washer
9. Bolt
10. Bellcrank assembly
11. Washer (4 ea)
12. Screw (4 ea)
13. Fitting assembly
14. Screw (2 ea)
15. Control tube
16. Clamp
17. Boot assembly
18. Screw (4 ea)
19. Fitting assembly
20. Bellcrank assembly
21. Cotter pin
22. Nut
23. Washer
24. Bolt
25. Washer (3 ea)
26. Bolt (3 ea)
27. Washer (2 ea)
28. Retaining ring
29. Washer (2 ea)
30. Shield
31. Slider plate
32. Retaining ring
33. Split bushing
34. Seal assembly
35. Bolt
36. Washer
37. Nut
38. Cotter pin
39. Cambox cam
40. Cotter pin
41. Nut
42. Washer
43. Slider housing
44. Slider
45. Bolt
46. Washer
47. Washer
48. Nut
49. Cotter pin
50. Control tube
51. Arm
52. Washer
53. Washer
54. Bell crank
55. Bearing
56. Pin
57. Washer
58. Rivet

ICN-412-A-761100-B-97499-00009-A-001-01

Figure 1. Droop Compensator Controls (S/N 36097 and Subsequent and S/N 33108 through 33213, S/N 36019 through 36096 Modified by TB 412-94-122) (Sheet 1 of 1)

Droop Compensator Controls - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Sealant	C-308	AR	
Sealing compound	C-353	AR	
Sealant	C-356	AR	

Procedure

1. If removed, install the bearings (55, Figure 1) as follows:

1.1. Apply retaining compound sealant (C-356) to the outside diameter of the bearings (55).

1.2. Install the bearings (55) in the bellcrank assembly (20).

1.3. Install the bellcrank (54) in the fitting (19).

1.4. Install the arm (51) with the washers (52 and 53) between the arm (51) and bearing (55) to remove the side play between the bellcrank (54) and the arm (51).

NOTE

It is permissible to have a lateral play of less than 0.0132 in (0.34 mm).

1.5. Install the new rivet (58) through the arm (51) and the bellcrank (54).

1.6. Install the washer (57) and pin (56).

2. Install the fitting assembly (19) and bellcrank assembly (20) using the washers (25) and bolts (26).

3. Move the boot assembly (17) with the large end toward the lower connection fitting assembly (19), retaining ring (28), and clamp (16) along the tube (15), (retaining ring goes on large end of the boot assembly and clamp goes on the small upper end of the boot), temporarily attach at midpoint of the control tube (15).

4. Install the control tube (15), adjustable rod end up, through the hole in engine deck and connect lower end of control tube to the outboard arm of the bellcrank assembly (20) with the bolt (24), washer (23), nut (22), and cotter pin (21).

5. Install the fitting assembly (13) to the engine service deck with the two screws (14) and two washers (29).

6. Install the bellcrank assembly (10) to the fitting assembly (13) with the washer (42), nut (41), and cotter pin (40).

7. Install the bellcrank assembly (10) to the cambox cam (39) with the bolt (1), washer (2), nut (3), and cotter pin (4). Put the arm of the bellcrank assembly approximately in middle of the adjustment slot of the cambox cam.

8. Install the bolt (9), washer (8), nut (7), and cotter pin (6).

9. Remove the retaining ring (32) and split bushing (33) from the seal assembly (34). Move the retaining ring along the control tube (15).

10. Install the seal assembly (34) on the engine deck using the screws (12) and washers (11). Apply a bead of sealing compound (C-353) to the mating surfaces of the seal assembly and engine deck prior to installation of the seal. Remove the excessive squeezeout.

11. Install the retaining ring (32) on the two part split bushing (33) located on the slider plate (31) in the seal assembly (34).

12. Connect the upper end of the control tube (15) to the bellcrank assembly (10). Check for free movement of the control system prior to the connecting linear-actuator to the slider housing (43).

13. Connect the control tube (50) to the bellcrank assembly (20). Install the bolt (45), washers (46 and 47), nut (48), and cotter pin (49).

14. Slide the boot assembly (17) with the large end down toward the lower connection fitting assembly (19) on the control tube (15). Apply a bead of sealant (C-308) to the mating surfaces of the boot assembly and shelf assembly and secure in place.

15. Install the clamp (16) and retaining ring (28) on the boot assembly (17). Clamp should be secured in the mid-travel point of the control tube (15).

NOTE

If hydraulic test unit is not available disconnect the control tube (19, Figure 2).

16. Connect the hydraulic test unit (412-A-29-00-00-00A-320A-A / 00616). Move the collective controls through the full travel and inspect for the binding and interference in droop compensator control system.

Applicable: SN: 33001-33213 or 36001-36019 or 412-570-001-103: Not Installed or (BHT-412-SI-74: Not Installed BHT-412-SI-97: Not Installed):

17. *Rig the modified droop compensator controls (*412-A-76-11-00-09A-276A-A / 01763*).*

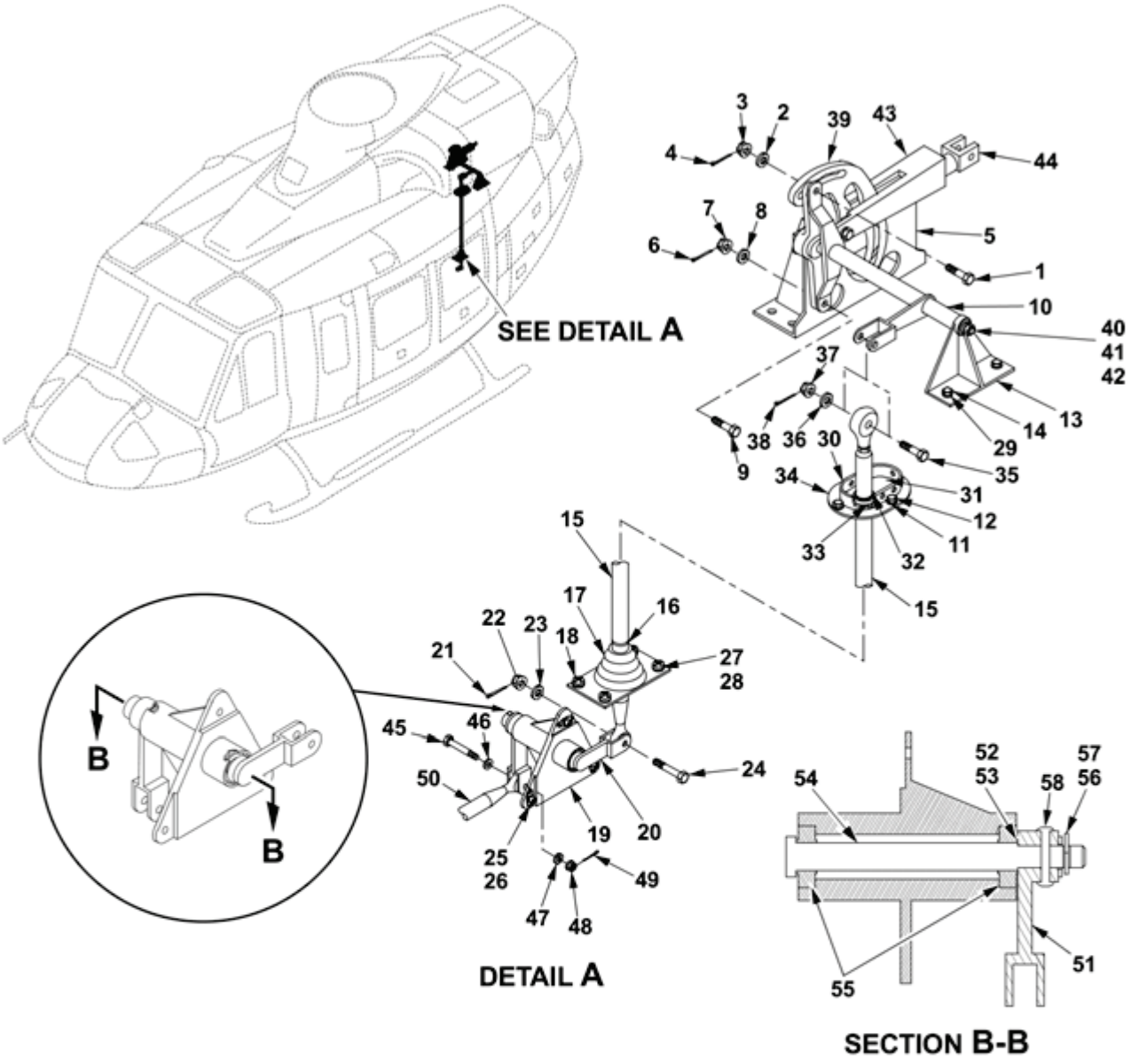
Applicable: SN: 36020-36096 or 412-570-001-103: Installed or BHT-412-SI-74: Installed or BHT-412-SI-97: Installed or (SN: 33108-33213 or 36001-36019 TB 412-94-122: Post):

18. *Rig the modified droop compensator controls (*412-A-76-11-00-09A-276B-A / 01764*).*

Applicable: SN: 36097-Subsequent:

19. Rig the droop compensator controls (412-A-76-11-00-09A-276B-A / 01764).

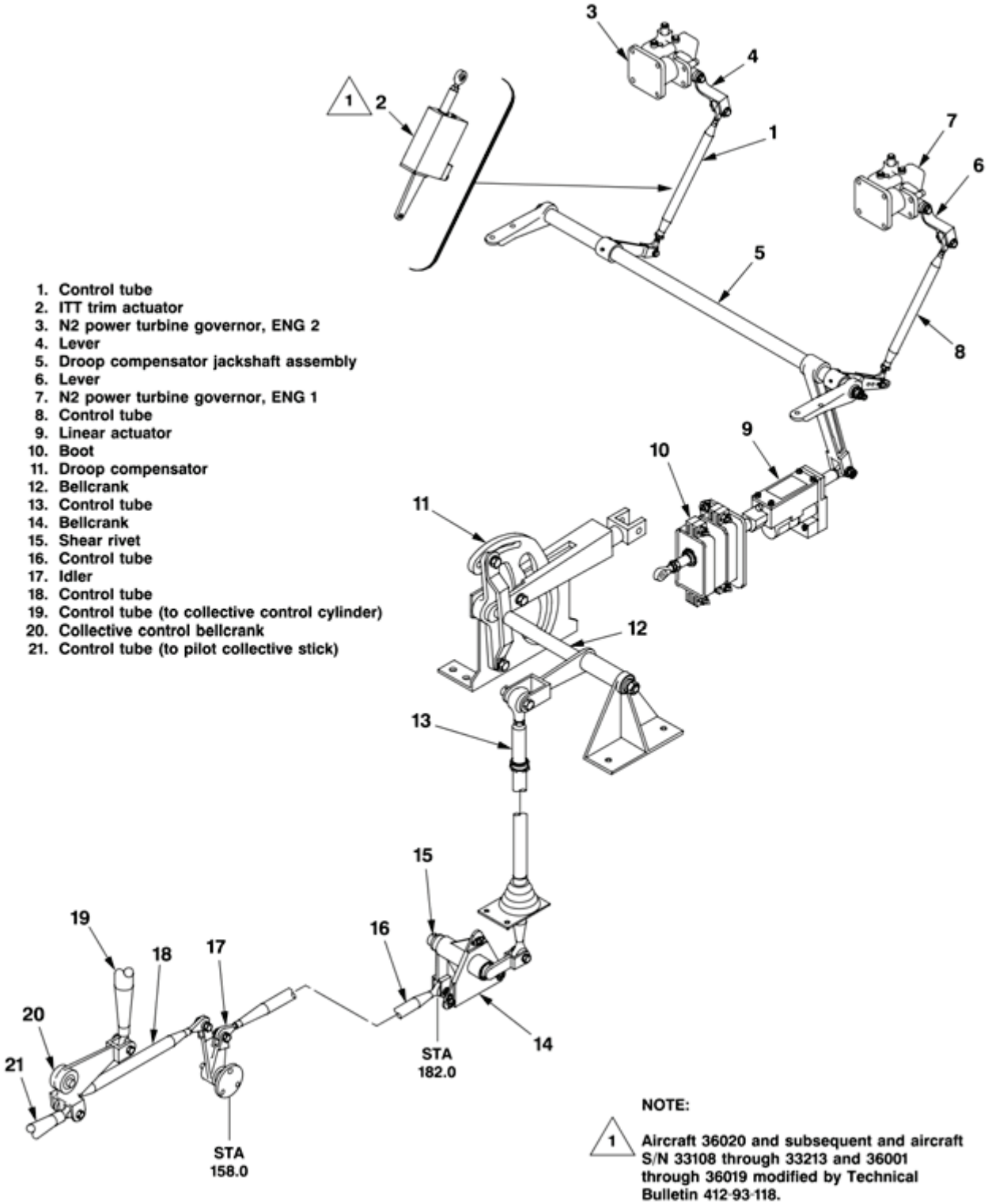
20. Perform the maintenance test flight.



1. Bolt
2. Washer
3. Nut
4. Cotter pin
5. Cambox assembly
6. Cotter pin
7. Nut
8. Washer
9. Bolt
10. Bellcrank assembly
11. Washer (4 ea)
12. Screw (4 ea)
13. Fitting assembly
14. Screw (2 ea)
15. Control tube
16. Clamp
17. Boot assembly
18. Screw (4 ea)
19. Fitting assembly
20. Bellcrank assembly
21. Cotter pin
22. Nut
23. Washer
24. Bolt
25. Washer (3 ea)
26. Bolt (3 ea)
27. Washer (2 ea)
28. Retaining ring
29. Washer (2 ea)
30. Shield
31. Slider plate
32. Retaining ring
33. Split bushing
34. Seal assembly
35. Bolt
36. Washer
37. Nut
38. Cotter pin
39. Cambox cam
40. Cotter pin
41. Nut
42. Washer
43. Slider housing
44. Slider
45. Bolt
46. Washer
47. Washer
48. Nut
49. Cotter pin
50. Control tube
51. Arm
52. Washer
53. Washer
54. Bell crank
55. Bearing
56. Pin
57. Washer
58. Rivet

ICN-412-A-761100-B-97499-00009-A-001-01

Figure 1. Droop Compensator Controls (S/N 36097 and Subsequent and S/N 33108 through 33213, S/N 36019 through 36096 Modified by TB 412-94-122) (Sheet 1 of 1)



ICN-412-A-760000-B-97499-00004-A-001-01

Figure 2. Droop Compensator Linkage (S/N 36097 and Subsequent and S/N 33108 through 33213 and 36001 through 36096 Modified by Technical Bulletin 412-94-122) (Sheet 1 of 1)

N2 Control - Rigging

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Corrosion preventive compound	C-101	AR	

Procedure

NOTE

Refer to 412-A-76-10-00-00A-276A-A / 01725 for N_1 (power lever control) rigging.

1. Comply with the following general procedures when rigging droop compensator controls.

1.1. When a rigging procedure requires a specific position to accomplish following steps, manually hold controls in position or secure controls in position with pilot friction adjustment.

1.2. Tolerance for rigging dimensions is ±0.030 inch (0.76 mm) unless otherwise stated.

1.3. All adjustable control tubes must have a maximum of 1.00 inch (25.4 mm) of exposed threads after adjustment. If control tubes have inspection holes, threads must be engaged sufficiently to cover holes.

1.4. Apply corrosion preventive compound (C-101) to threads of all adjustable rod end bearings and clevises after adjustment.

1.5. Torque control jamnuts 80 to 100 inch-pounds (9.0 to 11.3 Nm) after adjustment unless otherwise specified.

1.6. Control tubes must be free to rotate several degrees about longitudinal axis. If necessary, loosen jamnuts, adjust end fitting and retorque jamnuts.

2. Make sure collective pitch control system rigging has been accomplished (412-A-67-01-00-01A-276A-A / 01323).

3. Verify levers (2 and 6, Figure 1) are correctly installed on control shafts of power turbine governors (1 and 5). Nominal setting positions the centerline of levers 15° (one serration) clockwise from centerline of governor stop arm (18), viewed from left looking inboard (Detail A). In subsequent steps levers may be repositioned ±15° (one serration) from the nominal setting if required to achieve proper rigging and eliminate any fouling conditions.

4. Connect control tube (3) to governor lever (2) and droop compensator jackshaft assembly (4).

NOTE

The dimension given in Step 5. is approximate for initial rigging. Minor adjustments may be necessary during ground and flight test.

5. Verify adjustable control tube (7) is approximately 9.34 inches (237.23 mm) measured between attachment points (center of rod end bearings).

6. Connect control tube (7) to governor lever (6) and to the middle hole of jackshaft lever (2) (Detail A).

7. Disconnect linear actuator (8) from lever of droop compensator jackshaft assembly (4).

8. Disconnect linear actuator (8) rod end bearing from cambox slider (21).

9. Disconnect tube (13) from bellcrank (24).

NOTE

The dimension given in Step 10. is approximate for initial rigging. Minor adjustments may be necessary during ground and flight test.

10. Verify long arm of cambox bellcrank (23) is attached approximately in the middle of adjustment slot of cambox cam (20) (Detail B).

11. Use RPM INCR/DECR (beep) switch on collective switchbox to operate actuator. Verify linear actuator (8) is adjusted to a total stroke of approximately 0.62 inch (15.7 mm) from full extend (DECR) to full retract (INCR) positions. If not, refer to 412-A-76-11-00-01A-720A-A / 01743 for actuator rigging procedure.

12. Adjust rod end on linear actuator (8) to approximately 1.88 inches (47.8 mm) from center of bearing bolt hole to face of jamnut (Detail D).

13. Connect linear actuator (8) rod end bearing to cambox slider (21).

14. Connect linear actuator (8) to lever of droop compensator jackshaft assembly (4).

15. Connect control tube (13) to bellcrank (24).

16. Disconnect control tube (28) from arm (29) and rotate bellcrank (12) to ensure there is no fouling. Rotate cambox bellcrank (24) and ensure there is no fouling.

17. Connect control tube (28) to arm (29).

NOTE

If hydraulic test unit is not available disconnect control tube (27).

18. Connect hydraulic test unit (412-A-29-00-00-00A-009A-A / 00610).

19. With hydraulic boost ON, place collective stick full down. Adjust the length of control tube (13) to hold cambox cam (20) so lower end of cam slot is approximately 0.280 inch (7.11 mm) below slider housing (22).

20. With hydraulic boost ON, place collective stick full up. Verify cam slot is approximately 0.280 inch (7.11 mm) above slider housing (22).

21. With collective stick full up, hold RPM INCR/DECR switch to INCR until linear actuator (8) is fully retracted. Verify minimum clearance of 0.010 inch (0.25 mm) (Detail A) between stop arm (18) and maximum power stop screw (19) on both power turbine governors (1 and 5).

22. With hydraulic boost ON, place collective stick full down. Hold RPM INCR/DECR switch to DECR until linear actuator (8) is fully extended. Verify minimum clearance of 0.010 inch (0.25 mm) between stop arm (18) and minimum power stop screw (17) on both power turbine governors (1 and 5).

23. Disconnect hydraulic test unit or connect control tube (15) as applicable.



VERIFY ALL CONTROLS ARE PROPERLY CONNECTED PRIOR TO GROUND RUN AND FLIGHT TEST.

NOTE

Prior to accomplishing ground run verify Automatic Fuel Control Unit (AFCU) and Manual Fuel Control Unit (MFCU) maximum stops screws are being contacted with throttles full open (412-A-76-10-00-00A-276A-A / 01725). During ground run verify acceleration time difference between power sections is within 0.5 seconds (412-A-76-10-00-00A-280F-A / 01735, 412-A-76-10-00-00A-271E-A / 01736 and 412-A-76-10-00-00A-280G-A / 01737).

24. Start both power sections per BHT-412-FM and idle for 5 minutes.

25. Actuate RPM INCR/DECR switch to full DECR.

26. Place collective stick full down. Rotate ENG 2 throttle to full INCR. Record rotor RPM (N_R). The recorded value should be 95% rotor RPM. If rotor RPM is not 95% adjust as follows.



RETURN ENGINES TO IDLE FOR ANY ADJUSTMENTS TO CONTROLS WITH ENGINE RUNNING.

NOTE

Three turns of actuator (8) rod end is equal to 1% rotor RPM change.

26.1. If rotor RPM is below 95%, shorten rod end on actuator (8).

26.2. If rotor RPM is above 95%, lengthen rod end on linear actuator (8).

NOTE

The single engine beep range should be 95 to 99.5% rotor RPM.

27. Slowly actuate RPM INCR/DECR switch to full INCR and verify the rotor RPM is 99 to 99.5%. If the range is not within limits adjust linear actuator (8) stroke per 412-A-76-11-00-01A-720A-A / 01743.

28. Actuate RPM INCR/DECR switch to full DECR.

29. Return ENG 2 to IDLE, rotate ENG 1 throttle to full INCR. Record rotor RPM. The recorded value should be 95% rotor RPM. If rotor RPM is not 95% adjust as follows.



RETURN ENGINES TO IDLE FOR ANY ADJUSTMENTS TO CONTROLS WITH ENGINE RUNNING.

NOTE

Two turns of control tube (7) rod end is equal to 1% rotor RPM change.

29.1. If rotor RPM is below 95%, shorten rod end on control tube (7).

29.2. If rotor RPM is above 95%, lengthen rod end on control tube (7).

NOTE

The single engine beep range should be 95 to 99.5% rotor RPM.

30. Slowly actuate RPM INCR/DECR switch to full INCR and verify the rotor RPM is 99 to 99.5%.

NOTE

When performing the following steps helicopter will become airborne.

31. Return ENG 1 to IDLE, rotate ENG 2 to full INCR. Slowly actuate RPM INCR/DECR switch to 98% rotor RPM. Gradually increase collective from flat pitch to 60% engine torque and not static rotor RPM droop.

32. Return ENG 2 to IDLE, rotate ENG 1 to full INCR. Slowly actuate RPM INCR/DECR switch to 98% rotor RPM. Gradually increase collective from flat pitch to 60% engine torque and not static rotor RPM droop.

33. If more than 2% rotor RPM droop is encountered perform troubleshooting of engine pneumatic lines and N_2 power turbine governor per Pratt & Whitney Canada Maintenance Manual before proceeding with further adjustments.

34. Actuate RPM INCR/DECR switch to full DECR.

35. With collective stick full down, rotate ENG 1 and ENG 2 throttles to FULL INCR. Actuate RPM INCR/DECR switch to 100% rotor RPM.

NOTE

Engine torque difference (split) between ENG 1 and ENG 2 shall not exceed 4% from flat pitch to maximum power.

36. Gradually increase collective from flat pitch to full power (either transmission torque, ITT or N_1 limit) and note the difference in engine torque readings. If not within limits, adjust as follows:

36.1. If ENG 1 torque reading is greater than ENG 2 by more than 4% throughout the collective range, lengthen control tube (7).

36.2. If ENG 1 torque reading is less than ENG 2 by more than 4% throughout the collective range, shorten control tube (7).

36.3. If ENG 1 torque reading is greater than ENG 2 by more than 4% at high power range and too low at low power range, move control tube (7) rod end forward (DECR) on lever (25) (Detail C).

36.4. If ENG 1 torque reading is less than ENG 2 by more than 4% at high power range and/or too low at low power range, move control tube (7) rod end AFT (INCR) on lever (25) (Detail C).

37. Place collective stick full down; rotate both engine throttles to full INCR. Use RPM INCR/DECR switch to obtain 100% rotor RPM. Increase collective pitch slowly in a series of equal steps from flat pitch to full power (either transmission torque, engine ITT or N_1 limit). Rotor RPM should remain at 100 ±1% throughout power sweep.

38. If rotor RPM droops (decays below tolerance) or overspeeds (increases above tolerance) during power applications, make a cam rate adjustment.

NOTE

Make adjustments in small increments of approximately 0.125 inch (3.18 mm) measured on cam slot.

38.1. For increased cam compensation to correct droop, adjust cambox cam (20) counterclockwise relative to cambox bellcrank (23) (Detail B).

38.2. For reduced cam compensation to correct overspeed, adjust cambox cam (20) clockwise relative to cambox bellcrank (23).

38.3. After each cam adjustment, move the collective to full down and then full up positions to ensure there is no fouling between slider housing (22) and cambox slot. Verify that a minimum of 0.010 inch (0.25 mm) cam slot shows below and above slider housing. If required adjust control tube (13) for proper clearance.



RETURN ENGINES TO IDLE FOR ANY ADJUSTMENTS TO CONTROLS WITH ENGINE RUNNING.

NOTE

Adjustments to achieve proper twin-engine RPM range and adjustments to avoid torque needle split, will supersede the initial single engine beep ranges.

Three turns of actuator (8) rod end is equal to 1% rotor RPM change.

39. Actuate RPM INCR/DECR switch to full decrease with collective in full down position. Rotate both engine throttles to full INCR and record rotor RPM. The recorded rotor RPM should be 97%. If not adjust actuator (8) rod end.

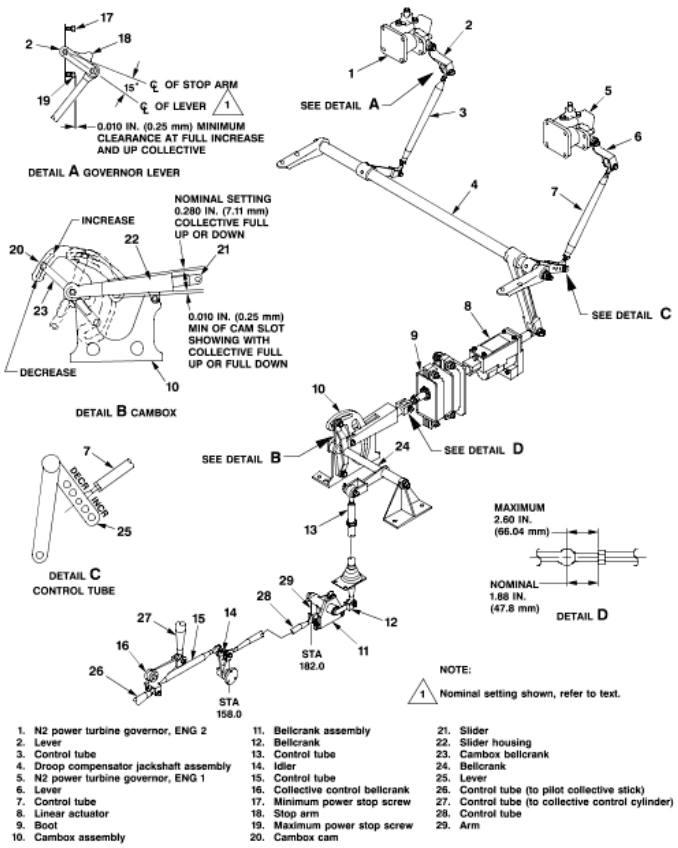
40. Actuate RPM INCR/DECR switch to full INCR and verify rotor RPM is 101.5%. The required range for twin-engine beep is 97 to 101.5% rotor RPM. If not, adjust linear actuator (8) stroke per 412-A-76-11-00-01A-720A-A / 01743.

NOTE

If hydraulic test unit is not available disconnect control tube (27).

41. With collective stick full up, hold RPM INCR/DECR switch to INCR until linear actuator (8) is fully retracted. Verify minimum clearance of 0.010 inch (0.25 mm) (Detail A) between stop arm (18) and maximum power stop screw (19) on both power turbine governors (1 and 5).

42. Check complete system for security of all parts and ensure no interference exists. Apply corrosion preventive compound (C-101) to all exposed threads.



ICN-412-A-761100-B-97499-00010-A-001-01

Figure 1. N2 Controls Rigging (S/N 33108 through 33213 and 36001 through 36019 Modified by TB 412-94-122) (Sheet 1 of 1)

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N2 Control - Rigging

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Corrosion preventive compound	C-101	AR	

Procedure

NOTE

Refer to 412-A-76-10-00-00A-276A-A / 01725 for N_1 (power lever control) rigging.

1. Obey the following general procedures when rigging droop compensator controls.

1.1. When a rigging procedure requires a specific position to accomplish following steps, manually hold controls in position or secure controls in its position with pilot friction adjustment.

1.2. Tolerance for rigging dimensions is ± 0.030 inch (0.76 mm) unless otherwise stated.

1.3. All adjustable controls tubes must have a maximum of 1.00 inch (25.4 mm) of exposed threads after adjustment. If control tubes have inspection holes, threads must be engaged sufficiently to cover holes.

1.4. Apply corrosion preventive compound (C-101) to threads of all adjustable rod end bearings and clevises after adjustment.

1.5. Torque control jamnuts 80 to 100 inch-pounds (9.0 to 11.3 Nm) after adjustment unless otherwise specified.

1.6. Control tubes must be free to rotate several degrees about its longitudinal axis. If necessary, loosen jamnuts, adjust the end fitting and torque jamnuts.

2. Make sure that the collective control system rigging has been accomplished (412-A-67-01-00-01A-276A-A / 01323).

3. Make sure that levers (2 and 6, Figure 1) are correctly installed on control shafts of power turbine governors (1 and 5). Nominal setting positions the centerline of levers 15° (one serration) clockwise from centerline of governor stop arm (18), viewed from left looking inboard (Detail A). In subsequent steps levers may be repositioned ± 15° (one serration) from the nominal setting if required to achieve proper rigging and eliminate any fouling conditions.

NOTE

(SN: 36020-36278 TB 412-01-176: Post) or SN: 36279-Subsequent:

Helicopters ITT actuator (3) total length is 9.455 inches (240.157 mm).

4. Connect the ENG 2 ITT actuator (3) to the governor lever (2) and the jackshaft (4). Make sure that the total stroke of the ITT actuator (3) is 0.150 inch (3.81 mm) and total length is 9.415 inches (239.14 mm) or 9.455 inches (240.157 mm) with the actuator fully extended. Refer to 412-A-76-11-00-02A-720A-A / 01746 for the installation and rigging procedure.

NOTE

(SN: 36020-36278 TB 412-01-176: Post) or SN: 36279-Subsequent:

The dimension given in Step 5. is approximate for initial rigging. Minor adjustments may be necessary during ground and flight test. For helicopters, the control tube (7) length is 9.38 inches (238.25 mm).

5. Make sure that the adjustable control tube (7) is approximately 9.34 or 9.38 inches (237.23 or 238.25 mm) measured between the attachment points (center of rod end bearings).

6. Connect the control tube (7) to the governor lever (6) and to the middle hole of the jackshaft lever (25) (Detail C).

7. Disconnect the linear actuator (8) from the lever of the droop compensator jackshaft assembly (4).

8. Disconnect the linear actuator (8) rod end bearing from the cambox slider (21).

9. Disconnect the control tube (13) from the bellcrank (24).

NOTE

The dimension given in Step 10. is approximate for initial rigging. Minor adjustments may be necessary during ground and the flight test.

10. Make sure that the long arm of the cambox bellcrank (23) is attached approximately in the middle of the adjustment slot of the cambox cam (20) (Detail B).

NOTE

(SN: 36020-36278 TB 412-01-176: Post) or SN: 36279-Subsequent:

For helicopters, the linear actuator (8) stroke is 0.850 inch (21.6 mm).

11. Use the RPM INCR/DECR (beep) switch on the collective switchbox to operate the actuator. Make sure that the linear actuator (8) is adjusted to a total stroke of approximately 0.62 or 0.850 inch (15.7 or 21.6 mm) from full extend (DECR) to full retract (INCR) positions. If not, refer to 412-A-76-11-00-01A-720A-A / 01743 for the actuator rigging procedure.

12. Adjust the rod end on the linear actuator (8) to approximately 1.88 inch (47.8 mm) from the center of the bearing bolt hole to the face of the jam nut (Detail D).

13. Connect the linear actuator (8) rod end bearing to the cambox slider (21).

14. Connect the linear actuator (8) to the lever of the droop compensator jackshaft assembly (4).

15. Connect the control tube (13) to the bellcrank (24).

16. Disconnect the control tube (28) from the arm (29) and rotate the bellcrank (12) to make sure that there is no fouling. Rotate the bellcrank (24) and make sure that there is no fouling.

17. Connect the control tube (28) to the arm (29).

NOTE

If a hydraulic test unit is not available, disconnect the control tube (27).

18. Connect a hydraulic test unit (412-A-29-00-00-00A-320A-A / 00616).

19. With the hydraulic boost ON, place the collective stick full down. Adjust the length of the control tube (13) to hold the cambox cam (20) so the lower end of the cam slot is approximately 0.280 inch (7.11 mm) below the slider housing (22).

20. With the hydraulic boost ON, place the collective stick full up. Make sure that the cam slot is approximately 0.280 inch (7.11 mm) above the slider housing (22).

21. With the collective stick full up, hold the RPM INCR/DECR switch to INCR until the linear actuator (8) is fully retracted. Hold the ENG 2 (+2/-2) switch to +2 until the ITT actuator (3) is fully retracted. Make sure that there is a minimum clearance of 0.010 inch (0.25 mm) (Detail A) between the stop arm (18) and the maximum power stop screw (19) on both power turbine governors (1 and 5).

22. With the hydraulic boost ON, place the collective stick full down. Hold the RPM INCR/DECR switch to DECR until the linear actuator (8) is fully extended. Hold the ENG 2 (+2/-2) switch to -2 until the ITT actuator (3) is fully extended. Make sure that there is a minimum clearance of 0.010 inch (0.25 mm) between the stop arm (18) and the minimum power stop screw (17) on both power turbine governors (1 and 5).

23. Disconnect the hydraulic test unit or connect the control tube (27), as necessary.



MAKE SURE THAT ALL CONTROLS ARE PROPERLY CONNECTED BEFORE THE GROUND RUN AND FLIGHT TEST.

NOTE

Before the ground run, make sure that the Automatic Fuel Control Unit (AFCU) and the Manual Fuel Control Unit (MFCU) maximum stops screws contact with the throttles full open (412-A-76-10-00-00A-276A-A / 01725). During the ground run, make sure that the acceleration time difference between the power sections is within 0.5 seconds (412-A-76-10-00-00A-280F-A / 01735, 412-A-76-10-00-00A-271E-A / 01736 and 412-A-76-10-00-00A-280G-A / 01737).

24. Start both power sections per BHT-412-FM and idle for five minutes.

25. Actuate the RPM INCR/DECR switch to full DECR.

NOTE

The positioning of the ENG 2 ITT actuator (3) to mid stroke can be done during the ground run by actuating the ENG 2 (+2/-2) switch to full +2, record the rotor RPM. Actuate the ENG 2 (+2/-2) switch to full -2, record the rotor RPM. Actuate the ENG 2 (+2/-2) switch to set the rotor RPM to the middle of the recorded range.

26. Make sure that the ENG 2 ITT actuator (3) is at the mid stroke position, 9.34 inches (237.2 mm), before proceeding to the next step.

27. Place the collective stick full down. Rotate the ENG 2 throttle to full INCR. Record the rotor RPM (N_R). Slowly actuate the RPM INCR/DECR switch to full INCR but not more than 101% RPM. Record the rotor RPM. Make sure that full INCR produces a 99% rotor RPM minimum.



RETURN ENGINES TO IDLE FOR ANY ADJUSTMENTS TO CONTROLS WITH ENGINE RUNNING.

NOTE

Three turns of the actuator (8) rod end is equal to 1% rotor RPM change.

28. If a minimum rotor RPM cannot be obtained, make an interim adjustment to the rod end of the linear actuator (8).

29. With the collective full down, rotate the ENG 2 throttle to full INCR. Actuate the RPM INCR/DECR switch to full DECR and the ENG 2 (+2/-2) switch to full +2. Record the rotor RPM. Actuate the ENG 2 (+2/-2) switch to full -2 and record the rotor RPM. The total spread between the rotor RPM must be 2 1/4 ± 1/4%. If the spread is not within the limits, adjust the ITT actuator (3) stroke per 412-A-76-11-00-02A-720A-A / 01746.

30. Return ENG 2 to IDLE, rotate the ENG 1 throttle to full INCR. Record the rotor RPM. Slowly actuate the RPM INCR/DECR switch to full INCR but not to exceed 101% rotor RPM. Record the rotor RPM. The minimum full INCR must be 99% rotor RPM.



RETURN ENGINES TO IDLE FOR ANY ADJUSTMENTS TO CONTROLS WITH ENGINE RUNNING.

NOTE

Two turns of the control tube (7) rod end is equal to 1% rotor RPM change.

31. If the minimum rotor RPM cannot be obtained, make an interim adjustment to the rod end of the tube (7).

NOTE

The helicopter will become airborne in the steps that follow.

32. Return ENG 1 to IDLE, rotate ENG 2 to full INCR. Slowly actuate the RPM INCR/DECR switch to 98% rotor RPM. Gradually increase collective from flat pitch to 60% engine torque and record the static rotor RPM droop.

33. Return ENG 2 to IDLE, rotate ENG 1 to full INCR. Slowly actuate the RPM INCR/DECR switch to 98% rotor RPM. Gradually increase collective from flat pitch to 60% engine torque and record the static rotor RPM droop.

34. If more than 2% rotor RPM droop is encountered, perform troubleshooting of the engine pneumatic lines and N_2 power turbine governor per the Pratt & Whitney Canada Maintenance Manual before proceeding with further adjustments.

35. With the rotor at flat pitch, increase both throttles to full INCR and adjust the rotor RPM to 100% using the RPM INCR/DECR switch.

36. Gradually increase collective from flat pitch to approximately 60% transmission torque or mast torque (as necessary). Using the +2/–2 switch, trim ENG 2 to minimum, and then to maximum and record the torque split. The torque split must be approximately equal above and below the ENG 1 torque indication. If the torque split difference is greater than 5% above or below ENG 1 torque, adjust the length of the control tube (7) as follows.

NOTE

A 1/2 turn of the rod end on the control tube (7) is equal to approximately 5% engine torque change.

36.1. To increase torque on ENG 1, shorten the rod end of the control tube (7).

36.2. To decrease torque on ENG 1, lengthen the rod end of the control tube (7).

NOTE

The engine torque difference (split) between ENG 1 and ENG 2 shall not exceed 4% during power application throughout this maintenance procedure. However, during normal operation, torque split may exceed 4%. The +2/-2 switch may be used to trim the engines for either torque matching or ITT matching.

37. At 100% rotor RPM, gradually increase collective to 60% transmission torque or mast torque (as necessary) and match the engines torques using the +2/-2 switch. Gradually decrease collective to flat pitch and record the torque split. Gradually increase collective to full power (either transmission torque or mast torque (as necessary), ITT, or N_1 limit) and record the torque split throughout the power range. Torque matching must remain constant within 4%. If the torque does not remain constant throughout the power range, adjust as follows.

37.1. If ENG 1 torque reading is greater than ENG 2 by more than 4%, move the control tube (7) rod end forward (DECR) on the lever (25) (Detail C).

37.2. If ENG 1 torque reading is less than ENG 2 by more than 4%, move the control tube (7) rod end aft (INCR) on the lever (25) (Detail C).

38. Put the collective stick full down, rotate both engine throttles to full INCR. Use the RPM INCR/DECR switch to obtain 100% rotor RPM and match engine torques using the +2/-2 switch. Increase collective pitch slowly in a series of equal steps from flat pitch to full power (either transmission torque or mast torque (as necessary), ITT or N_1 limit). Rotor RPM must remain at 100 ±1% throughout the power sweep.

39. If rotor RPM droops (decays below tolerance) or overspeeds (increases above tolerance) during power applications, make a cam rate adjustment.

NOTE

Make adjustments in small increments of approximately 0.125 inch (3.18 mm) measured on the cam slot.

39.1. For increased cam compensation to correct droop, adjust the cambox cam (20) counterclockwise relative to the cambox bellcrank (23) (Detail B).

39.2. For reduced cam compensation to correct overspeed, adjust the cambox cam (20) clockwise relative to the cambox bellcrank (23).

39.3. After each cam adjustment, move the collective to full down and then full up positions to make sure that there is no fouling between the slider housing (22) and the cambox slot. Make sure that a minimum of 0.010 inch (0.25 mm) cam slot shows below and above the slider housing. If necessary, adjust the control tube (13) for proper clearance.



RETURN ENGINES TO IDLE FOR ANY ADJUSTMENTS TO CONTROLS WITH ENGINE RUNNING.

NOTE

Three turns of the linear actuator (8) rod end is equal to 1% rotor RPM change.

40. Actuate the RPM INCR/DECR switch to full decrease with collective in the full down position. Rotate both engine throttles to full INCR and record the rotor RPM. The recorded rotor RPM must be 97%. If not, adjust the linear actuator (8) rod end.



DO NOT EXCEED 104% ROTOR RPM.

NOTE

TB 412-01-176: Pre: Twin-engine beep range is 97 to 101.5% rotor RPM.

(SN: 36020-36278 TB 412-01-176: Post) or SN: 36280-Subsequent: Twin-engine beep range is 97 to 103.5% rotor RPM for (PT6T-3D/DE/DF engine equipped) helicopters.

Applicable: (SN: 36020-36278 TB 412-01-176: Post) or SN: 36279-Subsequent:

41. **TB 412-01-176: Pre:** Actuate the RPM INCR/DECR switch to full INCR and make sure that rotor RPM is 101.5. The required range for twin-engine beep is 97 to 101.5% rotor RPM.

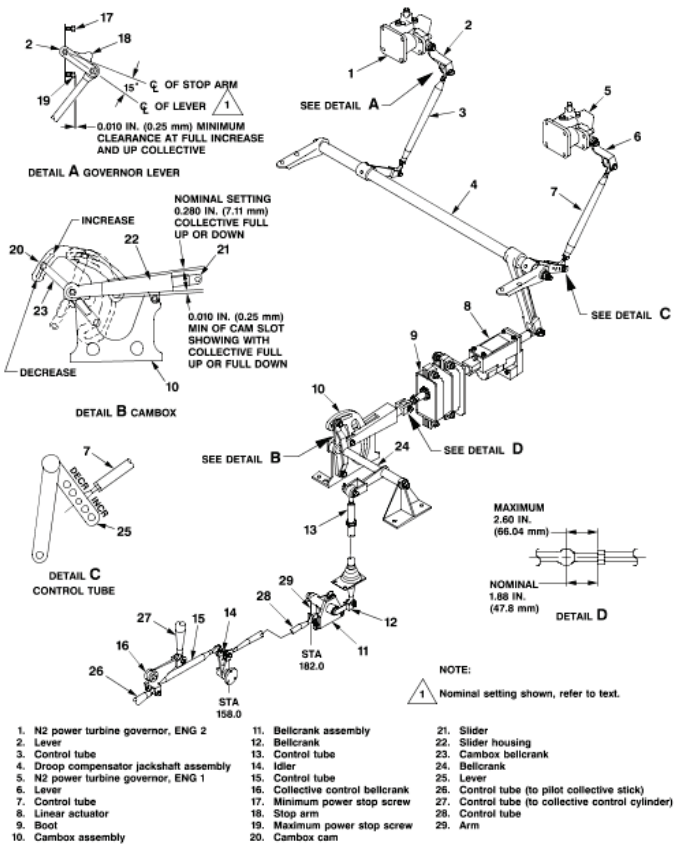
(SN: 36020-36278 TB 412-01-176: Post) or SN: 36280-Subsequent: Actuate the RPM INCR/DECR switch to full INCR and make sure that rotor RPM is 103.5%. The required range for twin-engine beep is 97 to 103.5% rotor RPM.

NOTE

If a hydraulic test unit is not available, disconnect the control tube (27).

42. With the collective stick full up, hold the RPM INCR/DECR switch to INCR until the linear actuator (8) is fully retracted. Hold the +2/-2 switch to +2 until the ITT actuator (3) is fully retracted. Make sure that there is a minimum clearance of 0.010 inch (0.25 mm) (Detail A) between the stop arm (18) and the maximum power stop screw (19) on both power turbine governors (1 and 5).

43. Check the complete system for security of all parts and make sure that no interference exists. Apply corrosion preventive compound (C-101) to all exposed threads.



1. N2 power turbine governor, ENG 2
2. Lever
3. Control tube
4. Droop compensator jackshaft assembly
5. N2 power turbine governor, ENG 1
6. Lever
7. Control tube
8. Linear actuator
9. Boot
10. Cambox assembly
11. Bellcrank assembly
12. Bellcrank
13. Control tube
14. Idler
15. Control tube
16. Collective control bellcrank
17. Minimum power stop screw
18. Stop arm
19. Maximum power stop screw
20. Cambox cam
21. Slider
22. Slider housing
23. Cambox bellcrank
24. Bellcrank
25. Lever
26. Control tube (to pilot collective stick)
27. Control tube (to collective control cylinder)
28. Control tube
29. Arm

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Figure 1. N2 Controls Rigging (S/N 36097 and Subsequent) (Sheet 1 of 1)

CHAPTER 79
ENGINE OIL SYSTEM

MAINTENANCE MANUAL

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List of Effective Data Modules

The listed documents are included in issue 007, dated 2025-11-18, of this publication.

C = Changed data module, N = New data module

Document title	Data module code / Seq number		Issue date	Applicable to
Engine Oil System - Description	DMC-412-A-79-00-00-00A-028A-A / 01766		2023-05-31	ALL
Engine Oil System - Troubleshooting	DMC-412-A-79-00-00-00A-422A-A / 01767	C	2025-11-18	ALL
Engine Oil Cooling System - Description	DMC-412-A-79-20-00-00A-042A-A / 01768		2023-05-31	ALL
Reduction (Combining) Gearbox Oil Temperature Switch - Removal	DMC-412-A-79-20-00-01A-520A-A / 01769		2023-05-31	ALL
Reduction (Combining) Gearbox Oil Temperature Switch - Installation	DMC-412-A-79-20-00-01A-720A-A / 01770		2023-05-31	ALL
Engine Oil Cooling System Check Valves - Description	DMC-412-A-79-20-00-02A-042A-A / 01771		2023-05-31	ALL
Engine Oil Cooling System Check Valves - Removal	DMC-412-A-79-20-00-02A-520A-A / 01772		2023-05-31	ALL
Engine Oil Cooling System Check Valves - Installation	DMC-412-A-79-20-00-02A-720A-A / 01773		2023-05-31	ALL
Engine Oil Cooling System Drain Valves - Description	DMC-412-A-79-20-00-03A-042A-A / 01774		2023-05-31	ALL
Engine Oil Cooling System Drain Valves - Removal	DMC-412-A-79-20-00-03A-520A-A / 01775		2023-05-31	ALL
Engine Oil Cooling System Drain Valves - Installation	DMC-412-A-79-20-00-03A-720A-A / 01776		2023-05-31	ALL
Combining Gearbox and Transmission Oil Cooler - Removal	DMC-412-A-79-20-00-04A-520A-A / 01777		2023-05-31	ALL
Combining Gearbox and Transmission Oil Cooler - Inspection	DMC-412-A-79-20-00-04A-280A-A / 01778		2023-05-31	ALL
Combining Gearbox and Transmission Oil Cooler - Installation	DMC-412-A-79-20-00-04A-720A-A / 01779		2023-05-31	ALL
Engine Power Section Oil Cooler - Removal	DMC-412-A-79-20-00-05A-520A-A / 01780		2023-05-31	ALL
Engine Power Section Oil Cooler - Inspection	DMC-412-A-79-20-00-05A-280A-A / 01781		2023-05-31	ALL
Engine Power Section Oil Cooler - Installation	DMC-412-A-79-20-00-05A-720A-A / 01782		2023-05-31	ALL
Engine and Transmission Oil Cooler Blower - Description	DMC-412-A-79-20-00-06A-042A-A / 01783		2023-05-31	ALL
Engine and Transmission Oil Cooler Blower - Removal	DMC-412-A-79-20-00-06A-520A-A / 01784		2023-05-31	ALL
Engine and Transmission Oil Cooler Blower - Inspection	DMC-412-A-79-20-00-06A-280A-A / 01785		2023-05-31	ALL
Engine and Transmission Oil Cooler Blower - Installation	DMC-412-A-79-20-00-06A-720A-A / 01786		2023-05-31	ALL

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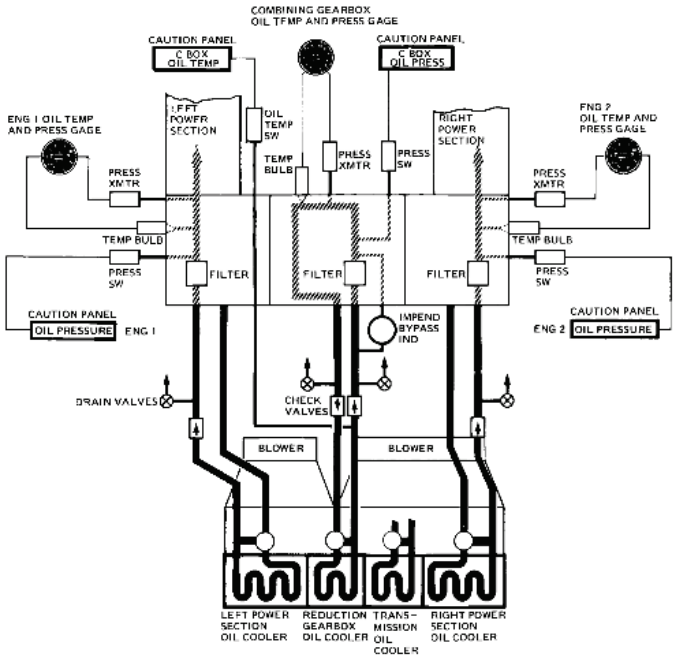
Engine Oil System - Description

NOTE

The terminology reduction gearbox and combining gearbox as used in this manual are one in the same. Bell Helicopter Textron (BHT) uses combining gearbox as evident on the annunciator light panel.

This chapter contains maintenance data for both engine (power sections) oil systems and reduction (combining) gearbox oil system. The oil systems of the power sections and reduction (combining) gearbox are separate systems and are connected by external hoses to three radiator-type coolers located aft of engine. The upper cooler is a dual unit, serving both the reduction (combining) gearbox and transmission oil system. The coolers are installed in a structural support, and are connected by air ducts to two blowers mounted on (and shaft-driven by) the reduction (combining) gearbox. Drain lines with manual valves are provided for all engine-to-cooler lines (Figure 1).

The oil supply for each power section is located in a tank in the accessory gearbox. The reduction (combining) gearbox has an internal oil supply. A temperature bulb and pressure transmitter in each system are connected to oil temperature and pressure gauges on the instrument panel. Pressure switches, temperature switches, and chip detectors in each system are connected to segments on the caution panel.



ICN-412-A-790000-B-97499-00001-A-001-01

Figure 1. Engine Oil System Schematic (Sheet 1 of 1)

Engine Oil System - Troubleshooting

Procedure

NOTE

If metallic pickup is found in the external lines of the engine, reduction (combining) gearbox, or transmission oil system, and if the engine, reduction (combining) gearbox, or transmission are being replaced, and if metal particles are suspected of being present in the oil cooler, replace the oil cooler.

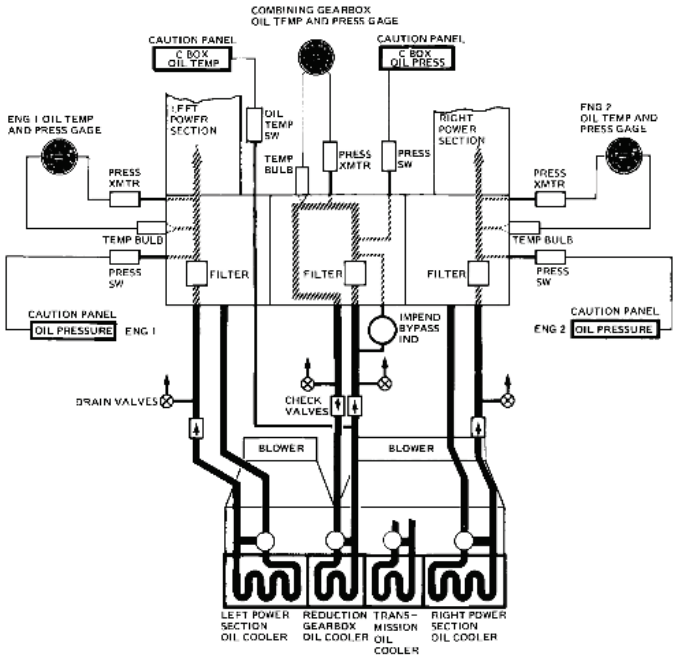
1. When troubleshooting the engine oil system, refer to Table 1.

Table 1. Engine Oil System - Troubleshooting

TROUBLE INDICATION	PROBABLE CAUSE	CORRECTIVE ACTION
ENG CHIP ENG1, ENG2, or C Box segment(s) illuminate.	Normal wear.	*Refer to Pratt and Whitney Engine Maintenance Manual Fault Isolation Chart.*
		DELETED
		DELETED
		DELETED
	Excessive wear.	*1. Refer to Pratt and Whitney Engine Maintenance Manual Fault Isolation Chart.*
		DELETED
		2. Check external lines of cooling system for evidence of metal chips. *a. If metallic pickup is found in external lines of engine oil system or if particles are suspected of being present in oil cooler, replace oil cooler with new cooler (412-A-79-20-00-05A-520A-A / 01780, 412-A-79-20-00-05A-280A-A / 01781 and 412-A-79-20-00-05A-720A-A / 01782).* *b. If it is necessary to replace engine because of metallic chip damage, oil cooler shall be replaced with new cooler (412-A-79-20-00-05A-520A-A / 01780, 412-A-79-20-00-05A-280A-A / 01781 and 412-A-79-20-00-05A-720A-A / 01782).*
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		DELETED
		DELETED
OIL PRESSURE ENG 1 or ENG 2 low	Fault in pressure gauge or caution panel circuits.	Replace faulty electrical components or repair circuit. A direct reading gauge can be connected at points shown in Figure 1 to check accuracy of indicating system.
	No oil supply.	Check oil level.
	Faulty oil pump or pressure relief valve.	Refer to Prate & Whiten Canada PT6T-3 Series engine maintenance manual.

Table 1. Engine Oil System - Troubleshooting (continued)

TROUBLE INDICATION	PROBABLE CAUSE	CORRECTIVE ACTION
Reduction gearbox impending bypass indicator extended.	Impending filter bypass.	1. Reset the indicator button and verify that it does not extend a second time by running the helicopter. 2. If indicator extends a second time, remove the filter and inspect for blockage. a. If filter is found serviceable, replace the impending bypass indicator (Section 71). b. If the filter is found obstructed or contaminated, refer to Pratt & Whitney Canada maintenance manual for corrective action.
Reduction gearbox (C-Box) oil pressure low.	Fault in indicating system.	Replace faulty electrical components or repair unit. A direct reading gauge can be connected at points shown in Figure 1 to check accuracy of indicating system.
	Low or no oil supply.	Check oil level.
	Faulty oil pressure regulating valve.	Refer to Pratt & Whitney Canada maintenance manual.
Reduction gearbox (C-Box) oil temperature high.	Fault in indicating system.	Replace faulty electrical components or repair unit.
	Faulty thermal bypass valve.	Repair or replace the valve.
	Partially blocked oil cooler.	Inspect and clean the oil cooler.
	Damaged oil cooler blower.	Repair or replace blower.



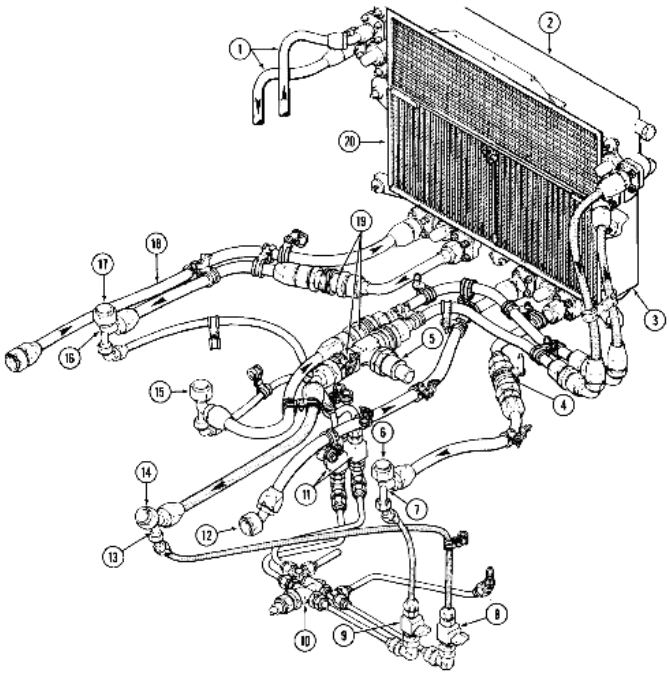
ICN-412-A-790000-B-97499-00001-A-001-01

Figure 1. Engine Oil System Schematic (Sheet 1 of 1)

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Engine Oil Cooling System - Description

The engine oil cooling system consists of engine power section and reduction (combining) gearbox oil coolers and lines, valves, and switch between engine and coolers.



1. Transmission oil lines
2. Reduction gearbox and transmission oil cooler
3. Left power section oil cooler
4. Check valve
5. Reduction gearbox oil temperature switch
6. Left power section oil inlet line
7. Direct reading gage attachment point
8. Drain valve
9. Drain valve
10. Drain manifold
11. Drain valves
12. Left power section oil outlet line
13. Direct reading gage attachment point
14. Reduction gearbox oil inlet line
15. Reduction gearbox oil outlet line
16. Direct reading gage attachment point
17. Right power section oil inlet line
18. Right power section oil outlet line
19. Check valves
20. Right power section oil cooler

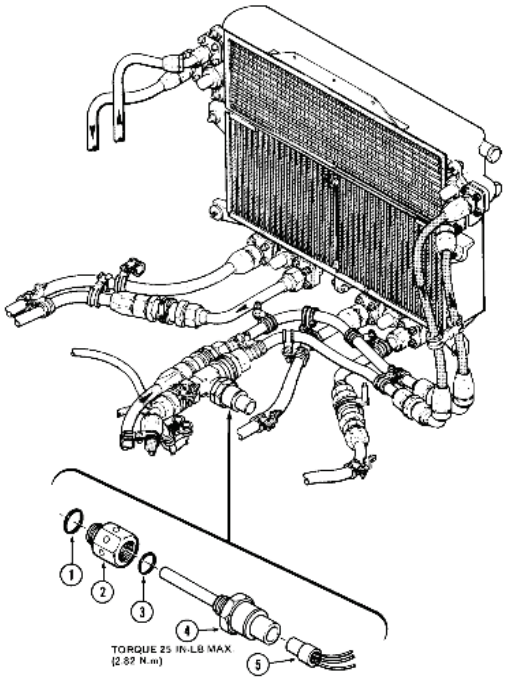
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Figure 1. Engine Oil Cooling System (Sheet 1 of 1)

Reduction (Combining) Gearbox Oil Temperature Switch - Removal

Procedure

1. Open left engine oil cooler support cowling.
2. Drain oil from combining gearbox oil cooler until oil level is below switch (4, Figure 1).
3. Disconnect connector (5).
4. Remove switch (4) with packing (3). Discard packing.
5. Remove bushing (2) with packing (1).



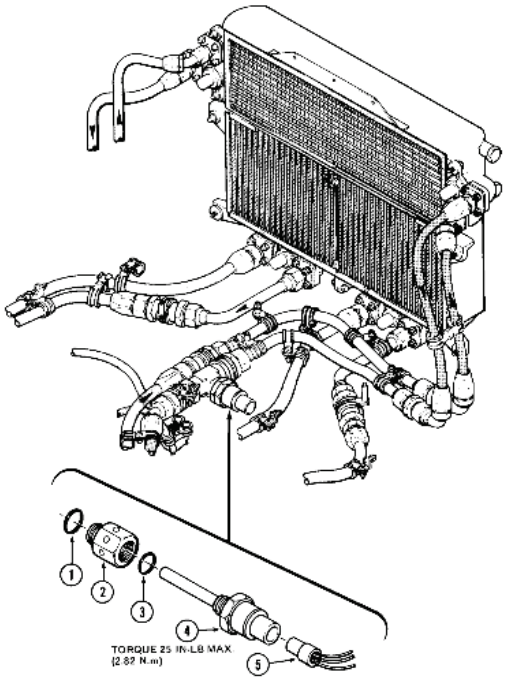
1. Packing
2. Bushing
3. Packing
4. Temperature switch
5. Electrical connector

ICN-412-A-792000-B-97499-00003-A-001-01

Figure 1. Reduction (Combining) Gearbox Oil Temperature Switch (Sheet 1 of 1)

Reduction (Combining) Gearbox Oil Temperature Switch - Installation Procedure

1. Install bushing (2, Figure 1) with new packing (1).
2. Install switch (4) with packing (3). Torque switch 25 in.-lbs. (2.82 N-m) maximum.
3. Connect connector (5).
4. Check gearbox oil level, service if required (412-A-12-00-00-00A-040A-A / 00126).
5. Ground run helicopter (BHT-412-FM). Ensure C BOX TEMP caution panel segment is extinguished.
6. Check for oil leaks.
7. Check combining gearbox oil level and service as required (412-A-12-00-00-00A-040A-A / 00126).
8. Close and secure cowling.



1. Packing
2. Bushing
3. Packing
4. Temperature switch
5. Electrical connector

ICN-412-A-792000-B-97499-00003-A-001-01

Figure 1. Reduction (Combining) Gearbox Oil Temperature Switch (Sheet 1 of 1)

Engine Oil Cooling System Check Valves - Description

NOTE

Removal and installation of all check valves is similar. The removal and installation procedures are typical for any valve, refer 412-A-79-20-00-02A-520A-A / 01772 and 412-A-79-20-00-02A-720A-A / 01773. Clamping arrangement however, may be different.

Four check valves are installed in engine oil cooling system. One check valve is used in each power section oil inlet line and one check valve is used in inlet and outlet oil lines of combining gearbox.

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Applicable to: ALL
ECCN EAR99

Engine Oil Cooling System Check Valves - Removal

Procedure

NOTE

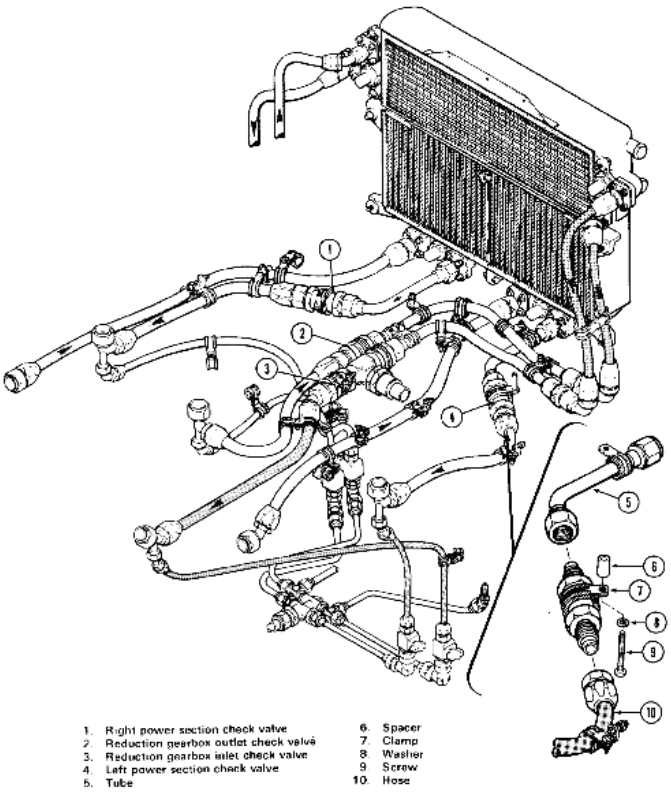
The removal procedures of all the check valves are similar. The following removal procedures are typical for any valve. Clamping arrangement however, may be different.

1. Open engine side cowling.

2. Drain oil from appropriate power section or from combining gearbox.

3. Disconnect tube (5, Figure 1) and hose (10).

4. Remove screw (9), washer (8), spacer (6), clamp (7), and check valve.



ICN-412-A-792000-B-97499-00004-A-001-01

Figure 1. Engine Oil Cooling System Check Valve (Sheet 1 of 1)

Engine Oil Cooling System Check Valves - Installation

Procedure

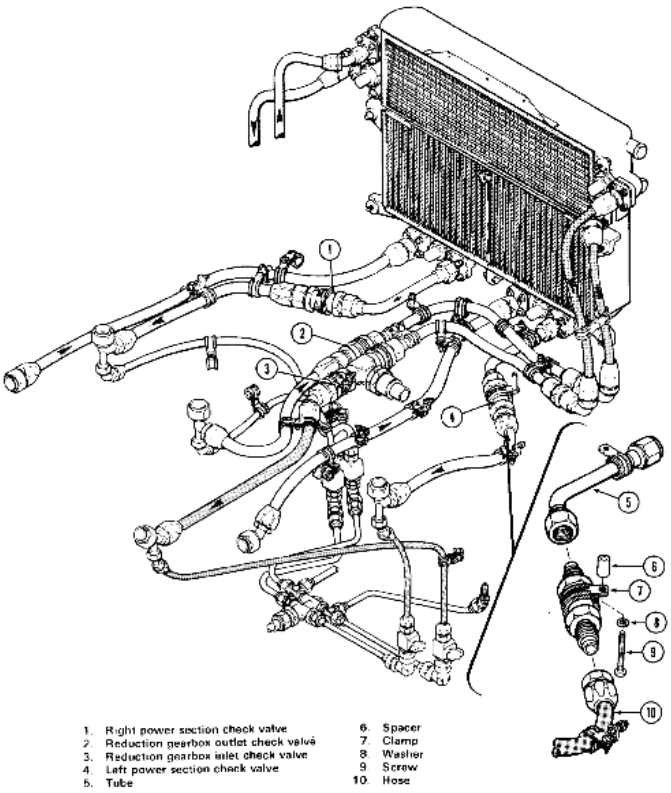


ENSURE CHECK VALVE IS INSTALLED WITH ETCHED ARROW IN DIRECTION OF FLOW.

NOTE

The installation procedures of all the check valves are similar. The following installation procedures are typical for any valve. Clamping arrangement however, may be different.

1. Position clamp (7, Figure 1) on check valve. Secure clamp with screw (9), washer (8), and spacer (6).
2. Connect tube (5) and hose (10).
3. Check oil level in appropriate power section or combining gearbox. Service as required (412-A-12-00-00-00A-040A-A / 00126).
4. Ground run helicopter (BHT-412-FM) and check for oil leaks.
5. Check oil level in appropriate power section or combining gearbox. Service as required (412-A-12-00-00-00A-040A-A / 00126).
6. Close and secure cowling.



ICN-412-A-792000-B-97499-00004-A-001-01

Figure 1. Engine Oil Cooling System Check Valve (Sheet 1 of 1)

Engine Oil Cooling System Drain Valves - Description

NOTE

Maintenance procedures for all drain valves are the same except for location.

Four drain valves are installed in engine oil system. Two drain valves are installed in combining gearbox oil system and one drain valve is installed in each power section oil system. The drain valves are mounted on engine aft center service deck.

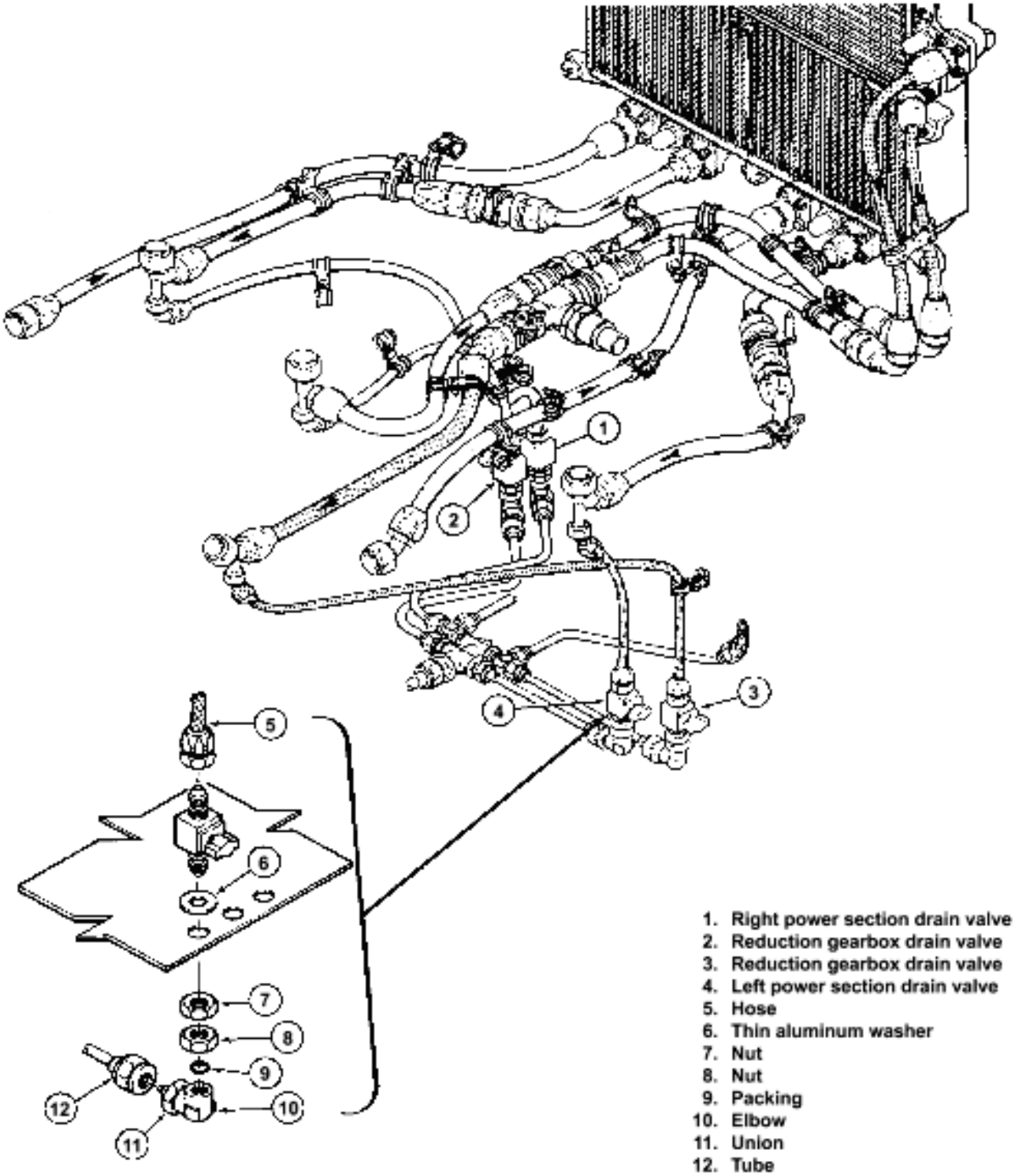
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Applicable to: ALL
ECCN EAR99

Engine Oil Cooling System Drain Valves - Removal

Procedure

1. Open engine side cowling where drain valve is located.
2. Remove access panel below side cowling.
3. Drain oil from appropriate power section or from combining gearbox.
4. Disconnect hose (5, Figure 1) and tube (12).
5. Loosen nut (8) and remove elbow (10) and union (11) as an assembly. Discard packing.
6. Remove nuts (8 and 7) and drain valve (4).



1. Right power section drain valve
2. Reduction gearbox drain valve
3. Reduction gearbox drain valve
4. Left power section drain valve
5. Hose
6. Thin aluminum washer
7. Nut
8. Nut
9. Packing
10. Elbow
11. Union
12. Tube

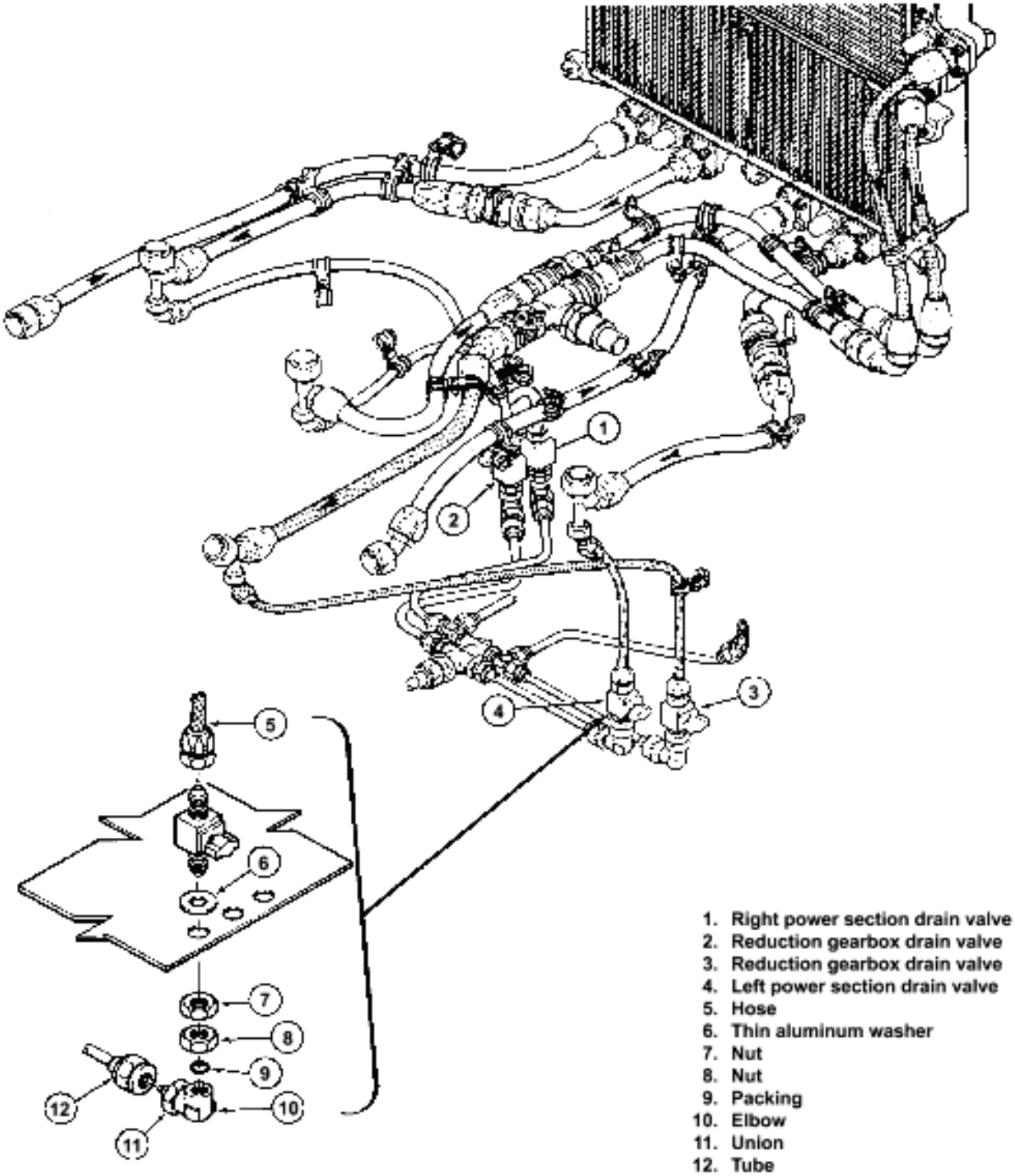
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Figure 1. Engine Oil Cooling System Drain Valve (Sheet 1 of 1)

Engine Oil Cooling System Drain Valves - Installation

Procedure

1. Position valve (3, Figure 1) and washer (6) on service deck and install nut (7).
2. Install nut (8), new packing (9), elbow (10), and union (11). Align elbow (10) with tube (12). Tighten nut (8) against elbow (10).
3. Connect hose (5) and tube (12).
4. Check oil level in appropriate power section or combining gearbox. Service if needed (412-A-12-00-00-00A-040A-A / 00126).
5. Ground run helicopter (BHT-412-FM) and check for oil leaks.
6. Check oil level in appropriate power section or combining gearbox, service if needed (412-A-12-00-00-00A-040A-A / 00126).
7. Install access panel. Close and secure cowling.



1. Right power section drain valve
2. Reduction gearbox drain valve
3. Reduction gearbox drain valve
4. Left power section drain valve
5. Hose
6. Thin aluminum washer
7. Nut
8. Nut
9. Packing
10. Elbow
11. Union
12. Tube

ICN-412-A-792000-B-97499-00005-A-001-01

Figure 1. Engine Oil Cooling System Drain Valve (Sheet 1 of 1)

Combining Gearbox and Transmission Oil Cooler - Removal

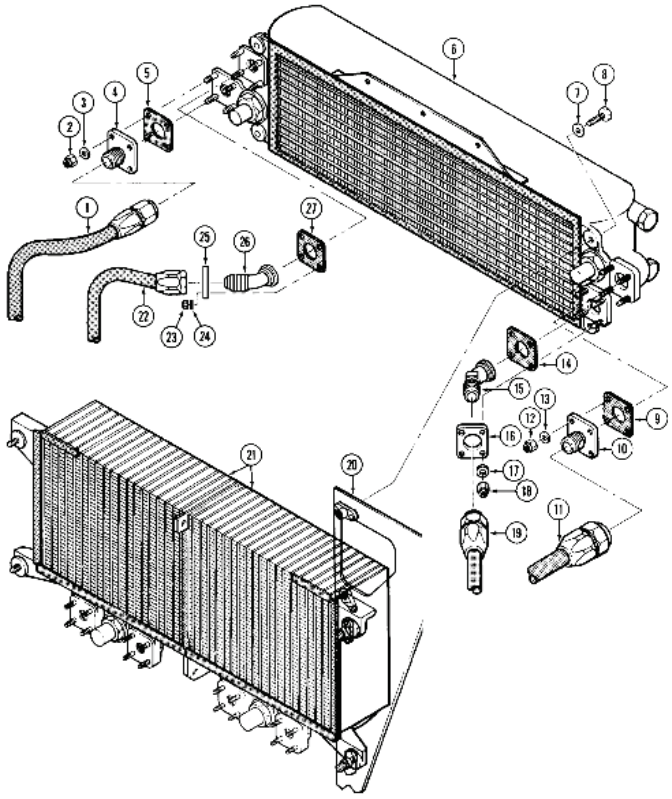
Procedure

1. Remove oil cooler support fairing.

NOTE

If oil hoses attached to cooler have enough slack, cooler can be removed by removing bolts (8, Figure 1) and washer (7) then moving cooler aft prior to disconnecting hoses. If this is possible, skip Step 3.. through Step 5..

2. Remove combining gearbox cowling.
3. Remove ejectors (412-A-71-61-00-01A-520A-A / 01701).
4. Remove oil cooler duct.
5. Remove access panel located below aft end of ejector.
6. Drain cooler.
7. Use suitable container to catch spilled oil. Disconnect hoses (1, 22, 19, and 11).
8. Remove bolts (8) and remove cooler (6).
9. Remove nuts (2), washers (3), connector (4), and gasket (5). Discard gasket.
10. Remove nuts (23), washers (24), flange (25), elbow (26), and gasket (27). Discard gasket.
11. Remove nuts (18), washers (17), flange (16), elbow (15), and gasket (14). Discard gasket.
12. Remove nuts (12), washers (13), connector (10), and gasket (9). Discard gasket.



ICN-412-A-792000-B-97499-00006-A-001-01

Figure 1. Combining Gearbox and Transmission Oil Cooler (Sheet 1 of 2)

1. Transmission oil hose
2. Nut
3. Thin aluminum washer
4. Connector
5. Gasket
6. Reduction gearbox and transmission cooler
7. Thin aluminum washer
8. Bolt
9. Gasket
10. Connector
11. Hose
12. Nut
13. Thin aluminum washer
14. Gasket
15. Elbow
16. Flange
17. Thin aluminum washer
18. Nut
19. Hose
20. Oil cooler support
21. Power section oil coolers
22. Hose
23. Nut
24. Thin aluminum washer
25. Flange
26. Elbow
27. Gasket

ICN-412-A-792000-B-97499-00007-A-001-01

Figure 1. Combining Gearbox and Transmission Oil Cooler (Sheet 2 of 2)

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Combining Gearbox and Transmission Oil Cooler - Inspection

Procedure

1. Inspect fitting, lines, and mountings for damage and general conditions.
2. Inspect air passages of oil cooler core for restrictions, bent core or cooling fins, and leaks.

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Combining Gearbox and Transmission Oil Cooler - Installation

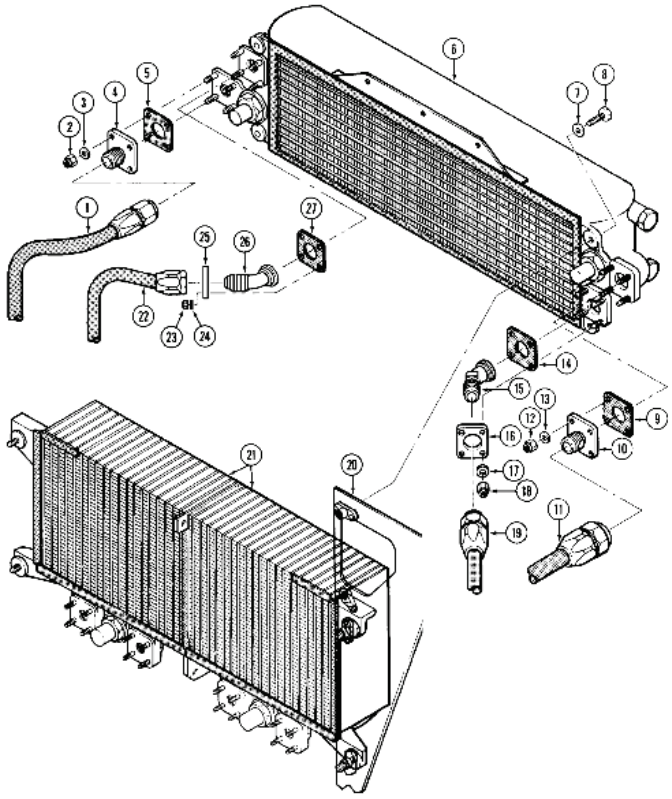
Procedure

1. Install gasket (5, Figure 1), connector (4), washers (3), and nuts (2).
2. Install gasket (27), elbow (26), flange (25), washers (24), and nuts (23).
3. Install gasket (14), elbow (15), flange (16), washers (17), and nuts (18).
4. Install gasket (9), connector (10), washers (13), and nuts (12).

NOTE

If oil cooler was removed without removing ejectors and oil cooler duct, connect hoses to cooler prior to positioning cooler to support.

5. Position cooler (6) in opening of support (20).
6. Install bolts (8) with washers (7).
7. Connect hoses (1, 22, 19, and 11).
8. Install access panel located below aft end of ejector.
9. Install oil cooler duct.
10. Install ejectors (412-A-71-61-00-01A-720A-A / 01704).
11. Check combining gearbox and transmission oil level, service if required (412-A-12-00-00-00A-040A-A / 00126).
12. Install cowling.
13. Install oil cooler support fairing.
14. Ground run helicopter (BHT-412-FM) and check for oil leaks.
15. Check combining gearbox and transmission oil level. Service if required (412-A-12-00-00-00A-040A-A / 00126).



ICN-412-A-792000-B-97499-00006-A-001-01

Figure 1. Combining Gearbox and Transmission Oil Cooler (Sheet 1 of 2)

1. Transmission oil hose
2. Nut
3. Thin aluminum washer
4. Connector
5. Gasket
6. Reduction gearbox and transmission cooler
7. Thin aluminum washer
8. Bolt
9. Gasket
10. Connector
11. Hose
12. Nut
13. Thin aluminum washer
14. Gasket
15. Elbow
16. Flange
17. Thin aluminum washer
18. Nut
19. Hose
20. Oil cooler support
21. Power section oil coolers
22. Hose
23. Nut
24. Thin aluminum washer
25. Flange
26. Elbow
27. Gasket

ICN-412-A-792000-B-97499-00007-A-001-01

Figure 1. Combining Gearbox and Transmission Oil Cooler (Sheet 2 of 2)

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Engine Power Section Oil Cooler - Removal

Procedure

1. Remove combining gearbox and transmission oil cooler (412-A-79-20-00-03A-720A-A / 01776).

NOTE

Either oil cooler can be removed individually or both coolers can be removed as a unit.

2. Drain oil from appropriate cooler.
3. If removing oil cooler (3, Figure 1), disconnect hose (23), tube (22), and remove bolts (2).
4. If removing oil cooler (8), disconnect tube (16), hose (17), and remove bolts (9).

NOTE

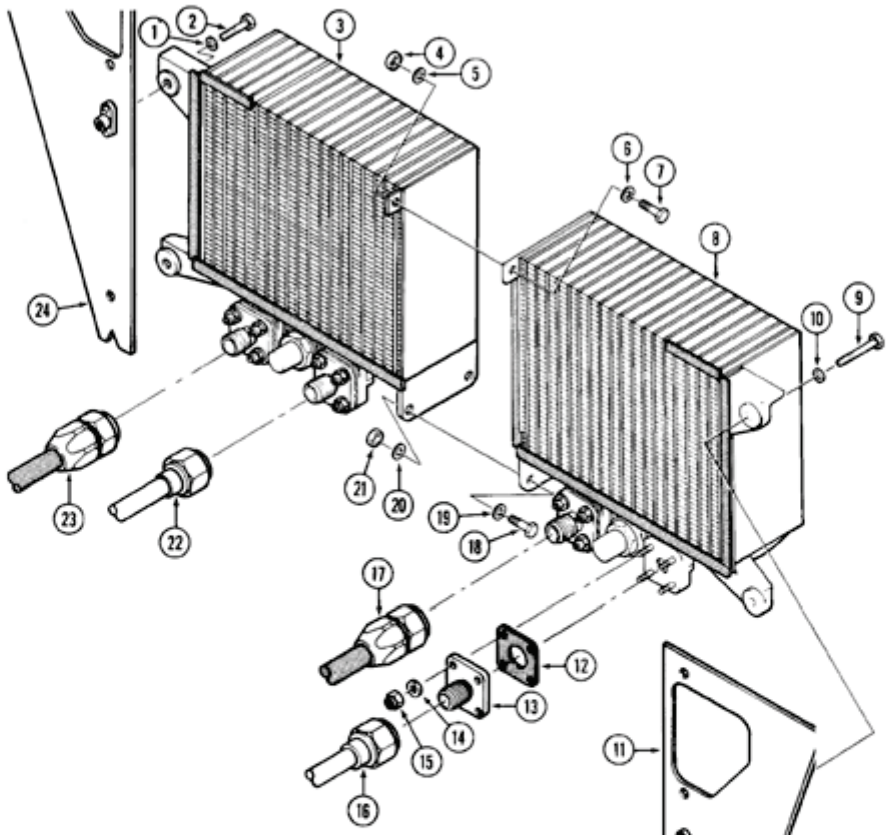
If removing oil coolers as a unit, bolts (7 and 18) can remain in place until coolers are removed.

5. Remove bolt (7).
6. Remove two bolts (18).
7. Remove cooler from oil cooler support.

NOTE

Removal procedures are the same for connectors on both coolers.

8. Remove nuts (15), washers (14), connector (13), and gasket (12). Discard gasket.



1. Thin aluminum washer
2. Bolt
3. Right power section oil cooler
4. Nut
5. Thin aluminum washer
6. Thin aluminum washer
7. Bolt
8. Left power section oil cooler
9. Bolt
10. Thin aluminum washer
11. Oil cooler support
12. Gasket
13. Connector
14. Thin aluminum washer
15. Nut
16. Tube
17. Hose
18. Bolt
19. Thin aluminum washer
20. Thin aluminum washer
21. Nut
22. Tube
23. Hose
24. Oil cooler support

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Figure 1. Engine Power Section Oil Cooler (Sheet 1 of 1)

Engine Power Section Oil Cooler - Inspection

Procedure

NOTE

If metallic pickup is found in external lines of engine, reduction (combining) gearbox, or transmission oil system, and if engine, reduction (combining) gearbox, or transmission are being replaced, and if metal particles are suspected of being present in oil cooler, replace oil cooler.

1. Inspect fittings, lines, and mountings for damage and general condition.

2. Inspect air passages of oil cooler core for restriction, bent core or cooling fins, and leaks.

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Engine Power Section Oil Cooler - Installation

Procedure

NOTE

Installation procedures are the same for connectors on both coolers.

1. Install gasket (12, Figure 1), connector (13), washers (14), and nuts (15).

NOTE

If oil coolers are being installed as a unit, install bolts (7 and 18) with washers and nuts prior to installing coolers.

2. Position cooler (3 or 8) in oil cooler support.

3. If you install the cooler (3), install the bolt (2) with washer (1). Connect tube (22) and hose (23).

4. If you install the cooler (8), install the bolt (9) with washer (10). Connect the hose (17) and tube (16).

5. Install the bolt (7), washers (6 and 5), and nut (4).

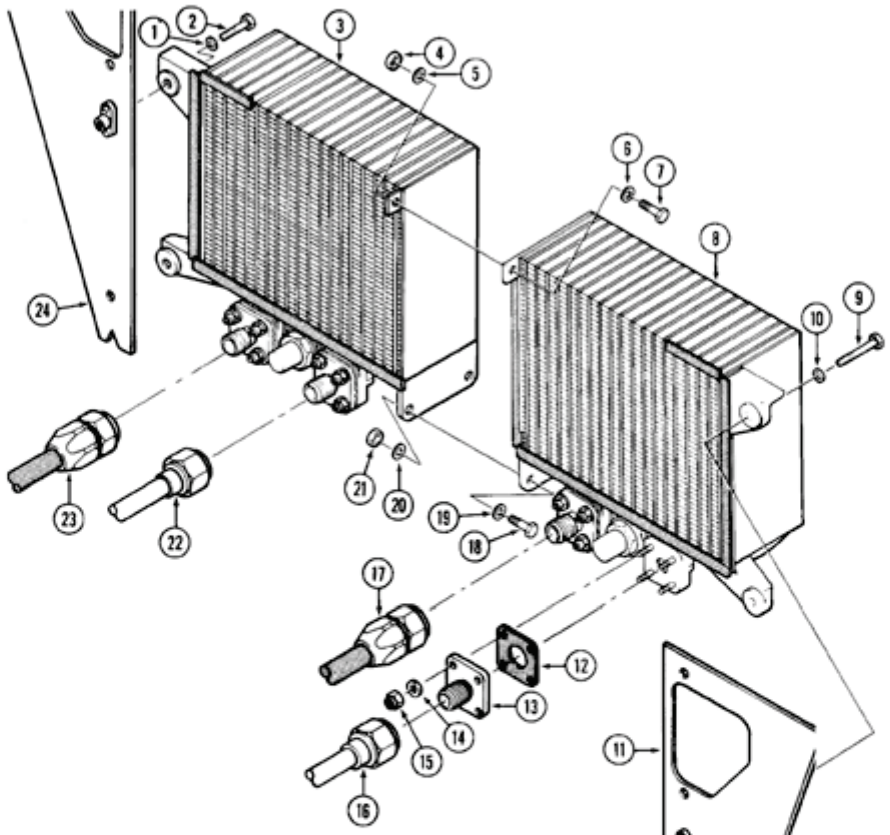
6. Install the bolts (18), washers (19 and 20), and nuts (21).

7. Install the combining gearbox and transmission oil cooler (412-A-79-20-00-04A-520A-A / 01777).

8. Examine the combining gearbox, transmission, and engine power sections oil levels. Service if required (412-A-12-00-00-00A-040A-A / 00126).

9. Ground run the helicopter (BHT-412-FM, Flight Manual) and examine for oil leaks.

10. Examine the combining gearbox, transmission, and engine power sections oil levels. Service if required (412-A-12-00-00-00A-040A-A / 00126).



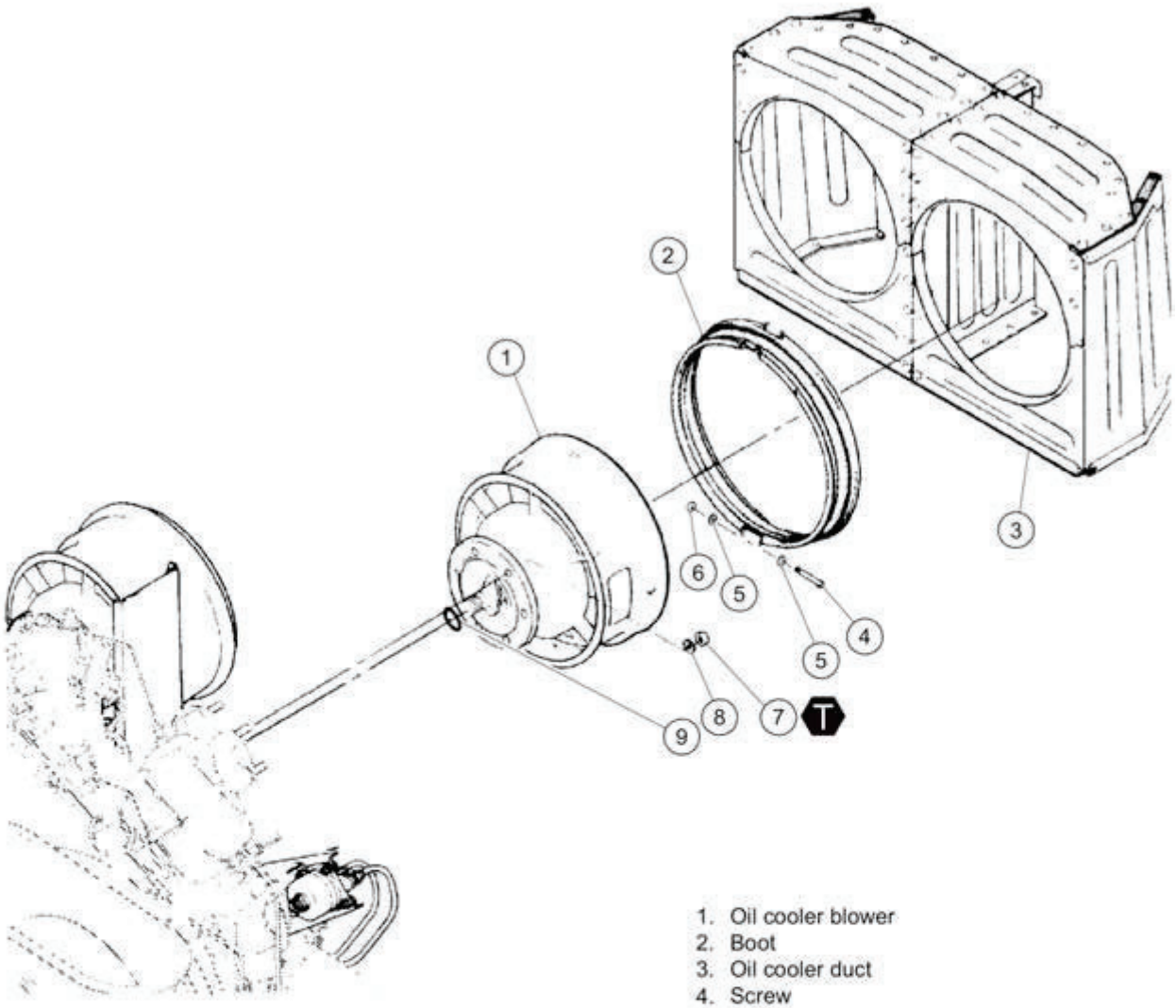
1. Thin aluminum washer
2. Bolt
3. Right power section oil cooler
4. Nut
5. Thin aluminum washer
6. Thin aluminum washer
7. Bolt
8. Left power section oil cooler
9. Bolt
10. Thin aluminum washer
11. Oil cooler support
12. Gasket
13. Connector
14. Thin aluminum washer
15. Nut
16. Tube
17. Hose
18. Bolt
19. Thin aluminum washer
20. Thin aluminum washer
21. Nut
22. Tube
23. Hose
24. Oil cooler support

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Figure 1. Engine Power Section Oil Cooler (Sheet 1 of 1)

Engine and Transmission Oil Cooler Blower - Description

Two shaft driven blowers are mounted on upper aft pads of the combining gearbox. The blowers provide air through connecting ducts to make sure that the air flow is correct through the four oil coolers (Figure 1).



1. Oil cooler blower
2. Boot
3. Oil cooler duct
4. Screw
5. Washer
6. Nut
7. Nut
8. Washer
9. Packing

T 275 TO 300 IN-LBS
(31.07 TO 33.90 Nm)

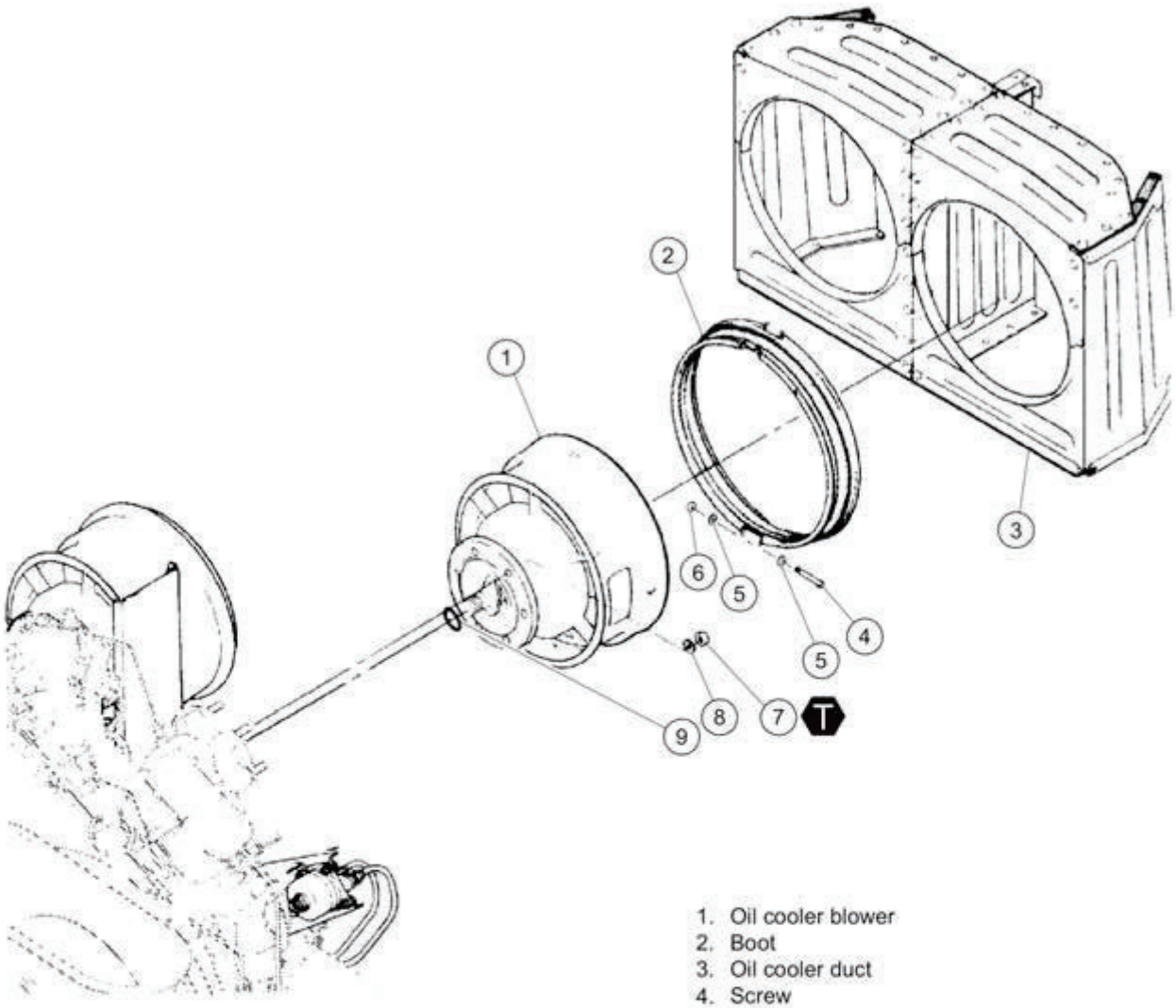
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Figure 1. Engine and Transmission Oil Cooler Blower (Sheet 1 of 1)

Engine and Transmission Oil Cooler Blower - Removal

Procedure

1. Open the engine top and upper cowl.
2. Open the oil cooler air scoop.
3. Remove the combining gearbox top cowl.
4. Remove ejector (412-A-71-61-00-01A-520A-A / 01701).
5. Loosen or remove the screw (4, Figure 1) and remove boot (2) from blower (1).
6. Remove the nuts (7) and blower (1).
7. Remove the opposite blower in the same manner.



1. Oil cooler blower
2. Boot
3. Oil cooler duct
4. Screw
5. Washer
6. Nut
7. Nut
8. Washer
9. Packing

T 275 TO 300 IN-LBS
(31.07 TO 33.90 Nm)

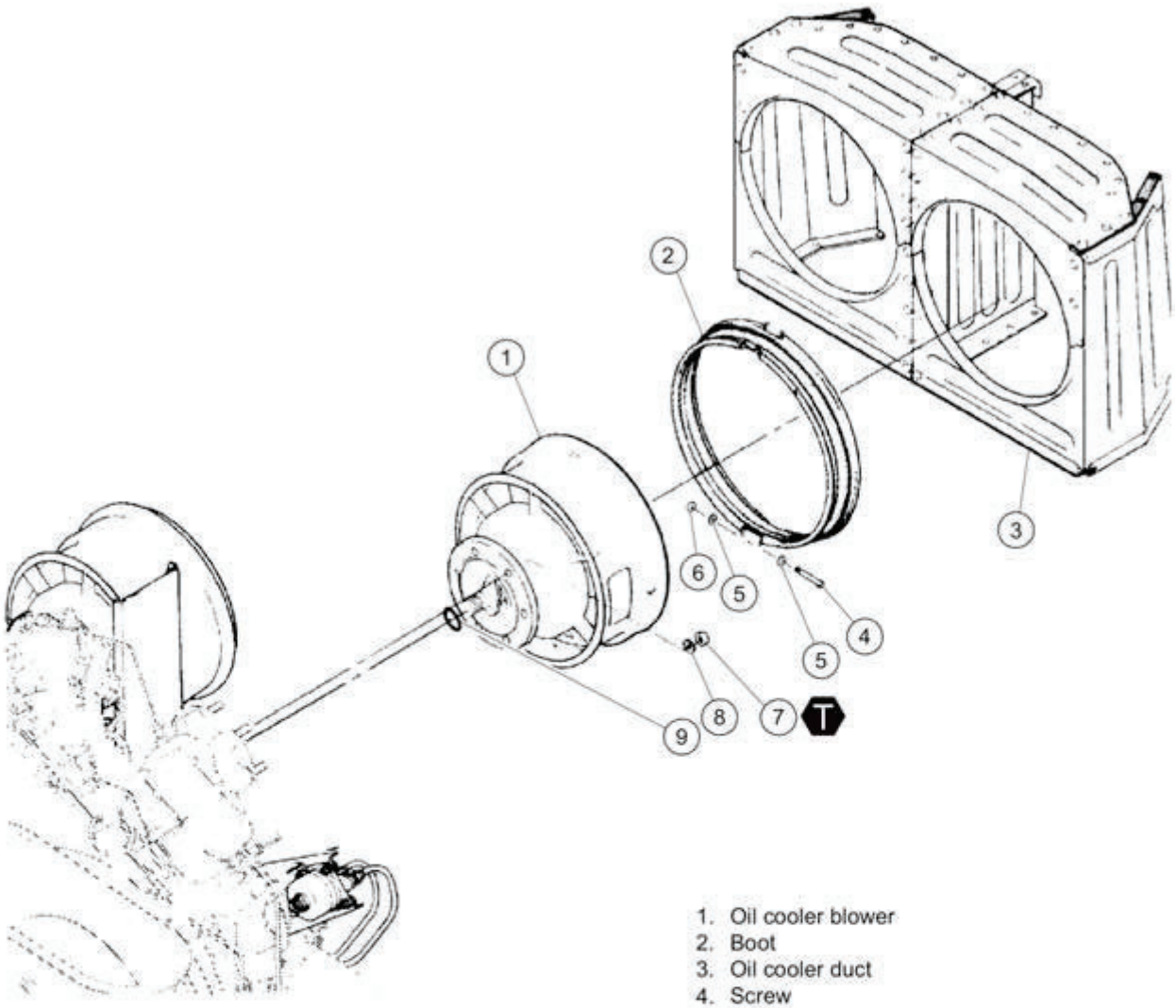
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Figure 1. Engine and Transmission Oil Cooler Blower (Sheet 1 of 1)

Engine and Transmission Oil Cooler Blower - Inspection

Procedure

1. Examine the boot (2, Figure 1) for holes and deterioration.
2. Examine the blades of impeller and guide vanes for cracks and nicks.
3. Turn the impeller and check bearing for roughness.



1. Oil cooler blower
2. Boot
3. Oil cooler duct
4. Screw
5. Washer
6. Nut
7. Nut
8. Washer
9. Packing

T 275 TO 300 IN-LBS
(31.07 TO 33.90 Nm)

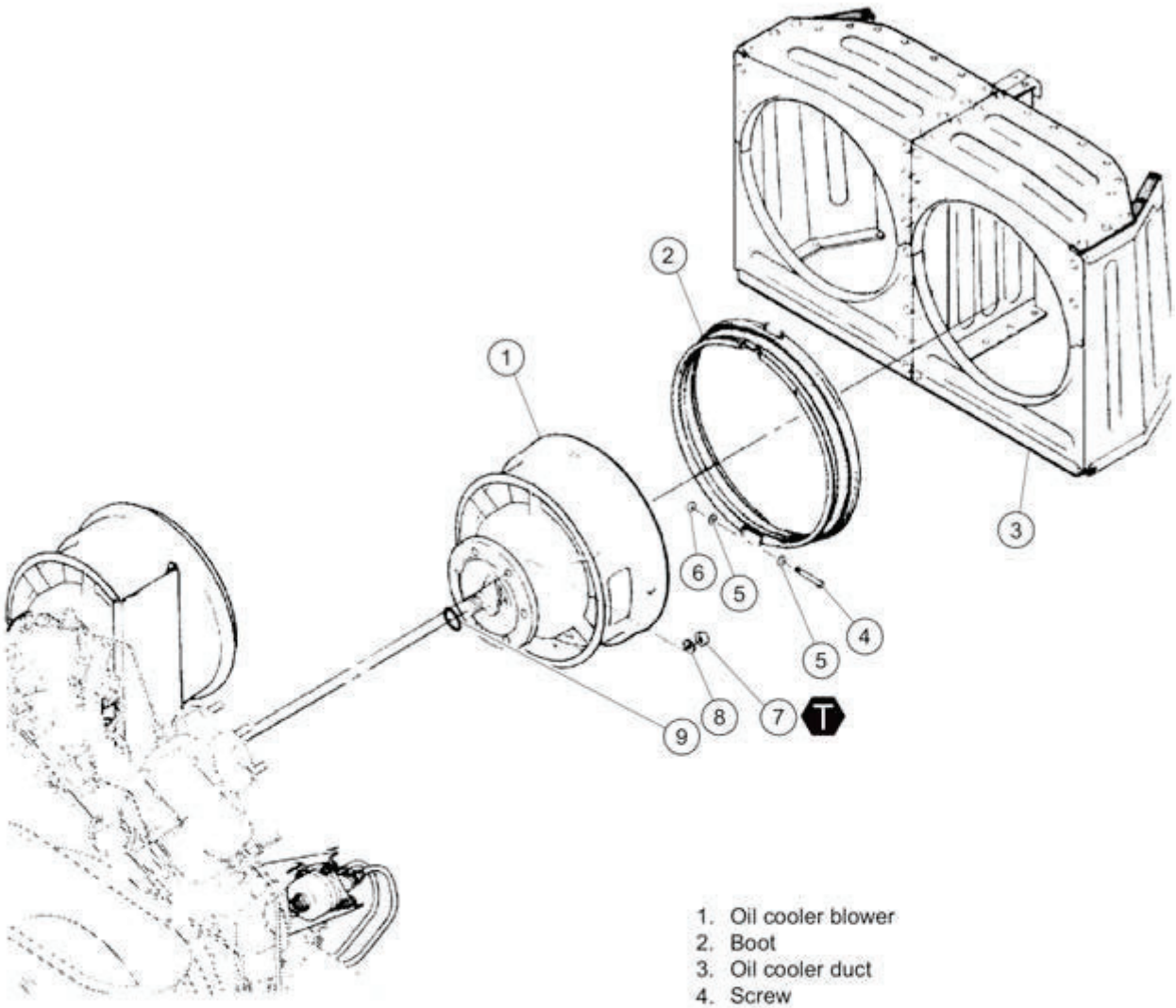
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Figure 1. Engine and Transmission Oil Cooler Blower (Sheet 1 of 1)

Engine and Transmission Oil Cooler Blower - Installation

Procedure

1. Put the packing (9, Figure 1) on the shaft of blower (1).
2. Put the blower (1) on the combining gearbox pad. Install the washers (8) and nuts (7). Torque the nuts (7) 275 to 300 inch-pounds (31.07 to 33.90 Nm).
3. Put the boot (2) on the blower (1). Install the screw (4), two washers (5), and nut (6).
4. Install the ejector (412-A-71-61-00-01A-720A-A / 01704).
5. Install the combining gearbox top cowl.
6. Close the open cowling.
7. Install the opposite blower in the same manner.



1. Oil cooler blower
2. Boot
3. Oil cooler duct
4. Screw
5. Washer
6. Nut
7. Nut
8. Washer
9. Packing

T 275 TO 300 IN-LBS
(31.07 TO 33.90 Nm)

ICN-412-A-792000-B-97499-00009-A-001-01

Figure 1. Engine and Transmission Oil Cooler Blower (Sheet 1 of 1)

CHAPTER 95
INSTRUMENT SYSTEM

MAINTENANCE MANUAL

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List of Effective Data Modules

The listed documents are included in issue 007, dated 2025-11-18, of this publication.

C = Changed data module, N = New data module

Document title	Data module code / Seq number	Issue date	Applicable to
Instrument System - General Description	DMC-412-A-95-00-00-00A-028A-A / 01788	2023-05-31	ALL
Non-Panel Mounted Instruments - Description	DMC-412-A-95-00-03-00A-042A-A / 01789	2023-05-31	ALL
Panel Mounted Instruments - Description	DMC-412-A-95-00-01-00A-042A-A / 01790	2023-05-31	ALL
Panel Mounted Instruments - Removal	DMC-412-A-95-00-01-00A-520A-A / 01791	2023-05-31	ALL
Panel Mounted Instruments - Installation	DMC-412-A-95-00-01-00A-720A-A / 01792	2023-05-31	ALL
Panel Mounted Instruments - Inspection	DMC-412-A-95-00-01-00A-280A-A / 01793	2023-05-31	ALL
Panel Mounted Instruments - Cleaning	DMC-412-A-95-00-01-00A-251A-A / 01794	2023-05-31	ALL
Panel Mounted Instruments - Repair or Replacement	DMC-412-A-95-00-01-00A-685A-A / 01795	2023-05-31	ALL
Instrument Panel - Description	DMC-412-A-95-00-02-00A-042A-A / 01796	2023-05-31	ALL
Instrument Panel - Removal	DMC-412-A-95-00-02-00A-520A-A / 01797	2023-05-31	ALL
Instrument Panel - Installation	DMC-412-A-95-00-02-00A-720A-A / 01798	2023-05-31	ALL
Instrument Panel - Adjustment	DMC-412-A-95-00-02-00A-271A-A / 01799	2023-05-31	ALL
Flight Instruments - Description	DMC-412-A-95-11-00-00A-042A-A / 01800	2023-05-31	ALL
Attitude Director Indicator (ADI) - Description	DMC-412-A-95-11-01-00A-042A-A / 01801	2023-05-31	ALL
Electronic Attitude Director Indicator (EADI) - Description	DMC-412-A-95-11-02-00A-042A-A / 01802	2023-05-31	ALL
Standby Attitude Indicator (When Installed) - Description	DMC-412-A-95-11-03-00A-042A-A / 01803	2023-05-31	ALL
Radar Altimeter Indicator - Description	DMC-412-A-95-11-04-00A-042A-A / 01804	2023-05-31	ALL
Pilot Pitot-Static System - Description	DMC-412-A-95-11-05-00A-042A-A / 01805	2023-05-31	ALL
Copilot Pitot-Static System (When Installed) - Description	DMC-412-A-95-11-05-00A-042B-A / 01806	2023-05-31	ALL
Pitot-Static System - Leak Check	DMC-412-A-95-11-05-00A-364A-A / 01807	2023-05-31	ALL
Pitot-Static System - Operational Check	DMC-412-A-95-11-05-00A-320A-A / 01808	2023-05-31	ALL
Pitot-Static System - Inspection	DMC-412-A-95-11-05-00A-280A-A / 01809	2023-05-31	ALL
Pitot-Static System Hardware - Repair or Replacement	DMC-412-A-95-11-05-00A-685A-A / 01810	2023-05-31	ALL

Document title	Data module code / Seq number	Issue date	Applicable to
Pitot and Static Lines - Draining Instructions	DMC-412-A-95-11-05-00A-220A-A / 01811	2023-05-31	ALL
Pitot Tube - Removal	DMC-412-A-95-11-06-00A-520A-A / 01812	2023-05-31	ALL
Pitot Tube - Installation	DMC-412-A-95-11-06-00A-720A-A / 01813	2023-05-31	ALL
Pitot Tube - Cleaning	DMC-412-A-95-11-06-00A-251A-A / 01814	2023-05-31	ALL
Pitot Tube and Static Port Heaters - Description	DMC-412-A-95-11-07-00A-042A-A / 01815	2023-05-31	ALL
Pitot Tube and Static Port Heaters - Operational Check	DMC-412-A-95-11-07-00A-320A-A / 01816	2023-05-31	ALL
Static Port Heater Elements - Removal	DMC-412-A-95-11-07-00A-520A-A / 01817	2023-05-31	ALL
Static Port Heater Elements - Installation	DMC-412-A-95-11-07-00A-720A-A / 01818	2023-05-31	ALL
Static Port Heater System - Inspection	DMC-412-A-95-11-07-00A-280A-A / 01819	2023-05-31	ALL
Static Port Heater System - Cleaning	DMC-412-A-95-11-07-00A-251A-A / 01820	2023-05-31	ALL
Static Port Heater System - Repair or Replacement	DMC-412-A-95-11-07-00A-685A-A / 01821	2023-05-31	ALL
Pitot-Static Instruments - Description	DMC-412-A-95-11-08-00A-042A-A / 01822	2023-05-31	ALL
Airspeed Indicator - Description	DMC-412-A-95-11-08-01A-042A-A / 01823	2023-05-31	ALL
Airspeed Indicator - Troubleshooting	DMC-412-A-95-11-08-01A-422A-A / 01824	2023-05-31	ALL
Altimeter - Description	DMC-412-A-95-11-08-02A-042A-A / 01825	2023-05-31	ALL
Altimeter - Troubleshooting	DMC-412-A-95-11-08-02A-422A-A / 01826	2023-05-31	ALL
Vertical Speed Indicator - Description	DMC-412-A-95-11-08-03A-042A-A / 01827	2023-05-31	ALL
Vertical Speed Indicator - Troubleshooting	DMC-412-A-95-11-08-03A-422A-A / 01828	2023-05-31	ALL
Navigation Instruments - Description	DMC-412-A-95-12-00-00A-042A-A / 01829	2023-05-31	ALL
Horizontal Situation Indicator (HSI) - Description	DMC-412-A-95-12-01-00A-042A-A / 01830	2023-05-31	ALL
Electronic Horizontal Situation Indicator (EHSI) - Description	DMC-412-A-95-12-02-00A-042A-A / 01831	2023-05-31	ALL
Distance Measuring Equipment (DME) Indicator - Description	DMC-412-A-95-12-03-00A-042A-A / 01832	2023-05-31	ALL
Course Select Indicator (CSI) (When Installed) - Description	DMC-412-A-95-12-04-00A-042A-A / 01833	2023-05-31	ALL
Marker Beacon Indicators (When Installed) - Description	DMC-412-A-95-12-05-00A-042A-A / 01834	2023-05-31	ALL
Standby Magnetic Compass - Description	DMC-412-A-95-12-06-00A-042A-A / 01835	2023-05-31	ALL

Document title	Data module code / Seq number		Issue date	Applicable to
Standby Magnetic Compass - Inspection	DMC-412-A-95-12-06-00A-280A-A / 01836		2023-05-31	ALL
Standby Magnetic Compass - Removal	DMC-412-A-95-12-06-00A-520A-A / 01837		2023-05-31	ALL
Standby Magnetic Compass - Installation	DMC-412-A-95-12-06-00A-720A-A / 01838		2023-05-31	ALL
Initial Setup - Compass Swing - Calibration	DMC-412-A-95-12-06-00A-273A-A / 01839		2023-05-31	ALL
Index Error Check - Calibration	DMC-412-A-95-12-06-00A-273B-A / 01840		2023-05-31	ALL
One-Cycle Error Check - Calibration	DMC-412-A-95-12-06-00A-273C-A / 01841		2023-05-31	ALL
Standby Magnetic Compass - Troubleshooting	DMC-412-A-95-12-06-00A-422A-A / 01842		2023-05-31	ALL
Propulsion and Hydraulic Instruments - Description	DMC-412-A-95-61-00-00A-028A-A / 01843		2023-05-31	ALL
Dual Torque Pressure Indicator - Description	DMC-412-A-95-61-01-00A-042A-A / 01844		2023-05-31	SN: 33001-36019
Triple Torque Pressure Indicator - Description	DMC-412-A-95-61-02-00A-042A-A / 01845	C	2025-11-18	SN: 36020-36086 or 36087-Subsequent or BHT-412-SI-74: Installed or BHT-412-SI-97: Installed or 412-570-001-103: Installed
Engine Oil Pressure and Temperature Indicators - Description	DMC-412-A-95-61-03-00A-042A-A / 01846		2023-05-31	ALL
Fuel Pressure Indicators - Description	DMC-412-A-95-61-04-00A-042A-A / 01847		2023-05-31	ALL
Fuel Quantity Indicator - Description	DMC-412-A-95-61-05-00A-042A-A / 01848		2023-05-31	ALL
Gas Producer Tachometer Indicators - Description	DMC-412-A-95-61-06-00A-042A-A / 01849		2023-05-31	ALL
Gearbox Oil Pressure and Temperature Indicator - Description	DMC-412-A-95-61-07-00A-042A-A / 01850		2023-05-31	ALL
Hydraulic Pressure and Temperature Indicators - Description	DMC-412-A-95-61-08-00A-042A-A / 01851		2023-05-31	ALL
Transmission Oil Pressure and Temperature Indicator - Description	DMC-412-A-95-61-09-00A-042A-A / 01852		2023-05-31	ALL
Triple Tachometer Indicator - Description	DMC-412-A-95-61-10-00A-042A-A / 01853		2023-05-31	ALL
Interturbine Temperature (ITT) Indicators - Description	DMC-412-A-95-61-11-00A-042A-A / 01854		2023-05-31	ALL
Miscellaneous Instruments - Description	DMC-412-A-95-63-00-00A-028A-A / 01855		2023-05-31	ALL
Digital Clock - Description	DMC-412-A-95-63-00-00A-042A-A / 01856		2023-05-31	SN: 36087-Subsequent

Document title	Data module code / Seq number	Issue date	Applicable to
Digital Clock - Functional Check	DMC-412-A-95-63-00-00A-340A-A / 01857	2023-05-31	ALL
Digital Clock - Operational Check	DMC-412-A-95-63-00-00A-320A-A / 01858	2023-05-31	ALL
GMT (Greenwich Mean Time) - Functional Check	DMC-412-A-95-63-00-00A-340B-A / 01859	2023-05-31	ALL
LT (Local Time) - Functional Check	DMC-412-A-95-63-00-00A-340C-A / 01860	2023-05-31	ALL
ET (Elapsed Time) Count-Up - Functional Check	DMC-412-A-95-63-00-00A-340D-A / 01861	2023-05-31	ALL
ET (Elapsed Time) Countdown - Functional Check	DMC-412-A-95-63-00-00A-340E-A / 01862	2023-05-31	ALL
Digital Clock - Troubleshooting	DMC-412-A-95-63-00-00A-422A-A / 01863	2023-05-31	ALL
Dual AC and DC Voltmeters - Description	DMC-412-A-95-63-01-00A-042A-A / 01864	2023-05-31	ALL
Dual DC Ammeter - Description	DMC-412-A-95-63-02-00A-042A-A / 01865	2023-05-31	ALL
Free Air Temperature Indicator - Description	DMC-412-A-95-63-03-00A-042A-A / 01866	2023-05-31	ALL
Engine Hourmeter - Description	DMC-412-A-95-63-04-00A-042A-A / 01867	2023-05-31	ALL
Air Data Command Display (DAFCS) - Description	DMC-412-A-95-63-05-00A-042A-A / 01868	2023-05-31	ALL
Flight Director Mode Selector (P/O DAFCS Flight Director Kit) - Description	DMC-412-A-95-63-06-00A-042A-A / 01869	2023-05-31	ALL

Instrument System - General Description

Maintenance information for flight, navigation, propulsion, hydraulic, and miscellaneous instruments installed in Models 412/412EP helicopter is presented in different sections. All instruments except standby magnetic compass, engine hourmeter, and free air temperature indicator are mounted in instrument panel.

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Non-Panel Mounted Instruments - Description

NOTE

Maintenance procedures are same as for panel mounted instruments.

The pilot standby magnetic compass is centered above right windshield.

The engine hourmeter is mounted on right side of pedestal near floor.

The bimetallic probe-type free air temperature indicator is mounted in the left forward portion of the upper right cabin roof window assembly.

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Panel Mounted Instruments - Description

The maintenance procedures in 412-A-95-00-01-00A-520A-A / 01791, 412-A-95-00-01-00A-720A-A / 01792, 412-A-95-00-01-00A-280A-A / 01793, 412-A-95-00-01-00A-251A-A / 01794 and 412-A-95-00-01-00A-685A-A / 01795, are applicable to all panel mounted instruments.

There are three types of instrument mountings used in this helicopter:

Front-mounted instruments that are secured by screws to instrument panel face with screws through front flange of instrument.

Front-mounted instruments that are secured with clamps, with clamp adjusting screws accessible on instrument panel face.

Back-mounted instruments that are secured to back of instrument panel with screws that mount through instrument panel face.

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Panel Mounted Instruments - Removal

Procedure



ENSURE ALL ELECTRICAL POWER IS OFF.



A GYRO MAY BE DAMAGED BY MOVING IT AFTER ELECTRICAL POWER IS REMOVED BUT BEFORE THE GYRO ROTOR STOPS. ALLOW APPROXIMATELY 15 MINUTES FOR INTERNAL GYRO IN STANDBY ATTITUDE INDICATOR TO WIND DOWN AFTER ELECTRICAL POWER IS REMOVED. WHEN REMOVING OR INSTALLING STANDBY ATTITUDE INDICATOR, DO SO GENTLY WITH NO QUICK MOTION.

1. Disconnect electrical leads and/or instrument piping, as applicable, from back of instrument. Necessary access may be gained through nose access door.

2. Protect ends with electrical tape and cap open piping, if applicable, and cap opening(s) on instrument.

3. Remove mounting screws or loosen mounting clamp screw. Carefully remove instrument from panel.

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Panel Mounted Instruments - Installation

Procedure



ENSURE ALL ELECTRICAL POWER IS OFF.



DO NOT TIGHTEN CLAMPS MORE THAN NECESSARY TO HOLD INSTRUMENT, AS EXCESSIVE TENSION MAY DEFORM INSTRUMENT CASE AND CAUSE ERRONEOUS READING OR CRACKED GLASS.

THE STANDBY ATTITUDE INDICATOR INCLUDES AN INTERNAL GYRO. WHEN INSTALLING STANDBY ATTITUDE INDICATOR, DO SO GENTLY WITH NO QUICK MOTION.

1. Carefully position instrument in panel and install mounting screws or tighten screw of mounting clamp.
2. Remove protective tape, caps and covers from electrical leads, piping and opening(s) on instruments.

NOTE

Leak check of pitot-static system (412-A-95-11-05-00A-364A-A / 01807) shall be performed, as applicable, whenever pitot/static piping is disconnected, then reconnected.

3. Connect electrical leads and/or instrument piping, as applicable, to back of instrument.

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Panel Mounted Instruments - Inspection

Procedure

1. Inspect for loose, missing, or improperly installed hardware, loose or cracked cover glasses, and proper mounting.
2. Inspect instrument for legibility of range markings and faulty decals.

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Panel Mounted Instruments - Cleaning

Procedure

1. Wipe off dust and clean instrument cover glasses with clean, soft, lint-free cloth.

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Panel Mounted Instruments - Repair or Replacement

Procedure

NOTE

All repairs must be conducted by an authorized instrument repair station.

1. Replace any inoperative, defective, or damaged instrument.

2. Replace instrument range markings as follows:

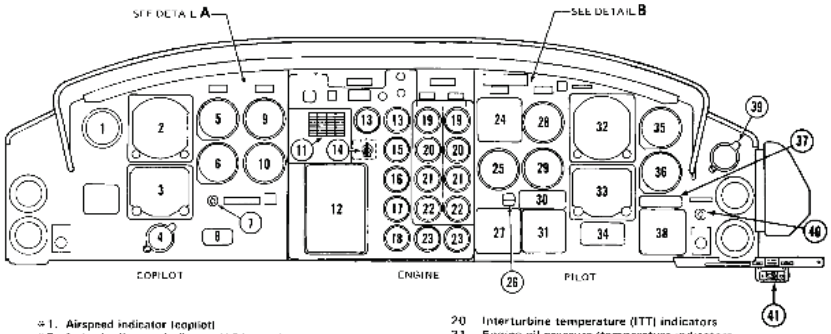
2.1. Refer to Flight Manual for range. Index cover glass to case with stripe of white lacquer.

2.2. Use prepared decals, suitable lacquer, or tape for markings and apply accurately to cover glass. Protect markings with a light coat of clear adhesive varnish or lacquer.

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Instrument Panel - Description

The instrument panel (Figure 1, Figure 2, or Figure 3) contains all flight, propulsion, navigation, and miscellaneous instruments except magnetic compass, engine hourmeter, and free air temperature indicator. For information on various switches, warning lights, and fire pull handles, refer to 412-A-96-00-02-00A-042A-A / 01887.

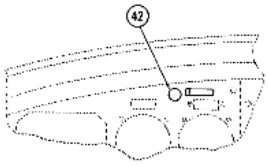


*1. Airspeed indicator (copilot)
*2. Attitude director indicator (ADII (copilot)
*3. Horizontal situatinn indicator (HSI) (copilot)
*4. Clock (copilot)
*5 Barometric altimeter (copilot)
*6 Vertical speed indicator (VSI) (copilot)
*7. VG FAST ERECT switch (copilot)
*8. COPILOT SEL switch (copilot)
*9. Triple tachometer (copilot)
*10. Torquemeter (copilot)
11. Airspeed limits (VNE)
12. Caution light panel
13. Hydraulic pressure/temperature indicators
14. Fuel quantity indicator switch
15. Fuel quantity indicator
16. Transmission oil pressure/temperature indicator
17. Gearbox oil pressure/temperature indicator
18. Dual DC ammeter
19. Gas producer tachometer (N1) indicators
20. Interturbine temperature (ITT) indicators
21. Engine oil pressure/temperature indicators
22. Fuel pressure indicators
23. Dual voltmeter
24. Standby attitude indicator
25. Triple tachometer (pilot)
26. Flight director indicator
27. Flight director
28. Airspeed indicator (pilot)
29. Torquemeter (pilot)
30. Distance measuring equipment (DME) indicator (pilot)
31. Course select indicator (pilot)
32. Attitude director indicator (ADI) (pilot)
33. Horizontal situation indicator (HSI) (pilot)
34. NAV AUDIO panel (pilot)
35. Barometric altimeter (pilot)
36. Vertical speed indicator (VSI) (pilot)
37. Marker beacon indicator (pilot)
38. Radar altimeter (pilot)
39. Clock (pilot)
40. VG FAST ERECT switch (pilot)
41. STATIC SOURCE switch

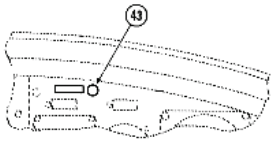
*If installed, refer to Service Instruction 412-10 for Maintenance data.

ICN-412-A-950002-B-97499-00001-A-001-01

Figure 1. Instrument Panel (S/N 33001 through 33107) (Sheet 1 of 2)



DETAIL A

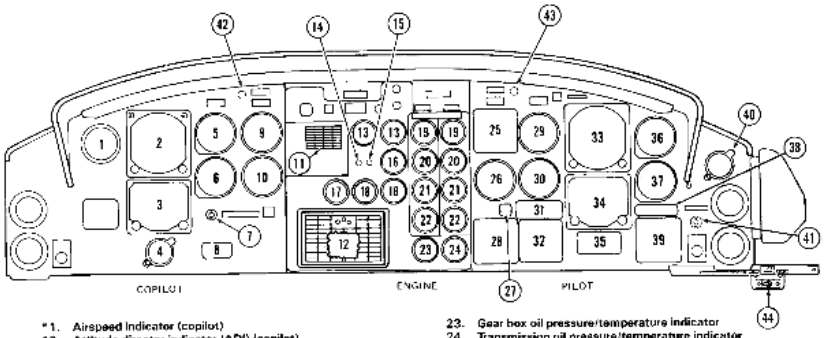


DETAIL B

42. Copilot indicator light
43. Pilot indicator light

ICN-412-A-950002-B-97499-00002-A-001-01

Figure 1. Instrument Panel (S/N 33001 through 33107) (Sheet 2 of 2)

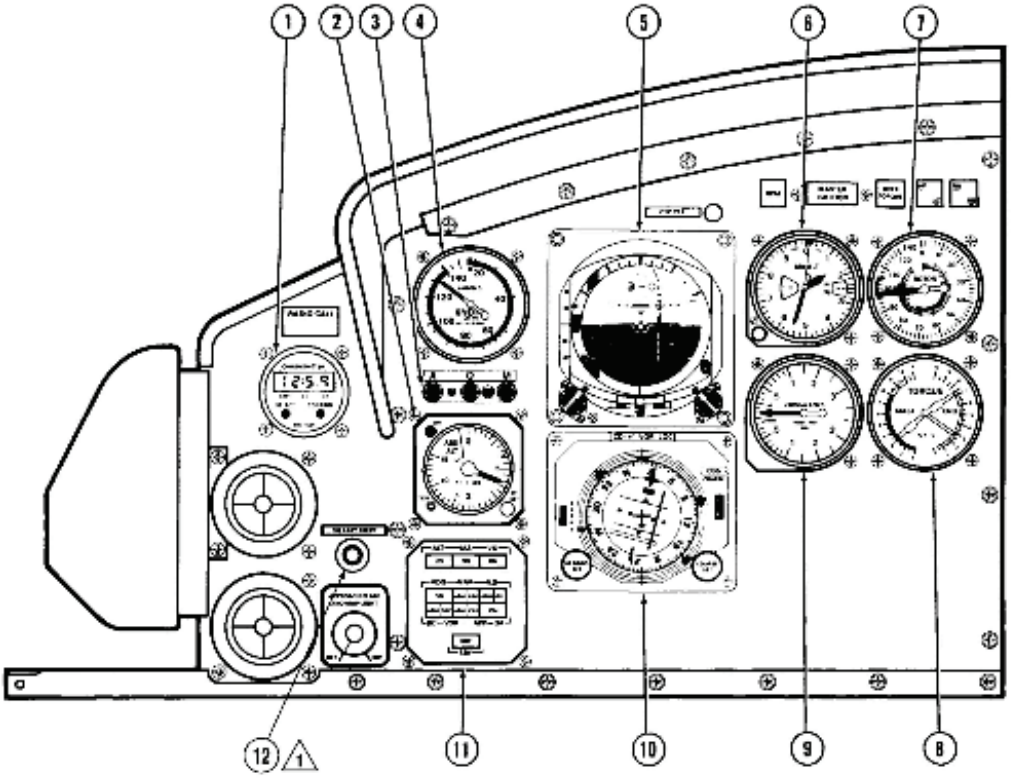


*1. Airspeed indicator (copilot)
*2. Attitude director indicator (ADI) (copilot)
*3. Horizontal situation indicator (HSI) (copilot)
*4. Clock (copilot)
*5. Barometric altimeter (copilot)
*6. Vertical speed indicator (VSI) (copilot)
*7. VG FAST ERECT switch (copilot)
*8. COPILOT SEL switch (copilot)
*9. Triple tachometer (copilot)
*10. Torquemeter (copilot)
11. Airspeed limits (VNE)
12. Caution light panel
13. Hydraulic pressure/temperature indicators
14. Fuel quantity selector switch
15. DIGITS TEST switch
16. Fuel quantity indicator
17. Dual DC ammeter
18. Dual voltmeters
19. Gas producer tachometer (N1) indicators
20. Interturbine temperature (ITT) indicators
21. Engine oil pressure/temperature indicators
22. Fuel pressure indicators
23. Gear box oil pressure/temperature indicator
24. Transmission oil pressure/temperature indicator
25. Standby attitude indicator
26. Triple tachometer (pilot)
27. Flight director indicator
28. Flight director
29. Airspeed indicator (pilot)
30. Torquemeter (pilot)
31. Distance measuring equipment (DME) indicator (pilot)
32. Course select indicator (pilot)
33. Attitude director indicator (ADI) (pilot)
34. Horizontal situation indicator (HSI) (pilot)
35. NAV AUDIO panel (pilot)
36. Barometric altimeter (pilot)
37. Vertical speed indicator (VSI) (pilot)
38. Marker beacon indicator (pilot)
39. Radar altimeter (pilot)
40. Clock (pilot)
41. VG FAST ERECT switch (pilot)
42. Copilot indicator light
43. Pilot indicator light
44. Static source switch

*If installed, refer to Service Instruction 412-10 for Maintenance data.

ICN-412-A-950002-B-97499-00003-A-001-01

Figure 2. Instrument Panel (S/N 33108 through 36086) (Sheet 1 of 1)



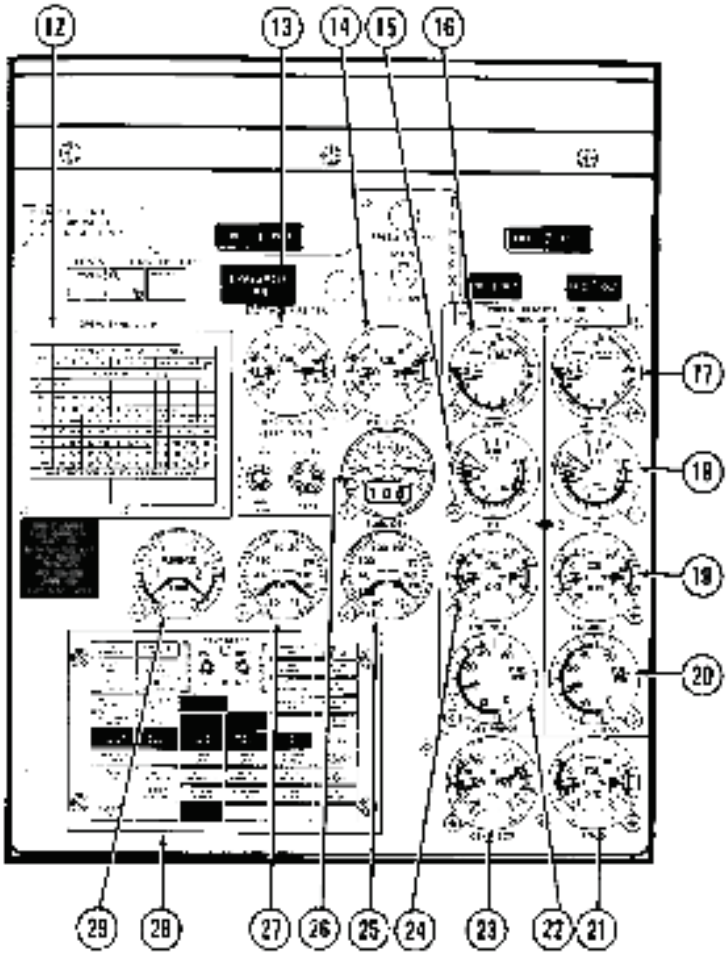
1. Digital chronometer (clock)
2. Radar altimeter
3. Marker beacon indicator
4. Airspeed indicator
5. Attitude director indicator (ADI)
6. Barometric altimeter
7. Triple tachometer
8. Triple torquemeter
9. Vertical speed indicator (VSI)
10. Horizontal situation indicator (HSI)
11. Flight director mode selector
12. VG Fast Erect switch

NOTE

1 Deleted S/N 36248 and subsequent.

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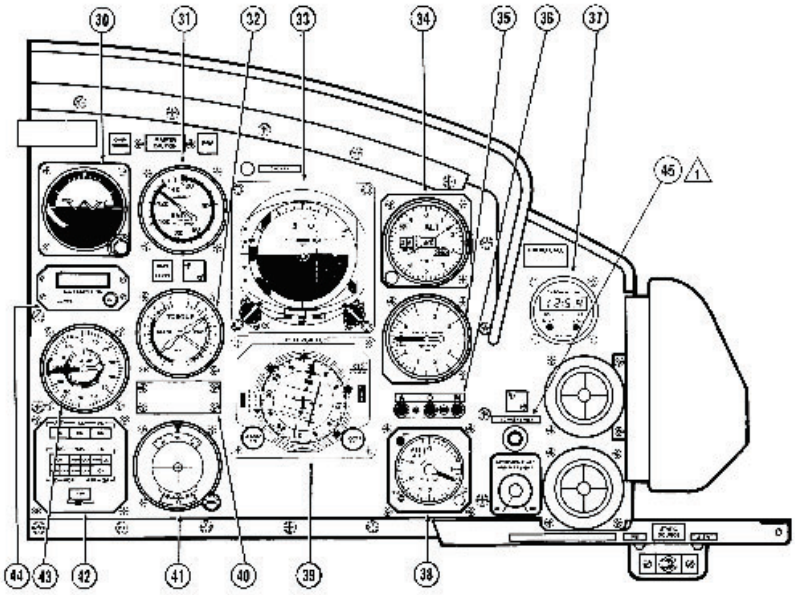
Figure 3. Instrument Panel (S/N 36087 through 36247 and S/N 36248 and Subsequent) (Sheet 1 of 3)



12. Airspeed limits (VNE)
13. Hydraulic pressure/temperature indicator
14. Hydraulic pressure/temperature indicator
15. Interturbine temperature (ITT) indicator
16. Gas producer tachometer (N1) indicator
17. Gas producer tachometer (N1) indicator
18. Interturbine temperature (ITT) indicator
19. Engine oil pressure/temperature indicator
20. Fuel pressure indicator
21. Transmission oil pressure/temperature indicator
22. Fuel pressure indicator
23. Gearbox oil pressure/temperature indicator
24. Engine oil pressure/temperature indicator
25. Dual voltmeter
26. Fuel quantity indicator
27. Dual voltmeter
28. Caution light panel (typical)
29. Dual dc ammeter

ICN-412-A-950002-B-97499-00005-A-001-01

Figure 3. Instrument Panel (S/N 36087 through 36247 and S/N 36248 and Subsequent) (Sheet 2 of 3)



30. Standby attitude indicator
31. Airspeed indicator
32. Triple torquemeter
33. Attitude director indicator (ADI)
34. Barometric altimeter
35. Vertical speed indicator (VSI)
36. Marker beacon indicator
37. Digital chronometer (clock)
38. Radar altimeter
39. Horizontal situation indicator (HSI)
40. Cover (DME indicator)
41. Course select indicator (CSI)
42. Flight director mode selector
43. Triple tachometer
44. Air data command display
45. VG Fast Erect switch

NOTE

1 Deleted S/N 36248 and subsequent.

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Figure 3. Instrument Panel (S/N 36087 through 36247 and S/N 36248 and Subsequent) (Sheet 3 of 3)

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Instrument Panel - Removal

Procedure

1. Disconnect battery quick-disconnect.
2. Disconnect instrument piping and all electrical connectors from back of panel and protect ends of electrical leads with electrical tape; and cap open piping, if applicable, and cap openings on instrument.
3. Disconnect instrument panel stabilizing braces from lower right and lower left corners of instrument panel.
4. Disconnect two adjustment tube assemblies from behind right side of panel.
5. Remove remaining mounting screws from panel and remove instrument panel.

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Instrument Panel - Installation

Procedure

1. Position instrument panel in place and install mounting screws.
2. Connect instrument panel stabilizing braces to instrument panel.
3. Connect two adjustment tube assemblies to right side of panel.

NOTE

Leak check of pitot-static system (412-A-95-11-05-00A-364A-A / 01807) shall be performed, as applicable, whenever pitot/static piping is disconnected, then reconnected.

4. Remove protective caps, covers, and tape and connect instrument piping and all electrical connectors, as applicable.
5. Connect battery quick-disconnect.

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Instrument Panel - Adjustment

Procedure

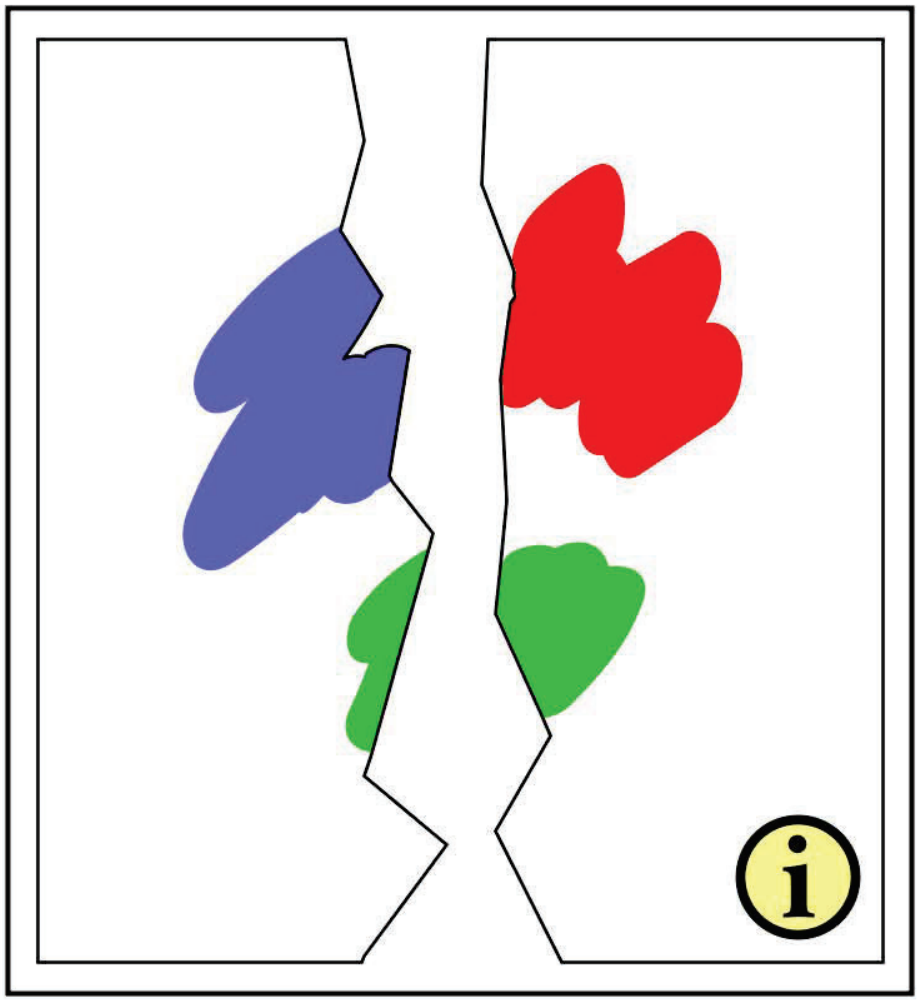
1. Instrument panel stabilizing braces and two adjustment tube assemblies located behind right side of instrument panel may be adjusted to eliminate or minimize instrument panel vibration.

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Flight Instruments - Description

Flight instruments include electrically operated Attitude Director Indicator (ADI) and pilot standby attitude indicator. Also included are the pitot-static instruments. When the optional 4-axis AFCS/Flight Director Kit is installed, the electromechanical ADI is replaced by an Electronic Attitude Director Indicator (EADI) as part of the Electronic Flight Instrument System (EFIS) .

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ICN-412-A-951100-B-97499-00001-A-001-01

Figure 1. Instrument Panel (Typical) 4-Axis/EFIS Configuration (36087 and Subsequent) (Sheet 1 of 1)

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Attitude Director Indicator (ADI) - Description

The Attitude Director Indicator (ADI) provides overall flight attitude information with the following displays:

The spheroid, which is gyro stabilized and is appropriately painted, provides an artificial horizon display. A miniature airplane symbol is provided to indicate helicopter position. Position of miniature aircraft with respect to artificial horizon provides an indication of pitch attitude when read against a dive/climb scale in degrees painted on spheroid.

A pitch trim knob provides for manual adjustment of position of miniature airplane to correspond with normal flight attitude of helicopter. A roll index pointer indicates roll, in degrees, when read against a roll scale provided on top part of indicator display.

The glide slope pointer displays helicopter deviation from glide slope beam center. Pointer represents position of beam center in relation to helicopter.

The rate-of-turn pointer indicates in which direction and at what rate helicopter is turning.

The inclinometer indicates when helicopter is in balance, either in a turn or in straight and level flight. If helicopter is yawing or slipping, ball will be off center.

The warning flags - attitude (ATT), glide slope (GS) and rate-of-turn (RT) provide visual indication of system malfunctions. A warning flag comes into view when a malfunction occurs in respective system circuit.

Operational check of ADI is contained in 412-A-96-12-00-03A-320A-A / 01989.

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Electronic Attitude Director Indicator (EADI) - Description

Refer to Honeywell, SPZ-7600 Integrated Flight Control System Bell 412 System Maintenance Manual, Publication Number A09-1169-024 for component description and operation.

The EADI provides the traditional sphere-type attitude display with lateral and vertical steering signals to assist the pilot in intercepting and maintaining a desired flight path.

The EADI (Figure 1) displays the following information:

- Attitude
- Pitch and Roll
- Command Bars
- Collective Command Cues
- Glidescope Deviation
- Expanded Localizer Deviation
- Marker Beacon Annunciator
- Radio Altitude
- - Rising Runway
- - Digital Readout
- Decision Height Setting/Warning
- Flight Director and SAR Mode Annunciators
- Rate-of-Turn

In addition to the familiar ADI information, the EADI displays the following amber caution and failure annunciations (Figure 2):

- Attitude Source Annunciator
- Flight Director Failure Annunciator
- Decision Height Warning Annunciator
- Flight Management System Message Annunciator
- Radio Altitude Failure Annunciator
- Comparison Monitor Annunciator
- Decision Height Failure Annunciator
- Common Symbol Generator Annunciator

The EADI also displays the following red failure annunciations (Figure 3):

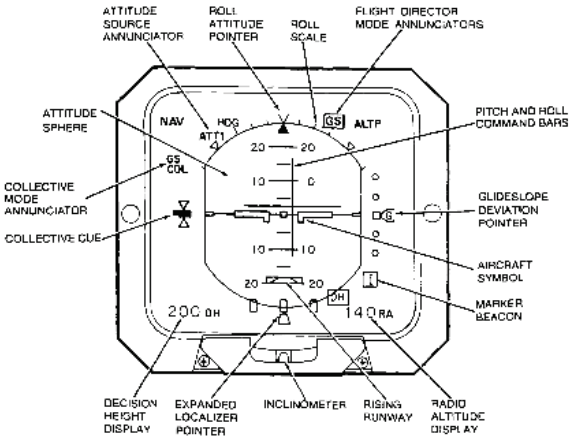
- Attitude Failure
- Glidescope, Glidepath or VNAV, Expanded Localizer, or Rate-of-Turn
- Internal Failure

Failure of the attitude source results in the pitch scale and roll pointer being removed, the sphere will be displayed blue, and a red ATT FAIL annunciator being displayed.

Failure of the glidescope, glidepath or VNAV, expanded localizer or azimuth, or rate-of-turn will result in the pointer being removed and a red X drawn through the scale. The letter G or V at the zero deviation position identifies the invalid information.

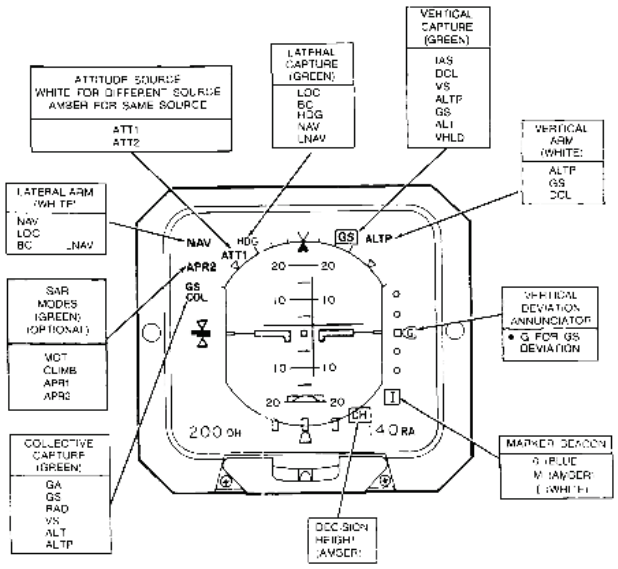
An extended failure of the symbol generator will cause the EADI to go blank. A failure of the input/processor results in a red X centered on the display, with annunciation SG FAIL.

For additional information on component displays/annunciation and operation of the EADI, refer to Honeywell, SPZ-7600 Integrated Flight Control System Bell 412 System Maintenance Manual, Publication Number A09-1169-024.



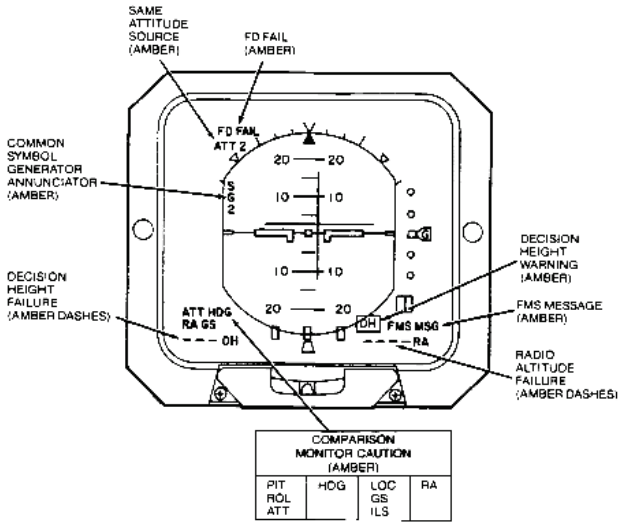
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Figure 1. EADI Displays and Annunciators (Sheet 1 of 2)



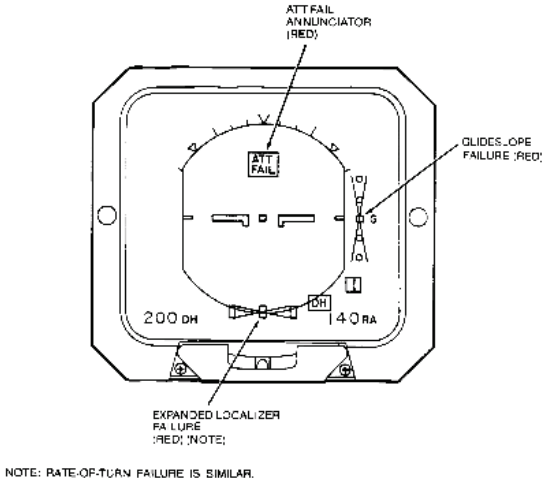
ICN-412-A-951102-B-97499-00002-A-001-01

Figure 1. EADI Displays and Annunciators (Sheet 2 of 2)



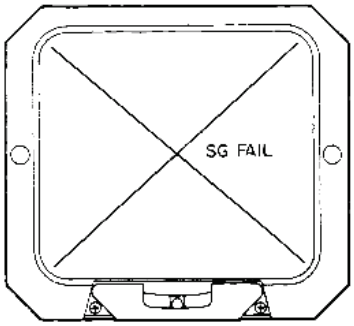
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Figure 2. EADI Cautions and Failure Annunciations (Amber) (Sheet 1 of 1)



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Figure 3. EADI Failure Annunciations (Red) (Sheet 1 of 1)



NOTE: COMPLETE FAILURE OF A SYMBOL GENERATOR WILL RESULT IN BLANK EADI AND EHSI DISPLAYS.

ICN-412-A-951102-B-97499-00005-A-001-01

Figure 4. Internal System Failure (Red) (Sheet 1 of 1)

Standby Attitude Indicator (When Installed) - Description

The standby attitude indicator is a self-contained instrument which comprises a vertical gyro and motor, a spheroid to indicate pitch and roll, a bank scale and index pointer to indicate roll, a miniature airplane symbol, a PULL TO CAGE (quick erect) knob and a striped power off flag. It serves as a backup for the Attitude Director Indicator (ADI) and is required for IFR flight.

The standby attitude indicator provides backup pitch and roll information in the event of a multiple power failure which disables emergency bus 2. In normal inflight and ground operating conditions the standby attitude indicator is powered by input from emergency bus 2. Under emergency conditions (no generator and battery output), the standby battery can be selected to power the standby attitude indicator by positioning STBY ATTD switch on overhead console to ON. A fully charged standby battery will operate the standby attitude indicator for approximately 90 minutes. Refer to DC Indicator Systems, 412-A-96-12-00-03A-320A-A / 01989 for standby attitude indicator system operational check and maintenance.

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Radar Altimeter Indicator - Description

The radar altimeter indicator provides altitude display of 0 to 1500 feet. The indicator has, in addition to the altitude display, a red power OFF warning flag, a TEST switch (button) for confidence testing, a Decision Height (DH) knob for setting a DH index, and a DH warning annunciator. Operational check (412-A-97-27-00-00A-340A-A / 02426) and troubleshooting functions are performed in Radar Altimeter System (412-A-97-27-00-00A-422A-A / 02427).

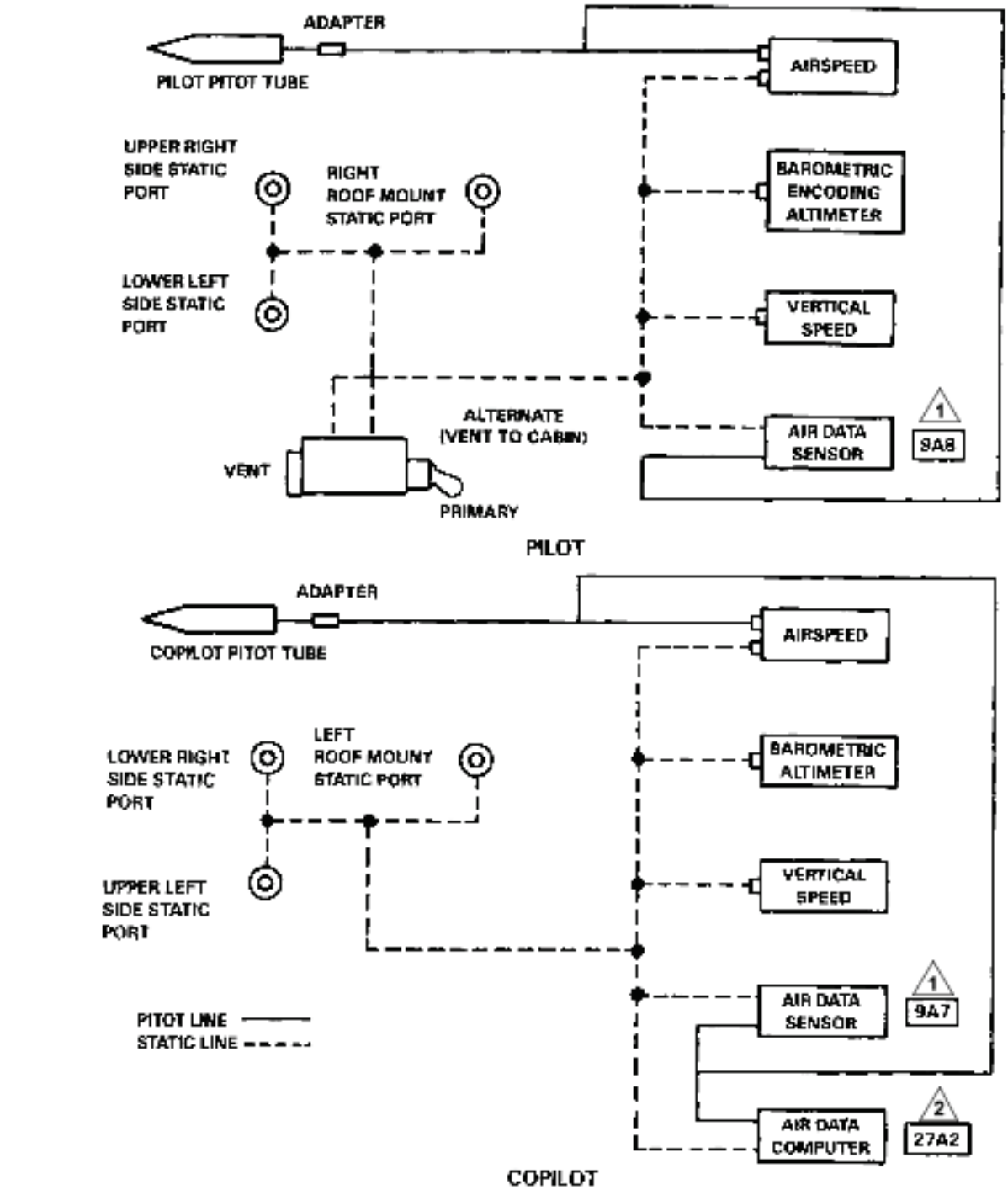
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Pilot Pitot-Static System - Description

The pilot pitot-static system (Figure 1) consists of right pitot tube, right and left static ports, roof mounted static port, alternate static air source, pilot airspeed indicator, barometric altimeter, vertical speed indicator, and appropriate tubing and hardware for interconnection of system components.

The right pitot tube supplies pitot pressure (impact air) directly to pilot airspeed indicator and air data sensor (P/O Flight Director kit). Right and left static ports are connected together and supply equalized static pressure to pilot airspeed indicator, altimeter, vertical speed indicator and air data sensor (P/O Flight Director kit). Right pitot tube and both static ports are equipped with electric heaters (optional for static ports) to provide all-weather capability.

An alternate static air source is provided by a pneumatic valve installed below the pilot instrument panel. Tubing and fittings are included to connect pneumatic valve to pilot static ports and instruments. With the toggle of the pneumatic valve set to PRI, static air is supplied to instruments from outside pilot static air ports. With the toggle set to ALTN, an alternate static air source (inside cabin air) is selected and outside static ports are shut off.



NOTES

1. Applicable to helicopters with flight director kit installed.
2. Applicable to helicopters with flight data recorder installed.

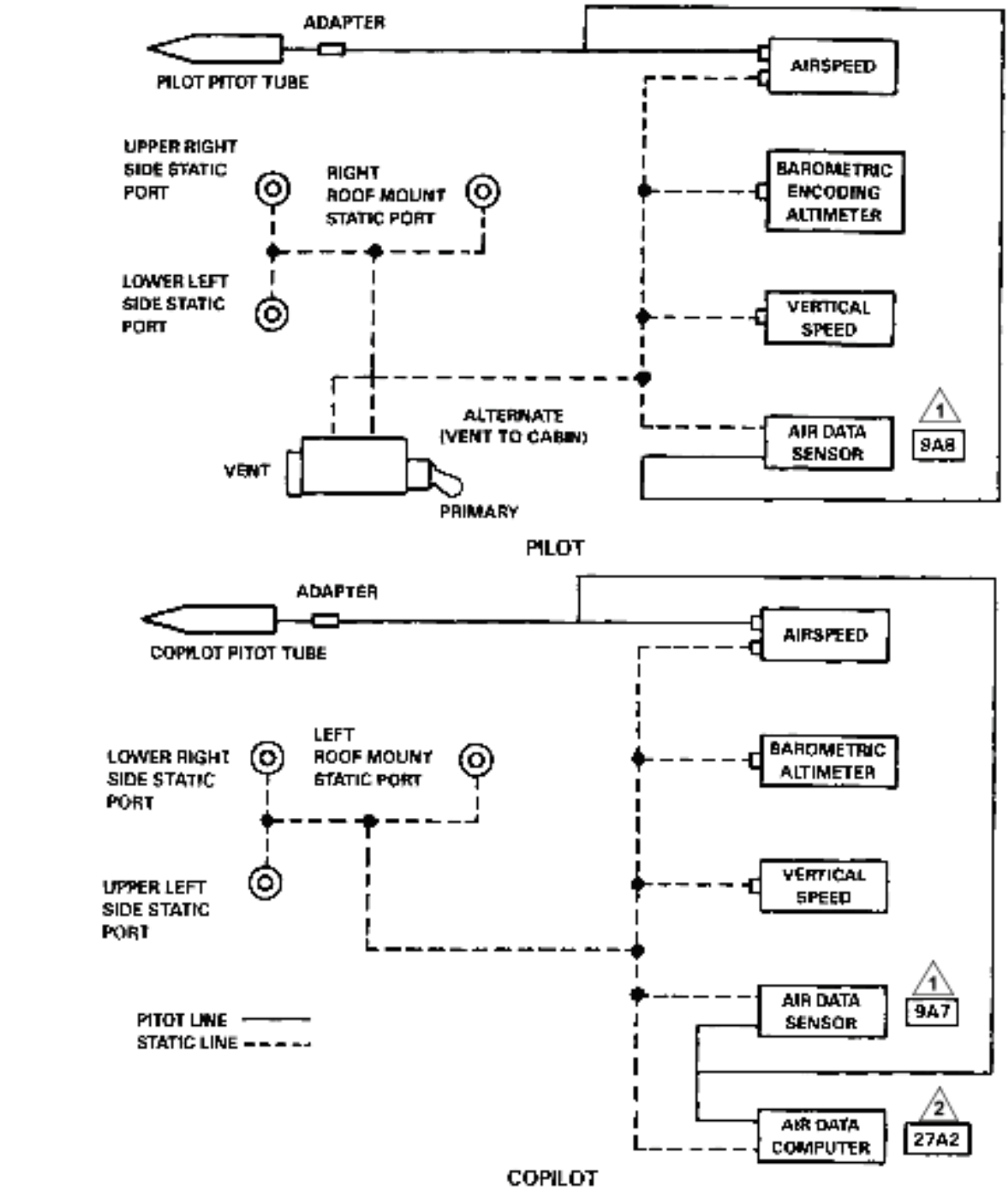
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Figure 1. Pitot-Static System Schematic (Sheet 1 of 1)

Copilot Pitot-Static System (When Installed) - Description

The copilot pitot-static system (Figure 1) consists of left pitot tube, left and right static ports, left roof-mounted static port, airspeed indicator, barometric altimeter, vertical airspeed indicator, and appropriate tubing and hardware for interconnection of system components.

The left pitot tube supplies pitot pressure (impact air) directly to copilot airspeed indicator, air data sensor (P/O Flight Director kit), and air data computer (P/O Flight Data Recorder kit). Upper left, lower right, and left roof-mounted static ports are connected together and supply equalized static pressure to copilot airspeed indicator, barometric altimeter, vertical speed indicator, air data sensor (P/O Flight Director kit) and air data computer (P/O Flight Data Recorder kit). Pitot tube and static ports are equipped with electric heaters (optional for static ports) to provide all-weather capability.



NOTES

1. Applicable to helicopters with flight director kit installed.
2. Applicable to helicopters with flight data recorder installed.

ICN-412-A-951105-B-97499-00001-A-001-01

Figure 1. Pitot-Static System Schematic (Sheet 1 of 1)

Pitot-Static System - Leak Check

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Static Field Tester	Tactair Type MB-1	1	or Equivalent
Pitot-Static Tester	Barfield Model 1811F-D	1	or Equivalent

Procedure

1. Leak Check - Pitot System

1.1. Connect appropriate hose of static field tester (Tactair Type MB-1) to pitot tube of system being checked, using appropriate adapter. Seal drain line(s) of system being checked and check all connections for tightness.

1.2. Open both vent valves to allow pressure in lines to equalize. Close all five valves securely on test set.

1.3. Pump pressure into pressure tank on test set until sufficient pressure for test is indicated on pressure dial (approximately 50 inches Hg or 170 kPa).



VALVES ON TEST SET ARE SENSITIVE AND SHOULD BE OPERATED SLOWLY AND WITH CARE TO AVOID POSSIBLE DAMAGE TO INSTRUMENTS. THESE INSTRUMENTS ARE EASILY "PEGGED" AND DAMAGED BY IMPROPER MANIPULATION OF THE VALVES ON TEST SET.

1.4. Open pressure vent valve approximately 1/4 turn. Slowly open pressure source valve. Close pressure vent valve.

1.5. Close pressure source valve with helicopter airspeed indicator(s) reading 130 ±4 knots. After 1 minute, check airspeed indication has not dropped more than 10 knots. Refer to troubleshooting 412-A-95-11-08-01A-422A-A / 01824 if limits are exceeded.

1.6. Slowly open pressure vent valve. Check airspeed indication returns to minimum limit of indicator.

1.7. Remove test set from helicopter. Remove seal from drain line(s).

2. Leak Check - Static System

NOTE

Ensure pilot STATIC SOURCE valve toggle is set to PRI when checking static system.

2.1. Connect appropriate hose of pitot-static tester (Barfield Model 1811F-D) to one static port of system being checked, using appropriate adapter. Seal all drain lines and remaining static ports of system being checked and check all connections for tightness.

2.2. Open both vent valves to allow pressure in lines to equalize. Close all five valves securely on test set.

2.3. Pump vacuum side until sufficient vacuum for test is indicated on vacuum dial (approximately 20 inches Hg or 68 kPa).



CLOSELY MONITOR BOTH HELICOPTER AND TEST SET INSTRUMENTS TO ENSURE NO INDICATORS BECOME "PEGGED OUT". VALVES OF TEST SET MUST BE MANIPULATED SLOWLY AND CAREFULLY TO AVOID POSSIBLE INSTRUMENT DAMAGE.

2.4. Gradually open vacuum source valve. Check altimeter indicator reading increases.

2.5. When helicopter altimeter reads 1000 ft. above helicopter elevation, close vacuum source valve. Tap gently on altitude indicator and check that, after 1 minute, loss in indicated altitude does not exceed 100 ft. Refer to troubleshooting 412-A-95-11-08-02A-422A-A / 01826 if limit is exceeded.

2.6. Open vacuum vent valve and observe the helicopter and test set altitude indicators return to approximate ground level readings.

2.7. Remove test set from helicopter. Remove seals from all drain lines and static ports.

Pitot-Static System - Operational Check

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Static Field Tester	Tactair Type MB-1	1	or Equivalent
Pitot-Static Tester	Barfield Model 1811F-D	1	or Equivalent

Procedure

NOTE

Perform following procedure for pilot pitot-static system. Repeat procedure for copilot pitot-static system, when installed.

1. Connect appropriate hoses of static field tester (Tactair Type MB-1) to pitot tube and one static port (Figure 1), using appropriate adapters. Seal all drain lines and remaining static ports and check all connections for tightness.

2. Open both vent valves to allow pressure in lines to equalize. Close all five valves securely on test set.

3. Set barometric pressure dials on helicopter and test set altitude indicators to 29.92 inches H_G. Gently tap on altimeters and check barometric altimeter installed in helicopter indicates within parameters of FAR 43, Appendix E of test set altitude indicator. If readings do not agree within this tolerance, replace barometric altimeter.

NOTE

Pressure in tank may be replenished as required for test.

4. Pump pressure into pressure tank on test set until sufficient pressure for test is indicated on pressure dial (approximately 50 inches H_G or 170 kPa).



VALVES ON TEST SET ARE SENSITIVE AND SHOULD BE OPERATED SLOWLY AND WITH CARE TO AVOID POSSIBLE DAMAGE TO INSTRUMENTS. THESE INSTRUMENTS ARE EASILY PEGGED AND DAMAGED BY IMPROPER MANIPULATION OF THE VALVES ON TEST SET.

5. Open pressure vent valve approximately 1/4 turn. Slowly open pressure source valve. Close pressure vent valve. Leave pressure source valve open until airspeed indicator on test set reads 130 knots, then close pressure source valve. Maintain this pressure by manipulation of pressure source valve as required. Gently tap on test set airspeed indicators and check that airspeed indicator reads 130 ±4 knots.

6. Close pressure source valve with helicopter airspeed indicator(s) reading 130 ±4 knots. After 1 minute, check airspeed indication has not dropped more than 10 knots. Refer to troubleshooting 412-A-95-11-07-00A-720A-A / 01818 if limit is exceeded.

7. Slowly open pressure vent valve. Check airspeed indication returns to minimum limit of indicator.

NOTE

Tank vacuum may be replenished as required for test.

8. Pump vacuum side until sufficient vacuum for test is indicated on vacuum dial (approximately 20 inches H_G or 68 kPa).



CLOSELY MONITOR BOTH HELICOPTER AND TEST SET INSTRUMENTS TO ENSURE NO INDICATORS BECOME "PEGGED" OUT. VALVES OF TEST SET MUST BE MANIPULATED SLOWLY AND CAREFULLY TO AVOID POSSIBLE INSTRUMENT DAMAGE.

9. Gradually open vacuum source valve. Check altimeter and airspeed indicator readings increase and Vertical Speed Indicator (VSI) shows a positive rate of climb.

10. Bring airspeed to approximately 130 knots and close vacuum source valve and pressure vent valve. Hold airspeed at 130 knots and increase altitude indications by manipulation of cross bleed and vacuum source valves. When

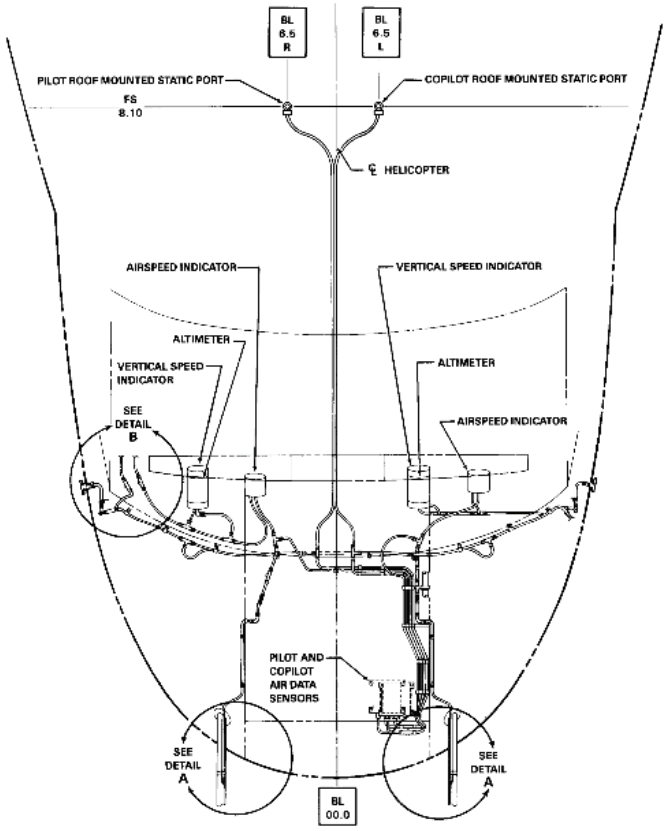
helicopter altimeter reads 1000 feet above helicopter elevation, close cross bleed and vacuum source valves. Tap gently on test set altitude indicators and check that, after 1 minute, loss in indicated altitude does not exceed 100 feet. Check VSI indication, if not zero, is negative and does not exceed 100 FPM.

11. Open vacuum vent valve momentarily to gradually decrease vacuum. Check VSI indicates a negative rate of climb.

12. Continue to decrease vacuum until test set altitude indicator reads 1000 feet. Maintain this reading by manipulation of vacuum source valve as required. Gently tap on altimeters and check barometric altimeter installed in helicopter indicates within 40 feet of test set indicator. If readings do not agree within this tolerance, replace barometric altimeter.

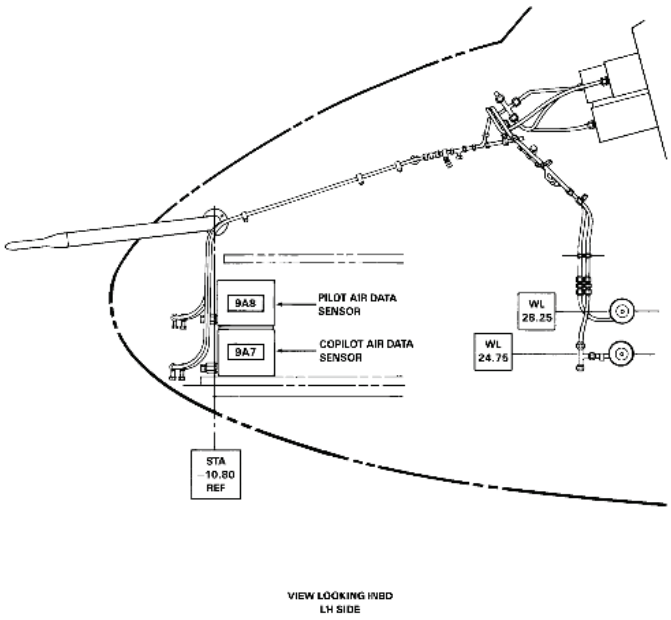
13. Open vacuum vent valve and observe helicopter and test set altitude indicators return to approximate ground level readings.

14. Open both pressure and vacuum vent valves. After system pressures are equalized, remove test set from helicopter. Remove seal placed on static port.



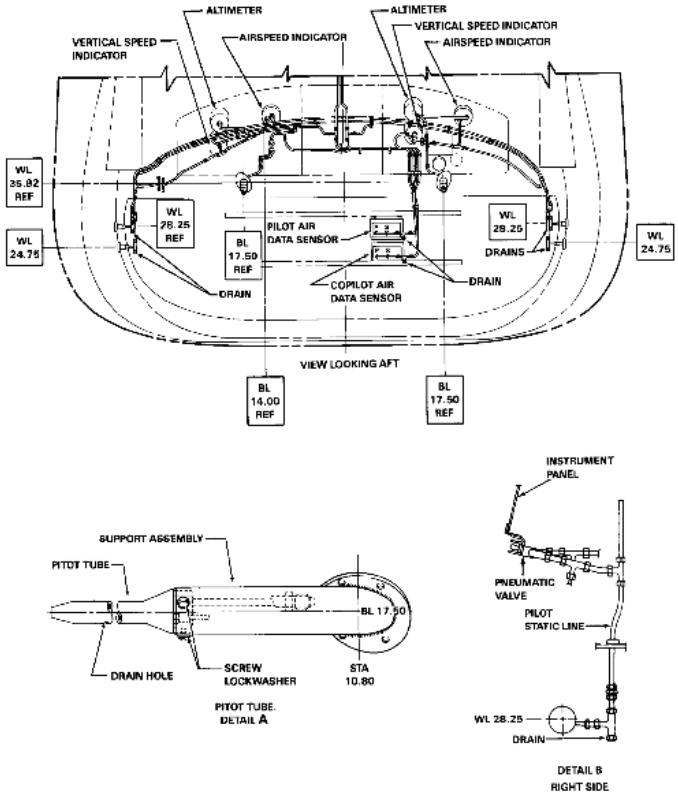
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Figure 1. Pitot-Static System (Sheet 1 of 3)



ICN-412-A-951105-B-97499-00003-A-001-01

Figure 1. Pitot-Static System (Sheet 2 of 3)



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Figure 1. Pitot-Static System (Sheet 3 of 3)

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Pitot-Static System - Inspection

Procedure

1. Inspect pitot tube for clogged or obstructed inlet opening.
2. Inspect pitot tube for clogged drain hole on bottom of tube.
3. Inspect pitot and static lines for leaks and chafing.
4. Visually inspect parts, pitot and static lines, accessories and instruments of pitot static system and replace defective components.

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Pitot-Static System Hardware - Repair or Replacement

Procedure

1. Tighten, install, or correct any loose, missing, or improperly installed hardware.

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Pitot and Static Lines - Draining Instructions

Procedure

1. Disconnect pitot and static lines from airspeed, altimeter, and vertical speed indicators and from air data sensors and computer, if installed. Cap openings in indicators, sensors, and computer.

2. Blow static lines clean with filtered, compressed air.

3. Uncap openings and reconnect lines.

4. Perform leak check of pitot-static system (412-A-95-11-05-00A-364A-A / 01807).

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Pitot Tube - Removal

Procedure



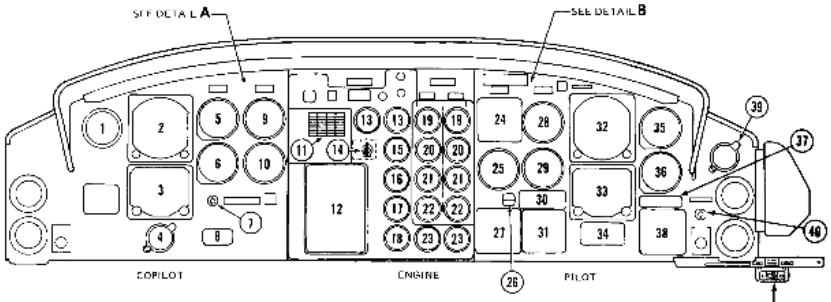
MAKE SURE ALL ELECTRICAL POWER IS OFF.

1. Disconnect pitot line from airspeed indicator (Figure 1). Cap opening on indicator and cap open piping.
2. Open clamps securing pitot line and create slack at base of pitot support by pulling line through grommet and clamps until connector is within 1 or 2 inches from grommet.
3. Remove screws and lockwashers attaching pitot tube (Detail A) to support and carefully pull tube from support until electrical connector is exposed. Disconnect and tape ends of connector to prevent entrance of foreign material.

NOTE

Make sure there is adequate slack in electrical wire before pulling pitot tube.

4. Continue pulling pitot tube until pitot line coupling is exposed. Disconnect and cap or cover openings in line.

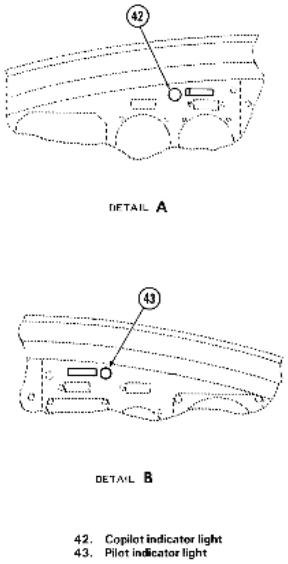


*1. Airspeed indicator (copilot)
*2. Attitude director indicator (ADI) (copilot)
*3. Horizontal situation indicator (HSI) (copilot)
*4. Clock (copilot)
*5. Barometric altimeter (copilot)
*6. Vertical speed indicator (VSI) (copilot)
*7. VG FAST ERECT switch (copilot)
*8. COPILOT SEL switch (copilot)
*9. Triple tachometer (copilot)
*10. Torquemeter (copilot)
11. Airspeed limits (VNE)
12. Caution light panel
13. Hydraulic pressure/temperature indicators
14. Fuel quantity indicator switch
15. Fuel quantity indicator
16. Transmission oil pressure/temperature indicator
17. Gearbox oil pressure/temperature indicator
18. Dual DC ammeter
19. Gas producer tachometer (N1) indicators
20. Interturbine temperature (ITT) indicators
21. Engine oil pressure/temperature indicators
22. Fuel pressure indicators
23. Dual voltmeter
24. Standby attitude indicator
25. Triple tachometer (pilot)
26. Flight director indicator
27. Flight director
28. Airspeed indicator (pilot)
29. Torquemeter (pilot)
30. Distance measuring equipment (DME) indicator (pilot)
31. Course select indicator (pilot)
32. Attitude director indicator (ADI) (pilot)
33. Horizontal situation indicator (HSI) (pilot)
34. NAV AUDIO panel (pilot)
35. Barometric altimeter (pilot)
36. Vertical speed indicator (VSI) (pilot)
37. Marker beacon indicator (pilot)
38. Radar altimeter (pilot)
39. Clock (pilot)
40. VG FAST ERECT switch (pilot)
41. STATIC SOURCE switch

*If installed, refer to Service Instruction 412-10 for Maintenance data.

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Figure 1. Instrument Panel (S/N 33001 through 33107) (Sheet 1 of 2)



42. Copilot indicator light
43. Pilot indicator light

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Figure 1. Instrument Panel (S/N 33001 through 33107) (Sheet 2 of 2)

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Pitot Tube - Installation

Procedure



MAKE SURE ALL ELECTRICAL POWER IS OFF.

1. Remove caps or covers from pitot lines and connect coupling.
2. Remove tape and connect electrical connector. While guiding pitot line and electrical wire through pitot base from inside of nose compartment, place pitot tube in pitot support.
3. Install screws and lockwashers that attach pitot tube to pitot support.
4. Pull pitot line through grommet and open clamps to connect line to airspeed indicator.
5. Remove caps on airspeed indicator and connect pitot line to indicator.
6. Adjust pitot line through grommet and open clamps to ensure adequate slack with no binding. Close and secure clamps to pitot line.
7. Perform leak check of pitot system (412-A-95-11-05-00A-364A-A / 01807).

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Pitot Tube - Cleaning

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Drycleaning solvent	C-304	AR	

Procedure

1. Clean pitot tube head and pitot support with a clean, lint-free cloth dampened with drycleaning solvent (C-304).

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Pitot Tube and Static Port Heaters - Description

Pitot tube heaters prevent pitot tube from icing over. Heating elements utilize 28 Vdc and are energized manually with PITOT HEATERS switch on overhead console. Static port heater elements are wired in parallel with respective pilot and copilot pitot heater system at PITOT HEATERS switch. Static port heater elements, designed to prevent static ports from icing over, utilize 28 Vdc power and are also energized by PITOT HEATERS switch.

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Pitot Tube and Static Port Heaters - Operational Check

Procedure



CHECKING PITOT TUBE FOR HEAT WITH BARE HANDS CAN CAUSE SERIOUS BURNS.

NOTE

Keep pitot tube heaters and static port heaters deenergized except for tests or when usage is necessary.

1. Close PITOT HTR PILOT circuit breaker. Place PITOT HEATERS switch on overhead console to ON. Check pilot pitot tube heater and static port heaters are energized and beginning to heat.

2. Set PITOT HEATERS switch to OFF. Check pilot pitot tube heater and static port heaters are deenergized and cooling. Open PITOT HTR PILOT circuit breaker.

3. Close PITOT HTR CPLT circuit breaker. Place PITOT HEATERS switch on overhead console to ON. Check copilot pitot tube heater and static port heaters are energized and beginning to heat.

4. Set PITOT HEATERS switch to OFF. Check copilot pitot tube heater and static port heaters are deenergized and cooling. Open PITOT HTR CPLT circuit breaker.

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Applicable to: ALL
ECCN EAR99

Static Port Heater Elements - Removal

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Sealant	C-308	AR	

Procedure

NOTE

The following removal procedures are applicable to all four static port heater elements.

1. Disconnect ground wire terminal, adjacent to static port.
2. Disconnect positive lead at splice.
3. Disconnect static tube from static port union.
4. Protect open end of static tube fitting with electrical tape.
5. Drill out two static port baffle rivets and remove baffle.
6. Remove burrs and metal particles around holes.
7. Apply a thin coat of sealant (C-308) to any bare metal.
8. Drill out six attaching rivets.
9. Remove static port heater element and heat shield.
10. Remove burrs and metal particles around holes.
11. Apply a thin coat of sealant (C-308) to any bare metal.

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Static Port Heater Elements - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Sealant	C-308	AR	

Procedure

NOTE

The following installation procedures are applicable to all four static port heater elements.

1. Align and position heat shield and static port heater element on mounting surface of inside cabin skin.
2. Install with six rivets, MS20426AD3-6, with heads flush outboard.

NOTE

Do not allow sealant to enter vent holes.

3. Seal around static port heater element and heat shield with sealant (C-308).
4. Remove protective tape from static port tube fitting.
5. Connect to static port.
6. Connect ground wire terminal, adjacent to static port.
7. Connect positive lead at splice.
8. Align and position static port baffle on mounting surface.
9. Install baffle with two rivets, MS20470AD3-6.
10. Perform leak check of static system (412-A-95-11-05-00A-364A-A / 01807).

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Static Port Heater System - Inspection

Procedure



USE EXTREME CARE WHEN CHECKING STATIC PORT HEATER ELEMENT WHILE IN OPERATION.

1. Inspect static tubing for leaks and chafing.
2. Inspect static tubing fittings, connectors, and hardware for proper installation.
3. Inspect wiring and leads for loose terminals, chafing, and damage.
4. Inspect static port heater element to ensure proper heating.

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Static Port Heater System - Cleaning

Procedure

1. Clean static port heater elements with a clean, lint-free cloth dampened with approved solvent.

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Static Port Heater System - Repair or Replacement

Procedure

1. Tighten or replace loose or defective tubing, fittings, connectors, or hardware.
2. Repair or replace defective wiring.
3. Remove and replace defective static port heater element (412-A-95-11-07-00A-520A-A / 01817 and 412-A-95-11-07-00A-720A-A / 01818).

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Pitot-Static Instruments - Description

Pilot pitot-static instruments are located on right side of instrument panel. If copilot instrument kit is installed, a duplicate set of instruments will be located on left side of instrument panel.

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Airspeed Indicator - Description

The airspeed indicator is a standard pitot-static instrument, single scaled and calibrated in knots. The indicator provides an indicated airspeed of helicopter at any time during forward flight by measuring difference between impact air pressure from pitot tube and static air pressure from static ports.

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Airspeed Indicator - Troubleshooting

Procedure

1. Refer to Table 1 and perform checks as necessary to isolate trouble.

Table 1. Troubleshooting - Airspeed Indicator

INDICATION OF TROUBLE	PROBABLE CAUSE	CORRECTIVE ACTION
Pointer fails to respond.	Pitot line not connected.	Connect line.
	Pitot lines clogged.	Disconnect pitot line(s) from airspeed indicator, air data sensors and/or air data computer, as required, and blow line(s) clear with clean, dry air.
Pointer indicates incorrectly.	Leak in pitot or static line.	Repair or replace line.
	Defective or leaking indicator.	Replace indicator.
Pointer vibrates.	Instrument loose in panel.	Tighten instrument mounting screws.
Pointer oscillates.	Leak in line.	Repair or replace line.
	Defective indicator.	Replace indicator.

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Altimeter - Description

The barometric altimeter furnishes direct reading of helicopter height in feet above sea level. Altimeter is connected by static lines to static pressure ports to sense atmospheric pressure. An external adjustment knob is provided to make compensation for variations of prevailing barometric pressure. Pilot altimeter contains an altitude encoder that provides digitized altitude signals to the transponder for transmission to the traffic control facilities. The encoding function of pilot altimeter is checked in IFF Transponder System (412-A-97-23-00-00A-028A-A / 02344, or 412-A-97-23-00-00A-028B-A / 02345).

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Applicable to: ALL
ECCN EAR99

Altimeter - Troubleshooting

Procedure

1. Refer to Table 1 and perform checks as necessary to isolate trouble.

Table 1. Troubleshooting - Altimeter

INDICATION OF TROUBLE	PROBABLE CAUSE	CORRECTIVE ACTION
Incorrect reading.	Leak in static pressure line.	Correct leak.
	Clogged static port or line.	Clean port or line.
	Defective indicator.	Replace indicator.

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Vertical Speed Indicator - Description

The vertical speed indicator registers ascent and descent speed of helicopter in feet per minute. Vertical speed indicator is connected to static air system to sense rate of atmospheric pressure change.

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Vertical Speed Indicator - Troubleshooting

Procedure

1. Refer to Table 1 and perform checks as necessary to isolate trouble.

Table 1. Troubleshooting - Vertical Speed Indicator

INDICATION OF TROUBLE	PROBABLE CAUSE	CORRECTIVE ACTION
Pointer off zero.	Mechanism shifted.	Return pointer to zero by turning adjustment screw on face of instrument; tap bezel of indicator lightly while adjusting.
Inaccurate readings.	Loose connections in static line.	Tighten connections.
	Indicator case leaks.	Replace indicator.
	Defective indicator.	Replace indicator.
Excessive pointer oscillation.	Leak in static line.	Tighten connections; replace leaky lines.
	Defective indicator.	Replace indicator.

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Navigation Instruments - Description

Navigation instruments are electrically powered, except for the standby magnetic compass. Navigation instruments include electromechanical Horizontal Situation Indicator (HSI) . When the optional 4-axis AFCS/Flight Director Kit is installed, the electromechanical HSI is replaced by an Electronic Horizontal Situation Indicator (EHSI) as part of the Electronic Flight Instrument System (EFIS) .

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Horizontal Situation Indicator (HSI) - Description

The horizontal situation indicator provides overall navigational information by means of the following individual displays:

The rotating azimuth dial (compass card) presents heading data received from either the Tarsyn Three-Axis Reference Sensor (S/N 33001 through 36247) or the Attitude Heading Reference System (AHRS) (S/N 36248 and Subsequent). Helicopter heading is read under upper lubber line.

The command heading marker is a movable reference point which indicates selected command heading. It is set to a selected heading by heading set knob. When the helicopter is flying selected heading, command heading marker will be aligned on lubber line. The marker moves with azimuth dial except when being mechanically positioned by heading set knob.

The command course pointer is a movable reference point indicating selected command radio course. It is set to a selected course with course set knob. When the helicopter is flying selected course, course pointer will be aligned on lubber line.

The digital course readout presented in course window is set by course set knob and corresponds to setting of command course pointer.

The deviation bar depicts relationship of selected radio beam centerline with respect to present navigation situation. Fixed aircraft symbol represents helicopter, and deviation bar depicts radio beam position with respect to helicopter. When helicopter is on selected radio course, deviation bar will be aligned with course pointer and fixed aircraft symbol.

The TO-FROM pointer depicts location of selected VOR station with respect to helicopter position. TO pointer is displayed at the nose of fixed aircraft symbol, and FROM pointer is displayed at tail of fixed aircraft symbol.

The glide slope pointer displays helicopter deviation from glide slope beam center. Pointer represents position of beam center in relation to helicopter.

Bearing pointer No. 1 indicates relative bearing of selected VOR radio station as determined by either VOR/LOC No. 1 receiver or VOR/LOC No. 2 receiver, depending upon selection of AFCS/HSI COMD and BRG PTR NO. 1 switches. The pointer is read against azimuth dial.

Bearing pointer No. 2 indicates relative bearing of received VOR radio station. Pointer is read/displays ADF information against azimuth dial.

Two warning flags - VOR/LOC warning flag and glideslope warning flag - provide a visual indication when the received signal is unreliably weak or the system is malfunctioning. Warning flag comes into view whenever the received signal is weak or a system malfunction occurs.

The OFF flag comes into view to provide a visual indication that the HSI or applicable compass system is not supplied with input power from respective buses.

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Electronic Horizontal Situation Indicator (EHSI) - Description

Refer to Honeywell, SPZ-7600 Integrated Flight Control System Bell 412 System Maintenance Manual, Publication Number A09-1169-024 for component description and operation.

The EHSI provides compass displays and a map-like display of the helicopter position relative to VOR radials, localizer, glidescope beam, or FMS waypoints.

The EHSI (Figure 1) provides the following displays and annunciators:

- Course select/desired
- Drift angle bug
- FMS approach message
- Heading source annunciator
- Heading select bug and heading select display
- Navigation source annunciators
- Distance display
- Preselected Nav source annunciator
- Course preselect pointer
- Compass card and fore and aft lubber lines
- Glideslope deviation pointer
- Groundspeed, time-to-go, or elapsed time display
- Bearing pointers and source annunciators
- Reciprocal course pointer
- Aircraft symbol
- Course deviation bar
- Weather radar mode and tilt annunciator
- To-From arrow
- Course select pointer
- Compass sync annunciator

In addition to the standard displays and annunciators, the EHSI also displays the following amber caution and failure annunciations (Figure 2):

- Waypoint alert annunciator
- Same heading or navigation source
- DME hold annunciator
- Weather radar (WX) failure
- Displays failures

  - amber dashes replace the digital display for the following systems when they fail:
  - - Groundspeed (GSPD)
  - - Time-to-Go (TTG)
  - - Elapsed Time (ET)
  - - Distance Measuring Equipment (DME)
  - - Heading Select (HDG)
  - - Course Select (CRS)
  - - Desired Track (DTRK)

The EHSI also displays the following red failure annunciations (Figure 3):

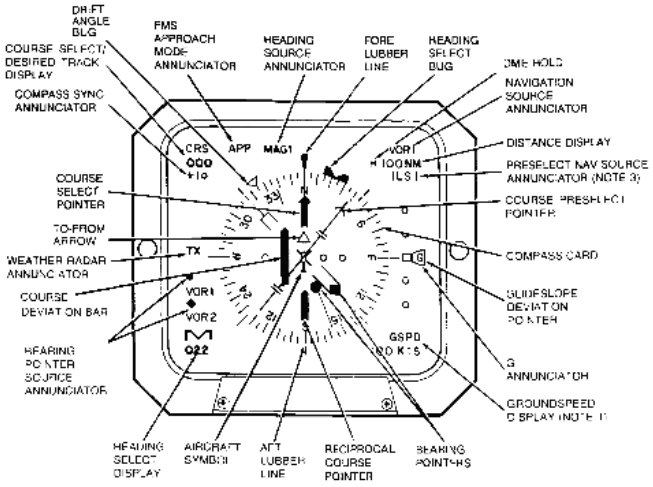
- Heading failure
- Course deviation
- Glidescope deviation

Failure of the heading system is indicated by the removal of the bearing pointers, TO/FROM arrow, select course pointer, course deviation bar, and course scale. The digital select course and digital heading select readouts are dashed and a red HDG FAIL is displayed. The heading source annunciator changes to a white HDG 1 through HDG 4 to include which heading source has failed.

A failure of the course deviation, or glidescope deviation results in the removal of the deviation pointer, and red X drawn through the scale.

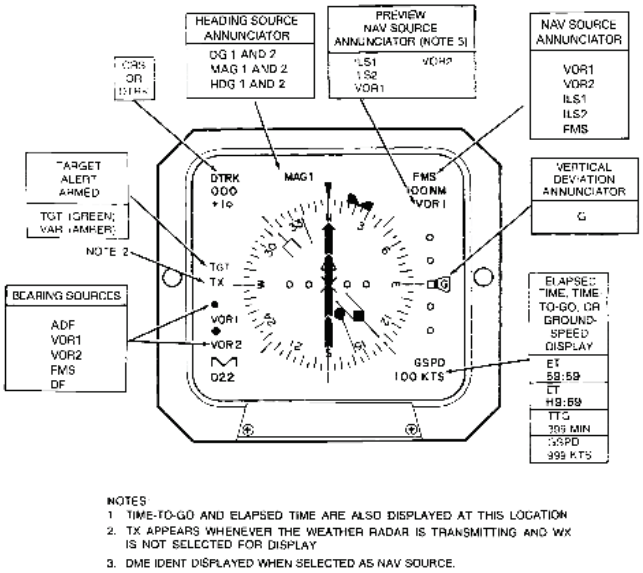
An internal failure of the symbol generator will cause the EHSI to go blank. A failure of the input/ processor results in a red X centered on the display, with annunciation SG FAIL.

For additional information on EHSI displays/annunciations and operation, refer to Honeywell, SPZ-7600 Integrated Flight Control System Bell 412 System Maintenance Manual, Publication Number A09-1169-024.



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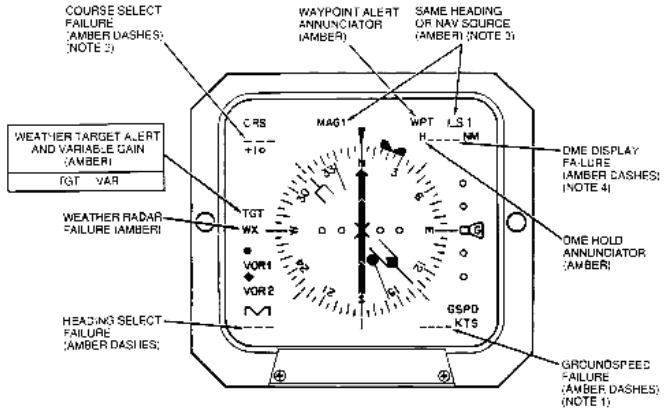
Figure 1. EHSI Full Compass Displays and Annunciators (Sheet 1 of 2)



NOTES:
1. TIME-TO-GO AND ELAPSED TIME ARE ALSO DISPLAYED AT THIS LOCATION
2. TX APPEARS WHENEVER THE WEATHER RADAR IS TRANSMITTING AND WX IS NOT SELECTED FOR DISPLAY
3. DME IDENT DISPLAYED WHEN SELECTED AS NAV SOURCE.

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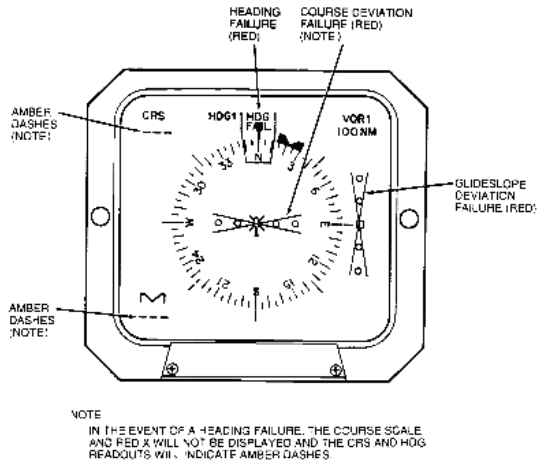
Figure 1. EHSI Full Compass Displays and Annunciators (Sheet 2 of 2)



NOTES:
1. TTG AND ET FAILURES ARE SIMILAR.
2. DTRK FAILURE IS SIMILAR.
3. IF BOTH THE PILOT AND THE COPILOT SELECT THE CROSS-SIDE NAV SOURCE, THE NAV ANNUNCIATORS ARE AMBER.
4. AN AMBER N/A IS DISPLAYED WHEN SELECTED DTN DISTANCE IS NOT AVAILABLE.
5. THIS DIAGRAM IS FOR ILLUSTRATION PURPOSES ONLY. IT DOES NOT NECESSARILY REPRESENT AN ACTUAL PRESENTATION.

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Figure 2. EHSI Full Compass Cautions and Failure Annunciations (Amber) (Sheet 1 of 1)



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Figure 3. EHSI Full Compass Display Annunciations (Red) (Sheet 1 of 1)

Distance Measuring Equipment (DME) Indicator - Description

The Distance Measuring Equipment (DME) indicator, used with DME navigation set, presents slant distance digital readout of helicopter to selected VOR station in nautical miles, groundspeed in knots, and time-to-station in minutes.

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Applicable to: ALL
ECCN EAR99

Course Select Indicator (CSI) (When Installed) - Description

The indicator provides navigation information by means of the following displays:

The course card with the use of the OBS knob selects the desired course beneath the course index.

The TO-FR indicator depicts location of VOR station with respect to the helicopters position.

The vertical pointer indicates deviation from the selected VOR radial. When the helicopter is on the selected VOR course, the pointer will be aligned with the course index.

The horizontal pointer indicates deviation above or below the glidepath. The pointer represents the position of the beam center in relation to the helicopter.

Two warning flags - NAV warning flag and GS flag - provide a visual indication of an unreliably weak VOR/localizer or glideslope signal, or a system malfunction.

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Marker Beacon Indicators (When Installed) - Description

The pilot and copilot marker beacon indicators (when installed), used with glide slope-marker beacon set, provide visual indication, in addition to audio tones, of passage over beacon stations located on airways and ILS approach courses.

Each indicator consists of three indicator lights (white, blue and amber). White indicator light (airways indicator) illuminates steady while helicopter is over fan marker beacon station. During ILS operation, blue indicator light will flash "on/off" at rate of two flashes per second when helicopter passes over outer marker beacon station. Amber light flashes "dot/dash" when the helicopter passes over middle marker beacon during ILS approach.

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Standby Magnetic Compass - Description

NOTE

Standby magnetic compass (or compasses) can be aligned simultaneously with compass system/compensation swing (412-A-97-21-00-00A-913A-A / 02304).

The pilot standby magnetic compass is mounted above windshield on center of panel. A second magnetic compass may be provided for copilot. Magnetic compass is used in conjunction with a magnetic compass correction card located on windshield center post.

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Standby Magnetic Compass - Inspection

Procedure

1. Inspect for loose, missing, or improperly installed hardware.

2. Inspect for loose, scratched, or cracked glass.

3. Inspect for legibility of markings.

4. Inspect compass card for excessive oscillation, sluggishness, or out-of-level condition.

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Standby Magnetic Compass - Removal

Procedure

1. Remove the four screws that attach compass to hinged bracket.
2. Carefully pull compass from hinged bracket far enough to disconnect light wire connector from compass.
3. Carefully remove compass from hinged bracket.

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Standby Magnetic Compass - Installation

Procedure

1. Position compass in hinged bracket and connect light wire connector.
2. Install the four screws that attach compass to hinged bracket.

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Initial Setup - Compass Swing - Calibration

Procedure

1. Before taking helicopter to compass rose, inspect compass in accordance with 412-A-95-12-06-00A-280A-A / 01836.
2. Secure any loose helicopter equipment constructed of magnetic material in the position it occupies in normal flight.
3. Position helicopter on compass rose.

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Index Error Check - Calibration

Procedure

NOTE

Any magnetic objects such as trucks, power carts, or other aircraft shall be removed from the compass rose swinging area to a point at which they will have no magnetic effect on the compass system.

1. Start the helicopter engines and operate at flight rpm. Turn on all electrical/avionics equipment normally used during flight, except landing lights and pitot-static heaters. Simulate actual flight conditions to the extent practical during compass calibration.

2. Rotate helicopter to the four cardinal headings: North, East, South, and West. At each heading record the difference in reading between standby compass card and compass rose. The difference will be plus if compass card reading is greater than that of compass rose, and minus if compass card reading is less than that of compass rose. Add the errors algebraically and divide by four. Using a nonmetallic screwdriver, align standby compass by adjusting position of the hinged bracket of compass mounting to satisfy resultant error.

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One-Cycle Error Check - Calibration

Procedure

NOTE

All corrections shall be made using a nonmetallic screwdriver. The compass should be lightly tapped after each adjustment and allowed to settle before taking readings.

1. With helicopter positioned to magnetic North heading on compass rose, compensate for all compass error by turning appropriate N-S screw.

2. With helicopter positioned to magnetic East heading on compass rose, compensate for all compass error by turning appropriate E-W screw.

3. With helicopter positioned to magnetic South heading on compass rose, take out one-half of the compass error by turning appropriate N-S screw.

4. With helicopter positioned to magnetic West heading on compass rose, take out one-half of the compass error by turning appropriate E-W screw.

5. Swing helicopter on 30 degree increments and note each indication. Correct errors on standby magnetic compass until error is 2 degrees or less at cardinal headings (0°, 90°, 180°, 270°) and 5 degrees or less at all other 30 degree points.

6. Record deviation errors on compass correction card.

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Standby Magnetic Compass - Troubleshooting

Procedure

1. Refer to Table 1 and perform checks as necessary to isolate trouble.

Table 1. Troubleshooting - Standby Magnetic Compass

INDICATION OF TROUBLE	PROBABLE CAUSE	CORRECTIVE ACTION
Excessive card error.	Improper compensation.	Compensate compass.
	External magnetic interference.	Locate magnetic influence and eliminate if possible.
	Air in bowl.	Replace compass.
	Insufficient liquid in bowl.	Replace compass.
Card element not level.	Leaking float chamber.	Replace compass.
	Card magnet is detached.	Replace compass.
Card sluggish.	Dirty pivots or jewels restricting rotation.	Replace compass.
	Weak magnetic card.	Replace compass.

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Propulsion and Hydraulic Instruments - Description

The propulsion and hydraulic instruments are electrically powered by AC or DC current. Operational checks for these instruments are provided in 412-A-96-05-00-02A-320A-A / 02134, or 412-A-96-05-00-02A-320C-A / 02136.

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Dual Torque Pressure Indicator - Description

The dual torque pressure indicator simultaneously displays torque output of each engine on inner dial and torque to transmission (combined torque of both engines) on outer dial. Indicator receives torque indications from engine No. 1 and engine No. 2 torque pressure transmitters. All indications are in percent of allowable torque.

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Triple Torque Pressure Indicator - Description

The triple torque pressure indicator simultaneously displays torque output of each engine on right side of indicator dial and mast torque on left side of indicator. The indicator receives signals from engine No. 1 and engine No. 2 torque sensors and a mast torque signal is generated by mast torque monopole sensor, located inside transmission. Indications are in percent of allowable torque.

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Engine Oil Pressure and Temperature Indicators - Description

No. 1 and No. 2 engine oil pressure and temperature indicators are dual-type indicators registering temperature in degrees Celsius and pressure in PSI for each engine. Temperature portion receives temperature indications from an electrical resistance-type bulb, and pressure portion receives pressure indications from engine oil pressure transmitter. Temperature portion is powered from 28 VDC emergency buses 1 and 2. Pressure portion is powered from 26 VAC buses 1 and 2.

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Fuel Pressure Indicators - Description

Engine No. 1 and No. 2 fuel pressure indicators register fuel pressure from indications received from fuel pressure transmitters. System is powered from 26 VAC buses 1 and 2.

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Fuel Quantity Indicator - Description

The fuel quantity indicator provides readings of fuel supply quantity in pounds as determined by position of fuel quantity selector switch (FWD TANK, TOTAL, or MID TANK). System is powered from 28 VDC emergency buses 1 and 2.

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Gas Producer Tachometer Indicators - Description

The engine No. 1 and No. 2 gas producer tachometer indicators register percent of RPM of engine No. 1 and No. 2 gas producer speeds. Gas producer tachometer generators, one on each engine, send voltage signals to respective indicator proportional to engine No. 1 and No. 2 gas producer speed. Gas producer tachometer indicator system is self-generating.

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Gearbox Oil Pressure and Temperature Indicator - Description

The gearbox oil pressure and temperature indicator is a dual-type indicator registering temperature in degrees Celsius and pressure in PSI of oil in combining gearbox. Temperature portion receives temperature indications from an electrical resistance-type bulb, and pressure portion receives pressure indications from oil pressure transmitter. Temperature portion is powered from 28 VDC emergency bus 1 and pressure portion is powered by 28 VAC bus 2.

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Hydraulic Pressure and Temperature Indicators - Description

The system No. 1 and system No. 2 hydraulic pressure and temperature indicators are dual type indicators registering hydraulic oil temperature in degrees Celsius and hydraulic pressure in psi. Pressure portion is powered from 26 VAC buses 1 and 2, and registers indications received from two hydraulic pressure transmitters. Temperature portion is powered from 28 VDC essential buses 1 and 2, and registers indications received from two electrical resistance-type temperature bulbs.

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Transmission Oil Pressure and Temperature Indicator - Description

The transmission oil pressure and temperature indicator is a dual type indicator registering temperature in degrees Celsius and pressure in psi. Temperature portion receives temperature indications from an electrical resistance-type bulb, and pressure portion receives pressure indications from oil pressure transmitter. Temperature portion is powered by 28 VDC emergency bus No. 2, and pressure portion is powered by 26 VAC bus No. 1.

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Triple Tachometer Indicator - Description

The triple tachometer indicator contains three pointers and simultaneously registers engine No. 1 and engine No. 2 power turbine and main rotor rpm in percent. Power is provided by three tachometer generators, mounted on engine No. 1, No. 2, and transmission respectively. These systems are self-generating.

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Interturbine Temperature (ITT) Indicators - Description

The engine No. 1 and No. 2 ITT indicators register power turbine inlet air temperature received from bayonet-type thermocouples mounted between gas producer turbine and power turbine in center engine compartments. Temperature indications are in degrees Celsius.

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Miscellaneous Instruments - Description

The miscellaneous instruments consist of the digital clock, dual AC and DC voltmeters, dual DC ammeter, free air temperature indicator, and engine hourmeter. Also included in this section are the air data command display (DAFCS) and flight director mode selector (DAFCS).

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Digital Clock - Description

The clock has a liquid crystal display, lighting system, and ultraviolet filter to provide uniform night illumination and protection from sunlight. The clocks two pushbuttons control the four displays.

- GMT - Greenwich Mean Time (24 hour format)
- LT - Local Time (12 or 24 hour format)
- ET - Elapsed Time (1 second to 99 hours and 59 minutes range)
- ET - Elapsed Time Countdown Timer (1 second to 1 hour range).

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Digital Clock - Functional Check

Procedure

1. Press and hold down SELECT button for 3 seconds. The display will read ’88.88' and GMT, LT, and ET will be illuminated.

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Digital Clock - Operational Check

Procedure

1. SELECT Button. The SELECT button sequentially pressed will illuminate GMT (Greenwich mean time), LT (local time), and ET (elapsed time).

2. CONTROL Button. The CONTROL button sequentially pressed (after pressing the SELECT and CONTROL button simultaneously) will change the display. Pressing the button momentarily will start and reset ET (elapsed time).

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GMT (Greenwich Mean Time) - Functional Check

Procedure

1. Press SELECT button momentarily until GMT is illuminated.
2. Press SELECT and CONTROL buttons once simultaneously to enter set mode (tens of hours digit starts flashing).
3. Press CONTROL button once to display first digit to be set.
4. Press CONTROL button momentarily to set digit. Each button push advances the digit.
5. Press SELECT button once to select the next digit to be set.
6. Perform Step 3., Step 4., and Step 5. above until remaining digits are set.
7. Press SELECT button once to exit the set mode (flashing display indicates clock is running).

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LT (Local Time) - Functional Check

Procedure

1. Press SELECT button momentarily until LT is illuminated.
2. Press SELECT and CONTROL buttons once simultaneously to enter set mode (tens of hours digit starts flashing).
3. Press CONTROL button once to display first digit to be set.
4. Press CONTROL button momentarily to set digit. Each button push advances the digit.
5. Press SELECT button once to select the next digit to be set.

NOTE

Minutes are synchronized with GMT and can not be changed in the LT (local time) setting mode. If minutes require changing, follow GMT procedures for setting minutes.

6. Perform Step 3., Step 4., and Step 5. above until remaining digits are set.
7. Press SELECT button once to exit the set mode (flashing display indicates clock is running).

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ET (Elapsed Time) Count-Up - Functional Check

Procedure

1. Press SELECT button momentarily until ET is illuminated.

2. Press CONTROL button once to start count-up. Counter will start counting in minutes and seconds up to 59 minutes and 59 seconds. Then the counter will start counting in hours and minutes up to 99 hours and 59 minutes.

3. Press CONTROL button once to reset the ET count-up setting to zero.

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ET (Elapsed Time) Countdown - Functional Check

Procedure

1. Press SELECT button momentarily until ET is illuminated.
2. Press SELECT and CONTROL buttons once simultaneously to enter set mode.
3. Press CONTROL button once to display first digit to be set.
4. Press CONTROL button momentarily to set digit. Each button push advances the digit up from 1 second to 59 minutes and 59 seconds.
5. Press SELECT button once to select the next digit to be set.
6. Perform Step 3., Step 4., and Step 5. above until remaining digits are set.
7. Press SELECT button once to exit the set mode.
8. Press CONTROL button once to start the countdown. Display will flash and an external alarm will be activated when the counter reaches zero.
9. Press SELECT or CONTROL button to deactivate the flashing display and external alarm.

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Digital Clock - Troubleshooting

Procedure

1. Refer to Table 1 for troubleshooting.

Table 1. Troubleshooting - Digital Clock

INDICATION OF TROUBLE	PROBABLE CAUSE	CORRECTIVE ACTION
Clock does not display time.	Defective instrument.	Replace clock.
	Defective wiring.	Repair wiring.
Clock does not illuminate GMT, LT, and ET when SELECT button is pressed.	SELECT button, or instrument defective.	Replace clock.
Clock display does not change when CONTROL button is pressed (after SELECT and CONTROL buttons were simultaneously pressed).	CONTROL button, or instrument defective.	Replace clock.

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Dual AC and DC Voltmeters - Description

The dual AC and DC voltmeters monitor and simultaneously indicate AC and DC bus voltage. No. 1 dual voltmeter indicates voltage present on 115 VAC and 28 VDC No. 1 buses, and No. 2 dual voltmeter indicates voltage present on the 115 VAC and 28 VDC No. 2 buses. For additional information and troubleshooting, refer to DC and AC power systems in 412-A-96-61-00-00A-028A-A / 01905 and 412-A-96-62-00-00A-028A-A / 01937.

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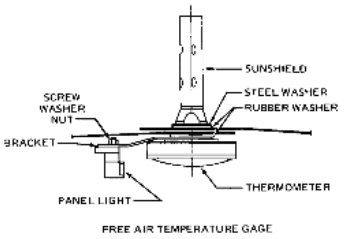
Dual DC Ammeter - Description

The dual DC ammeter indicates current output in amperes of starter-generators No. 1 and No. 2 by monitoring current flow at each respective shunt. Troubleshooting is covered as a portion of generator system in 412-A-96-61-00-04A-320C-A / 01925.

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Free Air Temperature Indicator - Description

The bimetallic probe-type free air temperature indicator (Figure 1) is mounted in the left forward portion of the upper right cabin roof window assembly. The indicator probe is exposed to outside temperatures through the window assembly and is protected by a sunshield. Indicator provides a direct reading of outside air temperature in degrees Celsius.



ICN-412-A-956303-B-97499-00001-A-001-01

Figure 1. Free Air Temperature Indicator (Sheet 1 of 1)

Engine Hourmeter - Description

The engine hourmeter is mounted on right side of pedestal near floor. Hourmeter clock mechanism is calibrated in hours and registers engine operating hours under flight power. Troubleshooting is covered as a portion of engine hourmeter electrical system in 412-A-96-12-00-07A-320A-A / 01998.

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Air Data Command Display (DAFCS) - Description

The air data command display is used to display air data parameter references. It also displays fault codes during DAFCS test cycle. The display decodes data from the digital flight control computer to illuminate the incandescent display. System is powered from 28 VDC essential bus 1. Refer to Digital Automatic Flight Control System (DAFCS) , 412-A-96-10-00-15A-340J-A / 02189 and 412-A-96-10-00-18A-422A-A / 02209 for operational check and troubleshooting procedures.

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Flight Director Mode Selector (P/O DAFCS Flight Director Kit) - Description

The mode selector is used to select and annunciate the various flight modes of the Digital Automatic Flight Control System (DAFCS) . The mode selector display consists of illuminated pushbutton switches for selecting DAFCS standby, roll axis, or pitch axis modes of operation. When a mode is selected, that mode is annunciated by a bright light level. The SBY switch, when held, tests all lamps in the mode selector and autopilot controller. Refer to 412-A-96-10-00-15A-340J-A / 02189 and DAFCS, 412-A-96-10-00-18A-422A-A / 02209 for operational check and troubleshooting.

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CHAPTER 96
ELECTRICAL SYSTEM

MAINTENANCE MANUAL

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List of Effective Data Modules

The listed documents are included in issue 007, dated 2025-11-18, of this publication.

C = Changed data module, N = New data module

Document title	Data module code / Seq number	Issue date	Applicable to
Electrical System - General Description	DMC-412-A-96-00-00-00A-028A-A / 01871	2023-05-31	ALL
Electrical Systems - Description	DMC-412-A-96-00-01-00A-042A-A / 01872	2023-05-31	ALL
Primary and Secondary Electrical Power Systems - Description	DMC-412-A-96-00-01-01A-042A-A / 01873	2023-05-31	ALL
Electrical Safety and Hazards - Description	DMC-412-A-96-00-01-02A-042A-A / 01874	2023-05-31	ALL
Standard Practices - Electrical Maintenance or Repairs	DMC-412-A-96-00-01-02A-913A-A / 01875	2023-05-31	ALL
Standard Practices - Electrical Work In Confined Spaces	DMC-412-A-96-00-01-02A-913B-A / 01876	2023-05-31	ALL
Standard Practices - After Electrical Maintenance or Repairs	DMC-412-A-96-00-01-02A-913C-A / 01877	2023-05-31	ALL
Miscellaneous Electrical Tools - Description	DMC-412-A-96-00-01-03A-042A-A / 01878	2023-05-31	ALL
Electrical Systems - Operational Checks and Functional Tests	DMC-412-A-96-00-01-00A-320A-A / 01879	2023-05-31	ALL
Electrical Systems - Troubleshooting	DMC-412-A-96-00-01-00A-422A-A / 01880	2023-05-31	ALL
Miscellaneous Electrical Components - Maintenance Practices	DMC-412-A-96-00-01-04A-042A-A / 01881	2023-05-31	ALL
Miscellaneous Electrical Components - Removal	DMC-412-A-96-00-01-04A-520A-A / 01882	2023-05-31	ALL
Miscellaneous Electrical Components - Cleaning	DMC-412-A-96-00-01-04A-251A-A / 01883	2023-05-31	ALL
Miscellaneous Electrical Components - Inspection	DMC-412-A-96-00-01-04A-280A-A / 01884	2023-05-31	ALL
Miscellaneous Electrical Components - Repair	DMC-412-A-96-00-01-04A-685A-A / 01885	2023-05-31	ALL
Miscellaneous Electrical Components - Installation	DMC-412-A-96-00-01-04A-720A-A / 01886	2023-05-31	ALL
Control Panels - Electrical - Description	DMC-412-A-96-00-02-00A-042A-A / 01887	2023-05-31	ALL
Control Panels - Removal	DMC-412-A-96-00-02-00A-520A-A / 01888	2023-05-31	ALL
Control Panels - Inspection	DMC-412-A-96-00-02-00A-280A-A / 01889	2023-05-31	ALL
Control Panels - Replacement	DMC-412-A-96-00-02-00A-921A-A / 01890	2023-05-31	ALL
Control Panels - Installation	DMC-412-A-96-00-02-00A-720A-A / 01891	2023-05-31	ALL
Circuit Breakers - Description	DMC-412-A-96-00-02-01A-042A-A / 01892	2023-05-31	ALL
Circuit Breakers - Removal	DMC-412-A-96-00-02-01A-520A-A / 01893	2023-05-31	ALL

Document title	Data module code / Seq number		Issue date	Applicable to
Circuit Breakers - Inspection	DMC-412-A-96-00-02-01A-280A-A / 01894		2023-05-31	ALL
Circuit Breakers - Replacement	DMC-412-A-96-00-02-01A-921A-A / 01895		2023-05-31	ALL
Circuit Breakers - Installation	DMC-412-A-96-00-02-01A-720A-A / 01896		2023-05-31	ALL
Semiconductor Devices - Description	DMC-412-A-96-00-02-02A-042A-A / 01897		2023-05-31	ALL
Semiconductor Devices - Maintenance Practices	DMC-412-A-96-00-02-02A-913A-A / 01898		2023-05-31	ALL
Track Mounted Relays - Description	DMC-412-A-96-00-02-03A-042A-A / 01899		2023-05-31	ALL
Track Mounted Relays - Removal	DMC-412-A-96-00-02-03A-520A-A / 01900		2023-05-31	ALL
Track Mounted Relays - Cleaning	DMC-412-A-96-00-02-03A-251A-A / 01901		2023-05-31	ALL
Track Mounted Relays - Inspection	DMC-412-A-96-00-02-03A-280A-A / 01902		2023-05-31	ALL
Track Mounted Relays - Installation	DMC-412-A-96-00-02-03A-720A-A / 01903		2023-05-31	ALL
Electrical and Avionics Equipment - Reference Designator, Nomenclature, and Location - Description	DMC-412-A-96-00-02-04A-042A-A / 01904	C	2025-11-18	ALL
DC Power Systems - Description	DMC-412-A-96-61-00-00A-028A-A / 01905		2023-05-31	ALL
Battery System - Description	DMC-412-A-96-61-00-01A-042A-A / 01906		2023-05-31	ALL
Battery System - Operational Check	DMC-412-A-96-61-00-01A-320A-A / 01907		2023-05-31	SN: 33001-33213 or 36001-36999
Battery - Description	DMC-412-A-96-61-00-02A-042A-A / 01908		2023-05-31	ALL
Battery - General Warnings	DMC-412-A-96-61-00-02A-012A-A / 01909		2023-05-31	ALL
Battery Vent - Description	DMC-412-A-96-61-00-02A-042C-A / 01910		2023-05-31	ALL
Battery - Maintenance Practices	DMC-412-A-96-61-00-02A-913A-A / 01911		2023-05-31	ALL
Battery Case and Battery Compartment - Cleaning	DMC-412-A-96-61-00-02A-251A-A / 01912		2023-05-31	ALL
Battery - Removal	DMC-412-A-96-61-00-02A-520A-A / 01913		2023-05-31	ALL
Battery - Inspection	DMC-412-A-96-61-00-02A-280A-A / 01914		2023-05-31	ALL
Battery - Installation	DMC-412-A-96-61-00-02A-720A-A / 01915		2023-05-31	ALL
External Power System - Description	DMC-412-A-96-61-00-03A-042A-A / 01916		2023-05-31	ALL
External Power System - Operational Check	DMC-412-A-96-61-00-03A-320A-A / 01917		2023-05-31	ALL

Document title	Data module code / Seq number	Issue date	Applicable to
External Power Receptacle - Removal	DMC-412-A-96-61-00-03A-520A-A / 01918	2023-05-31	ALL
External Power Receptacle - Installation	DMC-412-A-96-61-00-03A-720A-A / 01919	2023-05-31	ALL
External Power Receptacle - Inspection	DMC-412-A-96-61-00-03A-280A-A / 01920	2023-05-31	ALL
External Power Receptacle - Replacement	DMC-412-A-96-61-00-03A-921A-A / 01921	2023-05-31	ALL
Generator System - Description	DMC-412-A-96-61-00-04A-042A-A / 01922	2023-05-31	ALL
Generator No. 1 System - Operational Check	DMC-412-A-96-61-00-04A-320A-A / 01923	2023-05-31	ALL
Generator No. 2 System - Operational Check	DMC-412-A-96-61-00-04A-320B-A / 01924	2023-05-31	ALL
Generator No. 1 and Generator No. 2 System - Operational Check	DMC-412-A-96-61-00-04A-320C-A / 01925	2023-05-31	ALL
Voltage Control Paralleling Adjustment - Operational Check	DMC-412-A-96-61-00-04A-320D-A / 01926	2024-05-09	ALL
Non-essential Bus System - Operational Check	DMC-412-A-96-61-00-04A-320E-A / 01927	2023-05-31	ALL
Starter and Igniter Systems - Description	DMC-412-A-96-61-00-05A-042A-A / 01928	2023-05-31	ALL
Starter No. 1 Circuitry - Operational Check	DMC-412-A-96-61-00-05A-320A-A / 01929	2023-05-31	ALL
Starter No. 2 Circuitry - Operational Check	DMC-412-A-96-61-00-05A-320B-A / 01930	2023-05-31	ALL
Engine Ignition Unit No. 1 - Operational Check	DMC-412-A-96-61-00-06A-320B-A / 01931	2023-05-31	ALL
Engine Ignition Unit No. 2 - Operational Check	DMC-412-A-96-61-00-06A-320C-A / 01932	2023-05-31	ALL
Power Diodes - Functional Check	DMC-412-A-96-61-00-07A-340A-A / 01933	2023-05-31	ALL
Power Diodes - Removal	DMC-412-A-96-61-00-07A-520A-A / 01934	2023-05-31	ALL
Power Diodes - Inspection	DMC-412-A-96-61-00-07A-280A-A / 01935	2023-05-31	ALL
Power Diodes - Installation	DMC-412-A-96-61-00-07A-720A-A / 01936	2023-05-31	ALL
AC Power Systems - Description	DMC-412-A-96-62-00-00A-028A-A / 01937	2023-05-31	ALL
Inverter Systems - Description	DMC-412-A-96-62-00-01A-042A-A / 01938	2023-05-31	ALL
Inverter Systems - Operational Check	DMC-412-A-96-62-00-01A-320A-A / 01939	2023-05-31	ALL
Inverter No. 1 and Inverter No. 2 Circuitry - Operational Check	DMC-412-A-96-62-00-01A-320D-A / 01940	2023-05-31	SN: 33108-33213 or 36001-36999
Inverters - Removal	DMC-412-A-96-62-00-01A-520A-A / 01941	2023-05-31	ALL
Inverters - Inspection	DMC-412-A-96-62-00-01A-280A-A / 01942	2023-05-31	ALL

Document title	Data module code / Seq number	Issue date	Applicable to
Inverters - Repair	DMC-412-A-96-62-00-01A-685A-A / 01943	2023-05-31	ALL
Inverters - Installation	DMC-412-A-96-62-00-01A-720A-A / 01944	2023-05-31	ALL
AC Indicator Systems - Description	DMC-412-A-96-11-00-00A-028A-A / 01945	2023-05-31	ALL
Torque Pressure Indicator Systems - Description	DMC-412-A-96-11-00-01A-042A-A / 01946	2023-05-31	SN: 33001-33213 or 36001-36019 412-570-001-103: Not Installed
Torque Pressure Indicator Systems - Operational Check	DMC-412-A-96-11-00-01A-320A-A / 01947	2023-05-31	SN: 33001-33213 or 36001-36019 412-570-001-103: Not Installed
Engine Oil Pressure Indicator Systems - Description	DMC-412-A-96-11-00-02A-042A-A / 01948	2023-05-31	ALL
Engine Oil Pressure Indicator Systems - Operational Check	DMC-412-A-96-11-00-02A-320A-A / 01949	2023-05-31	ALL
Transmission Oil Pressure Indicator System - Description	DMC-412-A-96-11-00-03A-042A-A / 01950	2023-05-31	ALL
Transmission Oil Pressure Indicator System - Operational Check	DMC-412-A-96-11-00-03A-320A-A / 01951	2023-05-31	ALL
Hydraulic Pressure Indicator System - Description	DMC-412-A-96-11-00-04A-042A-A / 01952	2023-05-31	ALL
Hydraulic Pressure Indicator System - Operational Check	DMC-412-A-96-11-00-04A-320A-A / 01953	2023-05-31	ALL
Combining Gearbox (C Box) Oil Pressure Indicator Systems - Description	DMC-412-A-96-11-00-05A-042A-A / 01954	2023-05-31	ALL
Combining Gearbox (C Box) Oil Pressure Indicator Systems - Operational Check	DMC-412-A-96-11-00-05A-320A-A / 01955	2023-05-31	ALL
Fuel Pressure Indicator Systems - Description	DMC-412-A-96-11-00-06A-042A-A / 01956	2023-05-31	ALL
Fuel Pressure Indicator Systems - Operational Check	DMC-412-A-96-11-00-06A-320A-A / 01957	2023-05-31	ALL
Attitude Director Indicator Systems - Description	DMC-412-A-96-11-00-07A-042A-A / 01958	2023-05-31	ALL
Attitude Director Indicator Systems - Operational Check	DMC-412-A-96-11-00-07A-320A-A / 01959	2023-05-31	ALL
Rate Gyro - Maintenance	DMC-412-A-96-11-00-07A-913A-A / 01960	2023-05-31	SN: 33001-33213 or 36001-36247
Rate Gyro - Removal	DMC-412-A-96-11-00-07A-520A-A / 01961	2023-05-31	SN: 33001-33213 or 36001-36247

Document title	Data module code / Seq number	Issue date	Applicable to
Rate Gyro - Inspection	DMC-412-A-96-11-00-07A-280A-A / 01962	2023-05-31	SN: 33001-33213 or 36001-36247
Rate Gyro - Cleaning	DMC-412-A-96-11-00-07A-251A-A / 01963	2023-05-31	SN: 33001-33213 or 36001-36247
Rate Gyro - Repair or Replacement	DMC-412-A-96-11-00-07A-685A-A / 01964	2023-05-31	SN: 33001-33213 or 36001-36247
Rate Gyro - Installation	DMC-412-A-96-11-00-07A-720A-A / 01965	2023-05-31	SN: 33001-33213 or 36001-36247
Triple Tachometer Indicator Systems - Description	DMC-412-A-96-11-00-08A-042A-A / 01966	2023-05-31	ALL
Triple Tachometer Indicator Systems - Operational Check	DMC-412-A-96-11-00-08A-320A-A / 01967	2023-05-31	ALL
Gas Producer Tachometer Indicator Systems - Description	DMC-412-A-96-11-00-09A-042A-A / 01968	2023-05-31	ALL
Gas Producer Tachometer Indicator Systems - Operational Check	DMC-412-A-96-11-00-09A-320A-A / 01969	2023-05-31	ALL
Gas Producer Tachometer Indicator Systems - Maintenance	DMC-412-A-96-11-00-09A-913A-A / 01970	2023-05-31	ALL
Fuel Quantity Indicator System - Description	DMC-412-A-96-11-00-10A-042A-A / 01971	2023-05-31	SN: 33001-33107
Tank Unit And System Wiring Installation - Insulation Resistance Test	DMC-412-A-96-11-00-10A-366A-A / 01972	2023-05-31	SN: 33001-33107
Fuel Quantity Indicator System - Calibration Preferred Method (No Fuel In Tanks)	DMC-412-A-96-11-00-10A-273A-A / 01973	2025-05-13	SN: 33001-33107
Fuel Quantity Indicator System - Calibration Alternate Method (Fuel In Tanks)	DMC-412-A-96-11-00-10A-273B-A / 01974	2023-05-31	SN: 33001-33107
Fuel Quantity Coupler CP1/1CP1 - Maintenance	DMC-412-A-96-11-00-10A-913A-A / 01975	2023-05-31	SN: 33001-33107
Fuel Quantity Coupler CP1/1CP1 - Inspection	DMC-412-A-96-11-00-10A-280A-A / 01976	2023-05-31	SN: 33001-33107
Fuel Quantity Coupler CP1/1CP1 - Repair or Replacement	DMC-412-A-96-11-00-10A-685A-A / 01977	2023-05-31	SN: 33001-33107
Fuel Quantity Coupler CP1/1CP1 - Installation	DMC-412-A-96-11-00-10A-720A-A / 01978	2023-05-31	SN: 33001-33107
DC Indicator Systems - Description	DMC-412-A-96-12-00-00A-028A-A / 01979	2023-05-31	ALL

Document title	Data module code / Seq number		Issue date	Applicable to
Mast Torque Measurement System (412 Classic/SP to 412 HP Upgrade) - Description	DMC-412-A-96-12-00-01A-042A-A / 01980	C	2025-11-18	SN: 36020-36999 or 412-570-001-103: Installed or BHT-412-SI-74: Installed or BHT-412-SI-97: Installed
Mast Torque - Indication Check	DMC-412-A-96-12-00-01A-280A-A / 01981	C	2025-11-18	SN: 36020-36999 or 412-570-001-103: Installed or BHT-412-SI-74: Installed or BHT-412-SI-97: Installed
Mast Torque System - Operational Check	DMC-412-A-96-12-00-01A-320A-A / 01982	C	2025-11-18	SN: 36020-36999 or 412-570-001-103: Installed or BHT-412-SI-74: Installed or BHT-412-SI-97: Installed
Mast Torque System - Calibration Check	DMC-412-A-96-12-00-01A-273A-A / 01983	C	2025-11-18	SN: 36020-36999 or 412-570-001-103: Installed or BHT-412-SI-74: Installed or BHT-412-SI-97: Installed
Mast Torque System - Zero Torque Calibration	DMC-412-A-96-12-00-01A-273B-A / 01984	C	2025-11-18	SN: 36020-36999 or 412-570-001-103: Installed or BHT-412-SI-74: Installed or BHT-412-SI-97: Installed

Document title	Data module code / Seq number		Issue date	Applicable to
Mast Torque System - Flight Check	DMC-412-A-96-12-00-01A-280B-A / 01985	C	2025-11-18	SN: 36020-36999 or 412-570-001-103: Installed or BHT-412-SI-74: Installed or BHT-412-SI-97: Installed
Engine Torque Pressure Indicating System - Description	DMC-412-A-96-12-00-02A-042A-A / 01986	C	2025-11-18	SN: 36020-36999 or 412-570-001-103: Installed or BHT-412-SI-74: Installed or BHT-412-SI-97: Installed
Engine Torque Pressure Indicating System - Operational Check	DMC-412-A-96-12-00-02A-320A-A / 01987	C	2025-11-18	SN: 36020-36999 or 412-570-001-103: Installed or BHT-412-SI-74: Installed or BHT-412-SI-97: Installed
Standby Attitude Indicator - Description	DMC-412-A-96-12-00-03A-042A-A / 01988		2023-05-31	ALL
Standby Attitude Indicator - Operational Check	DMC-412-A-96-12-00-03A-320A-A / 01989		2023-05-31	ALL
Standby Battery - Maintenance/Testing	DMC-412-A-96-12-00-03A-913A-A / 01990		2025-05-13	ALL
Engine Oil Temperature Indicator System - Description	DMC-412-A-96-12-00-04A-042A-A / 01991		2023-05-31	ALL
Engine Oil Temperature Indicator System - Operational Check	DMC-412-A-96-12-00-04A-320A-A / 01992		2023-05-31	ALL
Transmission Oil Temperature Indicator System - Description	DMC-412-A-96-12-00-05A-042A-A / 01993		2023-05-31	ALL
Transmission Oil Temperature Indicator System - Operational Check	DMC-412-A-96-12-00-05A-320A-A / 01994		2023-05-31	ALL
Combining Gearbox (C Box) Oil Temperature Indicator System - Description	DMC-412-A-96-12-00-06A-042A-A / 01995		2023-05-31	ALL
Combining Gearbox (C Box) Oil Temperature Indicator System - Operational Check	DMC-412-A-96-12-00-06A-320A-A / 01996		2023-05-31	ALL
Engine Hourmeter System - Description	DMC-412-A-96-12-00-07A-042A-A / 01997		2023-05-31	ALL
Engine Hourmeter System - Operational Check	DMC-412-A-96-12-00-07A-320A-A / 01998		2024-05-09	ALL

Document title	Data module code / Seq number	Issue date	Applicable to
Turn and Slip (Part of Attitude Director Indicator) - Description	DMC-412-A-96-12-00-08A-042A-A / 01999	2023-05-31	ALL
Turn and Slip - Operational Check	DMC-412-A-96-12-00-08A-320A-A / 02000	2023-05-31	ALL
Interturbine Temperature (ITT) Indicator Systems - Description	DMC-412-A-96-12-00-09A-042A-A / 02001	2023-05-31	ALL
Interturbine Temperature (ITT) Indicator Systems - Functional Check	DMC-412-A-96-12-00-09A-340A-A / 02002	2023-05-31	ALL
Hydraulic Oil Temperature Indicator System - Description	DMC-412-A-96-12-00-10A-042A-A / 02003	2023-05-31	ALL
Hydraulic Oil Temperature Indicator System - Operational Check	DMC-412-A-96-12-00-10A-320A-A / 02004	2023-05-31	ALL
Fuel Quantity Indicator System - Description	DMC-412-A-96-12-00-11A-042A-A / 02005	2023-05-31	SN: 33108-33213 or 36001-36999
Fuel Quantity Indicator System - Testing and Calibration	DMC-412-A-96-12-00-11A-273A-A / 02006	2023-05-31	SN: 33108-33213 or 36001-36999
Fuel Quantity Indicator System - Thermistor Check	DMC-412-A-96-12-00-11A-280A-A / 02007	2023-05-31	SN: 33108-33213 or 36001-36999
Interior Lighting System - Description	DMC-412-A-96-01-00-00A-028A-A / 02008	2023-05-31	ALL
Cockpit Lights System - Description	DMC-412-A-96-01-00-01A-042A-A / 02009	2023-05-31	ALL
Cockpit Lights System - Operational Check	DMC-412-A-96-01-00-01A-320A-A / 02010	2023-05-31	ALL
Approach Plate and Map Lights System - Description	DMC-412-A-96-01-00-02A-042A-A / 02011	2023-05-31	ALL
Approach Plate and Map Lights System - Operational Check	DMC-412-A-96-01-00-02A-320A-A / 02012	2023-05-31	ALL
Dome Lights System - Description	DMC-412-A-96-01-00-03A-042A-A / 02013	2023-05-31	ALL
Dome Lights System - Operational Check	DMC-412-A-96-01-00-03A-320A-A / 02014	2023-05-31	ALL
Instrument Panel Lights System - Description	DMC-412-A-96-01-00-04A-042A-A / 02015	2023-05-31	ALL
Pilot Instruments and Passenger Warning Lighting Check - Operational Check	DMC-412-A-96-01-00-04A-320B-A / 02016	2023-05-31	ALL
Copilot Lighting Check - Operational Check	DMC-412-A-96-01-00-04A-320C-A / 02017	2023-05-31	ALL
Engine Instrument Lights System - Description	DMC-412-A-96-01-00-05A-042A-A / 02018	2023-05-31	ALL
Engine Instrument Lights System - Operational Check	DMC-412-A-96-01-00-05A-320A-A / 02019	2023-05-31	ALL
Instrument Secondary Lights System - Description	DMC-412-A-96-01-00-06A-042A-A / 02020	2023-05-31	ALL
Instrument Secondary Lights Systems - Operational Check	DMC-412-A-96-01-00-06A-320A-A / 02021	2023-05-31	ALL

Document title	Data module code / Seq number	Issue date	Applicable to
Pedestal Lights System - Description	DMC-412-A-96-01-00-07A-042A-A / 02022	2023-05-31	ALL
Pedestal Lights System - Operational Check	DMC-412-A-96-01-00-07A-320A-A / 02023	2023-05-31	ALL
Overhead Console Lights System - Description	DMC-412-A-96-01-00-08A-042A-A / 02024	2023-05-31	ALL
Overhead Console Lights System - Operational Check	DMC-412-A-96-01-00-08A-320A-A / 02025	2023-05-31	ALL
Baggage Compartment Lights System - Description	DMC-412-A-96-01-00-09A-042A-A / 02026	2023-05-31	ALL
Baggage Compartment Lights System - Operational Check	DMC-412-A-96-01-00-09A-320A-A / 02027	2023-05-31	ALL
Baggage Compartment Lights System - Maintenance	DMC-412-A-96-01-00-09A-913A-A / 02028	2023-05-31	ALL
Exterior Lighting Systems - Description	DMC-412-A-96-02-00-00A-028A-A / 02029	2023-05-31	ALL
Utility Lights - Description	DMC-412-A-96-02-00-01A-042A-A / 02030	2023-05-31	ALL
Utility Light System - Operational Check	DMC-412-A-96-02-00-01A-320A-A / 02031	2023-05-31	ALL
Position Lights System - Description	DMC-412-A-96-02-00-02A-042A-A / 02032	2023-05-31	ALL
Position Lights System - Operational Check	DMC-412-A-96-02-00-02A-320A-A / 02033	2023-05-31	ALL
Position Lights - Inspection	DMC-412-A-96-02-00-02A-280A-B / 02034	2023-05-31	SN: 36633-Subsequent
Anti-collision Lights System - Description	DMC-412-A-96-02-00-03A-042A-A / 02035	2023-05-31	ALL
Anti-collision Light - Removal	DMC-412-A-96-02-00-03B-520A-A / 02035.1	2023-08-28	SN: 33001-33100
Anti-collision Light - Removal	DMC-412-A-96-02-00-03A-520A-A / 02035.2	2023-08-28	SN: 33101-33213 or 36001-36999
Anti-collision Light - Inspection	DMC-412-A-96-02-00-03A-280A-A / 02035.3	2023-08-28	ALL
Anti-collision Light - Installation	DMC-412-A-96-02-00-03B-720A-A / 02035.4	2023-08-28	SN: 33001-33100
Anti-collision Light - Installation	DMC-412-A-96-02-00-03A-720A-A / 02035.5	2023-08-28	SN: 33101-33213 or 36001-36999
Anti-collision Lights System - Operational Check	DMC-412-A-96-02-00-03A-320A-A / 02036	2023-05-31	ALL
Searchlight System - Description	DMC-412-A-96-02-00-04A-042A-A / 02037	2023-05-31	ALL
Searchlight System - Operational Check	DMC-412-A-96-02-00-04A-320A-A / 02038	2023-05-31	ALL
Landing Light System - Description	DMC-412-A-96-02-00-05A-042A-A / 02039	2023-05-31	ALL
Landing Light System - Operational Check	DMC-412-A-96-02-00-05A-320A-A / 02040	2023-05-31	ALL

Document title	Data module code / Seq number	Issue date	Applicable to
Caution and Warning System - Description	DMC-412-A-96-03-00-00A-028A-A / 02041	2023-05-31	ALL
Caution and Warning System - Operation	DMC-412-A-96-03-00-00A-042A-A / 02042	2023-05-31	ALL
Master Caution Light System - Operational Test	DMC-412-A-96-03-00-00A-320A-A / 02043	2023-05-31	ALL
Caution Panel Light System - Operational Check	DMC-412-A-96-03-00-00A-320B-A / 02044	2023-05-31	ALL
DC Generator Caution Light Systems - Operational Check	DMC-412-A-96-03-01-00A-320A-A / 02045	2023-05-31	ALL
Generator No. 1 and No. 2 Overheat Caution Light Systems - Operational Check	DMC-412-A-96-03-01-01A-320A-A / 02046	2023-05-31	ALL
Inverter No. 1 and No. 2 Caution Light Systems - Operational Check	DMC-412-A-96-03-01-02A-320A-A / 02047	2023-05-31	ALL
Engine Oil Pressure Caution Light Systems - Operational Check	DMC-412-A-96-03-01-03A-320A-A / 02048	2023-05-31	ALL
Transmission Oil Pressure Caution Light System - Operational Check	DMC-412-A-96-03-01-04A-320A-A / 02049	2023-05-31	ALL
Transmission Oil Temperature Caution Light Systems - Operational Check	DMC-412-A-96-03-01-05A-320A-A / 02050	2023-05-31	ALL
Combining Gearbox Oil Pressure Caution Light System - Operational Check	DMC-412-A-96-03-01-06A-320A-A / 02051	2023-05-31	ALL
Combining Gearbox Oil Temperature Caution Light Systems - Operational Check	DMC-412-A-96-03-01-07A-320A-A / 02052	2023-05-31	ALL
Particle Separator Caution Light System - Operational Check	DMC-412-A-96-03-01-08A-320A-A / 02053	2023-05-31	ALL
Governor Manual Control Caution Light Systems - Operational Check	DMC-412-A-96-03-01-09A-320A-A / 02054	2023-05-31	ALL
Engine No. 1 Chip Detector Caution Light Circuitry - Operational Check	DMC-412-A-96-03-01-10A-320B-A / 02055	2023-05-31	ALL
Engine No. 2 Chip Detector Caution Light Circuitry - Operational Check	DMC-412-A-96-03-01-10A-320C-A / 02056	2023-05-31	ALL
Combining Gearbox Chip Detector Caution Light Systems - Operational Check	DMC-412-A-96-03-01-11A-320A-A / 02057	2023-05-31	ALL
Transmission Chip Detector Caution Light System - Operational Check	DMC-412-A-96-03-01-12A-320A-A / 02058	2023-05-31	ALL
42 degree/90 degree Gearbox Chip Detector Caution Light Systems - Operational Check	DMC-412-A-96-03-01-13A-320A-A / 02059	2023-05-31	ALL
Door Lock Caution Light System - Operational Check	DMC-412-A-96-03-01-14A-320A-A / 02060	2023-05-31	ALL
Door Lock Switches/Switch Levers - Adjustment Procedure	DMC-412-A-96-03-01-14A-271A-A / 02061	2023-05-31	ALL
Fuel Boost Caution Light System - Operational Check	DMC-412-A-96-03-01-15A-320A-A / 02062	2023-05-31	SN: 33001-33100
Fuel Boost Caution Light System - Operational Check	DMC-412-A-96-03-01-15A-320B-A / 02063	2023-05-31	SN: 33101-33107
Fuel Boost Caution Light System - Operational Check	DMC-412-A-96-03-01-15A-320C-A / 02064	2023-05-31	SN: 33108-33213 or 36001-36999
Fuel Filter Caution Light System - Operational Check	DMC-412-A-96-03-01-16A-320A-A / 02065	2023-05-31	ALL

Document title	Data module code / Seq number	Issue date	Applicable to
Procedure for Low Fuel Level (10% or Less) - Operational Check	DMC-412-A-96-03-01-17A-320B-A / 02066	2023-05-31	SN: 33001-33100
Procedure for Fuel in Tanks (10% or More) - Operational Check	DMC-412-A-96-03-01-17A-320C-A / 02067	2023-05-31	SN: 33001-33100
Procedure for Low Fuel Level (10% or Less) - Operational Check	DMC-412-A-96-03-01-17B-320B-A / 02068	2023-05-31	SN: 33101-33107
Procedure for Fuel in Tanks (10% or More) - Operational Check	DMC-412-A-96-03-01-17B-320C-A / 02069	2023-05-31	SN: 33101-33107
Fuel Low Caution Light Systems - Operational Check	DMC-412-A-96-03-01-17C-320A-A / 02070	2023-05-31	SN: 33108-33213 or 36001-36999
Engine No. 1 and No. 2 Fuel Valve Caution Light Systems - Operational Check	DMC-412-A-96-03-01-18A-320A-A / 02071	2023-05-31	ALL
No. 1 and No. 2 Fuel Transfer Caution Light Systems - Operational Check	DMC-412-A-96-03-01-19A-320A-A / 02072	2023-05-31	ALL
Fuel Interconnect Caution Light Systems - Operational Check	DMC-412-A-96-03-01-20A-320A-A / 02073	2023-05-31	ALL
Fuel Crossfeed Valve Caution Light System - Operational Check	DMC-412-A-96-03-01-21A-320A-A / 02074	2023-05-31	SN: 33001-33107
Fuel Crossfeed Valve Caution Light System - Operational Check	DMC-412-A-96-03-01-21A-320B-A / 02075	2023-05-31	SN: 33108-33213 or 36001-36999
Hydraulic Caution Light System - Operational Check	DMC-412-A-96-03-01-22A-320A-A / 02076	2023-05-31	ALL
External Power Caution Light System - Operational Check	DMC-412-A-96-03-01-23A-320A-A / 02077	2023-05-31	ALL
Battery Sensor Temperature Caution Light System - Operational Check	DMC-412-A-96-03-01-24A-320A-A / 02078	2023-05-31	SN: 33001-33120
Battery Sensor Temperature Caution Light System - Test Procedure	DMC-412-A-96-03-01-00A-320B-A / 02079	2023-05-31	SN: 33001-33120
Battery Sensor Temperature Caution Light System - Operational Check	DMC-412-A-96-03-01-24B-320A-A / 02080	2023-05-31	SN: 33121-33213 or 36001-36999
Battery Caution Light System - Operational Check	DMC-412-A-96-03-01-25A-320A-A / 02081	2023-05-31	ALL
Heater Air Line Caution Light System - Operational Check	DMC-412-A-96-03-01-26A-320A-A / 02082	2023-05-31	SN: 33001-33107
Heater Air Line Caution Light System - Operational Check	DMC-412-A-96-03-01-27A-320B-A / 02083	2023-05-31	SN: 33108-33213 or 36001-36999
AFCS Caution Light Systems - Operational Check	DMC-412-A-96-03-01-28A-320A-A / 02084	2023-05-31	SN: 33001-33213 or 36001-36086
No. 1 Autopilot and No. 2 Autopilot Caution Light Systems - Operational Check	DMC-412-A-96-03-01-29A-320A-A / 02085	2023-05-31	SN: 36087-36999
Autotrim Caution Light System - Operational Check	DMC-412-A-96-03-01-30A-320A-A / 02086	2023-05-31	SN: 36087-36999

Document title	Data module code / Seq number	Issue date	Applicable to
Rotor Brake Caution Light Systems - Operational Check	DMC-412-A-96-03-01-31A-320A-A / 02087	2023-05-31	ALL
Caution Panel Caution Light Systems - Operational Check	DMC-412-A-96-03-01-32A-320A-A / 02088	2023-05-31	ALL
Flight Data Recorder Caution Light System - Description	DMC-412-A-96-03-02-00A-042A-A / 02089	2023-05-31	SN: 36037-36999
Flight Data Recorder Caution Systems - Operational Check	DMC-412-A-96-03-02-00A-320A-A / 02090	2023-05-31	SN: 36037-36999
RPM Limit Warning Systems - Description	DMC-412-A-96-03-03-00A-042A-A / 02091	2023-05-31	ALL
RPM Limit Warning System - Operational Check (Without Engines Operating)	DMC-412-A-96-03-03-01A-320A-A / 02092	2023-05-31	ALL
RPM Limit Warning System - Operational Check and Alignment (With Engines Operating)	DMC-412-A-96-03-03-01A-320B-A / 02093	2023-05-31	ALL
Engine Low RPM Warning Check and Alignment - Operational Check	DMC-412-A-96-03-03-02A-320A-A / 02094	2023-05-31	ALL
RPM Limit Detector Repair and Bench Check - Operational Check	DMC-412-A-96-03-03-02A-320B-A / 02095	2023-05-31	ALL
Engine No. 1 Fire Detection Warning - Operational Check	DMC-412-A-96-03-03-04A-320A-A / 02096	2023-05-31	ALL
Engine No. 2 Fire Detection Warning - Operational Check	DMC-412-A-96-03-03-04A-320B-A / 02097	2023-05-31	ALL
Baggage Compartment Fire Detector System - Operational Check	DMC-412-A-96-03-03-04A-320C-A / 02098	2023-05-31	ALL
Hydraulic Filter Clogged Warning System - Description	DMC-412-A-96-03-03-05A-042A-A / 02099	2023-05-31	ALL
Hydraulic Filter Clogged Warning System - Operational Check	DMC-412-A-96-03-03-05A-320A-A / 02100	2023-05-31	ALL
Engine Control and Accessories Systems - Description	DMC-412-A-96-04-00-00A-028A-A / 02101	2023-05-31	ALL
Fuel Control Heater - Description	DMC-412-A-96-04-00-01A-042A-A / 02102	2023-05-31	ALL
Fuel Control Heater - Operational Check	DMC-412-A-96-04-00-01A-320A-A / 02103	2023-05-31	ALL
Fuel Boost Pump System - Description	DMC-412-A-96-04-00-02A-042A-A / 02104	2023-05-31	ALL
Engine 1 - Operational Check	DMC-412-A-96-04-00-02A-320B-A / 02105	2023-05-31	ALL
Engine 2 - Operational Check	DMC-412-A-96-04-00-02A-320C-A / 02106	2023-05-31	ALL
Fuel Transfer Pump Systems - Description	DMC-412-A-96-04-00-03A-042A-A / 02107	2023-05-31	ALL
Engine 1 - Operational Check	DMC-412-A-96-04-00-03A-320B-A / 02108	2023-05-31	ALL
Engine 2 - Operational Check	DMC-412-A-96-04-00-03A-320C-A / 02109	2023-05-31	ALL
Fuel Shutoff Valve Systems - Description	DMC-412-A-96-04-00-04A-042A-A / 02110	2023-05-31	ALL
Engine 1 - Operational Check	DMC-412-A-96-04-00-04A-320B-A / 02111	2023-05-31	ALL

Document title	Data module code / Seq number		Issue date	Applicable to
Engine 2 - Operational Check	DMC-412-A-96-04-00-04A-320C-A / 02112		2023-05-31	ALL
Fuel Control Valve System - Description	DMC-412-A-96-04-00-05A-042A-A / 02113		2023-05-31	ALL
Engine 1 - Operational Check	DMC-412-A-96-04-00-05A-320B-A / 02114		2023-05-31	ALL
Engine 2 - Operational Check	DMC-412-A-96-04-00-05A-320C-A / 02115		2023-05-31	ALL
Fuel Interconnect Valve System - Description	DMC-412-A-96-04-00-06A-042A-A / 02116		2023-05-31	ALL
Fuel Interconnect Valve System - Operational Check	DMC-412-A-96-04-00-06A-320A-A / 02117		2023-05-31	ALL
Fuel Crossfeed Valve System - Description	DMC-412-A-96-04-00-07A-042A-A / 02118		2023-05-31	ALL
Fuel Crossfeed Valve System - Operational Check	DMC-412-A-96-04-00-07A-320A-A / 02119		2023-05-31	ALL
Governor RPM Actuator System - Description	DMC-412-A-96-04-00-08A-042A-A / 02120	C	2025-11-18	SN: 33001-Subsequent or 36001-36019 or 412-570-001-103: Not Installed or (BHT-412-SI-74: Not Installed BHT-412-SI-97: Not Installed)
Governor RPM Actuator System - Operational Check	DMC-412-A-96-04-00-08A-320A-A / 02121	C	2025-11-18	SN: 33001-Subsequent or 36001-36019 or 412-570-001-103: Not Installed or (BHT-412-SI-74: Not Installed BHT-412-SI-97: Not Installed)
Governor RPM Actuator System - Description	DMC-412-A-96-04-00-08A-042B-A / 02122	C	2025-11-18	SN: 36020-36999 or 412-570-001-103: Installed or BHT-412-SI-74: Installed or BHT-412-SI-97: Installed

Document title	Data module code / Seq number		Issue date	Applicable to
Governor RPM Actuator System - Operational Check	DMC-412-A-96-04-00-08A-320B-A / 02123	C	2025-11-18	SN: 36020-36999 or 412-570-001-103: Installed or BHT-412-SI-74: Installed or BHT-412-SI-97: Installed
Sump Drain System - Description	DMC-412-A-96-04-00-09A-042A-A / 02124		2023-05-31	ALL
Sump Drain System - Operational Check	DMC-412-A-96-04-00-09A-320A-A / 02125		2023-05-31	ALL
Idle Stop Release Solenoid System - Description	DMC-412-A-96-04-00-10A-042A-A / 02126		2023-05-31	ALL
Idle Stop Release Solenoid System - Operational Check	DMC-412-A-96-04-00-10A-320A-A / 02127		2023-05-31	ALL
Engine Particle Separator System - Description	DMC-412-A-96-04-00-11A-042A-A / 02128		2023-05-31	ALL
Engine Particle Separator System - Operational Check	DMC-412-A-96-04-00-11A-320A-A / 02129		2023-05-31	ALL
Flight Control Systems - Description	DMC-412-A-96-05-00-00A-028A-A / 02130		2023-05-31	ALL
Force Trim System - Description	DMC-412-A-96-05-00-01A-042A-A / 02131		2023-05-31	ALL
Force Trim System - Operational Check	DMC-412-A-96-05-00-01A-320A-A / 02132		2023-05-31	ALL
Hydraulic Control Systems - Description	DMC-412-A-96-05-00-02A-042A-A / 02133		2023-05-31	ALL
Hydraulic Control Systems - Operational Check	DMC-412-A-96-05-00-02A-320A-A / 02134		2023-05-31	SN: 33001-33107
Hydraulic Control Systems - Systems Check	DMC-412-A-96-05-00-02A-320B-A / 02135		2023-05-31	SN: 33001-33107
Hydraulic Control Systems - Operational Check	DMC-412-A-96-05-00-02A-320C-A / 02136		2023-05-31	SN: 33108-33213 or 36001-36999
Hydraulic Control Systems - Clogged Filter Check	DMC-412-A-96-05-00-02A-320D-A / 02137		2023-05-31	SN: 33108-33213 or 36001-36999
Hydraulic Control Systems - Systems Check	DMC-412-A-96-05-00-02A-320E-A / 02138		2023-05-31	SN: 33108-33213 or 36001-36999
Heating Systems - Description	DMC-412-A-96-06-00-00A-028A-A / 02139		2023-05-31	ALL
Cabin Heater System - Description	DMC-412-A-96-06-00-01A-042A-A / 02140		2023-05-31	ALL

Document title	Data module code / Seq number	Issue date	Applicable to
Cabin Heater System - Operational Check	DMC-412-A-96-06-00-01A-320A-A / 02141	2023-05-31	SN: 33001-33213 or 36001-36999
Cabin Heater System - Operational Check	DMC-412-A-96-06-00-01A-320B-A / 02142	2023-05-31	SN: 37002-37999
Cabin Temperature Control - Rigging	DMC-412-A-96-06-00-01A-276A-A / 02143	2023-05-31	ALL
Cabin Temperature Control - Operational Check	DMC-412-A-96-06-00-01A-320C-A / 02144	2023-05-31	ALL
Vent Blower System - Description	DMC-412-A-96-06-00-02A-042A-A / 02145	2023-05-31	ALL
Vent Blower System - Operational Check	DMC-412-A-96-06-00-02A-320A-A / 02146	2023-05-31	ALL
Miscellaneous Systems - Description	DMC-412-A-96-07-00-00A-028A-A / 02147	2023-05-31	ALL
Windshield Wiper Systems - Description	DMC-412-A-96-07-00-01A-042A-A / 02148	2023-05-31	ALL
Windshield Wiper Systems - Operational Check	DMC-412-A-96-07-00-01A-320A-A / 02149	2023-05-31	ALL
Engine Fire Extinguisher Systems - Description	DMC-412-A-96-07-00-02A-042A-A / 02150	2023-05-31	ALL
Engine Fire Extinguisher Systems - Operational Check	DMC-412-A-96-07-00-02A-320A-A / 02151	2023-05-31	ALL
Weight-On-Gear Switch - Description	DMC-412-A-96-07-00-03A-042A-A / 02152	2023-05-31	SN: 36462-36999
Weight-On-Gear Annunciator - Description	DMC-412-A-96-07-00-03A-042B-A / 02153	2023-05-31	SN: 36528-36999
Weight-On-Gear Switch - Removal	DMC-412-A-96-07-00-03A-520A-A / 02154	2023-05-31	SN: 36462-36999
Weight-On-Gear - Installation	DMC-412-A-96-07-00-03A-720A-A / 02155	2023-05-31	SN: 36462-36999
Weight-On-Gear Switch - Adjustment	DMC-412-A-96-07-00-03A-271A-A / 02156	2023-05-31	SN: 36462-36999
Powered Passenger Step System (Kit) - Description	DMC-412-A-96-07-00-04A-042A-A / 02157	2023-05-31	ALL
Powered Passenger Step System - Operational Check	DMC-412-A-96-07-00-04A-320A-A / 02158	2023-05-31	ALL
Emergency Float (Kit) - Description	DMC-412-A-96-07-00-05A-042A-A / 02159	2023-05-31	ALL
Emergency Float - Operational Check	DMC-412-A-96-07-00-05A-320A-A / 02160	2023-05-31	ALL
Heated Windshield Kit (412-706-010/-138) - Description	DMC-412-A-96-07-00-06A-042A-A / 02161	2023-05-31	ALL
Controller Unit - Removal	DMC-412-A-96-07-00-06A-520A-A / 02162	2023-05-31	ALL
Controller Unit - Cleaning	DMC-412-A-96-07-00-06A-251A-A / 02163	2023-05-31	ALL
Controller Unit - Inspection	DMC-412-A-96-07-00-06A-280A-A / 02164	2023-05-31	ALL
Controller Unit - Installation	DMC-412-A-96-07-00-06A-720A-A / 02165	2023-05-31	ALL

Document title	Data module code / Seq number	Issue date	Applicable to
Voice/Flight Data Recorder System - Description	DMC-412-A-96-08-00-00A-028A-A / 02166	2023-05-31	ALL
Flight Data Recorder (FDR) System - Description	DMC-412-A-96-08-00-01A-042A-A / 02167	2023-05-31	SN: 36037-36333
Flight Data Recorder (FDR) System - Retrieval of Data	DMC-412-A-96-08-00-01A-131A-A / 02168	2023-05-31	SN: 36037-36333
Cockpit Voice Recorder (CVR) - Description	DMC-412-A-96-08-00-02A-042A-A / 02169	2023-05-31	SN: 36037-36333
Cockpit Voice Recorder (CVR) - Operational Check	DMC-412-A-96-08-00-02A-320A-A / 02170	2023-05-31	SN: 36037-36333
Integrated Flight Control System (IFCS) - Description	DMC-412-A-96-09-00-00A-028A-A / 02171	2023-05-31	ALL
Digital Automatic Flight Control System (DAFCS) - Description	DMC-412-A-96-10-00-00A-028A-A / 02172	2023-05-31	SN: 36087-36999
Attitude (ATT) Retention Mode - Description	DMC-412-A-96-10-00-00A-042A-A / 02173	2023-05-31	SN: 36087-36999
Stability Augmentation System (SAS) Mode - Description	DMC-412-A-96-10-00-00A-042B-A / 02174	2023-05-31	SN: 36087-36999
Digital Flight Control Computer - Description	DMC-412-A-96-10-00-01A-042A-A / 02175	2023-05-31	SN: 36087-36999
Autopilot Controller - Description	DMC-412-A-96-10-00-02A-042A-A / 02176	2023-05-31	SN: 36087-36999
Flight Director Mode Selector - Description	DMC-412-A-96-10-00-03A-042A-A / 02177	2023-05-31	SN: 36087-36999
Air Data Command Display - Description	DMC-412-A-96-10-00-04A-042A-A / 02178	2023-05-31	SN: 36087-36999
Rate Gyro and AHRS - Description	DMC-412-A-96-10-00-05A-042A-A / 02179	2023-05-31	SN: 36087-36247 or 36248-36999
Control Rod Assemblies - Description	DMC-412-A-96-10-00-06A-042A-A / 02180	2023-05-31	SN: 36087-36999
AFCS Indicator Panel - Description	DMC-412-A-96-10-00-07A-042A-A / 02181	2023-05-31	SN: 36087-36999
Air Data Sensor - Description	DMC-412-A-96-10-00-08A-042A-A / 02182	2023-05-31	SN: 36087-36999
System Operation - Description	DMC-412-A-96-10-00-09A-042A-A / 02183	2023-05-31	SN: 36087-36999
Self-Test - Description	DMC-412-A-96-10-00-12A-042A-A / 02184	2023-05-31	SN: 36087-36999
Flight Director Control Functions - Description	DMC-412-A-96-10-00-13A-042A-A / 02185	2023-05-31	SN: 36087-36999
Flight Director Mode Flow - Description	DMC-412-A-96-10-00-14A-042A-A / 02186	2023-05-31	SN: 36087-36999
Autopilot - Operational Description	DMC-412-A-96-10-00-10A-042A-A / 02187	2023-05-31	SN: 36087-36999
Autopilot - Functional Description	DMC-412-A-96-10-00-11A-042A-A / 02188	2023-05-31	SN: 36087-36999
DAFCS - Functional Test	DMC-412-A-96-10-00-15A-340J-A / 02189	2023-05-31	SN: 36087-36999
Preliminary Test Procedures - Functional Test	DMC-412-A-96-10-00-15A-340A-A / 02190	2023-05-31	SN: 36087-36999

Document title	Data module code / Seq number	Issue date	Applicable to
Lamp Test - Functional Test	DMC-412-A-96-10-00-15A-340B-A / 02191	2023-05-31	SN: 36087-36999
Force Trim Test - Functional Test	DMC-412-A-96-10-00-15A-340C-A / 02192	2023-05-31	SN: 36087-36999
Autopilot Engage Test - Functional Test	DMC-412-A-96-10-00-15A-340D-A / 02193	2023-05-31	SN: 36087-36999
ATT Gain Polarity Test - Functional Test	DMC-412-A-96-10-00-15A-340E-A / 02194	2023-05-31	SN: 36087-36999
ATT Mode Test - Functional Test	DMC-412-A-96-10-00-15A-340F-A / 02195	2023-05-31	SN: 36087-36999
Yaw Axis Test - Functional Test	DMC-412-A-96-10-00-15A-340G-A / 02196	2023-05-31	SN: 36087-36999
SAS Mode Test - Functional Test	DMC-412-A-96-10-00-15A-340H-A / 02197	2023-05-31	SN: 36087-36999
Flight Director-Logic Test - Operational Check	DMC-412-A-96-10-00-16A-320B-A / 02198	2023-05-31	SN: 36087-36999
Heading Mode Test - Operational Check	DMC-412-A-96-10-00-16A-320C-A / 02199	2023-05-31	SN: 36087-36999
VOR Mode Test - Operational Check	DMC-412-A-96-10-00-16A-320D-A / 02200	2023-05-31	SN: 36087-36999
ILS Mode Test - Operational Check	DMC-412-A-96-10-00-16A-320E-A / 02201	2023-05-31	SN: 36087-36999
VS Select Mode Test - Operational Check	DMC-412-A-96-10-00-16A-320F-A / 02202	2023-05-31	SN: 36087-36999
IAS Hold Mode Test - Operational Check	DMC-412-A-96-10-00-16A-320G-A / 02203	2023-05-31	SN: 36087-36999
ALT Hold Test - Operational Check	DMC-412-A-96-10-00-16A-320H-A / 02204	2023-05-31	SN: 36087-36999
Go Around Mode Test - Operational Check	DMC-412-A-96-10-00-16A-320J-A / 02205	2023-05-31	SN: 36087-36999
Flight Director Coupling Test (Pitch and Roll) - Operational Check	DMC-412-A-96-10-00-16A-320K-A / 02206	2023-05-31	SN: 36087-36999
Autotrim Test - Operational Check	DMC-412-A-96-10-00-16A-320L-A / 02207	2023-05-31	SN: 36087-36999
Airspeed System - Adjustment	DMC-412-A-96-10-00-17A-271A-A / 02208	2023-05-31	SN: 36087-36999
DAFCS - Troubleshooting	DMC-412-A-96-10-00-18A-422A-A / 02209	2023-05-31	SN: 36087-36999
Autopilot Controller - Removal	DMC-412-A-96-10-00-19A-520B-A / 02210	2023-05-31	SN: 36087-36999
Air Data Command Display (Altitude Preselector) - Removal	DMC-412-A-96-10-00-19A-520C-A / 02211	2023-05-31	SN: 36087-36999
Actuator Position Indicator Panel - Removal	DMC-412-A-96-10-00-19A-520D-A / 02212	2023-05-31	SN: 36087-36999
Digital Flight Control Computer (No. 1 or No. 2) - Removal	DMC-412-A-96-10-00-19A-520E-A / 02213	2023-05-31	SN: 36087-36999
Flight Director Mode Selector (No. 1 or No. 2) - Removal	DMC-412-A-96-10-00-19A-520F-A / 02214	2023-05-31	SN: 36087-36999
Air Data Sensor (No. 1 or No. 2) - Removal	DMC-412-A-96-10-00-19A-520G-A / 02215	2023-05-31	SN: 36087-36999

Document title	Data module code / Seq number	Issue date	Applicable to
Rate Gyro (No. 1 or No. 2) - Removal	DMC-412-A-96-10-00-19A-520H-A / 02216	2023-05-31	SN: 36087-36247
Relay Box - Removal	DMC-412-A-96-10-00-19A-520J-A / 02217	2023-05-31	SN: 36087-36999
Relay Box Assembly - Repair	DMC-412-A-96-10-00-20A-685A-A / 02218	2023-05-31	SN: 36087-36999
Autopilot Controller - Installation	DMC-412-A-96-10-00-20A-720A-A / 02218.1	2024-05-09	SN: 36087-36999
Air Data Command Display (Altitude Preselector) - Installation	DMC-412-A-96-10-00-20A-720B-A / 02218.2	2024-05-09	SN: 36087-36999
Actuator Position Indicator Panel - Installation	DMC-412-A-96-10-00-20A-720C-A / 02218.3	2024-05-09	SN: 36087-36999
Digital Flight Control Computer (No.1 or No.2) - Installation	DMC-412-A-96-10-00-20A-720D-A / 02218.4	2024-05-09	SN: 36087-36999
Flight Director Mode Selector (No. 1 or No. 2) - Installation	DMC-412-A-96-10-00-20A-720E-A / 02218.5	2024-05-09	SN: 36087-36999
Air Data Sensor (No. 1 or No. 2) - Installation	DMC-412-A-96-10-00-20A-720F-A / 02218.6	2024-05-09	SN: 36087-36999
Rate Gyro (No. 1 or No. 2) - Installation	DMC-412-A-96-10-00-20A-720G-A / 02218.7	2024-05-09	SN: 36087-36247
Relay Box - Installation	DMC-412-A-96-10-00-20A-720H-A / 02218.8	2024-05-09	SN: 36087-36999
Digital Automatic Flight Control System (DAFCS) - Inspection Procedures	DMC-412-A-96-10-00-00A-280A-A / 02219	2023-05-31	SN: 36087-36999
Digital Automatic Flight Control System (DAFCS) - Cleaning Procedures	DMC-412-A-96-10-00-00A-251A-A / 02220	2023-05-31	SN: 36087-36999
Digital Automatic Flight Control System (DAFCS) - Repair or Replacement	DMC-412-A-96-10-00-00A-685A-A / 02221	2023-05-31	SN: 36087-36999
Electronic Flight INstrument System (EFIS) - Description	DMC-412-A-96-13-00-00A-028A-A / 02222	2023-05-31	ALL
EFIS - Operational Check	DMC-412-A-96-13-00-01A-320A-A / 02223	2023-05-31	ALL
EFIS - Troubleshooting	DMC-412-A-96-13-00-01A-422A-A / 02224	2023-05-31	ALL
EFIS - Removal	DMC-412-A-96-13-00-01A-520A-A / 02225	2023-05-31	ALL
EFIS - Installation	DMC-412-A-96-13-00-01A-720A-A / 02226	2023-05-31	ALL
EFIS - Inspection Procedures	DMC-412-A-96-13-00-01A-280A-A / 02227	2023-05-31	ALL
EFIS - Cleaning Procedures	DMC-412-A-96-13-00-01A-251A-A / 02228	2023-05-31	ALL
EFIS - Repair or Replacement	DMC-412-A-96-13-00-01A-685A-A / 02229	2023-05-31	ALL
EFIS Cooling System - Description	DMC-412-A-96-13-00-02A-042A-A / 02230	2023-05-31	ALL
EFIS Cooling System - Operational Check	DMC-412-A-96-13-00-02A-320A-A / 02231	2023-05-31	ALL
EFIS Cooling System - Removal	DMC-412-A-96-13-00-02A-520A-A / 02232	2023-05-31	ALL

Document title	Data module code / Seq number	Issue date	Applicable to
EFIS Cooling System - Installation	DMC-412-A-96-13-00-02A-720A-A / 02233	2023-05-31	ALL
Electrical Fault Analysis - Description	DMC-412-A-96-14-00-00A-042A-A / 02234	2023-05-31	ALL

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Electrical System - General Description

This chapter provides the data necessary to understand the operation of the electrical systems and to do the necessary maintenance actions as follows:

- Electrical safety and hazard procedures
- Removal and installation procedures for the electrical components
- Inspection procedures for the electrical components
- Operational checks, functional checks, and troubleshooting charts for the electrical systems and components

For specific electrical wiring maintenance information and procedures, refer to the BHT-ELEC-SPM, Electrical Standard Practices Manual. This manual gives the data necessary to repair and replace the wires, cables, or the electrical components to the standards of performance and quality as specified by the manufacturer.

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Electrical Systems - Description

Maintenance of electrical circuitry, electrical equipment, and electrical components installed on the Model 412 helicopter is as follows:

Control panels for AC and DC electrical systems are located on the overhead console. All extended range fuel system circuit breakers referred to in this chapter are located in circuit breaker panels mounted on overhead console. Control relays, power relays, voltage regulators, and other equipment required to control, regulate, and effect power transfer and malfunction monitoring are located in respective compartments. For equipment locations, refer to 412-A-96-00-02-04A-042A-A / 01904. Refer to 412-A-98-00-00-00A-028A-A / 02455 for wiring diagrams.

Do operational checks to ensure no new malfunctions have been introduced when equipment is replaced and/or airframe wiring is repaired/replaced. When you do operational checks, external power should be used whenever possible. Using battery power may deplete the battery.

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Primary and Secondary Electrical Power Systems - Description

The Model 412 helicopter electrical power system provides 28 volts DC, 115 volts AC, and 26 volts AC for helicopter electrical equipment, avionics systems, and instruments. It also provides electrical engine starting capability and emergency battery power.

Primary 28 VDC electrical power is supplied by two 30 volt, 200 amp starter-generators, one mounted on each engine. Output voltage of each generator is monitored and regulated by a DC control unit. DC control units provide overvoltage and reverse current protection and control paralleled generator operation so the starter-generators share total load requirements within 20 amps. Each generator supplies power to a main DC bus and two interconnected non-essential buses. Each main DC bus, in turn, serves as a feeder for the two essential DC buses and two emergency buses.

Electrical separation between main buses and between starter-generators is accomplished through the use of circuit breakers and isolation diodes. If one starter-generator or engine should fail, both non-essential buses are automatically dropped and all essential and emergency loads are supplied by the remaining starter-generator. An override switch is installed so the pilot may manually restore power to the non-essential buses.

This arrangement allows automatic DC load shedding for 30 minutes powered by the battery. Should a flight of longer duration be required on battery only, the pilot may set the NORMAL EMERG LOAD switch to the EMERG LOAD position. This provides load shedding for a 76-minute flight.

Secondary power is supplied by two identical 250 volt-amp or 450 volt-amp, 115/26 volt, 400 Hz, single phase, solid state inverters. One inverter is energized by each essential DC bus. Inverter No. 1 supplies power to the 115 VAC and 26 VAC buses 1 and 2, and Inverter No. 2 supplies power to the 115 VAC and 26 VAC bus 3. The inverters are completely independent of each other. If an inverter fails, all AC loads are energized by the remaining inverter.

26 VAC and 115 VAC emergency power is supplied by inverters No. 1 and No. 2 when the inverter switches are in the ON position.

Starting power is normally supplied by a 24 volt battery. An external power receptacle located in the nose of the helicopter allows an external power source to be used for engine starting and ground check operations.

2. Emergency DC Power

If a starter-generator or engine fails, the remaining starter-generator can supply all essential DC loads. The battery provides a second source of emergency power if both starter-generators fail.

3. Emergency AC Power

If an inverter fails, all AC loads are powered by the remaining inverter.

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Electrical Safety and Hazards - Description



OBEY ALL THE SAFETY PRECAUTIONS WHEN YOU DO MAINTENANCE ON OR NEAR ELECTRICAL/ELECTRONIC EQUIPMENT.

High voltages in the electrical and electronic systems are hazardous to personnel. Electrical shocks may cause serious injuries. The generation of sparks in the presence of flammable vapors may cause explosions. Metallic objects that come in contact with the powered electrical conductors or the components can cause injuries to personnel and damage to the components.

1.1. Safety Practices - General



OBEY ALL THE SAFETY PRECAUTIONS WHEN YOU DO MAINTENANCE ON OR NEAR ELECTRICAL/ELECTRONIC EQUIPMENT.

When you do any maintenance or repairs to an electrical system, or to any system that requires electrical power, follow these precautions:

- Personnel must be trained for the specific task.
- Do not touch components that have electrical power. Direct contact with a high voltage circuit, or lower voltage circuit with a high current capability can be fatal.
- Proper tools and safety equipment must be available. Use them to ensure the safety of personnel. Use them to make sure the given task is done to the specified standards.
- Do not wear jewelry when you work on a system that is powered.
- Use recommended test equipment or their equivalents. Follow the equipment operation instructions. Additional test equipment, lights, etc., may be used as desired, provided the purpose and intent of the specified test is not compromised.
- The leads and the probes of the test equipment must be serviceable. They must be of a sufficient length to put the test equipment at a proper location. The location of the test equipment must give a full view of the indicator. Make sure that there is no tension on the test leads or the probes.
- Give special attention to the WARNINGS, CAUTIONS and NOTES before a maintenance procedure is initiated. Some maintenance procedures require the use of toxic chemicals, extreme temperatures, and flammable materials.

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Applicable to: ALL
ECCN EAR99

Standard Practices - Electrical Maintenance or Repairs

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Caps or plugs	C-136	AR	
Caps or plugs	C-156	AR	

Procedure



UNLESS POWER IS REQUIRED FOR A MAINTENANCE PROCEDURE, DISCONNECT THE BATTERY AND THE EXTERNAL POWER. PUT A TAG ON THE BATTERY SWITCH TO INFORM OTHER PERSONNEL THAT WORK IS BEING DONE.



INFORM ALL THE PERSONNEL THAT WORK ON THE HELICOPTER BEFORE YOU SUPPLY POWER TO THE ELECTRICAL SYSTEMS.

1. The procedure that follows will help to avoid injuries to personnel and damage to equipment.

1.1. If an electrical connector is to be left disconnected, install protective caps and/or plugs to protect the contacts from shorts and prevent FOD damage. Use either caps or plugs (C-136) for Electrostatic Discharge Sensitive (ESDS) equipment, or caps or plugs (C-156) for other non-ESDS sensitive equipment.

1.2. If a circuit breaker must remain open until a maintenance procedure is complete, install a clip or a plastic cable strap under the circuit breaker button. Put a tag on the circuit breaker to inform other personnel not to set the circuit breaker to the closed position.

1.3. If a switch must remain open or closed until a maintenance procedure is complete, put a tag on the switch to inform other personnel not to touch the switch.

1.4. Inform all of the personnel that work on the helicopter before you supply power to the electrical systems.

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Standard Practices - Electrical Work In Confined Spaces

Procedure

1. Some equipment installations are in confined areas and access to the test points can be difficult. When you do a test on a powered circuit in a confined area, follow these precautions:

1.1. Have the applicable fire safety equipment available when you do a maintenance procedure.

1.2. Before you start work in a confined area, look at the wiring diagrams (412-A-98-00-00-00A-028A-A / 02455) of the applicable electrical systems and at the location of the equipment you will test. Make a note of the location of the high voltage areas near the equipment you will test.

1.3. Be careful around any control cable or mechanism. Movement of engine or flight controls can cause movement of cables or control rods.

1.4. For systems not tested, set the applicable circuit breakers to the open position.

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Standard Practices - After Electrical Maintenance or Repairs

Procedure

1. After completing a task on any electrical system, make sure that you do the steps that follow before you return the helicopter to the flight configuration:

1.1. Remove all protective caps from equipment or electrical equipment.

1.2. Make sure all of the disconnected electrical connectors are properly reinstalled.

1.3. After troubleshooting any electrical system, reinstall any removed control panels or electrical components such as relays, terminal modules, switches, and other miscellaneous components.

1.4. Remove the tags or the cable straps from the switches and the circuit breakers. Close the circuit breakers.

NOTE

As applicable, refer to the engine shutdown procedures (BHT-412-FM, Section 2) to determine the switch and control positions.

1.5. Make sure that all the switches and the controls are in the correct position. This prevents accidental system operation when the power is turned on.

1.6. If applicable, connect the battery.

1.7. Do an operational check on the system or systems that were repaired.

1.8. Install all the applicable access panels and close all the doors (412-A-52-40-00-02A-720A-A / 00840).

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Miscellaneous Electrical Tools - Description

The majority of tools required to do basic electrical maintenance and repair practices are listed in the BHT-ELEC-SPM, Electrical Standard Practices Manual.

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Electrical Systems - Operational Checks and Functional Tests

Procedure

1. When you do operational checks, external power should be used whenever possible. Doing operational checks on battery power may deplete the charge in the battery.

2. Do operational checks to make sure that no new malfunctions have been introduced when components have been repaired or replaced, and/or airframe wiring has been repaired or replaced.

3. Operational checks for each applicable system are located with the system maintenance procedures. Also, use the system diagrams and wiring diagrams (412-A-98-00-00-00A-028A-A / 02455) when you do operational checks or functional tests.

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Electrical Systems - Troubleshooting

Procedure

1. The following troubleshooting procedures are general and may be used as a guide to find a malfunction in an electrical system. Make sure that the system and component operation is understood before you do any troubleshooting.

1.1. Make sure that electrical power is applied to the system being checked. Make sure that the ground connections meet the required bonding values, are tight, and properly installed. Refer to the BHT-ELEC-SPM, Chapter 8, Electrical Standard Practices Manual for bonding information, and to 412-A-98-00-00-00A-028A-A / 02455 of this manual for the applicable wiring diagrams.

1.2. Make sure that the switches, relays, diodes, and other electrical components are not defective.

1.3. Examine the wires, connectors, and terminal junctions. Make sure that they are in good condition. Make sure that the connections are properly connected and tightened.

1.4. Make sure that each component of a system operates correctly.

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Miscellaneous Electrical Components - Maintenance Practices

The miscellaneous electrical components include but are not limited to relays, contactors, solenoids, rheostats, switches, leads, connectors, wires, receptacles, shock mounts, capacitors, diodes, resistors, and panel lights.

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Applicable to: ALL
ECCN EAR99

Miscellaneous Electrical Components - Removal

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Caps or plugs	C-136	AR	
Caps or plugs	C-156	AR	

Procedure



BEFORE YOU REMOVE AN ELECTRICAL/ELECTRONIC COMPONENT, DISCONNECT THE ELECTRICAL POWER SOURCE.

1. Disconnect the battery.
2. Make sure that all the other sources of DC power are disconnected.

NOTE

To assist with reinstallation, it is recommended that the wires be identified for their installed location and orientation prior to removal.

3. Identify the connectors or wires connected to the component.
4. Disconnect these electrical connectors or wires from the component.
5. Install caps or plugs (C-136) or caps or plugs (C-156) on the connectors or apply tape on the end of the wires.
6. Remove the attaching hardware. If the attaching hardware is serviceable, keep it for reinstallation.
7. Remove the component.
8. Do a general visual inspection of the immediate area where the component was removed. Refer to 412-A-96-00-01-04A-280A-A / 01884 for the inspection requirements.

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Miscellaneous Electrical Components - Cleaning

Procedure

1. Remove all loose debris with a brush or clean cloth.



SOLVENTS CAN CAUSE DAMAGE TO SOME MATERIALS. THEY CAN REMOVE LUBRICANTS AND SOFTEN CERTAIN MATERIALS. REFER TO THE BHT-ALL-SPM, STANDARD PRACTICES MANUAL FOR THE APPROPRIATE SOLVENT FOR THE MATERIAL YOU ARE CLEANING.

2. Remove grease or dirt with an approved solvent depending on the material of the component (BHT-ALL-SPM, Chapter 5, Standard Practices Manual).

3. Make sure that the electrical component is totally dry before installation.

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Miscellaneous Electrical Components - Inspection

Procedure

NOTE

It is recommended to do a general visual inspection in the immediate area where the component was removed.

1. For the inspection requirements of miscellaneous electrical components, refer to Table 1.

2. Examine the attaching hardware for cracks, wear, mechanical damage, and corrosion. Replace as necessary.

Table 1. Miscellaneous Electrical Component Inspections - Electrical Wire Interconnection System (EWIS)

ITEM	DEFECT, DAMAGE, FAILURE, OR IRREGULARITY
Wire/wire harnesses	Wire bundle/wire bundle or wire bundle/structure contact/chafing
	Wire bundle sagging or improperly secured
	Wires damaged (obvious damage due to mechanical impact, overheat, localized chafing, etc.)
	Lacing tape and/or ties missing/incorrectly installed
	Wiring protection sheath/conduit deformity or incorrectly installed
	End of sheath rubbing on end attachment device
	Grommet missing or damaged
	Dust and lint accumulation
	Surface contamination by metal shavings/swarf
	Contamination by liquids
	Deterioration of previous repairs (e.g., splices)
	Deterioration of production splices
	Inappropriate repairs (e.g., incorrect splice)
	Inappropriate attachments to or separation from fluid lines
Connectors	External corrosion on receptacles
	Missing/loose or damaged dust cap
	Backshell tail broken
	Rubber pad or packing on backshell missing
	No backshell wire securing device
	Fool proofing chain broken
	Missing or broken safety wire
	Discoloration/evidence of overheating on terminal lugs/blocks
	Torque stripe misalignment
	Damaged or bent contacts
Switches/switch lights and annunciators	Rear protection cap damaged
	Corrosion
	Weak detents
Ground points	Corrosion and looseness
Bonding braid/bonding jumper	Braid broken or disconnected
	Multiple strands corroded or broken

Table 1. Miscellaneous Electrical Component Inspections - Electrical Wire Interconnection System (EWIS) (continued)

ITEM	DEFECT, DAMAGE, FAILURE, OR IRREGULARITY
Wiring clamps or brackets	Corroded Broken/missing Bent or twisted Faulty attachment (bad attachment or fastener missing) Debonded/detached Protection/cushion damaged
Supports (rails or tubes/conduit)	Broken Deformed Fastener missing Missing edge protection on rims of feed through holes Racetrack cushion damaged Obstructed drainage holes (in conduits)
Circuit breakers, contactors, relays, rheostats, or fuses	Signs of overheating Signs of arcing Looseness Corrosion

Miscellaneous Electrical Components - Repair

Procedure

1. Tighten loose terminal connections, mountings, and attachments of the electrical components.
2. Replace miscellaneous electrical components that fail to meet the inspection requirements.

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Miscellaneous Electrical Components - Installation

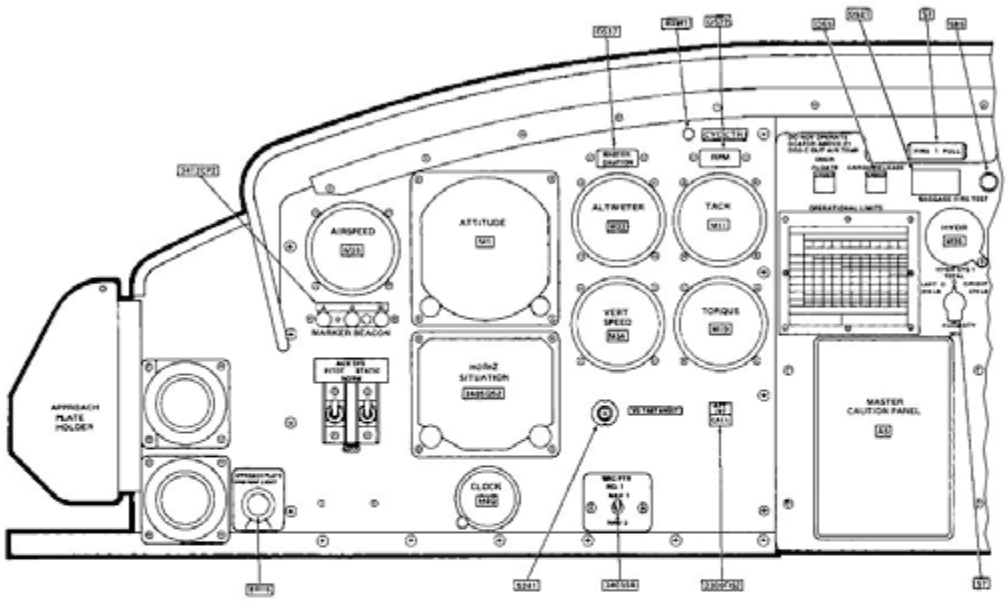
Procedure

1. Disconnect the battery.
2. Make sure that all the other sources of DC power are disconnected.
3. Do a general visual inspection of the immediate area where the component was removed. Refer to 412-A-96-00-01-04A-280A-A / 01884 for the inspection requirements.
4. Use the correct attaching hardware and install the component.
5. Remove the caps and/or plugs from the connectors and remove the tape from the ends of the wires. Use the applicable wiring diagram in 412-A-98-00-00-00A-028A-A / 02455 and attach the connectors or the wires to the proper location or terminals.
6. Do the applicable operational check of the system.
7. Return the helicopter to the flight configuration after electrical maintenance or repairs (412-A-96-00-01-02A-913C-A / 01877).

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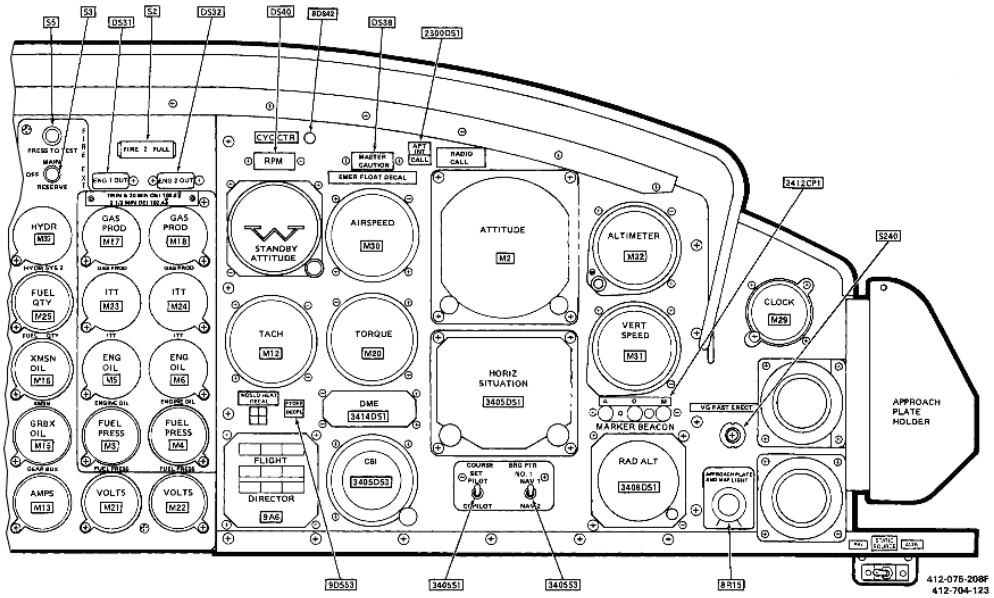
Control Panels - Electrical - Description

The controls and the control panels are mounted in the overhead console, pedestal, and instrument panel (Figure 1 through Figure 12).



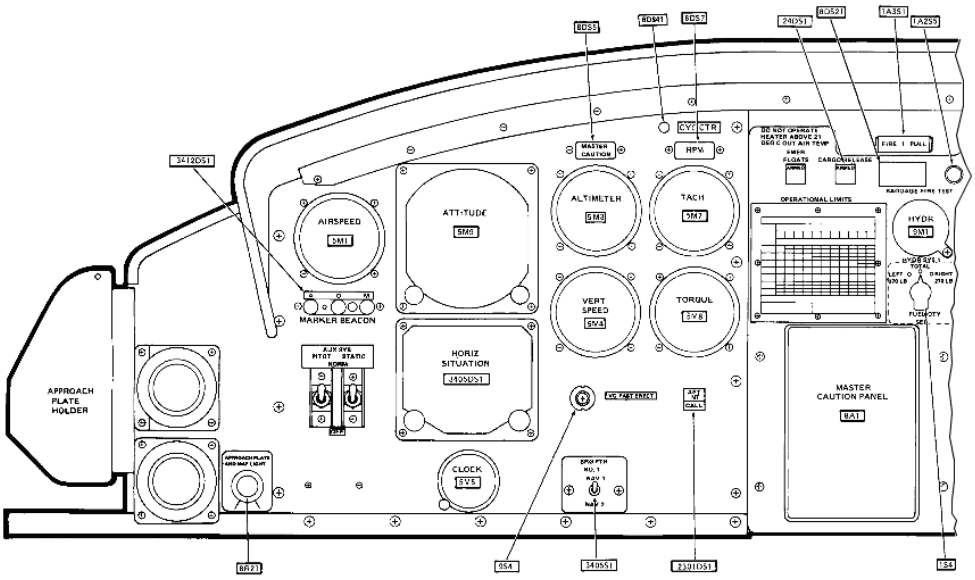
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Figure 1. Instrument Panel (S/N 33001 through 33100) (Post 412-704-115-101) (Sheet 1 of 2)



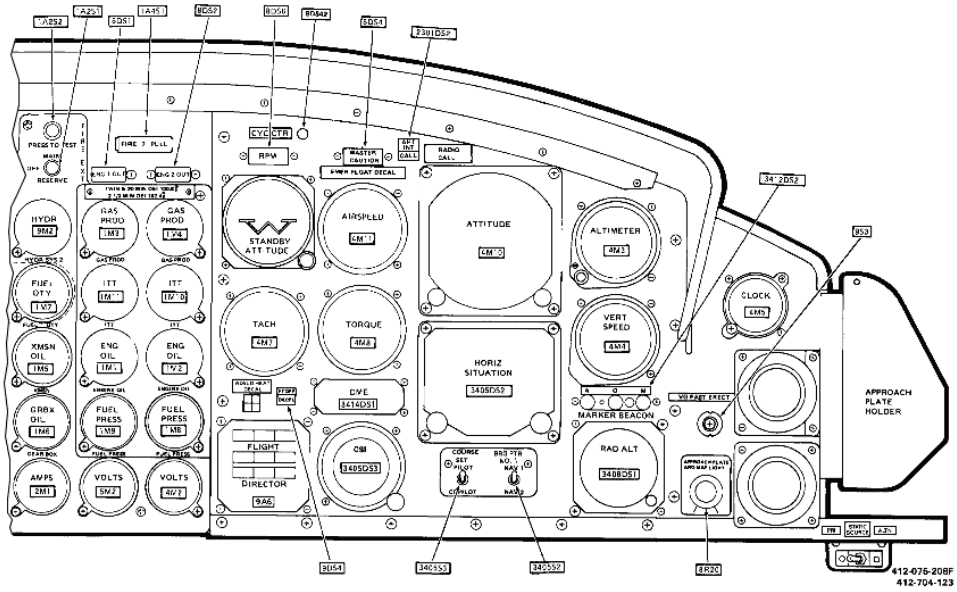
ICN-412-A-960002-B-97499-00083-A-001-01

Figure 1. Instrument Panel (S/N 33001 through 33100) (Post 412-704-115-101) (Sheet 2 of 2)



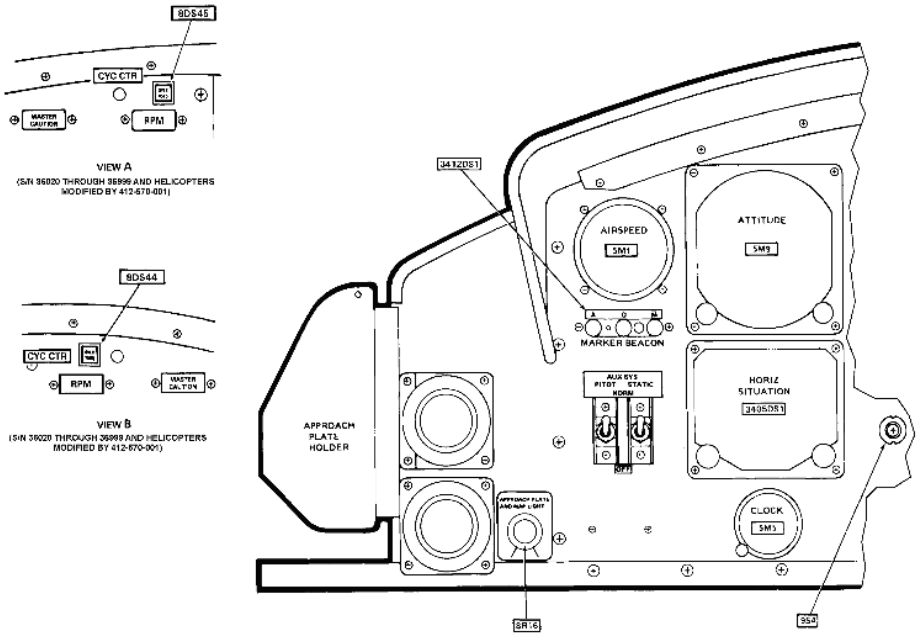
ICN-412-A-960002-B-97499-00084-A-001-01

Figure 2. Instrument Panel (S/N 33101 through 33107) (Sheet 1 of 2)



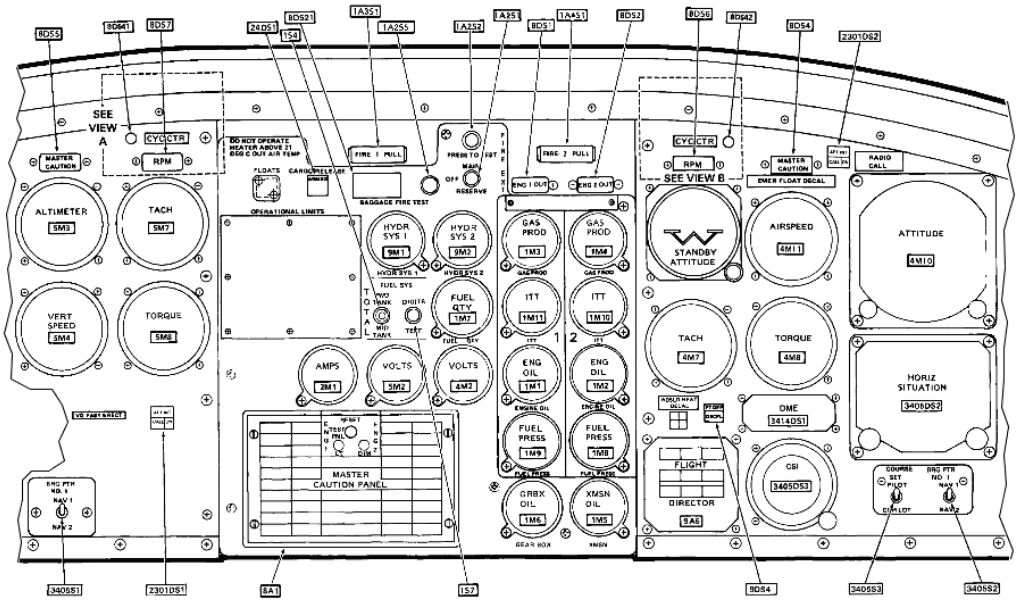
ICN-412-A-960002-B-97499-00085-A-001-01

Figure 2. Instrument Panel (S/N 33101 through 33107) (Sheet 2 of 2)



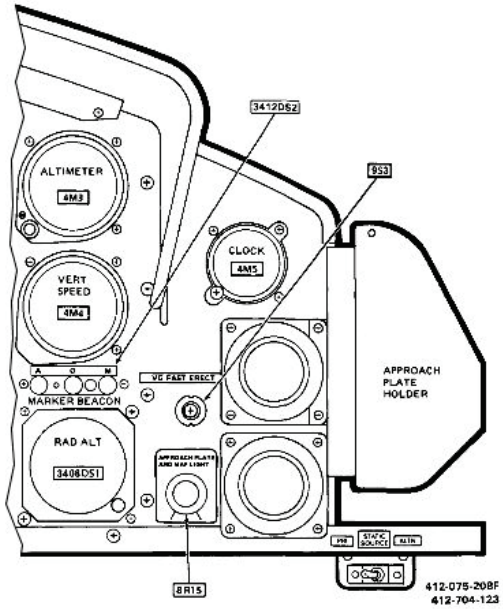
ICN-412-A-960002-B-97499-00086-A-001-01

Figure 3. Instrument Panel (S/N 33108 through 33213 and 36001 through 36086) (Sheet 1 of 3)



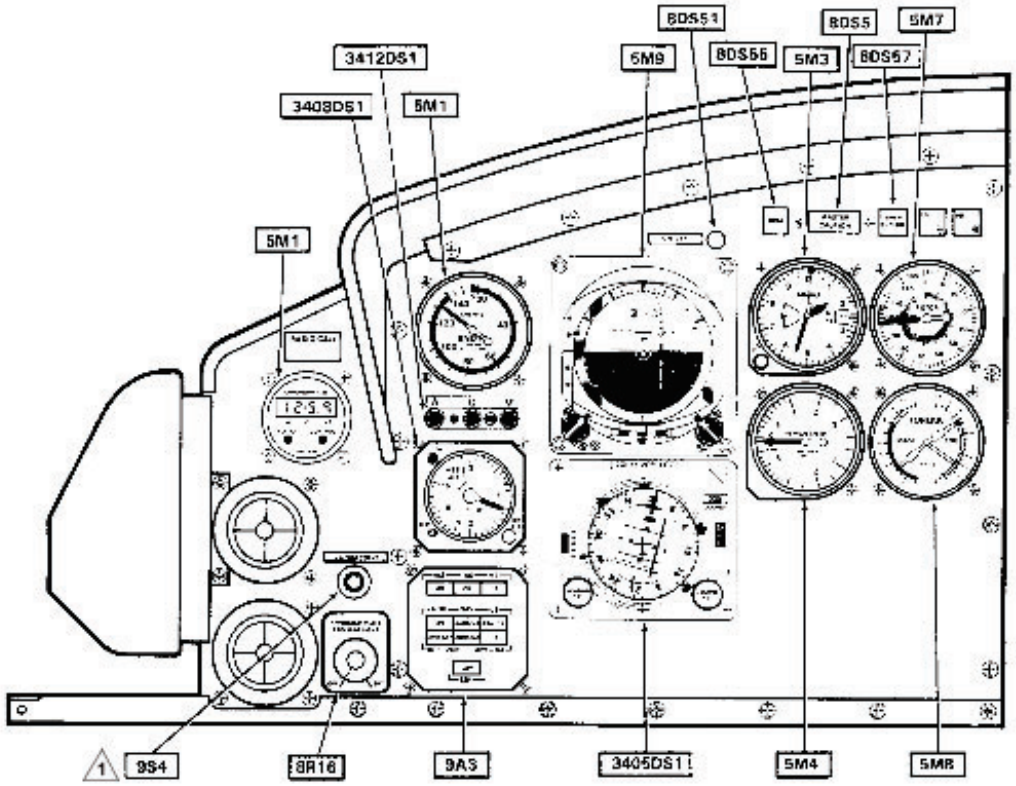
ICN-412-A-960002-B-97499-00087-A-001-01

Figure 3. Instrument Panel (S/N 33108 through 33213 and 36001 through 36086) (Sheet 2 of 3)



ICN-412-A-960002-B-97499-00088-A-001-01

Figure 3. Instrument Panel (S/N 33108 through 33213 and 36001 through 36086) (Sheet 3 of 3)

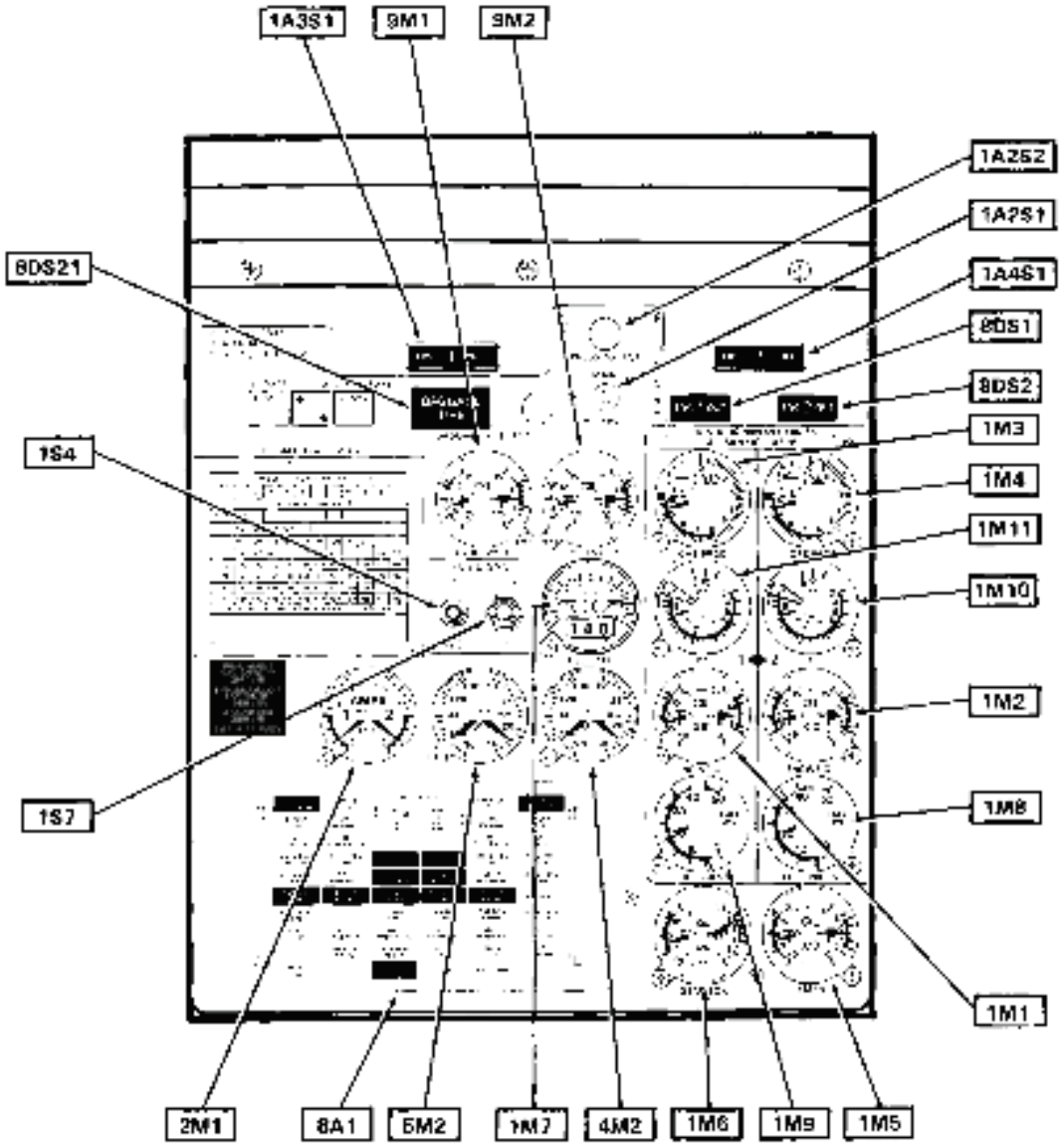


NOTE

1 Applicable to S/N 36087 through 36247.

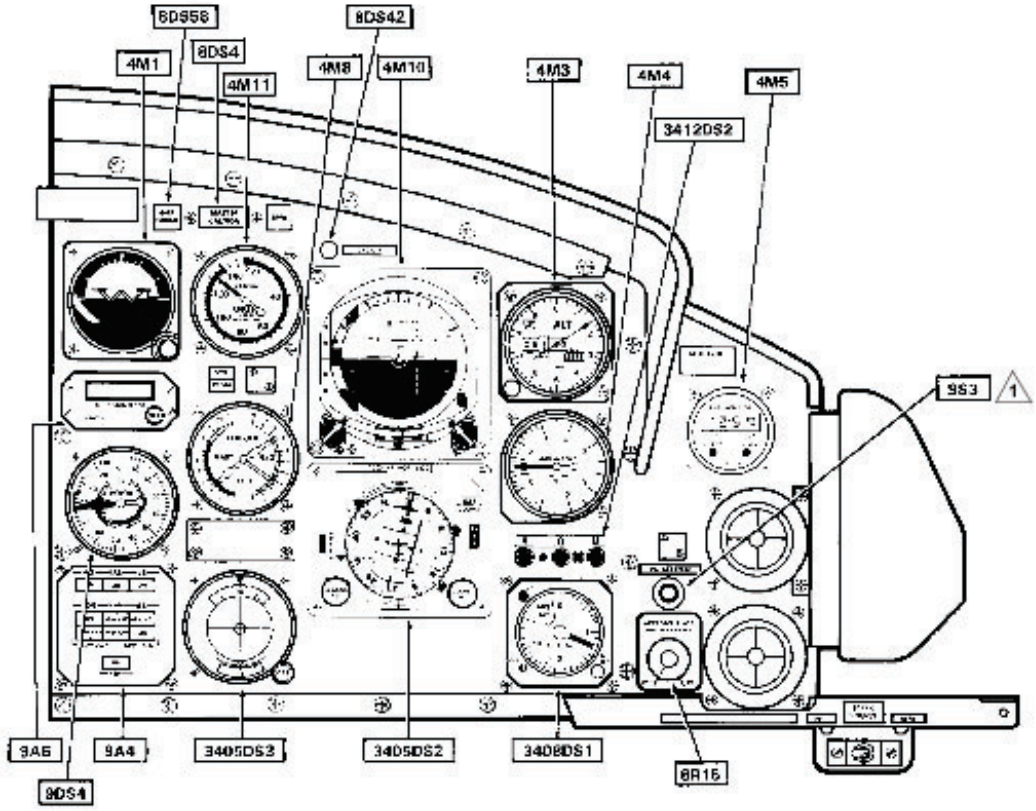
ICN-412-A-960002-B-97499-00089-A-001-01

Figure 4. Instrument Panel (S/N 36087 through 36999) (Sheet 1 of 3)



ICN-412-A-960002-B-97499-00090-A-001-01

Figure 4. Instrument Panel (S/N 36087 through 36999) (Sheet 2 of 3)

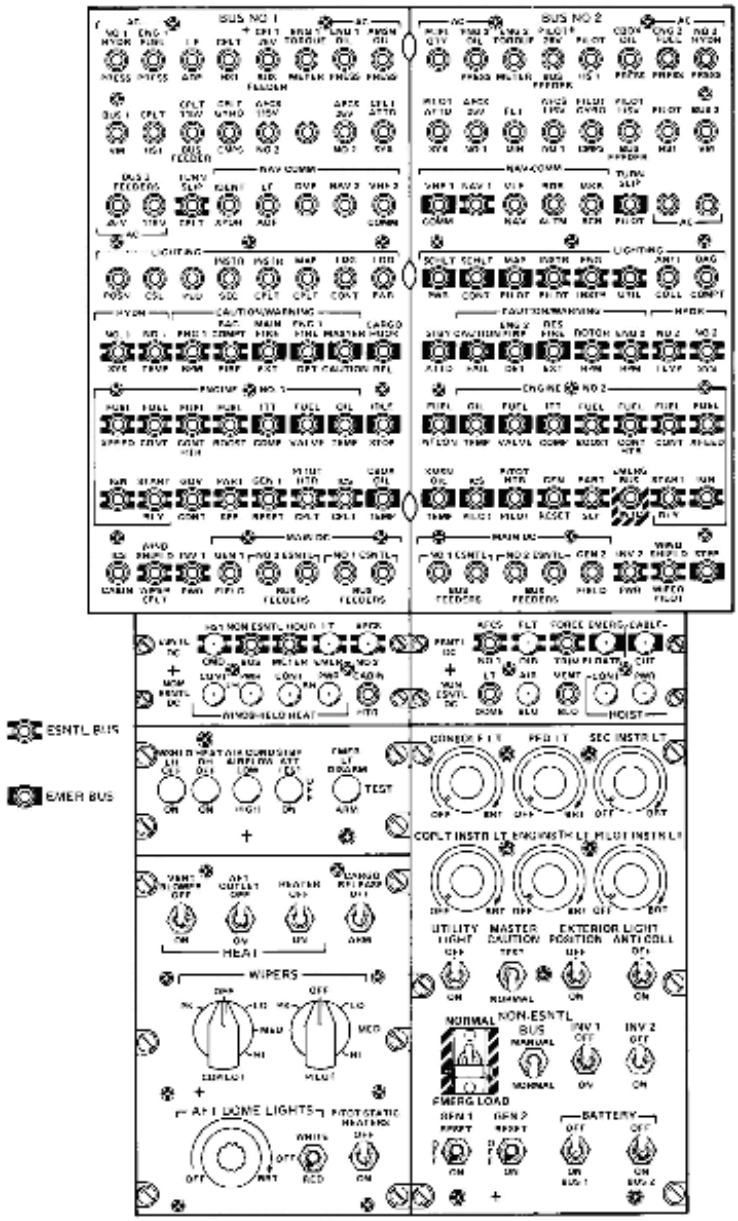


NOTE

1 Applicable to 36087 through 36247.

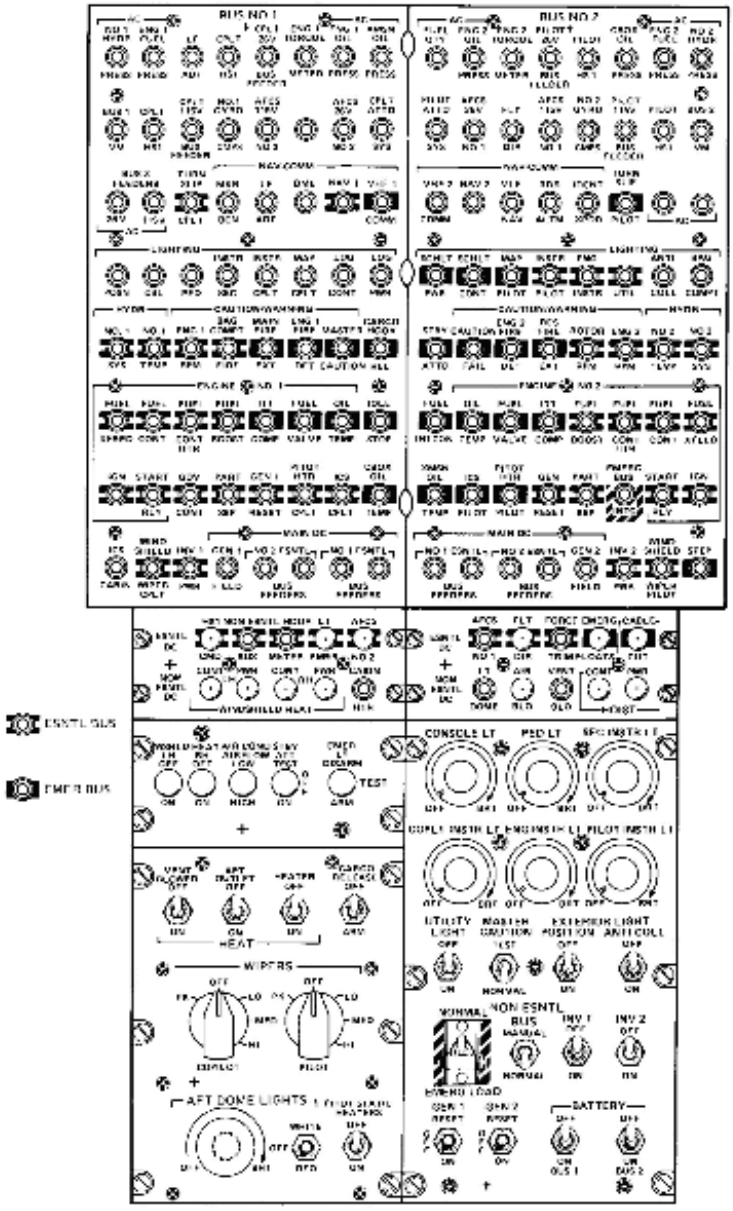
ICN-412-A-960002-B-97499-00091-A-001-01

Figure 4. Instrument Panel (S/N 36087 through 36999) (Sheet 3 of 3)



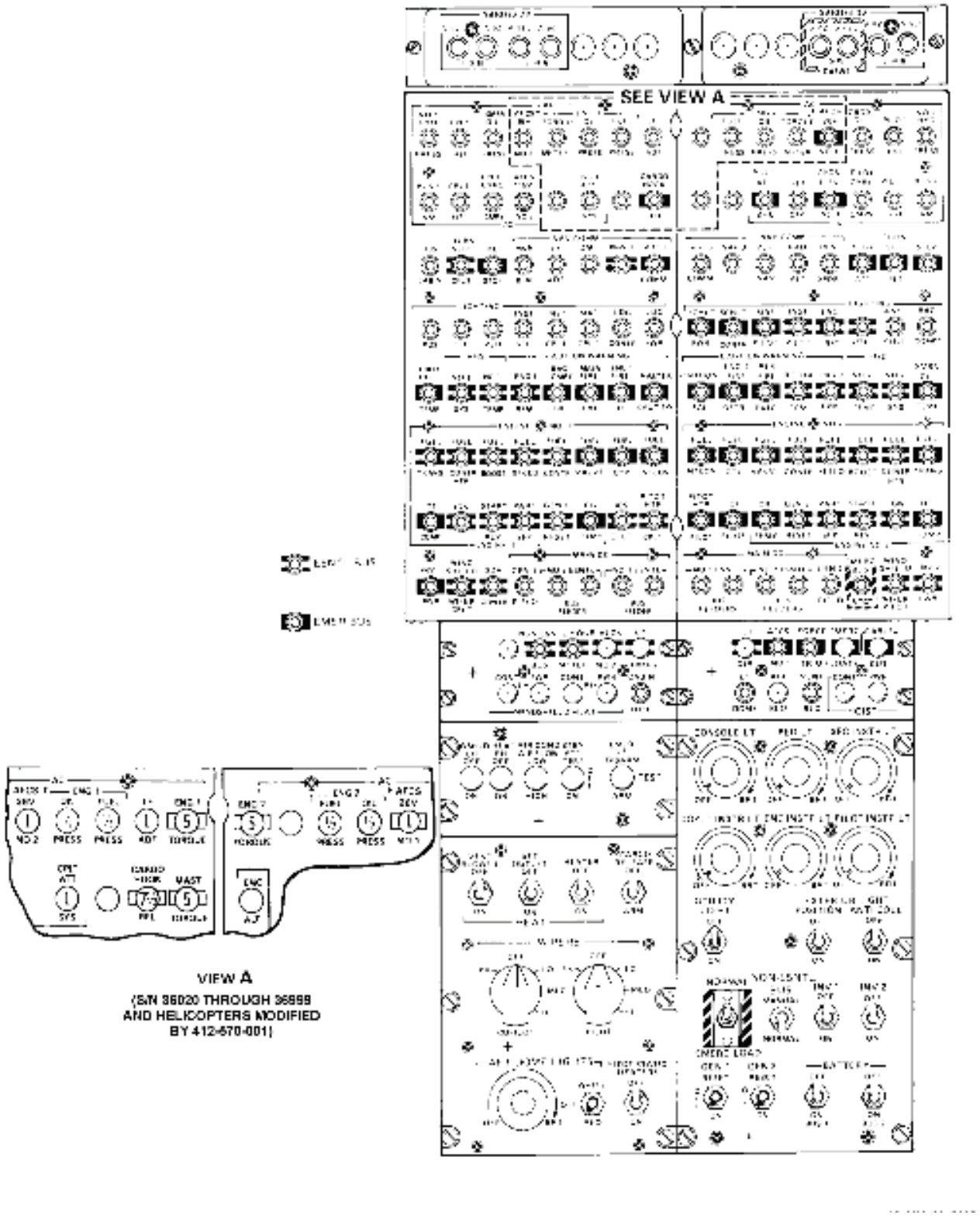
ICN-412-A-960002-B-97499-00092-A-001-01

Figure 5. Overhead Console (S/N 33001 through 33100) (Sheet 1 of 1)



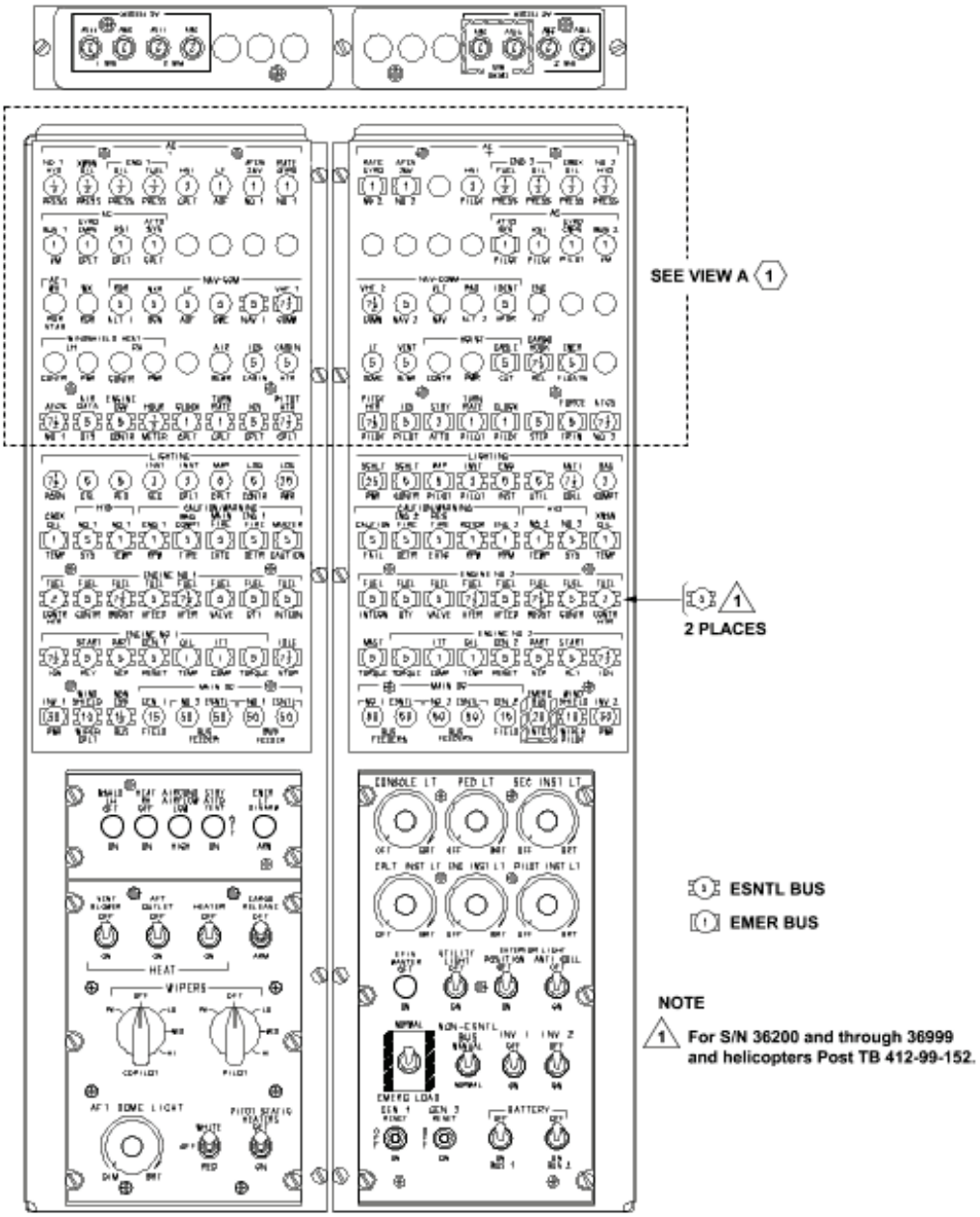
ICN-412-A-960002-B-97499-00093-A-001-01

Figure 6. Overhead Console (S/N 33101 through 33107) (Sheet 1 of 1)



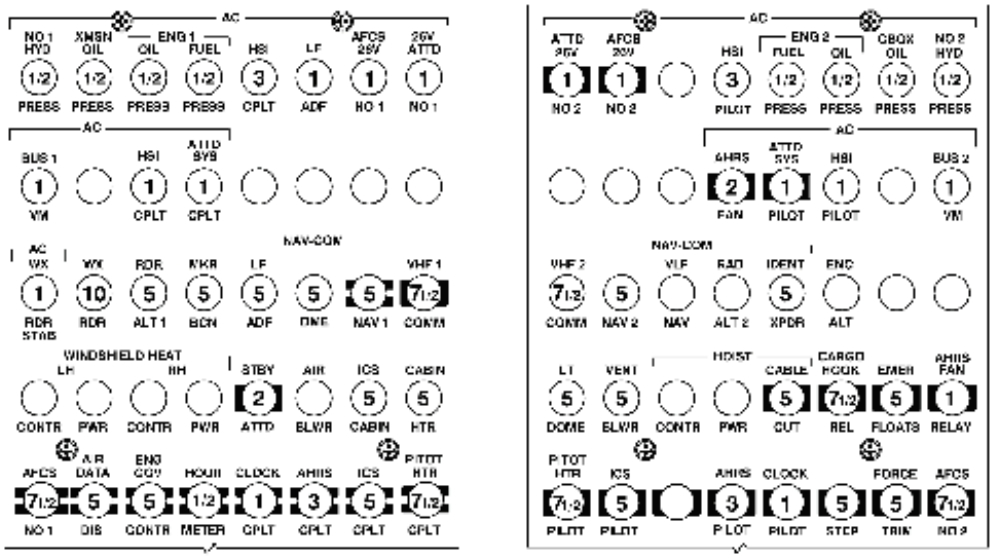
ICN-412-A-960002-B-97499-00094-A-001-01

Figure 7. Overhead Console (S/N 33108 through 36086) (Sheet 1 of 1)



ICN-412-A-960002-B-97499-00095-A-001-01

Figure 8. Overhead Console (S/N 36087 through 36999) (Sheet 1 of 2)



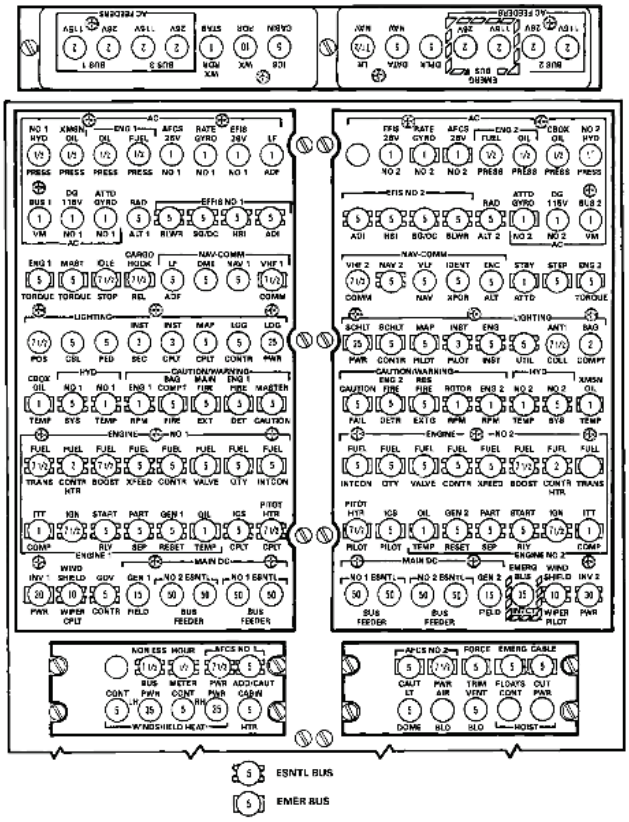
VIEW A (1)

EFFECTIVITY

(1) S/N 36248 THROUGH 36999.

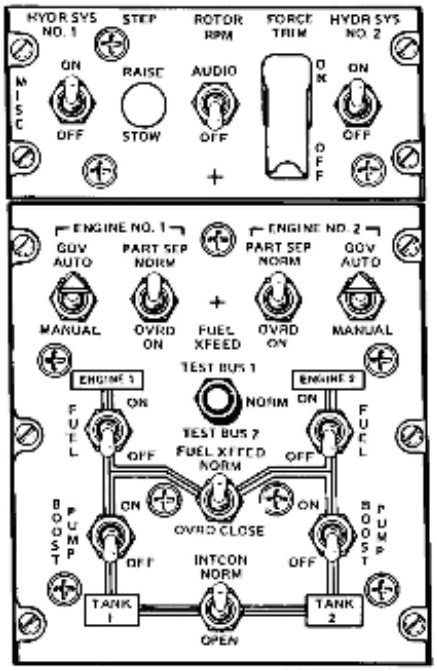
ICN-412-A-960002-B-97499-00096-A-001-01

Figure 8. Overhead Console (S/N 36087 through 36999) (Sheet 2 of 2)



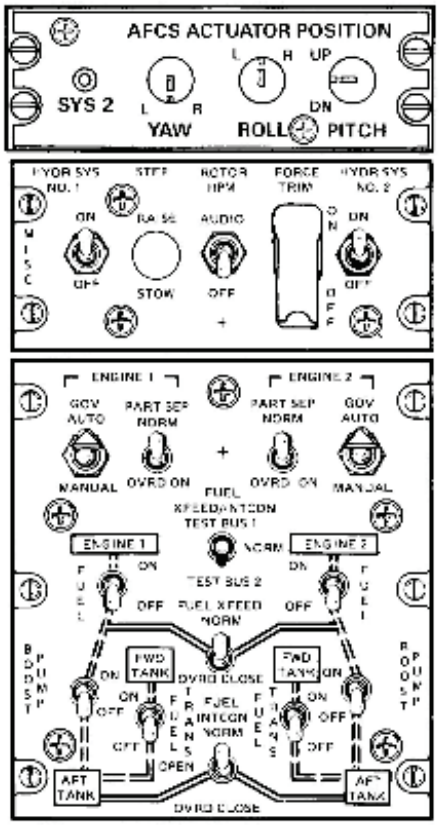
ICN-412-A-960002-B-97499-00097-A-001-01

Figure 9. Overhead Console (Typical 4-Axis Configuration) (Sheet 1 of 1)



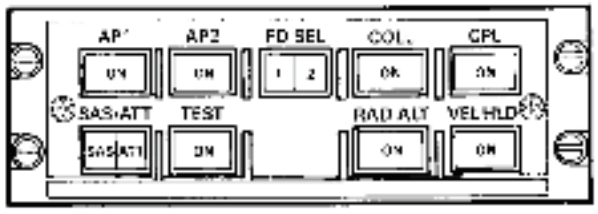
ICN-412-A-960002-B-97499-00098-A-001-01

Figure 10. Pedestal Panels (S/N 33001 through 33107) (Sheet 1 of 1)

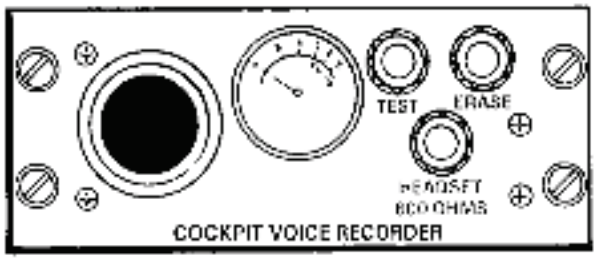


ICN-412-A-960002-B-97499-00099-A-001-01

Figure 11. Pedestal Panels (S/N 33108 through 33213 and 36001 through 36999) (Sheet 1 of 2)



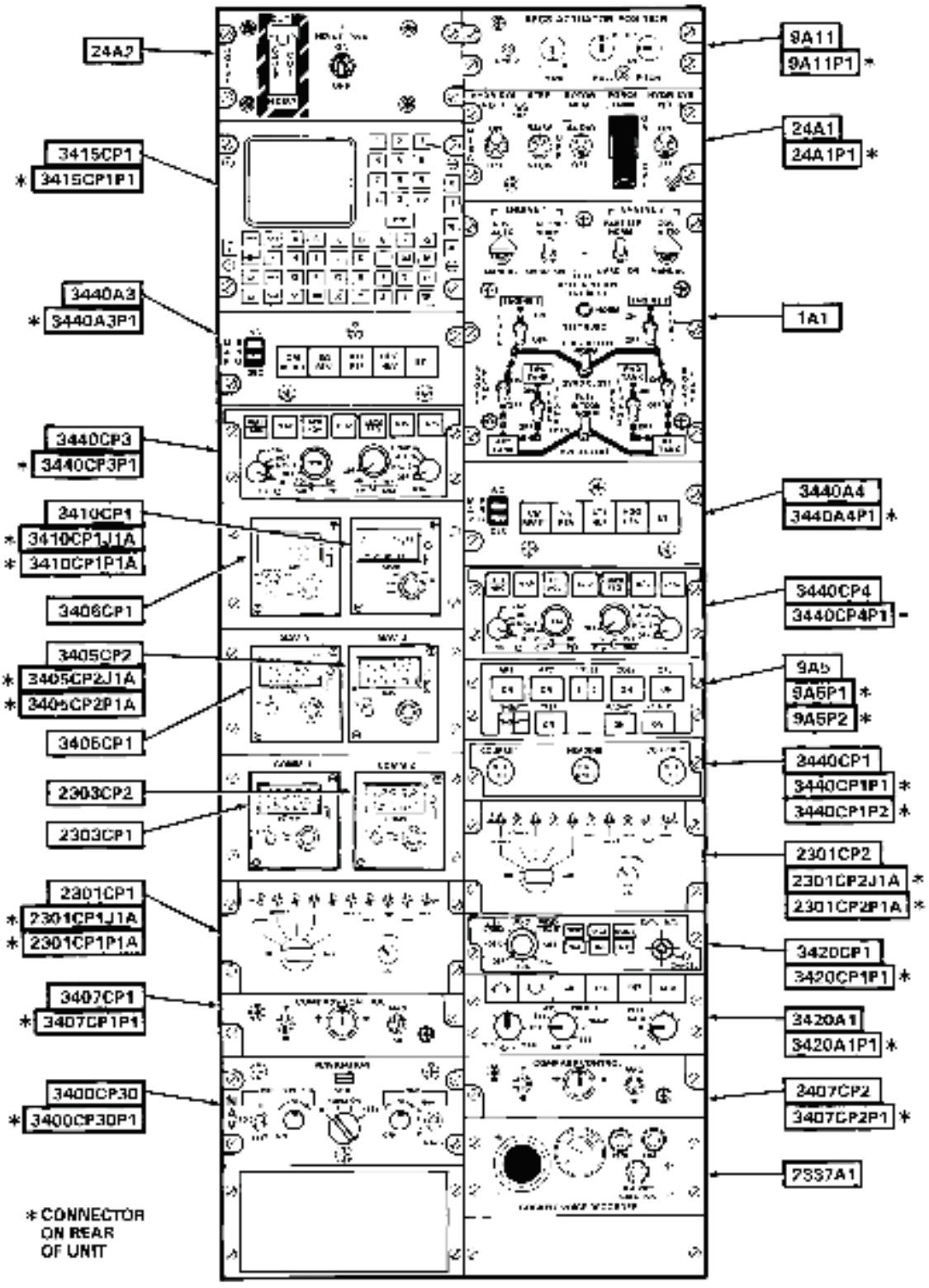
(HELICOPTERS S/N 36087 THROUGH 36999)



(HELICOPTERS S/N 36037 THROUGH 36999)

ICN-412-A-960002-B-97499-00100-A-001-01

Figure 11. Pedestal Panels (S/N 33108 through 33213 and 36001 through 36999) (Sheet 2 of 2)



ICN-412-A-960002-B-97499-00101-A-001-01

Figure 12. Pedestal (Typical) - 4-Axis Configuration (Sheet 1 of 1)

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Control Panels - Removal

Procedure

NOTE

The removal procedure for all panels is relatively the same. A single removal procedure may be used for any panel.

1. Remove all electrical power.

2. Disengage the fasteners that secure the panel to the mount.

3. Carefully lift the panel from the mount and disconnect the electrical connector.

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Control Panels - Inspection

Procedure

1. Visually inspect for scratched, chipped or broken edge and integrally lit panels, loose wiring connections, damaged connector, corrosion, and broken or missing mounting fasteners.

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Applicable to: ALL
ECCN EAR99

Control Panels - Replacement

Procedure

1. Replace items that fail to meet the inspection requirements.

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Control Panels - Installation

Procedure

1. Connect the electrical connector.
2. Position the panel in the mount being careful not to damage the wiring.
3. Engage the fasteners.

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Circuit Breakers - Description

There is one type of circuit breaker used on Model 412 helicopters, the fault clearing push-pull type. Circuit breakers are mounted in the overhead console, pedestal, nose compartment, and throughout the helicopter.

Circuit breakers are electromagnetic devices that protect the electrical circuits and equipment by automatically opening the electrical circuit when the current in the line exceeds a predetermined value.

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Circuit Breakers - Removal

Procedure



BEFORE YOU REMOVE A CIRCUIT BREAKER ON THE HELICOPTER, REMOVE ALL ELECTRICAL POWER AND DISCONNECT THE BATTERY. IF NOT ACCOMPLISHED, YOU CAN CAUSE ELECTRICAL SHOCK TO PERSONS AND/OR DAMAGE TO EQUIPMENT.

1. Remove all the electrical power.

NOTE

To assist with reinstallation, it is recommended that the wires and bus bars be identified for their installed location and orientation prior to removal.

2. Gain access to the circuit breaker wiring.
3. Tag the wires and disconnect.
4. Protect the wires.
5. Remove the screw that secures the circuit breaker to bus the bar, if required.
6. Remove the nut and the washers that secure the circuit breaker to the panel.
7. Remove the circuit breaker from the panel.

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Circuit Breakers - Inspection

Procedure

1. Inspect the circuit breakers for secure mounting, corrosion, general condition, power on and off actuation, and reset retention.

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Applicable to: ALL
ECCN EAR99

Circuit Breakers - Replacement

Procedure

1. Replace the circuit breaker if the inspection requirements can not be met.

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Circuit Breakers - Installation

Procedure



BEFORE YOU INSTALL A CIRCUIT BREAKER ON THE HELICOPTER, REMOVE ALL ELECTRICAL POWER AND DISCONNECT THE BATTERY. IF NOT ACCOMPLISHED, YOU CAN CAUSE ELECTRICAL SHOCK TO PERSONS AND/OR DAMAGE TO EQUIPMENT.

1. Position the circuit breaker on the structure and secure with the hardware.
2. Install the screw that secures the circuit breaker to the bus bar, if required.
3. Remove the tags and install the wires.
4. Do the circuit operational check.

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Semiconductor Devices - Description

The semiconductor devices used are as follows:

- Standard and zener diodes
- Transient suppression diode assemblies
- Transistors

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Semiconductor Devices - Maintenance Practices

Procedure

1. Remove the semiconductor device (412-A-96-00-01-04A-520A-A / 01882).

NOTE

Refer to the BHT-ELEC-SPM, Electrical Standard Practices Manual for detailed information on testing and troubleshooting semiconductor devices.

2. Inspect the semiconductor device (412-A-96-00-01-04A-280A-A / 01884).

NOTE

When you install a semiconductor device, refer to the associated wiring diagram in 412-A-98-00-00-00A-028A-A / 02455 to ensure the correct orientation and wiring attachment.

3. Install the semiconductor device (412-A-96-00-01-04A-720A-A / 01886).

4. Return the helicopter to the flight configuration (412-A-96-00-01-02A-913C-A / 01877).

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Track Mounted Relays - Description

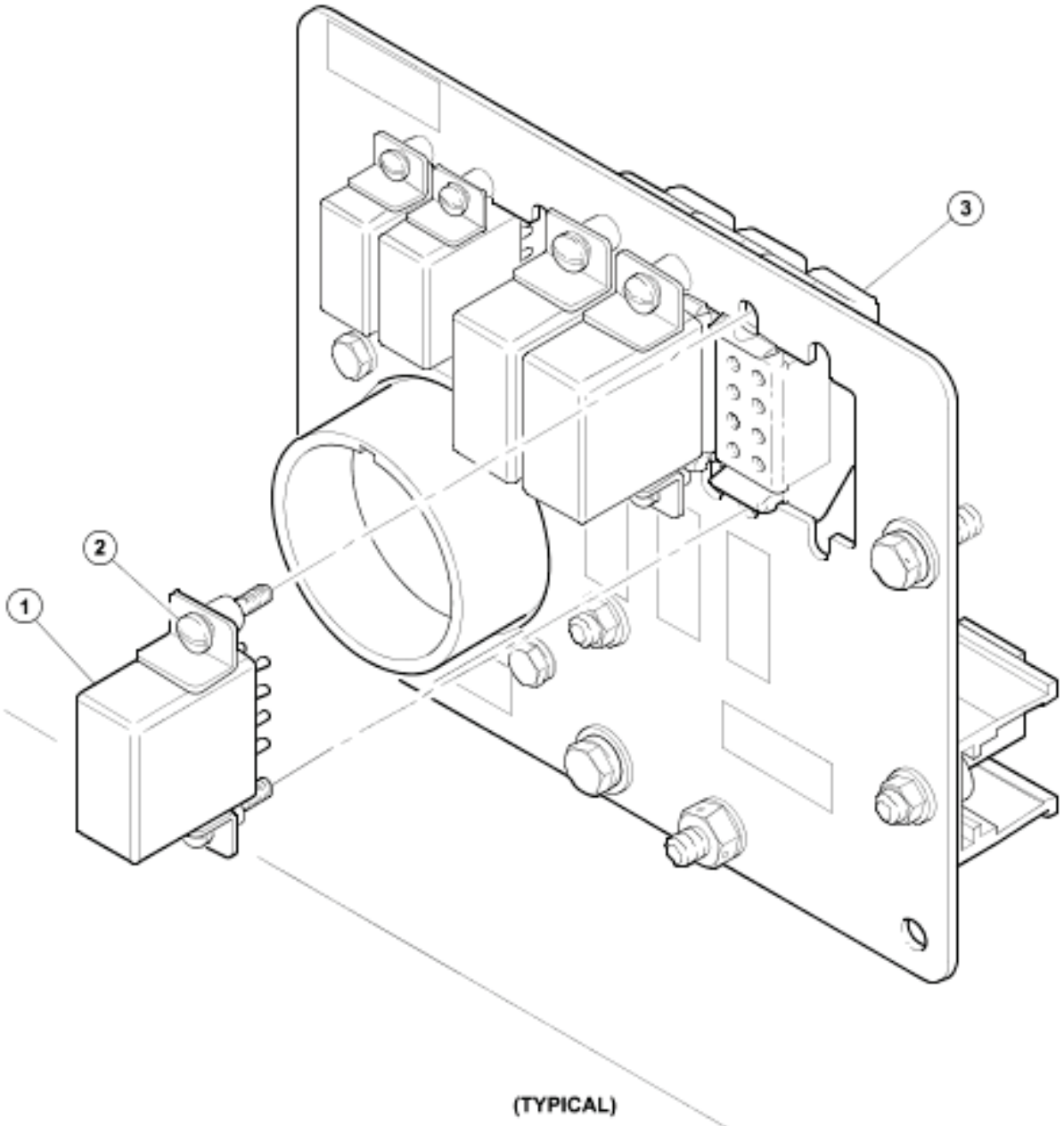
Track mounted relays are electromagnetic switches that open and close under the control of other components in the electrical circuit.

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Track Mounted Relays - Removal

Procedure

1. Disconnect the battery and the external power from the helicopter.
2. Loosen the screws (2, Figure 1).
3. Carefully pull the relay (1) out from the socket base (3).



1. Relay
2. Screw
3. Socket base

ICN-412-A-960002-B-97499-00001-A-001-01

Figure 1. Track Mounted Relay - Removal and Installation (Typical) (Sheet 1 of 1)

Track Mounted Relays - Cleaning

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Isopropyl alcohol	C-385	AR	
Clean cloth	C-516	AR	
Butyl acetate	C-552	AR	

Procedure

1. Clean the relay pins with isopropyl alcohol (C-385) and, if it is necessary to do a deep cleaning, use butyl acetate (C-552).

2. Dry the pins and the surfaces with a clean cloth (C-516). Turn the relay sockets down and let the solvent evaporate/dry for 20 minutes.

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Track Mounted Relays - Inspection

Procedure

1. For the inspection of the track mounted relay, refer to 412-A-96-00-01-04A-280A-A / 01884.

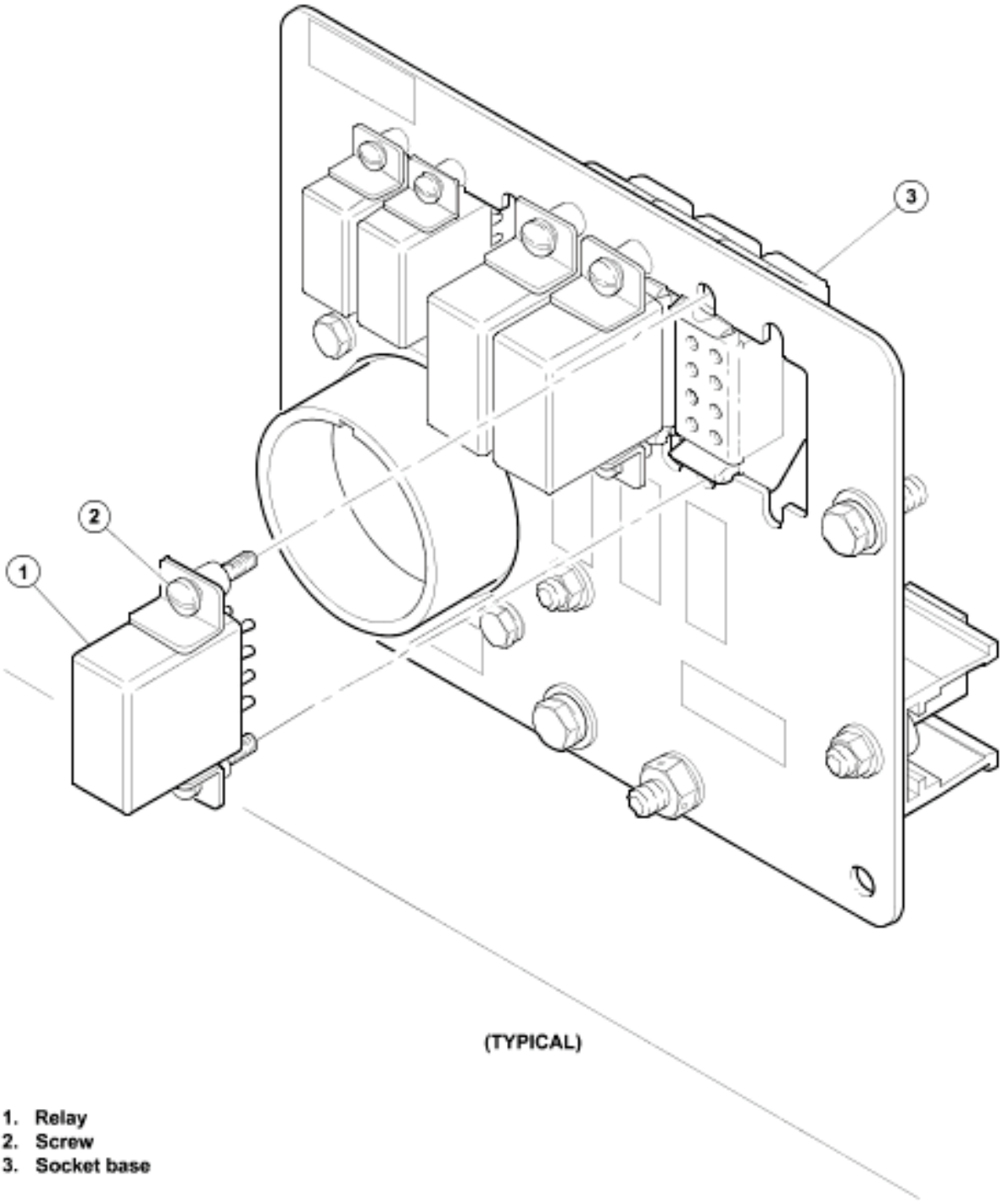
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Applicable to: ALL
ECCN EAR99

Track Mounted Relays - Installation

Procedure

1. Correctly position the relay (1, Figure 1) on the socket base (3) and apply finger pressure until it latches into place.
2. Tighten the screws (2).
3. Connect the battery and the external power.
4. Do the applicable operational check of the system.



1. Relay
2. Screw
3. Socket base

ICN-412-A-960002-B-97499-00001-A-001-01

Figure 1. Track Mounted Relay - Removal and Installation (Typical) (Sheet 1 of 1)

Electrical and Avionics Equipment - Reference Designator, Nomenclature, and Location - Description

To identify the installed location of electrical and avionics components, refer to Figure 1 through Figure 28.

To determine the nomenclature or general location of the reference designator items, refer to Table 1 through Table 5.

Table 1. Reference Designators (S/N 33001 through 33100)

REF DES	NOMENCLATURE
A3	PANEL, ENGINE AND FUEL
A5	PANEL, MASTER CAUTION
A10	PANEL, MISCELLANEOUS CONTROL
A13	PANEL, RESISTOR
A14	PANEL, RESISTOR
B1	SYNCHRO, TRANSMITTER TORQUE, LEFT
B2	SYNCHRO, TRANSMITTER TORQUE, RIGHT
B3	TRANSMITTER, ENGINE OIL PRESURE, LEFT
B4	TRANSMITTER, ENGINE OIL PRESSURE, RIGHT
B5	TRANSMITTER, FUEL PRESSURE, LEFT
B6	TRANSMITTER, FUEL PRESSURE, RIGHT
B7	TRANSMITTER, TRANSMISSION OIL PRESSURE
B8	MOTOR, GOVERNOR RPM ACTUATOR
B13	MOTOR, COPILOT WINDSHIELD WIPER
B14	MOTOR, PILOT WINDSHIELD WIPER
B15	MOTOR, PARTICLE SEPARATOR, LEFT ACTUATOR
B16	MOTOR, PARTICLE SEPARATOR, RIGHT ACTUATOR
B22	TRANSMITTER, C BOX OIL PRESSURE
B27	MOTOR, INVERTER BLOWER
BT1	BATTERY
CB5	CB, XMSN CHIP POS IND
CB5	CB, 28 VDC AUX RECEPT
CP1	TRANSMITTER, FUEL QUANTITY TRANSMITTER
CP2	CONN, TEE, FUEL QUANTITY
CP3	CONN, TEE FUEL QUANTITY
CP4	CONN, TEE, FUEL QUANTITY
CR9	DIODE, EXTERNAL POWER RELAY
CR19	DIODE, PARTICLE SEPARATOR RELAY
CR20	DIODE, PARTICLE SEPARATOR RELAY
CR47	DIODE, LIGHTING
CR71	DIODE, BATTERY INTERLOCK
CR73	DIODE, START RELAY 2
CR92	DIODE, RPM WARNING
DS5	RELEASE, CARGO
DS7	LIGHT, ANTI-COLLISION, LOWER
DS8	LIGHT, ANTI-COLLISION, UPPER
DS9	LIGHT, COPILOT UTILITY
DS10	LIGHT, PILOT UTILITY

Table 1. Reference Designators (S/N 33001 through 33100) (continued)

REF DES	NOMENCLATURE
DS11	LIGHT, DOME
DS12	LIGHT, SEARCH
DS13	LIGHT, LANDING
DS15	LIGHT, LEFT POSITION
DS16	LIGHT, RIGHT POSITION
DS17	LIGHT, TAIL
DS18	LIGHT, BAGGAGE COMPT
DS21	LIGHT, BAGGAGE FIRE TEST
DS28	INDICATOR, HYDRAULIC FILTER CLOG
DS29	LIGHT, LEFT UTILITY
DS30	LIGHT, RIGHT UTILITY
DS31	LIGHT, ENG 1 OUT
DS32	LIGHT, ENG 2 OUT
DS37	LIGHT, COPILOT MASTER CAUTION
DS38	LIGHT, PILOT MASTER CAUTION
DS39	LIGHT, COPILOT RPM CAUTION
DS40	LIGHT, PILOT RPM CAUTION
DS41	LIGHT, OUTSIDE AIR TEMPERATURE
DS50	INDICATOR, SUMP CHIP
DS51	INDICATOR, PLANETARY CHIP
DS52	INDICATOR, UPPER MAST CHIP
E1	DETECTOR, TRANSMISSION MAGNETIC CHIP
E2	DETECTOR, ENGINE GEARBOX
E3	DETECTOR, ENGINE 1 FORWARD
E4	DETECTOR, ENGINE 2 FORWARD
E5	DETECTOR, ENGINE 1 AFT
E6	DETECTOR, ENGINE 2 AFT
E7	DETECTOR, 42° GEARBOX MAGNETIC CHIP
E8	DETECTOR, 90° GEARBOX MAGNETIC CHIP
E17	DETECTOR, TRANSMISSION UPPER MAST
E26	DETECTOR, TRANSMISSION PLANETARY
G1	STARTER-GENERATOR, ENGINE 1
G2	STARTER-GENERATOR, ENGINE 2
G3	GENERATOR, GAS PRODUCER 1 TACH
G4	GENERATOR, GAS PRODUCER 2 TACH
G5	GENERATOR, POWER TURBINE 1 TACH
G6	GENERATOR, POWER TURBINE 2 TACH
G7	GENERATOR, ROTOR TACH
HR3	HEATER, PILOT LEFT STATIC PORT
HR4	HEATER, PILOT RIGHT STATIC PORT
J3	CONN, EXTERNAL POWER
J33	CONN, INSTRUMENT PANEL LEFT
J34	CONN, INSTRUMENT PANEL RIGHT

Table 1. Reference Designators (S/N 33001 through 33100) (continued)

REF DES	NOMENCLATURE
J41	CONN, TRANSMISSION
J46	CONN, PILOT CYCLIC STICK
J47	CONN, COPILOT CYCLIC STICK
J48	CONN, PILOT COLLECTIVE
J49	CONN, COPILOT COLLECTIVE
J56	CONN, 28V DC AUXILIARY RECEPTACLE
J57	CONN, PYLON LEFT
J58	CONN, PYLON RIGHT
J59	CONN, ENGINE 1 FORWARD COMPARTMENT
J60	CONN, ENGINE 2 FORWARD COMPARTMENT
J61	CONN, ROOF GENERATOR
J62	CONN, ROOF GENERATOR
J63	CONN, LEFT FIRE DETECT LEFT FWD ENG COMPT
J64	CONN, RIGHT FIRE DETECT LEFT FWD ENG COMPT
J67	CONN, LEFT FIRE DETECT RIGHT FWD ENG COMPT
J68	CONN, RIGHT FIRE DETECT RIGHT FWD ENG COMPT
J95	CONN, CENTER ENGINE COMPARTMENT
J96	CONN, CENTER ENGINE COMPARTMENT
J98	CONN, FIRE DETECT - EXT 2 AFT ENGINE COMPARTMENT
J99	CONN, ENGINE 1 AFT COMPARTMENT
J100	CONN, ENGINE 2 AFT COMPARTMENT
J117	CONN, TAILBOOM
J118	CONN, TAILBOOM
J129	CONN, CARGO SLING
J145	CONN, FUEL QUANTITY INLINE RIGHT
J146	CONN, FUEL QUANTITY INLINE RIGHT
J147	CONN, FUEL QUANTITY INLINE TEE
J148	CONN, FUEL QUANTITY INLINE TEE
J149	CONN, FUEL QUANTITY INLINE TEE
J168	CONN, HEATER COMPARTMENT
J195	CONN, STEP ACTUATOR
J355	CONN, ANTI-COLLISION LIGHT UPPER
J506	CONN, INSTRUMENT PANEL
J3018	CONN, PITCH SYSTEM 1 TRANSDUCER
J3020	CONN, YAW SYSTEM 1 TRANSDUCER
J3022	CONN, ROLL SYSTEM 1 TRANSDUCER
J3043	CONN, PYLON LEFT
K1	RELAY, EXTERNAL POWER
K2	RELAY, NON-ESSENTIAL BUS 2
K3	RELAY, STARTER GENERATOR 1
K4	RELAY, STARTER GENERATOR 2
K5	RELAY, STARTER 1
K6	RELAY, STARTER 2

Table 1. Reference Designators (S/N 33001 through 33100) (continued)

REF DES	NOMENCLATURE
K7	RELAY, STARTER CONTROL 1
K8	RELAY, STARTER CONTROL 2
K9	RELAY, BATTERY BUS 1
K10	RELAY, BATTERY BUS 2
K15	RELAY, IDLE STOP DELAY 1
K16	RELAY, IDLE STOP DELAY 2
K18	RELAY, HEATER
K26	RELAY, PARTICLE SEPARATOR 1
K27	RELAY, PARTICLE SEPARATOR 2
K30	RELAY, HOURMETER
K33	RELAY, NON-ESSENTIAL BUS
K34	RELAY, NON-ESSENTIAL BUS
K39	RELAY, NON-ESSENTIAL BUS
K42	RELAY, EMERGENCY FLOAT ARM
K52	RELAY, CAUTION PANEL FAIL
K57	RELAY, BATTERY INTERLOCK
L1	SOLENOID, FUEL CONTROL VALVE
L2	SOLENOID, FUEL CONTROL VALVE
L3	SOLENOID, HYDRAULIC BYPASS 1
L4	SOLENOID, HYDRAULIC BYPASS 2
L5	SOLENOID, IDLE STOP 1
L6	SOLENOID, IDLE STOP 2
L8	SOLENOID, MAGNETIC BRAKE ANTI TORQUE
L9	SOLENOID, MAGNETIC BRAKE FORWARD AND AFT
L10	SOLENOID, MAGNETIC BRAKE LATERAL
L14	SOLENOID, BLEED AIR VALVE 2
L15	SOLENOID, BLEED AIR VALVE 1
M1	INDICATOR, COPILOT ATTITUDE
M2	INDICATOR, PILOT ATTITUDE
M3	INDICATOR, ENGINE 1 FUEL PRESSURE
M4	INDICATOR, ENGINE 2 FUEL PRESSURE
M5	INDICATOR, ENGINE 1 OIL TEMPERATURE AND PRESSURE
M6	INDICATOR, ENGINE 2 OIL TEMPERATURE AND PRESSURE
M11	INDICATOR, COPILOT TRIPLE TACHOMETER
M12	INDICATOR, PILOT TRIPLE TACHOMETER
M13	INDICATOR, DUAL AMMETER
M15	INDICATOR, COMBINING GEARBOX OIL PRESSURE AND TEMPERATURE
M16	INDICATOR, TRANSMISSION OIL TEMPERATURE AND PRESSURE
M17	INDICATOR, ENGINE 1 GAS PRODUCER TACHOMETER
M18	INDICATOR, ENGINE 2 GAS PRODUCER TACHOMETER
M19	INDICATOR, COPILOT DUAL TORQUE
M20	INDICATOR, PILOT DUAL TORQUE
M21	INDICATOR, DUAL VOLTMETER 1

Table 1. Reference Designators (S/N 33001 through 33100) (continued)

REF DES	NOMENCLATURE
M22	INDICATOR, DUAL VOLTMETER 2
M23	INDICATOR, ENGINE NO. 1 INTERTURBINE TEMPERATURE
M24	INDICATOR, ENGINE NO. 2 INTERTURBINE TEMPERATURE
M25	INDICATOR, FUEL QUANTITY
M26	INDICATOR, HOURMETER
M28	INDICATOR, PILOT STANDBY COMPASS
M29	CLOCK, PILOT
M30	INDICATOR, PILOT AIRSPEED
M31	INDICATOR, PILOT VERTICAL SPEED
M32	ALTIMETER, PILOT
M33	INDICATOR, COPILOT ALTIMETER
M34	INDICATOR, COPILOT RATE OF CLIMB
M35	INDICATOR, COPILOT AIRSPEED
M36	INDICATOR, NO. 1 HYDRAULIC PRESSURE AND TEMPERATURE
M37	INDICATOR, NO. 2 HYDRAULIC PRESSURE AND TEMPERATURE
M40	CLOCK, COPILOT
P1	CONN, COPILOT PITOT HEATER
P2	CONN, PILOT PITOT HEATER
P10	CONN, DC CONTROL PANEL 2
P11	CONN, DC CONTROL PANEL 1
P15	CONN, COPILOT ATTITUDE
P27	CONN, COPILOT ROTOR TACH GEN
P29	CONN, COPILOT DUAL TORQUE
P35	CONN, ROTOR TACH GENERATOR
P36	CONN, TRANSMISSION OIL PRESSURE SWITCH
P37	CONN, TRANSMISSION OIL TEMPERATURE
P38	CONN, MISCELLANEOUS PANEL
P39	CONN, TRANSMISSION OIL PRESSURE
P41	CONN, TRANSMISSION
P42	CONN, CAUTION PANEL
P44	CONN, ENGINE-FUEL PANEL RIGHT
P45	CONN, ENGINE-FUEL PANEL LEFT
P51	CONN, COPILOT WINDSHIELD WASHER
P52	CONN, PILOT WINDSHIELD WASHER
P53	CONN, FIRE DETECT 1 AMPLIFIER
P54	CONN, FIRE DETECT 2 AMPLIFIER
P57	CONN, PYLON LEFT
P58	CONN, PYLON RIGHT
P59	CONN, ENGINE 1 FORWARD COMPARTMENT
P60	CONN, ENGINE 2 FORWARD COMPARTMENT
P61	CONN, ROOF GENERATOR 1
P62	CONN, ROOF GENERATOR 2
P63	CONN, LEFT FIRE DETECT LEFT FWD ENG COMPT

Table 1. Reference Designators (S/N 33001 through 33100) (continued)

REF DES	NOMENCLATURE
P64	CONN, RIGHT FIRE DETECT LEFT FWD ENG COMPT
P67	CONN, LEFT FIRE DETECT RIGHT FWD ENG COMPT
P68	CONN, RIGHT FIRE DETECT RIGHT FWD ENG COMPT
P73	CONN, ENGINE 1 OIL PRESSURE SWITCH
P74	CONN, ENGINE 2 OIL PRESSURE SWITCH
P77	CONN, ENGINE 1 OIL TEMPERATURE
P78	CONN, ENGINE 2 OIL TEMPERATURE
P79	CONN, ENGINE 1 OIL PRESSURE TRANSMITTER
P80	CONN, ENGINE 2 OIL PRESSURE TRANSMITTER
P81	CONN, FUEL PRESSURE TRANSMITTER
P82	CONN, FUEL PRESSURE TRANSMITTER
P83	CONN, LEFT OIL TANK CHIP DETECT
P84	CONN, RIGHT OIL TANK CHIP DETECT
P85	CONN, ENGINE FUEL HEAT 1
P86	CONN, ENGINE FUEL HEAT 2
P87	CONN, FUEL CONTROL VALVE 1
P88	CONN, FUEL CONTROL VALVE 2
P89	CONN, GAS PRODUCER TACH 1
P90	CONN, GAS PRODUCER TACH 2
P91	CONN, IGNITER 1
P92	CONN, IGNITER 2
P93	CONN, PARTICLE SEPARATOR ACTUATOR 1
P94	CONN, PARTICLE SEPARATOR ACTUATOR 2
P95	CONN, CENTER ENGINE COMPARTMENT
P96	CONN, CENTER ENGINE COMPARTMENT
P97	CONN, FIRE DETECT-EXT 1 AFT ENG COMPARTMENT
P98	CONN, FIRE DETECT-EXT 2 AFT ENG COMPARTMENT
P99	CONN, ENGINE 1 AFT COMPARTMENT
P100	CONN, ENGINE 2 AFT COMPARTMENT
P101	CONN, LEFT FIRE DETECT AFT ENG COMPARTMENT
P102	CONN, RIGHT FIRE DETECT AFT ENG COMPARTMENT
P103	CONN, LEFT FIRE DETECT AFT ENG COMPARTMENT
P104	CONN, RIGHT FIRE DETECT AFT ENG COMPARTMENT
P105	CONN, HYDRAULIC 2 TEMPERATURE
P106	CONN, HYDRAULIC 1 TEMPERATURE
P107	CONN, C BOX OIL TEMPERATURE BULB
P109	CONN, TORQUE TRANSMITTER 1
P110	CONN, TORQUE TRANSMITTER 2
P111	CONN, POWER TURBINE TACH 1
P112	CONN, POWER TURBINE TACH 2
P113	CONN, CHIP DETECTOR AFT 1
P114	CONN, CHIP DETECTOR AFT 2
P115	CONN, CHIP DETECTOR CENTER

Table 1. Reference Designators (S/N 33001 through 33100) (continued)

REF DES	NOMENCLATURE
P116	CONN, ANTI COLLISION LIGHT UPPER
P118	CONN, SMOKE DETECTOR
P119	CONN, SMOKE DETECTOR AMPLIFIER
P120	CONN, SMOKE DETECTOR AMPLIFIER
P121	CONN, FUEL SHUTOFF LEFT
P122	CONN, FUEL SHUTOFF RIGHT
P123	CONN, FORWARD FUEL INTERCONNECT VALVE
P124	CONN, AFT FUEL INTERCONNECT VALVE
P125	CONN, FUEL CROSSFEED VALVE
P126	CONN, HYDRAULIC BYPASS SOLENOID VALVE 1
P134A	CONN, FUEL QUANTITY TANK UNIT RIGHT
P135A	CONN, FUEL QUANTITY TANK UNIT RIGHT
P137A	CONN, FUEL QUANTITY TANK UNIT LEFT
P138A	CONN, FUEL QUANTITY TANK UNIT LEFT
P139A	CONN, FUEL QUANTITY TANK UNIT AFT
P140A	CONN, FUEL QUANTITY TANK UNIT AFT
P145A	CONN, FUEL QUANTITY INLINE RIGHT
P146A	CONN, FUEL QUANTITY INLINE RIGHT
P147A	CONN, FUEL QUANTITY INLINE TEE
P147B	CONN, FUEL QUANTITY INLINE TEE
P148A	CONN, FUEL QUANTITY INLINE TEE
P148B	CONN, FUEL QUANTITY INLINE TEE
P149A	CONN, FUEL QUANTITY INLINE TEE
P149B	CONN, FUEL QUANTITY INLINE TEE
P149C	CONN, FUEL QUANTITY INLINE TEE
P162	CONN, C BOX OIL PRESSURE
P165	CONN, COPILOT ENGINE 1 TACH GENERATOR
P166	CONN, COPILOT ENGINE 2 TACH GENERATOR
P167	CONN, DOOR POST ACTUATOR
P171	CONN, HYDRAULIC 2 PRESSURE SOLENOID
P173	CONN, HYDRAULIC 2 FILTER SWITCH
P174	CONN, HYDRAULIC 1 FILTER SWITCH
P175	CONN, IDLE STOP RELAY 1
P176	CONN, IDLE STOP RELAY 2
P177	CONN, BLEED AIR VALVE
P178	CONN, BLEED AIR VALVE
P181	CONN, C BOX OIL PRESSURE TRANSMITTER
P182	CONN, FUEL 1 DIFFERENCE PRESSURE SWITCH
P183	CONN, FUEL 2 DIFFERENCE PRESSURE SWITCH
P184	CONN, C BOX OIL TEMPERATURE SWITCH
P185	CONN, MISCELLANEOUS PANEL
P186	CONN, STANDBY COMPASS
P187	CONN, AFT OUTLET VALVE

Table 1. Reference Designators (S/N 33001 through 33100) (continued)

REF DES	NOMENCLATURE
P189	CONN, COPILOT ALTIMETER
P192	CONN, COPILOT AIRSPEED
P193	CONN, COPILOT RATE OF CLIMB
P196	CONN, COPILOT 5 VOLT POWER SUPPLY
P197	CONN, ENGINE INSTRUMENT 5 VOLT POWER SUPPLY
P198	CONN, PILOT 5 VOLT POWER SUPPLY
P212	CONN, HYDRAULIC 1 TRANSMITTER
P219	CONN, HYDRAULIC 2 TRANSMITTER
P222	CONN, WINTERIZATION VALVE
P223	CONN, HEATER CONTROL VALVE
P229	CONN, HYDRAULIC 2 TEMPERATURE BULB
P230	CONN, HYDRAULIC 1 TEMPERATURE BULB
P238	CONN, COPILOT DUAL TORQUE
P239	CONN, HEATER RELAY
P281	CONN, BATTERY TEMPERATURE SENSOR
P286	CONN, ITT COMPENSATOR 1
P287	CONN, ITT COMPENSATOR 1
P288	CONN, ITT COMPENSATOR 2
P289	CONN, ITT COMPENSATOR 2
P291	CONN, COPILOT MASTER CAUTION LIGHT
P293	CONN, COPILOT RPM CAUTION LIGHT
P294	CONN, ENGINE INSTRUMENT 5 VOLT POWER SUPPLY
P295	CONN, COPILOT CLOCK
P296	CONN, COPILOT STANDBY CMPS
P299	CONN, HYDRAULIC 2 PRESSURE SWITCH
P300	CONN, HYDRAULIC 1 PRESSURE SWITCH
P350	CONN, CABIN TEMPERATURE SERVO
P351	CONN, CABIN TEMPERATURE MOTOR
P3000	CONN, COPILOT THREE AXIS SENSOR
P3001	CONN, COPILOT RATE GYRO
P3002	CONN, PILOT THREE AXIS SENSOR
P3003	CONN, PILOT RATE GYRO
P3043	CONN, PYLON LEFT
P3052	CONN, RPM WARNING ENGINE 2
P3053	CONN, RPM WARNING ENGINE 1
PPS3	POWER SUPPLY, 5 VOLT COPILOT INSTRUMENT
PPS4	POWER SUPPLY, 5 VOLT ENGINE INSTRUMENT
PPS5	POWER SUPPLY, 5 VOLT PILOT INSTRUMENT
PPS6	POWER SUPPLY, 5 VOLT ENGINE INSTRUMENT
Q4	TRANSISTOR, SECONDARY INSTRUMENT LIGHT
Q5	TRANSISTOR, OVERHEAD CONSOLE LIGHT
Q6	TRANSISTOR, PEDESTAL LIGHT
Q7	TRANSISTOR, OVERHEAD CONSOLE LIGHT

Table 1. Reference Designators (S/N 33001 through 33100) (continued)

REF DES	NOMENCLATURE
R9	SHUNT, GENERATOR 1
R10	SHUNT, GENERATOR 2
R57	RESISTOR, TRANSFORMER
R58	RESISTOR, TRANSFORMER
S1	SWITCH, HANDLE NO. 1 EMERGENCY FIRE PULL
S2	SWITCH, HANDLE NO. 2 EMERGENCY FIRE PULL
S3	SWITCH, FIRE EXTINGUISHER
S5	SWITCH, FIRE DETECTOR PRESS TO TEST
S7	SWITCH, FUEL QUANTITY SELECT
S23	SWITCH, START
S26	SWITCH, TRANSMISSION OIL TEMPERATURE
S28	SWITCH, TRANSMISSION OIL PRESSURE
S31	SWITCH, AFT OUTLET LIMIT
S41	SWITCH, FUEL DRAIN VALVE 1
S42	SWITCH, FUEL DRAIN VALVE 2
S65	SWITCH, PILOT WINDSHIELD WIPER CONTROL
S71	SWITCH, LEFT PASSENGER DOOR
S72	SWITCH, RIGHT PASSENGER DOOR
S73	SWITCH, ENGINE 1 OIL PRESSURE
S74	SWITCH, ENGINE 2 OIL PRESSURE
S81	SWITCH, HEATER OVERHEAT
S82	SWITCH, BAGGAGE COMPARTMENT DOOR
S85	SWITCH, BAGGAGE FIRE TEST
S90	SWITCH, C BOX OIL PRESSURE
S94	SWITCH, HYDRAULIC FILTER
S95	SWITCH, HYDRAULIC FILTER
S100	SWITCH, FUEL 1 DIFFERENCE PRESSURE
S101	SWITCH, FUEL 2 DIFFERENCE PRESSURE
S102	SWITCH, C BOX OIL TEMPERATURE
S105	SWITCH, EXTERNAL POWER
S185	SWITCH, NONESSENTIAL BUS SENSOR
S186	SWITCH, NONESSENTIAL BUS SENSOR
S187	SWITCH, OVERHEAT, HEATER AIR LINE
S191	SWITCH, HEATER SENSOR
S206	SWITCH, NO. 1 HYDRAULIC PRESSURE
S207	SWITCH, NO. 2 HYDRAULIC PRESSURE
S209	SWITCH, NO. 1 GENERATOR OVERHEAT
S240	SWITCH, PILOT GYRO (VG FAST) ERECT
S241	SWITCH, COPILOT GYRO (VG FAST) ERECT
T1	TRANSFORMER, CURRENT
T2	TRANSFORMER, CURRENT
T3	TRANSFORMER, CURRENT
T4	TRANSFORMER, CURRENT

Table 1. Reference Designators (S/N 33001 through 33100) (continued)

REF DES	NOMENCLATURE
T5	TRANSFORMER, CURRENT
T6	TRANSFORMER, CURRENT
TB1	TB, NOSE RACK GROUND
TB6	TB, COCKPIT LIGHT RIGHT
TB7	TB, COCKPIT LIGHT LEFT
TB9	TB, AFT ROOF LIGHT
TB10	TB, RIGHT HAND FUEL CELL
TB11	TB, LEFT HAND FUEL CELL
TB12	TB, PEDESTAL LIGHTING
TB14	TB, SECONDARY INSTRUMENT LIGHT
TB15	TB, PEDESTAL LIGHTING
TB27	TB, 5 VOLT LIGHT POWER
TB28	TB, 5 VOLT LIGHT GROUND
TB29	TB, DC-AC GROUND
TB33	TB, RPM WARNING
TB50	TB, ATTITUDE SYSTEMS
TB51	TB, ATTITUDE SYSTEMS
TB52	TB, NONESSENTIAL
TB80	TB, TRANSMISSION CHIP DETECTOR
Z1	DETECTOR, RPM LIMIT
Z4	AMPLIFIER, FIRE DETECTOR, LEFT
Z5	BULB, TRANSMISSION OIL TEMPERATURE
Z6	AMPLIFIER, FIRE DETECTOR, RIGHT
Z7	BULB, ENGINE OIL TEMPERATURE LEFT
Z8	BULB, ENGINE OIL TEMPERATURE RIGHT
Z9	UNIT, IGNITION LEFT
Z10	UNIT, IGNITION RIGHT
Z11	AMPLIFIER, SMOKE DETECTOR
Z12	DETECTOR, BAGGAGE COMPT SMOKE
Z22	UNIT, DC CONTROL 2
Z23	UNIT, DC CONTROL 1
Z24	BULB, GEARBOX OIL TEMPERATURE
Z27	UNIT, ITT COMPENSATOR 1
Z28	UNIT, ITT COMPENSATOR 2
Z52	GYRO, RATE OF TURN
Z54	SENSOR, BATTERY TEMPERATURE
2CB1	CB, EMERGENCY BUS 1
2CB2	CB, EMERGENCY BUS 2
2K3	RELAY, EMERGENCY BUS INTERLOCK
2K4	RELAY, EMERGENCY BUS INTERLOCK
2K5	RELAY, EMERGENCY BUS INTERLOCK
2300DS1	ANNUNCIATOR, RIGHT AFT INT/CALL
2300DS2	ANNUNCIATOR, LEFT AFT INT/CALL

Table 1. Reference Designators (S/N 33001 through 33100) (continued)

REF DES	NOMENCLATURE
3K1	RELAY, AC VOLTAGE SENSOR NO. 1
3K2	RELAY, AC VOLTAGE SENSOR NO. 2
3K3	RELAY, AC BUS CONTROL
3K1P1	CONN, AC VOLTAGE SENSOR NO. 1
3K2P1	CONN, AC VOLTAGE SENSOR NO. 2
3PS1	INVERTER 1
3PS2	INVERTER 2
3PS1P1	CONN, INVERTER 1
3PS2P1	CONN, INVERTER 2
3405DS1	INDICATOR, PILOT HORIZONTAL SITUATION
3405DS2	INDICATOR, COPILOT HORIZONTAL SITUATION
3405DS3	INDICATOR, COURSE SELECT
3405S1	SWITCH, COURSE SET
3405S3	SWITCH, PILOT NAV BEARING POINTER
3405S4	SWITCH, COPILOT NAV BEARING POINTER
3408DS1	INDICATOR, RADAR ALTIMETER
3412CP1	INDICATOR, PILOT MARKER BEACON
3412CP2	INDICATOR, COPILOT MARKER BEACON
3414DS1	INDICATOR, DME MASTER
8DS41	LIGHT, COPILOT INDICATOR
8DS42	LIGHT, PILOT INDICATOR
8K1	RELAY, EMERGENCY FLOAT/CARGO RELEASE DIM
8R15	RESISTOR, PILOT MAP LIGHT
8R16	RESISTOR, COPILOT MAP LIGHT
8S31	SWITCH, FORE AND AFT PROXIMITY
8S32	SWITCH, LATERAL PROXIMITY
9A1P1	CONN, 3 AXIS HELIPILOT COMPUTER 1
9A2P1	CONN, 3 AXIS HELIPILOT COMPUTER 2
9A3P1	CONN, HELIPILOT CONTROLLER
9S3P2	CONN, HELIPILOT CONTROLLER
9A4P1	CONN, TRIM COMPUTER CONNECTOR
9A5P1	CONN, FLIGHT DIRECTOR COMPUTER
9A6	SELECTOR, FLIGHT DIRECTOR MODE
9A7P1	CONN, ALTITUDE SENSOR
9A8P1	CONN, AIRSPEED SENSOR
9A9J1	CONN, ROLL DETENT
9A10J1	CONN, PITCH DETENT
9A11P1	CONN, AFCS INDICATOR ASSEMBLY
9B1J1	CONN, PITCH SERIES ACTUATOR SYSTEM 1
9B2J1	CONN, ROLL SERIES ACTUATOR SYSTEM 1
9B3J1	CONN, YAW SERIES ACTUATOR SYSTEM 1
9B4J1	CONN, PITCH SERIES ACTUATOR SYSTEM 2
9B5J1	CONN, ROLL SERIES ACTUATOR SYSTEM 2

Table 1. Reference Designators (S/N 33001 through 33100) (continued)

REF DES	NOMENCLATURE
9B6P1	CONN, PITCH TRIM ACTUATOR
9B6P2	CONN, PITCH TRIM ACTUATOR
9B7P1	CONN, ROLL TRIM ACTUATOR
9B7P2	CONN, ROLL TRIM ACTUATOR
9DS53	INDICATOR, FT OFF/DCPL
9J2	CONN, RIGHT AFCS
9J3	CONN, RIGHT AFCS
9J4	CONN, LEFT AFCS
9J5	CONN, LEFT AFCS
9K1	RELAY, RELAY
9MT4J1	CONN, PITCH SYSTEM 2 TRANSDUCER
9MT5J1	CONN, ROLL SYSTEM 2 TRANSDUCER
9MT6P1	CONN, COLLECTIVE STICK TRANSDUCER
9P2	CONN, RIGHT AFCS
9P3	CONN, RIGHT AFCS
9P4	CONN, LEFT AFCS
9P5	CONN, LEFT AFCS
9TB1	TB, AFCS SYSTEM 1
9TB2	TB, AFCS SYSTEM 2
9TB3	TB, AFCS ACTUATOR AND TRANSDUCER
9TB4	TB, AFCS FLIGHT DIRECTOR
24J3	CONN, ROTOR TRACK
3405DS2P1	CONN, COPILOT HSI
3407L1	COMPENSATOR, DUAL REMOTE
3407MP1	GYRO, PILOT ATTITUDE
3407MP2	GYRO, COPILOT ATTITUDE
3407MP1P1	CONN, PILOT ATTITUDE GYRO
3407MP2P1	CONN, COPILOT ATTITUDE GYRO

Table 2. Reference Designators (S/N 33101 through 33107)

REF DES	NOMENCLATURE
S23	SWITCH, START
S100	AUXILIARY POWER START SWITCH
TB14	TB, SECONDARY INSTRUMENT LIGHT
1A1	PANEL, ENGINE AND FUEL CONTROL
1A1K1	RELAY, FUEL CROSSFEED
1A1P1	CONN, ENGINE - FUEL PANEL LEFT
1A1P2	CONN, ENGINE - FUEL PANEL RIGHT
1A1S1	SWITCH, ENGINE 1 FUEL
1A1S2	SWITCH, ENGINE 2 FUEL
1A1S3	SWITCH, ENGINE 1 GOV
1A1S4	SWITCH, ENGINE 2 GOV
1A1S5	SWITCH, ENGINE 1 BOOST PUMP

Table 2. Reference Designators (S/N 33101 through 33107) (continued)

REF DES	NOMENCLATURE
1A1S6	SWITCH, ENGINE 2 BOOST PUMP
1A1S7	SWITCH, INTCON
1A1S8	SWITCH, FUEL XFEED
1A1S9	SWITCH, FUEL XFEED TEST
1A2S1	SWITCH, FIRE EXTINGUISHER
1A2S2	SWITCH, FIRE DETECTOR PRESS TO TEST
1A2S5	SWITCH, BAGGAGE FIRE TEST
1A3S1	SWITCH, HANDLE NO. 1 EMERGENCY FIRE PULL
1A4S1	SWITCH, HANDLE NO. 2 EMERGENCY FIRE PULL
1B1	TRANSMITTER, ENGINE 1 OIL PRESSURE
1B1P1	CONN, ENGINE 1 OIL PRESSURE TRANSMITTER
1B2	TRANSMITTER, ENGINE 2 OIL PRESSURE
1B2P1	CONN, ENGINE 2 OIL PRESSURE TRANSMITTER
1B3	SYNCHRO, TRANSMITTER TORQUE, ENGINE 1
1B3P1	CONN, ENGINE 1 TRANSMITTER TORQUE SNYCHRO
1B4	SYNCHRO, TRANSMITTER TORQUE, ENGINE 2
1B4P1	CONN, ENGINE 2 TRANSMITTER TORQUE SYNCHRO
1B5	TRANSMITTER, TRANSMISSION OIL PRESSURE
1B5P1	CONN, TRANSMISSION OIL PRESSURE TRANSMITTER
1B6	TRANSMITTER, C BOX OIL PRESSURE
1B6P1	CONN, C BOX OIL PRESSURE TRANSMITTER
1B7	MOTOR, GOVERNOR RPM ACTUATOR
1B8	TRANSMITTER, FUEL PRESSURE, ENGINE 2
1B8P1	CONN, ENGINE 2 FUEL PRESSURE TRANSMITTER
1B9	TRANSMITTER, FUEL PRESSURE, ENGINE 1
1B9P1	CONN, ENGINE 1 FUEL PRESSURE TRANSMITTER
1B10	MOTOR, FUEL CROSSFEED VALVE
1B10P1	CONN, FUEL CROSSFEED VALVE MOTOR
1B11P1	CONN, FORWARD FUEL INTERCONNECT VALVE MOTOR
1B12	MOTOR, FUEL INTERCONNECT VALVE, AFT
1B12P1	CONN, AFT FUEL INTERCONNECT VALVE MOTOR
1B13	MOTOR, NO. 1 FUEL SHUTOFF VALVE
1B13P1	CONN, NO. 1 FUEL SHUTOFF VALVE MOTOR
1B14	MOTOR, NO. 2 FUEL SHUTOFF VALVE
1B14P1	CONN, NO. 2 FUEL SHUTOFF VALVE MOTOR
1B15	MOTOR, PARTICLE SEPARATOR, ACTUATOR NO. 1
1B15J1	CONN, PARTICLE SEPARATOR ACTUATOR NO. 1 MOTOR
1B15P1	CONN, PARTICLE SEPARATOR ACTUATOR NO. 1 MOTOR
1B16	MOTOR, PARTICLE SEPARATOR, ACTUATOR NO. 2
1B16J1	CONN, PARTICLE SEPARATOR ACTUATOR NO. 2 MOTOR
1B16P1	CONN, PARTICLE SEPARATOR ACTUATOR NO. 2 MOTOR
1CB36	CB, XMSN CHIP POSN IND
1CP1	COUPLER, FUEL QUANTITY

Table 2. Reference Designators (S/N 33101 through 33107) (continued)

REF DES	NOMENCLATURE
1CP2	TEE, FUEL QUANTITY INLINE
1CP2J3	CONN, FUEL QUANTITY INLINE TEE
1CP2P1	CONN, FUEL QUANTITY INLINE TEE
1CP2P2	CONN, FUEL QUANTITY INLINE TEE
1CP2P3	CONN, FUEL QUANTITY INLINE TEE
1CP3	TEE, FUEL QUANTITY INLINE
1CP3J3	CONN, FUEL QUANTITY INLINE TEE
1CP3P1	CONN, FUEL QUANTITY INLINE TEE
1CP3P2	CONN, FUEL QUANTITY INLINE TEE
1CP4	TEE, FUEL QUANTITY INLINE
1CP4J3	CONN, FUEL QUANTITY INLINE TEE
1CP4P1	CONN, FUEL QUANTITY INLINE TEE
1CP4P2	CONN, FUEL QUANTITY INLINE TEE
1CP5	RECEPTACLE, FUEL QUANTITY INLINE
1CP5J2	CONN, FUEL QUANTITY INLINE RECEPTACLE
1CP5P1	CONN, FUEL QUANTITY INLINE RECEPTACLE
1CP6	RECEPTACLE, FUEL QUANTITY INLINE
1CP6J2	CONN, FUEL QUANTITY INLINE RECEPTACLE
1CP6P1	CONN, FUEL QUANTITY INLINE RECEPTACLE
1G1	GENERATOR, POWER TURBINE 1 TACH
1G1P1	CONN, POWER TURBINE 1 TACH GENERATOR
1G2	GENERATOR, POWER TURBINE 2 TACH
1G2P1	CONN, POWER TURBINE 2 TACH GENERATOR
1G3	GENERATOR, GAS PRODUCER 1 TACH
1G3P1	CONN, GAS PRODUCER 1 TACH GENERATOR
1G4	GENERATOR, GAS PRODUCER 2 TACH
1G4P1	CONN, GAS PRODUCER 2 TACH GENERATOR
1G7	GENERATOR, ROTOR TACH
1G7P1	CONN, ROTOR TACH GENERATOR
1HR1	HEATER, FUEL CONTROL NO. 1
1HR1P1	CONN, FUEL CONTROL NO. 1 HEATER
1HR2	HEATER, FUEL CONTROL NO. 2
1HR2P1	CONN, FUEL CONTROL NO. 2 HEATER
1J1	CONN, ENGINE 1 FORWARD COMPARTMENT
1J2	CONN, ENGINE 2 FORWARD COMPARTMENT
1J3	CONN, CENTER ENGINE COMPARTMENT
1J4	CONN, CENTER ENGINE COMPARTMENT
1J5	CONN, ENGINE 1 AFT COMPARTMENT
1J6	CONN, ENGINE 2 AFT COMPARTMENT
1J7	CONN, PYLON LEFT
1J8	CONN, PYLON RIGHT
1J9	CONN, TRANSMISSION
1J11	CONN, FIRE DETECT - EXT 1 AFT ENGINE COMPARTMENT

Table 2. Reference Designators (S/N 33101 through 33107) (continued)

REF DES	NOMENCLATURE
1J12	CONN, FIRE DETECT - EXT 2 AFT ENGINE COMPARTMENT
1K1	RELAY, START CONTROL NO. 1
1K2	RELAY, START CONTROL NO. 2
1K3	RELAY, IDLE STOP DELAY ENG 1
1K3P1	CONN, IDLE STOP DELAY 1 RELAY
1K4	RELAY, IDLE STOP DELAY ENG 2
1K4P1	CONN, IDLE STOP DELAY 2 RELAY
1K5	RELAY, PARTICLE SEPARATOR 1
1K6	RELAY, PARTICLE SEPARATOR 2
1K7	RELAY, HOURMETER
1L1	SOLENOID, FUEL CONTROL VALVE NO. 1
1L1P1	CONN, FUEL CONTROL VALVE NO. 1 SOLENOID
1L2	SOLENOID, FUEL CONTROL VALVE NO. 2
1L2P1	CONN, FUEL CONTROL VALVE NO. 2 SOLENOID
1L5	SOLENOID, IDLE STOP 1
1L6	SOLENOID, IDLE STOP 2
1M1	INDICATOR, ENGINE 1 OIL TEMPERATURE AND PRESSURE
1M2	INDICATOR, ENGINE 2 OIL TEMPERATURE AND PRESSURE
1M3	INDICATOR, ENGINE 1 GAS PRODUCER TACHOMETER
1M4	INDICATOR, ENGINE 2 GAS PRODUCER TACHOMETER
1M5	INDICATOR, TRANSMISSION OIL TEMPERATURE AND PRESSURE
1M6	INDICATOR, COMBINING GEARBOX OIL PRESSURE AND TEMPERATURE
1M7	INDICATOR, FUEL QUANTITY
1M8	INDICATOR, ENGINE 2 FUEL PRESSURE
1M9	INDICATOR, ENGINE 1 FUEL PRESSURE
1M10	INDICATOR, ENGINE 2 INTERTURBINE TEMPERATURE
1M11	INDICATOR, ENGINE 1 INTERTURBINE TEMPERATURE
1M14	HOURMETER
1P1	CONN, ENGINE 1 FORWARD COMPARTMENT
1P2	CONN, ENGINE 2 FORWARD COMPARTMENT
1P3	CONN, CENTER ENGINE COMPARTMENT
1P4	CONN, CENTER ENGINE COMPARTMENT
1P5	CONN, ENGINE 1 AFT COMPARTMENT
1P6	CONN, ENGINE 2 AFT COMPARTMENT
1P7	CONN, PYLON LEFT
1P8	CONN, PYLON RIGHT
1P9	CONN, TRANSMISSION
1P11	CONN, FIRE DETECT - EXT 1 AFT ENG COMPARTMENT
1P12	CONN, FIRE DETECT EXT 2 AFT ENG COMPARTMENT
1S1	SWITCH, SUMP DRAIN VALVE 1
1S2	SWITCH, SUMP DRAIN VALVE 2
1S4	SWITCH, FUEL QUANTITY SELECT
1Z1	BULB, ENGINE 1 OIL TEMPERATURE

Table 2. Reference Designators (S/N 33101 through 33107) (continued)

REF DES	NOMENCLATURE
1Z1P1	CONN, ENGINE 1 OIL TEMPERATURE BULB
1Z2	BULB, ENGINE 2 OIL TEMPERATURE
1Z2P1	CONN, ENGINE 2 OIL TEMPERATURE BULB
1Z3P1	CONN, LEFT FORWARD FUEL CELL TANK UNIT
1Z3P2	CONN, LEFT FORWARD FUEL CELL TANK UNIT
1Z4P1	CONN, RIGHT FORWARD FUEL CELL TANK UNIT
1Z4P2	CONN, RIGHT FORWARD FUEL CELL TANK UNIT
1Z5	BULB, TRANSMISSION OIL TEMPERATURE
1Z5P1	CONN, TRANSMISSION OIL TEMPERATURE BULB
1Z6	BULB, COMBINING GEARBOX OIL TEMPERATURE
1Z6P1	CONN, COMBINING GEARBOX OIL TEMPERATURE
1Z9	UNIT, IGNITION, ENGINE 1
1Z9P1	CONN, ENGINE 1 IGNITION UNIT
1Z10	UNIT, IGNITION, ENGINE 2
1Z10P1	CONN, ENGINE 2 IGNITION UNIT
1Z11	UNIT, ITT COMPENSATOR 1
1Z11P1	CONN, ITT COMPENSATOR 1 UNIT
1Z11P2	CONN, ITT COMPENSATOR 1 UNIT
1Z12	UNIT, ITT COMPENSATOR 2
1Z12P1	CONN, ITT COMPENSATOR 2 UNIT
1Z12P2	CONN, ITT COMPENSATOR 2 UNIT
1Z13P1	CONN, CENTER AFT FUEL CELL TANK UNIT
1Z13P2	CONN, CENTER AFT FUEL CELL TANK UNIT
1Z14TB1	TB, RIGHT FORWARD FUEL CELL
1Z15TB1	TB, LEFT FORWARD FUEL CELL
2BT1	BATTERY
2CB13	CB, 28 V DC AUX RECPT
2CB14	CB, EMERGENCY BUS 2
2CB15	CB, EMERGENCY BUS 1
2CB16	CB, NO. 2 BUS BATTERY SWITCH
2CB17	CB, NO. 1 BUS BATTERY SWITCH
2CR23	DIODE, BATTERY INTERLOCK
2CR24	DIODE, START RELAY 2
2CR25	DIODE, START RELAY 1
2CR28	DIODE, EXTERNAL POWER RELAY
2G1	STARTER-GENERATOR, ENGINE 1
2G2	STARTER-GENERATOR, ENGINE 2
2J1	CONN, EXTERNAL POWER
2J2	CONN, 28 V DC AUX RECPT
2J3	CONN, ROOF GENERATOR 1
2J4	CONN, ROOF GENERATOR 2
2K1	RELAY, EXTERNAL POWER
2K2	RELAY, NONESSENTIAL BUS 2

Table 2. Reference Designators (S/N 33101 through 33107) (continued)

REF DES	NOMENCLATURE
2K3	RELAY, NO. 1 GENERATOR
2K4	RELAY, NO. 2 GENERATOR
2K5	RELAY, NO. 1 START-GEN
2K6	RELAY, NO. 2 START-GEN
2K7	RELAY, NO. 1 BUS/BAT CONTROL
2K8	RELAY, NO. 2 BUS/BAT CONTROL
2K9	RELAY, NONESSENTIAL BUS
2K10	RELAY, NONESSENTIAL BUS
2K11	RELAY, NONESSENTIAL BUS
2K12	RELAY, BATTERY INTERLOCK
2K13	RELAY, EMERGENCY BUS INTERLOCK
2K14	RELAY, EMERGENCY BUS INTERLOCK
2K15	RELAY, EMERGENCY BUS INTERLOCK
2M1	INDICATOR, DUAL AMMETER
2P3	CONN, ROOF GENERATOR 1
2P4	CONN, ROOF GENERATOR 2
2R1	SHUNT, NO. 1 GENERATOR
2R2	SHUNT, NO. 2 GENERATOR
2R7	RESISTOR, TRANSFORMER
2R8	RESSISTOR, TRANSFORMER
2S3	SWITCH, BATTERY BUS 1
2S4	SWITCH, BATTERY BUS 2
2S5	SENSOR, FEEDER POWER
2S6	SENSOR, FEEDER POWER
2T1	XFMR, NO. 1 CURRENT SENSING
2T2	XFMR, NO. 2 CURRENT SENSING
2T3	XFMR, NO. 1 CURRENT SENSING
2T4	XFMR, NO. 2 CURRENT SENSING
2T5	XFMR, NO. 1 CURRENT SENSING
2T6	XFMR, NO. 2 CURRENT SENSING
2TB1	TB, EMERGENCY BUS INTERLOCK
2TB2	TB, NONESSENTIAL
2Z1	PANEL, NO. 1 DC CONTROL
2Z1P1	CONN, DC CONTROL PANEL 1
2Z2	PANEL, NO. 2 DC CONTROL
2Z2P1	CONN, DC CONTROL PANEL 2
3K1	RELAY, AC VOLTAGE SENSOR NO. 1
3K1P1	CONN, AC VOLTAGE SENSOR RELAY NO. 1
3K2	RELAY, AC VOLTAGE SENSOR NO. 2
3K2P1	CONN, AC VOLTAGE SENSOR RELAY NO. 2
3K3	RELAY, AC BUS CONTROL
3PS1	INVERTER 1
3PS1P1	CONN, INVERTER 1

Table 2. Reference Designators (S/N 33101 through 33107) (continued)

REF DES	NOMENCLATURE
3PS2	INVERTER 2
3PS2P1	CONN, INVERTER 2
4A1J1	CONN, PILOT CYCLIC STICK
4A2J1	CONN, PILOT COLLECTIVE
4M2	INDICATOR, DUAL VOLTMETER 2
4M3	INDICATOR, PILOT ALTIMETER
4M4	INDICATOR, PILOT VERTICAL SPEED
4M5	CLOCK, PILOT
4M6	INDICATOR, PILOT STANDBY COMPASS
4M6P1	CONN, PILOT STANDBY COMPASS
4M7	INDICATOR, PILOT TRIPLE TACHOMETER
4M8	INDICATOR, PILOT DUAL TORQUE
4M10	INDICATOR, PILOT ATTITUDE
4M11	INDICATOR, PILOT AIRSPEED
4TB1	TB, RIGHT COCKPIT LIGHT
4Z1	GYRO, PILOT RATE SWITCHING
4Z1P1	CONN, PILOT RATE SWITCHING GYRO
5A1J1	CONN, COPILOT CYCLIC STICK
5A2J1	CONN, COPILOT COLLECTIVE
5M1	INDICATOR, COPILOT AIRSPEED
5M1P1	CONN, COPILOT AIRSPEED INDICATOR
5M2	INDICATOR, DUAL VOLTMETER 1
5M3	INDICATOR, COPILOT ALTIMETER
5M3P1	CONN, COPILOT ALTIMETER
5M4	INDICATOR, COPILOT VERTICAL SPEED
5M4P1	CONN, COPILOT VERTICAL SPEED INDICATOR
5M5P1	CONN, COPILOT CLOCK
5M5	CLOCK, COPILOT
5M6P1	CONN, COPILOT STANDBY COMPASS
5M7	INDICATOR, COPILOT TRIPLE TACHOMETER
5M7P1	CONN, COPILOT TRIPLE TACHOMETER
5M7P2	CONN, COPILOT TRIPLE TACHOMETER
5M7P3	CONN, COPILOT TRIPLE TACHOMETER
5M8	INDICATOR, COPILOT DUAL TORQUE
5M8P1	CONN, COPILOT DUAL TORQUE INDICATOR
5M8P2	CONN, COPILOT DUAL TORQUE INDICATOR
5M9	INDICATOR, COPILOT ATTITUDE
5M9P1	CONN, COPILOT ATTITUDE INDICATOR
5TB1	TB, LEFT COCKPIT LIGHT
5Z1P1	CONN, COPILOT RATE SWITCHING GYRO
8A1	PANEL, MASTER CAUTION
8A1P1	CONN, MASTER CAUTION PANEL
8A1P2	CONN, MASTER CAUTION PANEL

Table 2. Reference Designators (S/N 33101 through 33107) (continued)

REF DES	NOMENCLATURE
8CR11	DIODE, RPM WARNING
8DS1	LIGHT, ENG 1 OUT
8DS2	LIGHT, ENG 2 OUT
8DS3	LIGHT, ANTI-COLLISION, LOWER
8DS4	LIGHT, PILOT MASTER CAUTION
8DS5	LIGHT, COPILOT MASTER CAUTION
8DS5P1	CONN, COPILOT MASTER CAUTION LIGHT
8DS6	LIGHT, PILOT RPM CAUTION
8DS7	LIGHT, COPILOT RPM CAUTION
8DS7P1	CONN, COPILOT RPM CAUTION LIGHT
8DS8	LIGHT, ANTICOLLISION, UPPER
8DS8P1	CONN, UPPER ANTICOLLISION LIGHT
8DS9	LIGHT, COPILOT MAP
8DS10	LIGHT, PILOT MAP
8DS11	LIGHT, OUTSIDE AIR TEMPERATURE
8DS12	LIGHT, SEARCH
8DS13	LIGHT, LANDING
8DS14	LIGHT, RIGHT POSITION, UPPER
8DS15	LIGHT, LEFT POSITION, UPPER
8DS16	LIGHT, RIGHT POSITION, LOWER
8DS17	LIGHT, LEFT POSITION, LOWER
8DS19	LIGHT, BAGGAGE COMPARTMENT
8DS20	LIGHT, BAGGAGE COMPARTMENT
8DS21	LIGHT, BAGGAGE FIRE WARNING
8DS27	LIGHT, LEFT TAIL
8DS28	LIGHT, RIGHT TAIL
8DS29	LIGHT, LEFT UTILITY
8DS30	LIGHT, RIGHT UTILITY
8DS31	LIGHT, LEFT DOME
8DS32	LIGHT, CENTER DOME
8DS33	LIGHT, RIGHT DOME
8DS34	INDICATOR, SUMP CHIP
8DS35	INDICATOR, PLANETARY CHIP
8DS36	INDICATOR, UPPER MAST CHIP
8DS41	LIGHT, COPILOT INDICATOR
8DS42	LIGHT, PILOT INDICATOR
8E1	DETECTOR, TRANSMISSION MAGNETIC CHIP
8E2	DETECTOR, ENGINE GEARBOX MAGNETIC CHIP
8E2P1	CONN, ENGINE GEARBOX MAGNETIC CHIP DETECTOR
8E3	DETECTOR, ENGINE 1 FORWARD MAGNETIC CHIP
8E3P1	CONN, ENGINE 1 FORWARD MAGNETIC CHIP DETECTOR
8E4	DETECTOR, ENGINE 2 FORWARD MAGNETIC CHIP
8E4P1	CONN, ENGINE 2 FORWARD MAGNETIC CHIP DETECTOR

Table 2. Reference Designators (S/N 33101 through 33107) (continued)

REF DES	NOMENCLATURE
8E5	DETECTOR, ENGINE 1 AFT MAGNETIC CHIP
8E5P1	CONN, ENGINE 1 AFT MAGNETIC CHIP DETECTOR
8E6	DETECTOR, ENGINE 2 AFT MAGNETIC CHIP
8E6P1	CONN, ENGINE 2 AFT MAGNETIC CHIP DETECTOR
8E7	DETECTOR, 42° GEARBOX MAGNETIC CHIP
8E8	DETECTOR, 90° GEARBOX MAGNETIC CHIP
8E9	DETECTOR, TRANSMISSION PLANETARY MAGNETIC CHIP
8E10	DETECTOR, TRANSMISSION UPPER MAST MAGNETIC CHIP
8J1	CONN, LEFT FIRE DETECT LEFT FWD ENG COMPT
8J2	CONN, RIGHT FIRE DETECT RIGHT FWD ENG COMPT
8J3	CONN, RIGHT FIRE DETECT LEFT FWD ENG COMPT
8J4	CONN, LEFT FIRE DETECT RIGHT FWD ENG COMPT
8J5	CONN, ANTICOLLISION LIGHT, UPPER
8J6	CONN, TAILBOOM
8J7	CONN, PYLON, LEFT
8K1	RELAY, CAUTION PANEL FAIL
8K2	RELAY, EMERGENCY FLOAT/CARGO RELEASE DIM
8P1	CONN, LEFT FIRE DETECT LEFT FWD ENG COMPT
8P2	CONN, RIGHT FIRE DETECT RIGHT FWD ENG COMPT
8P3	CONN, RIGHT FIRE DETECT LEFT FWD ENG COMPT
8P4	CONN, LEFT FIRE DETECT RIGHT FWD ENG COMPT
8P6	CONN, TAILBOOM
8P7	CONN, PYLON, LEFT
8PS1	POWER SUPPLY, 5 V DC COPILOT INSTR LT
8PS1P1	CONN, 5 V DC COPILOT INSTR LT POWER SUPPLY
8PS2	POWER SUPPLY, 5 V DC ENG INSTR LT
8PS2P1	CONN, 5 V DC ENG INSTR LT POWER SUPPLY
8PS3	POWER SUPPLY, 5 V DC PILOT INSTR LT
8PS3P1	CONN, 5 V DC PILOT INSTR LT POWER SUPPLY
8PS4	POWER SUPPLY, 5 V DC ENG INSTR LT
8PS4P1	CONN, 5 V DC ENGINE INSTR LT POWER SUPPLY
8Q1	TRANSISTOR, 28 V DC CONSOLE LTS
8Q2	TRANSISTOR, 28 V DC CONSOLE LTS
8Q3	TRANSISTOR, 28 V DC PEDESTAL LTS
8Q4	TRANSISTOR, 28 V DC SEC INST LTS
8R20	RHEOESTAT, PILOT MAP LIGHT
8R21	RHEOSTAT, COPILOT MAP LIGHT
8S1	SWITCH, TRANSMISSION OIL TEMPERATURE
8S2	SWITCH, TRANSMISSION OIL PRESSURE
8S2P1	CONN, TRANSMISSION OIL
8S3	SWITCH, LEFT PASSENGER DOOR
8S4	SWITCH, RIGHT PASSENGER DOOR
8S5	SWITCH, ENGINE 1 OIL PRESSURE

Table 2. Reference Designators (S/N 33101 through 33107) (continued)

REF DES	NOMENCLATURE
8S5P1	CONN, ENGINE 1 OIL PRESSURE SWITCH
8S6	SWITCH, ENGINE 2 OIL PRESSURE
8S6P1	CONN, ENGINE 2 OIL PRESSURE SWITCH
8S7	SWITCH, BAGGAGE COMPARTMENT DOOR
8S8	SWITCH, COMBINING GEARBOX OIL PRESSURE
8S8P1	CONN, COMBINING GEARBOX OIL PRESSURE SWITCH
8S9	SWITCH, FUEL 1 DIFFERENTIAL PRESSURE
8S9P1	CONN, FUEL 1 DIFFERENTIAL PRESSURE SWITCH
8S10	SWITCH, FUEL 2 DIFFERENTIAL PRESSURE
8S10P1	CONN, FUEL 2 DIFFERENTIAL PRESSURE SWITCH
8S11	SWITCH, COMBINING GEARBOX OIL TEMPERATURE
8S11P1	CONN, COMBINING GEARBOX OIL TEMPERATURE SWITCH
8S12	SWITCH, EXTERNAL POWER
8S13P1	CONN, NO. 1 HYDRAULIC TEMPERATURE SWITCH
8S14P1	CONN, NO. 2 HYDRAULIC TEMPERATURE SWITCH
8S15	SWITCH, NO. 1 HYDRAULIC PRESSURE
8S15P1	CONN, NO. 1 HYDRAULIC PRESSURE SWITCH
8S16	SWITCH, NO. 2 HYDRAULIC PRESSURE
8S16P1	CONN, NO. 2 HYDRAULIC PRESSURE SWITCH
8S17	SWITCH, NO. 1 GENERATOR OVERHEAT
8S18	SWITCH, NO. 2 GENERATOR OVERHEAT
8S31	SWITCH, LATERAL PROXIMITY
8S32	SWITCH, FORE/AFT PROXIMITY
8TB1	TB, PILOT LIGHTING
8TB2	TB, MISCELLANEOUS LIGHTING
8TB3	TB, COPILOT LIGHTING
8TB4	TB, AFT ROOF LIGHT
8TB8	TB, TRANSMISSION CHIP DETECTOR
8TB10	TB, PROXIMITY SWITCH
8Z1	DETECTOR, RPM LIMIT
8Z1P1	CONN, RPM LIMIT DETECTOR
8Z1P2	CONN, RPM LIMIT DETECTOR
8Z2	DETEC, NO. 2 CONTROL FIRE
8Z2P1	CONN, RIGHT FIRE DETECTOR AMPLIFIER
8Z3	DETEC, NO. 1 CONTROL FIRE
8Z3P1	CONN, LEFT FIRE DETECTOR AMPLIFIER
8Z4	AMPLIFIER, SMOKE DETECTOR CONTROL
8Z4P1	CONN, SMOKE DETECTOR AMPLIFIER
8Z4P2	CONN, SMOKE DETECTOR AMPLIFIER
8Z5	DETECTOR, BAGGAGE COMPT SMOKE
8Z5J1	CONN, BAGGAGE COMPT SMOKE DETECTOR
8ZRP1	CONN, BAGGAGE COMPT SMOKE DETECTOR
8Z8P1	CONN, RIGHT FIRE DETECT AFT ENG COMPARTMENT

Table 2. Reference Designators (S/N 33101 through 33107) (continued)

REF DES	NOMENCLATURE
8Z8P2	CONN, LEFT FIRE DETECT AFT ENG COMPARTMENT
8Z9P1	CONN, RIGHT FIRE DETECT AFT ENG COMPARTMENT
8Z9P2	CONN, LEFT FIRE DETECT AFT ENG COMPARTMENT
8Z10	SENSOR, BATTERY TEMPERATURE
8Z10P1	CONN, BATTERY TEMPERATURE SENSOR
9A1	COMPUTER, 3 AXIS HELIPILOT NO. 1
9A1P1A	CONN, 3 AXIS HELIPILOT COMPUTER NO. 1
9A1P1B	CONN, 3 AXIS HELIPILOT COMPUTER NO. 1
9A2	COMPUTER, 3 AXIS HELIPILOT NO. 2
9A2P1A	CONN, 3 AXIS HELIPILOT COMPUTER NO. 1
9A2P1B	CONN, 3 AXIS HELIPILOT COMPUTER NO. 2
9A3P1	CONN, HELIPILOT CONTROLLER
9A3P2	CONN, HELIPILOT CONTROLLER
9A4	COMPUTER, TRIM
9A4P1	CONN, TRIM COMPUTER
9A5P1A	CONN, FLIGHT DIRECTOR COMPUTER
9A5P1B	CONN, FLIGHT DIRECTOR COMPUTER
9A6	SELECTOR, FLIGHT DIRECTOR MODE
9A7P1	CONN, ALTITUDE SENSOR
9A8P1	CONN, AIRSPEED SENSOR
9A9J1	CONN, ROLL DETENT
9A10J1	CONN, PITCH DETENT
9A11P1	CONN, AFCS INDICATOR ASSEMBLY
9A14	AMPLIFIER, PILOT ISOLATION
9A15	AMPLIFIER, PILOT ISOLATION
9B1J1	CONN, PITCH SERIES ACTUATOR SYSTEM 1
9B2J1	CONN, ROLL SERIES ACTUATOR SYSTEM 1
9B3J1	CONN, YAW SERIES ACTUATOR SYSTEM 1
9B4J1	CONN, PITCH SERIES ACTUATOR SYSTEM 2
9B5J1	CONN, ROLL SERIES ACTUATOR SYSTEM 2
9B6	ACTUATOR, PITCH TRIM
9B6P1	CONN, PITCH TRIM ACTUATOR
9B6P2	CONN, PITCH TRIM ACTUATOR
9B7	ACTUATOR, ROLL TRIM
9B7P1	CONN, ROLL TRIM ACTUATOR
9B7P2	CONN, ROLL TRIM ACTUATOR
9B8P1	CONN, HYDRAULIC PRESSURE TRANSMITTER NO. 2
9B9P1	CONN, HYDRAULIC PRESSURE TRANSMITTER NO. 1
9C1	CAPACITOR, PITCH TRIM
9C2	CAPACITOR, ROLL TRIM
9DS1	INDICATOR, HYDRAULIC FILTER CLOG
9DS4	INDICATOR, FT OFF/DCPL
9J2	CONN, RIGHT AFCS

Table 2. Reference Designators (S/N 33101 through 33107) (continued)

REF DES	NOMENCLATURE
9J3	CONN, RIGHT AFCS
9J4	CONN, LEFT AFCS
9J5	CONN, LEFT AFCS
9K1	RELAY, ANNUN DIMMING
9K2	RELAY, TRIM FAIL ANNUN
9K3	RELAY, CLTV WARNING ANNUN
9K24	RELAY, YAW TRIM RELEASE
9K25	RELAY, COLL TRIM RELEASE
9L1	SOLENOID, HYDRAULIC BYPASS, 1
9L1P1	CONN, HYDRAULIC BYPASS 1 SOLENOID
9L2	SOLENOID, HYDRAULIC BYPASS 2
9L2P1	CONN, HYDRAULIC BYPASS 2 SOLENOID
9L8	SOLENOID, MAGNETIC BRAKE ANTITORQUE
9L8J1	CONN, MAGNETIC BRAKE ANTITORQUE SOLENOID
9L8P1	CONN, MAGNETIC BRAKE ANTITORQUE SOLENOID
9M1	INDICATOR, NO. 1 HYDRAULIC PRESSURE AND TEMPERATURE
9M2	INDICATOR, NO. 2 HYDRAULIC PRESSURE AND TEMPERATURE
9MT1J1	CONN, PITCH SYSTEM 1 TRANSDUCER
9MT2J1	CONN, ROLL SYSTEM 1 TRANSDUCER
9MT3J1	CONN, YAW SYSTEM 1 TRANSDUCER
9MT4J1	CONN, PITCH SYSTEM 2 TRANSDUCER
9MT5J1	CONN, ROLL SYSTEM 2 TRANSDUCER
9MT6P1	CONN, COLLECTIVE STICK TRANSDUCER
9P2	CONN, RIGHT AFCS
9P3	CONN, RIGHT AFCS
9P4	CONN, LEFT AFCS
9P5	CONN, LEFT AFCS
9S1	SWITCH, HYDRAULIC 1 FILTER
9S1P1	CONN, HYDRAULIC 1 FILTER SWITCH
9S2	SWITCH, HYDRAULIC 2 FILTER SWITCH
9S3	SWITCH, PILOT GYRO (VG FAST) ERECT
9S4	SWITCH, COPILOT GYRO (VG FAST) ERECT
9TB1	TB, AFCS SYSTEM 1
9TB2	TB, AFCS SYSTEM 2
9TB3	TB, AFCS ACTUATOR AND TRANSDUCER
9TB4	TB, AFCS FLIGHT DIRECTOR
9TB6	TB, AFCS ACTUATOR AND TRANSDUCER
9Z1P1	CONN, HYDRAULIC 1 TEMPERATURE BULB
9Z2P1	CONN, HYDRAULIC 2 TEMPERATURE BULB
10A1	PANEL, RESISTOR
10A2	PANEL, RESISTOR
10B1	MOTOR, COPILOT WINDSHIELD WIPER
10B1P1	CONN, COPILOT WINDSHIELD WIPER MOTOR

Table 2. Reference Designators (S/N 33101 through 33107) (continued)

REF DES	NOMENCLATURE
10B2	MOTOR, PILOT WINDSHIELD WIPER
10B2P1	CONN, PILOT WINDSHIELD WIPER MOTOR
10B5P1	CONN, CABIN TEMPERATURE MOTOR
10HR1P1	CONN, COPILOT PITOT HEATER
10HR2P1	CONN, PILOT PITOT HEATER PORT
10HR3	HEATER, PILOT LEFT STATIC PORT
10HR4	HEATER, PILOT RIGHT STATIC PORT
10K1	RELAY, HEATER
10K1P1	CONN, HEATER RELAY
10L1P1	CONN, VARIABLE MIXING VALVE
10L2P1	CONN, DOOR POST OUTLET VALVE
10L3	SOLENOID, BLEED AIR VALVE 1
10L3P1	CONN, BLEED AIR VALVE 1 SOLENOID
10L4	SOLENOID, BLEED AIR VALVE 2
10L4P1	CONN, BLEED AIR VALVE 2 SOLENOID
10L5P1	CONN, AFT OUTLET VALVE
10L6P1	CONN, WINTERIZATION KIT VALVE
10S1	SWITCH, AFT OUTLET LIMIT
10S2	SWITCH, FORWARD OVERHEAT
10S3	SWITCH, AFT OVERHEAT
10S4	SWITCH, OVERHEAT, HEATER AIR LINE
10S4P1	CONN, HEATER AIR LINE OVERHEAT SWITCH
10Z1P1	CONN, CABIN TEMPERATURE SERVO
22J1	CONN, LEFT NOSE
22J2	CONN, RIGHT NOSE
22J3	CONN, LEFT NOSE
22J4	CONN, RIGHT NOSE
22J5	CONN, LEFT NOSE
22J6	CONN, RIGHT NOSE
22J7	CONN, LEFT BEAM
22J8	CONN, RIGHT BEAM
22J9	CONN, LEFT BEAM
22J10	CONN, RIGHT BEAM
22J12	CONN, RIGHT WINDSHIELD POST
22J14	CONN, LEFT WINDSHIELD POST
22J16	CONN, RIGHT INSTRUMENT PANEL
22J17	CONN, LEFT INSTRUMENT PANEL
22WT1	MODULE, GROUND
22WT2	MODULE, GROUND
22WT3	MODULE, GROUND
22WT4	MODULE, GROUND
22WT5	MODULE, GROUND
22WT6	MODULE, GROUND

Table 2. Reference Designators (S/N 33101 through 33107) (continued)

REF DES	NOMENCLATURE
22WT7	MODULE, GROUND
22WT8	MODULE, DC-AC GROUND
22WT9	MODULE, DC-AC GROUND
22WT10	MODULE, 5 VOLT LIGHT GROUND
22WT11	MODULE, GROUND
22WT12	MODULE, GROUND
22WT13	MODULE, GROUND
24A1	PANEL, MISCELLANEOUS CONTROL
24A1P1	CONN, MISCELLANEOUS CONTROL PANEL
24A1P2	CONN, MISCELLANEOUS CONTROL PANEL
24A6J1	CONN, BALANCER
24DS1	LIGHT, CARGO RELEASE ARM
24J2	CONN, STEP ACTUATOR
24K1	RELAY, CARGO HOOK RELEASE
24K10	RELAY, STEP LIGHT LATCH
24S10	SWITCH, PILOT WINDSHIELD WIPER CONTROL
24Z2J1	CONN, CARGO SLING
2301DS1	ANNUNCIATOR, COPILOT AFT INT/CALL
2301DS2	ANNUNCIATOR PILOT AFT INT/CALL
3405DS1	INDICATOR, COPILOT HORIZONTAL SITUATION
3405DS1P1	CONN, COPILOT HORIZONTAL SITUATION INDICATOR
3405DS2	INDICATOR, PILOT HORIZONTAL SITUATION
3405DS3	INDICATOR, COURSE SELECT
3405S1	SWITCH, COPILOT NAV BEARING POINTER
3405S2	SWITCH, PILOT NAV BEARING POINTER
3405S3	SWITCH, COURSE SET
3407L1	COMPENSATOR, DUAL REMOTE COMPASS
3407MP1	SENSOR, THREE-AXIS REFERENCE NO. 1 (COPILOT)
3407MP1P1	CONN, THREE-AXIS REFERENCE SENSOR NO. 1
3407MP1P2	CONN, THREE-AXIS REFERENCE SENSOR NO. 1
3407MP2	SENSOR, THREE-AXIS REFERENCE NO. 2 (PILOT)
3407MP2P1	CONN, THREE-AXIS REFERENCE SENSOR NO. 2
3407MP2P2	CONN, THREE-AXIS REFERENCE SENSOR NO. 2
3408DS1	INDICATOR, RADAR ALTIMETER
3412DS1	INDICATOR, COPILOT MARKER BEACON
3412DS2	INDICATOR, PILOT MARKER BEACON
3414DS1	INDICATOR, DME

Table 3. Reference Designators (S/N 33108 through 33213 and 36001 through 36086)

REF DES	NOMENCLATURE
TB14	TB, SECONDARY INSTRUMENT LIGHT
1A1	PANEL, ENGINE AND FUEL CONTROL
1A1K1	RELAY, FUEL CROSSFEED

Table 3. Reference Designators (S/N 33108 through 33213 and 36001 through 36086) (continued)

REF DES	NOMENCLATURE
1A1P1	CONN, ENGINE - FUEL PANEL LEFT
1A1P2	CONN, ENGINE - FUEL PANEL RIGHT
1A1S1	SWITCH, ENGINE 1 FUEL
1A1S2	SWITCH, ENGINE 2 FUEL
1A1S3	SWITCH, ENGINE 1 GOV
1A1S4	SWITCH, ENGINE 2 GOV
1A1S5	SWITCH, ENGINE 1 BOOST PUMP
1A1S6	SWITCH, ENGINE 2 BOOST PUMP
1A1S7	SWITCH, FUEL INTCON
1A1S8	SWITCH, FUEL XFEED
1A1S9	SWITCH, FUEL XFEED/INTCON
1A1S12	SWITCH, ENGINE 1 FUEL TRANS
1A1S13	SWITCH, ENGINE 2 FUEL TRANS
1A2S1	SWITCH, FIRE EXTINGUISHER
1A2S2	SWITCH, FIRE EXTINGUISHER PRESS TO TEST
1A2S5	SWITCH, BAGGAGE FIRE TEST
1A3S1	SWITCH, FIRE 1 PULL
1A4S1	SWITCH, FIRE 2 PULL HANDLE
1A5	SIGNAL CONDITIONER NO. 1
1A5P1	CONN, SIGNAL CONDITIONER NO. 1
1A6	SIGNAL CONDITIONER NO. 2
1A6P1	CONN, SIGNAL CONDITIONER NO. 2
1A7	SIGNAL CONDITIONER, MAST TORQUE
1A7P1	CONN, MAST TORQUE SIGNAL CONDITIONER
1A8	SENSOR, MAST TEMPERATURE
1A8P1	CONN, MAST TEMPERATURE SENSOR
1A9	SENSOR, TRANSMISSION MAST TORQUE
1A9P1	CONN, TRANSMISSION MAST TORQUE SENSOR
1A10	SIGNAL FILTER, TORQUE
1A10P1	CONN, TORQUE SIGNAL FILTER
1B1	TRANSMITTER, ENGINE 1 OIL PRESSURE
1B1P1	CONN, ENGINE 1 OIL PRESSURE TRANSMITTER
1B2	TRANSMITTER, ENGINE 2 OIL PRESSURE
1B2P1	CONN, ENGINE 2 OIL PRESSURE TRANSMITTER
1B3	SYNCHRO, TRANSMITTER TORQUE, ENGINE 1
1B3P1	CONN, ENGINE 1 TRANSMITTER TORQUE SYNCHRO
1B4	SYNCHRO, TRANSMITTER TORQUE, ENGINE 2
1B4P1	CONN, ENGINE 2 TRANSMITTER TORQUE SYNCHRO
1B5	TRANSMITTER, TRANSMISSION OIL PRESSURE
1B5P1	CONN, TRANSMISSION OIL PRESSURE TRANSMITTER
1B6	TRANSMITTER, C BOX OIL PRESSURE
1B6P1	CONN, C BOX OIL PRESSURE TRANSMITTER
1B7	ACTUATOR, GOVERNOR RPM

Table 3. Reference Designators (S/N 33108 through 33213 and 36001 through 36086) (continued)

REF DES	NOMENCLATURE
1B8	TRANSMITTER, FUEL PRESSURE, ENGINE 2
1B8P1	CONN, ENGINE 2 FUEL PRESSURE TRANSMITTER
1B9	TRANSMITTER, FUEL PRESSURE, ENGINE 1
1B9P1	CONN, ENGINE 1 FUEL PRESSURE TRANSMITTER
1B10	VALVE, FUEL CROSSFEED
1B10P1	CONN, FUEL CROSSFEED VALVE
1B12	VALVE, AFT FUEL INTERCONNECT
1B12P1	CONN, AFT FUEL INTERCONNECT VALVE MOTOR
1B13	MOTOR, NO. 1 FUEL SHUTOFF VALVE
1B13P1	CONN, NO. 1 FUEL SHUTOFF VALVE MOTOR
1B14	MOTOR, NO. 2 FUEL SHUTOFF VALVE
1B14P1	CONN, NO. 2 FUEL SHUTOFF VALVE MOTOR
1B15	ACTUATOR NO. 1, PARTICLE SEPARATOR
1B15J1	CONN, PARTICLE SEPARATOR ACTUATOR NO. 1
1B15P1	CONN, PARTICLE SEPARATOR ACTUATOR NO. 1
1B16	ACTUATOR NO. 2 PARTICLE SEPARATOR
1B16J1	CONN, PARTICLE SEPARATOR ACTUATOR NO. 2
1B16P1	CONN, PARTICLE SEPARATOR ACTUATOR NO. 2
1B17	PUMP, LEFT LOWER FORWARD CELL TRANSFER
1B18	PUMP, RIGHT LOWER FORWARD CELL TRANSFER
1B19	PUMP, NO. 1 (LEFT) BOOST
1B20	PUMP, NO. 2 (RIGHT) BOOST
1B22	ACTUATOR, ENGINE 2 RPM GOVERNOR
1CB36	CB, XMSN CHIP POSN IND
1CB50	CB, MAST TORQUE
1CR3	DIODE, IDLE STOP
1CR4	DIODE, IDLE STOP
1G1	GENERATOR, POWER TURBINE 1 TACH
1G1P1	CONN, POWER TURBINE 1 TACH GENERATOR
1G2	GENERATOR, POWER TURBINE 2 TACH
1G2P1	CONN, POWER TURBINE 2 TACH GENERATOR
1G3	GENERATOR, GAS PRODUCER 1 TACH
1G3P1	CONN, GAS PRODUCER 1 TACH GENERATOR
1G4	GENERATOR, GAS PRODUCER 2 TACH
1G4P1	CONN, GAS PRODUCER 2 TACH GENERATOR
1G7	GENERATOR, ROTOR TACH
1G7P1	CONN, ROTOR TACH GENERATOR
1HR1	HEATER, FUEL CONTROL NO. 1
1HR1P1	CONN, FUEL CONTROL NO. 1 HEATER
1HR2	HEATER, FUEL CONTROL NO. 2
1HR2P1	CONN, FUEL CONTROL NO. 2 HEATER
1J1	CONN, ENGINE 1 FORWARD COMPARTMENT
1J2	CONN, ENGINE 2 FORWARD COMPARTMENT

Table 3. Reference Designators (S/N 33108 through 33213 and 36001 through 36086) (continued)

REF DES	NOMENCLATURE
1J3	CONN, CENTER ENGINE COMPARTMENT
1J4	CONN, CENTER ENGINE COMPARTMENT
1J5	CONN, ENGINE 1 AFT COMPARTMENT
1J6	CONN, ENGINE 2 AFT COMPARTMENT
1J7	CONN, PYLON LEFT
1J8	CONN, PYLON RIGHT
1J9	CONN, TRANSMISSION
1J10	CONN, AUXILIARY FUEL
1J11	CONN, FIRE DETECT - EXT 1 AFT ENGINE COMPARTMENT
1J12	CONN, FIRE DETECT - EXT 2 AFT ENGINE COMPARTMENT
1J13	CONN, AUXILIARY FUEL
1K1	RELAY, START CONTROL NO. 1
1K2	RELAY, START CONTROL NO. 2
1K3	RELAY, IDLE STOP DELAY ENG 1
1K3P1	CONN, IDLE STOP DELAY 1 RELAY
1K4	RELAY, IDLE STOP DELAY ENG 2
1K4P1	CONN, IDLE STOP DELAY 2 RELAY
1K5	RELAY, PARTICLE SEPARATOR 1
1K6	RELAY, PARTICLE SEPARATOR 2
1K7	RELAY, HOURMETER
1K8	RELAY, FUEL TRANSFER 2
1K9	RELAY, FUEL TRANSFER 1
1K10	RELAY, INTERCONNECT TEST 2
1K11	RELAY, INTERCONNECT TEST 1
1K12	RELAY, TRANSFER TEST 2
1K13	RELAY, TRANSFER TEST 1
1K14	RELAY, FUEL INTERCONNECT NORMAL
1K15	RELAY, FUEL INTERCONNECT CLOSE
1K16	RELAY, FUEL INTERCONNECT OPEN
1L1	SOLENOID, FUEL CONTROL VALVE NO. 1
1L1P1	CONN, FUEL CONTROL VALVE NO. 1 SOLENOID
1L2	SOLENOID, FUEL CONTROL VALVE NO. 2
1L2P1	CONN, FUEL CONTROL VALVE NO. 2 SOLENOID
1L5	SOLENOID, IDLE STOP 1
1L6	SOLENOID, IDLE STOP 2
1L7	SOLENOID, LOWER LEFT MAIN CELL SUMP DRAIN VALVE
1L8	SOLENOID, LOWER RIGHT MAIN
1A1S7	CELL SUMP DRAIN VALVE
1MT1	TRANSDUCER, ENGINE 1 TORQUE
1MT1P1	CONN, ENGINE 1 TORQUE TRANSDUCER
1MT2	TRANSDUCER, ENGINE 2 TORQUE
1MT2P1	CONN, ENGINE 2 TORQUE TRANSDUCER
1M1	INDICATOR, ENGINE 1 OIL TEMPERATURE AND PRESSURE

Table 3. Reference Designators (S/N 33108 through 33213 and 36001 through 36086) (continued)

REF DES	NOMENCLATURE
1M2	INDICATOR, ENGINE 2 OIL TEMPERATURE AND PRESSURE
1M3	INDICATOR, ENGINE 1 GAS PRODUCER TACHOMETER
1M4	INDICATOR, ENGINE 2 GAS PRODUCER TACHOMETER
1M5	INDICATOR, TRANSMISSION OIL TEMPERATURE AND PRESSURE
1M6	INDICATOR, COMBINING GEARBOX OIL PRESSURE AND TEMPERATURE
1M7	INDICATOR, FUEL QUANTITY
1M8	INDICATOR, ENGINE 2 FUEL PRESSURE
1M9	INDICATOR, ENGINE 1 FUEL PRESSURE
1M10	INDICATOR, ENGINE 2 INTERTURBINE TEMPERATURE
1M11	INDICATOR, ENGINE 1 INTERTURBINE TEMPERATURE
1M14	HOURMETER
1P1	CONN, ENGINE 1 FORWARD COMPARTMENT
1P2	CONN, ENGINE 2 FORWARD COMPARTMENT
1P3	CONN, CENTER ENGINE COMPARTMENT
1P4	CONN, CENTER ENGINE COMPARTMENT
1P5	CONN, ENGINE 1 AFT COMPARTMENT
1P6	CONN, ENGINE 2 AFT COMPARTMENT
1P7	CONN, PYLON LEFT
1P8	CONN, PYLON RIGHT
1P9	CONN, TRANSMISSION
1P11	CONN, FIRE DETECT - EXT 1 AFT ENG COMPARTMENT
1P12	CONN, FIRE DETECT - EXT 2 AFT ENG COMPARTMENT
1S1	SWITCH, SUMP DRAIN VALVE 1
1S2	SWITCH, SUMP DRAIN VALVE 2
1S4	SWITCH, FUEL QUANTITY SELECT
1S5	SWITCH, NO. 1 FUEL PRESSURE
1S6	SWITCH, NO. 2 FUEL PRESSURE
1S7	SWITCH, DIGITS TEST
1TB1	TB, FUEL SYSTEM
1XK7	SOCKET, HOURMETER
1Z1	BULB, ENGINE 1 OIL TEMPERATURE
1Z1P1	CONN, ENGINE 1 OIL TEMPERATURE BULB
1Z2	BULB, ENGINE 2 OIL TEMPERATURE
1Z2P1	CONN, ENGINE 2 OIL TEMPERATURE BULB
1Z5	BULB, TRANSMISSION OIL TEMPERATURE
1Z5P1	CONN, TRANSMISSION OIL TEMPERATURE BULB
1Z6	BULB, COMBINING GEARBOX OIL TEMPERATURE
1Z6P1	CONN, COMBINING GEARBOX OIL TEMPERATURE
1Z9	UNIT, IGNITION, ENGINE 1
1Z9P1	CONN, ENGINE 1 IGNITION UNIT
1Z10	UNIT, IGNITION, ENGINE 2
1Z10P1	CONN, ENGINE 2 IGNITION UNIT
1Z11	UNIT, ITT COMPENSATOR 1

Table 3. Reference Designators (S/N 33108 through 33213 and 36001 through 36086) (continued)

REF DES	NOMENCLATURE
1Z11P1	CONN, ITT COMPENSATOR 1 UNIT
1Z11P2	CONN, ITT COMPENSATOR 1 UNIT
1Z12	UNIT, ITT COMPENSATOR 2
1Z12P1	CONN, ITT COMPENSATOR 2 UNIT
1Z12P2	CONN, ITT COMPENSATOR 2 UNIT
1Z14TB1	TB, RIGHT HAND FUEL CELL
1Z15P1	CONN, LEFT HAND AFT FUEL QUANTITY TANK UNIT
1Z15TB1	TB, LEFT HAND FUEL CELL
1Z16P1	CONN, RIGHT HAND AFT FUEL QUANTITY TANK UNIT
1Z17P1	CONN, LEFT HAND UPPER OUTER FUEL QUANTITY TANK UNIT
1Z18P1	CONN, RIGHT HAND UPPER OUTER FUEL QUANTITY TANK UNIT
2BT1	BATTERY
2BT1P1	BATTERY DISCONNECT
2BT1P2	CONN, BATTERY
2CB13	CB, 28 V DC AUX RECPT
2CB14	CB, EMERGENCY BUS 2
2CB15	CB, EMERGENCY BUS 1
2CB16	CB, NO. 2 BUS BATTERY SWITCH
2CB17	CB, NO. 1 BUS BATTERY SWITCH
2CR23	DIODE, BATTERY INTERLOCK
2CR24	DIODE, START RELAY 2
2CR25	DIODE, START RELAY 1
2CR28	DIODE, EXTERNAL POWER RELAY
2CR33	DIODE, EMERGENCY DC BUS 1
2CR34	DIODE, EMERGENCY DC BUS 2
2G1	STARTER-GENERATOR, ENGINE 1
2G2	STARTER-GENERATOR, ENGINE 2
2J1	RECPT, EXTERNAL POWER
2J2	CONN, 28 V DC AUX RECPT
2J3	CONN, ROOF GENERATOR 1
2J4	CONN, ROOF GENERATOR 2
2K1	RELAY, EXTERNAL POWER
2K2	RELAY, NONESSENTIAL BUS 2
2K3	RELAY, NO. 1 GENERATOR
2K4	RELAY, NO. 2 GENERATOR
2K5	RELAY, NO. 1 START-GEN
2K6	RELAY, NO. 2 START-GEN
2K7	RELAY, NO. 1 BUS/BAT CONTROL
2K8	RELAY, NO. 2 BUS/BAT CONTROL
2K9	RELAY, NONESSENTIAL BUS
2K10	RELAY, NONESSENTIAL BUS
2K11	RELAY, NONESSENTIAL BUS
2K12	RELAY, BATTERY INTERLOCK

Table 3. Reference Designators (S/N 33108 through 33213 and 36001 through 36086) (continued)

REF DES	NOMENCLATURE
2K13	RELAY, EMERGENCY BUS INTERLOCK
2K14	RELAY, EMERGENCY BUS INTERLOCK
2K15	RELAY, EMERGENCY BUS INTERLOCK
2K16	RELAY, NO. 2 EMERGENCY BUS POWER
2K17	RELAY, NO. 1 EMERGENCY BUS POWER
2M1	INDICATOR, DUAL AMMETER
2P3	CONN, ROOF GENERATOR 1
2P4	CONN, ROOF GENERATOR 2
2R1	SHUNT, NO. 1 GENERATOR
2R2	SHUNT, NO. 2 GENERATOR
2R7	RESISTOR, TRANSFORMER
2R8	RESISTOR, TRANSFORMER
2S5	SENSOR, FEEDER POWER
2S6	SENSOR, FEEDER POWER
2S11	AUXILIARY POWER START SWITCH
2T1	XFMR, NO. 1 CURRENT SENSING
2T2	XFMR, NO. 2 CURRENT SENSING
2T3	XFMR, NO. 1 CURRENT SENSING
2T4	XFMR, NO. 2 CURRENT SENSING
2T5	XFMR, NO. 1 CURRENT SENSING
2T6	XFMR, NO. 2 CURRENT SENSING
2TB1	TB, EMERGENCY BUS INTERLOCK
2TB2	TB, NONESSENTIAL
2Z1	PANEL, NO. 1 DC CONTROL
2Z1P1	CONN, DC CONTROL UNIT 1
2Z2	PANEL, NO. 2 DC CONTROL
2Z2P1	CONN, DC CONTROL UNIT 2
3CB11	CB, INVERTER 1 EMERGENCY INTERLOCK
3CB12	CB, INVERTER 1 EMERGENCY INTERLOCK
3K1	RELAY, AC VOLTAGE SENSOR NO. 1
3K1P1	CONN, AC VOLTAGE SENSOR RELAY NO. 1
3K2	RELAY, AC VOLTAGE SENSOR NO. 2
3K2P1	CONN, AC VOLTAGE SENSOR RELAY NO. 2
3K4	RELAY, INVERTER 2 INTERLOCK
3K5	RELAY, INVERTER 1 EMERGENCY BUS
3PS1	INVERTER 1
3PS1P1	CONN, INVERTER 1
3PS2	PRODUCER
3PS2P1	CONN, INVERTER 2
4A1J1	CONN, PILOT CYCLIC STICK
4A2J1	CONN, PILOT COLLECTIVE
4M2	INDICATOR, DUAL VOLTMETER 2
4M3	INDICATOR, PILOT ALTIMETER

Table 3. Reference Designators (S/N 33108 through 33213 and 36001 through 36086) (continued)

REF DES	NOMENCLATURE
4M4	INDICATOR, PILOT VERTICAL SPEED
4M5	CLOCK, PILOT
4M6	INDICATOR, PILOT STANDBY COMPASS
4M6P1	CONN, PILOT STANDBY COMPASS
4M7	INDICATOR, PILOT TRIPLE TACHOMETER
4M8	SN: 33108-33214 or 36001-36019: INDICATOR, PILOT DUAL TORQUE
4M8	SN: 36020-36086 or 412-570-001-103: Installed or BHT-412-SI-74: Installed or BHT-412-SI-97: Installed: *INDICATOR, PILOT TRIPLE TORQUE (412Classic/SP TO 412HP UPGRADE)*
4M8P1	SN: 36020-36086 or 412-570-001-103: Installed or BHT-412-SI-74: Installed or BHT-412-SI-97: Installed: *CONNECTOR, PILOT TRIPLE TORQUE INDICATOR (412Classic/SP TO 412HP UPGRADE)*
4M8P2	SN: 36020-36086 or 412-570-001-103: Installed: CONNECTOR, PILOT TRIPLE TORQUE INDICATOR
4M8P3	SN: 36020-36086 or 412-570-001-103: Installed: CONNECTOR, PILOT TRIPLE TORQUE INDICATOR
4M10	INDICATOR, PILOT ATTITUDE
4M11	INDICATOR, PILOT AIRSPEED
4TB1	TB, RIGHT COCKPIT LIGHT
4Z1	GYRO, PILOT RATE SWITCHING
4Z1P1	CONN, PILOT RATE GYRO
5A1J1	CONN, COPILOT CYCLIC STICK
5A2J1	CONN, COPILOT COLLECTIVE
5M1	INDICATOR, COPILOT AIRSPEED
5M1P1	CONN, COPILOT AIRSPEED INDICATOR
5M2	INDICATOR, DUAL VOLTMETER 1
5M3	INDICATOR, COPILOT ALTIMETER
5M3P1	CONN, COPILOT ALTIMETER
5M4	INDICATOR, COPILOT VERTICAL SPEED
5M4P1	CONN, COPILOT VERTICAL SPEED INDICATOR
5M5	CLOCK, COPILOT
5M6P1	CONN, COPILOT STANDBY COMPASS
5M7	INDICATOR, COPILOT TRIPLE TACHOMETER
5M7P1	CONN, COPILOT TRIPLE TACHOMETER
5M7P2	CONN, COPILOT TRIPLE TACHOMETER
5M7P3	CONN, COPILOT TRIPLE TACHOMETER
5M8	SN: 33108-33214 or 36001-36019: INDICATOR, COPILOT DUAL TORQUE
5M8	SN: 36020-36086 or 412-570-001-103: Installed or BHT-412-SI-74: Installed or BHT-412-SI-97: Installed: *INDICATOR, COPILOT TRIPLE TORQUE (412Classic/SP TO 412HP UPGRADE)*
5M8P1	CONN, COPILOT DUAL TORQUE INDICATOR
5M8P1	SN: 36020-36086 or 412-570-001-103: Installed or BHT-412-SI-74: Installed or BHT-412-SI-97: Installed: *CONNECTOR, COPILOT TRIPLE TORQUE INDICATOR (412Classic/SP TO 412HP UPGRADE)*

Table 3. Reference Designators (S/N 33108 through 33213 and 36001 through 36086) (continued)

REF DES	NOMENCLATURE
5M8P2	CONN, COPILOT DUAL TORQUE INDICATOR
5M8P2	SN: 36020-36086 or 412-570-001-103: Installed or BHT-412-SI-74: Installed or BHT-412-SI-97: Installed: *CONNECTOR, COPILOT TRIPLE TORQUE INDICATOR (412Classic/SP TO 412HP UPGRADE)*
5M8P3	SN: 36020-36086 or 412-570-001-103: Installed or BHT-412-SI-74: Installed or BHT-412-SI-97: Installed: *CONNECTOR, COPILOT TRIPLE TORQUE INDICATOR (412Classic/SP TO 412HP UPGRADE)*
5M9P1	CONN, COPILOT ATTITUDE INDICATOR
5TB1	TB, LEFT COCKPIT LIGHT
5Z1P1	CONN, COPILOT RATE GYRO
5Z18A1	PANEL, MASTER CAUTION
8A1P2	CONN, MASTER CAUTION PANEL
8CR8	DIODE, TRANSMISSION CHIP
8CR9	DIODE, TRANSMISSION CHIP
8CR10	DIODE, TRANSMISSION CHIP
8CR12	DIODE, SEARCHLIGHT
8CR13	DIODE, SEARCHLIGHT
8CR14	DIODE, SEARCHLIGHT
8CR15	DIODE, SEARCHLIGHT
8DS1	LIGHT, ENG 1 OUT
8DS2	LIGHT ENG 2 OUT
8DS3	LIGHT, ANTI-COLLISION, LOWER
8DS4	LIGHT, PILOT MASTER CAUTION
8DS5	LIGHT, COPILOT MASTER CAUTION
8DS5P1	CONN, COPILOT MASTER CAUTION LIGHT
8DS6	LIGHT, PILOT RPM CAUTION
8DS7	LIGHT, COPILOT RPM CAUTION
8DS7P1	CONN, COPILOT RPM CAUTION LIGHT
8DS8	LIGHT, ANTICOLLISION, UPPER
8DS8P1	CONN, UPPER ANTICOLLISION LIGHT
8DS9	LIGHT, COPILOT UTILITY
8DS10	LIGHT, PILOT UTILITY
8DS11	LIGHT, OUTSIDE AIR TEMPERATURE
8DS12	LIGHT, SEARCH
8DS13	LIGHT, LANDING
8DS15	LIGHT, LEFT POSITION, UPPER
8DS16	LIGHT, RIGHT POSITION, LOWER
8DS17	LIGHT, LEFT POSITION, LOWER
8DS19	LIGHT, BAGGAGE COMPARTMENT
8DS20	LIGHT, BAGGAGE COMPARTMENT
8DS21	LIGHT, BAGGAGE FIRE TEST
8DS27	LIGHT, LEFT TAIL
8DS29	LIGHT, LEFT UTILITY
8DS30	LIGHT, RIGHT UTILITY

Table 3. Reference Designators (S/N 33108 through 33213 and 36001 through 36086) (continued)

REF DES	NOMENCLATURE
8DS31	LIGHT, LEFT DOME
8DS32	LIGHT, CENTER DOME
8DS33	LIGHT, RIGHT DOME
8DS34	INDICATOR, SUMP, XMSN CHIP
8DS35	INDICATOR PLANETARY, XMSN CHIP
8DS36	INDICATOR, UPPER MAST, XMSN CHIP
8DS41	LIGHT, COPILOT INDICATOR
8DS42	LIGHT, PILOT INDICATOR
8DS44	LIGHT, PILOT OVERTORQUE WARNING
8DS45	LIGHT, COPILOT OVERTORQUE WARNING
8E1	DETECTOR, TRANSMISSION MAGNETIC CHIP
8E2	DETECTOR, ENGINE GEARBOX MAGNETIC CHIP
8E2P1	CONN, ENGINE GEARBOX MAGNETIC CHIP DETECTOR
8E3	DETECTOR, ENGINE 1 FORWARD MAGNETIC CHIP
8E3P1	CONN, ENGINE 1 FORWARD MAGNETIC CHIP DETECTOR
8E4	DETECTOR, ENGINE 2 FORWARD MAGNETIC CHIP
8E4P1	CONN, ENGINE 2 FORWARD MAGNETIC CHIP DETECTOR
8E5	DETECTOR, ENGINE 1 AFT MAGNETIC CHIP
8E5P1	CONN, ENGINE 1 AFT MAGNETIC CHIP DETECTOR
8E6	DETECTOR, ENGINE 2 AFT MAGNETIC CHIP
8E6P1	CONN, ENGINE 2 AFT MAGNETIC CHIP DETECTOR
8E7	DETECTOR, 42° GEARBOX MAGNETIC CHIP
8E8	DETECTOR, 90° GEARBOX MAGNETIC CHIP
8E9	DETECTOR, TRANSMISSION PLANETARY MAGNETIC CHIP
8E10	DETECTOR, TRANSMISSION UPPER MAST MAGNETIC CHIP
8E11	MONITOR/DETECTOR, TRANSMISSION OIL
8E11P1	CONN, TRANSMISSION OIL MONITOR/DETECTOR
8J1	CONN, LEFT FIRE DETECT LEFT FWD ENG COMPT
8J2	CONN, RIGHT FIRE DETECT RIGHT FWD ENG COMPT
8J3	CONN, RIGHT FIRE DETECT LEFT FWD ENG COMPT
8J4	CONN, LEFT FIRE DETECT RIGHT FWD ENG COMPT
8J5	CONN, ANTICOLLISION LIGHT, UPPER CONN, TAILBOOM
8J7	CONN, PYLON LEFT
8K1	RELAY, CAUTION PANEL FAIL
8K2	RELAY, EMERGENCY FLOAT/CARGO RELEASE DIM
8K3	RELAY, AFCS CAUTION LIGHT
8P1	CONN, LEFT FIRE DETECT LEFT FWD ENG COMPT
8P2	CONN, RIGHT FIRE DETECT RIGHT FWD ENG COMPT
8P3	CONN, RIGHT FIRE DETECT LEFT FWD ENG COMPT
8P4	CONN, LEFT FIRE DETECT RIGHT FWD ENG COMPT
8P7	CONN, PYLON LEFT
8PS1	POWER SUPPLY, 5 V DC COPILOT INSTR LT
8PS1P1	CONN, 5-VOLT COPILOT INSTRUMENT POWER SUPPLY

Table 3. Reference Designators (S/N 33108 through 33213 and 36001 through 36086) (continued)

REF DES	NOMENCLATURE
8PS2	POWER SUPPLY, 5 V DC ENG INSTR LT
8PS2P1	CONN, 5-VOLT ENGINE INSTRUMENT POWER SUPPLY
8PS3	POWER SUPPLY, 5 V DC PILOT INSTR LT
8PS3P1	CONN, 5-VOLT PILOT INSTRUMENT POWER SUPPLY
8PS4	POWER SUPPLY, 5 V DC ENG INSTR LT
8PS4P1	CONN, 5 VOLT ENGINE INSTRUMENT POWER SUPPLY
8Q1	TRANSISTOR, 28 V DC CONSOLE LTS
8Q2	TRANSISTOR, 28 V DC CONSOLE LTS
8Q3	TRANSISTOR, 28 V DC PEDESTAL LTS
8Q4	TRANSISTOR, 28 V DC SEC INST LTS
8R15	RESISTOR, PILOT MAP LIGHT
8R16	RESISTOR, COPILOT MAP LIGHT
8S1	SWITCH, TRANSMISSION OIL TEMPERATURE
8S1P1	CONN, TRANSMISSION OIL TEMPERATURE SWITCH
8S2	SWITCH, TRANSMISSION OIL PRESSURE
8S2P1	CONN, TRANSMISSION OIL PRESSURE SWITCH
8S3	SWITCH, LEFT PASSENGER DOOR
8S4	SWITCH, RIGHT PASSENGER DOOR
8S5	SWITCH, ENGINE 1 OIL PRESSURE
8S5P1	CONN, ENGINE 1 OIL PRESSURE SWITCH
8S6	SWITCH, ENGINE 2 OIL PRESSURE
8S6P1	CONN, ENGINE 2 OIL PRESSURE SWITCH
8S7	SWITCH, BAGGAGE COMPARTMENT DOOR
8S8	SWITCH, COMBINING GEARBOX OIL PRESSURE
8S8P1	CONN, COMBINING GEARBOX OIL PRESSURE SWITCH
8S9	SWITCH, FUEL 1 DIFFERENTIAL PRESSURE
8S9P1	CONN, FUEL 1 DIFFERENTIAL PRESSURE SWITCH
8S10	SWITCH, FUEL 2 DIFFERENTIAL PRESSURE
8S10P1	CONN, FUEL 2 DIFFERENTIAL PRESSURE SWITCH
8S11	SWITCH, COMBINING GEARBOX OIL TEMPERATURE
8S11P1	CONN, COMBINING GEARBOX OIL TEMPERATURE SWITCH
8S12	SWITCH, EXTERNAL POWER
8S13P1	CONN, NO. 1 HYDRAULIC TEMPERATURE SWITCH
8S14P1	CONN, NO. 2 HYDRAULIC TEMPERATURE SWITCH
8S15	SWITCH, NO. 1 HYDRAULIC PRESSURE
8S15P1	CONN, NO. 1 HYDRAULIC PRESSURE SWITCH
8S16	SWITCH, NO. 2 HYDRAULIC PRESSURE
8S16P1	CONN, NO. 2 HYDRAULIC PRESSURE SWITCH
8S17	SWITCH, NO. 1 GENERATOR OVERHEAT
8S31	SWITCH, FORE AND AFT PROXIMITY
8S32	SWITCH, LATERAL PROXIMITY
8TB1	TB, PILOT LIGHTING
8TB2	TB, RPM WARNING

Table 3. Reference Designators (S/N 33108 through 33213 and 36001 through 36086) (continued)

REF DES	NOMENCLATURE
8TB3	TB, 5 VOLT LIGHT POWER
8TB4	TB, AFT ROOF LIGHT
8TB8	TB, TRANSMISSION CHIP DETECTOR
8TB9	TB, FIRE DETECTOR TEST
8TB10	TB, PROXIMITY SWITCH
8XK1	SOCKET, HOURMETER
8Z1	DETECTOR, RPM LIMIT
8Z1P1	CONN, RPM LIMIT DETECTOR
8Z1P2	CONN, RPM LIMIT DETECTOR
8Z2	DETEC, NO. 2 CONTROL FIRE
8Z2P1	CONN, RIGHT FIRE DETECTOR AMPLIFIER
8Z3	DETEC, NO. 1 CONTROL FIRE
8Z3P1	CONN, LEFT FIRE DETECTOR AMPLIFIER
8Z4	AMPLIFIER, SMOKE DETECTOR CONTROL
8Z4P1	CONN, SMOKE DETECTOR AMPLIFIER
8Z4P2	CONN, SMOKE DETECTOR AMPLIFIER
8Z5	DETECTOR, BAGGAGE COMPT SMOKE
8Z5J1	CONN, BAGGAGE COMPT SMOKE DETECTOR
8Z5P1	CONN, BAGGAGE COMPT SMOKE DETECTOR
8Z8P1	CONN, RIGHT FIRE DETECT AFT ENG COMPARTMENT
8Z8P2	CONN, LEFT FIRE DETECT AFT ENG COMPARTMENT
8Z9P1	CONN, RIGHT FIRE DETECT AFT ENG COMPARTMENT
8Z9P2	CONN, LEFT FIRE DETECT AFT ENG COMPARTMENT
8Z10	SENSOR, BATTERY TEMPERATURE
8Z10P1	CONN, BATTERY TEMPERATURE SENSOR
9A1	COMPUTER, 3 AXIS HELIPILOT NO. 1
9A2	COMPUTER, 3 AXIS HELIPILOT NO. 2
9A3P1	CONN, HELIPILOT CONTROLLER
9A3P2	CONN, HELIPILOT CONTROLLER
9A4	COMPUTER, TRIM
9A4P1	CONN, TRIM COMPUTER
9A5P1	CONN, FLIGHT DIRECTOR COMPUTER
9A6	SELECTOR, FLIGHT DIRECTOR MODE
9A7P1	CONN, ALTITUDE SENSOR
9A8P1	CONN, AIRSPEED SENSOR
9A9J1	CONN, ROLL DETENT
9A10J1	CONN, PITCH DETENT
9A11P1	CONN, AFCS INDICATOR ASSEMBLY
9A14	AMPLIFIER, ISOLATION
9A15	AMPLIFIER, ISOLATION
9B1J1	CONN, PITCH SERIES ACTUATOR SYSTEM 1
9B2J1	CONN, ROLL SERIES ACTUATOR SYSTEM 1
9B3J1	CONN, YAW SERIES ACTUATOR SYSTEM 1

Table 3. Reference Designators (S/N 33108 through 33213 and 36001 through 36086) (continued)

REF DES	NOMENCLATURE
9B4J1	CONN, PITCH SERIES ACTUATOR SYSTEM 2
9B5J1	CONN, ROLL SERIES ACTUATOR SYSTEM 2
9B6	ACTUATOR, PITCH TRIM
9B6P1	CONN, PITCH TRIM ACTUATOR
9B6P2	CONN, PITCH TRIM ACTUATOR
9B7	ACTUATOR, ROLL TRIM
9B7P1	CONN, ROLL TRIM ACTUATOR
9B7P2	CONN, ROLL TRIM ACTUATOR
9B8P1	CONN, HYDRAULIC PRESSURE TRANSMITTER NO. 2
9B9P1	CONN, HYDRAULIC PRESSURE TRANSMITTER NO. 1
9C1	CAPACITOR, PITCH TRIM
9C2	CAPACITOR, ROLL TRIM
9DS1	INDICATOR, HYDRAULIC FILTER CLOG
9DS4	INDICATOR, FT OFF/DCPL
9DS53	INDICATOR, FT OFF/DCPL
9J2	CONN, RIGHT AFCS
9J3	CONN, RIGHT AFCS
9J4	CONN, LEFT AFCS
9J5	CONN, LEFT AFCS
9K1	RELAY, HYDRAULIC INTERLOCK
9L1	SOLENOID, HYDRAULIC BYPASS 1
9L1P1	CONN, HYDRAULIC BYPASS 1 SOLENOID
9L2	SOLENOID, HYDRAULIC BYPASS 2
9L2P1	CONN, HYDRAULIC BYPASS 2 SOLENOID
9L8	SOLENOID, MAGNETIC BRAKE ANTITORQUE
9L8J1	CONN, MAGNETIC BRAKE ANTITORQUE SOLENOID
9L8P1	CONN, MAGNETIC BRAKE ANTITORQUE SOLENOID
9M1	INDICATOR, NO. 1 HYDRAULIC PRESSURE AND TEMPERATURE
9M2	INDICATOR, NO. 2 HYDRAULIC PRESSURE AND TEMPERATURE
9MT1J1	CONN, PITCH SYSTEM 1 TRANSFER
9MT2J1	CONN, ROLL SYSTEM 1 TRANSDUCER
9MT3J1	CONN, YAW SYSTEM 1 TRANSDUCER
9MT4J1	CONN, PITCH SYSTEM 2 TRANSDUCER
9MT5J1	CONN, ROLL SYSTEM 2 TRANSDUCER
9MT6P1	CONN, COLLECTIVE STICK TRANSDUCER
9P2	CONN, RIGHT AFCS
9P3	CONN, RIGHT AFCS
9P4	CONN, LEFT AFCS
9P5	CONN, LEFT AFCS
9S1	SWITCH, HYDRAULIC 1 FILTER
9S1P1	CONN, HYDRAULIC 1 FILTER SWITCH
9S2	SWITCH, HYDRAULIC 2 FILTER SWITCH
9S3	SWITCH, PILOT GYRO (VG FAST) ERECT

Table 3. Reference Designators (S/N 33108 through 33213 and 36001 through 36086) (continued)

REF DES	NOMENCLATURE
9S4	SWITCH, COPILOT GYRO (VG FAST) ERECT
9TB1	TB, AFCS SYSTEM 1
9TB2	TB, AFCS SYSTEM 2
9TB3	TB, AFCS ACTUATOR AND TRANSDUCER
9TB4	TB, AFCS FLIGHT DIRECTOR
9TB6	TB, AFCS ACTUATOR AND TRANSDUCER
9Z1P1	CONN, HYDRAULIC 1 TEMPERATURE BULB
9Z2P1	CONN, HYDRAULIC 2 TEMPERATURE BULB
10A1	PANEL, RESISTOR
10A2	PANEL, RESISTOR
10B1	MOTOR, COPILOT WINDSHIELD WIPER
10B1P1	CONN, COPILOT WINDSHIELD WIPER MOTOR
10B2	MOTOR, PILOT WINDSHIELD WIPER
10B2P1	CONN, PILOT WINDSHIELD WIPER MOTOR
10B5P1	CONN, CABIN TEMPERATURE MOTOR
10HR1P1	CONN, COPILOT PITOT HEATER
10HR2P1	CONN, PILOT PITOT HEATER PORT
10K1	RELAY, HEATER
10K1P1	CONN, HEATER RELAY
10L2P1	CONN, DOOR POST OUTLET VALVE
10L3	SOLENOID, BLEED AIR VALVE 1
10L3P1	CONN, BLEED AIR VALVE 1 SOLENOID
10L4	SOLENOID, BLEED AIR VALVE 2
10L4P1	CONN, BLEED AIR VALVE 2 SOLENOID
10L5P1	CONN, AFT OUTLET VALVE
10S1	SWITCH, AFT OUTLET LIMIT
10S2	SWITCH, OVERHEAT, HEATER AIR LINE
10S3	SWITCH, HEATER OVERHEAT
10Z1P1	CONN, CABIN TEMPERATURE CONTROLLER
22J1	CONN, LEFT NOSE
22J2	CONN, RIGHT NOSE
22J3	CONN, LEFT NOSE
22J4	CONN, RIGHT NOSE
22J5	CONN, LEFT NOSE
22J6	CONN, RIGHT NOSE
22J7	CONN, LEFT BEAM
22J8	CONN, RIGHT BEAM
22J9	CONN, LEFT BEAM
22J10	CONN, RIGHT BEAM
22J11	CONN, LEFT DOOR POST
22J12	CONN, RIGHT WINDSHIELD POST
22J13	CONN, LEFT DOOR POST
22J14	CONN, LEFT WINDSHIELD POST

Table 3. Reference Designators (S/N 33108 through 33213 and 36001 through 36086) (continued)

REF DES	NOMENCLATURE
22J15	CONN, LEFT DOOR POST
22J16	CONN, RIGHT INSTRUMENT PANEL
22J17	CONN, LEFT INSTRUMENT PANEL
22J19	CONN, LEFT BEAM
22P8	CONN, RIGHT BEAM
22P10	CONN, RIGHT BEAM
22P16	CONN, RIGHT INSTRUMENT PANEL
22P17	CONN, LEFT INSTRUMENT PANEL
22P18	CONN, RIGHT BEAM
22WT1	MODULE, GROUND
22WT2	MODULE, GROUND
22WT3	MODULE, GROUND
22WT4	MODULE, GROUND
22WT5	MODULE, GROUND
22WT6	MODULE, GROUND
22WT7	MODULE, GROUND
22WT8	MODULE, DC-AC GROUND
22WT9	MODULE, DC-AC GROUND
22WT10	MODULE, 5 VOLT LIGHT GROUND
22WT11	MODULE, GROUND
22WT12	MODULE, GROUND
22WT13	MODULE, GROUND
22WT14	MODULE, GROUND
2301DS1	ANNUNCIATOR, COPILOT AFT INT/CALL/ON
2301DS2	ANNUNCIATOR RIGHT AFT INT/CALL/ON
2337A1	SN: 36037-36086: RECORDER, COCKPIT VOICE
2337A1P1	SN: 36037-36086: CONNECTOR, COCKPIT VOICE RECORDER
2337CB1	SN: 36037-36086: CIRCUIT BREAKER, COCKPIT VOICE
2337CP1	SN: 36037-36086: CONTROL, COCKPIT VOICE RECORDER
2337CP1P1	SN: 36037-36086: CONNECTOR, COCKPIT VOICE RECORDER CONTROL
2337S1P1	SN: 36037-36086: SWITCH, INERTIA
24A1	PANEL, MISCELLANEOUS CONTROL
24A1P1	CONN, MISCELLANEOUS CONTROL PANEL
24A1P2	CONN, MISCELLANEOUS CONTROL PANEL
24A6J1	CONN, BALANCER
24DS1	LIGHT, CARGO RELEASE ARM
24J2	CONN, STEP ACTUATOR
24K1	RELAY, CARGO HOOK RELEASE
24K10	RELAY, EMERGENCY FLOAT ARM

Table 3. Reference Designators (S/N 33108 through 33213 and 36001 through 36086) (continued)

REF DES	NOMENCLATURE
24S10	SWITCH, WIPERS PILOT
27A1	SN: 36037-36086: RECORDER, FLIGHT DATA
27A2	SN: 36037-36086: COMPUTER, AIR DATA
27AR1	SN: 36037-36086: ACQUISITION UNIT, FLIGHT DATA
27A1P1	SN: 36037-36086: CONNECTOR, FLIGHT DATA RECORDER
27A2P1	SN: 36037-36086: CONNECTOR, AIR DATA COMPUTER
27AR1P1A	SN: 36037-36086: CONNECTOR, FLIGHT DATA ACQUISITION UNIT
27AR1P1B	SN: 36037-36086: CONNECTOR, FLIGHT DATA ACQUISITION UNIT
27CB1	SN: 36037-36086: CIRCUIT BREAKER, FLIGHT DATA ACQUISITION UNIT
27CB2	SN: 36037-36086: CIRCUIT BREAKER, FLIGHT DATA RECORDER
27E1	SN: 36037-36086: GAUGE, TEMPERATURE OUTSIDE AIR
27E1P1	SN: 36037-36086: CONNECTOR, OUTSIDE AIR TEMPERATURE
27K1	SN: 36037-36086: RELAY, FLIGHT DATA RECORDER
27XK1	SN: 36037-36086: MOUNT, RELAY
27MT1	SN: 36037-36086: ACCELEROMETER
27MT1P1	SN: 36037-36086: CONNECTOR, ACCELEROMETER
3405DS1	INDICATOR, COPILOT HORIZONTAL SITUATION
3405DS1P1	CONN, COPILOT HORIZONTAL SITUATION INDICATOR
3405DS2	INDICATOR, PILOT HORIZONTAL SITUATION
3405DS3	INDICATOR, COURSE SELECT
3405S1	SWITCH, COPILOT NAV BEARING POINTER
3405S2	SWITCH, PILOT NAV BEARING POINTER
3405S3	SWITCH, COURSE SET
3407L1	COMPENSATOR, DUAL REMOTE COMPASS
3407MP1	SENSOR, THREE-AXIS REFERENCE NO. 1 (COPILOT)
3407MP1P1	CONN, SENSOR, THREE-AXIS REFERENCE NO. 1 (COPILOT)
3407MP1P2	CONN, SENSOR, THREE-AXIS REFERENCE NO. 1 (COPILOT)
3407MP2	SENSOR, THREE-AXIS REFERENCE NO. 2 (PILOT)
3407MP2P1	CONN, SENSOR, THREE-AXIS REFERENCE NO. 2 (PILOT)
3407MP2P2	CONN, SENSOR, THREE-AXIS REFERENCE NO. 2 (PILOT)
3408DS1	INDICATOR, RADAR ALTIMETER
3414DS1	INDICATOR, COPILOT MARKER BEACON
3414DS2	INDICATOR, PILOT MARKER BEACON

Table 4. Reference Designators (S/N 36087 through 36999)

REF DES	NOMENCLATURE
1A1	ENGINE AND FUEL CONTROL PANEL
1A1P1	CONNECTOR
1A1P2	CONNECTOR
1A1K1	FUEL CROSSFEED RELAY
1A1S1	ENGINE 1 FUEL SWITCH
1A1S2	ENGINE 2 FUEL SWITCH
1A1S3	ENGINE 1 GOV SWITCH
1A1S4	ENGINE 2 GOV SWITCH
1A1S5	ENGINE 1 BOOST PUMP SWITCH
1A1S6	ENGINE 2 BOOST PUMP SWITCH
1A1S7	FUEL INTCON SWITCH
1A1S8	FUEL CROSSFEED SWITCH
1A1S9	FUEL INTCON/CROSSFEED SWITCH
1A1S12	ENGINE 1 FUEL TRANSFER SWITCH
1A1S13	ENGINE 1 FUEL TRANSFER SWITCH
1A2	FIRE EXTINGUISHER CONTROL PANEL
1A2S1	FIRE EXTINGUISHER SWITCH
1A2S2	FIRE EXTINGUISHER TEST SWITCH
1A2S5	BAGGAGE FIRE TEST SWITCH
1A3	FIRE 1 PULL HANDLE
1A3DS1	FIRE 1 PULL LIGHT
1A3S1	FIRE 1 PULL SWITCH
1A4S1	FIRE 2 PULL HANDLE
1A4DS1	FIRE 2 PULL LIGHT
1A4S1	FIRE 2 PULL SWITCH
1A5	NO. 1 SIGNAL CONDITIONER
1A5P1	CONNECTOR
1A6	NO. 2 SIGNAL CONDITIONER
1A6P1	CONNECTOR
1A7	MAST TORQUE SIGNAL CONDITIONER
1A7P1	CONNECTOR
1A8	MAST TEMPERATURE SENSOR
1A8P1	CONNECTOR
1A9	MAST TORQUE SENSOR
1A9P1	CONNECTOR
1A10	TORQUE SIGNAL FILTER
1A10P1	CONNECTOR
1A11	FIRE EXTINGUISHER PANEL
1A11J1	CONNECTOR
1A12	FUEL SYSTEM PANEL
1A12J1	CONNECTOR
1B1	ENGINE 1 OIL PRESSURE TRANSMITTER

Table 4. Reference Designators (S/N 36087 through 36999) (continued)

REF DES	NOMENCLATURE
1B1P1	CONNECTOR
1B2	ENGINE 2 OIL PRESSURE TRANSMITTER
1B2P1	CONNECTOR
1B3	ENGINE 1 TRANSMITTER TORQUE SYNCHRO
1B3P1	CONNECTOR
1B4	ENGINE 2 TRANSMITTER TORQUE SYNCHRO
1B4P1	CONNECTOR
1B5	TRANSMISSION OIL PRESSURE TRANSMITTER
1B5P1	CONNECTOR
1B6	C BOX OIL PRESSURE TRANSMITTER
1B6P1	CONNECTOR
1B7	GOVERNOR RPM ACTUATOR
1B8	ENGINE 2 FUEL PRESSURE TRANSMITTER
1B8P1	CONNECTOR
1B9	ENGINE 1 FUEL PRESSURE TRANSMITTER
1B9P1	CONNECTOR
1B10	FUEL CROSSFEED VALVE MOTOR
1B10P1	CONNECTOR
1B12	AFT FUEL INTERCONNECT VALVE
1B12P1	CONNECTOR
1B13	NO. 1 FUEL SHUTOFF VALVE MOTOR
1B13P1	CONNECTOR
1B14	NO. 2 FUEL SHUTOFF VALVE MOTOR
1B14P1	CONNECTOR
1B15	NO. 1 PART SEP ACTUATOR
1B15P1	CONNECTOR
1B16	NO. 2 PART SEP ACTUATOR
1B16P1	CONNECTOR
1B17	LEFT LOWER FORWARD CELL PUMP
1B18	RIGHT LOWER FORWARD CELL PUMP
1B19	NO. 1 BOOST PUMP
1B20	NO. 2 BOOST PUMP
1B22	ENGINE NO. 2 RPM GOVERNOR ACTUATOR
1CB36	TRANSMISSION CHIP DET INDICATOR CB
1CB50	MAST TORQUE
1CR3	IDLE STOP DIODE
1CR4	IDLE STOP DIODE
1G1	NO. 1 POWER TURBINE TACH GENERATOR
1G1P1	CONNECTOR
1G2	NO. 2 POWER TURBINE TACH GENERATOR
1G2P1	CONNECTOR
1G3	NO. 1 GAS PRODUCER TACH GENERATOR
1G3P1	CONNECTOR

Table 4. Reference Designators (S/N 36087 through 36999) (continued)

REF DES	NOMENCLATURE
1G4	NO. 2 GAS PRODUCER TACH GENERATOR
1G4P1	CONNECTOR
1G7	ROTOR TACH GENERATOR
1G7P1	CONNECTOR
1HR1	NO. 1 FUEL CONTROL HEATER
1HR1P1	CONNECTOR
1HR2	NO. 2 FUEL CONTROL HEATER
1HR2P1	CONNECTOR
1J/P1	NO. 1 ENGINE FORWARD DISCONNECT
1J/P2	NO. 2 ENGINE FORWARD DISCONNECT
1J/P3	CENTER ENGINE COMPARTMENT DISCONNECT
1J/P4	CENTER ENGINE COMPARTMENT DISCONNECT
1J/P5	NO. 1 ENGINE AFT DISCONNECT
1J/P6	NO. 2 ENGINE AFT DISCONNECT
1J/P7	LEFT PYLON DISCONNECT
1J/P8	RIGHT PYLON DISCONNECT
1J/P9	TRANSMISSION DISCONNECT
1J/P10	AUX FUEL DISCONNECT
1J/P11	NO. 1 FIRE DET-EXT DISCONNECT
1J/P12	NO. 2 FIRE DET-EXT DISCONNECT
1J/P13	AUX FUEL DISCONNECT
1K1	NO. 1 START CONTROL RELAY
1K2	NO. 2 START CONTROL RELAY
1K3	ENGINE 1 IDLE STOP DELAY RELAY
1K3P1	CONNECTOR
1K4	ENGINE 2 IDLE STOP DELAY RELAY
1K5	NO. 1 PART SEP RELAY
1K6	NO. 2 PART SEP RELAY
1K7	HOURMETER RELAY
1XK7	SOCKET
1K8	NO. 2 FUEL TRANSFER RELAY
1K9	NO. 1 FUEL TRANSFER RELAY
1K10	NO. 2 INTERCONNECT TEST RELAY
1K11	NO. 1 INTERCONNECT TEST RELAY
1K12	NO. 2 TRANSFER TEST RELAY
1K13	NO. 1 TRANSFER TEST RELAY
1K14	NORMAL FUEL INTERCONNECT RELAY
1K15	CLOSE FUEL INTERCONNECT RELAY
1L1	NO. 1 FUEL CONTROL VALVE SOLENOID
1L1P1	CONNECTOR
1L2	NO. 2 FUEL CONTROL VALVE SOLENOID
1L2P1	CONNECTOR
1L5	IDLE STOP 1 SOLENOID

Table 4. Reference Designators (S/N 36087 through 36999) (continued)

REF DES	NOMENCLATURE
1L6	IDLE STOP 2 SOLENOID
1L7	LOWER LEFT MAIN CELL SUMP DRAIN VALVE SOLENOID
1L8	LOWER RIGHT MAIN CELL SUMP DRAIN VALVE SOLENOID
1MT1	ENGINE 1 TORQUE TRANSDUCER
1MT1P1	CONNECTOR
1MT2	ENGINE 2 TORQUE TRANSDUCER
1MT2P1	CONNECTOR
1M1	ENGINE 1 OIL TEMPERATURE AND PRESSURE INDICATOR
1M2	ENGINE 2 OIL TEMPERATURE AND PRESSURE INDICATOR
1M3	ENGINE 1 GAS PRODUCER TACH INDICATOR
1M4	ENGINE 2 GAS PRODUCER TACH INDICATOR
1M5	TRANSMISSION OIL TEMPERATURE AND PRESSURE INDICATOR
1M6	CBOX OIL TEMPERATURE AND PRESSURE INDICATOR
1M7	FUEL QUANTITY INDICATOR
1M8	ENGINE 2 FUEL PRESSURE INDICATOR
1M9	ENGINE 1 FUEL PRESSURE INDICATOR
1M10	ENGINE 2 ITT INDICATOR
1M11	ENGINE 1 ITT INDICATOR
1M14	HOURMETER
1S1	NO. 1 SUMP DRAIN VALVE SWITCH
1S2	NO. 2 SUMP DRAIN VALVE SWITCH
1S4	FUEL QUANTITY SELECT
1S5	NO. 1 FUEL PRESSURE SWITCH
1S6	NO. 2 FUEL PRESSURE SWITCH
1S7	DIGITS TEST SWITCH
1TB1	FUEL SYSTEM TERM BOARD
1Z1	ENGINE 1 OIL TEMPERATURE BULB
1Z1P1	CONNECTOR
1Z2	ENGINE 2 OIL TEMPERATURE BULB
1Z2P1	CONNECTOR
1Z5	TRANSMISSION OIL TEMPERATURE BULB
1Z5P1	CONNECTOR
1Z6	CBOX OIL TEMPERATURE BULB
1Z6P1	CONNECTOR
1Z9	ENGINE 1 IGNITION UNIT
1Z9P1	CONNECTOR
1Z10	ENGINE 2 IGNITION UNIT
1Z10P1	CONNECTOR
1Z11	ENGINE 1 ITT COMPENSATOR UNIT
1Z11P1	CONNECTOR
1Z12	ENGINE 2 ITT COMPENSATOR UNIT
1Z12P1	CONNECTOR
1Z12P2	CONNECTOR

Table 4. Reference Designators (S/N 36087 through 36999) (continued)

REF DES	NOMENCLATURE
1Z14TB1	RIGHT HAND FUEL CELL TERM BOARD
1Z15	LEFT HAND FUEL CELL FUEL QUANTITY UNIT
1Z15P1	CONNECTOR
1Z15TB1	TERM BOARD
1Z16	RIGHT AFT FUEL CELL FUEL QUANTITY UNIT
1Z16P1	CONNECTOR
1Z17	LEFT HAND UPPER OUTER FUEL QUANTITY UNIT
1Z17P1	CONNECTOR
1Z18	RIGHT HAND UPPER OUTER FUEL QUANTITY UNIT
1Z18P1	CONNECTOR
2BT1	BATTERY
2BT1P1	BATTERY DISCONNECT
2BT1P2	SENSOR CONNECTOR
2CB14	EMER BUS 2
2CB15	EMER BUS 1
2CB16	BAT SWITCH BUS NO. 2
2CB17	BAT SWITCH BUS NO. 1
2G1	ENGINE 1 STARTER-GENERATOR
2G2	ENGINE 2 STARTER-GENERATOR
2J1	EXTERNAL POWER RECEPTACLE
2J/P3	GEN 1 ROOF DISCONNECT
2J/P4	GEN 2 ROOF DISCONNECT
2K1	EXT POWER RELAY
2K2	NON ESNTL BUS 2 RELAY
2K5	NO. 1 START-GEN RELAY
2K6	NO. 2 START-GEN RELAY
2K7	NO. 1 BUS/BAT RELAY
2K8	NO. 2 BUS/BAT RELAY
2K9	NON ESNTL BUS 2 RELAY
2K10	NON ESNTL BUS 1 RELAY
2K11	NON ESNTL BUS RELAY
2K12	BATTERY INTLK RELAY
2K13	EMERG BUS INTLK RELAY
2K14	EMERG BUS INTLK RELAY
2K15	EMERG BUS INTLK RELAY
2K16	NO. 2 EMERG BUS POWER RELAY
2K17	NO. 1 EMERG BUS POWER RELAY
2M1	DUAL AMMETER INDICATOR
2R1	NO. 1 GEN SHUNT
2R2	NO. 2 GEN SHUNT
2R7	TRANSFORMER RESISTOR
2R8	TRANSFORMER RESISTOR
2S5	NON ESNTL BUS SENSOR

Table 4. Reference Designators (S/N 36087 through 36999) (continued)

REF DES	NOMENCLATURE
2S6	NON ESNTL BUS SENSOR
2S11	AUX PWR START SWITCH
2T1	NO. 1 CURRENT SENSING TRANSFORMER
2T2	NO. 2 CURRENT SENSING TRANSFORMER
2T3	NO. 1 CURRENT SENSING TRANSFORMER
2T4	NO. 2 CURRENT SENSING TRANSFORMER
2T5	NO. 1 CURRENT SENSING TRANSFORMER
2T6	NO. 2 CURRENT SENSING TRANSFORMER
2TB1	EMERG BUS INTLK TERM BOARD
2TB2	NON ESNTL TERM BOARD
2Z1	NO. 1 DC CONTROL PANEL
2Z1P1	CONNECTOR
2Z2	NO. 2 DC CONTROL PANEL
2Z2P1	CONNECTOR
3K1	INV NO 1 VOLT SENSING RELAY
3K1P1	CONNECTOR
3K2	INV NO 2 VOLT SENSING RELAY
3K2P1	CONNECTOR
3K4	INV 2 INTLK RELAY
3XK4	SOCKET
3K5	INV 1 EMER BUS RELAY
3XK5	SOCKET
3PS1	INV NO 1
3PS1P1	CONNECTOR
3PS2	INV NO 2
3PS2P1	CONNECTOR
4A1	PILOT CYCLIC
4A1J1	CONNECTOR
4A2	PILOT CLTV HEAD
4A2CR2	LNDG LT BLNK DIODE
4A2CR3	LNDG LT BLNK DIODE
4CB11	PILOT ATTD SYS
4M2	DUAL VOLTMETER 2 INDICATOR
4M3	PILOT ALTIMETER
4M3P1	CONNECTOR
4M4	PILOT VERT SPEED INDICATOR
4M5	PILOT CLOCK
4M6	PILOT STANDBY COMPASS
4M6P1	CONNECTOR
4M7	PILOT TRIPLE TACH INDICATOR
4M7P1	CONNECTOR
4M8	PILOT DUAL TORQUE INDICATOR
4M8P1	CONNECTOR

Table 4. Reference Designators (S/N 36087 through 36999) (continued)

REF DES	NOMENCLATURE
4M8P2	CONNECTOR
4M8P3	CONNECTOR
4M9	PILOT ATTITUDE INDICATOR
4M9P1	CONNECTOR
4M11	PILOT HORZ SIT INDICATOR
4TB1	RIGHT COCKPIT LIGHT TERM BOARD
5A1	COPILOT CYCLIC
5A1J1	CONNECTOR
5A2	COPILOT COLLECTIVE
5A2J1	CONNECTOR
5M1	COPILOT AIRSPEED INDICATOR
5M1P1	CONNECTOR
5M2	DUAL VOLTMETER 2 INDICATOR
5M2P1	CONNECTOR
5M3	COPILOT ALTIMETER
5M3P1	CONNECTOR
5M4	COPILOT VERT SPEED INDICATOR
5M4P1	CONNECTOR
5M5	COPILOT CLOCK
5M6	COPILOT STANBY COMPASS
5M6P1	CONNECTOR
5M7	COPILOT TRIPLE TACH INDICATOR
5M7P1	CONNECTOR
5M7P2	CONNECTOR
5M7P3	CONNECTOR
5M8	COPILOT TRIPLE TORQUE INDICATOR
5M8P1	CONNECTOR
5M8P2	CONNECTOR
5M8P3	CONNECTOR
5M9	COPILOT ADI
5M9P1	CONNECTOR
5M12	COPILOT HSI
5M12P1	CONNECTOR
5TB1	LEFT COCKPIT LIGHT TERM BOARD
5Z1	COPILOT RATE GYRO
5Z1P1	CONNECTOR
8A1	MASTER CAUTION PANEL
8A1P1	CONNECTOR
8CR1	DIODE MODULE
8CR2	DIODE MODULE
8CR3	DIODE MODULE
8CR4	DIODE MODULE
8CR5	DIODE MODULE

Table 4. Reference Designators (S/N 36087 through 36999) (continued)

REF DES	NOMENCLATURE
8CR6	DIODE MODULE
8CR8	TRANSMISSION CHIP DIODE
8CR9	TRANSMISSION CHIP DIODE
8CR10	TRANSMISSION CHIP DIODE
8CR22	DIODE
8CR23	DIODE
8CR25	DIODE ASSY
8CR26	DIODE ASSY
8CR27	DIODE ASSY
8CR34	DIODE ASSY
8CR35	DIODE ASSY
8CR36	DIODE ASSY
8DS1	ENGINE 1 OUT LIGHT
8DS2	ENGINE 2 OUT LIGHT
8DS3	LOWER ANTICOLLISION LIGHT
8DS4	PILOT MASTER CAUTION LIGHT
8DS4P1	CONNECTOR
8DS5	COPILOT MASTER CAUTION LIGHT
8DS5P1	CONNECTOR
8DS6	PILOT RPM CAUTION LIGHT
8DS6P1	CONNECTOR
8DS7	COPILOT RPM CAUTION LIGHT
8DS7P1	CONNECTOR
8DS8	UPPER ANTICOLLISION LIGHT
8DS9	COPILOT UTILITY LIGHT
8DS10	PILOT UTILITY LIGHT
8DS11	OUTSIDE AIR TEMPERATURE GAUGE LIGHT
8DS12	SEARCH LIGHT
8DS13	LANDING LIGHT
8DS15	UPPER LEFT POSITION LIGHT
8DS16	LOWER RIGHT POSITION LIGHT
8DS17	LOWER LEFT POSITION LIGHT
8DS18	UPPER RIGHT POSITION LIGHT
8DS19	BAGGAGE COMPARTMENT LIGHT
8DS20	BAGGAGE COMPARTMENT LIGHT
8DS21	BAGGAGE COMPARTMENT FIRE TEST LIGHT
8DS27	LEFT TAIL POSITION LIGHT
8DS28	RIGHT TAIL POSITION LIGHT
8DS31	LEFT DOME LIGHT
8DS32	CENTER DOME LIGHT
8DS33	RIGHT DOME LIGHT
8DS34	TRANSMISSION SUMP CHIP INDICATOR
8DS35	TRANSMISSION PLANETARY CHIP INDICATOR

Table 4. Reference Designators (S/N 36087 through 36999) (continued)

REF DES	NOMENCLATURE
8DS36	TRANSMISSION UPPER MAST CHIP INDICATOR
8DS44	COPILOT OVERTORQUE WARNING LIGHT
8DS45	PILOT OVERTORQUE WARNING LIGHT
8DS51	ANNUN-COPILOT CYCLIC POS
8DS51P1	CONNECTOR
8DS52	ANNUN, PILOT CYCLIC POS
8DS52P1	CONNECTOR
8DS53	ANNUN, COPILOT DCPL
8DS53P1	CONNECTOR
8DS54	ANNUN, PILOT DCPL
8DS54P1	CONNECTOR
8DS55	ANNUN, COPILOT RPM
8DS55P1	CONNECTOR
8DS56	ANNUN, PILOT RPM
8DS56P1	CONNECTOR
8DS57	ANNUN, COPILOT OVER TRQ
8DS57P1	CONNECTOR
8DS58	ANNUN, PILOT OVER TRQ
8DS58P1	CONNECTOR
8E1	TRANSMISSION SUMP MAG CHIP DETECTOR
8E1P1	CONNECTOR
8E2	ENGINE GEARBOX MAG CHIP DETECTOR
8E2P1	CONNECTOR
8E3	ENGINE 1 FWRD MAG CHIP DETECTOR
8E3P1	CONNECTOR
8E4	ENGINE 2 FWRD MAG CHIP DETECTOR
8E4P2	CONNECTOR
8E5	ENGINE 1 AFT MAG CHIP DETECTOR
8E5P1	CONNECTOR
8E6	ENGINE 2 AFT MAG CHIP DETECTOR
8E6P1	CONNECTOR
8E7	42° GEARBOX MAG CHIP DETECTOR
8E8	90° GEARBOX MAG CHIP DETECTOR
8E9	TRANSMISSION PLANETARY GEAR MAG CHIP DETECTOR
8E10	TRANSMISSION UPPER MAST MAG CHIP DETECTOR
8E11	TRANSMISSION OIL MON/DET
8E11P1	CONNECTOR
8J/P1	LEFT FWD ENGINE FIRE DET DISCONNECT
8J/P2	RIGHT FWD ENGINE FIRE DET DISCONNECT
8J/P3	LEFT FWD ENGINE FIRE DET DISCONNECT
8J/P4	RIGHT FWD ENGINE FIRE DET DISCONNECT
8J/P5	UPPER ANTI-COLLISION LIGHT DISCONNECTOR
8J/P7	LEFT PYLON DISCONNECT

Table 4. Reference Designators (S/N 36087 through 36999) (continued)

REF DES	NOMENCLATURE
8K1	CAUT PANEL FAIL RELAY
8XK1	SOCKET
8K2	CARGO RELEASE RELAY
8XK2	SOCKET
8K3	AFCS CAUTION LIGHT RELAY
8K21	ANN LT TEST RELAY
8XK21	SOCKET
8K21	ANN LT TEST RELAY
8XK21	SOCKET
8PS1	COPILOT INSTR PWR SUPPLY
8PS1P1	CONNECTOR
8PS2	ENGINE INSTR PWR SUPPLY
8PS2P1	CONNECTOR
8PS3	PILOT INSTR PWR SUPPLY
8PS3P1	CONNECTOR
8PS4	ENGINE INSTR PWR SUPPLY
8PS4P1	CONNECTOR
8Q1	CONSOLE LT TRANSISTOR
8Q2	CONSOLE LT TRANSISTOR
8Q3	PEDESTAL LT TRANSISTOR
8Q4	INST SECONDARY LT TRANSISTOR
8R15	PILOT MAP LT RESISTOR
8R16	COPILOT MAP LT RESISTOR
8R22	RESISTOR ASSY
8R23	RESISTOR ASSY
8R24	RESISTOR ASSY
8R25	RESISTOR ASSY
8R26	RESISTOR ASSY
8S1	TRANSMISSION OIL TEMPERATURE SWITCH
8S1P1	CONNECTOR
8S2	TRANSMISSION OIL PRESSURE SWITCH
8S2P1	CONNECTOR
8S3	LEFT PASSENGER DOOR SWITCH
8S4	RIGHT PASSENGER DOOR SWITCH
8S5	ENGINE 1 OIL PRESSURE SWITCH
8S5P1	CONNECTOR
8S6	ENGINE 2 OIL PRESSURE SWITCH
8S6P1	CONNECTOR
8S7	BAGGAGE COMPARTMENT DOOR SWITCH
8S8	CBOX OIL PRESSURE SWITCH
8S8P1	CONNECTOR
8S9	NO. 1 FUEL DIFFERENTIAL PRESSURE SWITCH
8S9P1	CONNECTOR

Table 4. Reference Designators (S/N 36087 through 36999) (continued)

REF DES	NOMENCLATURE
8S10	NO. 2 FUEL DIFFERENTIAL PRESSURE SWITCH
8S10P1	CONNECTOR
8S11	CBOX OIL TEMPERATURE SWITCH
8S11P1	CONNECTOR
8S13	NO. 1 HYD TEMPERATURE SWITCH
8S13P1	CONNECTOR
8S14	NO. 2 HYD TEMPERATURE SWITCH
8S14P1	CONNECTOR
8S15	NO. 1 HYD PRESSURE SWITCH
8S15P1	CONNECTOR
8S16	NO. 2 HYD PRESSURE SWITCH
8S16P1	CONNECTOR
8S17	NO. 1 GEN OVERHEAT SWITCH
8S17P1	CONNECTOR
8S18	NO. 2 GEN OVERHEAT SWITCH
8S18P1	CONNECTOR
8S31	FORE AND AFT PROXIMITY SWITCH
8S32	LATERAL PROXIMITY SWITCH
8TB1	PILOT WARNING TERM BOARD
8TB2	RPM WARNING TERM BOARD
8TB3	5 VOLT LT PWR TERM BOARD
8TB4	AFT ROOF TERM BOARD
8TB6	TERM BOARD
8TB7	TERM BOARD
8TB8	TRANSMISSION CHIP DET TERM BOARD
8TB9	FIRE DETECTOR TERM BOARD
8TB10	PROXIMITY SWITCH TERM BOARD
8Z1	RPM LIMIT DETECTOR
8Z1P1	CONNECTOR
8Z2	NO. 2 FIRE DET AMPLIFIER
8Z2P1	CONNECTOR
8Z3	NO. 1 FIRE DET AMPLIFIER
8Z3P1	CONNECTOR
8Z4	SMOKE DET AMPLIFIER
8Z4P1	CONNECTOR
8Z5	BAGGAGE COMPARTMENT SMOKE DETECTOR
8Z5P1	CONNECTOR
8Z5P2	CONNECTOR
8Z6	RIGHT ENGINE FIRE DETECTOR
8Z6P1	CONNECTOR
8Z6P2	CONNECTOR
8Z7	LEFT ENGINE FIRE DETECTOR
8Z7P1	CONNECTOR

Table 4. Reference Designators (S/N 36087 through 36999) (continued)

REF DES	NOMENCLATURE
8Z7P2	CONNECTOR
8Z8	RIGHT ENGINE FIRE DETECTOR
8Z8P1	CONNECTOR
8Z8P2	CONNECTOR
8Z9	LEFT ENGINE FIRE DETECTOR
8Z9P1	CONNECTOR
8Z9P2	CONNECTOR
8Z10	BATTERY TEMPERATURE SENSOR
8Z10P1	CONNECTOR
9A1	DFC COMPUTER NO. 1
9A1P1A	CONNECTOR
9A1P1B	CONNECTOR
9A1P1C	CONNECTOR
9A2	DFC COMPUTER NO. 2
9A2P1A	CONNECTOR
9A2P1B	CONNECTOR
9A2P1C	CONNECTOR
9A3	COPILOT MODE SELECTOR
9A3P1	CONNECTOR
9A4	PILOT MODE SELECTOR
9A4P1	CONNECTOR
9A5	AUTOPILOT CONTROLLER
9A5P1	CONNECTOR
9A5P2	CONNECTOR
9A6	ALTD PRESELECT
9A6P1	CONNECTOR
9A7	NO. 1 AIR DATA SENSOR
9A7P1	CONNECTOR
9A8	NO. 2 AIR DATA SENSOR
9A8P1	CONNECTOR
9A8P2	CONNECTOR
9A9	ROLL DETENT SWITCH
9A9P1	CONNECTOR
9A10	PITCH DETENT SWITCH
9A10P1	CONNECTOR
9A11	AFCS ACTUATOR POS INDICATOR PANEL
9A11P1	CONNECTOR
9A14	PILOT ISOLATION AMPLIFIER
9A14P1	CONNECTOR
9A15	COPILOT ISOLATION AMPLIFIER
9A15P1	CONNECTOR
9A30	AFCS RELAY CONTROL BOX
9A30P1	CONNECTOR